

Dyadic divisor products and rectangular multiplication tables

Tseng

Abstract

For every fixed integer $k \geq 1$, we determine the order of magnitude of $H^{(k+1)}(x, \mathbf{y}, 2\mathbf{y})$, the number of integers $n \leq x$ divisible by a product $d_1 \cdots d_k$ with $y_i < d_i \leq 2y_i$. The estimate is uniform for $3 \leq y_1 \leq \cdots \leq y_k$ and $x \geq 2^k y_1 \cdots y_k y_k$. It yields a uniform formula for the number of distinct products in every rectangular multiplication table with real ordered side lengths $1 \leq N_1 \leq \cdots \leq N_{k+1}$. The formula covers coincident cutoffs, bounded sides, and rectangles in which several saddle constraints are active. Its main exponential term comes from an ordered saddle; the remaining factors measure count windows, divisor overlap, and a terminal bridge. The overlap factors involve fractional moments of conditional Ornstein–Uhlenbeck survival probabilities. We also establish a locally $C^{1,1}$ positive-order persistence spectrum, a strict annealing gap, and a full large-deviation principle for the scalar conditional survival cost.

Contents

1	Introduction	5
1.1	Earlier results and the rectangular obstruction	5
1.2	Divisor overlap and the proof architecture	6
1.3	Notation	7
2	Main results and the harmonic representation	8
2.1	Divisor boxes and the harmonic sum	9
2.2	Size truncation of the harmonic volume sum	9
3	The structural profile	11
3.1	The saddle and the prime ranges	11
3.2	The endpoint costs	12
3.3	Scalar persistence and canonical edge factors	13
3.4	The ordinary terminal suffix	14
3.5	A heuristic for the profile	14
3.6	Reading and evaluating the profile	16
3.7	Two worked profiles	17
4	Geometry of the ordered saddle	19
4.1	The objective and the main conclusions	19
4.2	Concavity and the outer boundaries	20
4.3	Full suffix curvature and uniqueness	22
4.4	The finite optimization and its product bounds	23
4.5	Exact node margins and root centering	25
4.6	Complete unit-step chains: sign and effective time	26
4.7	A hierarchy of plateau lengths	29
4.8	A conditional root count	31

5	Conditional survival and the persistence spectrum	33
5.1	The Gaussian exponent and effective finite-time bounds	33
5.2	Tilted environments and the private endpoint filter	35
5.3	Parameter stability	38
5.4	The positive persistence spectrum	38
5.5	A strict annealing gap	42
5.6	Scalar deviations and the quenched exponent	43
5.7	Integrable walls and Brownian reserves	43
5.8	Independent exponential gaps: relative coupling	45
5.9	Exact Gamma conditioning and relative local regularity	48
5.10	Projection to complete cells	51
5.11	Positive drift: an effective prefix and uniform continuation	55
5.12	Physical Poisson time and weighted insertion	57
6	Conditioned marked bridges	59
6.1	The exact binary bridge	60
6.2	Whole inherited measures and endpoint boxes	63
6.3	Allocating a fixed fraction of the endpoint budget	65
6.4	Ordinary bridges with physical curves	66
6.5	One root count and the ordinary terminal suffix	68
7	Count constraints for the lower bound	71
7.1	Physical data and compatible endpoint pairs	71
7.2	The whole coupled count measure and its mixed moment	73
7.3	One prefix-measurable root with a fixed small endpoint	74
8	Transport of the selected measure	75
8.1	Adaptive targets and exact density	75
8.2	Complete cells and the mass of the selected measure	77
9	Intersections of divisor boxes and the lower bound	78
9.1	A weighted sum over intersection histories	79
9.2	Squarefree supply and the harmonic lower bound	81
10	Discretization at the fixed saddle	82
11	Ordered exponential sums and ordinary root estimates	83
11.1	The tail of the ordinary rectangle	86
12	Ordered covers and the upper bound	88
12.1	Notation and preliminary estimates	88
12.2	Routing capacities and two left covers	90
12.3	Conditional half-line estimates	92
12.4	Lower cutoffs and composition of the covers	95
12.5	One unread ordinary root and the arithmetic conclusion	97
13	Prime clocks and the upper bound	98
13.1	Prime ranges and harmonic clocks	99
13.2	Bounded effective banks	100
13.3	Comparison with the Poisson law	100
13.4	A bounded total cell deficit and the physical rectangle	101
13.5	The ordinary suffix integral	101
13.6	Proof of the upper bound for positive lengths	102
14	Stability and the uniform arithmetic estimates	103
14.1	Stability of the original harmonic sum	103

14.2	Completion at one fixed lift	104
14.3	Classical special cases	104
15	Rectangular multiplication tables	105
16	Further questions	107
A	Saddle and endpoint geometry	108
A.1	Two reference-cost identities	108
A.2	Complete-reference geometry and original endpoint boxes	110
A.3	The optimizer and endpoint costs	137
A.4	Selection of binary constraints and unrestricted intervals	139
B	Conditional probability and bridge estimates	144
B.1	Auxiliary probability estimates	144
B.2	Uniform scalar estimates at final complete types	148
B.3	Coupling of the rank channels	157
B.4	Explicit exact-clock and exact-type middle continuation	161
B.5	Numerical OU endpoint, wall and multiplicativity constants	164
B.6	Numerical flat preparation, relative failures and whole-block independence	168
B.7	Numerical ordinary bridge estimates	172
C	Count types and the selected measure	179
C.1	The joint lower measure and its transport	180
C.2	Exact count-source disintegration and accepted-prefix retention	197
D	Comparison histories and overlap sums	202
D.1	Comparison histories and their geometric costs	202
D.2	Endpoint forests and summability of original pivots	221
D.3	Partial-child, root-separated and weighted source-star comparisons	226
D.4	Recoloring and weighted history costs	234
D.5	Conditional label laws and comparison sums	239
E	Prime grids and harmonic mass	245
E.1	Prime supply, collisions, and the arithmetic grid	245
E.2	Rectangles, root counts, and arithmetic multiplicities	255
E.3	Complete cells, frozen physical optimizer and actual root counts	262
E.4	Numerical harmonic estimates, complete cells and squarefree supply	266
F	Covers and conditional upper estimates	268
F.1	Geometric covers and conditional estimates for the upper bound	268
F.2	Conditional plateau tower, normalized capacities and shot moments	281
G	Scalar large deviations of conditional survival	286
G.1	Pressure and the rate function	286
G.2	The inverse-moment domain	287
G.3	A covariance interpolation for existence	287
G.4	Pressure regularity	289
G.5	Both pressure boundaries	289
G.6	The full LDP bounds	290
G.7	Quenched slope, scalar duality and entropy	291
H	Constants and the length threshold	292
H.1	Coefficient bounds, scalar sensitivity and pivot sums	292
H.2	An explicit length threshold	299
H.3	Harmonic stability and equivalent profiles	304
I	Evaluation of the ordered saddle	305

I.1	The scalar operation	306
I.2	Initial elementary data	306
I.3	The finite recursive rule	307
I.4	Monotonicity of the scalar equations	307
I.5	Exact reconstruction of the ordered maximum	308
I.6	The finite evaluator and the length threshold	309

1 Introduction

The size of a multiplication table measures how much distinct integer factorizations overlap. In a rectangular table, the different side lengths impose constraints at different prime scales. Our aim is to retain all of these constraints and obtain an estimate uniform in the shape of the rectangle, for every fixed number of factors.

For an ordered vector of side lengths, put

$$A_{k+1}(\mathbf{N}) = \#\{n_1 \cdots n_{k+1} : n_i \in \mathbb{N}, n_i \leq N_i\}.$$

The dyadic divisor count associated with this problem is

$$H^{(k+1)}(x, \mathbf{y}, 2\mathbf{y}) = \#\{n \leq x : \exists d_1, \dots, d_k \in \mathbb{N}, y_i < d_i \leq 2y_i, d_1 \cdots d_k \mid n\}.$$

We determine its order of magnitude uniformly throughout

$$3 \leq y_1 \leq \cdots \leq y_k, \quad x \geq 2^k(y_1 \cdots y_k)y_k,$$

for every fixed k . The resulting profile also determines $A_{k+1}(\mathbf{N})$ uniformly for every real ordered vector $1 \leq N_1 \leq \cdots \leq N_{k+1}$. The conclusions are stated in Theorems 2.1 and 2.2. All comparison constants depend only on k ; no separation between adjacent cutoffs or bounds on their ratios are imposed.

1.1 Earlier results and the rectangular obstruction

The multiplication-table problem asks for the number of distinct integers among the products ab , with $1 \leq a, b \leq N$. Erdős [9] proved that this number is $o(N^2)$. Thus the multiplicity of integer factorizations changes the scale of the problem, even though there are N^2 ordered pairs. Ford [12, Corollary 3] determined its order of magnitude:

$$\#\{ab : 1 \leq a, b \leq N\} \asymp \frac{N^2}{(\log N)^\delta (\log \log N)^{3/2}}, \quad \delta = 1 - \frac{1 + \log \log 2}{\log 2}.$$

Erdős [10] had earlier established $N^2(\log N)^{-\delta+o(1)}$ as the scale of the square table. Tenenbaum [26] developed estimates for the more general count $H^{(2)}(x, y, z)$ of integers with a divisor in $(y, z]$. Ford's interval theorem [12, Corollary 2] gives

$$H^{(2)}(x, y, 2y) \asymp \frac{x}{(\log y)^\delta (\log \log y)^{3/2}} \quad (3 \leq y \leq \sqrt{x}).$$

His separate treatment of the dyadic case [13] exposes the divisor-clustering and order-statistics estimates used in the later generalizations. The small exponent δ reflects the balance between the scarcity of integers with sufficiently many prime factors and the overlap among their divisors.

For higher-dimensional balanced tables, Koukoulopoulos [17, Corollary 1] established, for every fixed $k \geq 2$,

$$A_{k+1}(N, \dots, N) \asymp_k \frac{N^{k+1}}{(\log N)^{Q(k/\log(k+1))} (\log \log N)^{3/2}}, \quad Q(u) = u \log u - u + 1,$$

for $N \geq 3$. His subsequent work [19] treats rectangular tables through a comparison with dyadic divisor counts [19, Theorem 1.1]. In conjunction with Ford's dyadic estimate, this comparison

also gives all two-dimensional rectangles with sides at least 3. For $2 \leq k \leq 5$, Theorem 1.3 of [19] gives a uniform estimate for arbitrary ordered cutoffs and hence tables with three through six factors. For $k \geq 6$, Theorem 1.4 gives a matching estimate when $\log y_k \leq (\log y_1)^{1+\varepsilon_k}$ for a suitable $\varepsilon_k > 0$. Theorem 1.5 gives an upper bound for every fixed $k \geq 2$, and a matching lower bound under a restriction on the saddle parameter.¹ His analysis of the additional constraints [19, Section 2.2, Lemma 2.3 and Remark 2.4] explains why the single-parameter expression can overestimate some higher-dimensional rectangles. The remaining unrestricted dimensions are also discussed in [18, Chapter 6, pp. 87–88]. The ordered saddle below retains distinct constraints on successive prime ranges.

Related work makes the subset-sum structure of divisor overlap particularly transparent. Eberhard, Ford and Green [8, Theorem 1.1] proved that the probability that a uniformly random permutation in S_n has an invariant j -element set is $\asymp j^{-\delta}(1 + \log j)^{-3/2}$, uniformly for $1 \leq j \leq n/2$. Ford, Green and Koukoulopoulos [15] developed higher-dimensional optimization methods for equal subset sums and the concentration of divisors. More recently, Green and Sawhney [16, Theorem 1.1] proved an asymptotic for the limiting invariant-set probability, with a smooth positive periodic factor in $\log_2 j$. They conjecture that this factor is nonconstant. Sawhney’s report of joint work with Green [25] announces an integer square-table asymptotic with a periodic amplitude; the permutation preprint describes the integer treatment as forthcoming. These refinements are distinct from the uniform comparison estimates sought here. In a different regime, Sabuncu and Táfula [24, Theorem 1.2] determine the asymptotic leading coefficient of the eventual polynomial in the dimension for tables with a common side length. Their result applies as the side length tends to infinity and the dimension is sufficiently large in terms of that side length. Section 16 discusses the extension to variable-dimensional rectangles.

1.2 Divisor overlap and the proof architecture

The arithmetic starting point is Koukoulopoulos’s reduction [19, Theorem 1.7] to a harmonic sum of volumes of divisor-box unions. A prime in $(y_{i-1}, y_i]$ can be used in divisor coordinate i or any later coordinate, or left unused. An assignment of all primes to these categories is a routing word. Its divisor vector, on the logarithmic scale, determines a box of fixed side length. The volume of the union counts the space covered by these words, with overlaps counted only once. Estimating that volume is the geometric form of the original counting problem.

Two effects determine its size. The prime counts must be large enough to provide sufficiently many routing words, but the words can cluster if too many small primes accumulate before a coordinate’s cutoff. The saddle balances the cost of the counts against the available volume. For rectangles, this balance has to respect the order of the coordinates. Consecutive equal saddle coordinates form a plateau; a strict drop between plateaux gives a separate count-window cost. Within a plateau, comparisons between words give path constraints whose endpoint reserves are calculated from a finite list of legal contexts. The ordered saddle and these endpoint costs keep track of the constraints that a single saddle parameter can lose.

The resulting profile has four parts. The saddle gives the main exponential term. Strict drops give count-window factors. The path constraints within each plateau give binary persistence factors. A final ordinary bridge and its exact count atom give the terminal factor. Section 3 defines each part and combines them in (3.12). In dimension one there are no binary factors: the

¹In the notation of [19], let $h_i = k - i + 2$ and $\ell_i = \log(3 \log y_i / \log y_{i-1})$, with $y_0 = 3$. The parameter α is determined by $\sum_i h_i^\alpha (\log h_i) \ell_i = \sum_i (h_i - 1) \ell_i$. For fixed $\varepsilon > 0$, the matching lower bound in Theorem 1.5 requires $\alpha \geq 1 + \varepsilon - (\log(k+1))^{-1} \log(((k+1) \log(k+1) - 2 \log 2) / (k-1))$; its constants may depend on k and ε .

saddle and the ordinary factor recover Ford’s dyadic estimate. The balanced case is recovered in Section 14.3. Two worked profiles in Section 3.7 show how the endpoint and suffix rules change the polynomial factors, including a family with a strict saddle separation.

The binary factors require a conditional probability. Prime positions are common to all routing words, whereas their labels vary from word to word. We therefore fix the positions, sum the admissible label mass, take its fractional power, and only then average over positions. Averaging the labels and positions together at the outset would lose the dependence relevant to divisor overlap. This distinction leads to the exponent $\chi(\eta, \rho)$, defined by fractional moments of conditional Ornstein–Uhlenbeck survival. The use of fractional moments builds on [17, Section 1]; the separation of the common positions from the private labels is what determines the exponent needed here. Section 3.5 explains how a count model predicts the saddle cost, where that prediction loses projected volume, and why a conditional bridge produces the endpoint factor with exponent 2χ .

The lower and upper bounds use this profile in different ways. For the lower bound, we construct a measure on complete count types, including the exact terminal count, and then impose the path constraints conditional on each type. Spatial clocks are independent at this stage. Transport of the label measure places the surviving words at compatible endpoints; a fractional bound for their intersections converts their mass into a lower bound for the divisor-box volume. For the upper bound, ordered covers successively restrict the available divisor coordinates. Their estimates preserve the plateau powers and leave one terminal count unread. Its atom and path constraint are integrated jointly, producing the same ordinary factor as in the lower bound.

Here is the order of the argument. Section 2 states the arithmetic theorems and the harmonic representation. Section 3 defines the profile, and Section 4 proves the saddle and endpoint properties on which it rests. Section 5 develops conditional survival estimates; Section 6 transfers them to marked bridges under exact conditioning. Sections 7–9 construct the count measure, transport it, and prove the volume lower bound. The deterministic and arithmetic upper bounds culminate in Section 13. Harmonic stability in Section 14 then extends both estimates to coincident cutoffs, and Section 15 proves the rectangular theorem, including bounded sides.

The persistence theory also has consequences beyond the arithmetic estimate. Mallein and Miloš [20] established almost-sure Brownian and Ornstein–Uhlenbeck survival exponents above an independent random wall. For the fractional-moment exponent used here, Section 5 proves uniform finite-time comparisons, a positive-order spectrum with local $C^{1,1}$ regularity, and a strict annealing gap. Appendix G gives the scalar large-deviation theory and identifies the almost-sure exponent with the derivative at order zero. That appendix is independent of the arithmetic assembly.

The remaining appendices supply the detailed endpoint geometry, count and intersection estimates, prime discretization, and numerical choices of constants. Each begins with the role of its estimates in the main proof. Numbering in citations to [19] refers throughout to arXiv:1102.3236v3; numbering for [17] refers to the author version linked in the bibliography. Where the published and revised versions of [13] differ, the version used is specified at the citation.

1.3 Notation

All logarithms are natural, and $\mathbb{N} = \{1, 2, \dots\}$. The dimension k is fixed. We write $f \ll_k g$ if $f \leq C_k g$ for a constant depending only on k , and $f \asymp_k g$ if both inequalities hold. Empty products equal 1, and a minimum over an empty list is $+\infty$. We put $(u)_+ = \max(u, 0)$ and use Lebesgue measure for volumes. Probability laws and conditioning fields are specified where they enter the proof. Within a section, or a numbered appendix subsection, an equation reference may

omit its prefix.

2 Main results and the harmonic representation

We first state the two arithmetic conclusions. Their common profile is specified by the finite saddle and endpoint rules in Section 3; the universal scalar exponent used there is defined probabilistically and has effective finite-time approximations. The proofs estimate one harmonic volume sum, which is introduced after the theorem statements.

For the dyadic count, let

$$H^{(k+1)}(x, \mathbf{y}, 2\mathbf{y}) = \#\{n \in \mathbb{N} : n \leq x, \exists d_1, \dots, d_k, y_i < d_i \leq 2y_i, d_1 \cdots d_k \mid n\}.$$

Put $H = k + 1$, $y_0 = 3$ and

$$L_i = \log \frac{\log y_i}{\log y_{i-1}} \geq 0, \quad \ell_i = L_i + \log 3.$$

The lengths L_i measure the widths of the successive prime ranges on the log log scale. A coincident pair of cutoffs gives a zero length. To state a single formula covering this boundary, let Λ_k be the fixed constant specified in Appendix H.2. For $L_i^* = \max(L_i, \Lambda_k)$ use the finite rules in Section 3 and the scalar inverse evaluator in Appendix I to compute $\Psi_k(L^*)$. Define

$$\boxed{F_k^{\text{dyad}}(\mathbf{y}) = \Phi_k(\ell) = \Psi_k(L^*)}. \quad (2.1)$$

The fixed lift changes the harmonic sum by a factor depending only on the dimension; this is proved in Section 14. It is used to evaluate the profile, while the divisor count retains the given cutoffs and its stated range of x . The scalar recursion in Appendix I also covers saddle coordinates that coincide.

Theorem 2.1 (Uniform dyadic formula). *For each fixed $k \geq 1$ there are positive constants c_k, C_k such that for every $3 \leq y_1 \leq \dots \leq y_k$ and every $x \geq 2^k(y_1 \cdots y_k)y_k$,*

$$c_k F_k^{\text{dyad}}(\mathbf{y}) \leq \frac{H^{(k+1)}(x, \mathbf{y}, 2\mathbf{y})}{x} \leq C_k F_k^{\text{dyad}}(\mathbf{y}).$$

The estimate includes equal adjacent cutoffs.

For real ordered side lengths $1 \leq N_1 \leq \dots \leq N_{k+1}$, let

$$A_{k+1}(\mathbf{N}) = \#\{n_1 \cdots n_{k+1} : n_i \in \mathbb{N}, n_i \leq N_i\}.$$

Each distinct product is counted once.

Theorem 2.2 (Uniform multiplication-table formula). *For each fixed integer $k \geq 1$, uniformly for all real $1 \leq N_1 \leq \dots \leq N_{k+1}$,*

$$A_{k+1}(\mathbf{N}) \asymp_k \left(\prod_{i=1}^{k+1} N_i \right) \Phi_k(\ell(\mathbf{Y})), \quad Y_i = \max\{3, N_i/2\}.$$

Equivalently, with the same dimension-dependent lift as in the dyadic theorem, the profile here is

$$\Phi_k(\ell(\mathbf{Y})) = \Psi_k \left(\left(\max \left\{ \log \frac{\log Y_i}{\log Y_{i-1}}, \Lambda_k \right\} \right)_{i=1}^k \right), \quad Y_0 = 3.$$

The normalization by the product of the side lengths makes the table formula comparable to the density in Theorem 2.1. The largest side enters this product but does not add a cutoff to the profile. Section 15 proves the transfer and treats the bounded sides by a comparison with tables of lower dimension.

2.1 Divisor boxes and the harmonic sum

The arithmetic reduction replaces existence of a divisor product by the volume covered by its possible logarithmic divisor vectors. Let $\mathcal{A}_k(\mathbf{y})$ consist of squarefree pairwise coprime tuples $\mathbf{a} = (a_1, \dots, a_k)$, whose primes in a_1 are at most y_1 and whose primes in a_i for $i > 1$ lie in $(y_{i-1}, y_i]$. Empty selections are allowed, and 2, 3 are in the first prime range. Set

$$\mathcal{L}^{(k+1)}(\mathbf{a}) = \bigcup_{\substack{d_1, \dots, d_k \in \mathbb{N} \\ d_1 \cdots d_j | a_1 \cdots a_j \ (1 \leq j \leq k)}} \prod_{j=1}^k [\log(d_j/2), \log d_j], \quad (2.2)$$

$$L^{(k+1)}(\mathbf{a}) = \text{vol}_k \mathcal{L}^{(k+1)}(\mathbf{a}),$$

$$K_k(\ell) = e^{-\sum_i h_i L_i} \sum_{\mathbf{a} \in \mathcal{A}_k(\mathbf{y})} \frac{L^{(k+1)}(\mathbf{a})}{a_1 \cdots a_k}, \quad h_i = k - i + 2. \quad (2.3)$$

Each prime in group i has h_i possible labels: one for each eligible divisor coordinate and one for nonuse. Thus the prefix divisibility conditions in (2.2) express precisely the ordered routing rule. The reciprocal weights give the harmonic prime measure, and the exponential factor in (2.3) provides its normalization. The empty tuple has unnormalized contribution $(\log 2)^k$.

The arithmetic comparison of Koukoulopoulos [19, Theorem 1.7] states, with the harmless normalization above,

$$H^{(k+1)}(x, \mathbf{y}, 2\mathbf{y})/x \asymp_k K_k(\ell) \quad (x \geq 2^k(y_1 \cdots y_k)y_k). \quad (2.4)$$

The comparison is applied at the given cutoffs $\mathbf{y}$ and with the stated range of x .

Theorem 2.3 (Harmonic formula). *With (H.2.1), $K_k(\ell) \asymp_k \Psi_k(L^*)$ on $[\log 3, \infty)^k$.*

The lower bound is established in Section 7, the upper bound in Section 13, and the passage to all nonnegative block lengths in Section 14.

2.2 Size truncation of the harmonic volume sum

The geometric covers used for the upper bound need an a priori bound on the logarithmic width available to each coordinate. Such a bound holds when the tuple is truncated in size, whereas the harmonic sum in (2.3) includes every squarefree tuple. The next lemma shows that this truncation retains a fixed fraction of the weighted volume. Its proof uses only the effect of inserting one prime, so it is available before either sharp bound is proved.

Fix $H = k + 1$ and $3 \leq y_1 \leq \cdots \leq y_k$. Write $V(\mathbf{a}) = L^{(k+1)}(\mathbf{a})$ and

$$S(\mathbf{y}) = \sum_{\mathbf{a} \in \mathcal{A}_k(\mathbf{y})} \frac{V(\mathbf{a})}{a_1 \cdots a_k}.$$

All prime sets here are finite; in particular all following rearrangements are identities between finite nonnegative sums. Choose one absolute constant C_W for the elementary weighted prime estimate

$$\sum_{p \leq y} \frac{\log p}{p} \leq C_W \log y \quad (y \geq 3). \quad (2.5)$$

For example the usual Chebyshev bound for $\sum_{p \leq y} \log p$ and partial summation give this estimate; see [23, Theorem 9]. Put

$$\tau_k = 2kHC_W + 1, \quad S_{\text{tr}}(\mathbf{y}) = \sum_{\substack{\mathbf{a} \in \mathcal{A}_k(\mathbf{y}) \\ a_i \leq y_i^{\tau_k} \ (1 \leq i \leq k)}} \frac{V(\mathbf{a})}{a_1 \cdots a_k}.$$

Lemma 2.4 (Uniform positive weighted truncation). *Uniformly in the ordered cutoffs,*

$$S_{\text{tr}}(\mathbf{y}) \leq S(\mathbf{y}) \leq 2S_{\text{tr}}(\mathbf{y}). \quad (2.6)$$

The same comparison holds after multiplication by the exact normalizing factor defining K_k . Every truncated tuple has, for each i ,

$$\log(a_1 \cdots a_i) \leq \tau_k \sum_{g \leq i} \log y_g \leq k\tau_k \log y_i. \quad (2.7)$$

Proof. Let p be a prime in original block i , and remove it from that coordinate of a squarefree tuple. Inserting p back permits at most $k - i + 2 \leq H$ eligible routing labels, including the unused label. For each label the new divisor-box union is a translate of the old one. Therefore, if p does not already divide the tuple,

$$V(b_1, \dots, pb_i, \dots, b_k) \leq HV(\mathbf{b}). \quad (2.8)$$

Using the squarefree identity $\log a_i = \sum_{p|a_i} \log p$ gives the exact decomposition

$$\begin{aligned} \sum_{\mathbf{a} \in \mathcal{A}_k(\mathbf{y})} \frac{V(\mathbf{a}) \log a_i}{a_1 \cdots a_k} &= \sum_{p \text{ in block } i} \frac{\log p}{p} \sum_{\substack{\mathbf{b} \in \mathcal{A}_k(\mathbf{y}) \\ p \nmid b_i}} \frac{V(b_1, \dots, pb_i, \dots, b_k)}{b_1 \cdots b_k} \\ &\leq HS(\mathbf{y}) \sum_{p \text{ in block } i} \frac{\log p}{p} \leq HC_W \log y_i S(\mathbf{y}). \end{aligned} \quad (2.9)$$

The prime blocks are disjoint, so removal and reinsertion in this identity preserve every original prime-range restriction. In the inequality we only enlarged the inner nonnegative sum.

For the bad set $a_i > y_i^{\tau_k}$, multiply its indicator by $\log a_i / (\tau_k \log y_i) \geq 1$ and use (2.9). A union bound over i yields

$$S(\mathbf{y}) - S_{\text{tr}}(\mathbf{y}) \leq \frac{kHC_W}{\tau_k} S(\mathbf{y}) < \frac{1}{2} S(\mathbf{y}).$$

This proves (2.6). The common normalizing factor does not change its ratio. Finally (2.7) follows from the individual size bounds and the ordering of the cutoffs. $\square$

For the covers, (2.7) supplies the required baseline rectangle: every coordinate using only primes from the first i groups has logarithmic width at most $C_k \log y_i$. The contribution of earlier groups to a later coordinate obeys the corresponding bound at the last earlier cutoff. Here

$C_k = k\tau_k$; its fixed powers are absorbed into the cover constants and reserves. The lengths, saddle coordinates, and plateaux remain those of the given vector.

The order of application is useful to keep explicit. We apply the covers to the truncated squarefree sum, where the size bounds hold, and then enlarge the positive sums to allow repeated primes and cell approximations. Lemma 2.4 transfers the resulting upper bound to the full harmonic sum.

3 The structural profile

Throughout, $k \geq 1$ is fixed, $H = k + 1$, and $V_i = \sum_{g \leq i} L_g$, with $V_0 = 0$. We define the profile for $L_i \geq 1$; the fixed lift in (2.1) then extends it to $L_i \geq 0$.

A fine routing word assigns each prime from range i either to a divisor coordinate $c \geq i$ or to the unused category. Its logarithmic divisor vector is a sum of vectors $(\log p)e_c$, where e_c is the c th coordinate vector. Different assignments can produce overlapping boxes of side length $\log 2$, so the problem depends on the overlap of these ordered subset sums as well as on the number of assignments.

In the Poisson model, the uncolored prime positions form a common environment. We sum the conditional mass of admissible routing words before raising it to a fractional power, and then average over this environment. This order retains the dependence among the count types and the path constraints.

The definitions below follow the four sources of cost described in Section 1.2. The ordered saddle fixes the tilted prime intensities and the plateaux. Endpoint costs then specify how much reserve is available to each binary path comparison. Conditional survival converts these reserves into persistence factors. Finally, a terminal ordinary bridge supplies the root factor. All enumerations have size bounded in terms of the fixed dimension k . The constants in the construction depend only on k .

The distinction between a group and a plateau is used throughout. A group is a prescribed prime range and keeps its own spatial clock. A plateau is a run of equal saddle coordinates and determines which label comparisons share a fractional power. Several groups can therefore belong to one plateau without being combined into one clock.

3.1 The saddle and the prime ranges

For a prime in group i , one choice uses coordinate i and the remaining choices continue to later coordinates or to nonuse. If the current fractional power is s_i , a tail with effective weight Z_{i+1} contributes Z_{i+1}^{1/s_i} before that power is taken. Adding the current choice therefore gives $(1 + Z_{i+1}^{1/s_i})^{s_i}$. These successive sums explain the nested functions below. A common power gives $Z_i = h_i^{s_i}$; different powers retain the separate coordinate capacities. The ordered maximum determines where the powers must coincide.

On the closed ordered simplex $1 \geq s_1 \geq \dots \geq s_k \geq 0$ put

$$G_k = s_k \log 2, \quad G_i = s_i \log(1 + \exp(G_{i+1}/s_i)), \quad (3.1)$$

$$Z_i = \exp G_i, \quad J(L; s) = \sum_i L_i + \sum_j s_j V_j - \sum_i L_i Z_i. \quad (3.2)$$

At zero products use the continuous perspective extension. Let $s = s(L)$ be the unique maximizer of $J(L; s)$ on the ordered simplex. Its existence and uniqueness are proved in Section 4; Appendix I

gives a scalar recursion for its evaluation. The geometric theorem supplies

$$\frac{1}{2} < s_k \leq s_1 < 1 - \epsilon_H, \quad \epsilon_H = \frac{2 \log 2 - 1}{2H(\log H)^2}. \quad (3.3)$$

In particular $1 \leq Z_i \leq H$. A maximal interval $[a, b]$ on which s_i is constant is called a plateau; denote its common value by σ . A nonterminal plateau has $b < k$ and $z = \exp(G_{b+1}/\sigma)$; the terminal plateau has $b = k$ and $z = 1$. The exterior names of this plateau are $a, \dots, b$. Its local groups are the original groups $i \in [a, b]$, of ranks $r_i = b - i + 1$. Original groups $g < a$ are its earlier groups, of rank $b - a + 1$. Each original group retains its own random clock.

In the divisor model, name c is coordinate d_c : primes in group i may be assigned to $c \geq i$, and V_c is its deadline. Assignments to coordinates beyond b , together with the unused primes, form the plateau's compound child.

For $l < k$ define

$$\beta_l = \frac{\exp(G_{l+1}/s_l)}{1 + \exp(G_{l+1}/s_l)}.$$

The reference intensity per eligible exterior name is

$$\kappa_g = \begin{cases} Z_g \prod_{l=g}^{a-1} \beta_l / (b - a + 1 + z), & g < a, \\ (b - g + 1 + z)^{\sigma-1}, & a \leq g \leq b. \end{cases} \quad (3.4)$$

The coefficients for the earlier groups are the intensities transported along their routes into the plateau. We keep them separately to retain the constraints at each deadline. Set

$$E_q(z) = (q + z) \log(q + z) - z \log z, \quad g_g(q) = q - \kappa_g E_q(z).$$

3.2 The endpoint costs

A binary comparison removes some exterior names and retains the others, together with the compound child. The two ends of its time interval have different available reserves. To calculate them, we enumerate the legal choices for all the surrounding cuts; these are the outside contexts. Only a finite list of contexts is needed because each cut occurs at a group boundary.

For every local group i enumerate the edges $r_i \rightarrow q$, where $0 \leq q < r_i$ for a nonterminal plateau, and $1 \leq q < r_i$ for the terminal plateau. The latter exclusion removes the degenerate ordinary edge. The retained exterior names are $B = \{b - q + 1, \dots, b\}$; B is empty when $q = 0$. The compound child is always retained.

For each edge, enumerate the outside contexts as follows. For each name $c \in \{a, \dots, i - 1\}$ choose $\tau_c \in \{V_0, \dots, V_c\}$ independently. For each $c \in B$ choose $\tau_c \in \{V_i, \dots, V_c\}$ independently. In the left completion set $\tau_c = V_{i-1}$ for all $i \leq c \leq b - q$; in the right completion set those same cuts to V_i . All other cuts are identical in the two completions. This gives a finite set of lawful pairs (v^-, v^+) . Each cut lies before its coordinate's deadline, and the compound child remains intact.

For either completion v and original group $1 \leq g \leq b$, let

$$A_{v,g} = \{c \in \{a, \dots, b\} : \tau_c \geq V_g\}, \quad q_{v,g} = |A_{v,g}|, \quad t_v = \max_{a \leq c \leq b} \tau_c.$$

In an earlier group the shallowest eligible name is a , and in local group g it is g . Call g good when $0 < q_{v,g} < r_g$ and this shallowest eligible name is absent from $A_{v,g}$. Define

$$\overline{C}_v = \sum_{g \leq b} L_g g_g(q_{v,g}), \quad S_{\text{good}}(v) = \sum_{g \text{ good}} \sqrt{L_g}. \quad (3.5)$$

At the optimizer, the endpoint geometry gives $\overline{C}_v \geq 0$, with the coefficients and lengths of the earlier groups included.

For a nonterminal plateau put $T = \sum_{a \leq i \leq b} L_i$, $d = \sigma/s_{b+1} - 1$, and

$$\hat{R} = 1 + \frac{\sqrt{1+T}}{1+d\sqrt{1+T}}.$$

For the terminal plateau set $\hat{R} = 0$. Define

$$S_v = \begin{cases} 1 + \hat{R} + \overline{C}_v, & t_v \leq V_b/2, \\ 1 + \overline{C}_v + S_{\text{good}}(v), & t_v > V_b/2, \end{cases} \quad X_e = \min_v S_{v-}, \quad Y_e = \min_v S_{v+}. \quad (3.6)$$

The early score includes the uncertainty of a proper exit-count window. At the late endpoint the incoming contribution cancels from the intrinsic cost; the additional square-root terms instead measure the fluctuation needed to remove an eligible shallow name while retaining deeper names. The two minima are taken separately over the same finite set of paired lawful contexts. The lower bound makes the two scores compatible by keeping the actual type increment Δ fixed and choosing the legal pair $x = \max(a_* X_e, a_* Y_e - \Delta)$, $y = x + \Delta$; see (7.5).

3.3 Scalar persistence and canonical edge factors

The path comparison has two sources of fluctuation: the prime positions shared by all words and the private labels of the word being summed. The following Gaussian model preserves this distinction. The parameter ρ is the fraction of variance supplied by the common environment, and η is the power at which its conditional survival mass is averaged.

Let E, W be independent stationary standard Ornstein–Uhlenbeck processes, with covariance $e^{-|u-v|/2}$. For $0 < \eta, \rho < 1$ set

$$F_{\eta,\rho}(t) = \mathbb{E}_E \left[\mathbb{P}_W \{ \sqrt{\rho} E_u + \sqrt{1-\rho} W_u \geq 0 \mid 0 \leq u \leq t \mid E \}^\eta \right], \quad (3.7)$$

$$\chi(\eta, \rho) = \lim_{t \rightarrow \infty} -t^{-1} \log F_{\eta,\rho}(t). \quad (3.8)$$

Theorem 5.4 and Propositions 5.1 and 5.5 give existence, local Lipschitz continuity, and $\eta/2 \leq \chi \leq 1/2$. With $b(t) = -\log F_{\eta,\rho}(t)$, the finite-time comparison gives a constant C_Q on each compact parameter set such that

$$\frac{b(t) - \log C_Q}{t} \leq \chi(\eta, \rho) \leq \frac{b(t)}{t}. \quad (3.9)$$

Thus χ is characterized by $C_Q^{-1} e^{-\chi t} \leq F_{\eta,\rho}(t) \leq e^{-\chi t}$ for all $t \geq 0$.

For evaluating this function, Proposition 5.1 gives the finite Gaussian grid rule, displaced thresholds, crossing control and Gaussian translation error. Combine its finite-time enclosures with (3.9); Section B.5 specifies C_Q on a compact parameter set. Taking t large and then refining

the Gaussian grid determines this two-variable function to arbitrary prescribed accuracy. For $k = 1$ the edge product is empty, so χ is never evaluated at a degenerate parameter.

For $e = (i, r, q)$ put $a = \log(r + z)$, $b_e = \log(q + z)$, $p = (q + z)/(r + z)$, and

$$m = a - b_e p, \quad \rho_e = \frac{m^2}{m^2 + b_e^2 p(1 - p)}, \quad T_e = \frac{(1 + X_e)(1 + Y_e)}{1 + (r + z)^\sigma L_i}, \quad P_e = \min(1, T_e)^{2\chi(\sigma, \rho_e)}. \quad (3.10)$$

The binary laws are nondegenerate: a nonterminal zero-output edge has $z > 1$, and the terminal zero-output edge is excluded. The product in (3.12) includes every canonical edge.

3.4 The ordinary terminal suffix

The terminal comparison leaves no compound child beyond the last plateau. Its zero-output edge is therefore an ordinary one-channel constraint, handled separately from the binary factors. The suffix chosen below contains a fixed proportion of the total time and enough drift to account for the preceding groups. Its exact count and path constraint together give one factor.

Here $h_i = k - i + 2$ on the complete original chain. If the selected suffix starts at group j , then $V_{\text{pre}} = V_{j-1}$ includes all preceding plateaux as well as the earlier terminal groups. For $2 \leq h \leq H$ define

$$\alpha_h = \frac{\log((h-1)/\log h)}{\log h}, \quad c_w = \frac{1}{4H^2}, \quad \tau = \min \left\{ \frac{1}{4}, \frac{c_w}{2H(\log H)^2}, \frac{1}{4} \min_{2 \leq h < H} (\alpha_{h+1} - \alpha_h) \right\}.$$

The thresholds α_h are strictly increasing. If the terminal product σ lies in $[\alpha_{h_i} - \tau, \alpha_{h_i} + \tau]$ for a terminal local group, begin the suffix at that group; at most one qualifies. Otherwise begin at the first terminal local group with $e_i = h_i^\sigma \log h_i - (h_i - 1) > 0$. The KKT theorem ensures existence. Let U be the suffix length, M its largest original block, and A, B the total lengths before and after that block within the suffix. Break largest-block ties at the earliest index. Put

$$D = \sum_{i \text{ in suffix}} e_i L_i, \quad \mathcal{B}(L) = \frac{1}{\sqrt{1+U}} \min \left\{ 1, \frac{(1+B)(1+D+A)}{1+M} \right\}. \quad (3.11)$$

Here $D \geq 0$, and the suffix theorem gives $D \geq c_k V_{\text{pre}}$ and $U \geq c_k V_k$. This is the sole ordinary root factor.

The positive-length raw expression is

$$\Psi_k(L) = e^{-J(L; s(L))} \mathcal{B}(L) \prod_{j \text{ proper}} (1 + d_j \sqrt{1 + T_j})^{-\sigma_j} \prod_{e \text{ canonical}} P_e. \quad (3.12)$$

The saddle, plateaux and endpoint costs in this expression are evaluated at L . The final formula uses L^* as in (2.1). There are at most H^{2H} contexts for each of at most kH canonical edges, and the scalar evaluator has depth k .

3.5 A heuristic for the profile

We now explain why both count and path costs enter (3.12). The discussion concerns the positive-length Poisson core. The Gaussian model below predicts the binary exponent; the exact bridge and arithmetic estimates are needed to make that prediction uniform.

A count model and its missing constraints. In the Poisson model of the harmonic prime measure, take independent counts $n_i \sim \text{Pois}(L_i)$. There are $\prod_i h_i^{n_i}$ routing words, whereas the available logarithmic volume has scale $\exp(\sum_j V_j)$. A first model treats the words as if they covered this volume without clustering. Dividing the word count by this capacity and capping at 1 gives its normalized volume. The bound $\min(1, t) \leq t^\alpha$ yields

$$\mathbb{E} \min \left\{ 1, e^{\sum_i n_i \log h_i - \sum_j V_j} \right\} \leq \exp \left\{ - \sum_i L_i - \alpha \sum_j V_j + \sum_i L_i h_i^\alpha \right\}, \quad 0 \leq \alpha \leq 1.$$

The exponent on the right is $-J(L; \alpha, \dots, \alpha)$. Optimizing it balances the supply of words against volume and tilts the means to $L_i h_i^\alpha$. The full recursion (3.2) makes this balance at successive coordinates. At its optimizer, Proposition 4.10 identifies J with the Poisson entropy $\sum_i L_i \mathcal{Q}(Z_i)$.

The count model loses two kinds of information. Even with the right total number of words, too many choices may lie in the same coordinate projection. Moreover, the prime positions can force clustering at an intermediate cut despite admissible endpoint counts. Ford's dyadic heuristic [13] already explains the second loss. Koukoulopoulos's first rectangular heuristic includes an ordinary clustering correction; his further analysis [19, Section 2.2, Lemma 2.3 and Remark 2.4] identifies additional projected-volume constraints.

The first loss can change the exponential scale. A stationary common-power candidate only balances the sum of the coordinate derivatives, whereas ordered optimality requires every proper prefix sum to be nonpositive; see (4.20). For example, for $k \geq 2$ take $L_1 = T$, $L_k = cT$ and fixed intermediate lengths, with $c > 0$ chosen so that the common-power stationary parameter tends to a fixed α as $T \rightarrow \infty$. At this candidate,

$$\sum_{j < k} \partial_j J = T \left[(k-1) - (k+1)^{\alpha-1} \{ (k+1) \log(k+1) - 2 \log 2 \} \right] + o(T).$$

This is the coefficient appearing in the obstruction [19, inequality (2.13)], with fixed shifts of the lengths irrelevant to the leading comparison. When the coefficient is strictly positive, increasing the first $k-1$ powers improves J for large T . Thus a single power overestimates the leading volume, and the ordered saddle must split. The endpoint corrections below address the path losses that remain after the saddle has been chosen.

The count windows and endpoint reserves. On a proper plateau of length T , central counts fluctuate on scale $\sqrt{T}$. The exit likelihood penalizes a positive carry on scale d^{-1} ; see (A.2.9). Up to the fixed reserve floor, the useful width is therefore $\min(\sqrt{T}, d^{-1})$. Its fraction of the central count mass is

$$w \asymp (1 + d\sqrt{1+T})^{-1}.$$

The mixed-moment calculation contributes w from count acceptance and $w^{\sigma-1}$ from the inverse weight, giving w^σ ; see (7.7)–(7.10). The transition is controlled by $d\sqrt{1+T}$, so a small gap alone does not remove the window cost.

The reference cost $\bar{C}_v$ is the margin between a lawful cut's logarithmic capacity and its reference word entropy. This margin supplies a path reserve. At an early context of a proper plateau, the exit-count uncertainty supplies an additional reserve; at a late end, removing a shallow name while retaining deeper names can use a square-root label fluctuation. These are the extra terms in (3.6). Taking the most restrictive lawful cuts gives X_e, Y_e . They are separately minimized scores; the compatible-pair construction in (7.5) is needed to realize a legal endpoint pair at those scales on one exact count fiber.

Why the binary power is 2χ . For an edge, use a, b_e, p, m from (3.10). A near-critical contrast has two fluctuation channels: the common prime positions, with variance scale m^2 , and the private retained-label count, with variance scale $b_e^2 p(1-p)$. Their Gaussian variance ratio predicts ρ_e . This is a Gaussian separation of the channels, rather than an assertion of independence of the two physical-time counting processes.

Normalize the total variance and model the contrast as a linear chord from $x > 0$ to $y > 0$ plus two independent centered Brownian bridges of length n , with weights $\sqrt{\rho}, \sqrt{1-\rho}$. A centered bridge has the representation in law $\beta(t) = (n-t)B_{t/(n-t)}/\sqrt{n}$, where B is standard Brownian motion. The time change $v = t/(n-t)$, followed by $u = \log v$, turns its nonnegative-path condition into

$$\sqrt{\rho} X_u + \sqrt{1-\rho} Y_u \geq -\frac{x}{\sqrt{n}} e^{-u/2} - \frac{y}{\sqrt{n}} e^{u/2},$$

where X, Y are independent stationary standard OU processes. The two boundary terms become of order one at $u = 2 \log(x/\sqrt{n})$ and $u = 2 \log(\sqrt{n}/y)$. When $xy < n$, the effective OU interval between these points has length $2 \log(n/(xy))$. The exponential boundary terms have bounded OU action on this interval, so Lemma 5.11 changes the zero-wall moment only by a constant factor. Thus conditional persistence predicts

$$\exp\{-\chi(\sigma, \rho)[2 \log(n/(xy))]_+\} = \min\{1, (xy/n)^{2\chi(\sigma, \rho)}\}.$$

The endpoint product and the factor 2 come from this time change; very unequal reserves shift the interval without changing its length. Putting n on scale $(r+z)^\sigma L_i$ and restoring fixed reserve floors gives P_e . Theorem 6.1 proves the discrete comparison, and (8.11) justifies the use of the reference parameter ρ_e on the rare branch.

The order of averaging is essential here. In this Gaussian model, let q be the private survival mass conditional on the environment. The plateau power gives $\mathbb{E}q^\sigma$, with rate $\chi(\sigma, \rho)$. Replacing it by $(\mathbb{E}q)^\sigma$ would give the ordinary OU rate $\sigma/2$. The strict gap in Proposition 5.9 shows that this substitution loses a power when the effective interval grows. The canonical factors are assembled by the conditional cover and intersection estimates; their product does not assert independence of all cut events.

One terminal atom and one ordinary bridge. The terminal constant-mark constraint has no private binary channel. Its count atom costs $(1+U)^{-1/2}$. Conditional on that same count, the ordered drift supplies two ballot reserves, of scales $1+B$ and $1+D+A$. With M the longest block, the remaining path cost is $\min\{1, (1+B)(1+D+A)/(1+M)\}$. The minimum also covers suffixes whose outward drift makes survival a constant cost. These two costs give $\mathcal{B}(L)$; Theorem 6.9 estimates them jointly. There is one root atom, rather than one atom for each group. For $k=1$, $\mathcal{B}(L) = (1+L_1)^{-3/2}$: a count-only prediction sees the square-root atom but misses the additional inverse-length loss from clustering. This is the ordinary barrier loss in Ford's dyadic formula. In higher dimensions, the binary factors retain the analogous losses in the conditional label mass.

3.6 Reading and evaluating the profile

Formula (3.12) separates the leading count cost from the corrections caused by overlap. The exponential of the saddle value pays for the tilted counts. Each proper plateau boundary contributes its count window, with the power of the plateau from which it exits. Each canonical edge contributes the survival cost calculated from its two endpoint scores. The factor $\mathcal{B}(L)$ contains

both the ordinary count atom and the terminal path constraint; neither is inserted a second time elsewhere in the profile.

The calculation proceeds from the lengths to the ordered saddle, then to its plateaux, the finite endpoint lists, and the displayed factors. The scalar recursion in Appendix I evaluates the saddle without a separate search over plateau patterns. The only universal function to approximate is χ ; its finite-time enclosure is (3.9). Thus the definition is the same for all rectangles in a fixed dimension, including those whose active constraints change as the lengths vary.

For reference, the principal quantities are collected here. Their precise definitions and hypotheses are in the indicated sections.

Quantity	Meaning and definition
L_i, V_i	Length of prime group i on the logarithmic clock, and its cumulative deadline; Sections 2 and 3.
$s(L), \sigma$	Ordered saddle coordinates, and the common coordinate on a plateau; (3.2).
Z_i, κ_g	Tilted group intensity, and reference intensity per exterior name reaching a plateau; (3.2) and (3.4).
$\bar{C}_v, S_v$	Reference cost and endpoint score of a lawful context; (3.5) and (3.6).
X_e, Y_e	The left and right endpoint scores used for canonical edge e ; (3.6).
$\chi(\eta, \rho), P_e$	Conditional survival exponent and its canonical binary factor; (3.8) and (3.10).
$\mathcal{B}(L)$	Ordinary suffix factor, including its exact count atom; (3.11).
$K_k(\ell)$	Normalized harmonic volume sum at the given cutoffs; (2.3).
$\Psi_k(L), \Phi_k(\ell)$	Positive-length profile and its extension by the fixed lift; (3.12) and (2.1).

3.7 Two worked profiles

The following examples evaluate the factors in (3.12) in two different regimes. The first has comparable prime ranges and a positive ordinary drift. The second has a strict saddle separation and a binary factor which changes the power of the logarithmic clock. All lengths are at least the fixed lift threshold. They correspond to actual ordered cutoffs by taking $y_i = \exp\{(\log 3)e^{V_i}\}$; hence the lift is inactive and Theorem 2.1 gives $H^{(k+1)}(x, \mathbf{y}, 2\mathbf{y}) \asymp_k x\Psi_k(L)$ for $x \geq 2^k(y_1 \cdots y_k)y_k$. Likewise, Theorem 2.2 gives $A_{k+1}(\mathbf{N}) \asymp_k (\prod_i N_i)\Psi_k(L)$ when $N_i = 2y_i$ for $i \leq k$ and $N_{k+1} \geq 2y_k$.

Example 3.1 (Two comparable prime ranges). Let $k = 2$ and $L_1 = L_2 = T \rightarrow \infty$. The saddle has one plateau, $s_1 = s_2 = \sigma$, where

$$3^\sigma \log 3 + 2^\sigma \log 2 = 3, \quad \alpha_2 < \sigma < \alpha_3.$$

Indeed, this equation gives stationarity along the equality face. Its proper prefix derivative is $T\{1 - 3^{\sigma-1}(3 \log 3 - 2 \log 2)\} < 0$, so the ordered KKT condition holds. The sign follows already at α_2 from $3^{\alpha_2-1}(3 \log 3 - 2 \log 2) > 1$ and is preserved as σ increases. Strict concavity identifies the saddle.

Put $c_1 = 1 - 2 \cdot 3^{\sigma-1} \log 2$ and $c_2 = 2 - 3^\sigma \log 3 = 2^\sigma \log 2 - 1$. The only binary edge is $2 \rightarrow 1$ in group 1. Its retained name 2 has cut T or $2T$: the former is early, at $t_v = V_2/2$, and the latter

is late. The corresponding left scores are $1 + c_1 T$ and $1 + (c_1 - c_2)T + \sqrt{T}$; the right scores are $1 + c_2 T$ and 1. Thus

$$X = 1 + \min\{c_1 T, (c_1 - c_2)T + \sqrt{T}\}, \quad Y = 1.$$

Here $c_1 > c_2 > 0$. For example, rigorous scalar brackets give $0.54133 < \sigma < 0.54134$, $0.1624 < c_1 < 0.1625$, and $0.0087 < c_2 < 0.0088$. In (3.10), $a = \log 3$, $b_e = \log 2$, $p = 2/3$, and $\eta = \sigma$. Consequently $T_e \rightarrow 2(c_1 - c_2)/3^\sigma > 0$ and $P_e \asymp 1$: the linear left reserve compensates for the bounded right reserve.

For $H = 3$, the thresholds satisfy $\alpha_2 < 0.52877$, $\alpha_3 > 0.54532$, and $\tau < 0.00384$; thus σ is outside both bands in the ordinary-suffix rule. The suffix is therefore group 2, with $U = M = T$, $A = B = 0$, and $D = c_2 T$. There is no proper exit window. Writing $I_2 = 2 + 3\sigma - 3^\sigma - 2^\sigma$, we obtain

$$\mathcal{B}(L) = \frac{1 + c_2 T}{(1 + T)^{3/2}}, \quad \Psi_2(T, T) \asymp e^{-I_2 T} T^{-1/2}.$$

Positive drift makes the ordinary path cost bounded away from zero; only its single count atom supplies a power of T . The zero-drift balanced case in Section 14.3 has the additional inverse-length path cost.

Example 3.2 (A strict separation with a conditional power). Let $k = 6$, fix $\lambda \geq \max\{1, \Lambda_6\}$, and take $L = (T, \lambda, \lambda, \lambda, \lambda, 100T)$ as $T \rightarrow \infty$. On the face $s_1 = \dots = s_5$, J/T tends to the objective of the two-node chain $(h_1, h_2) = (7, 2)$, with multiplicities $(5, 1)$ and lengths $(1, 100)$. It has products $\sigma_0 > \tau_0$, characterized by

$$\begin{aligned} z_0 &= 2^{\tau_0/\sigma_0}, & 5 &= (5 + z_0)^{\sigma_0-1} E_5(z_0), \\ 101 &= \log 2 \{100 \cdot 2^{\tau_0} + z_0(5 + z_0)^{\sigma_0-1}\}. \end{aligned} \tag{3.13}$$

To verify the strict separation, set $\theta = 1 + \log(5/E_5(2))/\log 7$. Then $\alpha_2 < \theta < \alpha_7$ and

$$c_* := \frac{6 - 7^\theta \log 7}{2^\theta \log 2 - 1} \quad \text{satisfies} \quad 54 < c_* < 55.$$

Since $100 > c_*$, the stationary common power α is below θ . Its first node derivative $5 - 7^{\alpha-1} E_5(2)$ is then positive, violating ordered KKT. The unique two-node saddle therefore splits and satisfies (3.13). The global bounds give $\alpha_2 < \tau_0 < \sigma_0 < 1$.

Put $g(q) = q - (5 + z_0)^{\sigma_0-1} E_q(z_0)$. The first equation gives $g(0) = g(5) = 0$; strict concavity gives $g(q) > 0$ for $0 < q < 5$. The complete-chain prefix derivatives, divided by T , tend to $-g(5 - j) < 0$ at $j = 1, \dots, 4$. These KKT signs identify the full limiting saddle: all four internal constraints are active and the gap after group 5 is strict. The nonsingular two-variable Hessian gives face-critical products $\sigma_T = \sigma_0 + O_\lambda(T^{-1})$ and $\tau_T = \tau_0 + O_\lambda(T^{-1})$. The strict signs and gap persist, so these are the actual saddle products for large T , with plateaux $[1, 5]$ and $[6, 6]$.

The critical binary edge is $5 \rightarrow 0$ in group 1. It has one context pair: all five cuts are 0 on the left and T on the right. For $T > 4\lambda$, the left context is early and the right context is late, with no good groups. With $z_T = 2^{\tau_T/\sigma_T}$ and $g_{1,T}(q) = q - (5 + z_T)^{\sigma_T-1} E_q(z_T)$, the exact scores are

$$X = 1 + \widehat{R}, \quad Y = 1 + T g_{1,T}(5), \quad T g_{1,T}(5) = -\lambda \sum_{r=1}^4 \{r - (r + z_T)^{\sigma_T-1} E_r(z_T)\} = O_\lambda(1).$$

Here $\widehat{R}$ uses the plateau length $T + 4\lambda$. The last identity is the top plateau balance; reference nonnegativity also gives $Y \geq 1$. Since $d_T = \sigma_T/\tau_T - 1 \rightarrow d_0 > 0$, both scores are bounded. Hence $T_e \asymp_\lambda T^{-1}$ and $P_e \asymp_\lambda T^{-2\chi(\sigma_0, \rho_0)}$, where

$$p_0 = \frac{z_0}{5 + z_0}, \quad m_0 = \log(5 + z_0) - p_0 \log z_0, \quad \rho_0 = \frac{m_0^2}{m_0^2 + (\log z_0)^2 p_0 (1 - p_0)} \in (0, 1).$$

Local Lipschitz continuity of χ justifies replacing its T -dependent parameters by these limits. For each other edge in group 1, its left reference cost is $Tg(q) + O_\lambda(1)$ with $1 \leq q < 5$, so its factor is bounded below. Every remaining binary edge has the fixed length λ and factor at least $\min\{1, 4/(1 + 7\lambda)\}$. All these factors are therefore comparable to 1.

The proper window contributes $T^{-\sigma_0/2}$. The ordinary suffix consists of group 6: $U = M = 100T$, $A = B = 0$, and $D = (2^{\tau_T} \log 2 - 1)100T$. Its drift is strictly positive because $\tau_0 > \alpha_2$, so $\mathcal{B}(L) \asymp T^{-1/2}$. Finally, with $I_{100} = Q((5 + z_0)^{\sigma_0}) + 100Q(2^{\tau_0})$, the entropy identity gives $J = I_{100}T + O_\lambda(1)$ and

$$\boxed{\Psi_6(L) \asymp_\lambda e^{-I_{100}T} T^{-1/2 - \sigma_0/2 - 2\chi(\sigma_0, \rho_0)}}. \quad (3.14)$$

The three powers record the ordinary atom, the proper exit window, and the conditional binary bridge. Proposition 5.9 gives $2\chi(\sigma_0, \rho_0) > \sigma_0$; averaging the private survival before taking its plateau power would therefore miss a strictly positive power of T in this family.

4 Geometry of the ordered saddle

The ordered saddle determines both the exponential cost of supplying prime factors and the constraints that remain after this cost has been paid. Equal consecutive products form plateaux. Each strict separation between plateaux introduces a count window whose width depends on the product gap; within a plateau, the stationarity margins provide the reserves for the path estimates. We therefore need more than existence of an optimizer: the argument requires uniform control of its curvature, its local drifts, and the relative lengths of its plateaux.

The proof has three stages. First, the recursive objective is concave, and each term has enough curvature to control all later products. Boundary variations then place the unique optimizer in a fixed compact region. Second, the stationarity equations identify the node margins and center the uncolored root exactly. Third, a positive entropy surplus across each strict separation forces a length hierarchy. This hierarchy supplies a long block in which a single Poisson count can impose the root condition, even when the final original block is short.

We treat a general finite rank chain before specializing to the complete chain of the divisor problem. The generality is needed when equalities are imposed during changes of parameters.

4.1 The objective and the main conclusions

Fix integers

$$H = h_1 > h_2 > \cdots > h_m \geq 2,$$

where the one-element chain is allowed. Set

$$p_i = h_i - h_{i+1} \quad (i < m), \quad p_m = h_m - 1, \quad V_j = \sum_{i=1}^j L_i, \quad V_0 = 0,$$

with arbitrary $L_i > 0$. The symbols p_i always denote these positive integers. Child probabilities will instead be denoted by β_i .

On the ordered simplex

$$\mathcal{D} = \{s \in \mathbb{R}^m : 1 \geq s_1 \geq \dots \geq s_m \geq 0\}$$

define, first at positive products,

$$\begin{aligned} G_m(s) &= s_m \log h_m, \\ G_i(s) &= s_i \log(p_i + \exp(G_{i+1}(s)/s_i)) \end{aligned} \quad (i < m), \quad (4.1)$$

$$\begin{aligned} Z_i(s) &= \exp G_i(s), \\ J(L; s) &= \sum_i L_i + \sum_j s_j p_j V_j - \sum_i L_i Z_i(s). \end{aligned} \quad (4.2)$$

Use the continuous extension at zero products. Indeed, induction gives $0 \leq G_i(s) \leq s_i \log h_i$: if $G_{i+1} \leq s_{i+1} \log h_{i+1}$, ordering gives $\exp(G_{i+1}/s_i) \leq h_{i+1}$. Thus when $s_i = 0$ its whole suffix is zero and $G_i = 0$, $Z_i = 1$. In particular,

$$1 \leq Z_i(s) \leq h_i \quad (s \in \mathcal{D}). \quad (4.3)$$

For $E \subseteq \{1, \dots, m-1\}$, let

$$\mathcal{D}_E = \{s \in \mathcal{D} : s_j = s_{j+1} \text{ for every } j \in E\}.$$

Write $s^E(L)$ for the maximizer on $\mathcal{D}_E$ and $J_E^*(L) = J(L; s^E(L))$. The constraints defining $\mathcal{D}$ remain in force, and additional equalities may occur at the maximizer. The next theorem gives existence and uniqueness.

Throughout, $c, C > 0$ may change from line to line and depend only on the fixed chain. In particular they do not depend on L , its ratios, the face E , or the sizes of product gaps and multipliers. The notation $A \asymp B$ means $cB \leq A \leq CB$, with this convention.

Theorem 4.1 (Global bounds). *Put*

$$c_0 = 2 \log 2 - 1, \quad \varepsilon_H = \frac{c_0}{2H(\log H)^2}. \quad (4.4)$$

For every positive length vector and every E , there is a unique maximizer on $\mathcal{D}_E$, and it satisfies

$$\frac{1}{2} < s_m^E(L) \leq s_1^E(L) < 1 - \varepsilon_H. \quad (4.5)$$

4.2 Concavity and the outer boundaries

Continuity on the compact ordered simplex gives existence. To prove uniform interiority, we vary the entire top or bottom plateau, so that every prescribed equality is respected. The bottom variation compares the linear budget with the cost of increasing the final alphabets; the top variation uses a definite entropy excess near product one.

The map $x \mapsto \log(p + e^x)$ is convex and increasing. Its perspective $(s, y) \mapsto s \log(p + e^{y/s})$ is convex for $s > 0$ and is increasing in y . Induction in (4.1) therefore shows that each G_i , and hence each Z_i , is convex. The continuous extension gives convexity on $\mathcal{D}$. Thus J is concave and continuous on the compact convex set $\mathcal{D}_E$, so a maximum exists.

4.2.1 The bottom plateau

Lemma 4.2. *Every maximizer on $\mathcal{D}_E$ satisfies $s_m > 1/2$.*

Proof. If the maximal bottom block $a, \dots, m$ has common product $\sigma \leq 1/2$, we can raise the whole block. When $a > 1$, maximality makes the preceding gap strict, so this variation respects all prescribed equalities. Inside the block the recursion telescopes exactly:

$$G_i = \sigma \log h_i, \quad Z_i = h_i^\sigma \quad (i \geq a).$$

For $i < a$, set

$$\beta_i = \frac{\exp(G_{i+1}/s_i)}{p_i + \exp(G_{i+1}/s_i)}.$$

Along this bottom-block direction,

$$\frac{dG_i}{d\sigma} = \left(\prod_{i \leq b < a} \beta_b \right) \log h_a.$$

The identity

$$Z_i \beta_i = Z_{i+1} \beta_i^{1-s_i}$$

implies

$$Z_i \prod_{i \leq b < a} \beta_b = Z_a \prod_{i \leq b < a} \beta_b^{1-s_b} \leq Z_a.$$

When $\sigma = 0$, all preceding products, if present, are positive, and these formulas give the right derivative. There are no preceding terms if $a = 1$.

The derivative of the linear budget is

$$\sum_{j \geq a} p_j V_j = (h_a - 1) V_{a-1} + \sum_{i \geq a} (h_i - 1) L_i.$$

Consequently the directional derivative of J is at least

$$V_{a-1}(h_a - 1 - h_a^\sigma \log h_a) + \sum_{i \geq a} L_i(h_i - 1 - h_i^\sigma \log h_i). \quad (4.6)$$

For every real $h > 1$, one has $\sqrt{h} \log h < h - 1$. Indeed, with $t = \sqrt{h}$, the difference $t^2 - 1 - 2t \log t$ vanishes at 1 and has positive derivative $2(t - 1 - \log t)$ for $t > 1$. Every bracket in (4.6) is thus positive. Since the lengths are positive, the derivative is strictly positive. This contradicts maximality, including at $\sigma = 0$. $\square$

4.2.2 The top plateau

Lemma 4.3. *Every maximizer on $\mathcal{D}_E$ satisfies $s_1 < 1 - \varepsilon_H$.*

Proof. Let $1, \dots, b$ be the maximal top equal-product block, with common value σ . If $b < m$, put $C = h_{b+1}$ and $z = \exp(G_{b+1}/\sigma)$; otherwise put $C = z = 1$. For $i \leq b$, let $n_i = h_i - C \geq 1$. Telescoping yields

$$G_i = \sigma \log(n_i + z).$$

Here $1 \leq z \leq C$ and $n_i + z \leq H$, by the ordered recursion. Varying the top block while keeping its child products fixed gives

$$\frac{dZ_i}{d\sigma} = (n_i + z)^\sigma A(n_i, z), \quad A(n, z) = \log(n + z) - \frac{z \log z}{n + z}. \quad (4.7)$$

This formula includes the variation of $z = \exp(G_{b+1}/\sigma)$. The budget derivative is $\sum_{i \leq b} L_i n_i$.

Write

$$E_n(z) = (n + z) \log(n + z) - z \log z = (n + z) A(n, z).$$

It is increasing in z , since $\partial_z E_n = \log(1 + n/z) > 0$. Moreover

$$E_n(z) \geq E_n(1) = (n + 1) \log(n + 1) \geq 2n \log 2.$$

For the last inequality, $(n + 1) \log(n + 1)/n$ is increasing for $n > 0$, with derivative $[n - \log(n + 1)]/n^2 > 0$, and $n \geq 1$. Also $0 < A(n, z) \leq \log H$. With n, z fixed at their values at the point under consideration, for $0 \leq \tau \leq 1$,

$$\frac{d}{d\tau} ((n + z)^\tau A(n, z)) \leq H(\log H)^2.$$

This auxiliary comparison keeps z fixed. If $\sigma \geq 1 - \varepsilon_H$, it follows that

$$(n + z)^\sigma A(n, z) - n \geq n(2 \log 2 - 1) - \varepsilon_H H(\log H)^2 \geq c_0/2 > 0.$$

Thus raising the top block has strictly negative derivative for J . Lowering it a sufficiently small amount is feasible and increases J , a contradiction. This also covers $\sigma = 1$. $\square$

Let $\varepsilon = \varepsilon_H$, which lies in $(0, 1/2)$, and put

$$\mathcal{D}_\varepsilon = \{s : \varepsilon \leq s_m \leq \dots \leq s_1 \leq 1 - \varepsilon\}.$$

The preceding lemmas put every maximizer on every prescribed face in this single compact set. We will use differential estimates on the slightly larger convex set $\mathcal{D}_{\varepsilon/2}$.

4.3 Full suffix curvature and uniqueness

Convexity alone would allow flat maximizing directions. The stronger fact needed here is that a single recursive term controls every coordinate on which it depends. The derivative recursion below separates the motion at the current node from the motion transmitted by its child; neither can vanish unless the whole suffix direction vanishes. Summing these estimates with the original length weights gives the natural cumulative-length scale for the Hessian.

For $i < m$, introduce

$$a_i = G_i/s_i = \log S_i, \quad u_i = s_{i+1}/s_i, \quad \beta_i = \frac{e^{G_{i+1}/s_i}}{S_i}, \quad A_i = a_i - u_i \beta_i a_{i+1},$$

where $S_i = p_i + e^{G_{i+1}/s_i}$; set $a_m = A_m = \log h_m$. All these functions and the derivatives needed below are bounded on $\mathcal{D}_{\varepsilon/2}$. The quantities $s_i, a_i, \beta_i, 1 - \beta_i$ have positive chain-dependent lower bounds. The same holds for A_i : it is the entropy of the distribution consisting of p_i named probabilities $1/S_i$ and the one compound probability β_i . These assertions use compactness but no separation of adjacent products.

Lemma 4.4 (One perspective term controls its entire suffix). *For any $s \in \mathcal{D}_{\varepsilon/2}$, direction $z \in \mathbb{R}^m$, and i ,*

$$c \sum_{j \geq i} z_j^2 \leq D^2 Z_i(s)[z, z] \leq C \sum_{j \geq i} z_j^2. \quad (4.8)$$

Proof. We prove the lower and upper curvature bounds together by propagating the first and second variations down the recursion. Write $g_i = DG_i[z]$ and $Q_i = D^2 G_i[z, z]$. Differentiation gives

$$\begin{aligned} g_m &= (\log h_m) z_m, & Q_m &= 0, \\ g_i &= A_i z_i + \beta_i g_{i+1}, \\ Q_i &= \beta_i Q_{i+1} + \frac{\beta_i(1 - \beta_i)}{s_i} (g_{i+1} - u_i a_{i+1} z_i)^2. \end{aligned} \quad (4.9)$$

In particular $Q_i \geq 0$. Put $E_i = g_i^2 + Q_i$ and $w_i = g_{i+1} - u_i a_{i+1} z_i$. Then

$$g_i = a_i z_i + \beta_i w_i, \quad Q_i \geq c w_i^2 + c Q_{i+1}.$$

Thus $z_i^2 \leq C E_i$, and then $g_{i+1}^2 \leq C E_i$, so that $E_{i+1} \leq C E_i$. Iteration over the fixed number of indices, starting with $E_m = (\log h_m)^2 z_m^2$, proves

$$E_i \geq c \sum_{j \geq i} z_j^2.$$

The same recursion gives the upper bound, since all coefficients in (4.9) are bounded. Passing from the logarithm to its exponential uses

$$D^2 Z_i[z, z] = Z_i E_i,$$

and Z_i is bounded above and below. □

Define the negative objective Hessian by $\mathcal{H}(L; s) = -D_s^2 J(L; s)$. Summing (4.8) with weights L_i gives

$$c \sum_j V_j z_j^2 \leq z^\top \mathcal{H}(L; s) z \leq C \sum_j V_j z_j^2. \quad (4.10)$$

Furthermore,

$$|\mathcal{H}_{ij}(L; s)| \leq C V_{\min(i, j)}, \quad (4.11)$$

because only terms $L_\ell Z_\ell$ with $\ell \leq \min(i, j)$ depend on both coordinates, and their second derivatives are bounded. These estimates hold for every positive length vector, without stationarity or a length hierarchy.

To conclude uniqueness, suppose two distinct points maximize on $\mathcal{D}_E$. Concavity makes their connecting segment maximizing. The boundary lemmas place this segment in $\mathcal{D}_\varepsilon$, where (4.10) gives strict concavity along every nonzero direction. The contradiction completes Theorem 4.1.

4.4 The finite optimization and its product bounds

We now express the same optimization in node notation, which makes the routing probabilities and stationarity equations explicit. Retain

$$H = h_1 > h_2 > \cdots > h_m \geq 2, \quad p_j = h_j - h_{j+1} \ (j < m), \quad p_m = h_m - 1.$$

Let $L_i > 0$, $V_j = \sum_{i \leq j} L_i$, and $V_0 = 0$. Write the ordered simplex as

$$\mathcal{S} = \{1 \geq s_1 \geq s_2 \geq \cdots \geq s_m \geq 0\}.$$

For positive products define recursively

$$G_m = s_m \log h_m, \quad G_j = s_j \log(p_j + \exp(G_{j+1}/s_j)) \quad (j < m), \quad (4.12)$$

$$Z_j = e^{G_j}, \quad J(\mathbf{s}) = \sum_i L_i + \sum_j s_j p_j V_j - \sum_i L_i Z_i. \quad (4.13)$$

The extension to $\mathcal{S}$ is the one used above: when $s_j = 0$, the whole suffix has zero product and $G_j = 0$. The bound $0 \leq G_j \leq s_j \log h_j$ proves continuity on these faces. Compactness gives existence. The convex, increasing perspective of $y \mapsto \log(p + e^y)$ makes each G_j and then each Z_j convex on $\mathcal{S}$, so J is concave. Henceforth the optimizer is the unique one of Theorem 4.1.

When products are positive, put

$$S_m = h_m, \quad S_j = p_j + S_{j+1}^{u_j}, \quad u_j = s_{j+1}/s_j, \quad (4.14)$$

$$\beta_j = S_{j+1}^{u_j}/S_j, \quad A_j = \log S_j - u_j \beta_j \log S_{j+1} \quad (j < m), \quad A_m = \log h_m. \quad (4.15)$$

Then $G_j = s_j \log S_j$, $Z_j = S_j^{s_j}$, and $2 \leq S_j \leq h_j \leq H$. We repeatedly use the entropy increment

$$E_n(z) = (n+z) \log(n+z) - z \log z, \quad n > 0, \quad z \geq 1. \quad (4.16)$$

For a nonterminal node, $A_j = E_{p_j}(S_{j+1}^{u_j})/S_j > 0$.

Lemma 4.5 (Uniform upper product bound). *Every maximizer satisfies $s_1 < 1 - \epsilon_H$, where $\epsilon_H = (2 \log 2 - 1)/(2H(\log H)^2)$.*

Proof. This is the upper bound in Theorem 4.1, applied with no prescribed equalities. $\square$

Lemma 4.6 (Canonical thresholds). *For real $h > 1$ define*

$$\alpha_h = \frac{\log((h-1)/\log h)}{\log h} = 1 - \frac{\log(h \log h/(h-1))}{\log h}.$$

Then $\alpha_h > 1/2$ and α_h is strictly increasing. In particular, $2^{\alpha_2} \log 2 = 1$.

Proof. Write $y = \log h$ and $z = y/2$. Direct calculation gives

$$\alpha_h = \frac{1}{2} + \frac{\log(\sinh z/z)}{2z}.$$

The second term is positive. Its derivative has the sign of

$$N(z) = z \coth z - 1 - \log(\sinh z/z).$$

Here $N(0+) = 0$ and $N'(z) = 1/z - z/\sinh^2 z > 0$. This proves both assertions without a discrete approximation. $\square$

Lemma 4.7 (Transported natural intensities). *For $i < j$ at positive products,*

$$Z_i \prod_{q=i}^{j-1} \beta_q = Z_j \prod_{q=i}^{j-1} \beta_q^{1-s_q} \leq Z_j. \quad (4.17)$$

If $s_1 < 1$, the inequality is strict.

Proof. The one-edge identity $Z_q \beta_q = Z_{q+1} \beta_q^{1-s_q}$ follows directly from (4.12)–(4.15). Multiplication telescopes. Since $0 < \beta_q < 1$, the conclusions follow. $\square$

Theorem 4.8 (Global product compactness). *For $H \geq 3$ and every positive length vector, every maximizer satisfies*

$$\alpha_2 < s_m \leq s_1 < 1 - \epsilon_H. \quad (4.18)$$

For the exceptional one-node chain $H = 2$, the unique maximizing product is α_2 . In particular all products, local ratios and finite positive mark probabilities used below lie in compact sets determined by H .

Proof. Lemma 4.5 already gives the upper bound. Let $a, \dots, m$ be the bottom maximal plateau, with product $\sigma \leq \alpha_2$. Raising its common product is feasible. Put

$$I_{\text{in}} = \sum_{i < a} L_i Z_i \prod_{q=i}^{a-1} \beta_q.$$

The derivative in that direction is

$$(h_a - 1)V_{a-1} - I_{\text{in}} \log h_a + \sum_{i \geq a} L_i (h_i - 1 - h_i^\sigma \log h_i). \quad (4.19)$$

The derivative remains valid from the right at $\sigma = 0$, since all older products are positive and the bottom recursion is $G_i = \sigma \log h_i$. Transport bounds the incoming term by $I_{\text{in}} \leq h_a^\sigma V_{a-1}$. Lemma 4.6 then makes every bracket in (4.19) nonnegative, and every bracket with $h_i > 2$ positive. Any such rank in the bottom plateau therefore makes (4.19) strictly positive along a feasible variation. Otherwise the plateau is the single rank $h_m = 2$, with $a = m > 1$ when $H \geq 3$. In that case $I_{\text{in}} < 2^\sigma V_{m-1}$ by Lemma 4.5 and the strict transport identity, so the derivative is again positive, even at $\sigma = \alpha_2$. Both alternatives contradict maximality. For the chain $H = 2$, directly differentiating $L(1 + s - 2^s)$ gives its unique maximizer α_2 . $\square$

4.5 Exact node margins and root centering

The first derivatives have two different uses. Their individual stationarity equations record the reserve at each order constraint; their weighted sum centers the one uncolored root condition. The latter identity is exact because the recursive logarithms are homogeneous, including on a face where several products coincide.

We now work at a maximizer. The preceding result removes the upper and lower faces of the ordered simplex. There are multipliers

$$\lambda_0 = \lambda_m = 0, \quad \lambda_j \geq 0, \quad \partial_j J = \lambda_{j-1} - \lambda_j, \quad \lambda_j (s_j - s_{j+1}) = 0 \quad (j < m). \quad (4.20)$$

Write $m_j = \lambda_j - \lambda_{j-1}$. A maximal plateau is a maximal interval $[a, b]$ on which the products are equal. Its exterior multipliers satisfy $\lambda_{a-1} = \lambda_b = 0$.

Lemma 4.9 (Node equations and Euler root identity). *For every j ,*

$$A_j \sum_{i \leq j} L_i Z_i \prod_{q=i}^{j-1} \beta_q = p_j V_j + m_j. \quad (4.21)$$

Let $r_j = s_j/s_1$, $a_j = \log S_j$ and $\mu_j = L_j Z_j$. Then

$$T_1 := \sum_j r_j p_j V_j = \sum_j c_j \mu_j, \quad c_j = r_j a_j. \quad (4.22)$$

Moreover $0 < c_-(H) \leq c_j, Z_j \leq c_+(H) < \infty$.

Proof. Differentiation of the recursion gives $\partial G_j / \partial s_j = A_j$ and $\partial G_i / \partial G_j = \prod_{q=i}^{j-1} \beta_q$ for $i < j$. Substitution into $\partial_j J$ and (4.20) proves (4.21). Every G_i is homogeneous of degree one in the products. Consequently

$$\sum_j s_j \partial_j J = \sum_j s_j p_j V_j - \sum_i L_i Z_i G_i.$$

The left side is zero because $\sum_j s_j m_j = \sum_{j < m} \lambda_j (s_j - s_{j+1}) = 0$. Division by s_1 proves (4.22). Finally $2 \leq S_j \leq H$ and (4.18) bound r_j above and below, which gives the stated constants. $\square$

Proposition 4.10 (Poisson entropy at the saddle). *For positive original lengths, let $s = s(L)$ be the ordered optimizer in (3.2). With $\mathcal{Q}(z) = z \log z - z + 1$,*

$$J(L; s) = \sum_{i=1}^k L_i \mathcal{Q}(Z_i). \quad (4.23)$$

Equivalently, J is the relative entropy of independent Poisson counts with means $L_i Z_i$ with respect to independent counts with means L_i .

Proof. The upper and lower simplex faces are inactive, so dilation of all products is feasible in both directions, even on an equality face. Homogeneity of the G_i and stationarity give

$$0 = \sum_j s_j \partial_{s_j} J = \sum_j s_j V_j - \sum_i L_i Z_i G_i.$$

Substitute $G_i = \log Z_i$ into (3.2). For a single count, $D_{\text{KL}}(\text{Pois}(Lz) \parallel \text{Pois}(L)) = L \mathcal{Q}(z)$; relative entropy adds over independent coordinates. $\square$

This entropy supplies the exponential factor of the profile. The persistence factors account for the additional restrictions on the conditional bridges.

4.6 Complete unit-step chains: sign and effective time

For the divisor problem the alphabet loses one name at each step. This additional structure makes the local drift nonnegative and identifies where a long persistence constraint can occur. Away from the first node of a plateau the drift has a fixed positive lower bound. At the first node it can be small, and its scale is determined by the incoming length and the stationarity multiplier. The secant estimate below is what propagates the first-node sign through a plateau.

In this section only, assume

$$H \geq 3, \quad m = H - 1, \quad h_j = H - j + 1, \quad p_j = 1 \quad (1 \leq j \leq m).$$

The *natural local node drift* is

$$e_j = Z_j A_j - 1. \quad (4.24)$$

The rank here is one. It must not be replaced by the larger rank of an entire equal-product plateau with its child left unsplit.

Lemma 4.11 (A secant threshold with one valley). *For $x \geq 2$ put*

$$D(x) = x \log x - (x-1) \log(x-1), \quad t(x) = 1 - \frac{\log D(x)}{\log x}.$$

For $2 \leq x \leq y$,

$$t(x) \leq \max\{\alpha_2, t(y)\}, \quad (4.25)$$

strictly when $2 < x < y$. More quantitatively, if $y \leq H$, $x \geq 2 + \eta$ and $y \geq x + \eta$, then

$$\max\{\alpha_2, t(y)\} - t(x) \geq \Delta_H(\eta) := \frac{\eta^2 \log(2 \log 2)}{2H^2(1 + \log H)(\log H)^2}. \quad (4.26)$$

Proof. Set $l(x) = \log(x/(x-1))$. Direct differentiation yields

$$t'(x) = \frac{F(x)}{xD(x)(\log x)^2}, \quad (4.27)$$

$$F(x) = D(x) \log D(x) - x \log x l(x), \quad (4.28)$$

$$F'(x) = l(x) \log D(x) + \log x(1/(x-1) - l(x)) > 0. \quad (4.29)$$

Here $D(x) \geq 2 \log 2 > 1$, and $\log(1+u) < u$ proves the second term positive. Thus t first decreases and then increases, allowing either monotone case, with no flat interval. Its maximum on $[2, y]$ is attained at an endpoint and $t(2) = \alpha_2$. This proves (4.25) and its strict form.

For the quantitative assertion, throughout $[2, H]$,

$$F'(v) \geq \frac{\log(2 \log 2)}{H}, \quad vD(v)(\log v)^2 \leq H(1 + \log H)(\log H)^2.$$

The latter bound follows from $D(v) = \int_{v-1}^v (1 + \log w) dw \leq 1 + \log H$. If $F(x) \geq 0$, integrate t' over $[x, x + \eta]$, using the linear lower bound for $F(v)$ based at x , and then use monotonicity up to y . If $F(x) \leq 0$, integrate backwards over $[x - \eta, x]$; F is nonpositive on $[2, x]$, so $t(2) \geq t(x - \eta)$. Both integrations give exactly (4.26). $\square$

Theorem 4.12 (Nonnegative local node drift). *On the complete unit-step chain, every positive length vector has*

$$e_j \geq 0 \quad (1 \leq j \leq m), \quad e_m > 0. \quad (4.30)$$

If a is the first index of a maximal plateau, then $e_a > 0$ when $a > 1$. In addition, for every nonterminal index that is not first in its plateau,

$$e_j \geq c_H := 2^{\Delta_H(\sqrt{2}-1)} - 1 > 0. \quad (4.31)$$

There is no assumed lower bound on length ratios, gaps between plateau products, or nonzero multipliers.

Proof. Let $[a, b]$ be a maximal plateau with product σ . At its first node $m_a = \lambda_a \geq 0$. With $I_{\text{in}} = \sum_{i < a} L_i Z_i \prod_{q=i}^{a-1} \beta_q$, the exact node equation and transport give

$$\lambda_a = A_a(L_a Z_a + I_{\text{in}}) - V_a \leq (Z_a A_a - 1)V_a = e_a V_a. \quad (4.32)$$

This proves $e_a \geq 0$. If $a > 1$, strict transport makes the inequality strict, so $e_a > 0$.

For $b < m$ put $C = h_{b+1}$ and $z = \exp(G_{b+1}/\sigma)$; for $b = m$ put $C = z = 1$. Within the plateau $S_i = h_i - C + z$ decreases by one at each step. At every node its immediate child weight is $S_i - 1$, including the last node of the plateau. Therefore

$$A_i = D(S_i)/S_i, \quad e_i = S_i^{\sigma-1} D(S_i) - 1 = \exp((\sigma - t(S_i)) \log S_i) - 1. \quad (4.33)$$

The first-node sign gives $\sigma \geq t(S_a)$; global compactness gives $\sigma > \alpha_2$. Lemma 4.11 propagates $\sigma \geq t(S_i)$ through the plateau. At the terminal node, $e_m = 2^{s_m} \log 2 - 1 > 0$.

For $i > a$ with $i < m$, we have $S_a \geq S_i + 1$. If $b = m$, then $S_i = h_i \geq 3$. Otherwise $u_b = s_{b+1}/\sigma > 1/2$ and $S_{b+1} \geq 2$, whence $z > \sqrt{2}$ and $S_i \geq 1 + z > 1 + \sqrt{2}$. Thus $S_i \geq 2 + (\sqrt{2} - 1)$ in both cases. Apply (4.26) with $\eta = \sqrt{2} - 1$ and then (4.33), using $S_i \geq 2$, to obtain (4.31). $\square$

Theorem 4.13 (Exact first-node drift scale). *For the first index a of any maximal plateau on the complete chain,*

$$e_a \asymp_H \frac{\lambda_a + V_{a-1}}{V_a}. \quad (4.34)$$

For $a = 1$ this is the exact equality $e_1 = \lambda_1/L_1$. When $a > 1$, set

$$q_- = H^{-H}, \quad q_+ = (1 - 1/H)^{\epsilon_H} < 1.$$

Then (4.34) holds with explicit lower and upper comparison constants $1 - q_+$ and q_-^{-1} . Consequently, with the infinite-quotient convention,

$$1 + \min\{L_a, e_a^{-2}\} \asymp_H 1 + \min\left\{L_a, \left(\frac{V_a}{\lambda_a + V_{a-1}}\right)^2\right\}. \quad (4.35)$$

For a non-first, nonterminal node, the left effective time is at most $1 + c_H^{-2}$.

Proof. On the unit-step chain $1/H \leq \beta_q \leq 1 - 1/H$. Also $\epsilon_H < 1 - s_q < 1$. Formula (4.17) implies, when $a > 1$,

$$q_- V_{a-1} \leq I_{\text{in}}/Z_a \leq q_+ V_{a-1}.$$

There are fewer than H factors in each transported term; the upper bound uses at least one factor. Write $w = I_{\text{in}}/(Z_a V_{a-1}) \in [q_-, q_+]$. The exact equality in (4.32), before using transport, is

$$\lambda_a = e_a(L_a + wV_{a-1}) - (1 - w)V_{a-1}, \quad (4.36)$$

$$e_a = \frac{\lambda_a + (1 - w)V_{a-1}}{L_a + wV_{a-1}}. \quad (4.37)$$

Its numerator lies between $(1 - q_+)(\lambda_a + V_{a-1})$ and $\lambda_a + V_{a-1}$; its denominator lies between $q_- V_a$ and V_a . This proves the stated comparison. At $a = 1$ there is no incoming intensity and (4.21) gives the exact equality directly. Taking inverse squares and truncating by L_a preserves comparison by fixed constants, including the zero-drift case. The final assertion follows from (4.31). $\square$

Remark 4.14. The effective time in (4.35) governs the one-end persistence scale. The reverse reserve, terminal band, and incoming private state are treated by the conditional bridge estimates below; their dependence must be retained when combining local constructions.

4.7 A hierarchy of plateau lengths

A root count requires a block with a substantial fraction of the total length. Such a block need not be the final original block. We obtain it by varying a whole plateau: its total drift balances the incoming budget, and an entropy surplus keeps that incoming budget from being absorbed at zero cost. This argument works for a sparse chain and remains uniform as a strict product gap tends to zero.

We return to an arbitrary sparse chain $H = h_1 > \dots > h_m \geq 2$. No sign assertion for its individual local node drifts is needed. For a maximal equal-product plateau $[a, b]$, let its common product be σ . Define

$$(C, z) = (h_{b+1}, S_{b+1}^{u_b}) \quad (b < m), \quad (4.38)$$

$$(C, z) = (1, 1) \quad (b = m), \quad (4.39)$$

$$n_i = h_i - C \quad (a \leq i \leq b), \quad n = n_a, \quad S = S_a = n + z, \quad (4.40)$$

$$A_* = E_n(z)/S, \quad \mathbf{e}_i = (n_i + z)^{\sigma-1} E_{n_i}(z) - n_i. \quad (4.41)$$

The symbol $\mathbf{e}_i$ is a *whole-plateau* drift: it uses the entire exterior rank n_i , with the child intact. It is different from e_i in (4.24).

Lemma 4.15 (Exact plateau balance). *With $I_{\text{in}} = \sum_{i < a} L_i Z_i \prod_{q=i}^{a-1} \beta_q$,*

$$\sum_{i=a}^b L_i \mathbf{e}_i = n V_{a-1} - A_* I_{\text{in}}. \quad (4.42)$$

If $a > 1$, then

$$I_{\text{in}} = \frac{\beta_{a-1}}{A_{a-1}} (p_{a-1} V_{a-1} - \lambda_{a-2}) \leq \frac{\beta_{a-1} p_{a-1}}{A_{a-1}} V_{a-1}. \quad (4.43)$$

Proof. The sum of budget derivatives over $[a, b]$ is $n V_{a-1} + \sum_{i=a}^b n_i L_i$. On the plateau, $G_i = \sigma \log(n_i + z)$ and its derivative is $E_{n_i}(z)/(n_i + z)$, including the variation in z . The derivative for an older block is $A_* \prod_{q=i}^{a-1} \beta_q$. The sum of the KKT derivatives on the plateau is $\lambda_{a-1} - \lambda_b = 0$. Substituting these derivatives proves (4.42).

To bound the incoming intensity, use the node $a - 1$ immediately before the plateau. Its margin is $m_{a-1} = \lambda_{a-1} - \lambda_{a-2} = -\lambda_{a-2} \leq 0$. Multiplying (4.21) at that node by the routing factor β_{a-1} gives (4.43), with all older intensities still included in their actual mixture. $\square$

Lemma 4.16 (Whole-plateau entropy surplus). *If $a > 1$, then*

$$n - A_* \frac{\beta_{a-1} p_{a-1}}{A_{a-1}} \geq \frac{n p_{a-1}}{2H^2(1 + \log H)^2} > 0. \quad (4.44)$$

This estimate is independent of the size of the strict product gap before the plateau.

Proof. For real $r > 0$ and $x \geq 1$ let $D(r, x) = E_r(x)/r$. Integrating $1 + \log(x + t)$ gives

$$D(r, x) = 1 + \log x + \frac{1}{r} \int_0^r \log(1 + t/x) dt, \quad (4.45)$$

$$1 + \log x + \frac{r}{2(x + r)} \leq D(r, x) \leq 1 + \log(x + r). \quad (4.46)$$

For the lower bound use $\log(1 + t/x) \geq t/(x + t) \geq t/(x + r)$. Write $p = p_{a-1}$, $u = u_{a-1}$ and $w = S^u$. Then $1 \leq w \leq S$, $p + w \leq H$, and

$$\frac{\beta_{a-1}p}{A_{a-1}} = \frac{w}{D(p, w)}, \quad \frac{n}{A_*} = \frac{S}{D(n, z)} \geq \frac{S}{1 + \log S}.$$

Since $x/(1 + \log x)$ is nondecreasing for $x \geq 1$,

$$\begin{aligned} \frac{n}{A_*} - \frac{\beta_{a-1}p}{A_{a-1}} &\geq \frac{w}{1 + \log w} - \frac{w}{D(p, w)} \\ &= \frac{w[D(p, w) - (1 + \log w)]}{(1 + \log w)D(p, w)} \geq \frac{p}{2H(1 + \log H)^2}. \end{aligned} \quad (4.47)$$

Finally (4.46) implies $E_n(z) \geq n$, so $A_* \geq n/H$. Multiplication of (4.47) by A_* proves (4.44). Strict separation of the products was used only for the zero exterior multiplier, not to obtain its numerical constant. $\square$

Theorem 4.17 (Maximal-plateau length hierarchy). *At every global ordered KKT maximizer on an arbitrary sparse chain, each maximal plateau $[a, b]$ satisfies*

$$V_{a-1} \leq C_H \sum_{i=a}^b L_i, \quad C_H = 2H^3(1 + \log H)^3. \quad (4.48)$$

For $a > 1$, its signed incoming whole-plateau drift satisfies

$$A_* I_{\text{in}} - nV_{a-1} \leq -\frac{np_{a-1}}{2H^2(1 + \log H)^2} V_{a-1}. \quad (4.49)$$

The estimates require no positive length-ratio or product-gap bound.

Proof. Combining (4.43) and (4.44) gives (4.49), and then (4.42) gives

$$\sum_{i=a}^b L_i \mathfrak{e}_i \geq \frac{np_{a-1}}{2H^2(1 + \log H)^2} V_{a-1}.$$

For every summand, $1 \leq n_i + z \leq H$, $0 < \sigma < 1$ and (4.46) give $|\mathfrak{e}_i| \leq H(1 + \log H)$. Hence

$$V_{a-1} \leq \frac{2H^3(1 + \log H)^3}{np_{a-1}} \sum_{i=a}^b L_i \leq C_H \sum_{i=a}^b L_i.$$

For $a = 1$, $V_0 = 0$, so (4.48) is immediate. $\square$

Corollary 4.18 (A long block in the final plateau). *Let $[a, m]$ be the final maximal plateau, and select a largest block d within it. Then*

$$L_d \geq \frac{V_m}{(H - 1)(1 + C_H)}. \quad (4.50)$$

Proof. Set $W = \sum_{i=a}^m L_i$. Theorem 4.17 gives $V_m \leq (1 + C_H)W$. The plateau has at most $H - 1$ blocks, so $L_d \geq W/(H - 1)$. $\square$

Remark 4.19. The hierarchy compares the total length of a plateau with its preceding length. Individual blocks within it may be short. Their drifts can cancel, and their incoming routing words retain the dependence needed in the conditional estimates below.

4.8 A conditional root count

The long block just found supplies an unread reservoir. After the outside data have been fixed in their admitted central windows, exact root centering places the required reservoir count in a central Poisson window. Its atom has the usual inverse square-root mass, pointwise in the outside data. Pointwise uniformity is essential: the outside path weights may already be very small, and the root condition must multiply those weights without changing their scale.

We estimate the conditional mass of a root count in the Poisson model at the optimizer. The path restrictions will be imposed subsequently. In block i take an independent homogeneous Poisson point process of intensity Z_i over length L_i , giving $N_i \sim \text{Pois}(\mu_i)$ with $\mu_i = Z_i L_i$. Any independent private marks can be attached to these points. Let

$$V = V_m, \quad X_1 = \sum_i c_i N_i - T_1, \quad c_i = r_i \log S_i.$$

The centering $T_1 = \sum_i c_i \mu_i$ is exact by Lemma 4.9.

Choose deterministically a block d satisfying $L_d \geq \kappa_H V$ for some fixed $\kappa_H > 0$. Either a globally largest block or the block in Corollary 4.18 does so. In block d , choose a deterministic subinterval of length τL_d , where $0 < \tau_0 \leq \tau \leq \tau_1 < 1$ are fixed. Denote its uncolored count by N_R and its mean by $\mu_R = \tau \mu_d$.

Let $\mathcal{O}$ be the σ -field of all points and private marks *outside* that interval. Then N_R is independent of $\mathcal{O}$. Write $N_i^{\text{out}} = N_i$ for $i \neq d$ and $N_d^{\text{out}} = N_d - N_R$, with corresponding means $\mu_i^{\text{out}} = \mu_i$ and $\mu_d^{\text{out}} = (1 - \tau)\mu_d$. The target x must be measurable in the uncolored part of $\mathcal{O}$. Assume, for fixed $A, B < \infty$,

$$|x| \leq A \log(2 + V), \quad \mathcal{C}_B = \bigcap_i \{|N_i^{\text{out}} - \mu_i^{\text{out}}| \leq B\sqrt{1 + L_i}\}. \quad (4.51)$$

The reservoir and its count have not been read by the target or by any previously selected outside cost.

Theorem 4.20 (Uniform conditional root atom). *Define the $\mathcal{O}$ -measurable integer*

$$n_*(\mathcal{O}) = \left\lfloor \frac{T_1 + x - \sum_i c_i N_i^{\text{out}}}{c_d} \right\rfloor. \quad (4.52)$$

For $L_d \geq L_0(H, A, B, \kappa_H, \tau_0, \tau_1)$, on $\mathcal{C}_B$ it is nonnegative and

$$|n_* - \mu_R| \leq C\sqrt{L_d}, \quad (4.53)$$

$$\mathbb{P}(N_R = n_* \mid \mathcal{O}) \asymp (1 + V)^{-1/2}. \quad (4.54)$$

The constants depend only on the displayed fixed parameters. On $N_R = n_*$,

$$x - c_d < X_1 \leq x. \quad (4.55)$$

For every nonnegative $\mathcal{O}$ -measurable random variable F , including arbitrarily rare outside private-word weights,

$$\mathbb{E}[F \mathbf{1}_{\mathcal{C}_B} \mathbf{1}_{\{N_R = n_*\}}] \asymp (1 + V)^{-1/2} \mathbb{E}[F \mathbf{1}_{\mathcal{C}_B}]. \quad (4.56)$$

Proof. Subtract $c_d\mu_R$ inside (4.52). Exact centering then gives

$$n_* - \mu_R = \frac{x - \sum_i c_i(N_i^{\text{out}} - \mu_i^{\text{out}})}{c_d} + \varepsilon, \quad -1 < \varepsilon \leq 0.$$

There are at most $H - 1$ blocks. On $\mathcal{C}_B$,

$$\sum_i \sqrt{1 + L_i} \leq C_H \sqrt{1 + V} \leq C_{H, \kappa_H} \sqrt{L_d}, \quad \log(2 + V) \leq C_{H, \kappa_H} \sqrt{L_d}$$

when $L_d \geq 1$. Uniform bounds for c_i prove (4.53). Since $\mu_R \geq c_H \tau_0 L_d$, taking L_0 sufficiently large makes n_* positive.

For completeness, if $n = \mu_R + v$ with $|v| \leq C\sqrt{\mu_R}$, then

$$n \log(n/\mu_R) - n + \mu_R = O_C(1), \quad n! \asymp \sqrt{n} (n/e)^n.$$

Stirling's inequalities now bound the Poisson atom $e^{-\mu_R} \mu_R^n / n!$ above and below by fixed multiples of $\mu_R^{-1/2}$. The comparison $\mu_R \asymp L_d \asymp V$ proves (4.54) on each admitted outside fiber. The rounding convention in (4.52) gives (4.55). Conditioning on $\mathcal{O}$ then multiplies any nonnegative outside weight by this same atom bound, which is (4.56). $\square$

Proposition 4.21 (Compatibility with exact margin-carry bands). *Assume $m \geq 2$. Suppose, in addition to the hypotheses of Theorem 4.20, that a subsequent retained word satisfies the exact recursions*

$$X_{j+1} = (X_j - Y_j)/u_j \quad (j < m) \tag{4.57}$$

and the bands are represented exactly as

$$Y_j = m_j + y_j + \xi_j - \xi_{j-1}/u_{j-1}, \quad j < m, \quad \xi_0 = 0, \tag{4.58}$$

where the initial term with u_0 is absent. Choose the uncolored root target $x = \sum_{j < m} r_j y_j$. Then

$$X_m = \frac{X_1 - x}{r_m} - \lambda_{m-1} - \frac{\xi_{m-1}}{u_{m-1}}. \tag{4.59}$$

The identity is unaffected by which block supplies the root count.

Proof. Since $r_{j+1} = r_j u_j$, recursion (4.57) gives $r_m X_m = X_1 - \sum_{j < m} r_j Y_j$. Complementarity in (4.20) gives $\sum_{j \leq m} r_j m_j = 0$ and $m_m = -\lambda_{m-1}$, hence $\sum_{j < m} r_j m_j = r_m \lambda_{m-1}$. The carry terms telescope:

$$\sum_{j < m} r_j (\xi_j - \xi_{j-1}/u_{j-1}) = r_{m-1} \xi_{m-1}.$$

Substitute these equalities and $x = \sum_{j < m} r_j y_j$ to obtain (4.59). The argument contains no assumption about the position of the selected reservoir. $\square$

Remark 4.22 (Events within the reservoir). After conditioning on $N_R = n_*$, the reservoir points are uniform order statistics on the fixed interval and their private marks retain their original law. A later event P that reads that reservoir needs a further pointwise conditional estimate at the prescribed count. For example, if on $\mathcal{C}_B$,

$$c G(\mathcal{O}) \leq \mathbb{E}[P \mid \mathcal{O}, N_R = n_*(\mathcal{O})] \leq C G(\mathcal{O}),$$

then conditioning extends (4.56), with P on the left and G on the right. The required estimate is pointwise at the selected count. For the path events used below, these pointwise estimates are furnished by the bridge theorems.

5 Conditional survival and the persistence spectrum

The binary factors in the structural profile measure how much private label mass survives a path constraint on a given spatial clock. Their scale is governed by Gaussian persistence against a random environment. We begin with that Gaussian problem, establish estimates up to constant factors, and then transfer them to exponential gaps, exact totals, and complete cells. The order of integration is common to all these models: the marks and any permitted count window are integrated conditionally on the gap or Gaussian environment before the fractional power is taken.

There are two distinct issues in this transfer. Endpoint estimates must hold under the law already weighted by survival, so that surviving segments can be joined at constant cost. Approximation failures must also be small relative to the surviving mass, which may itself tend to zero. The endpoint filter and the relative coupling estimates address these issues in turn. The positive-order spectrum and its scalar deviations describe the Gaussian object beyond the fractional orders needed for the arithmetic argument.

Write $\mathbb{P}$ and $\mathbb{E}$ for probability and expectation. Comparison constants are uniform on the compact parameter sets explicitly specified below. They may depend on fixed curve amplitudes, endpoint boxes, density margins, and regularity tolerances. None depends on the length of the process, its allowed reserve, or a count in the stated central range.

5.1 The Gaussian exponent and effective finite-time bounds

The environmental channel is held fixed while the private channel is tested for survival. Let X, Y be independent stationary standard Ornstein–Uhlenbeck processes with covariance $\mathbb{E}X_s X_t = e^{-|s-t|/2}$. For $0 < \lambda, \rho < 1$ put

$$\begin{aligned} A_t &= \{\sqrt{\rho}X_s + \sqrt{1-\rho}Y_s \geq 0 : 0 \leq s \leq t\}, \\ q_t(X) &= \mathbb{P}_Y(A_t \mid X), \quad F_{\lambda, \rho}(t) = \mathbb{E}_X q_t(X)^\lambda. \end{aligned} \tag{5.1}$$

We suppress the parameters when they are fixed. The conditional moment retains fluctuations in the environment that an ordinary survival probability would average out. The proposition below first identifies its exponential rate and supplies a finite-time enclosure of that rate. Finite-grid assertions pass to continuous paths through decreasing events on nested dense grids and bounded convergence.

Proposition 5.1 (Existence and effective bounds). *The limit*

$$\chi(\lambda, \rho) = \lim_{t \rightarrow \infty} -t^{-1} \log F_{\lambda, \rho}(t)$$

exists. If $b(t) = -\log F(t)$ and $p_g = (1 + e^{-g/2})/(1 - e^{-g/2})$, then, for $t, g > 0$,

$$\frac{b(t)}{p_g(t+g)} \leq \chi(\lambda, \rho) \leq \frac{b(t)}{t}, \quad \frac{\lambda}{2} \leq \chi(\lambda, \rho) \leq \frac{1}{2}. \tag{5.2}$$

Thus $g = 2 \log t$, $t > 1$, gives an effective interval of width $O(\log t/t)$. Finite Gaussian integrals give arbitrarily accurate bounded values of the endpoints of this interval.

Proof. Association gives one product inequality, while separated blocks and Gaussian Hölder give the reverse inequality with an explicit loss. Together they determine the limiting rate. We then control the error in approximating each finite-time moment by a Gaussian grid integral.

The private survival indicators on two adjacent time intervals are increasing functions of a Gaussian vector with nonnegative covariances. Gaussian association [21], in the finite form proved in Lemma B.1, first in Y and then in X , gives

$$F(s+t) \geq F(s)F(t). \quad (5.3)$$

The function b is nonnegative, locally bounded and subadditive. Writing $T = nt + r$, $0 \leq r < t$, proves that its limiting slope exists and equals $\inf_{t>0} b(t)/t$.

For the separated-block estimate we use the following finite-dimensional Gaussian Hölder inequality: if a centered Gaussian block vector has covariance bounded above by $\text{diag}(p_i \Sigma_{ii})$, then $\mathbb{E} \prod_i f_i(Z_i) \leq \prod_i (\mathbb{E} f_i(Z_i)^{p_i})^{1/p_i}$ for nonnegative functions. This is [6, Theorem 1(i)]. We apply it first to the private Gaussian blocks and then to the environmental blocks. Consider N intervals of length t , separated by gaps g . The OU Markov property and conditional expectation at their endpoints imply that the covariance of linear functionals on blocks i, j is bounded in absolute value by $r^{|i-j|}$ times their standard deviations, where $r = e^{-g/2}$. Consequently, by $2ab \leq a^2 + b^2$,

$$\text{Var} \left(\sum_i L_i \right) \leq \left(1 + 2 \sum_{k \geq 1} r^k \right) \sum_i \text{Var}(L_i) = p_g \sum_i \text{Var}(L_i).$$

This is covariance domination, uniformly in the grids and N . Drop survival in the intervening gaps. Private Gaussian Hölder gives $q_{\text{joint}} \leq \prod_i q_i^{1/p_g}$. Applying the same inequality to the environmental expectation gives

$$F(Nt + (N-1)g) \leq \mathbb{E} \prod_i q_i^{\lambda/p_g} \leq F(t)^{N/p_g}.$$

The second application acts on the powers produced by the first, so only one factor p_g remains. Taking $N \rightarrow \infty$ proves the first inequality of (5.2).

The composite process in (5.1) is itself a standard OU process. Brownian time change and reflection give its ordinary survival probability

$$a(t) = \pi^{-1} \arcsin(e^{-t/2}). \quad (5.4)$$

The inequalities $q^\lambda \geq q$ and Jensen give $a(t) \leq F(t) \leq a(t)^\lambda$, proving the range of χ and also $b(t) \leq t/2 + \log \pi$. This last bound makes the width assertion explicit.

It remains to make the finite-time enclosure computable. The grid approximation has two errors: crossings between grid points and the positive threshold used to suppress those crossings. At mesh size $\Delta = t/n$, let G_h denote the fractional grid survival moment with common threshold h . Conditional on two composite endpoints $x, y \geq h$, the intervening crossing probability is $\exp[-xy/\sinh(\Delta/2)]$. Therefore

$$\begin{aligned} F(t) &\leq G_0, \\ F(t) &\geq G_h - \{n \exp[-h^2/\sinh(\Delta/2)]\}^\lambda. \end{aligned} \quad (5.5)$$

Indeed the conditional error probability is first raised to λ ; Jensen bounds its expectation by its annealed probability to that power. Write $g_h(X)$ for the conditional private grid probability, so $G_h = \mathbb{E}_X g_h(X)^\lambda$. For $h \geq 0$ these grid events are nested for each fixed environment, so concavity and Jensen give $|G_h - G_0| \leq \mathbb{E}_X (g_0(X) - g_h(X))^\lambda \leq (\mathbb{E}_X (g_0(X) - g_h(X)))^\lambda$. The last expectation

is the difference of the two composite grid probabilities, bounded by the total variation under translation; for $h < 0$ reverse the nested events. Translation of the composite Gaussian grid gives

$$|G_h - G_0| \leq \left\{ \frac{|h|}{\sqrt{2\pi}} \sqrt{1 + n \tanh(\Delta/4)} \right\}^\lambda. \quad (5.6)$$

The squared translation norm is $h^2 \mathbf{1}^T \Sigma^{-1} \mathbf{1} = h^2 [1 + n \tanh(\Delta/4)]$, where $\Sigma_{ij} = e^{-|i-j|\Delta/2}$. Equal-covariance Gaussian total variation is at most that norm divided by $\sqrt{2\pi}$. Taking, for example, $h = \Delta^{1/4}$ makes both errors vanish. On a bounded cube the Gaussian density is uniformly continuous and the orthant integrand is bounded and monotone; displaced-threshold upper and lower cube sums, followed by Gaussian tail bounds, give bounded finite quadratures. Together with (5.5) and (5.6), these approximate $F(t)$ to arbitrary accuracy. $\square$

5.2 Tilted environments and the private endpoint filter

To join surviving segments, we must control the endpoints under the measure that their survival moments actually induce. Write $\nu_{t,\lambda}$ for the environmental probability measure with density $q_t^\lambda / F(t)$ relative to the OU law. We first compare these environmental tilts by stochastic order. We then control the private endpoint conditioned on survival against the fixed environment. This second estimate supplies the mixed endpoint bound needed at a junction.

Lemma 5.2 (Order of the tilted laws). *For $0 \leq \lambda \leq 1$,*

$$\nu_{t,0} \preceq_{\text{st}} \nu_{t,\lambda} \preceq_{\text{st}} \nu_{t,1}. \quad (5.7)$$

For each $a < 1/2$,

$$\sup_{t \geq 0, 0 \leq u \leq t} \mathbb{E}_{\nu_{t,\lambda}} e^{aX_u^2} < \infty. \quad (5.8)$$

A nonnegative decreasing environmental path functional has tilted expectation at most its ordinary expectation.

Proof. The order comparison follows from association of each intermediate tilt. The endpoint moment then follows from an explicit description of the fully annealed tilt.

On a finite grid, the nonpositive off-diagonal entries of the OU precision matrix make its density multivariate totally positive of order two (MTP₂). Set $Z = -Y$ and $a_\rho = \sqrt{\rho/(1-\rho)}$. The constraint $\{z_i \leq a_\rho x_i \ \forall i\}$ is closed under coordinatewise minimum and maximum, so its indicator is MTP₂. Multiply by the MTP₂ Gaussian density and marginalize over Z . This proves that $q(x)$ is MTP₂. To make this joint integral finite, one may insert an independent product Gaussian density in x ; its factors cancel from the MTP₂ inequality after marginalization.

It follows that $\varphi_\Sigma(x)q(x)^\lambda$ is MTP₂ and hence associated. We use here the closure and association properties in [11, Proposition 3.4(ii), equation (3.5) and Section 4.1]. Under this tilted law, association of an increasing test function with $q^{1-\lambda}$ gives the right comparison in (5.7). Ordinary Gaussian association gives the left comparison, which also yields the assertion for decreasing functionals.

Under $\nu_{t,1}$ rotate to $Z = \sqrt{\rho}X + \sqrt{1-\rho}Y$ and an independent standard OU process V . Only Z is conditioned to be positive. Its density at time u is

$$\frac{\varphi(z)\mathbb{P}_z(\tau_0 > u)\mathbb{P}_z(\tau_0 > t-u)}{a(t)} \mathbf{1}_{\{z>0\}}, \quad \mathbb{P}_z(\tau_0 > v) = 2\Phi\left(\frac{z}{\sqrt{e^v-1}}\right) - 1.$$

For each side of length at least one the latter probability is at most $Cze^{-v/2}$; a shorter side contributes at most one. Separating the cases with zero, one or two long sides and using (5.4) bounds the density by $C(1+z)^2\varphi(z)$, uniformly in u, t . The rotation gives uniform Gaussian exponential moments for X_u under $\nu_{t,1}$. Apply (5.7) to the increasing function $e^{a(x_+)^2}$ and the decreasing function $e^{a(x_-)^2}$ separately. This proves (5.8); dense grids give the path assertions. $\square$

Lemma 5.3 (Uniform private filter estimate). *On compact subsets of $0 < \lambda, \rho < 1$ there exist $a > 0$ and $C < \infty$ such that*

$$\mathbb{E}_X \left[e^{aX_t^2} \left(\mathbb{E}_Y [\mathbf{1}_{A_t} e^{aY_t^2} \mid X] \right)^\lambda \right] \leq CF(t). \quad (5.9)$$

The same assertion holds with simultaneous Gaussian weights at both initial and terminal endpoints, after decreasing a .

Proof. We control the private endpoint through its mode. Strong log-concavity controls fluctuations around the mode; contraction of the OU evolution controls the mode itself by a decreasing functional of the environment. The stochastic order from Lemma 5.2 then bounds that functional under the survival tilt.

Fix the environmental path and let the private wall be $Y_s \geq -a_\rho X_s$. The filtered endpoint density has the form $\mu_v(y) \propto \varphi(y)g_v(y)$, where g_v is increasing and log-concave. Indeed the conditional OU bridge mean is increasing in its endpoint, and its constraints are convex; marginal log-concavity is Prékopa's marginalization theorem [22, Theorem 6, p. 342], first on a finite grid and then by a limit. Thus μ_v is 1-strongly log-concave and its mode m_v is nonnegative. For any such density with mode m ,

$$|Z - m| \preceq_{\text{st}} |N(0, 1)|, \quad |\mathbb{E}Z - m| \leq \sqrt{2/\pi}. \quad (5.10)$$

To verify this directly, both functions $\mu(m+r)e^{r^2/2}$ and $\mu(m-r)e^{r^2/2}$ decrease for $r \geq 0$. Their sum therefore gives likelihood-ratio domination of the radial half-normal density after normalization. The density is also dominated in monotone likelihood-ratio order by $m + |N|$: on $y \geq m$ its ratio to $e^{-(y-m)^2/2}$ decreases, while on $y < m$ the comparison density is zero. OU transition kernels are TP_2 ; composition with killing indicators and integration preserve this property. The likelihood-ratio convention allowing zero densities agrees with [7, Theorem 2.1 and Remark 3.6]; kernel composition here follows from the MTP_2 marginalization just used. Consequently this order propagates through a killed OU segment. Raising a wall also increases its normalized endpoint law, by association under the original truncated MTP_2 path measure.

Consider a segment of length $v \in [1, 2]$, and set $r = e^{-v/2}$ and $B = \max(0, \sup_{\text{segment}}(-a_\rho X))$. By the propagated likelihood-ratio order and wall monotonicity, it suffices to use the larger incoming law $m + |N|$ and constant wall B .

If $rm \geq B + 2$, an OU-noise tube of radius one has fixed positive probability and enforces survival. The unconditional endpoint above rm has a uniform Gaussian tail; dividing by that fixed probability and integrating bounds the conditional mean by $rm + C$. If $rm < B + 2$, first retain the initial value in $[e(B+3), e(B+3)+1]$, an event of probability at least $ce^{-C(B+1)^2}$, and then use the same tube. Dividing the unconditional endpoint tail by this lower probability gives conditional mean at most $rm + C(B+1)$. Combining the two cases with (5.10) yields

$$m_{\text{new}} \leq e^{-v/2}m + C(B+1). \quad (5.11)$$

An initial segment of length at most one is handled only once: raise its wall to its supremum and retain a fixed initial interval above $e(B + 3)$. The same tube and tail argument bounds its mode by $C(B + 1)$. Thus the iteration uses intervals of uniformly positive length; the single short initial interval contributes only an initial bound.

Partition backwards from the terminal time into intervals I_j of length at most two. Iterating (5.11), including the initial truncation, gives

$$m_t(X) \leq C \left(1 + \sum_{j \geq 0} e^{-j/2} \sup_{s \in I_j} (-a_\rho X_s)_+ \right) =: \mathcal{M}_t(X). \quad (5.12)$$

OU suprema on bounded intervals have uniform Gaussian tails. Weighted Cauchy–Schwarz and Jensen imply $\sup_t \mathbb{E} e^{c\mathcal{M}_t^2} < \infty$ for some $c > 0$; independence of the intervals is not needed. The majorant is decreasing in X , so Lemma 5.2 transfers this bound to $\nu_{t,\lambda}$. Finally, for a small, (5.10) gives

$$\mathbb{E}[e^{aY_t^2} \mid A_t, X] \leq (1 - 4a)^{-1/2} e^{2am_t(X)^2}.$$

Multiply by q_t^λ , use (5.8) and the preceding mode bound, and apply Cauchy–Schwarz under $\nu_{t,\lambda}$. This proves (5.9). Reversibility treats the other endpoint; Cauchy–Schwarz first inside the private law and then in the environmental law treats both endpoints simultaneously. Throughout, the endpoint law is the private law conditioned on the original survival event. $\square$

The mixed filter estimate turns the rate enclosure into a constant-factor asymptotic. It also supplies positive endpoint boxes in which two surviving pieces can be connected.

Theorem 5.4 (Constant-factor Gaussian survival). *Uniformly on compact subsets of $0 < \lambda, \rho < 1$,*

$$F(s)F(t) \leq F(s+t) \leq CF(s)F(t), \quad C^{-1}e^{-\lambda t} \leq F(t) \leq e^{-\lambda t}. \quad (5.13)$$

The second comparison is valid for $t \geq 1$. There exist compatible, sufficiently wide fixed positive endpoint boxes for both Gaussian channels that can be imposed at a constant cost. These are the boxes constructed below, or fixed boxes containing them; this assertion does not prescribe an arbitrary narrow endpoint box.

Proof. The Mehler density relative to two independent standard Gaussians is

$$k_r(x, y) = \frac{1}{\sqrt{1-r^2}} \exp \left(\frac{2rxy - r^2(x^2 + y^2)}{2(1-r^2)} \right) \leq \frac{1}{\sqrt{1-r^2}} \exp \left(\frac{r(x^2 + y^2)}{2(1+r)} \right).$$

Choose a fixed gap g so large that $r = e^{-g/2}$ makes the exponent smaller than the coefficient allowed in Lemma 5.3. Drop survival across the gap. Apply this density bound inside the private conditional integral and then to the environmental transition. The two reference blocks are independent; (5.9) at their adjacent endpoints gives $F(s+g+t) \leq CF(s)F(t)$. On the other hand, (5.3) gives $F(s+g+t) \geq F(s+t)F(g)$. This proves the first comparison. Iteration at nt and Proposition 5.1 yield the second one.

By (5.9), excluding any of the four Gaussian endpoints from $[-R, R]$ costs at most $Ce^{-cR^2}F(t)$ in fractional mass. Choose fixed R so that their union costs at most $F(t)/2$. Add to each channel $A(e^{-s/2} + e^{-(t-s)/2})$, with any fixed $A \geq 2R$. This preserves survival and moves all retained endpoints into the fixed positive box $[A-R, A(1+e^{-1/2})+R]$, uniformly for $t \geq 1$. The Cameron–Martin translation formula [3], in the form including an independent Gaussian initial value given in [1, Theorem 2.1], applies after the OU-to-Brownian time change. The density of this shift

depends only on those endpoints and is bounded above and below on the box, uniformly for $t \geq 1$. Change variables first in the private integral and then in the environmental integral. The positive-box event therefore retains a fixed fraction of $F(t)$. A required fixed positive lower endpoint level can be accommodated by choosing A first and then retaining this sufficiently wide box (or a fixed containing box). $\square$

5.3 Parameter stability

Exact count conditioning changes the binary proportions and gap means slightly. The next estimate shows that these changes can be made in the persistence exponent without losing uniform comparisons later. Its proof is independent of the size of the time grid.

Proposition 5.5 (Local Lipschitz continuity). *On each compact subset of $(0, 1)^2$,*

$$|\chi(\lambda, \rho) - \chi(\lambda', \rho')| \leq C(|\lambda - \lambda'| + |\rho - \rho'|).$$

Proof. Whiten a finite Gaussian grid. Its survival indicator is a fixed function f , and $q_\rho = T_s f$, where T_s is the Gaussian noise semigroup and $e^{-s} = \sqrt{\rho}$. In any unit direction Gaussian integration by parts and the one-dimensional rearrangement inequality give

$$|\nabla q_\rho| \leq \sqrt{\frac{\rho}{1-\rho}} I(q_\rho), \quad I(v) = \varphi(\Phi^{-1}v).$$

For clarity, the rearrangement step fixes the value $v = \mathbb{E}f$ and maximizes $|\mathbb{E}[Z_1 f(Z)]|$ over $0 \leq f \leq 1$; the maximizer is a half-space in the first coordinate, with value $I(v)$. Gaussian tail bounds give $I(v)^2 \leq C v^2 (1 - \log v)$, including both endpoints. Differentiating under the Gaussian integral and integrating by parts,

$$\partial_s \log \mathbb{E} q_s^\lambda = \lambda(1 - \lambda) \frac{\mathbb{E} q_s^{\lambda-2} |\nabla q_s|^2}{\mathbb{E} q_s^\lambda}.$$

Convexity of $K(\lambda) = \log \mathbb{E} q^\lambda$, with $K(0) = 0$, gives $\mathbb{E}_{q^\lambda}[-\log q] \leq -K(\lambda)/\lambda$. Since $\mathbb{E} q^\lambda \geq a(t) \geq \pi^{-1} e^{-t/2}$, both the λ derivative and, on a compact ρ interval, the ρ derivative are bounded in absolute value by $C(t+1)$, independently of the grid dimension. Smooth approximation justifies the derivatives when f is an indicator. Integrating these bounds, passing to dense grids, dividing by t , and taking $t \rightarrow \infty$ proves the assertion. $\square$

5.4 The positive persistence spectrum

The preceding estimates concern fractional orders. Allowing every positive order reveals the structure of the same conditional survival cost: higher orders increasingly favor environments with larger private survival probability. We first extend the constant-factor estimate, then obtain regularity from a dimension-free curvature inequality.

For the same conditional probability q_t in (5.1), define, for every $p > 0$ and $0 < \rho < 1$,

$$F_{p,\rho}(t) = \mathbb{E} q_t^p, \quad b_t(p, \rho) = -\log F_{p,\rho}(t).$$

Every continuous environmental path is bounded on a compact time interval. A sufficiently high private OU tube therefore has positive probability and satisfies its survival constraint. The initial constraint also gives $q_t \leq \Phi(\beta X_0) < 1$, where $\beta = \sqrt{\rho/(1-\rho)}$. Thus $0 < q_t < 1$ almost surely and $S_t = -\log q_t$ is finite.

Proposition 5.6 (Positive orders and finite-time error). *The persistence exponent*

$$\chi(p, \rho) = \lim_{t \rightarrow \infty} b_t(p, \rho)/t$$

exists for every $p > 0$ and $0 < \rho < 1$. If $p_g = (1 + e^{-g/2})/(1 - e^{-g/2})$, then

$$\frac{b_t(p, \rho)}{p_g(t + g)} \leq \chi(p, \rho) \leq \frac{b_t(p, \rho)}{t}, \quad \frac{\min(p, 1)}{2} \leq \chi(p, \rho) \leq \frac{\max(p, 1)}{2}.$$

On every compact subset of $(0, \infty) \times (0, 1)$,

$$b_t(p, \rho) = t\chi(p, \rho) + O(1), \quad t \geq 1. \quad (5.14)$$

Proof. The association proof of (5.3) uses only positivity of the moment order. It gives $F_{p,\rho}(s+t) \geq F_{p,\rho}(s)F_{p,\rho}(t)$ for every $p > 0$. For $p \leq 1$, $a(t) \leq F_{p,\rho}(t) \leq a(t)^p$; for $p \geq 1$, $a(t)^p \leq F_{p,\rho}(t) \leq a(t)$, by Jensen and $0 \leq q_t \leq 1$. Consequently

$$0 \leq b_t(p, \rho) \leq \max(p, 1)[t/2 + \log \pi]. \quad (5.15)$$

This proves the subadditive limit and its range. The two Gaussian Hölder applications in Proposition 5.1 likewise apply to all $p > 0$: their private bound is $q_{\text{joint}} \leq \prod_i q_i^{1/p_g}$ and the environmental bound is $\mathbb{E} \prod_i q_i^{p/p_g} \leq F_{p,\rho}(t)^{N/p_g}$. This proves the effective lower bound without an additional factor p_g .

The remaining task is the endpoint estimate for orders above one. Integer orders admit a useful representation by several independent private channels; stochastic order then covers the intervening orders. On a finite OU grid, q_t is increasing and MTP₂; hence the environmental law $d\nu_{t,p} \propto q_t^p d\mathbb{P}_X$ is associated. If n is an integer with $n \geq p$, association under $\nu_{t,p}$ applied to an increasing test function and q_t^{n-p} gives

$$\nu_{t,0} \preceq_{\text{st}} \nu_{t,p} \preceq_{\text{st}} \nu_{t,n}. \quad (5.16)$$

Dense grids pass these inequalities to the path law, including nonnegative monotone endpoint and short-interval supremum functions by truncation and monotone convergence.

To control $\nu_{t,n}$, take independent stationary standard OU processes $U = (X, Y_1, \dots, Y_n)$ and the cone

$$C = \{(x, y_1, \dots, y_n) : y_j + \beta x \geq 0 \text{ for all } j\}.$$

Write $p_C(t) = \mathbb{P}(U_s \in C \text{ for } 0 \leq s \leq t)$. The environmental marginal under this conditioning is exactly $\nu_{t,n}$. For $0 < a < 1/2$, let $v = (1 - 2a)^{-1}$ and $f_v(h) = \log(1 + (e^h - 1)/v)$. Stationarity, the Markov property and the OU representation $U_h = e^{-h/2}(U_0 + B_{e^h-1})$ give the exact identity

$$\mathbb{E}[e^{a|U_s|^2}; U_{[0,t]} \subset C] = v^{(n+1)/2} p_C(f_v(s) + f_v(t-s)), \quad 0 \leq s \leq t. \quad (5.17)$$

Indeed the weight changes the variance of U_s from 1 to v ; the past and future are conditionally independent OU branches. Dividing each branch by $\sqrt{v}$ and changing Brownian time yields a standard OU branch of duration $f_v(h)$. The remaining deterministic multiplier is positive and does not affect cone membership.

The constraint signals $Y_j + \beta X$ have nonnegative covariances. Gaussian association therefore gives $p_C(s+t) \geq p_C(s)p_C(t)$. Since $0 \leq h - f_v(h) \leq \log v$, the ratio of the survival probabilities

in (5.17) is at most $1/p_C(2\log v)$. The positive orthant is contained in C , so $p_C(2\log v) \geq a(2\log v)^{n+1} > 0$. In particular,

$$\sup_{t \geq 0, 0 \leq s \leq t} \mathbb{E}_{\nu_{t,n}} e^{aX_s^2} < \infty.$$

Apply (5.16) separately to $e^{a(X_s^+)^2}$ and $e^{a(X_s^-)^2}$, the latter being decreasing. We obtain the same endpoint bound under $\nu_{t,p}$. The dominating integer n can be fixed uniformly on a compact positive order set.

The deterministic private filter proof of Lemma 5.3 does not use $p < 1$: its endpoint mode majorant is a decreasing environmental functional and hence has, by (5.16), its ordinary Gaussian exponential bound under every $\nu_{t,p}$. Combining that bound with the preceding environmental endpoint bound and Cauchy–Schwarz gives, for sufficiently small $a > 0$,

$$\mathbb{E}_X \left[e^{a(X_0^2 + X_t^2)} \left(\mathbb{E}_Y [\mathbf{1}_{A_t} e^{a(Y_0^2 + Y_t^2)} \mid X] \right)^p \right] \leq C F_{p,\rho}(t).$$

All constants are locally uniform. The Mehler gap argument in Theorem 5.4 now gives $F_{p,\rho}(s+t) \leq C F_{p,\rho}(s) F_{p,\rho}(t)$ for all positive orders. Iterating the two product bounds and taking the limiting slope proves (5.14). $\square$

The extension above identifies the exponential rate at every positive order. To differentiate that rate, we need a variance bound for the conditional cost under its tilted laws. The next lemma provides this bound for any convex Gaussian constraint.

Lemma 5.7 (Convex Gaussian probability curvature). *Let $C \subset \mathbb{R}^d$ be convex with positive Gaussian measure, $Z \sim N(0, I_d)$, and $A(x) = -\log \mathbb{P}(Z + \beta x \in C)$, where $\beta > 0$. Then*

$$0 \preceq D^2 A \preceq \beta^2 I_d, \quad |\nabla A|^2 \leq 2\beta^2 A. \quad (5.18)$$

For $d\nu_p \propto e^{-pA} d\gamma_d$, $p > 0$,

$$\text{Var}_{\nu_p}(A) \leq 2\beta^2 \mathbb{E}_{\nu_p} A. \quad (5.19)$$

Proof. The Gaussian convolution is positive and smooth. If U_x is $N(\beta x, I_d)$ conditioned to lie in C , differentiation gives $D^2 A = \beta^2(I_d - \text{Cov}(U_x))$. The Brascamp–Lieb variance inequality [2], in the form [4, equation (1.3)], gives $\text{Cov}(U_x) \preceq I_d$. For completeness, the hard convex restriction follows by first using smooth convex approximations to the potential $|u - \beta x|^2/2 + N \text{dist}(u, C)^2$ and then letting $N \rightarrow \infty$. The Hessian is at least I_d before taking the limit, and Gaussian domination passes means and covariances to the conditioned law. A convex set of positive Gaussian measure has nonempty interior and its boundary has Lebesgue measure zero, so replacing it by its closure does not change these probabilities. This proves both Hessian bounds. Since $A \geq 0$ and its Hessian is bounded above by $\beta^2 I_d$,

$$0 \leq A(x - \beta^{-2} \nabla A(x)) \leq A(x) - |\nabla A(x)|^2 / (2\beta^2).$$

Finally the potential $|x|^2/2 + pA(x)$ has Hessian at least I_d . Brascamp–Lieb, followed by this gradient bound, proves (5.19). $\square$

Theorem 5.8 (Structure and regularity of the spectrum). *For each $0 < \rho < 1$, the function $p \mapsto \chi(p, \rho)$ extends continuously to $p = 0$ with $\chi(0, \rho) = 0$, is concave, and belongs to $C_{\text{loc}}^{1,1}((0, \infty))$. It is strictly increasing, with*

$$\partial_p \chi(p, \rho) > 0, \quad \chi(1, \rho) = \frac{1}{2}, \quad \lim_{p \rightarrow \infty} \chi(p, \rho) = \infty, \quad \lim_{p \rightarrow \infty} \frac{\chi(p, \rho)}{p} = \lim_{p \rightarrow \infty} \partial_p \chi(p, \rho) = 0.$$

The function χ is jointly locally Lipschitz in (p, ρ) , and $\partial_p \chi$ is jointly continuous. On compact positive parameter sets,

$$|\partial_p \chi(p, \rho) - \partial_p \chi(p, \rho')| \leq C|\rho - \rho'|^{1/2}, \quad (5.20)$$

$$\partial_p b_t(p, \rho) = t \partial_p \chi(p, \rho) + O(\sqrt{t}), \quad t \geq 1. \quad (5.21)$$

For fixed $p < 1$, $\chi(p, \rho)$ is nondecreasing in ρ ; for fixed $p > 1$ it is nonincreasing in ρ .

Proof. The curvature lemma gives uniform control of the first two order derivatives on finite grids. We pass those bounds to continuous paths and to the limiting rate, determine the order boundaries, and finally treat the dependence on the correlation parameter.

For a finite time grid with covariance Σ , whiten both Gaussian channels. The survival cone becomes $C = \Sigma^{-1/2} \mathbb{R}_+^d$, and the conditional negative log probability is the A of Lemma 5.7. For $b = -\log \mathbb{E} e^{-pA}$,

$$b' = \mathbb{E}_{\nu_p} A \geq 0, \quad -b'' = \text{Var}_{\nu_p}(A) \leq 2\beta^2 b'.$$

Concavity and $b(0) = 0$ give $b'(p) \leq b(p)/p$. The annealed grid survival is at least $a(t)$, so (5.15) also bounds its b . Thus for $t \geq 1$ the functions b_t/t have bounded first derivatives and Lipschitz first derivatives, uniformly on compact positive parameter sets and independently of the grid dimension.

Refine nested dense grids that include both endpoints. Their conditional probabilities decrease to q_t , and the negative logarithms increase to S_t . On every compact positive p interval, $x^j e^{-px}$ is bounded uniformly for $j = 0, 1, 2$ and $x \geq 0$. Dominated convergence therefore passes the first two order derivatives and the variance inequality to continuous paths, with positive normalizers throughout.

The first derivatives of b_t/t are equicontinuous. Any convergent subsequence of them can be integrated over a compact order interval; the pointwise limiting slope then identifies its limit as the derivative of the same limiting function. Hence the limit is C^1 , its derivative is locally Lipschitz, and the derivatives converge locally uniformly. Concavity and monotonicity pass to the limit as well.

For a fixed $t > 0$, $b_t(p, \rho) \rightarrow 0$ as $p \downarrow 0$, so the upper effective bound proves $\chi(p, \rho) \rightarrow 0$. For fixed $t, g > 0$, $q_t < 1$ almost surely gives $b_t(p, \rho) \rightarrow \infty$ as $p \rightarrow \infty$ by dominated convergence. The lower effective bound then proves $\chi(p, \rho) \rightarrow \infty$. Concavity and nondecreasingness imply $\partial_p \chi > 0$: a zero derivative at a finite positive order would force the function to be constant thereafter. For fixed t , the essential supremum of q_t is 1. Indeed, the environmental event $\inf_{[0,t]} X_s \geq M$ has positive probability and on it $q_t \geq \mathbb{P}(\inf_{[0,t]} Y_s \geq -\beta M) \rightarrow 1$ as $M \rightarrow \infty$. Hence $b_t(p, \rho)/p \rightarrow 0$; explicitly, for each $\epsilon > 0$, $F_{p,\rho}(t) \geq \mathbb{P}(S_t \leq \epsilon) e^{-p\epsilon}$. The upper effective bound proves $\chi(p, \rho)/p \rightarrow 0$. Concavity gives $0 \leq \partial_p \chi \leq \chi/p$, proving the last limit. The identity at $p = 1$ follows from (5.4).

For dependence on ρ , the whitened grid probability is $q_\rho = T_u \mathbf{1}_C$ with $u = -\frac{1}{2} \log \rho$; the cone is unchanged because it is invariant under positive scaling. The semigroup calculation in Proposition 5.5, with p in place of λ , gives

$$\partial_u \log \mathbb{E} q_\rho^p = p(1-p) \frac{\mathbb{E} q_\rho^{p-2} |\nabla q_\rho|^2}{\mathbb{E} q_\rho^p}.$$

The dimension-free gradient bound and Gaussian tail estimate used there show that its absolute value is at most $C(1 + \mathbb{E}_{\nu_p} S_t)$ on compact positive parameter sets. Since $\mathbb{E}_{\nu_p} S_t = b'_t(p) \leq b_t(p)/p$, (5.15) bounds this by $C(t+1)$. All approximation limits can be justified with a slightly smaller positive order, which dominates the logarithmic factors. Integration of these parameter bounds

proves joint local Lipschitz continuity after taking limiting slopes. The displayed sign also proves the stated monotonicity in ρ .

By (5.14), b_t/t approximates χ with error $O(t^{-1})$. Comparing derivatives with difference quotients of length $t^{-1/2}$, using their uniform Lipschitz bounds in p , gives (5.21). Finally compare difference quotients of χ at ρ and ρ' with a common order step $|\rho - \rho'|^{1/2}$. Joint Lipschitz continuity of χ and the uniform Lipschitz bound on its order derivative give (5.20), hence joint continuity. $\square$

5.5 A strict annealing gap

Jensen bounds the fractional exponent below by the fractional multiple of the ordinary exponent. The next proposition shows that this bound is always strict. The mechanism is a separation between the typical environment and the environment selected by ordinary survival: their time averages differ by an amount proportional to the horizon.

Proposition 5.9 (Strict annealing gap). *For $0 < p < 1$ and $0 < \rho < 1$,*

$$\chi(p, \rho) - \frac{p}{2} \geq \frac{\rho}{16\pi} \min\{p, 1 - p\} > 0.$$

In particular the gap is uniformly positive on every compact subset of $(0, 1)^2$.

Proof. Let P_t be the law of the environmental path X on $[0, t]$, and let Q_t be its law conditional on A_t . Its density is

$$f_t = \frac{dQ_t}{dP_t} = \frac{q_t}{a(t)}.$$

For $H_t = \int_0^t X_s ds$, direct integration of the covariance gives

$$v_t = \text{Var}_{P_t} H_t = 4t - 8(1 - e^{-t/2}) \leq 4t.$$

Rotate the two independent OU channels to $Z = \sqrt{\rho}X + \sqrt{1 - \rho}Y$ and an independent standard OU channel V , so that $X = \sqrt{\rho}Z + \sqrt{1 - \rho}V$. Only Z is conditioned. By stationarity, reversibility and the Markov property, its density at $s \in [0, t]$ conditional on A_t is proportional to

$$\varphi(z)h_s(z)h_{t-s}(z)\mathbf{1}_{\{z>0\}}, \quad h_u(z) = \mathbb{P}_z(\tau_0 > u).$$

For $u > 0$ the Brownian time change gives $h_u(z) = 2\Phi(z/\sqrt{e^u - 1}) - 1$; set $h_0(z) = 1$ for $z > 0$. Both factors are increasing in z . Weighting a positive half-normal by their increasing product increases its mean: this follows by integrating $(z - z')[h_s(z)h_{t-s}(z) - h_s(z')h_{t-s}(z')] \geq 0$ against two independent copies of that half-normal law. Hence

$$\mathbb{E}_{Q_t} H_t \geq \sqrt{2\rho/\pi} t =: m_0 t.$$

To turn this separation of means into an exponential separation of the two laws, we use concentration under the convex survival constraint. If G is a standard finite-dimensional Gaussian vector, C is a convex set of positive Gaussian measure, and $L = \ell \cdot G$, completion of the square gives

$$\log \mathbb{E}[e^{\theta L} \mid G \in C] = \frac{\theta^2 |\ell|^2}{2} + \log \mathbb{P}(G + \theta \ell \in C) - \log \mathbb{P}(G \in C).$$

The translated probability is log-concave in θ by Prékopa's marginalization theorem [22, Theorem 6, p. 342]. Its tangent at zero therefore proves

$$\mathbb{E}[e^{\theta(L - \mathbb{E}[L|G \in C])} \mid G \in C] \leq e^{\theta^2 \text{Var}(L)/2}, \quad \theta \in \mathbb{R}.$$

Apply this fact to the jointly Gaussian vector comprising H_t and a finite survival grid; if its covariance is singular, represent it on its Gaussian linear support before whitening. The grid constraint is convex. Along nested dense grids its probabilities decrease to $a(t) > 0$. Gaussian integrability passes the means and exponential moments to the limit. Consequently H_t under $\mathbf{Q}_t$ is sub-Gaussian about its mean with variance proxy v_t .

Set $B_t = \{H_t \geq m_0 t/2\}$. Chernoff's inequality under both laws then gives, for $t > 0$,

$$\mathbf{P}_t(B_t) \leq e^{-\rho t/(16\pi)}, \quad \mathbf{Q}_t(B_t^c) \leq e^{-\rho t/(16\pi)}.$$

Hölder's inequality, separately on B_t and its complement, yields

$$\begin{aligned} \int f_t^p d\mathbf{P}_t &\leq \mathbf{Q}_t(B_t)^p \mathbf{P}_t(B_t)^{1-p} + \mathbf{Q}_t(B_t^c)^p \mathbf{P}_t(B_t^c)^{1-p} \\ &\leq 2 \exp \left[-\frac{\rho t}{16\pi} \min\{p, 1-p\} \right]. \end{aligned}$$

Since $F_{p,\rho}(t) = a(t)^p \int f_t^p d\mathbf{P}_t$ and $-\log a(t) = t/2 + O(1)$, taking limiting logarithmic slopes proves the proposition. $\square$

Remark 5.10. For $0 < p < 1$, define $D_p(\mathbf{Q}_t \parallel \mathbf{P}_t) = (p-1)^{-1} \log \int f_t^p d\mathbf{P}_t$. The exact identity

$$-\log F_{p,\rho}(t) = -p \log a(t) + (1-p) D_p(\mathbf{Q}_t \parallel \mathbf{P}_t)$$

expresses the annealing gap as a positive information cost per unit time for the change in environmental law induced by survival.

5.6 Scalar deviations and the quenched exponent

The structural theory also describes the distribution of the conditional cost $S_t = -\log q_t$. Appendix G proves a full large-deviation principle for S_t/t , with a good convex rate I_ρ that is finite precisely on $(0, \infty)$. Its limiting cumulant $\Lambda_\rho(\theta)$ is finite precisely for $\theta < (1-\rho)/\rho$ and is locally $C^{1,1}$ there. In particular,

$$\frac{S_t}{t} \longrightarrow \gamma_\rho \quad \text{almost surely,} \quad \gamma_\rho = \Lambda'_\rho(0) = \partial_p \chi(0+, \rho), \quad \frac{1}{2} + \frac{\rho}{16\pi} \leq \gamma_\rho \leq \frac{1}{2(1-\rho)}.$$

The extended spectrum and the rate function are convex duals, and positive-order environmental tilts select the scalar value $\partial_p \chi(p, \rho)$. This gives a probabilistic interpretation of the spectrum: its order derivative is the conditional cost selected by the corresponding environmental tilt. The assertions here concern the stationary OU conditional law. The arithmetic arguments below use the fractional-order estimates, with their own separately defined accepting measures.

5.7 Integrable walls and Brownian reserves

In the marked walks, an initial reserve, a small drift, and a curved wall all perturb the flat Gaussian constraint. After Lamperti time change, these perturbations decay from one or both endpoints. We record a wall comparison in a form that treats all three through their Cameron–Martin action and coefficient variation.

For an absolutely continuous shift h on $[0, t]$ whose first derivative has bounded variation, define

$$\|h\|_{\mathcal{H}}^2 = h(0)^2 + \int_0^t (h' + h/2)^2 ds,$$

$$J_t(h) = |h(0)/2 - h'(0)| + |h'(t) + h(t)/2| + \|(h/4) ds - dh'\|_{\text{TV},(0,t)}.$$

Endpoint derivatives are one-sided traces. In expressions containing h'' , use the distributional derivative of h' : the last term is the total variation of the signed measure $(h/4) ds - dh'$ on $(0, t)$. Thus jumps of the first derivative are included, whereas jumps of h itself are excluded. The corresponding linear integral below is a signed-measure integral against the continuous OU path.

Lemma 5.11 (Bounded-action walls). *For $b/\sqrt{\rho}$ in this class, replacing the zero composite OU wall by a deterministic wall b changes $F(t)$ by a factor bounded above and below whenever $J_t(b/\sqrt{\rho}) + \|b/\sqrt{\rho}\|_{\mathcal{H}}^2$ is bounded. An endpoint-restricted version holds with compatible sufficiently wide boxes chosen after the wall bound, or with the endpoint boxes supplied by Theorem 5.4 translated together with the path.*

Proof. The Cameron–Martin linear functional is

$$L_h(X) = (h(0)/2 - h'(0))X_0 + (h'(t) + h(t)/2)X_t + \int_{(0,t)} X_s((h(s)/4) ds - dh'(s)).$$

Environmental translation alone gives exactly $F_b(t)/F(t) = \mathbb{E}_{\nu_{t,\lambda}} \exp[-L_h(X) - \|h\|_{\mathcal{H}}^2/2]$, with $h = b/\sqrt{\rho}$ and the consistent translation sign. The absolute variation of the coefficients of L_h is bounded. Jensen with its normalized absolute coefficient measure and (5.8) bounds both exponential signs uniformly. Jensen also supplies a positive lower bound, or equivalently use the bound for $\mathbb{E}_{\nu}|L_h|$. Mixed endpoint moments, followed by Cauchy–Schwarz and truncation, give the same conclusion for compatible sufficiently wide boxes chosen after the fixed wall bound. Equivalently, translate the endpoint boxes in the same environmental change of variables. A fixed feasible interior margin and a finite initial or terminal tube permit the sufficiently wide compatible central boxes used below; their constants are chosen after the wall and drift bounds. The comparison holds at the original power λ . $\square$

Corollary 5.12 (Brownian one-end estimate). *Let independent Brownian channels have environmental variance σ_E^2 and private variance σ_C^2 , bounded above and below by positive constants. Put $\rho = \sigma_E^2/(\sigma_E^2 + \sigma_C^2)$. From reserve $h \geq 1$, over duration n , with $|\delta|\sqrt{n} \leq D$, the conditional private survival probability Q satisfies*

$$\mathbb{E}Q^\lambda \asymp B_n(h) := \min\{1, ((1+h)/\sqrt{n})^{2\chi(\lambda,\rho)}\}. \quad (5.22)$$

The upper bound allows a fixed soft wall $-h - gs^\gamma$, $0 < \gamma < 1/2$. For a sufficiently large fixed minimum reserve, the lower bound permits a strong wall $-h + gs^\gamma$, positive central terminal risk, and compatible sufficiently wide central terminal boxes of the separate channels. Choose these boxes after the compact-law, wall, and drift bounds; their locations and widths are then fixed uniformly in n, h .

Proof. For $h^2 < n$, scale from time h^2 and use Lamperti time $u = \log(s/h^2)$. Its length is $\log(n/h^2)$. Reserve, power wall and bounded scaled drift become bounded multiples of

$$e^{-u/2}, \quad e^{-(1/2-\gamma)u}, \quad e^{-(\log(n/h^2)-u)/2}.$$

Their J and Hilbert norms are bounded. Drop the initial time interval for the upper bound and apply Lemma 5.11 and Theorem 5.4. For the lower bound prepare, on a fixed multiple of h^2 , both channels in positive central boxes using a fixed Brownian tube. Brownian scaling makes

its probability and its endpoint transition density bounds uniform. Thereafter use the endpoint version of those lemmas. The preparation has fixed cost before the fractional power as well as after it. A large fixed minimum reserve makes the normalized strong-wall amplitude as small as required. If h is comparable to or larger than $\sqrt{n}$, a fixed-scale positive tube and monotonicity give a constant lower bound; the upper bound is one. Reflection of both channels proves the same assertions for the opposite risk sign. $\square$

5.8 Independent exponential gaps: relative coupling

We now transfer the Gaussian estimate to a rank walk. The gap sequence is the environment and the marks form the private channel. A coupling with small absolute failure probability is insufficient when survival is rare. We therefore establish preparation and reserve comparisons first, use them to bound each coupling failure relative to survival, and only then insert the Gaussian wall estimate.

Let C_i be iid marks in a fixed finite alphabet, with probabilities in a compact subset of its open simplex. Their weights have two values,

$$w(C_i) = l - bB_i \in [w_*, l], \quad p = \mathbb{P}(B_i = 1), \quad \omega = l - bp,$$

where all parameters range over compact sets, $b, w_* > 0$, and p stays away from zero and one. Independent gaps E_i have exponential mean τ in a positive compact interval. Write

$$S_j = \sum_{i \leq j} (w(C_i) - E_i), \quad \delta = \omega - \tau, \quad \rho_\tau = \frac{\tau^2}{\tau^2 + b^2 p(1-p)}, \quad F_n(h) = \mathbb{E}\mathbb{P}_C\{\min_{j \leq n} S_j \geq -h \mid E\}^\lambda. \quad (5.23)$$

The following arguments also apply to $-S$, with all monotonicities reversed together. Assume initially $|\delta|\sqrt{n} \leq D$.

Lemma 5.13 (Preparation and reserve comparisons). *Uniformly for $h \geq 1$, $u \geq 0$,*

$$F_n(h) \geq c_D n^{-1/2}, \quad F_n(h+u) \leq C_D(1+u)F_n(h), \quad F_{2n}(h) \geq c_D F_n(h). \quad (5.24)$$

The drift bound for a doubled interval is enlarged by a fixed factor. The preparation comparison also applies to a negative soft wall $-gj^\gamma$. Positive walls are treated using flat survival to bound the exceptional events and the strong Gaussian lower estimate; see Section B.2.

Proof. The preparation raises the walk to a prescribed height while keeping both channel totals central. Optional stopping supplies the entrance probability, and a positive scaled tube supplies the continuation. For the centered walk stopped on leaving $[-h_0, B]$, upper overshoots are bounded and lower overshoots have exponential tails; for the reversed walk these roles are interchanged. Optional stopping of the walk gives an upper-exit probability comparable to $1/B$, from a fixed reserve $h_0 > 0$. Stopping its square minus the variance times time bounds the mean exit time by CB : on an upper exit its squared displacement is $O(B^2)$ and that exit has probability $O(1/B)$; on a lower exit the second moment is bounded. Truncation first justifies both stopping identities. By Markov's inequality, a sufficiently large fixed multiple of B^2 retains a fixed fraction of the upper exits. From height B , a positive scaled tube over a further comparable time has probability bounded below by the functional central limit theorem, uniformly on our compact laws. It can end above any prescribed fixed multiple of B . The lower-overshoot memoryless law and bounded upper overshoot provide uniformity at entrance. The stopped centered gap and

mark martingales have second moments at most $C\mathbb{E}\tau_{\text{exit}} \leq CB$. Their probability of exceeding RB before the retained entrance is therefore at most $C/(R^2B)$. Choosing fixed R large retains a fixed fraction of the B^{-1} entrance probability with both channel sums central; the subsequent tube preserves centrality at the final time. Small drift is handled by changing the exponential mean on the retained central gap-sum tube: if $m = O(B^2)$ and $|\delta|\sqrt{m} = O_D(1)$, its log density ratio is $O_D(1)$ there. The same proof, or its sign reversal, therefore gives a preparation of length $O((1+u)^2)$ from fixed reserve to height at least u , of annealed probability at least $c_D/(1+u)$.

For $u \leq \sqrt{n}$, let p_u be its conditional private probability. Concatenating with an independent n -step survival from $h+u$, and using $\mathbb{E}p_u^\lambda \geq \mathbb{E}p_u$, gives $F_{n+m}(h) \geq c_D(1+u)^{-1}F_n(h+u)$. Time monotonicity gives the middle inequality. The same preparation and tube give ordinary survival at least $c_D/\sqrt{n}$, hence the first inequality. For $u > \sqrt{n}$ these two inequalities and $F_n(h+u) \leq 1$ suffice.

For doubling, discard environments whose centered prefix gap sum is unfavorable by more than $R\sqrt{n}$. Survival has the opposite coordinate monotonicity, so product association bounds the discarded F_n -weighted mass by the unweighted probability times F_n . Choose R large enough to retain a fixed fraction.

On each retained environment, impose the favorable private count tail above $(R+D+c)\sqrt{n}$, or below its negative threshold in the reversed orientation. A fixed central subinterval of this tail gives a positive probability lower bound. The event retained is the whole one-sided tail, with no upper truncation. Its monotonicity agrees with survival, so association preserves the same fraction of the conditional survival probability. Its ending reserve is at least $c\sqrt{n}$.

From this height, an independent further n steps survive with annealed probability bounded below by a constant. If f is the prefix mark probability and Z is the completed probability at the future environment, then $0 \leq Z \leq f$ and $\mathbb{E}Z \geq cf$. Consequently $\mathbb{E}Z^\lambda \geq cf^\lambda$, and integration proves doubling. Section B.6 supplies the finite conventions and explicit constants.

The same preparation works for a negative soft wall: after a prefix of length m , the future condition at $-gj^\gamma$ is stronger than the required condition at $-g(m+j)^\gamma$. For a positive strong wall, use the flat bad-event bound and absorb its loss against the bounded-action Gaussian lower comparison, as shown in Section B.2. All preliminary preparation constants are fixed before any value of χ is used. $\square$

Lemma 5.14 (Relative cost of a bad block). *Generate the environmental and private couplings separately on independent deterministic blocks. A block's sigma-field includes both original channels, their coupled Gaussian paths, and every coupling auxiliary. For the early estimate, let $\mathcal{D}$ be measurable with respect to whole blocks contained in the first $b \leq n/2$ slots; the sigma-field of these blocks is independent of all variables used by the remaining slots. Without coupling auxiliaries, it is enough that $\mathcal{D}$ depend on the first b original slots. Then*

$$\mathbb{E}\mathbb{P}_C(\text{survival}, \mathcal{D} \mid \text{extended environment})^\lambda \leq C_D(1+b)^2\mathbb{P}(\mathcal{D})^{\lambda/4}F_n(h). \quad (5.25)$$

For the late estimate, assume that the whole block sigma-field read by $\mathcal{D}$ is independent of the whole block sigma-field used by survival on the first $\lfloor n/16 \rfloor$ slots, including every auxiliary on both sides. Then its relative cost is at most $C_D\mathbb{P}(\mathcal{D})^\lambda F_n(h)$. In particular this applies when every block read by $\mathcal{D}$ starts after $n/4$. Fixed fractions in these statements may be changed, with corresponding constants. The same bounds hold for the Gaussian comparison walk and for both risk signs.

Proof. If positive jumps are bounded, discard prefix survival and use the endpoint bound $S_b \leq lb$. Independence of the remainder, Jensen over the prefix, and Lemma 5.13 give the stronger upper

bound $C(1+b)\mathbb{P}(\mathcal{D})^\lambda F_n(h)$. For unbounded positive jumps divide positive prefix endpoints into unit bins $[k, k+1)$. Their probabilities are bounded both by $\mathbb{P}(\mathcal{D})$ and by $C \exp[-c \min(k^2/b, k)]$, with the fixed $O_D(\sqrt{b})$ drift displacement absorbed in the constants and a first group of bins. Apply the reserve comparison bin by bin. Using the geometric mean of the two probability bounds and summing the Gaussian/exponential tail proves (5.25); the displayed exponent $\lambda/4$ and quadratic polynomial are deliberately weaker than necessary. Negative endpoints can only decrease the available reserve. The Gaussian case has the same tail estimate. For the late estimate retain only survival on the first $a = \lfloor n/16 \rfloor$ slots. By the stated sigma-field independence, its fractional moment factors from that of the conditional private probability of $\mathcal{D}$. Jensen bounds the latter by $\mathbb{P}(\mathcal{D})^\lambda$, and finitely many doublings compare $F_a(h)$ with $F_n(h)$. Bounded $n < 32$ uses $\mathbb{E}\mathbb{P}_C(\mathcal{D} \mid \text{extended environment})^\lambda \leq \mathbb{P}(\mathcal{D})^\lambda$ and the crude lower bound. If a fixed initial coupling block overlaps both read regions, the lengths below sixteen times that block are also bounded; the same argument, with the corresponding fixed comparison constant, treats them without asserting independence. $\square$

Lemma 5.15 (Relative control of the initial coupling block). *Fix the compact laws and the bounded scaled-drift range in Lemma 5.14, and let the fractional exponent satisfy $\lambda \in [\lambda_*, 1]$, where $\lambda_* > 0$ is fixed. Fix an integer $b \geq 1$. Couple the centered exponential and binary channels separately to their scalar Gaussian channels on the first b slots, independently of all later blocks. The environmental coupling is outside the private conditional probability, and the private coupling is inside it. For every prescribed $\varepsilon > 0$ there is a finite H_b , depending only on these fixed data and ε , such that, whenever $n \geq 2b$ and $h \geq H_b$, the following assertion holds for each marginal survival functional covered by Lemma 5.14. There is an initial good event $\mathcal{G}_b$ on which the discrepancy of the two initial risk paths is at most $H_b/8$, and*

$$\mathbb{E}[\mathbb{P}_{\text{private}}(\text{survival}, \mathcal{G}_b^c \mid \text{extended environment})^\lambda] \leq \varepsilon F_n(h).$$

The same H_b works for both risk signs. A prescribed fixed strong or soft curve on this initial finite interval may also be absorbed in $H_b/8$ by increasing H_b before lengths vary.

Proof. Let M_b be the maximum of the absolute centered partial sums of both original channels on the first b slots and of the absolute Gaussian suprema on the corresponding time interval. Enlarge its definition by the fixed compact coefficients needed to reconstruct the risk. A union bound using exponential moments of the gaps, boundedness of the binary increments, and the Gaussian maximal tail gives constants $C_b, c_b > 0$ such that

$$\mathbb{P}\{M_b > v\} \leq C_b e^{-c_b v} \quad (v \geq C_b).$$

The constants are independent of n and h . The deterministic drift on the first b slots is bounded in terms of the fixed data, because $n \geq 2b$ and the full-horizon scaled drift is bounded. Choose a fixed compact-data constant C_0 so that, on $\mathcal{G}_b = \{M_b \leq H_b/C_0\}$, the two initial risk paths differ by at most $H_b/8$; make the same choice large enough to pay the initial drift. A fixed curve on this finite interval has a finite maximum, which is paid by the final increase of H_b .

The bad event reads only the first b slots and their coupling auxiliaries. Lemma 5.14 therefore gives

$$\mathbb{E}[\mathbb{P}_{\text{private}}(\text{survival}, \mathcal{G}_b^c \mid \text{extended environment})^\lambda] \leq C(1+b)^2 (C_b e^{-c_b H_b/C_0})^{\lambda/4} F_n(h).$$

After b has been fixed, choose H_b so that its coefficient is at most ε , uniformly for $\lambda \geq \lambda_*$. This choice does not depend on a persistence exponent or on the eventual reserve h . Since $h \geq H_b$, a

change by $H_b/8$ lies between the reserves $7h/8$ and $9h/8$. The discrepancy bound is restricted to $\mathcal{G}_b$; the complement is paid by its relative tail estimate, allowing every exponential gap to retain its full support. The finitely many experiments $n < 2b$ are outside this lemma and are handled by their explicitly supported finite cases. $\square$

Theorem 5.16 (One-end persistence). *For $|\delta|\sqrt{n} \leq D$, $h \geq 1$, and either sign of S ,*

$$F_n(h) \asymp_D \min\{1, ((1+h)/\sqrt{n})^{2\chi(\lambda, \rho_\tau)}\}. \quad (5.26)$$

The upper estimate allows a fixed soft power wall of exponent below $1/2$. For $h \geq H_0$ the lower estimate may retain a fixed strong curve, a fixed positive central unreserved endpoint, and central gap, binary, and full-color endpoints. The central boxes are chosen compatible and sufficiently wide after the compact laws, wall, D , and D' are fixed, before the coupling accuracy and minimum reserve are chosen. For this endpoint version take $h \leq D'\sqrt{n}$; larger reserves are not needed for the endpoint assertion.

Proof. The construction is given in Section B.2; its quantitative inputs are proved in Sections B.3, B.6 and B.5. Couple the two channels separately on independent dyadic blocks. The relative early and late failure estimates, with their full auxiliary sigma-fields, first give comparison with flat Gaussian survival. The accumulated good-event error becomes a bounded-action wall after Lamperti time change.

For a positive strong curve, bound the failed-coupling mass by the flat survival majorant just established and absorb it against the Gaussian strong-wall lower bound. For a negative weak wall, use its reserve comparison in the relative error argument. The resulting path inclusions give the displayed bounds while retaining regularity and compatible separate-channel boxes. Below the fixed coupling scale, a favorable supported gap/mark event supplies the finite convention. $\square$

5.9 Exact Gamma conditioning and relative local regularity

Uniform spatial positions correspond to exponential gaps with a fixed total. We make this conditioning through one joint density for every selected set of gaps. For a lower bound, a central prefix is followed by an exact-total continuation of constant ordinary probability; that continuation is averaged before the fractional power. The same density comparison then lets us delete local regularity failures at relative cost.

Let $T_n = \sum_{i=1}^n E_i$ and condition on $T_n = A > 0$, where $|A - n\tau| \leq D\sqrt{n}$ and $n \geq 1$. Write $0 < \tau_- \leq \tau \leq \tau_+ < \infty$ for the fixed gap-mean range. All such conditioning is by a regular conditional density. Write $\mathbb{E}_A$ for the resulting environmental expectation. The private marks remain independent of the gaps and of one another.

Lemma 5.17 (Selected-gap density). *For a deterministic proper set I of $m < n$ coordinates, the conditional density relative to independent mean- τ exponentials is exactly*

$$\frac{f_{n-m}(A - \sum_{i \in I} E_i)}{f_n(A)}, \quad f_k(x) = \frac{x^{k-1} e^{-x/\tau}}{\tau^k \Gamma(k)} \mathbf{1}_{\{x > 0\}}. \quad (5.27)$$

There is a fixed threshold $N_\Gamma = N_\Gamma(D, \tau_-, \tau_+) \geq \max\{1, (2D/\tau_-)^2\}$ such that the following two upper bounds hold for $n \geq N_\Gamma$: if $m \leq cn$, $c < 1$, the ratio is bounded above uniformly on its whole support; for any proper selected set it has the weaker upper bound $C_D\sqrt{n}$. If $m \leq cn$, it is bounded below by a positive constant on $|\sum_I E_i - m\tau| \leq R\sqrt{n}$ for $n \geq n_0(c, D, R) \geq N_\Gamma$, with constants depending on c, D, R and the compact gap-mean range. This threshold ensures that the

displayed central set lies inside the support of the remaining Gamma density. In the exact-mean case $A = n\tau$, both upper bounds hold for every $n \geq 1$. The central lower bound still requires its stated threshold. No uniform upper bound is asserted for a near-central bounded- n window that permits $A \downarrow 0$.

Proof. Independence and convolution of Gamma densities give (5.27). Stirling's formula, uniformly for τ in its compact interval, gives

$$\sup_x f_k(x) \leq Ck^{-1/2}, \quad f_k(k\tau + z\sqrt{k}) \asymp_R k^{-1/2} \quad (|z| \leq R, k \geq N(R, \tau_-)).$$

For $k = 1$ the supremum is $1/\tau$ and the same upper bound holds. For $n \geq N_\Gamma$, enlarge the threshold for the preceding central estimate with $R = D$; the denominator in (5.27) is then comparable to $n^{-1/2}$, and $A \geq n\tau/2$. If $A = n\tau$, the explicit density and Stirling's bounds instead give $f_n(n\tau) \asymp n^{-1/2}$ for every $n \geq 1$, uniformly in τ . The asserted bounds follow; on the central selected set the remaining total has displacement $O(\sqrt{n})$ from a mean with shape comparable to n . The comparison applies to the joint law of all selected coordinates. Independence is used only in the reference law before exact conditioning. $\square$

Proposition 5.18 (One-end exact-total bridge). *For fixed D, D' , the compact laws, and a fixed curve as below, there are N_{one}, H_0 such that the following holds for $n \geq N_{\text{one}}$. Suppose $|\omega - \tau|\sqrt{n} \leq D$, $|A - n\tau| \leq D\sqrt{n}$, and $H_0 \leq h \leq D'\sqrt{n}$. The flat conditional survival probability has λ moment at most $CB_n(h)$, with the exponent of (5.26). A stronger event has moment at least $cB_n(h)$ and can retain a fixed curve $g \min(j, n - j)^\gamma$, $0 < \gamma < 1/6$, positive unreserved terminal risk in $[a_0\sqrt{n}, a_1\sqrt{n}]$, and central full-color windows. The constants and windows can be fixed uniformly on the displayed parameter range. One may enlarge H_0 after fixing N_{one} so that $H_0 > D'\sqrt{N_{\text{one}}}$; the reserve range then itself enforces the length threshold. The flat upper bound alone also holds for bounded n , by increasing its constant.*

Proof. All density comparisons below are made after taking $N_{\text{one}} \geq N_\Gamma$ and the required central-support thresholds. Keeping just the first $\lfloor n/4 \rfloor$ survival constraints and using Lemma 5.17 proves the upper bound. For the lower bound first assume $h \leq \varepsilon\sqrt{n}$. On $m = \lfloor n/4 \rfloor$ slots retain the strong reference family in Theorem 5.16, with

$$a_0\sqrt{n} \leq S_m \leq a_1\sqrt{n}, \quad |T_m - m\tau| \leq R\sqrt{n}, \quad |N_c(m) - \pi_c m| \leq R\sqrt{n} \quad \text{for every color } c. \quad (5.28)$$

Its conditional mark probability f has $\mathbb{E}f^\lambda \geq cB_n(h)$. The selected-gap density has a positive lower bound on these boxes. Fix one retained prefix environment and one retained prefix label word. For $k = n - m$, the remaining exact gap total $A - T_m$ is central. Represent these gaps as

$$E'_i = (A - T_m) \frac{W_i}{W_1 + \cdots + W_k}, \quad W_i \text{ iid exponential of mean one.}$$

The centered polygonal partial gap sums, divided by $\sqrt{k}$, converge to τ times a Brownian bridge plus their prescribed central linear drift. Independently, centered color counts converge to multinomial Brownian motion on the zero-sum hyperplane. Its weight projection has positive variance. Both convergences are uniform over the central boxes and compact laws: every parameter sequence has a convergent subsequence; characteristic-function expansion identifies its Gaussian finite-dimensional limits, and the uniform fourth-moment bound $\mathbb{E}|\sum_{i=1}^k Z_i|^4 \leq C(k^2 + k)$ gives tightness, including polygonal intervals shorter than $1/k$. Normalizing by $k^{-1} \sum W_i$ preserves convergence by the law of large numbers.

After scaling, the starting risk lies in a fixed positive compact interval. Choose a continuous path in the zero-sum color hyperplane whose weight projection stays positive and reaches a fixed positive final window, and choose the zero centered gap bridge. The multinomial covariance is nondegenerate on its hyperplane, so a sufficiently small tube about each chosen pair of paths has positive limiting probability.

A finite collection of these tubes covers all possible retained prefix endpoints and central remaining drifts. Their probabilities have a positive minimum; uniform convergence transfers a fixed fraction of that minimum to the finite bridges. Enlarge the final color windows once to absorb the prefix deviations. The retained risk is at least $c\sqrt{n}$ throughout the continuation and hence exceeds gn^γ for large n . The first quarter already has the required global curve from its one-sided strong event. Thus every retained prefix word has joint remaining-environment-and-mark continuation probability at least $c > 0$.

For each prefix environment let Z be the full successful mark probability as a function of the remaining environment. Exactly

$$0 \leq Z \leq f, \quad \mathbb{E}_{\text{remainder}} Z \geq cf, \quad \mathbb{E}_{\text{remainder}} Z^\lambda \geq cf^\lambda. \quad (5.29)$$

The last inequality follows from $z^\lambda \geq f^{\lambda-1}z$ for $0 \leq z \leq f$. Integrate against the central lower density in Lemma 5.17. This proves the lower bound. When h is comparable to $\sqrt{n}$, a fixed positive tube directly gives a constant lower bound; alternatively use a smaller fixed multiple of $\sqrt{n}$ and monotonicity. The continuation lower bound averages the remaining environment for each retained prefix word. It retains central full-type windows; it neither gives a lower bound for every remaining environment nor assigns constant mass to an individual exact type. The length threshold also includes the uniform tube-convergence and endpoint-support thresholds after the windows are fixed. Indeed $l - \omega = bp$ is bounded below by some $\beta > 0$, whereas $A \leq n\omega + 2D\sqrt{n}$. Thus $nl - A \geq \beta n - 2D\sqrt{n}$, which exceeds $a_1\sqrt{n}$ for sufficiently large n . For the opposite sign the corresponding available endpoint is $A - n(l - b) \geq b(1 - p)n - 2D\sqrt{n}$, and $b(1 - p)$ has the same kind of positive compact lower bound. Thus a positive terminal window is not demanded beyond the possible mark total in either orientation. The retained central color tubes above give its actual support. The lower construction therefore uses its stated large-length threshold, including the endpoint-support thresholds; excluded small n receive no compactness lower bound. For the flat upper bound in the bounded range, use $q \leq 1$ and $B_n(h) \geq n^{-1/2}$ for $h \geq 1$. $\square$

Fix a small $\epsilon > 0$, $0 < \gamma < 1/6$, and put $d_n(i) = \min(i, n - i)$. Let REG_K consist of the following inequalities for every $0 \leq i < j \leq n$ and every color c :

$$\begin{aligned} |T_j - T_i - \tau(j - i)| &\leq \epsilon(j - i) + K[1 + d_n(i)^\gamma + d_n(j)^\gamma], \\ |N_c(i, j) - \pi_c(j - i)| &\leq \epsilon(j - i) + K[1 + d_n(i)^\gamma + d_n(j)^\gamma]. \end{aligned} \quad (5.30)$$

Lemma 5.19 (Relative regularity under the exact total). *Assume $n \geq 1$, $A > 0$, $|A - n\tau| \leq D\sqrt{n}$, $|\omega - \tau|\sqrt{n} \leq D$, and $K \geq 0$. If $Q_{\text{bad}}(E)$ is the conditional mark probability of flat survival and REG_K^c , then*

$$\mathbb{E}_A Q_{\text{bad}}^\lambda \leq Ce^{-cK} B_n(h), \quad h \geq 1. \quad (5.31)$$

The constants are uniform in the stated near-centered regime.

Proof. For $0 \leq K < 1$, the flat survival upper bound already gives the assertion after increasing C : for large n , retain the first $\lfloor n/4 \rfloor$ slots and apply the selected-gap upper density bound and Theorem 5.16; bounded n uses $Q_{\text{bad}} \leq 1$. Thus it suffices to prove the summable interval

estimate for $K \geq 1$. Choose a fixed $N_{\text{REG}} \geq \max\{32, N_{\Gamma}, (D/\epsilon)^2\}$, enlarging it for the fixed-fraction comparisons below, and first take $n \geq N_{\text{REG}}$. For a single bad interval $(i, j]$, exponential moments and Chernoff give under the independent reference law

$$\mathbb{P}(\mathcal{D}_{i,j}) \leq C \exp\{-c\epsilon^2(j-i) - c\epsilon K[1 + d_n(i)^\gamma + d_n(j)^\gamma]\}. \quad (5.32)$$

This follows by choosing a fixed tilt z with $\log \mathbb{E}e^{z(E-\tau)} \leq Cz^2$ and $C|z| \leq \epsilon/2$; the finite color variables are even bounded. We divide the bad intervals according to which short survival prefix can be retained. This preserves the survival factor in every region while leaving a proper selected gap set for the density comparison.

For an early interval, $j \leq n/4$, retain survival only to $n/2$. The relaxed event uses at most half the gaps. Lemma 5.17 compares its conditional moment with the independent reference moment, and Lemma 5.14 bounds the latter by $C(1+j)^2 \mathbb{P}(\mathcal{D}_{i,j})^{\lambda/4} B_n(h)$.

For $i \geq n/8$, retain survival only on the first $\lfloor n/16 \rfloor$ slots. This prefix is disjoint from the bad interval, and their union omits a fixed fraction of the gaps. One joint density comparison therefore gives independent reference variables and the bound $C\mathbb{P}(\mathcal{D}_{i,j})^\lambda B_n(h)$.

Every remaining interval has $i < n/8$, $j > n/4$, and length at least $n/8$. For a proper gap interval, use the weaker $C\sqrt{n}$ density bound in Lemma 5.17. For the full gap set, the total is fixed and satisfies its linear tolerance for large n . The gap conditioning leaves the mark law unchanged. The macroscopic interval-length tail consequently makes all these errors exponentially small, including after their λ powers.

In the early region, the polynomial in j is summable against e^{-cKj^γ} and the interval-length factor. In the late region near the terminal end sum in $n-i$; away from both ends the additional e^{-cKn^γ} absorbs the number of intervals. There are at most n^2 macroscopic intervals, and their exponential errors are smaller than $e^{-cK} B_n(h)$ since $B_n(h) \geq cn^{-1/2}$. Subadditivity of the λ power proves (5.31). For $1 \leq n < N_{\text{REG}}$, every gap sub-sum is at most $A \leq N_{\text{REG}}\tau_+ + D\sqrt{N_{\text{REG}}}$, and every color count is at most N_{REG} . Consequently all of (5.30) hold deterministically whenever

$$K \geq K_{\text{fin}} := 1 + 2N_{\text{REG}}\tau_+ + D\sqrt{N_{\text{REG}}} + N_{\text{REG}}.$$

For $0 \leq K < K_{\text{fin}}$ use $Q_{\text{bad}} \leq 1$ and $B_n(h) \geq N_{\text{REG}}^{-1/2}$; increasing C to at least $e^{cK_{\text{fin}}} \sqrt{N_{\text{REG}}}$ proves the same estimate. This handles the entire bounded- n range even when $A \downarrow 0$, without a Gamma density comparison. $\square$

5.10 Projection to complete cells

The arithmetic model reads complete cell occupancies rather than the positions within a cell. We therefore transfer the continuous path family to a family invariant under every internal ordering. The projection below changes path caps by only a fixed reserve. A separate regularity projection controls the two partial cells at the ends of any rank interval. Take r independent positions U_i uniform on $[0, v]$, with v an integer. Fix an integer $1 \leq N < v$, and set

$$J_i = N + \left\lfloor \frac{v-N}{v} U_i \right\rfloor, \quad J_i \in \{N, \dots, v-1\}. \quad (5.33)$$

These are independent uniform retained cells and $U_i - 1 < J_i \leq U_i + N$. Let the slope per cell be $c > 0$. With reverse coordinates $\tau_i = v - U_i$, $\xi_i = v - J_i$, use strict before-jump deficits

$$R_{\text{full}}(t) = ct - \sum_{\tau_i < t} w(C_i), \quad R_{\text{cell}}(t) = ct - \sum_{\xi_i < t} w(C_i).$$

The displacement $\tau_i - N \leq \xi_i < \tau_i + 1$ and positivity of all weights give exactly

$$R_{\text{cell}}(t) \leq c + R_{\text{full}}(t - 1), \quad R_{\text{full}}(t) \leq cN + R_{\text{cell}}(t - N). \quad (5.34)$$

The inequalities apply respectively for $t \geq 1$ and $t \geq N$; earlier times have bounds c and cN . Integer before-jump tests control the entire cell path since its deficit increases linearly between jumps. In particular a cell flat cap implies a continuous flat cap with only cN extra reserve.

There are $n = r + 1$ spacings between the r positions and both endpoints. In physical units their law is that of n exponential gaps conditioned on total cv . Introduce one independent dummy mark at the last spacing. Auxiliary after-jump and before-jump risks differ by at most l at each tested time; removing the dummy changes each color total by at most one. Fixed reserve and central-window adjustments therefore transfer Proposition 5.18 to these actual positions. This transfer uses $n \geq N_{\text{one}}$ after those adjustments; it makes no endpoint-support assertion for excluded short clocks. The last empty gap is retained throughout. The resulting assertion retains central windows. Prescribing an individual exact type or the dummy color would require an additional type-mass estimate.

Lemma 5.20 (Permutation-invariant regularity projection). *There is a cell event depending only on the occupancy and the color counts in each cell, implied by (5.30), such that its occurrence implies the required color interval bounds, with a larger fixed coefficient, for every ordering within cells. It also bounds each cell occupancy by a fixed constant times one plus its distance to the nearer physical endpoint raised to γ .*

Proof. We formulate the projection using bounded displacement of ordered positions. The gap regularity first compares rank depth with physical depth, then bounds cell occupancy and the errors in the partial endpoint cells. Let ordered physical positions $x_i = T_i$, $x_0 = 0, x_n = A$, be mapped to nondecreasing cells J_i satisfying $|aJ_i - x_i| \leq \Delta_0$, with fixed a, Δ_0 ; this includes the reverse version of (5.33). Put $D_A(t) = \min(t, A - t)$. Applying the gap inequalities in (5.30) to $(0, i]$ and $(i, n]$ gives

$$(\tau - \epsilon)d_n(i) - K[1 + d_n(i)^\gamma] \leq D_A(x_i) \leq (\tau + \epsilon)d_n(i) + K[1 + d_n(i)^\gamma].$$

The inequality $Kx^\gamma \leq \eta x + C_{\eta, \gamma}K^{1/(1-\gamma)}$ therefore implies

$$C^{-1}d_n(i) - B_K \leq D_A(x_i) \leq Cd_n(i) + B_K. \quad (5.35)$$

The single-gap bound gives $x_{i+1} - x_i \leq \tau + \epsilon + K[1 + d_n(i)^\gamma + d_n(i+1)^\gamma]$. If a physical cut t is within Δ_0 of $[x_i, x_{i+1}]$, subtract this bound from the lower bound in (5.35) and absorb its sublinear term. Thus

$$d_n(i)^\gamma \leq CD_A(t)^\gamma + B'_K. \quad (5.36)$$

This includes $i = 0, n - 1$.

Let $I(s) = \#\{i : J_i \leq s\}$, $g_s = I(s) - I(s - 1)$, $D(s) = D_A(as)$, and let $N_c(s, t)$ count color c in cells $(s, t]$. A cell cut as is within Δ_0 of $[x_{I(s)}, x_{I(s)+1}]$. Equations (5.30) and (5.36) therefore imply

$$\begin{aligned} |N_c(s, t) - \pi_c[I(t) - I(s)]| &\leq \epsilon[I(t) - I(s)] + K_1[1 + D(s)^\gamma + D(t)^\gamma], \\ g_s &\leq K_1[1 + D(s)^\gamma]. \end{aligned} \quad (5.37)$$

For the second assertion, the first and last positions of an occupied cell are separated by at most $a + 2\Delta_0$; the gap lower bound then bounds $g_s - 1$, using (5.36) at the adjacent cell cuts. Furthermore, for each rank u belonging to that cell,

$$D(s) \leq C d_r(u) + B_K'' . \quad (5.38)$$

This follows from (5.35) and the bounded displacement; $|d_n(u) - d_r(u)| \leq 1$. It can be tested solely from occupancy counts, since the smallest $d_r(u)$ in a cell's consecutive rank range occurs at one of its rank endpoints.

Define the cell regularity event by (5.37) and (5.38). For an arbitrary rank interval in any internal ordering, remove its two partial endpoint cells. Apply the first bound of (5.37) to the remaining complete-cell interval. The two omitted parts have total error at most their occupancies, bounded by the second assertion. Convert their physical endpoint distances back to rank distances by (5.38). Adjacent cuts differ by a fixed amount, and $|d_r(u+1) - d_r(u)| \leq 1$. Consequently, for every order,

$$|N_c(u, v) - \pi_c(v - u)| \leq \epsilon(v - u) + K_2[1 + d_r(u)^\gamma + d_r(v)^\gamma].$$

If both endpoints are in one cell its entire contribution is controlled by that cell's occupancy. The same argument covers the first and last occupied cells and empty cuts. All constants are fixed before n varies. $\square$

Theorem 5.21 (Critical complete-cell family). *Suppose $n = r + 1 \asymp T$, $|cv - n\tau| \leq D\sqrt{n}$, $|\omega - \tau|\sqrt{n} \leq D$, and the retained-cell density is bounded strictly below a fixed $\zeta < 1$. Fix $0 < \gamma < 1/6$. For $H_0 \leq h \leq D'\sqrt{T}$ there is a family depending only on complete colored cell occupancies whose conditional mark probability f satisfies*

$$\mathbb{E}f^\lambda \geq c \min\{1, ((1+h)/\sqrt{T})^{2\chi(\lambda, \rho_\tau)}\}. \quad (5.39)$$

*The same family has the following properties: the ordinary forward occupancy bounds $G_s \leq s + 1$ and $G_s \leq \zeta s + C_0$; a reverse flat cap with reserve $h + O(1)$; a prescribed fixed strong physical curve; the complete-cell regularity of Lemma 5.20, hence color regularity for all internal orderings; a fixed positive central window for the unreserved final risk; and central full-color windows. The constants C_0, N, H_0 and all window widths are fixed. The assertion uses $n \geq N_{\text{cell}}$, with a fixed threshold chosen after these data and containing the adjusted N_{one} and central-support thresholds. If $n \geq c_*T$ is the fixed comparability bound, one may finally enlarge $H_0 > D'\sqrt{N_{\text{cell}}}/c_*$ so that its reserve range already enforces this length restriction. More explicitly, the two reverse restrictions may be written $R_{\text{cell}}(k) \leq h + C_1$ and $R_{\text{cell}}(k) \leq h + C_g - g \min(k, v - k)^\gamma$ at every integer cut. The flat extra reserve C_1 is independent of the selected amplitude g .*

If a necessary competitor event skips at most $J_d \leq C(1+d)^a$ slots and retains a deterministic free block of a fixed positive fraction of n , of length at most $c_1 n$ with $c_1 < 1$, its cell conditional probability q_d obeys

$$\mathbb{E}q_d^\lambda \leq C(1+d)^C \min\{1, ((1+h)/\sqrt{T})^{2\chi(\lambda, \rho_\tau)}\}. \quad (5.40)$$

When a skip is too large to leave that block, the same bound holds after increasing the fixed polynomial exponent.

Proof. There are three stages. We impose the ordinary occupancy conditions by product association, transfer the strong path to complete cells, and delete the two regularity losses relative to the original survival scale. The ordering matters because the event used for association must be monotone in the independent labeled positions.

First the ordinary occupancy conditions have fixed positive probability. For independent retained cells with density at most $\zeta_0 < \zeta$, a binomial exponential bound gives $\mathbb{P}(G_s > \zeta s + C_0) \leq C e^{-cC_0 - cs}$. Choose C_0 large. For $G_s \leq s + 1$, the first N cells are empty, and the same negative-drift bound summed over $s \geq N$ has arbitrarily small total when fixed N is large. After deleting these fixed cells the density still has a uniform margin for all sufficiently large v . Thus their intersection has probability $c_{\text{occ}} > 0$. Both events are increasing when any labeled forward position is moved rightward.

Begin with the stronger rank curve in Proposition 5.18 and subtract the relative loss in (5.31). On the retained family, (5.35)–(5.36) convert the rank curve to the desired physical curve, while the single-gap bound controls the before-jump interpolation.

For the association step, enlarge this family by dropping regularity and the auxiliary prefix boxes. Keep the physical strong and flat conditions, total-type windows, and final-risk windows. The windows are position independent, and the physical conditions are monotone in every independent original forward position for each fixed mark word. The enlargement preserves the lower fractional moment. If its conditional mark probability is $Q(U)$, Harris’s product-measure inequality gives

$$\mathbb{E}[\mathbf{1}_{\text{occ}} Q(U)^\lambda] \geq \mathbb{P}(\text{occ}) \mathbb{E} Q(U)^\lambda. \quad (5.41)$$

It is applied to independent labeled positions, not to the dependent Dirichlet spacings. For one coordinate the inequality is the identity $2 \text{Cov}(f, g) = \mathbb{E}[(f(Z) - f(Z'))(g(Z) - g(Z'))] \geq 0$; induction proves the product version. Total-type restrictions do not disturb monotonicity because they do not depend on the positions.

Project by (5.33). Equation (5.34) and the Hölder modulus of $\min(t, v - t)^\gamma$ give a cell strong event and a separate cell flat cap. The latter has only the fixed extra reserve c , independent of the subsequent curve amplitude. By the converse in (5.34), a bad cell family lies, for every compatible latent position vector, in continuous flat survival with $cN + O(1)$ extra reserve and failure of a stronger continuous regularity envelope. Lemma 5.19 bounds its fractional mass. Subtract this second REG loss and use Lemma 5.20. Pointwise comparisons are taken before fractional powers: if $Q_{\text{full}}(U) \leq Q_{\text{cell}}(J(U))$, then integration gives the lower direction immediately; the converse bad-event containment gives the upper direction for the error. Thus both transfers follow from pointwise path containment before the fractional power is taken.

Choose the constants in the following order. Fix the retained-cell deletion, density margin, and c_{occ} . Fix central endpoint boxes and a small fraction of their flat comparison constant. Choose regularity tolerances and K so that (5.31) pays for both REG losses and the occupancy factor. Then choose the rank-curve amplitude needed in the cell projection. Choose the relative KMT accuracy and finally enlarge H_0 to dominate its fixed initial error and make the normalized curve cost small. This is possible because $\gamma < 1/2$; its normalized amplitude decreases as H_0 grows. Relative error estimates are uniform for reserves at least one, so the earlier choices do not change. Finally fix N_{cell} above all length thresholds, and, if desired, make the final reserve enlargement stated in the theorem. This proves (5.39).

For a competitor, discarding J_d positive bounded jumps increases its necessary reserve by at most lJ_d . At the same rank index, replacing latent positions by cell centers changes only the two endpoint times by a fixed amount, not by an amount proportional to the rank. Its free block is therefore majorized by reference survival with reserve $h + C + lJ_d$. Since $n \geq N_{\text{cell}} \geq N_\Gamma$, Lemma 5.17 and Theorem 5.16 give (5.40). If the skip exceeds a fixed fraction of T , use $q_d \leq 1$; a larger fixed power of $1 + d$ absorbs one since $B_T(h) \geq cT^{-1/2}$. This completes the same-family assertion. $\square$

In the critical application $h = H_0 + M/4$, $0 \leq M \leq \sqrt{T}$, $\lambda = \theta + O_D(T^{-1/2})$, and $\rho_\tau = \rho_c + O_D(T^{-1/2})$. Proposition 5.5 shows that the factors in (5.39) and (5.40) are uniformly comparable to

$$\min\{1, ((1+M)/\sqrt{T})^{2\chi(\theta, \rho_c)}\}.$$

Indeed the error in its logarithm is $O_D(\log T/\sqrt{T})$. The assertions hold for every fixed central first count; mixing a fixed central Poisson slab of positive probability preserves the lower bound. Here the original length and each retained central count are taken above the fixed N_{cell} threshold, as required by the lower construction.

5.11 Positive drift: an effective prefix and uniform continuation

A positive drift changes the relevant survival horizon from the full length to the time at which drift and fluctuations have comparable size. We isolate a prefix at that effective scale and retain a positive inner height. Exact conditioning on the remaining gaps and color type then permits a uniform continuation above a linear margin, which also pays the strong curve.

Continue with (5.23), now $0 < \delta = \omega - \tau \leq \delta_0$, and condition on $T_n = n\tau$. Put

$$B_{n,\delta}(h) = \min\{1, [(1+h)(\delta + n^{-1/2})]^{2\chi(\lambda, \rho_\tau)}\}. \quad (5.42)$$

Theorem 5.22 (Positive-drift complete-cell family). *For fixed $\gamma \in (0, 1/6)$ and fixed curve and occupancy requirements, there are $\delta_0, H_0, D_0 > 0$ such that, for $0 < \delta \leq \delta_0$, $\delta\sqrt{n} \geq D_0$, $h \geq H_0$, the flat exact-total survival moment is at most $CB_{n,\delta}(h)$, and a complete-cell family with the properties of Theorem 5.21 has moment at least $cB_{n,\delta}(h)$. Its final unreserved risk can be restricted to $[\delta n - C\sqrt{n}, \delta n + C\sqrt{n}]$, and its full color counts are central. Deterministic competitor skips satisfy the corresponding upper bound $C(1+J)^C B_{n,\delta}(h)$.*

Proof. Fix constants A_0, K_0 , and set $m = \lfloor A_0/\delta^2 \rfloor$. Choose D_0 later so $m \leq n/8$, and decrease δ_0 to absorb rounding. On this effective prefix $\delta\sqrt{m}$ is bounded. Apply Theorem 5.16, retaining

$$S_m \in [K_0/\delta, C_{A_0, K_0}/\delta], \quad |T_m - m\tau| \leq C_{A_0, K_0}/\delta, \quad |N_c(m) - \pi_c m| \leq C_{A_0, K_0}/\delta. \quad (5.43)$$

Its conditional probability f has moment at least $c \min(1, [(1+h)\delta]^{2\chi})$. For $h\delta$ bounded below, a fixed tube at the effective scale gives this constant lower bound directly; large reserves help. Lemma 5.17 bounds the prefix density above and below on (5.43). The upper survival bound already follows by retaining only this prefix. For this application the complete total is exactly $n\tau$, so the upper comparison uses the all- n exact-mean clause. The central lower comparison uses its length threshold, enforced by the eventual choices of large D_0 and small δ_0 .

Fix its environment and a retained label word. Let $k = n - m$ and $R = n\tau - T_m$. Retain remaining full-color totals within fixed $C_1\sqrt{k}$ windows around $k\pi_c$. Their joint multinomial probability is bounded below. Conditional on any such type, use independent reference parameters equal to its proportions π'_c and to $\tau' = R/k$. The binary sequence is the uniform fixed-count permutation and the gaps are a Dirichlet bridge. The new parameters satisfy

$$\pi'_c = \pi_c + O(C_1/\sqrt{n}), \quad \tau' = \tau + O(C_{A_0, K_0}/(\delta n)), \quad \delta' := \sum_c \pi'_c w(c) - \tau' \geq \delta/2,$$

after D_0 is large in terms of the constants already fixed. Also $\delta' \leq C\delta$. The complete endpoint then belongs to the stated $\delta n + O(\sqrt{n})$ window and is at least $c\delta n$.

We verify the required continuation estimates under these two exact conditions. A selected initial or terminal block of at most $k/2$ gaps has density bounded by a constant relative to the

new reference law. Indeed its full total is exactly $R = k\tau'$ with τ' in a positive compact interval, so this is again the all- k exact-mean upper bound. The corresponding binary density ratio is

$$\frac{\mathbb{P}\{\text{Bin}(k-j, p') = u - U_j\}}{\mathbb{P}\{\text{Bin}(k, p') = u\}}, \quad u = kp'.$$

Its denominator is comparable to $k^{-1/2}$ and numerator at most $C(k-j)^{-1/2}$, so it too is bounded. The two ratios multiply by independence. Specifying all colors leaves this same binary bridge law.

For independent reference increments $X = w(C) - E$, uniform exponential moments and Taylor expansion give, for a fixed small $a > 0$,

$$\mathbb{E} \exp[-a\delta(X - \delta'/2)] \leq 1.$$

The exponential supermartingale therefore gives, from $x \geq K_0/\delta$,

$$\mathbb{P}\{\exists j \geq 0 : x + \sum_{i \leq j} X_i < x/2 + \delta'j/2\} \leq e^{-c\delta x}. \quad (5.44)$$

The bounded initial-half density transfers this estimate to the first half of the conditioned continuation. For its late half, the maximal Bernstein estimate for each independent centered channel is

$$\mathbb{P}\{\max_{j \leq k/2} |\sum_{i \leq j} Z_i| > z\} \leq C \exp[-c \min(z^2/k, z)].$$

Apply the same bounded density to the initial half and to the reversed terminal half. Both centered totals are fixed to zero; these halves cover the complete bridge, giving

$$\mathbb{P}\{\max_{j \leq k} |S'_j - \delta'j| > c_2\delta k\} \leq Ce^{-c\delta^2 k}. \quad (5.45)$$

Using a maximal estimate on each half avoids a factor k from a union over individual times.

The global strong curve on the continuation is at most $g(m+j)^\gamma$. At its beginning $gm^\gamma = o(\delta^{-1})$, and its derivative is $O(\delta^{2-2\gamma}) = o(\delta)$ as $\delta_0 \rightarrow 0$. Thus the positive linear margin in (5.44) dominates it on the first half. On the late half $gn^\gamma = o(\delta n)$ uniformly when $\delta\sqrt{n} \geq D_0$, so (5.45) suffices. Choose K_0 so $Ce^{-cK_0} < 1/4$, then D_0 so the late failure is less than $1/4$ and all parameter comparisons hold. Every retained prefix word and remaining type now has conditional continuation probability at least $1/2$, averaged over its remaining exact gaps. The central type window has fixed positive mass. Thus the inequalities (5.29) hold again, proving the strong exact-total lower bound after integrating the central prefix density.

We still need interval regularity in addition to the central endpoint windows. It follows from the same relative deletion argument as in the near-centered regime. Before exact conditioning, the same effective-prefix argument and (5.44) prove flat reference survival comparable to (5.42): use the near-centered theorem when $\delta\sqrt{n}$ is bounded, and the effective prefix otherwise. Hence time doubling and polynomial reserve comparison hold uniformly throughout this positive-drift layer. Repeat the three-region proof of Lemma 5.19 with the unchanged exponential interval tails. It gives a relative bad moment at most $Ce^{-cK} B_{n,\delta}(h)$ under the exact total. The ordinary occupancy, physical projection, and two-stage REG/Harris argument are now precisely those in Theorem 5.21. The proof-only effective-prefix boxes are dropped before Harris. The final endpoint and total-type windows are position independent and remain.

Constants can be fixed in the following order: occupancy/deletion requirements; effective-prefix constants A_0, K_0 and their endpoint fraction; central type widths; REG tolerance and K ; required curve amplitude and relative coupling accuracy; minimum reserve H_0 ; then large D_0 ,

small δ_0 , and the length threshold. The reference relative errors are uniform in $h \geq 1$. Enlarging H_0 and shrinking δ_0 make normalized curve costs small without changing the earlier fixed success fraction. The two displayed parameter inequalities enforce $n \geq (D_0/\delta_0)^2$; take these final choices so that this exceeds every required length threshold. Competitor blocks have enlarged reserve $h + C + lJ$ and bounded selected-gap density. Use the reference upper bound; if J is a fixed fraction of the effective time $\min(n, \delta^{-2})$, the trivial bound one is absorbed by a larger polynomial. This proves all assertions. $\square$

In the ordinary boundary application, $n \asymp T$ and the count-specific drift equals $d/\kappa + O(T^{-1/2})$, with κ bounded above and below. The reference parameters differ from their critical values by $O(d + T^{-1/2})$. Proposition 5.5 permits the critical exponent in the scale

$$\min\{1, [(1+h)(d + T^{-1/2})]^{2\chi_c}\}.$$

Indeed $d \log(1/d)$ and $T^{-1/2} \log T$ are bounded. Theorem 5.21 covers the region of bounded $d\sqrt{T}$, so the two estimates meet across the transition.

5.12 Physical Poisson time and weighted insertion

The upper arithmetic cover uses constraints at physical times and a crowding weight depending on nearby Poisson points. We first derive a physical-time upper bound from a deterministic rank prefix. Inserting points changes the required reserve by a bounded amount per point; the Mecke identity then controls the crowding weight while retaining the same survival factor.

Let Π be a homogeneous Poisson process of rate κ on $[0, s]$, where κ ranges over a fixed positive compact interval. Its points carry independent marks of the preceding law. For fixed $c > 0$ write

$$S(t) = \sum_{u \in \Pi, u \leq t} w(C_u) - ct, \quad \tau = c/\kappa, \quad \rho_\tau = \frac{\tau^2}{\tau^2 + \text{Var}(w)}.$$

Before-jump and after-jump conventions differ by at most l and will always be allowed that fixed reserve adjustment.

Theorem 5.23 (Physical one-end upper bound). *Fix $0 < \gamma < 1/2$, $g \geq 0$, $H \geq 1$. Given Π , let q be the conditional mark probability that either $S(t)$ or $-S(t)$ stays above $-H - gt^\gamma$ for every $0 \leq t \leq s$. If $|\kappa\omega - c|\sqrt{s} \leq D$, then*

$$\mathbb{E}q^\lambda \leq C_{D,g,\gamma} \min\{1, [(1+H)/\sqrt{s}]^{2\chi(\lambda, \rho_\tau)}\}. \quad (5.46)$$

The correlation is the arrival-gap correlation ρ_τ displayed above. The bound also permits a fixed logarithmic wall in place of the power wall.

Proof. We first note why the soft-wall form of Theorem 5.16 has its stated uniformity. After a bad prefix of length b , $(b+j)^\gamma \leq b^\gamma + j^\gamma$ changes reserve by only a fixed polynomial in b . Thus the relative bad-prefix proof is unchanged up to its fixed polynomial. Under Lamperti transformation from time H^2 , the wall becomes $gH^{2\gamma-1}e^{-(1/2-\gamma)u}$, whose action and coefficient variation are bounded for $H \geq 1$. The relative KMT proof and its absorption therefore hold for both signs; no fixed unconditional REG error is subtracted to prove this upper bound.

Take $m = \lfloor \kappa s/2 \rfloor$; bounded s is absorbed in constants. The Poisson lower-tail probability $\mathbb{P}(N_s < m)$ is at most $e^{-c_0 s}$. Extend the process beyond s if necessary. Its first m scaled arrival

gaps $E_i = c(t_i - t_{i-1})$ are independent exponentials of mean τ . The extension does not alter the conditional mark probability on $[0, s]$.

On $N_s \geq m$, physical survival imposes necessary inequalities at these first m arrivals. For the sign S , at a relevant test time the accumulated weight is at most lj ; hence survival implies $ct \leq lj + H + gt^\gamma$. Since $\gamma < 1$, it follows that $t \leq C(j + H + 1)$ and consequently $gt^\gamma \leq C(j^\gamma + H + 1)$. Thus the necessary rank event has reserve $C(H + 1)$ and a fixed soft-power coefficient. For the sign $-S$, no arrival-time upper bound is needed when $-S(t) \geq 0$: its rank lower wall is automatic. When $-S(t) < 0$, one has $ct < \sum w \leq lj$, which again bounds the allowed physical soft wall by a fixed rank power wall. This gives the needed rank-wall bound in the second orientation as well, by treating only the times when its rank lower wall is active.

At a fixed full environment the mark probability of this necessary rank event depends only on the first m gaps. Dropping the restriction $N_s \geq m$ in its expectation gives

$$\mathbb{E}q^\lambda \leq \mathbb{E}q_{\text{rank}}^\lambda + e^{-c_0 s}.$$

Here $|\omega - \tau|\sqrt{m} \leq C_D$. Apply Theorem 5.16 with its soft wall to the iid arrival gaps. Since $m \asymp s$ this gives (5.46). Finally the displayed scale is at least $cs^{-1/2}$, because $\chi \leq 1/2$ and $H \geq 1$, so the exponential tail is absorbed. A logarithm is bounded by a fixed constant plus any fixed positive power t^γ , proving the last assertion. $\square$

Corollary 5.24 (Effective physical time). *If the physical mean $d = \kappa\omega - c \geq 0$ is sufficiently small, the same full-horizon necessary event has fractional moment at most*

$$C \min\{1, [(1 + H)(d + s^{-1/2})]^{2\chi(\lambda, \rho\tau)}\}.$$

The bound is valid for either risk sign and the preceding soft walls.

Proof. Retain only the deterministic interval of length $s_0 = c_0 \min(s, d^{-2})$, with $d^{-2} = \infty$ when $d = 0$. Then $d\sqrt{s_0}$ is bounded. Apply Theorem 5.23; $s_0^{-1/2} \asymp d + s^{-1/2}$. When s_0 is bounded the conclusion follows from the trivial upper bound one. Full survival implies survival on this deterministic interval, which is all that the upper bound requires. $\square$

Proposition 5.25 (Palm bounds with the survival weight retained). *Let $T \geq 1$ and let $q_H(\Pi)$ be a physical conditional survival probability of the preceding kind on $[0, T]$. Suppose its proved upper bound is $CB_T(H)$, where either*

$$B_T(H) = \min\{1, [(1 + H)/\sqrt{T}]^{2\chi}\} \quad \text{or} \quad B_T(H) = \min\{1, [(1 + H)(d + T^{-1/2})]^{2\chi}\}.$$

For $t \in [0, T]$ set

$$Z(t) = 1 + \sum_{u \in \Pi, u \geq t} e^{-(u-t)}, \quad \mathcal{A} = 1 + C_0 \sup_{0 \leq t \leq T} \frac{Z(t)^5}{(2+t)^4}, \quad N_T = |\Pi|.$$

For each integer $m \geq 0$ and every fixed $a > 1/4$,

$$\begin{aligned} \mathbb{E}[Z(t)^m q_H^\lambda] &\leq C_m B_T(H), \\ \mathbb{E}[N_T Z(t)^m q_H^\lambda] &\leq C_m T B_T(H), \\ \mathbb{E}[(T + N_T) \mathcal{A}^a q_H^\lambda] &\leq C_a T B_T(H). \end{aligned} \tag{5.47}$$

Moreover for $k \geq 0$,

$$\mathbb{E}[(T + N_T) q_{H+k+\lceil \log \mathcal{A} \rceil}^\lambda] \leq C(1+k)^C T B_T(H). \tag{5.48}$$

Proof. Adding j deterministic points changes every risk by at most jl , regardless of their marks. Integrating the new marks therefore gives the pointwise inequality

$$q_H(\Pi + \{u_1, \dots, u_j\}) \leq q_{H+jl}(\Pi). \quad (5.49)$$

It holds for either sign. The proved survival upper bounds imply $\mathbb{E}q_{H+jl}^\lambda \leq C(1+j)^C B_T(H)$, since $H \geq 1$ and $2\chi \leq 1$.

Expand $Z(t)^m$ and group monomials by their distinct Poisson points. The multivariate Mecke identity for j distinct points says that

$$\mathbb{E} \sum_{u_1, \dots, u_j \in \Pi}^{\neq} G(u_1, \dots, u_j, \Pi) = \kappa^j \int_{[0, T]^j} \mathbb{E} G(u_1, \dots, u_j, \Pi + \{u_1, \dots, u_j\}), d\mathbf{u}.$$

For a Poisson process this follows directly by expanding its factorial probabilities $e^{-\kappa T}(\kappa T)^n/n!$, choosing the j ordered distinct points, and reindexing $n - j$, with the weight evaluated in the resulting environment. In each monomial the insertion integrals have bounded mass, since every point carries at least one factor $e^{-(u-t)} \mathbf{1}_{\{u \geq t\}}$. Use (5.49) after insertion to obtain the first bound in (5.47). One additional Mecke point without a decaying factor gives a factor κT and proves the second; coincidences with existing monomial points only reduce the number of integrations.

For $j \leq t < j+1$ one has $Z(t) \leq eZ(j)$. Hence $\mathcal{A}^a \leq C_a[1 + \sum_{j \geq 0} (2+j)^{-4a} Z(j)^{5a}]$. Bound the noninteger moment by a larger integer moment and sum the convergent series, since $4a > 1$. This proves the last assertion of (5.47). Finally, on $\lceil \log \mathcal{A} \rceil = j$ one has $\mathcal{A}^a \geq e^{a(j-1)}$. Apply the weighted estimate with reserve $H + k + j$, separately in each bin, to get

$$\mathbb{E}[(T + N_T)q_{H+k+\lceil \log \mathcal{A} \rceil}^\lambda] \leq CT \sum_{j \geq 0} e^{-a(j-1)} B_T(H + k + j) \leq C(1+k)^C T B_T(H).$$

The estimate is applied with the reserve corresponding to each bin j , so the crowding weight and survival mass remain in the same expectation. The global crowding majorant also controls arbitrary stopped depths: deletion of unseen points need not preserve the location of the stop. $\square$

We have now obtained the conditional mark estimates required by the profile. To use them for divisor-box volumes, the lower argument constructs a positive complete-cell kernel and the upper argument uses a geometric cover with a joint terminal-count estimate. Those steps connect the conditional probabilities to the actual divisor union; the present section makes no pointwise identification between the two quantities.

6 Conditioned marked bridges

Fixing a complete count type turns the marked path into a bridge with prescribed spatial total and label count. Its two endpoint reserves need not be comparable: one may remain bounded while the other grows with the block length. We prove a single estimate covering these regimes.

For comparable reserves, one joint likelihood ratio exposes both ends and a central tube completes the middle. For unequal reserves, a shorter effective prefix is followed by positive-drift continuation. In either case the remaining clock is integrated before taking the fractional power. This order also allows the new bridge to be applied to a whole measure of inherited private states on one common clock. Finally we treat the ordinary terminal bridge and its root count together.

6.1 The exact binary bridge

Fix a compact family of marks $w_+ > w_- > 0$, their difference $b = w_+ - w_- > 0$, speed $c > 0$, gap mean $\tau > 0$, and $\eta, \pi \in (0, 1)$, with all positive quantities and complementary probabilities bounded away from zero. Let $n \geq 1$, $k = n\pi$ be an integer, and $P = (n + 1)\tau$. The clock G consists of n ordered uniform points in $[0, P]$. Independently, the low marks occupy a uniform k -subset of its slots. For either orientation $\varepsilon \in \{-1, 1\}$, write

$$Z(t) = x + \varepsilon\{S(t) - ct\}, \quad y - x = \varepsilon(nw_+ - bk - cP), \quad (6.1)$$

where $S(t)$ is the sum of the marks at or before t . Both one-sided values at a jump are included in every wall condition. Put

$$\rho = \frac{(c\tau)^2}{(c\tau)^2 + b^2\pi(1 - \pi)}, \quad \chi = \chi(\eta, \rho), \quad Q_n(x, y) = \min \left\{ 1, \left(\frac{(1+x)(1+y)}{n} \right)^{2\chi} \right\}. \quad (6.2)$$

All $n + 1$ gaps, including the final unmarked gap, are present.

For a deterministic grid with bounded boundary density, including $0, P$, write $d_P(t) = \min(t, P - t)$. Given a slope $\vartheta > 0$ and $0 < \zeta < 1/2$, the condition REG_K is

$$|N_a((s, t]) - \lambda_a(t - s)| \leq \vartheta(t - s) + K\{1 + d_P(s)^\zeta + d_P(t)^\zeta\}, \quad \lambda_- = k/P, \quad \lambda_+ = (n - k)/P, \quad (6.3)$$

for both colors and every grid pair $s \leq t$.

Theorem 6.1 (Exact binary bridges with all endpoint reserves). *Fix $g \geq 0$ and $\zeta < \gamma < 1/2$. There are fixed h_0, K such that, for every n , both signs, and all legal $x, y \geq h_0$,*

$$\mathbb{E}_G \mathbb{P}_{\text{marks}} \{Z(t) \geq gd_P(t)^\gamma \ (0 \leq t \leq P), \text{REG}_K \mid G\}^\eta \asymp Q_n(x, y). \quad (6.4)$$

The upper bound holds after dropping the curve and regularity, and the comparison is uniform for arbitrarily large reserves. One may replace the rates in (6.3) by prescribed rates differing from them by at most $C/\sqrt{n}$, after changing the fixed constants.

The proof uses strong one-end families that retain regularity at the same survival scale. We record this input before separating the endpoint regimes.

Lemma 6.2 (Relative one-end regularity). *For the independent exponential/binary reference law with either risk sign and bounded scaled drift on m steps, set*

$$F_m(h) = \min\{1, ((1 + h)/\sqrt{1 + m})^{2\chi}\}.$$

The one-end lower family from Theorem 5.16, with its strong curve, compatible central separate totals, and positive inner endpoint, can also retain two-ended local gap and color regularity, at a fixed fraction of its η -moment, uniformly for $h_{\text{pre}} \leq h \leq C_{\text{res}}\sqrt{m}$. After a fixed reserve adjustment its curve and regularity hold in physical time.

Proof. Section B.2 gives the relative coupling and regularity deletion, followed by its exact-total/type comparisons. The joint Gamma/binomial density compares exposed ends with independent reference ends, and the exact middle tube completes them with its private mass averaged before the fractional power. Endpoint differences remain the signed exact-count increment; every $n + 1$ gap, including the final unmarked gap, is retained. For one large reserve the construction uses an effective short prefix and two-channel continuation. For two large reserves it uses a lower chord with the same centered fluctuation. The legal middle tube and its constants are proved in Section B.4. These estimates retain the stated regularity variants and uniform factor. $\square$

Proof of Theorem 6.1. The upper bounds follow by exposing only the end segments on which survival is still restrictive. For the lower bounds we complete strong end families by an exact middle bridge or a positive-drift tail. We handle comparable reserves, two large reserves, and the asymmetric regime in that order.

Upper bounds. Write $B(G; x)$ for the flat exact-type mass. If both reserves exceed $\sqrt{n}$, use $B \leq 1$. If both are at most $2\sqrt{n}$, expose the first and reversed last $m = \lfloor n/8 \rfloor$ marks and their distinct gaps. The two reference drifts are opposite, with

$$\delta = \varepsilon(w_+ - b\pi - c\tau) = \frac{y - x + \varepsilon c\tau}{n}, \quad (6.5)$$

and bounded scaled magnitude. The type part of the joint density (B.1.1) bounds the private mass pointwise by Cq_Lq_R . Its gap part transfers the η -moment to independent reference ends. Theorem 5.16 gives

$$\mathbb{E}B^\eta \leq C \min\{1, ((1+x)/\sqrt{n})^{2\chi}\} \min\{1, ((1+y)/\sqrt{n})^{2\chi}\} \leq CQ_n(x, y).$$

If $x \leq \sqrt{n}$ and $y > 2\sqrt{n}$, then $\delta \asymp y/n > 0$, apart from bounded n . Expose only a prefix of length $m = \lfloor c_0 n^2 / (1+y)^2 \rfloor$. When this exceeds a fixed scalar threshold, it is at most $n/4$ and $\delta\sqrt{m} \leq C$. Dropping all subsequent tests and using (B.1.1) and the one-end upper gives

$$\mathbb{E}B^\eta \leq C \min\{1, ((1+x)/\sqrt{m})^{2\chi}\} \leq CQ_n(x, y).$$

If m is bounded, $y \gg n$, and $Q_n(x, y) \gg 1$; the trivial upper suffices. Reverse the entire bridge for the opposite asymmetric case, reversing all $n+1$ gaps and exchanging the signs and reserves. All remaining bounded n are immediate.

Lower bound for comparable reserves. First restrict $h_0 \leq x, y \leq C_{\text{res}}\sqrt{n}$, where C_{res} is any fixed constant. On the same two exposed ends use Lemma 6.2 and the strong one-end lower. In the positive-jump direction the rank reserve $x - w_+$ enforces every forward pre-jump value; in the reverse direction minima occur after the negative marks. Choose the endpoint boxes wide and compatible after C_{res} , the laws, and the curves. The two actual inner heights are in $[a\sqrt{n}, A\sqrt{n}]$; separate outer gap sums and color counts are central.

For every retained complete outer state the middle has $r = n - 2m$ marks and $r+1$ gaps, a compact gap mean and interior type. If its duration is T' and its endpoints are x_L, y_R , its mean pre-jump height before mark $j+1$ is exactly

$$(1 - j/r)x_L + (j/r)y_R - \varepsilon \frac{cT'(r-j)}{r(r+1)}, \quad 0 \leq j < r. \quad (6.6)$$

The final endpoint has height y_R . The correction is $O(1)$ for both signs. Lemma B.2, separately for its $r+1$ gaps and r colors, gives a fixed positive probability of remaining above $a\sqrt{n}/2$, which pays the global curve gP^γ for large n .

An ordinary exact-type bridge has uniformly small regularity failure:

$$\mathbb{P}(\text{REG}_K^c) \leq Ce^{-cK}. \quad (6.7)$$

Indeed its two colors are independent sets of uniform points with fixed respective counts. Each interval count is binomial of mean at most $C|I|$. Bernstein gives a bound $C \exp\{-c|I| - cK[1 + d(s)^\zeta + d(t)^\zeta]\}$ for its failure. Bounded grid density and $\sum_{l \geq 0} e^{-cl^\zeta} < \infty$ sum all pairs. Add the piece endpoints to the grid if necessary. Choose K so the failure is less than half the middle's

constant success probability. Piecewise empirical centers differ from the full centers by $O(n^{-1/2})$; start with a smaller slope, absorb this difference for large n , and split a tested interval at the at most two junctions. Piece depths are no larger than global depths, so the finitely many junction terms are absorbed by the intercept.

Conditional on the outer clocks put $a = q_L q_R$, their retained reference private mass. The complete private success mass U , as a function of the middle clock, obeys

$$0 \leq U \leq Ca, \quad \mathbb{E}_{\text{middle}} U \geq ca.$$

The first inequality uses the all-support type density upper. The second uses its central lower and the constant ordinary middle success for each complete outer word. Therefore

$$\mathbb{E}_{\text{middle}} U^\eta \geq (Ca)^{\eta-1} \mathbb{E}_{\text{middle}} U \geq c'a^\eta, \quad (6.8)$$

with zero interpreted separately. Transfer the one joint outer gap density and average the independent reference ends. The result is $\gg F_n(x)F_n(y) \asymp Q_n(x, y)$ in this fixed central box.

Two large reserves. If both reserves exceed $\sqrt{n}$, a small tube for the whole exact bridge remains above half its chord, pays the global curve, and has constant ordinary probability. Delete ordinary regularity failure using (6.7). Its fractional moment is at least its ordinary probability, proving the saturated case.

Asymmetric reserves. After reflection, the remaining regime is $x < \sqrt{n}$ and $y > K_{\text{asym}}\sqrt{n}$. The drift (6.5) is positive and comparable to y/n . For small δ take

$$m = \lfloor c_1 \delta^{-2} \rfloor, \quad x' = \min(x, C_2 \sqrt{m}), \quad y' = y - x + x'.$$

Lowering the entrance this way lowers the whole path. Choose K_{asym} large enough that $m \leq n/8$. Only the one-end threshold h_{pre} is required for x' . The regular strong one-end family retains

$$T_{\text{pre}} \asymp m, \quad a\sqrt{m} \leq H \leq A\sqrt{m}, \quad |T_{\text{pre}} - m\tau| + |K_{\text{pre}} - m\pi| \leq C\sqrt{m},$$

and has moment

$$\mathbb{E} q_{\text{pre}}^\eta \gg F_m(x') \asymp Q_n(x, y). \quad (6.9)$$

The comparison follows from $m^{-1/2} \asymp \delta \asymp y/n$; when entrance clipping occurs both sides are bounded below. For each retained complete prefix set

$$\begin{aligned} n' &= n - m, & P' &= P - T_{\text{pre}}, & k' &= k - K_{\text{pre}}, \\ M' &= (n' - k')w_+ + k'w_-, & v' &= (y' - H)/P' = \varepsilon(M'/P' - c). \end{aligned}$$

The errors are $O(\delta^{-1})$, whereas $y' \asymp n\delta$. Increasing K_{asym} makes $n\delta^2$ uniformly large and yields

$$v' \asymp \delta, \quad Hv' \geq a_0 > 0, \quad P'(v')^2 \geq R_{\text{tail}}. \quad (6.10)$$

Residual type and gap parameters stay compact. These bounds hold for every complete prefix retained by the construction, which is needed before its private mass is integrated.

Use the positive compact speed $c_{\text{half}} = c + \varepsilon v'/2 = (c + M'/P')/2$. On the same remaining word,

$$H/2 + \varepsilon(S_{\text{rem}}(s) - c_{\text{half}}s) = Z_{\text{rem}}(s) - (H + v's)/2.$$

Lemma B.3, with entrance $H/2$, drift $v'/2$ and fixed scaled reserve $a_0/4$, gives constant ordinary probability of $Z_{\text{rem}}(s) \geq (H + v's)/2$. By (6.10),

$$(H + v's)/2 \geq c_* \sqrt{m}(1 + s/m), \quad d_P(T_{\text{pre}} + s)^\gamma \leq C_* m^\gamma (1 + s/m)^\gamma.$$

For m above a fixed threshold this pays the entire remaining curve, uniformly even when n/m is arbitrarily large. Retain ordinary tail regularity by (6.7), and join it to the prefix. Residual rates differ from full rates by $O(\sqrt{m}/n + 1/n) = O(n^{-1/2})$; the prefix's center conversion uses its fixed large effective-time threshold. Apply the single prefix density (B.1.1). The full mass again satisfies $0 \leq U \leq Cq_{\text{pre}}$ and $\mathbb{E}_{\text{rem}}U \geq cq_{\text{pre}}$, where the whole exact remaining clock and type are integrated. Equation (6.8) and (6.9) prove the lower.

If this effective m is bounded, $\delta \geq \delta_0 > 0$ and, for large n , the physical drift $v = (y - x)/P \geq v_0 > 0$. Apply the half-chord construction to the whole bridge. Choose the final h_0 so $h_0v_0/4 \geq 1$ and

$$h_0/2 + (v_0/2)t \geq gt^\gamma \quad (t \geq 0).$$

The finite supremum of $gt^\gamma - v_0t/2$ permits this choice. Continuation and ordinary regularity have constant joint probability; $Q_n(x, y)$ is also of constant order. Thus the bounded effective-prefix regime is handled directly by continuation from the final fixed reserve.

Choice of constants and finite horizons. First fix the regular strong one-end family, its own threshold, and c_1, C_2 . Its endpoint boxes fix a_0 . Then choose continuation constants and K_{asym} , then the central-box constants, then a large effective-time threshold exceeding the curve and prefix thresholds and $(h_{\text{pre}}/C_2)^2$. This fixes v_0 . Only now choose the final h_0 , the ordinary middle/tail regularity intercepts, and a large horizon threshold. For the remaining bounded n , enlarge h_0, K so the bounded possible excursions, curve and counts are covered. This last enlargement changes no internal prefix construction, which required only h_{pre} . Finally, prescribed rate differences $O(n^{-1/2})$ are absorbed into an initially smaller slope at large n , and into K at bounded n . $\square$

6.2 Whole inherited measures and endpoint boxes

A local bridge is read after earlier blocks have produced a measure of entrance states and types. These states share the next spatial clock. The next theorem estimates their whole private mass on that clock, including the old private words and the complete terminal type window in one integral. The inherited measure is fixed before the new clock is read.

Theorem 6.3 (A whole inherited measure on one clock). *Fix common $n, P, w_+, w_-, c, \varepsilon$ as above. Let μ be a finite nonnegative measure on old states and exact types, independent of the new clock G . Each state has legal endpoints satisfying*

$$X \leq x \leq C_{\text{box}}X, \quad Y \leq y \leq C_{\text{box}}Y, \quad \epsilon \leq k/n \leq 1 - \epsilon,$$

where X, Y exceed a sufficiently large fixed threshold. Let $W = \mu(1)$ and integrate the conditional private masses over this whole measure, before taking the power. Put $T = (1 + X)(1 + Y)/n$. For a sufficiently small fixed $t_0 > 0$, define

$$\Psi_{\text{box}} = \begin{cases} 1, & T \geq t_0, \\ T^{2\chi(\eta, \rho_0)}, & T < t_0, \end{cases} \quad q_0 = \frac{w_+ - cP/n}{b}, \quad \rho_0 = \frac{(c\tau)^2}{(c\tau)^2 + b^2q_0(1 - q_0)}. \quad (6.11)$$

In the second branch with $W > 0$, q_0 lies in a fixed compact subinterval of $(0, 1)$. The whole flat mass $q(G)$ satisfies

$$\mathbb{E}q(G)^\eta \asymp W^\eta \Psi_{\text{box}}. \quad (6.12)$$

The same comparison holds when every constituent bridge retains the strong curve and regularity of Theorem 6.1. For $W = 0$ both sides are zero.

Proof. Time reversal reduces to $\varepsilon = 1$. In the rare regime, lawful bridges enclosing the common endpoint boxes will bound the entire measure pointwise on one clock. We first prove the coordinate monotonicity that makes this enclosure possible. If two lawful pairs satisfy $x_2 \geq x_1$ and $y_2 \geq y_1$, couple their low subsets using the first k_1 and first k_2 indices of one uniform permutation. The endpoint relation gives $b(k_2 - k_1) = (x_2 - x_1) - (y_2 - y_1)$. If $k_2 \geq k_1$, the path difference lies between the two nonnegative endpoint differences. If $k_2 < k_1$, it is the entrance difference plus b times the number of changed low marks already seen. Thus

$$B_{k_1, x_1}(G) \leq B_{k_2, x_2}(G) \quad \text{whenever } x_1 \leq x_2, y_1 \leq y_2. \quad (6.13)$$

It holds at every physical time.

In the saturated branch $q \leq W$. With $\nu = \mu/W$, concavity gives

$$\mathbb{E}q^\eta \geq W^\eta \int \mathbb{E}B_{k,x}^\eta d\nu.$$

Each integrand is bounded below by Theorem 6.1, since its endpoint product is at least t_0 . This also proves the strong/regular lower there.

In the rare branch the endpoint identity gives

$$|k/n - q_0| \leq C(X + Y)/n \leq C'T.$$

Choose t_0 so this forces $q_0 \in [3\epsilon/4, 1 - 3\epsilon/4]$. Set $U = nw_+ - cP$ and choose lawful enclosing bridges

$$\begin{aligned} x_- = X/2, \quad k_- &= \left\lfloor \frac{x_- + U - Y/2}{b} \right\rfloor, \quad y_- = x_- + U - bk_- \in [Y/2, Y/2 + b), \\ x_+ = 2C_{\text{box}}X, \quad k_+ &= \left\lfloor \frac{x_+ + U - 2C_{\text{box}}Y}{b} \right\rfloor, \quad y_+ = x_+ + U - bk_+ \in [2C_{\text{box}}Y, 2C_{\text{box}}Y + b). \end{aligned}$$

Choose the reserve threshold so $X/2, Y/2$ meet the bridge theorem and $b \leq Y/2$. Then $x_- \leq x \leq x_+$ and $y_- \leq y \leq y_+$. Their type proportions differ from q_0 by $O((X + Y)/n + 1/n) = O(T)$; after decreasing t_0 and increasing the fixed length threshold they lie in $[\epsilon/2, 1 - \epsilon/2]$. The floor choices are therefore admissible types. Equation (6.13) gives, on the one actual clock,

$$WB_-(G) \leq q(G) \leq WB_+(G).$$

Their endpoint products are comparable to T . The local Lipschitz estimate for χ gives $|\chi_\pm - \chi_0| \leq CT$. Since $T|\log T|$ is bounded, both bridge moments are comparable to Ψ_{box} . Taking the power proves the flat comparison. The same exponent argument makes every constituent $Q_n(x, y)$ comparable to Ψ_{box} . Normalized-measure concavity and Theorem 6.1 give the strong/regular lower; the upper drops these extra conditions. Regularity enters through the individual strong lower bounds and concavity; the subset coupling is used only for the flat upper and lower enclosure. $\square$

Corollary 6.4 (A central terminal band). *Let an old-state measure κ be independent of G and have mass m_κ . In state ω , let the admissible private counts form an integer interval J_ω containing $\asymp r$ integers, $r \geq 1$, all with $|k - n\beta| \leq C_{\text{type}}\sqrt{n}$. Assume the common endpoint boxes of Theorem 6.3. Give type k its original mass $p_k = \mathbb{P}\{\text{Bin}(n, \beta) = k\}$ and let q be the entire old-state and terminal-band mass. Then, with*

$$\rho_\beta = \frac{(c\tau)^2}{(c\tau)^2 + b^2\beta(1 - \beta)},$$

one has

$$\mathbb{E}q^\eta \asymp m_\kappa^\eta \min(1, r/\sqrt{n})^\eta \min(1, T^{2\chi(\eta, \rho\beta)}).$$

This also holds with the strong curve and regularity.

Proof. Uniform central Stirling gives $\sum_{k \in J_\omega} p_k \asymp \min(1, r/\sqrt{n})$ in every state. Thus $W \asymp m_\kappa \min(1, r/\sqrt{n})$. In the rare branch, $|q_0 - \beta| \leq C(T + n^{-1/2})$ and $T \geq c/n$. The exponent change times $|\log T|$ is bounded by $C(T|\log T| + n^{-1/2} \log n)$. In the other branch the factors are of constant order. Apply Theorem 6.3. $\square$

The following count-resolved formulation shows explicitly where the terminal type mass enters the local kernel. On a local block of rate ν , duration L , high mark a , low mark $a - b$, and child probability β , put $A = a - b\beta$ and $e = \nu A - c$. Given $N = n$ and entrance x , let

$$v_k = x + an - bk - cL, \quad M_n(x, I) = \sum_{k: v_k \in I} \mathbb{P}\{\text{Bin}(n, \beta) = k\}.$$

The private terminal kernel has the exact expansion

$$K_G(x, I) = \sum_{k: v_k \in I} \mathbb{P}\{\text{Bin}(n, \beta) = k\} B_k(G; x). \quad (6.14)$$

For a nonempty band $I = [y, y + \Delta]$, with $2b \leq \Delta \leq C(1 + y)$, central selected types, and $h_0 \leq x \leq C_{\text{res}}\sqrt{n}$ and $h_0 \leq y \leq y + \Delta \leq C_{\text{res}}\sqrt{n}$, monotonicity sandwiches this mass between $M_n B_{k_{\max}}$ and $M_n B_{k_{\min}}$. Consequently

$$\mathbb{E}_{G|n} K_G(x, I)^\eta \asymp \min(1, \Delta/\sqrt{n})^\eta \min\{1, ((1+x)(1+y)/n)^{2\chi^*}\}. \quad (6.15)$$

The exponent may be evaluated at either extreme selected type. Their proportions differ by $O(n^{-1/2})$, so its Lipschitz change times $\log n$ is bounded. Sufficient count-resolved conditions are

$$|n - \nu L| \leq C_N \sqrt{L}, \quad |eL| \leq C_e \sqrt{L}, \quad h_0 \leq x, y, \quad x, y + \Delta \leq C_{\text{res}} \sqrt{n}.$$

Indeed $k - \beta n = [x + A(n - \nu L) + eL - v_k]/b = O(\sqrt{L})$. Then $\tau = L/(n + 1) = \nu^{-1} + O(L^{-1/2})$ and $\pi = \beta + O(L^{-1/2})$; one may use $\rho = A^2/(A^2 + b^2\beta(1 - \beta))$ in (6.15). The displayed identity verifies centrality for every selected private type, in addition to centrality of the uncolored count.

6.3 Allocating a fixed fraction of the endpoint budget

The geometric lower construction spends part of each endpoint reserve on other constraints. The next comparison allows that fixed allocation without changing the centered word, its type probability, or the scale of the binary bridge factor.

Corollary 6.5 (Contracted chords). *On the same exact bridge write*

$$Z(t) = \ell(t) + B(t), \quad \ell(t) = (1 - t/P)x + (t/P)y, \quad B(t) = \varepsilon\{S(t) - (t/P)M\}, \quad M = nw_+ - bk.$$

For $\theta \in [\theta_0, 1]$, $\theta_0 > 0$, replacing the wall condition in (6.4) by $\theta\ell(t) + B(t) \geq g_{d_P}(t)^\gamma$ preserves its comparison to the original $Q_n(x, y)$, after increasing the fixed reserve threshold. The whole-measure version of Theorem 6.3 also holds, with θ allowed to depend on the old state and type, provided that measure and selection remain independent of G .

Proof. The exact same-word identity is

$$\theta\ell(t) + B(t) = \theta x + \varepsilon\{S(t) - c_\theta t\}, \quad c_\theta = \theta c + (1 - \theta)M/P.$$

This speed is a convex combination of two positive compact rates. Theorem 6.1 applies with endpoints $\theta x, \theta y$ and unchanged type, clock and regularity centers. If $T = (1 + x)(1 + y)/n$ is bounded below, both its factor and the original factor are bounded below. Otherwise

$$|c_\theta - c| = (1 - \theta)|y - x|/P \leq C(x + y)/n \leq CT.$$

Lipschitz continuity gives $|\chi_\theta - \chi| \leq CT$, while $\theta_0^2 T \leq (1 + \theta x)(1 + \theta y)/n \leq T$. Boundedness of $T|\log T|$ proves the factor comparison. For the whole measure, the contracted event is a subset of the original event since $\ell \geq 0$, proving the upper. Normalize the measure, apply concavity, and integrate the individual lower for the lower. Thus contraction changes the auxiliary path constraint while keeping the original type probabilities and likelihoods. $\square$

There is also a complete-cell version of the flat wall and regularity statement. Requiring them only at cell boundaries enlarges the continuous lower event. Conversely, if widths are bounded by $\Delta_{\max}$, a boundary wall implies continuous survival after increasing both reserves by $c\Delta_{\max}$: use the preceding boundary for positive jumps and the following boundary for negative jumps. The new Q is comparable to the old one. The boundary tests read only cell/color counts; their conditional private mass depends only on cell occupancies. Projection of uniform positions to these occupancies therefore preserves that mass and its fractional moment exactly.

6.4 Ordinary bridges with physical curves

The terminal ordinary factor has constant marks and no private binary variance. Its flat bridge estimate is a ballot bound. We then retain the strong physical curve and regularity by a separate one-channel construction, including asymmetric and arbitrarily large reserves.

Lemma 6.6 (Flat constant-mark bridge). *Let n uniform points lie in $[0, P]$, and let $Z(t) = x + \varepsilon(wN(t) - ct)$ with legal endpoint $y = x + \varepsilon(nw - cP)$. For w, c in fixed positive compact ranges,*

$$\mathbb{P}\{Z(t) \geq 0 \ (0 \leq t \leq P)\} \asymp \min\{1, xy/(1 + n)\}, \quad x, y \geq \max(w, 1).$$

For arbitrary nonnegative endpoints its upper holds with $(1 + x)(1 + y)$ in place of xy .

Proof. It suffices to take $\varepsilon = 1$. Scale height by w and time by w/c , and put $l = cP/w$, $a = x/w$, $b' = y/w = a + n - l$. Reversing the ordered uniform points gives precisely the constraints $\xi_j \geq (j - b')/l$. By [19, Lemma 8.3, pp. 53–54], for integer $n \geq 1$ and real $u, v \geq 1$ with $u + v - n \geq 1$,

$$n! \text{vol}\{0 \leq \xi_1 \leq \dots \leq \xi_n \leq 1 : \xi_j \geq (j - u)/v\} \asymp \min\{1, u(u + v - n)/n\}.$$

Use $u = b'$, $v = l$, so the other endpoint is a . When $l < 1$ and $a, b' \geq 1$, survival is automatic and the displayed factor is bounded below. The case $n = 0$ is deterministic. Raising both reserves by w proves the upper for arbitrary nonnegative endpoints, with the same elementary small-duration cases. Reflection proves the negative-jump assertion and reverses all $n + 1$ gaps. $\square$

Theorem 6.7 (Ordinary strong bridge with all reserves). *Fix positive compact w, c, τ , put $P = (n+1)\tau$, and take $n \geq 1$ uniform points in $[0, P]$. Let $Z(t) = x + \varepsilon(wN(t) - ct)$ have legal endpoint y . Fix $g > 0$, $0 < \zeta < \gamma < 1/2$, a regularity slope $\vartheta > 0$, and a grid of bounded boundary density. There are fixed h_0, K such that, for all legal $x, y \geq h_0$ and both signs, let $\mathcal{A}_{x,y}$ be the joint event*

$$\begin{aligned} Z(t) &\geq gd_P(t)^\gamma \quad (0 \leq t \leq P), \\ |N(t) - N(s) - (n/P)(t-s)| &\leq \vartheta(t-s) + K[1 + d_P(s)^\zeta + d_P(t)^\zeta] \quad \text{for all grid pairs.} \end{aligned}$$

Then

$$\mathbb{P}(\mathcal{A}_{x,y}) \asymp \beta_n(x, y) := \min\{1, (1+x)(1+y)/(1+n)\}. \quad (6.16)$$

The regularity may hold at all physical pairs. Rates within $C/\sqrt{1+n}$ of n/P are also permitted. The same result holds for a fixed contracted chord $\theta\ell + B$, with $\theta \in [\theta_0, 1]$.

Proof. Section B.7 proves the result with one exponential channel. A stopped quadratic potential gives the drifted flat upper bound. The OU reflection density, numerical coupling, and relative flat-error estimate give the strong one-end lower bound.

For comparable reserves, one joint two-end Gamma density exposes the reference ends and an exact middle tube connects them. For asymmetric reserves, an effective prefix is followed by exact continuation; the reversed late-half estimate uses its factor-four threshold. For large common reserves, endpoint boxes are imposed on a capped surrogate. The actual chord adds a common reserve and preserves the centered fluctuation, so the actual terminal height has no upper box restriction. The construction retains the physical strong curve and uncolored regularity for every legal endpoint pair. $\square$

Corollary 6.8 (Fixed entrance and a nonnegative terminal surplus). *Fix $w > 0$, $g \geq 0$, $0 < \zeta < \gamma < 1/2$ and $\vartheta > 0$. There are fixed $h, K, H, \epsilon_0 > 0$ such that, for n uniform points on $[0, U]$ and $v = nw - U$ with $0 \leq v \leq \epsilon_0 U$, the joint event*

$$\begin{aligned} h + wN(t) - t &\geq gd_U(t)^\gamma, \\ |N(t) - N(s) - (n/U)(t-s)| &\leq \vartheta(t-s) + K[1 + d_U(s)^\zeta + d_U(t)^\zeta], \\ |N(t) - (n/U)t| &\leq H + \vartheta d_U(t) \end{aligned}$$

has probability $\asymp (1+v)/(1+U)$, for all sufficiently large U . The same holds on a fixed complete-cell mesh of bounded width.

Proof. Use Theorem 6.7 with endpoints $h, h+v$. First fix h , then choose ϵ_0 so $\epsilon_0 h$ is sufficiently small. Since $n \asymp U$, its factor is $\asymp (1+v)/(1+U)$. Take regularity slope initially below $\vartheta/2$. Setting $s = 0$ or $t = U$ and using the exact total bounds the endpoint discrepancy by $(\vartheta/2)d_U(t) + K'[1 + d_U(t)^\zeta]$. The inequality $K'z^\zeta \leq (\vartheta/2)z + H$ gives the third condition at no probability cost. A boundary-only wall implies a continuous flat wall after a fixed reserve increase, which proves the mesh upper; the continuous lower already implies the mesh lower. $\square$

The all-physical regularity also yields, for $T_0 = 0, T_{n+1} = P$,

$$|T_j - T_i - \frac{P}{n+1}(j-i)| \leq \vartheta_{\text{gap}}(j-i) + K_{\text{gap}}[1 + d_n(i)^\zeta + d_n(j)^\zeta], \quad d_n(i) = \min(i, n+1-i).$$

Indeed the rank-count difference at arrival endpoints is at most one. End-interval instances of physical regularity show that time and rank depths are comparable up to an additive constant,

by absorbing Kz^ζ into a small linear term. Solve the resulting interval inequality; replacing P/n by $P/(n+1)$ costs a fixed constant. This includes both boundary gaps.

Optional named regularity is compatible with the ordinary lower. Conditional on each retained uncolored regular clock, refine its slots by a uniform exact full type in a compact simplex. A category count in an interval is hypergeometric with mean $p_a N(I)$. Start with uncolored slope and intercept K_0 fixed. Then $N(I) \leq C|I| + CK_0[1 + d(s)^\zeta + d(t)^\zeta]$. Hypergeometric Bernstein, with a named intercept K_1 sufficiently larger than K_0 , bounds each failure by

$$C \exp\{-c|I| - cK_1[1 + d(s)^\zeta + d(t)^\zeta]\}.$$

Summation over grid pairs makes its probability as small as desired uniformly for every retained uncolored clock. Thus this refinement retains a fixed fraction before any outer mixture; it does not assert retention of an arbitrary additional private future test.

6.5 One root count and the ordinary terminal suffix

The ordinary factor in the profile includes both a root count window and survival along the terminal suffix. These two requirements use the same reservoir count. We expose the other counts, select one central reservoir atom that places the total in its root window, and then estimate the path conditional on the complete count vector. The largest block supplies the atom; a block near its drift threshold, when present, supplies the only nonconstant ballot factor.

Theorem 6.9 (Joint ordinary root and path estimate). *Fix H , a compact product interval $[\sigma_-, \sigma_+] \subset (0, 1)$, $C_\zeta < \infty$, and $W \geq \log H$. Let the occupied alphabets be $H \geq a_1 > \dots > a_m \geq 2$, with positive lengths L_i , and put*

$$\nu_i = a_i^\sigma, \quad w_i = \log a_i, \quad c_i = a_i - 1, \quad e_i = \nu_i w_i - c_i, \quad V = \sum_i L_i, \quad D = \sum_i e_i L_i \geq 0.$$

Read independent rate- ν_i Poisson clocks in reverse block order $m, \dots, 1$, with jumps w_i and downward speed c_i , obtaining $R(0) = 0$ and $X = R(V) = \sum_i w_i N_i - \sum_i c_i L_i$. Let $E = D + \zeta$, where $|\zeta| \leq C_\zeta \sqrt{1 + D}$ is fixed independently of these clocks. For a largest block i_ , with a fixed tie rule, set*

$$M = L_{i_*}, \quad A = \sum_{i < i_*} L_i, \quad B = \sum_{i > i_*} L_i, \quad \beta_D(h) = \min \left\{ 1, \frac{(1 + h + B)(1 + h + D + A)}{1 + M} \right\}.$$

There are fixed V_0, h_0 such that, uniformly for $V \geq V_0$ and all $h \geq h_0$,

$$\mathbb{P}\{h + R(t) \geq 0 \ (0 \leq t \leq V), E - W \leq X \leq E\} \asymp (1 + V)^{-1/2} \beta_D(h). \quad (6.17)$$

The upper holds without the lower length threshold. The same estimate holds conditionally on any complete external state independent of the local clocks. That state may set h using private prefix labels; when the target is required to be uncolored, E is set by its uncolored data. The root count selected in the proof does not read h .

Proof. The all-count Poisson estimates are

$$\sup_n p_\mu(n) \ll (1 + \mu)^{-1/2}, \quad \sup_n \frac{p_\mu(n)}{1 + n} \ll (1 + \mu)^{-3/2}. \quad (6.18)$$

The first is Stirling. For the second, use $n \geq \mu/2$ to retain its denominator, and the exponential lower tail otherwise. At a fixed central distance from a sufficiently large mean, $p_\mu(n) \gg (1 + \mu)^{-1/2}$.

Condition on all counts except $N_* = N_{i_*}$ and write

$$B_{\text{act}} = \sum_{i > i_*} (w_i N_i - c_i L_i), \quad A_{\text{act}} = - \sum_{i < i_*} (w_i N_i - c_i L_i).$$

The largest block's actual entrance and exit reserves are $x = h + B_{\text{act}}$ and $y = h + X + A_{\text{act}}$. If either is negative there is no survival. Otherwise discard all other path tests and use Lemma 6.6. At most $1 + W/\log 2$ reservoir counts lie in the root window. The two bounds (6.18) separately give

$$C(1 + M)^{-1/2}, \quad C(1 + M)^{-3/2}[1 + h + (B_{\text{act}})_+][1 + h + D + |\zeta| + (A_{\text{act}})_+].$$

The side sums are independent and $\mathbb{E}|B_{\text{act}}| \ll 1 + B$, $\mathbb{E}|A_{\text{act}}| \ll 1 + A$ by their Poisson means and variances. Also $1 + D + |\zeta| \ll 1 + D$. Average the bounds separately, then take their minimum. Since $M \leq V \leq mM$, this is the upper in (6.17), with constants independent of h . The upper thus integrates all side counts, including those outside the central windows used below for the lower.

For the lower retain all other counts in the independent windows

$$|N_i - \nu_i L_i| \leq C_0 \sqrt{1 + L_i}, \quad i \neq i_*. \quad (6.19)$$

Choose C_0 so their joint mass is bounded below, using Chebyshev and the fixed number of blocks. For each retained vector select one count

$$N_* = \left\lfloor \frac{\sum_i c_i L_i + D + \zeta - \sum_{i \neq i_*} w_i N_i}{w_*} \right\rfloor. \quad (6.20)$$

Then $E - w_* < X \leq E$, hence the root window holds. The exact mean identity $\sum_i w_i \nu_i L_i = \sum_i c_i L_i + D$ gives

$$|N_* - \nu_* M| \leq C \left(1 + |\zeta| + \sum_{i \neq i_*} \sqrt{1 + L_i} \right) \leq C' \sqrt{1 + M},$$

since $D \ll V \ll M$. Choose V_0 so this is a nonnegative central count. Its pointwise probability is $\gg (1 + M)^{-1/2}$. On the complete selected vector,

$$|w_i(N_i - \nu_i L_i)| \leq C_1 \sqrt{1 + L_i} \quad \text{for all } i. \quad (6.21)$$

Conditioned on this vector, the original uniform positions of the blocks are independent. Neither the windows nor the selected root read h .

With this complete count vector fixed, we turn to the path. The ordered drift thresholds separate outward side blocks from a possible single central block where the ballot factor remains restrictive. The thresholds $\alpha_a = \log((a - 1)/\log a)/\log a$ strictly increase with a , by Lemma 4.6. Choose fixed disjoint small threshold neighborhoods. At most one occupied block i_0 is near its threshold; if present, one can fix $g_0 > 0$ so

$$e_i \leq -g_0 \quad (i < i_0), \quad e_i \geq g_0 \quad (i > i_0), \quad |e_{i_0}| \leq g_0/8.$$

Away from all occupied thresholds, every present drift has magnitude at least some fixed $g_1 > 0$. These choices follow from finite threshold separation and $|e'_a(\sigma)| \leq H(\log H)^2$. A single occupied alphabet is included by its one near-threshold case.

For a block with outward positive drift at least g , sufficiently large L_i and (6.21) imply actual endpoint increment at least $gL_i/2$. From a fixed reserve, Lemma 6.6 gives constant survival probability, since $N_i \ll 1 + L_i$. For bounded L_i , both selected count and duration are bounded, so the possible loss is deterministically bounded. Concatenating at most H blocks gives an outward side path bounded below by a fixed $-K_{\text{side}}$ with constant conditional probability. Read positive groups forward and negative groups backward. This event is independent of the chosen $h \geq h_0$. Since $E \geq D - C_\zeta \sqrt{1 + D} \geq -C$, choosing h_0 once makes both endpoint reserves sufficient.

If i_0 is present, write L_0, A_0, B_0 for its length and the lengths on its two sides, and

$$A_{\text{mean}} = -\sum_{i < i_0} e_i L_i, \quad B_{\text{mean}} = \sum_{i > i_0} e_i L_i, \quad D = -A_{\text{mean}} + B_{\text{mean}} + e_0 L_0.$$

Here $A_{\text{mean}} \asymp A_0$, $B_{\text{mean}} \asymp B_0$. Their actual values differ by $O(\sqrt{1 + A_0})$ and $O(\sqrt{1 + B_0})$. The actual central junction heights are

$$x = h + B_{\text{act}}, \quad y = h + D + \zeta + \epsilon_{\text{root}} + A_{\text{act}}, \quad -W \leq \epsilon_{\text{root}} \leq 0.$$

For all $h \geq h_0$, choose h_0 so that

$$x \asymp 1 + h + B_0, \quad y \asymp 1 + h + D + A_0, \quad x, y \geq \log H. \quad (6.22)$$

Indeed $g_0 t - C\sqrt{1+t} \geq g_0 t/2 - C'$ and $D - C_\zeta \sqrt{1+D} \geq D/2 - C''$; the fixed remainders are absorbed while retaining at least $h/2$. This is uniform in an unbounded h .

The central count is at most $C(1 + L_0)$. Its conditional ballot probability is therefore at least

$$c\beta_0(h), \quad \beta_0(h) = \min\{1, (1 + h + B_0)(1 + h + D + A_0)/(1 + L_0)\}.$$

At count zero the positive endpoints suffice directly. The central bridge and two outward side events use disjoint positions after all counts are fixed; their junctions are the actual values above. Their joint conditional success is at least $c\beta_0(h)$.

If $i_* = i_0$, this is the desired factor. If $i_* < i_0$, then $A_{\text{mean}} \geq g_0 M$ and $L_0 \leq M$, whence $B_{\text{mean}} \geq 7g_0 M/8 + D$ and $B_0 \gg M$. If $i_* > i_0$, similarly $A_{\text{mean}} + D \geq 7g_0 M/8$ and $D + A_0 \gg M$. In either case $\beta_0(h_0) \gg 1$, so monotonicity in h supplies constant success, which dominates $\beta_D(h) \leq 1$. These cases also cover ties. If there is no near-threshold block, the negative group followed in reverse and the positive group followed forward have constant joint success. Their junction agrees because their exact increments add to X . All-negative nonempty groups are excluded by $D \geq 0$; empty one-sided groups cause no difficulty. Multiplying the conditional success by the constant outside-count mass and the one atom in (6.20) proves the lower. $\square$

With a fixed reserve $h = h_0$, Theorem 6.9 gives the intrinsic scales $(1 + B, 1 + D + A)$. Its $D = \zeta = 0$ specialization gives

$$(1 + V)^{-1/2} \min\{1, (1 + A)(1 + B)/(1 + M)\}.$$

For a single occupied block this is $(1 + V)^{-3/2}$. Thus root centering and ordinary survival use the same count even in the most restrictive case.

More generally, if $\mathcal{F}_{\text{pre}}$ contains the full old word and all its clocks, the theorem applies conditionally whenever the local clocks are still independent with the displayed laws. For every nonnegative $\mathcal{F}_{\text{pre}}$ -measurable weight A_0 ,

$$\mathbb{E}[A_0 \mathbf{1}_{\{\text{root, survival}\}}] \asymp \mathbb{E}[A_0 (1 + V)^{-1/2} \beta_D(h)].$$

This applies the local estimate to each full inherited state and then integrates its weight. The rare shared prefix measure and its actual reserve h are thereby retained in their original order, including any earlier fractional power. The lower threshold $V \geq V_0$ cannot be removed for arbitrary allowed translates: at vanishing length a fixed negative translate can put the root window between all attainable counts. The fixed width $W \geq \log H$ ensures that the root selection always hits its lattice at large length.

7 Count constraints for the lower bound

The lower bound must supply enough divisor boxes while controlling how often they overlap. We build a bounded weight on routing words whose first mass has the predicted profile and whose intersection row is uniformly bounded. The construction passes through three measures: the categorical law of labels, the Poisson law of uncolored positions, and the squarefree harmonic measure on prime tuples.

The order of conditioning is central. We first select a joint measure on complete count types. At each fixed type the original block clocks are independent, so the local bridge estimates can be multiplied with their respective plateau powers. We then integrate the entire count measure, select one root count in an unread ordinary suffix, and transport the label weight to targets determined by its uncolored costs. The transport preserves first mass up to a fixed factor; its targets pay the exceptional overlap classes. Finally, a fractional intersection estimate converts this first mass into volume of the original divisor union.

Section A.2 proves the finite reference and endpoint geometry, Section C.1 constructs the measure, and Section D.1 bounds the full history sums. The scalar bridge upper estimates used for the local cost moments come from the preceding sections and do not use the arithmetic upper bound.

Theorem 7.1 (Lower bound at a fixed lift). *For each fixed k , with $L_* = \Lambda_k$ given by (H.2.1), there is a constant $c_k > 0$ such that, for every $\ell_i \geq \log 3$, putting*

$$L_i = \ell_i - \log 3, \quad L_i^+ = \max(L_i, L_*),$$

one has

$$K_k(\ell) \geq c_k \Psi_k(L^+). \tag{7.1}$$

Here Ψ_k is defined by (3.12), with its saddle and endpoint costs evaluated at L^+ . The constant $L_* = \Lambda_k$ depends only on k , and the estimate is uniform in the length ratios, product gaps, and equality faces.

7.1 Physical data and compatible endpoint pairs

The deterministic saddle chooses the count widths and endpoint scales. The actual local count then fixes the difference between the two endpoints. These two pieces of data must be used together: choosing left and right minima independently would generally produce an incompatible bridge. We normalize each plateau cost by its incoming exit before forming the common endpoint box. For the moment all original physical lengths are large. Their unique ordered saddle is s , their maximal products are $\sigma_1 > \dots > \sigma_m$, and we set

$$u_j = \frac{\sigma_{j+1}}{\sigma_j}, \quad d_j = u_j^{-1} - 1, \quad r_j = \frac{\sigma_j}{\sigma_1}.$$

These ratios and the original fine-label probabilities lie in fixed compact ranges. A plateau is a deterministic group of indices whose original physical blocks remain separate clocks. Let T_j be plateau j 's local total. On a nonterminal plateau use the width convention

$$R_j = \max\{R_0, \min(\sqrt{T_j}, d_j^{-1})\}, \quad w_j = R_j/\sqrt{T_j}, \quad j < m. \quad (7.2)$$

The large-length threshold makes $w_j \leq 1$. Also $d_j R_j \leq C_k$ and $w_j \asymp_k (1 + d_j \sqrt{1 + T_j})^{-1}$. The fixed lower width ensures that each admissible band contains a count value.

For a plateau write its actual normalized history cost as $C_v = a_0 - I_v$ and its output as $Y = a_0 - b_0$. The source uses the *intrinsic* cost

$$K_v = C_v - b_0 = Y - I_v. \quad (7.3)$$

On the displaced full-name count box, local completed endpoint histories have finite scales

$$S_v^I = \begin{cases} h + R_j + \overline{C}_v, & t_v \leq T_{\text{old+local}}/2, \quad j < m, \\ h + \overline{C}_v, & t_v \leq T_{\text{old+local}}/2, \quad j = m, \\ h + \overline{C}_v + S_{\text{good}}(v), & t_v > T_{\text{old+local}}/2. \end{cases} \quad (7.4)$$

The terminal early score is constructed at zero common root target. The early comparison uses the positive reference budget to pay the incoming strict-exit subtraction; in the late row it cancels exactly in (7.3). All contexts are lawful named contexts of the original chain, including the earlier groups.

For the lower source choose every canonical weak edge

$$\mathcal{I} = \{e : |\nu_e| \leq \epsilon_w\}, \quad \epsilon_w = M_1^{-100},$$

including weak edges whose numerical factor is saturated. The same-input-rank strong-concavity argument in Section A.4 gives at most one such edge per original physical block. Every source weak wall is imposed. The omitted strongly drifting edges have factors bounded below, by the exact reference-increment inequality $|\nu_e|L_i \leq X_e + Y_e$; see Section A.4. The upper uses the separately fixed reserve-independent rarity threshold proved there. For $i \in \mathcal{I}$ its original risk is

$$c_i t - a_i N_i(t) + b_i N_{B_i}(t), \quad a_i = \log S_i, \quad b_i = \log B_i, \quad c_i > 0.$$

The original marks $a_i, a_i - b_i$ are positive and nondegenerate. Let X_i^I, Y_i^I be the minima of (7.4) over its left and right completed contexts. Conditional on its actual total n_i and actual subset count k_i , put

$$\Delta_i = c_i L_i - a_i n_i + b_i k_i, \quad x_i^- = \max\{a_* X_i^I, a_* Y_i^I - \Delta_i\}, \quad y_i^- = x_i^- + \Delta_i. \quad (7.5)$$

Here $a_* > 0$ is fixed and small. The count-box theorem makes this a legal pair in a common positive box of scales X_i^I, Y_i^I . Its difference is the actual type increment, while the deterministic scales specify the common box.

The finite scalar theorem gives the factor

$$\Psi_i^I(\eta) = \begin{cases} \left(\frac{(1 + X_i^I)(1 + Y_i^I)}{1 + n_i} \right)^{2\chi(\eta, \rho_i^0)}, & \text{below the fixed rare threshold,} \\ 1, & \text{otherwise,} \end{cases}$$

$$\rho_i^0 = \frac{(c_i L_i / (n_i + 1))^2}{(c_i L_i / (n_i + 1))^2 + b_i^2 p_i^0 (1 - p_i^0)}, \quad (7.6)$$

where $p_i^0 = (a_i - c_i L_i / n_i) / b_i$ on the rare branch. There p_i^0 is in a fixed interior interval; bounded supported experiments are treated separately and contribute a fixed factor. The conditional clock has all $n_i + 1$ gaps. Constant-mark bridges are handled by the ordinary bridge theorem.

The endpoint construction is proved in Section A.2. The bridge estimates apply with compatible endpoint boxes chosen after the fixed wall constants. Both near-central Gamma upper bounds require $n \geq N_\Gamma$, and the central lower bound also requires $n \geq n_0(c, D, R)$ and feasible remaining support; at the exact reference mean the upper bounds hold for every sample size. The fixed lift is chosen so that every original-block count meets the thresholds required here.

7.2 The whole coupled count measure and its mixed moment

A fractional bridge moment becomes useful for the first mass through a mixed moment with its dominating uncolored cost. This converts the local survival estimate into a quantity that can be multiplied before the coupled type measure is integrated. The calculation also explains why a count-window width appears to the product of its own plateau. Count admission can couple all blocks, although their clocks will be independent at a fixed complete type. Conditional on every original uncolored total in its own central window, diagonal strict-exit bands and endpoint-count adjustment give a single count-only measure μ on complete prefix types with

$$W := \mu(\text{all types}) \asymp_k \prod_{j < m} w_j. \quad (7.7)$$

The diagonal splitting variables are independent two-name splits before conditioning their bands. The subsequent acceptance may couple every block. The entire band is summed in (7.7). Before spatial acceptance, fill all terminal names in the retained prefix by their uniform conditional law. Every accepted prefix label subsequently stays fixed. After the actual root count is selected, refine only previously unread suffix labels; this completion has a fixed positive conditional mass for each retained prefix and admitted root vector. It changes neither the uncolored root nor any proper coarse word. The exact disintegration is proved in Section C.2.

For each complete prefix type ω and selected block define its conditional private source mass $q_i(\omega, G_i)$ using the allocated strong curve, binary regularity and full named regularity at the pair (7.5). Conditional on the binary word, uniform refinement of both mark lists supplies named REG. There is a nonnegative uncolored cost $D_i(G_i)$, chosen before any target is read, such that for $\eta = \sigma_{\text{plateau}(i)}$,

$$q_i \leq C D_i, \quad \mathbb{E}_{G_i} q_i^\eta \geq c \Psi_i^I(\eta), \quad \mathbb{E}_{G_i} D_i^\eta \leq C \Psi_i^I(\eta), \quad (2 + n_i)^{-D_0} \leq D_i \leq 1. \quad (7.8)$$

The upper cost is the joint original-class free-bank envelope, uniform in the complete prefix type and in every permitted logarithmic recoloring. For depths beyond its active cutoff the bank test is one, and the summable coefficient tail pays its contribution to the last moment in (7.8).

The local bounds immediately give the mixed moment which will be needed with different plateau powers:

$$\mathbb{E}_{G_i} [q_i D_i^{\eta-1}] \asymp_k \Psi_i^I(\eta). \quad (7.9)$$

Indeed $q_i D_i^{\eta-1} \leq C D_i^\eta$ gives the upper bound. Writing $q_i^\eta = (q_i D_i^{\eta-1})^\eta (D_i^\eta)^{1-\eta}$ and applying Hölder gives

$$\mathbb{E}[q_i D_i^{\eta-1}] \geq \frac{(\mathbb{E} q_i^\eta)^{1/\eta}}{(\mathbb{E} D_i^\eta)^{(1-\eta)/\eta}},$$

which gives the lower bound in (7.9).

Fix a *complete prefix type*. The original local blocks then have independent clocks and uniform multiset words, so we can multiply (7.9) using each block's own plateau power. Tonelli integrates this product over μ afterwards; the type coordinates, exits, and old lists retain their coupling throughout that final integral. Define

$$D_j^{\text{own}} = w_j \prod_{i: \text{plateau}(i)=j} D_i \quad (j < m), \quad D_m^{\text{own}} = \prod_{i: \text{plateau}(i)=m} D_i. \quad (7.10)$$

The width powers multiply (7.7) as $\prod w_j \prod w_j^{\sigma_j-1} = \prod w_j^{\sigma_j}$. This is the source of the nonterminal-plateau count exponents.

Section C.1 proves the joint type construction and these local bounds, including the uniform refinement of both mark lists. The count measure is integrated after the complete prefix type has been fixed in each path estimate. The unread ordinary suffix is integrated first through the pointwise kernel below.

7.3 One prefix-measurable root with a fixed small endpoint

The terminal contribution combines one count atom with an ordinary path constraint. Both are imposed on the same suffix experiment. The prefix fixes its target before any suffix clock is read; all accepted prefix labels are retained. This gives a kernel estimate pointwise in the complete prefix, which can subsequently be combined with its mixed moments. All selected terminal binary edges lie in the negative-drift head. The remaining unread suffix is ordinary. Let its total length be U , its largest original block have length M , and let A, B be its preceding and following lengths within the suffix. Its mean is $D \geq 0$. The hierarchy gives the macro reservoir and the required mean comparison with the full prefix length V_{pre} . Set

$$\mathcal{B}_I = (1 + U)^{-1/2} \min \left\{ 1, \frac{(1 + B)(1 + D + A)}{1 + M} \right\}. \quad (7.11)$$

This is the ordinary factor in (3.11).

Let $x = x(G_{\text{pre}})$ be prefix-uncolored measurable with $0 \leq x \leq A_{\text{tar}}(k) \log(2 + V_{\text{pre}})$. The constant $A_{\text{tar}}(k)$ is fixed by the floors and cascade range below. The prefix paths and their pairs are fixed before evaluating x . Conditional on the complete prefix word, keep the other suffix counts central and select the one reservoir count

$$n_*^x = \left\lfloor \frac{\sum_{i \in \text{suffix}} c_i L_i + E_x - \sum_{i \neq *} w_i n_i}{w_*} \right\rfloor, \quad (7.12)$$

$$E_x = D + \frac{x - \Delta_{\text{pre}}}{r_m}, \quad w_i = \log(r_i + 1), \quad c_i = r_i.$$

The symbol Δ_{pre} is the actual uncolored prefix offset in the exact root identity. On the count support, $|E_x - D| \leq C\sqrt{1 + D}$. The count in (7.12) is nonnegative and central after the fixed threshold is chosen; its probability is comparable to $(1 + U)^{-1/2}$. In the Poisson experiment, this selects the event that the sampled count equals (7.12). On that event the root is $X_1 = x + \epsilon$, where $-C_k \leq \epsilon \leq 0$.

The terminal intrinsic endpoint theorem supplies the ordinary local scales $X_0^I \asymp 1 + P$ and $Y_0^I \asymp 1 + B$, where P is the full length preceding its weak block. The old incoming strict exit is absent from Y_0^I . If the weak block is the reservoir, the new count changes its increment by $-w_*(n_*^x - n_*^0)$:

the smaller legal left representative grows by at most $O(1+x)$ and the right representative decreases while retaining its fixed positive lower bound. The inequality $\log(2+V_{\text{pre}}) \leq CX_0^I$ pays the left change. If it is not the reservoir, its local count is unchanged. The zero-root completion used in this argument only establishes that the prefix types have legal boxes; the actual experiment uses (7.12) anew.

At the selected counts, impose the ordinary allocated strong wall and REG at the intrinsic pair and then perform the true fine-name refinement. Strong suffix blocks use their fixed signed drift and REG; the unique nearly centered block uses the ordinary exact bridge. This gives the following joint statement, before any private prefix integration:

$$\mathbb{E}_{\text{suffix}} F_0^{I,x}(\omega_{\text{pre}}, G_{\text{suffix}}) \asymp_k \mathcal{B}_I F_{\text{pre}}^I(\omega_{\text{pre}}). \quad (7.13)$$

The upper bound follows from the intrinsic full-stop context's fixed reserve ordinary wall and its all-count ballot estimate. The lower constructs the type box and strong wall jointly on the same root fiber. The one root atom occurs once in (7.13). Combining it with (7.9) gives

$$\mathbb{E} \left[f_0^{I,x}(G) \prod_{j=1}^m (D_j^{\text{own}})^{\sigma_j-1} \middle| \mathbf{n}_{\text{pre}} \right] \asymp_k \mathcal{B}_I \prod_{j < m} w_j^{\sigma_j} \prod_{i \in \mathcal{I}} \Psi_i^I(\sigma_{\text{plateau}(i)}). \quad (7.14)$$

Here the original prefix totals $\mathbf{n}_{\text{pre}}$ are fixed in their central windows; the uniform comparison permits their later integration. The private mass $f_0^{I,x} = \mathbb{E}_{\pi} F_0^{I,x}$ retains all coupled types and the true root acceptance.

The ordinary bridge theorem and the same-count construction in Section C.1 prove (7.13) uniformly in the displayed range of prefix-measurable targets.

8 Transport of the selected measure

The local costs specify how far each plateau output should be shifted. A triangular cascade of recolorings realizes these shifts with bounded rounding errors and a bounded density relative to the original fine law. The construction must retain two properties simultaneously: enough first mass and the path support needed to control the target's divisor box intersections. We verify the density and support separately, then compute the first mass by an exact Poisson tilt.

8.1 Adaptive targets and exact density

The targets are logarithms of uncolored costs and are therefore fixed on each environment before the private word is read. An interior bank in each nonterminal plateau contains enough occurrences of two actual fine names to implement the cascade. Uniform subset recoloring makes the output density explicit, even though the accepting measure itself may have coupled all count types. Put

$$y_j = -\log D_j^{\text{own}} \geq 0, \quad x = \sum_{j=1}^m r_j y_j. \quad (8.1)$$

The polynomial floors bound y_j by $C_k \log(2+T_j)$. The hierarchy gives the same bound for the transfers below. We choose envelopes uniformly over this range and then enlarge the source threshold, keeping the floors fixed.

In nonterminal plateau j choose its admitted local macro bank and the two actual fine names: the immediate child's first shallow name c_j and that block's eligible shallow exterior destination

t_j . Write A_{j+1} for the logarithm of the child's first effective alphabet and $b_j = u_j A_{j+1}$. With $e_0 = 0$, set

$$e_j = \left\lceil \frac{y_j + A_j e_{j-1}}{b_j} \right\rceil, \quad \delta_j = b_j e_j - A_j e_{j-1} - y_j \in [0, b_j). \quad (8.2)$$

Uniformly choose e_j occurrences of c_j in that bank and recolor them t_j . The banks are disjoint original blocks. Relative broad bank admission, performed inside the local plateau-power norm, ensures enough source occurrences while retaining a fixed proportion of that norm.

For target bank counts k'_c, k'_t , the exact unrestricted output density relative to the original fine law is

$$\left(\frac{p_c}{p_t}\right)^e \frac{\binom{k'_t}{e}}{\binom{k'_c + e}{e}}, \quad \frac{p_c}{p_t} = \exp((u_j - 1)A_{j+1}). \quad (8.3)$$

On the logarithmically broad source bank its logarithm is bounded by $C(e/\log(2 + T_j) + e^2/N_{\text{bank}}) \leq C_k$. Discarding preimages preserves the upper density bound. If ν_y is the push-forward of the actual accepted source measure, define

$$F_y^I = \frac{1}{C_R} \frac{d\nu_y}{d\pi}, \quad 0 \leq F_y^I \leq 1, \quad \mathbb{E}_\pi F_y^I(G, \cdot) = f_0^{I,x}(G)/C_R. \quad (8.4)$$

The last equality is pointwise in the full uncolored environment, including its actual root indicator.

The necessary path support is reconstructed from the inverse source type. For a flipped selected block of length P , every output type has the same inverse type and hence the same pair x_0, y_0 . If $L_{\text{flip}}(t)$ counts its recolorings, exactly

$$\tilde{Z}(t) = x_0 + (c + be/P)t - aN(t) + bN_{B,\text{target}}(t) = Z_{\text{source}}(t) - b[L_{\text{flip}}(t) - et/P]. \quad (8.5)$$

Both endpoints remain x_0, y_0 . Macro interiority gives $|L_{\text{flip}}(t) - et/P| \leq CeP^{-\gamma} \min(t, P - t)^\gamma$. The preassigned spare source curve absorbs this at the final threshold. Thus the transported necessary banks retain the original source scales. Earlier recolorings may change a later old list while preserving its local points. The same-word geometry below transfers full fine-history support.

The exact grouped outputs satisfy

$$\begin{aligned} Z_j^y &= y_j + \eta_j^y - \eta_{j-1}^y / u_{j-1}, & j < m, \\ Z_m^y &= y_m + \epsilon / r_m - \eta_{m-1}^y / u_{m-1}, \\ \eta_j^y - \eta_j^0 &= \sum_{l \leq j} (\sigma_l / \sigma_j) \delta_l = O_k(1), & 0 \leq E_y := \sum_{j < m} d_j \eta_j^y \leq C_k. \end{aligned} \quad (8.6)$$

The already chosen common root supplies the terminal target, and the root suffix is left unrecolored. For two retained fine words in the same environment,

$$\frac{\pi(\omega)}{\pi(\omega')} = \exp(E_y(\omega) - E_y(\omega')). \quad (8.7)$$

This bounded likelihood ratio applies to the full chain.

The exact recoloring and density calculations, with their support conditions, are given in Section C.1.

8.2 Complete cells and the mass of the selected measure

Arithmetic realization uses slots in prime cells. We therefore define the acceptance and its costs on complete cells from the outset. Boundary walls extend to every positioning of the slots inside their cells with a fixed allowance. This preserves the mixed moments and lets the selected weight be used later without reading individual prime values. The exact tilt then identifies the exponential cost and every remaining profile factor. The preceding construction is made directly on the final complete-cell grid. For total n in a block of length L , its exact uncolored occupancy law is

$$\mathbb{P}(g \mid n) = n! \prod_c \frac{(|c|/L)^{g_c}}{g_c!}. \quad (8.8)$$

This is the image of the n uniform points and their $n + 1$ gaps. The count acceptance, walls, regularity conditions, banks, costs, targets and root count are defined on these cell words.

For either one-sided jump convention the loss in extending a boundary wall into a cell is at most the linear speed times its fixed mesh. It does not depend on the number of points in the cell. For REG, round each interval endpoint inward and outward: the length changes by at most two meshes, the reference mean by a fixed amount, and $|d(t)^\zeta - d(t')^\zeta| \leq |t - t'|^\zeta$. Thus boundary regularity gives continuous REG for every lift of the same word, with fixed extra constants. The exact-type bank argument applies directly to these boundary events. Its functions q_i, D_i are constant on each uncolored cell fiber, so integration by (8.8) preserves q_i^η and $q_i D_i^{\eta-1}$ exactly. Uniform subset recoloring commutes with within-cell slot permutations. Thus F_y^I is a cell-invariant weight that can be assigned to slots before their prime values are chosen.

Let $g = (g_{i,c})$, $h_i = k - i + 2$, and

$$W_g = \prod_{i,c} \frac{|c|^{g_{i,c}}}{g_{i,c}!}, \quad Z_F(g) = \sum_\omega F_y^I(g, \omega).$$

The normalized nominal counting mass is $\mathcal{N}(F) = e^{-\sum_i h_i L_i} \sum_g W_g Z_F(g)$. Expanding the independent natural Poisson cell law of intensity Z_i and then the original fine categorical law gives the exact tilt identity

$$\mathcal{N}(F) = e^{-J(L;s)} \mathbb{E}_{\text{nat}} \mathbb{E}_\pi \left[F_y^I e^{-E_y} \prod_{j=1}^m (D_j^{\text{own}})^{\sigma_j-1} e^{(1/r_m - \sigma_1)\epsilon} \right]. \quad (8.9)$$

The expression is zero off the accepted root/count support. To check the target powers, with $q_j = \sigma_j/\sigma_m - 1$ use $q_j - q_{j+1}/u_j = d_j$ and $(1/r_m - \sigma_1)r_j - q_j = 1 - \sigma_j$. The deterministic part of the Poisson expansion is the current $J(L; s)$; no limiting rate is substituted. The identity is algebraic at fixed products and will remain valid on the frozen grid. The root error and E_y in (8.9) are bounded, whereas the logarithmic targets are not discarded as constants. Equations (7.14) and (8.4) now give

$$\mathcal{N}(F) \asymp_k e^{-J(L;s)} \mathcal{B}_I \prod_{j < m} w_j^{\sigma_j} \prod_{i \in \mathcal{I}} \bar{\Psi}_i^I(\sigma_{\text{plateau}(i)}). \quad (8.10)$$

Here the natural-count factors are obtained as follows. If $\mu_i = Z_i L_i$ and $|n_i - \mu_i| \leq C\sqrt{1 + \mu_i}$, then on the compact rare branch the changes in p_i^0, ρ_i^0 are $O(\mu_i^{-1/2})$. The compact Lipschitz bound for χ gives

$$|\log \Psi_i^I - \log \bar{\Psi}_i^I| \leq C \frac{\log(2 + \mu_i)}{\sqrt{\mu_i}} \leq C.$$

A threshold crossing is harmless since both bases are then comparable to the fixed threshold. The finitely many original Poisson central windows have constant joint mass. Integrating them pays no new type atom and no extra root atom.

We finally compare the centered natural correlation with the original mark correlation in (3.10). This comparison is made only on the compact rare branch. Fix a selected edge $r \rightarrow q$, write $L = L_i$, and put

$$\lambda = (r+z)^\sigma, \quad c = r-q, \quad a = \log(r+z), \quad b_e = \log(q+z), \quad p = \frac{q+z}{r+z}, \quad m = a - b_e p.$$

Its centered natural type and finite-mean correlation are

$$\bar{p} = \frac{a - c/\lambda}{b_e}, \quad \bar{\rho} = \frac{(cL/(\lambda L + 1))^2}{(cL/(\lambda L + 1))^2 + b_e^2 \bar{p}(1 - \bar{p})}. \quad (8.11)$$

In general $\bar{p} \neq p$. For every paired context in (3.5), its right and left reference costs differ by

$$\bar{C}_{v+} - \bar{C}_{v-} = L\{c - \lambda(a - b_e p)\} = L\nu, \quad \nu = c - \lambda m.$$

Both reference costs are nonnegative. Hence $Y_e \geq L\nu$ if $\nu \geq 0$, and $X_e \geq L|\nu|$ if $\nu < 0$, even though the endpoint minima are taken separately. Since $L \geq 1$ and the natural intensity is bounded above and below, the positive score floors give

$$|\nu| \leq CT_e, \quad T_e \geq \frac{c_k}{1+L}, \quad \bar{p} - p = -\frac{\nu}{b_e \lambda}, \quad |\bar{\rho} - \rho_e| \leq C(|\nu| + L^{-1}).$$

The last estimate applies on the compact rare parameter range; the ordinary degenerate law is handled by its separate bridge theorem. Proposition 5.5 therefore gives, below a fixed rare threshold for T_e ,

$$|\chi(\sigma, \bar{\rho}) - \chi(\sigma, \rho_e)| |\log T_e| \leq C(T_e + L^{-1}) |\log T_e| \leq C.$$

The L^{-1} term retains the finite denominator $\lambda L + 1$. Thus the two powers of T_e are comparable by fixed constants. Replacing the fixed positive score constants in (7.4) by those in (3.6) also changes each factor by fixed constants. Branch crossings and the finitely many saturated edges have bounded factors. This proves that the product in (8.10) is precisely the binary profile product up to the required dimension-dependent constants.

Section C.1 gives the complete-cell construction and verifies the count and slot invariances used in this calculation.

9 Intersections of divisor boxes and the lower bound

The first mass counts weighted boxes with multiplicity. To obtain the volume of their union, we need a uniform bound for the total weighted intersection seen from any retained box. A comparison history records which divisor labels can differ and when their partitions separate. Its geometric cost controls complicated histories; its conditional label mass controls the remaining free intervals. The targets make these two bounds complementary, so that every plateau cost is paid exactly once.

9.1 A weighted sum over intersection histories

The row estimate is pointwise in a retained anchor and sums all original histories, including distinct pivots that round to the same cell boundary. Its proof first reduces a history to endpoint geometry, then estimates labels under its actual comparison class, and finally sums the remaining movement penalties. Keeping these stages separate preserves the conditioning on full types and the root coefficient. Fix a cell occupancy g , its fine-word space Ω_g , a retained anchor ω , and a comparison history h . Write $Q_h(\omega)$ for the corresponding class and $\Phi_g(h)$ for its geometric multiplier. The history sum includes all permutations, zero pivots, ties and distinct positive pivot choices. Define

$$\begin{aligned} I_h &= \log \Phi_g(h) + \log \frac{\pi(Q_h(\omega))}{\pi(\omega)}, & R_y &= \sum_{j=1}^m Z_j^y, & C_y(h) &= R_y - I_h, \\ q_h^{\text{rel}} &= \mathbb{E}_\pi \left[F_y^I(\omega') \frac{\pi(\omega)}{\pi(\omega')} \middle| Q_h(\omega) \right]. \end{aligned} \quad (9.1)$$

The arithmetic class uses the slot thresholds $I_j < q$ of (E.1.4). For the geometric argument, replace each positive I_j by the last slot $\bar{I}_j$ of its assigned cell, leaving zero unchanged, and write $\bar{Q}_h(\omega)$ for the resulting class. This activates each boundary after its whole cell. The original history h remains the summation index; rounded pivots may coincide. The rounded data are an auxiliary geometric history, not necessarily a member of $\mathfrak{H}_\mathbf{r}$. They remain deadline-legal, and

$$Q_h(\omega) \subseteq \bar{Q}_h(\omega), \quad E_g(\bar{I}_j) = E_g(I_j), \quad \bar{\Phi}_g(h) = \Phi_g(h).$$

The block assignments and $J_i(h)$ are unchanged because rounding stays inside each pivot's original block. Neither zero pivots nor distinct positive original pivots are removed from the summation. Define $\bar{I}_h, \bar{C}_y(h), \bar{q}_h^{\text{rel}}$ using $\bar{Q}_h$ and this unchanged multiplier. The finite class identity gives

$$e^{-C_y(h)} q_h^{\text{rel}} = e^{-R_y} \Phi_g(h) \sum_{\omega' \in Q_h(\omega)} F_y^I(\omega') \leq e^{-\bar{C}_y(h)} \bar{q}_h^{\text{rel}}.$$

It therefore suffices to estimate the complete-cell classes while retaining every original pivot and its multiplicity. There is a fixed $0 < \delta \leq 1$, chosen from the finite geometry and scalar ranges and independently of roughness, such that

$$\sum_{h \text{ original}} e^{-\delta C_y(h)} (q_h^{\text{rel}})^\delta \leq C_k \left(\prod_j D_j^{\text{own}} \right)^\delta. \quad (9.2)$$

The proof of this row has three parts. The first identifies positive costs from branching and delayed cuts. The second converts the surviving cuts to endpoint constraints. The third estimates their actual label classes and sums all pivot multiplicities. Section D.1 proves the finite estimates used in these steps.

First, legal parent/child projection gives the exact residual-entropy identity and a nonnegative rank-delay correction. A positive mixed entropy loss is charged only when at least two child-touching cells survive and one is mixed with an outside name. Nonterminal-plateau branching normal form pays pure-cell lifetimes. The terminal plateau uses a floating last nonsingleton spine, retains the expired-first and ghost terms, and then makes a lawful same-time permanent-0 comparison. Its zero-mean named discrepancy is bounded by regularity on the actual floating interval. The reference fraction and the spread/depth budgets reserved by the global star-comparison theorem pay this discrepancy; this comparison is applied with those reserved budgets.

Second, synchronize cuts only within an original physical block, using the ordered simplex when several plateau cuts share it. Strong entropy concavity pays spread; actual empirical contrasts are supported on the moved intervals. Endpoint interpolation retains the reference, good-count and mixed-time budgets and allocates the available endpoint reserve once over the finitely many local coordinates. Old weak, proper-input and wrong-name directions are paid by their reference charges. Each remaining selected local direction uses its allocated intrinsic wall. A maximum-weight lawful endpoint vertex pays the root flag without contracting its coefficient. The same-word target identity (8.5) and exact carry (8.6) then give, for the appropriate endpoint signature of an arbitrary fine history,

$$\bar{C}_y(h) \geq \sum_{j \in \mathcal{E}(h)} y_j + c_k \Lambda(h) - C_k.$$

Here $\mathcal{E}(h)$ is the set of early-completion plateaux and Λ retains all positive movement, delay, pure/mixed lifetime, reference/good-count and strong-wall depth penalties required below.

Third, fix the complete-cell class $\bar{Q}_h$. On a block which is FULL away from its endpoint strips, condition on the complete target type and the revealed assignments. The remaining word has its exact original multiset law, and its two necessary bank events admit one joint Gamma/type comparison. Their conditional probability is bounded by the corresponding all-transfer bank envelope. Blocks which are fully exposed pay only their own count and floor polynomials; a comparator-FULL block which is not original-class FULL forces a pure, off-spine or mixed penalty. Late non-FULL endpoint groups have the displaced square-root charge. Early completion keeps the entire y_j instead of attempting to pay a huge exposed suffix by a short prefix budget. These are precisely the signatures preserved in $\Lambda(h)$.

For each class we sum the full inverse-type acceptance, with its paths and future child weights, before taking the δ power. At active depths use $q_{i,d} \leq D_i \exp(b_* d^\gamma)$; at exposed or far depths use the polynomial floor. Choosing b_* below a fixed fraction of the geometric depth coefficient leaves a positive depth charge that sums every original pivot multiplicity. Erased projected times are matched to actual physical deadlines and keep their nonnegative distance penalty. Added deadline representatives may repeat in an enlarged comparison row; original full histories keep their distinct pivots. Finally (8.7) supplies the true relative likelihood in (9.1). This proves the complete-cell row; the displayed class inclusion then proves (9.2) for the same transported density.

By summing (8.6),

$$R_y = \sum_j y_j + \epsilon/r_m - E_y, \quad e^{R_y} \prod_j D_j^{\text{own}} = e^{\epsilon/r_m - E_y} \leq 1. \quad (9.3)$$

Write $M_h(F; \omega) = \sum_{\omega' \in Q_h(\omega)} F_y^I(\omega')$. There is the exact identity

$$\Phi_g(h) M_h(F; \omega) = e^{I_h} q_h^{\text{rel}}. \quad (9.4)$$

Multiplying (9.2) by $e^{\delta R_y}$ and using (9.3) proves the unnormalized arithmetic row

$$\sum_h \Phi_g(h)^\delta M_h(F; \omega)^\delta \leq C_k \quad (F_y^I(\omega) > 0). \quad (9.5)$$

Bounded comparisons between physical and cell endpoint positions cost one fixed factor per original cut. By monotonicity, the same row holds for a nonnegative density dominated by a fixed multiple of F_y^I , with that multiple absorbed into the constant.

Section D.1 proves the projection, synchronization, and full-class estimates just used. In particular, the terminal spine's same-time comparison is paid by the global reference and depth budgets; it does not assert a conditional closure for an arbitrary child weight. The exact class identity also appears in the collision calculation of Section E.1.

9.2 Squarefree supply and the harmonic lower bound

The probabilistic construction is now complete. We assign distinct primes to its slots and compare their harmonic mass with the nominal cell mass. The occupancy tube prevents repeated-prime exclusions from accumulating a significant loss. The bounded intersection row then controls a higher moment of the weighted box count, and Hölder's inequality recovers the nominal first mass as a lower bound for union volume. Summing the named boundary regularity gives an uncolored occupancy tube

$$g_{i,t} \leq A_0 + A_1 \min(t, v_i - t + 1) \quad (9.6)$$

with constants fixed before roughness and the prime grid. Recoloring does not change g . Choose a fixed rough-prime cutoff and greedy complete prime cells of harmonic mass at most their fixed mesh $\eta_i = \log(k - i + 2)/(k - i + 1)$. For a cell D with g slots, the successive-distinct-drawing bound is

$$1 \geq \frac{g! e_g(D)}{\eta_i^g} \geq \exp \left[-\frac{2g^2}{\eta_* \min D} \right]. \quad (9.7)$$

The roughness choice makes every elementary factor positive. Cell minima grow at least exponentially in their index while (9.6) grows linearly. Thus for every retained occupancy,

$$c_k W_g \leq S_g := \sum_{\mathbf{a} \in \mathcal{A}(g)} \frac{1}{a_1 \cdots a_k} \leq W_g. \quad (9.8)$$

The reverse tube and Mertens' difference estimate also put every retained tuple inside one fixed allowed size truncation. Its auxiliary size exponent is fixed from the tube.

Let $B_\omega(\mathbf{a})$ be the original divisor box of side $b = \log 2$. On the same product of harmonic tuple measure and Lebesgue measure define

$$f(\mathbf{a}, t) = \sum_{\omega \in \Omega_g} F_y^I(\omega) \mathbf{1}_{B_\omega(\mathbf{a})}(t), \quad Z_F = \sum_{\omega} F_y^I(\omega).$$

Then $\int f = b^k S_g Z_F \geq c W_g Z_F$. The correlated prime-cell kernel applies to this cell-invariant bounded weight. Its finite class identity is

$$\sum_h \Phi_g(h)^\delta \sum_{\omega} F_y^I(\omega) M_h(F; \omega)^\delta = \sum_{\omega} F_y^I(\omega) \sum_h \Phi_g(h)^\delta M_h(F; \omega)^\delta.$$

By (9.5) and the kernel theorem, $\int f^{1+\delta} \leq C W_g Z_F$. Hölder on $\{f > 0\}$ gives

$$\text{meas}\{f > 0\} \geq \frac{(\int f)^{(1+\delta)/\delta}}{(\int f^{1+\delta})^{1/\delta}} \geq c_k W_g Z_F. \quad (9.9)$$

This retains the nominal first mass to the *first* power. The set $\{f > 0\}$ lies in the full original divisor union, so summing (9.9) over disjoint occupancies and using (8.10) gives the desired actual lower on consistent large grid data.

The prime-cell kernel and the pointwise supply estimates are proved in Section E.1, for the same bounded cell-invariant weight used above.

10 Discretization at the fixed saddle

The lower construction has so far used the physical block lengths. To obtain an arithmetic family, we must realize its accepted words by squarefree prime selections. Complete harmonic cells provide this realization, but their lengths differ slightly from the physical lengths. The point of the comparison is to absorb this bounded change while retaining the saddle and all the conditional laws already used.

There are three steps. We choose the grid before choosing the lift, compare reference costs at the fixed physical optimizer, and then transfer the count boxes and root completion to the grid. The two endpoint templates whose costs can be small are treated by exact identities; all other templates have enough intrinsic charge to absorb the grid error.

Fix the finite geometric, scalar, count, regularity and occupancy constants. These determine the roughness cutoff and complete-cell grid; let R_k bound the total length loss. Choose the lift $L_* > C_k(1 + R_k)^2$ above the remaining source thresholds. With this choice made, lift L to L^+ , solve its KKT problem once, and denote the optimizer by s^+ . For final cell lengths L' one has

$$\sum_i |L'_i - L_i^+| \leq R_k.$$

The products, plateaux, categorical laws, selected local edges and root suffix are those of (L^+, s^+) . An affine map from each physical block to its grid block preserves the order of legal cuts and deadlines. Since the reference integrands are bounded, the comparison is uniform over the entire original history space:

$$|\overline{C}_{L',s^+}(h') - \overline{C}_{L^+,s^+}(h)| \leq C_k R_k, \quad |J(L'; s^+) - J(L^+; s^+)| \leq C_k R_k. \quad (10.1)$$

Thus each reference lower bound acquires a fixed intercept. The same-word, class and likelihood identities are count identities and remain exact on the grid.

Two exceptional endpoint templates must be handled by their exact outputs. If a plateau's grid natural mean residual is $m_{\text{ref}} = O_k(R_k)$ and $\zeta_g = H_g - \mathbb{E}H_g$, then

$$\begin{aligned} C_v &= a_0 - \sum_{g \leq l} \zeta_g + \overline{C}'_v + \sum_{\text{proper}} \nu_{v,g}, \\ &= b_0 + m_{\text{ref}} + \sum_{g > l} \zeta_g + \overline{C}'_v + \sum_{\text{proper}} \nu_{v,g}. \end{aligned} \quad (10.2)$$

For the all-zero template the row gives $C_v = a_0$ exactly. For the complete-deadline template, $\nu = 0$ and $\overline{C}'_v = -m_{\text{ref}}$; the residual cancels, giving $C_v = b_0$ and zero intrinsic cost. Thus both small endpoints retain their original values. All remaining templates have reference plus good-count charge at least $c_k \sqrt{\min_i L_i^+} - C_k$. At the chosen lift this charge absorbs (10.1).

We next transfer the count selection. Bounded integer translations of the band centers preserve the widths of the complete strict count bands; at the final threshold their Stirling ratios are bounded. The local pair uses its actual grid increment. Frozen physical endpoint scales and all grid scalar parameters are comparable, with bounded exponent-times-log error. The newly selected grid root uses the exact grid budget. If its old prefix is nonempty, its positive incoming mean survives the bounded defect. If that prefix is empty, use the legal exact ordinary pair and its bounded root floor directly. The root/error/carry identities and (9.3) remain exact.

It follows that the direct cell family, full row and arithmetic transfer above apply on (L', s^+) and give

$$K_k(\log 3 + L^+) \geq c_k e^{-J(L'; s^+)} \mathcal{P}_{\text{frozen}}^I(L'; s^+) \geq c_k e^{-J(L^+; s^+)} \mathcal{P}_I(L^+). \quad (10.3)$$

Here $\mathcal{P}_I$ is (8.10)'s nonexponential factor, comparable to the factors of (3.12). The frozen likelihood comparison above applies off stationarity on the grid.

To return from the lift to the given cutoffs, apply Theorem 14.1 to the harmonic sum for $\ell_i \geq \log 3$. Since the lift changes each coordinate by at most the fixed L_* ,

$$K_k(\ell) \geq c_k K_k(\log 3 + L^+).$$

Combining this with (10.3) proves Theorem 7.1. The occupancy constants remain fixed after choosing roughness, and the size of L_* affects only the theorem's constants.

Section E.1 gives the detailed grid comparison, including the exceptional endpoints and the exact root count. Section 14 proves the parameter stability used here directly for the original arithmetic sum.

The arithmetic comparison is ultimately applied at the original vector $\ell(\mathbf{y})$ under (2.4). The auxiliary lift of K_k does not alter the cutoffs in that hypothesis. The same observation applies to the original table axes in Section 15.

11 Ordered exponential sums and ordinary root estimates

A threshold rectangle inside the ordinary root block has unresolved widths given by sums of exponentials of ordered prime positions. Consequently its large-volume event is controlled by an ordered-sum condition, rather than by a flat order-statistics wall. We prove the estimate needed for this event and then integrate it against the root count in Section 12.5.

The argument follows the band decomposition in [17, Lemma 5.3, pp. 31–33], with explicit integer rounding to obtain the $A = 0$ case of [19, Lemma 5.5]. A large departure below the flat wall forces many points to crowd near the minimizing index; the factorial cost of that crowding supplies the strong tail. We first state the two order-statistics inputs. Constants may depend on the fixed parameters $C > 1$ and $\mu > 1$.

Put

$$\begin{aligned} \Delta_r &= \{0 \leq \xi_1 \leq \cdots \leq \xi_r \leq 1\}, \\ S_r(u, v) &= \{\xi \in \Delta_r : \xi_j \geq (j - u)/v \ (1 \leq j \leq r)\}, \\ Q_r(u, v) &= r! \operatorname{vol}(S_r(u, v)). \end{aligned}$$

Lemma 11.1 (Flat-wall upper bound). *For an integer $r \geq 1$ and real $u, v > 0$ with $w = u + v - r > 0$,*

$$Q_r(u, v) \ll \frac{(u + 1)(w + 1)}{r}.$$

This is the upper assertion of [17, Lemma 5.1, p. 30]. Koukoulopoulos derives this statement from Ford's estimates [14, Theorem 1] and [12, Lemma 11.1]. The lower assertion of Lemma 5.1 is not used here.

Lemma 11.2 (Crowding estimate). *Fix $C > 1$. Suppose g, r, s, u, v are integers satisfying*

$$2 \leq g \leq r/2, \quad s \geq 0, \quad r \leq Cv, \quad u \geq 0, \quad u + v \geq r + 1.$$

Let R consist of the points of $S_r(u, v)$ for which there is an integer $g + 1 \leq l \leq r$ such that

$$\frac{l - u}{v} \leq \xi_l \leq \frac{l - u + 1}{v}, \quad \xi_{l-g} \geq \frac{l - u - s}{v}.$$

Then

$$\text{vol}(R) \ll_C \frac{[C(s+1)]^g}{(g-1)!} \frac{(u+1)(u+v-r)}{(r+1)!}.$$

This is [17, Lemma 5.2, p. 30, author version], derived there from the 2008 version of [13, Lemma 4.3] using Lemma 5.1 in place of Ford's Lemma 4.1. In that derivation the constant 10 becomes C , $(u+v-r)^2$ becomes $u+v-r$, and $(g-2)!$ becomes $(g-1)!$.

For any $\gamma \geq 0$, define

$$\mathcal{T}_\mu(r, v, \gamma) = \left\{ \xi \in \Delta_r : \sum_{i \leq j} \mu^{v\xi_i} \geq \mu^{j-\gamma} \ (1 \leq j \leq r) \right\}.$$

Lemma 11.3 (An ordered-sum estimate). *Fix $C > 1$ and $\mu > 1$. For integers r, v, γ satisfying $\gamma \geq 0$ and $1 \leq r \leq Cv$, set $b = r - v$, $d = b - \gamma$, and*

$$Y = \begin{cases} b, & d \geq 1, \\ (1-d)(\gamma+1), & d \leq 0. \end{cases}$$

Then

$$\text{vol}(\mathcal{T}_\mu(r, v, \gamma)) \ll_{C, \mu} \frac{Y}{\mu^{\mu^d} (r+1)!}. \quad (11.1)$$

Proof. We divide the event according to its greatest departure below the linear wall. Small departures are covered by the flat-wall estimate. For a large departure, the ordered-sum condition forces a crowded cluster; we apply the crowding estimate and sum over departure bands.

The flat-wall part. Choose the integer threshold

$$B = \left\lceil 2 + \frac{\log 32}{\log \mu} \right\rceil, \quad t = \max\{d, B\}. \quad (11.2)$$

Both d and t are integers. For $\xi \in \mathcal{T}_\mu(r, v, \gamma)$ let $D = \max_{j \leq r} (j - \gamma - v\xi_j)$. If $D < t$, all coordinates satisfy $\xi_j > (j - \gamma - t)/v$; call this part V_1 . Otherwise choose the integer $h = \lfloor D \rfloor + 1 \geq t + 1$ and an index l attaining D . Then

$$\min_{j \leq r} \left(\xi_j - \frac{j - \gamma}{v} \right) = \xi_l - \frac{l - \gamma}{v} \in [-h/v, (-h+1)/v]. \quad (11.3)$$

Call the union of these bands V_2 . This also covers $D = t$; overlaps at endpoints have measure zero.

The flat-wall part V_1 is empty when $d \geq B$: then $t = d$, and the inequality at $j = r$ would force $\xi_r > 1$. When $d < B$ we have $t = B$, and Lemma 11.1 applies with $u = \gamma + B > 0$, $w = B - d \geq 1$. It gives

$$\text{vol}(V_1) \ll \frac{(\gamma + B + 1)(B - d + 1)}{(r + 1)!}. \quad (11.4)$$

Here $1/(rr!) \leq 2/(r+1)!$. For $d \leq 0$ the numerator is at most $(B+1)^2 Y$ and $\mu^{\mu^d} \leq \mu$. For $1 \leq d < B$ it is at most $B(B+2)b$, while $\mu^{\mu^d} \leq \mu^{\mu^{B-1}}$. Since B depends only on μ , both cases give the required bound for V_1 , including its double exponential denominator.

A crowded cluster in each remaining band. Fix an integer band $h \geq t + 1$. Write

$$a = \left\lceil \frac{\log 4}{\log \mu} \right\rceil, \quad m_0 = 1 + a, \quad h_0 = h - m_0.$$

These are integers, and

$$h_0 \geq t - \left(\frac{\log 4}{\log \mu} + 1 \right) \geq 1 + \frac{\log 8}{\log \mu}. \quad (11.5)$$

We claim that there is an integer $m \geq h_0$ with $g_m = \lfloor \mu^m \rfloor < l/2$ such that

$$\xi_{l-g_m} \geq \frac{l - \gamma - 2m}{v}. \quad (11.6)$$

To see the cluster, take the last half of the first l summands and group them by backward index in $[g_m, g_{m+1})$. The first group is bounded using μ^{h_0} terms; each subsequent group is bounded using $g_{m+1} - g_m$ terms. Monotonicity of the ordered positions gives

$$\sum_{j \leq l} \mu^{v\xi_j} \leq 2 \left(\mu^{h_0} \mu^{v\xi_l} + \sum_{\substack{m \geq h_0 \\ g_m < l/2}} (g_{m+1} - g_m) \mu^{v\xi_{l-g_m}} \right).$$

If (11.6) failed for all candidates, the band condition would therefore give

$$\sum_{j \leq l} \mu^{v\xi_j} < 2\mu^{l-\gamma} \left(\mu^{-a} + \sum_{m \geq h_0} (g_{m+1} - g_m) \mu^{-2m} \right) \leq \mu^{l-\gamma}.$$

Indeed, $\mu^{-a} \leq 1/4$, and summation by parts with $\mu^m - 1 \leq g_m \leq \mu^m$ gives

$$\sum_{m \geq h_0} (g_{m+1} - g_m) \mu^{-2m} \leq \mu^{1-h_0} + \mu^{-2h_0} \leq 2\mu^{1-h_0} \leq 1/4.$$

When there are no candidates, the first group alone bounds the sum by $\mu^{l-\gamma}/2$, so no strict-inequality argument is needed. Either case contradicts membership in $\mathcal{T}_\mu$ and proves (11.6).

Summing the cluster bounds. Apply Lemma 11.2 with $u = \gamma + h$, $s = 2m$, $g = g_m$. The minimal-band condition places ξ in $S_r(u, v)$ and gives the required interval for ξ_l ; moreover

$$\xi_{l-g} \geq (l - \gamma - 2m)/v \geq (l - u - s)/v.$$

All parameters are integers, $g \geq 2$ by (11.5), $g < l/2 \leq r/2$, $l \geq g+1$, $s, u \geq 0$, $u+v-r = h-d \geq 1$, and $r \leq Cv$. The crowding lemma includes the union over l , so there is no extra factor r . We therefore sum only over h, m , obtaining

$$\text{vol}(V_2) \ll_C \frac{1}{(r+1)!} \sum_{h \geq t+1} \sum_{m \geq h-m_0} (\gamma + h + 1)(h - d) \frac{[C(2m+1)]^{\lfloor \mu^m \rfloor}}{(\lfloor \mu^m \rfloor - 1)!}. \quad (11.7)$$

For fixed C, μ , Stirling's lower bound implies

$$\frac{[C(2m+1)]^{\lfloor \mu^m \rfloor}}{(\lfloor \mu^m \rfloor - 1)!} \ll_{C, \mu} \mu^{-\mu^{m+m_0}}.$$

Indeed its logarithm is $-(\log \mu)m\mu^m + O_{C, \mu}(\mu^m \log(m+1) + m)$. For large m this is smaller than $-(\log \mu)\mu^{m+m_0}$; the remaining finite range is included in the constant depending on C, μ . The ratios of successive terms $\mu^{-\mu^q}$, for $q \geq t+1 \geq B+1$, are bounded by a fixed number less than

one, so the m tail is $\ll_{\mu} \mu^{-\mu^h}$. To sum the bands, write $h = t + 1 + j$. Each linear numerator factor increases by at most $j + 1$, and geometric decay also sums the resulting $(j + 1)^2$. We obtain

$$\text{vol}(V_2) \ll_{C,\mu} \frac{(\gamma + t + 2)(t + 1 - d)}{\mu^{\mu^{t+1}}(r + 1)!}. \quad (11.8)$$

If $d \geq B$, its numerator is $b + 2 \ll b = Y$. If $d \leq 0$, it is at most $(B + 2)(B + 1)Y$. If $1 \leq d < B$, it is at most $B(B + 3)b$. In all cases $t + 1 \geq d$, which proves the desired bound for V_2 . Together with (11.4) and the covering, this proves (11.1). $\square$

Corollary 11.4 (The $A = 0$ estimate for every count). *Fix $\mu > 1$. For all positive integers r, v and real $\gamma \geq \max\{0, r - v\}$,*

$$r! \text{vol}(\mathcal{T}_{\mu}(r, v, \gamma)) \leq C_{\mu} \min \left\{ 1, \frac{(\gamma - r + v + 1)(\gamma + 1)}{r} \right\}. \quad (11.9)$$

Proof. For $1 \leq r \leq 2v$, set $b = r - v$ and $\Gamma = \lceil \gamma \rceil$. Monotonicity in γ gives $\mathcal{T}_{\mu}(r, v, \gamma) \subseteq \mathcal{T}_{\mu}(r, v, \Gamma)$. We can apply Lemma 11.3 with $C = 2$ and $d = b - \Gamma \leq 0$. Dropping the denominator $\mu^{\mu^d} \geq 1$ and using

$$(\Gamma - b + 1)(\Gamma + 1) \leq 4(\gamma - b + 1)(\gamma + 1),$$

gives (11.9), after also using the simplex bound $\text{vol}(\mathcal{T}_{\mu}) \leq 1/r!$. If $r > 2v$, then $b > r/2$, $\gamma \geq b$, and $(\gamma - b + 1)(\gamma + 1)/r > 1/2$. The simplex bound alone proves the assertion with constant two in this case. The two cases cover every r/v . The case $r = 0$ will be treated directly when the estimate is applied, so the expression $1/r$ is used only for positive counts. $\square$

11.1 The tail of the ordinary rectangle

We now put the ordered-sum estimate into the form used by the ordinary root rectangle. Its two numerator factors will become the two endpoint reserves. The denominator will be kept with the Poisson atom at the same count when the root is integrated.

Lemma 11.5 (Ordered-sum estimate in the required $A = 0$ range). *Fix $\mu > 1$. For positive integers n, v and $\gamma \geq 0$ with $\gamma \geq n - v$, let*

$$\mathcal{T}_{\mu}(n, v, \gamma) = \left\{ 0 \leq \xi_1 \leq \cdots \leq \xi_n \leq 1 : \sum_{\ell \leq j} \mu^{v\xi_{\ell}} \geq \mu^{j-\gamma} \ (1 \leq j \leq n) \right\}.$$

Then

$$n! \text{vol}(\mathcal{T}_{\mu}(n, v, \gamma)) \leq C_{\mu} \min \left\{ 1, \frac{(\gamma - n + v + 1)(\gamma + 1)}{n} \right\}. \quad (11.10)$$

Proof. This is Corollary 11.4, with $r = n$. $\square$

This is the $A = 0$ case of [19, Lemma 5.5]. The preceding proof includes the integer rounding and the passage to real γ , and covers every n/v . The factor $n!$ converts ordered-simplex volume to the probability for the order statistics of n independent uniform points. The case $A > 0$ is not needed below.

Lemma 11.6 (All-count tail of the ordinary rectangle). *Fix an integer $2 \leq h \leq H$ and $C_0 \geq 1$. Set $c = h - 1$, $w = \log h$ and $\mu = h^{1/c}$. For $v \geq 1$ an integer, let $\xi_1 \leq \dots \leq \xi_n$ be the ordered independent uniform points on $[0, 1]$, and put*

$$Z = (n - v) \log h, \quad F_n = \min_{0 \leq j \leq n} h^{-j} \left(1 + \sum_{\ell \leq j} \mu^{v \xi_\ell} \right)^c, \quad W_n = \min\{1, C_0 e^Z F_n\}. \quad (11.11)$$

There exist constants C, C_1 depending only on H, C_0 such that, for every $n \geq 0$ and $\ell \geq 0$,

$$\mathbb{P}\{W_n > e^{-\ell} \mid n\} \leq \min \left\{ 1, \frac{C(1 + \ell)(C_1 + Z + \ell)_+}{1 + n} \right\}. \quad (11.12)$$

For $\ell < 0$ the event is empty.

Proof. For positive counts, a large rectangle forces the ordered-sum event at a translate determined by its endpoint exponent. The exact count-budget relation makes this translate fall in the range of the preceding lemma. We then treat count zero directly.

Suppose first $n \geq 1$ and the event is nonempty. The term $j = 0$ gives $F_n \leq 1$, hence $Z + \ell + \log C_0 > 0$. Define

$$\gamma_0 = \frac{Z + \ell + \log C_0}{\log h}, \quad \gamma = \gamma_0 + \frac{\log 2}{\log \mu}.$$

For each $1 \leq j \leq n$, the event in question implies

$$1 + \sum_{a \leq j} \mu^{v \xi_a} > \mu^{j - \gamma_0}.$$

The sum without the initial 1 is at least 1. Consequently

$$\sum_{a \leq j} \mu^{v \xi_a} > \frac{1}{2} \mu^{j - \gamma_0} = \mu^{j - \gamma}.$$

Here $\gamma \geq 0$, and the exact identity $Z = (n - v) \log h$ gives

$$\gamma - n + v = \frac{\ell + \log C_0}{\log h} + \frac{\log 2}{\log \mu} \geq 0.$$

Lemma 11.5 therefore applies in its proved $A = 0$ range. Furthermore

$$\gamma - n + v + 1 \leq C(1 + \ell), \quad \gamma + 1 \leq C(C_1 + Z + \ell)_+$$

on the nonempty event, after choosing C_1 large in terms of H, C_0 . There are only finitely many $\mu = h^{1/(h-1)}$, $2 \leq h \leq H$; the constants from Corollary 11.4 may therefore be replaced by their maximum over this fixed finite set. Its probability is at most the right side of (11.12), since $1 + n \leq 2n$.

If $n = 0$, then $F_0 = 1$ and $W_0 = \min(1, C_0 e^Z)$ is deterministic. On its nonempty event $Z + \ell > -\log C_0$. Increasing C_1 once makes the displayed right side equal to one. This proves the assertion also at count zero. Finally $W_n \leq 1$ makes the event empty for $\ell < 0$. $\square$

In the actual root application v is the number of completed harmonic cells, of width $\eta = \log h/(h-1)$, and their budget is $U = \eta v$. Uniform latent points in the cells give exactly the variables used in (11.11); rounding changes the geometric constant C_0 by a fixed factor. Thus $Z = n \log h - (h-1)U$ exactly. A bounded change of physical budget is kept in the external root translate, as in Section 13. If there are no cells, the completed Poisson count is identically zero and is covered directly. The subsequent Poisson maximum-atom and layer integrals use the same count.

12 Ordered covers and the upper bound

The upper bound measures how much of the full rectangle can remain once the routing constraints have been imposed. We reveal the divisor coordinates plateau by plateau and stop a branch when its projected rectangle becomes small. A branch that continues carries a positive half-line weight; a branch that stops carries a negative half-line weight. These two weights allow the conditional estimates to be composed in the original routing filtration.

Most local transitions are controlled by a short necessary one-end test. The exceptional case is a full-to-zero transition with two small endpoint reserves. There we use two left covers and a rising right boundary, obtaining a bound both for complete survivors and for internal right stops. After composition, all prefix costs are placed in one uncolored coefficient, and the unread ordinary suffix supplies the root atom and ballot factor. This division of the proof keeps each plateau at its own fractional power.

12.1 Notation and preliminary estimates

Fix the dimension bound H . Work at one physical positive length vector and take its ordered optimizer. Its maximal equal-product plateaux have products

$$0 < \sigma_* \leq \sigma_m < \cdots < \sigma_1 \leq \sigma^* < 1.$$

All constants below depend on H and the fixed conventions in the previously established estimates. Write

$$s = \sigma_1, \quad r_j = \frac{\sigma_j}{s}, \quad u_j = \frac{\sigma_{j+1}}{\sigma_j}, \quad d_j = \frac{1}{u_j} - 1 \quad (j < m). \quad (12.1)$$

The compound child weight of a nonterminal plateau is denoted by z_j . It satisfies $1 < z_* \leq z_j \leq H$, and a local slot with r eligible exterior names has original coarse law

$$\pi_j(c) = \frac{1}{r + z_j} \quad (c \text{ exterior}), \quad \pi_j(\text{child}) = \frac{z_j}{r + z_j}. \quad (12.2)$$

The following proposition records the local input in the order in which it is used. Rank separation identifies disjoint test groups; the endpoint geometry selects the bank lengths; scalar survival and maximum-atom estimates bound their conditional masses. The final assertion permits shots and overlapping count reserves to be included in those bounds. Section F.1 proves these facts.

Proposition 12.1 (Geometric and scalar estimates). *The following assertions hold with constants depending only on H .*

1. *Every retained local transition is canonical $r \rightarrow q$, with full input and deepest-suffix output. Within a plateau, sufficiently rare transitions have disjoint rank intervals: if one precedes*

another, then $q_1 \geq r_2$. Thus a zero-output transition, if present, is last. Its actual intrinsic endpoint scores are positive, and its persistence factor is

$$\Psi_i = \min \left\{ 1, \left(\frac{(1 + X_i)(1 + Y_i)}{1 + \mu_i} \right)^{2\chi_i} \right\}.$$

The exponents and rates are in fixed compact ranges. Factors above any fixed positive rarity threshold are bounded below by a fixed positive constant.

2. A retained nonzero-output transition has a necessary test on an original short local bank. The test reads that bank's coarse labels and the uncolored counts in its intermediate local groups. The corresponding failure cover has volume at most $S_i e^{-h}$. Its private probability $p_i(h)$ satisfies

$$\mathbb{E} p_i(h)^{\sigma_{o(i)}} \leq C(1 + h)^K \Psi_i.$$

The same assertion holds for a proper zero-output transition when $1 + X_i \geq c_X \sqrt{1 + L_i}$. In that case its intermediate groups are all the later local groups of its plateau. These current-plus- intermediate groups are pairwise disjoint within each plateau and across plateaux. All these estimates allow fixed point insertions at the same power. Original count middles are disjoint from the short label banks.

3. For any proper full-to-zero transition which remains rare, let M be its local block length, A, B the preceding and following local lengths of that plateau, and V_{old} its complete earlier physical length. Then

$$V_{\text{old}} + A + B \leq C\tau M, \quad T_j \asymp M, \quad (12.3)$$

$$L = 1 + B, \quad R = 1 + V_{\text{old}} + A + \varrho,$$

$$\varrho = \max \left\{ \varrho_0, \frac{c_0 \sqrt{1 + M}}{1 + d_j \sqrt{1 + M}} \right\},$$

$$X_i \asymp R, \quad Y_i \asymp L, \quad |e|M \leq C(V_{\text{old}} + A + B), \quad (12.4)$$

where $e = (r + z_j)^{\sigma_j} \mathbb{E} W - r$ and $W = \log(r + z_j) - \log z_j \mathbf{1}_{\text{child}}$. The length V_{old} refers to the earlier original groups, whose transported coefficients enter the endpoint costs.

4. On disjoint original end banks and a middle bank in the M block, the signed logarithmically lowered one-end estimates and the whole-band binomial maximum atom give

$$\mathbb{E} B_w(h, n)^{\sigma_j} \leq C(1 + h)^K (1 + |n|)^K F_j^0, \quad F_j^0 = \left(\frac{\varrho}{\sqrt{1 + M}} \right)^{\sigma_j} \min \left\{ 1, \left(\frac{LR}{1 + M} \right)^{2\chi_i} \right\}. \quad (12.5)$$

Here $w \geq M/2$; a completed-word band has width ϱ , while a first-hit band may have fixed width. The left reserve is at most $h + C(1 + N_B)$ and the right reserve is at most $C(1 + h + V_{\text{old}} + A + (|n| + 1)\varrho)$. The estimates sum all original counts and keep all $n + 1$ conditional gaps. If the larger reserve is superdiffusive, only the smaller-reserve bank of length $cM^2 / \max(L, R)^2$ is retained. In the small-gap case $d_j \sqrt{M} < 1$, a left bank alone gives the same factor, since $\varrho / \sqrt{M} \asymp 1$.

5. Fixed-dimensional original shot sums have all fixed moments. Their insertion estimates are obtained by a finite Poisson–Mecke partition before summing bank locations. A count middle occurring in an intermediate reserve group is handled by the conditional inequality

$$\mathbb{E}[f(N)g(N)] \leq \mathbb{E}f(N) \mathbb{E}g(N) \quad (f \text{ nonincreasing, } g \text{ nondecreasing}), \quad (12.6)$$

with all other groups fixed. This includes its count weight $(1+d_j\sqrt{1+N})^{-1}$ and an increasing necessary-bank reserve.

The bank lengths explain both the drift control and the profile factor in (12.5). By (12.4), two banks of length $M/16$ have bounded scaled drift when $\max(L, R) \leq C\sqrt{M}$. Otherwise use a bank of length $cM^2/\max(L, R)^2$. The original correlation and the centered scalar correlation differ by $O(LR/M)$; replacing one exponent by the other costs $\exp\{O((LR/M)|\log(LR/M)|)\}$, a bounded factor. Consequently (12.5) has the same intrinsic local factor as the lower bound.

We collect the local estimates into the factors that will appear after averaging the cover. Let F_j be the nonterminal plateau's count factor, at power σ_j , times all its retained local persistence factors. For the terminal plateau it is just the product of its retained nondegenerate factors. Consequently

$$\prod_{j=1}^m F_j = \mathcal{C}_{\text{count}} \mathcal{P}_{\text{ret}}. \quad (12.7)$$

Each F_j has a dimension-dependent polynomial lower bound in the total length of the original local groups it reads. The terminal ordinary factor is denoted by $\mathcal{B}_I$ and is kept separate.

12.2 Routing capacities and two left covers

We first prove the capacity bounds on actual divisor tuples. By Lemma 2.4, the full squarefree harmonic volume sum is at most twice its restriction to $a_i \leq y_i^{T_k}$ for all i . The prefix bounds (2.7) supply the dimension-constant full-size rectangles below. Fix such an original squarefree tuple and its complete uncolored data G . Its original product fine-label measure is π_G . Put $B_{\text{geo}} = \sum_i (h_i - 1)L_i$. For a measurable covering set E in the original divisor coordinates, use the normalized volume

$$\text{Vol}(E) = e^{-B_{\text{geo}}} \text{vol}_k(E). \quad (12.8)$$

For an original label class Q , $\text{Vol}(Q)$ denotes the normalized volume of its associated cover. Its probability is always the absolute original class probability $\pi_G(Q)$, including any revealed ancestor route. Let $\mathcal{Q}_j$ reveal the complete first j plateau routes and leave the descendant alphabet compounded. Write Z_j for the actual grouped output and put

$$P_j = \sum_{\ell \leq j} Z_\ell, \quad \eta_j = \sum_{\ell \leq j} \frac{\sigma_\ell}{\sigma_j} Z_\ell, \quad X = \sum_{j=1}^m r_j Z_j. \quad (12.9)$$

The common root X is uncolored. For $j < m$ the exact identities are

$$P_j = \eta_j - \sum_{\ell < j} d_\ell \eta_\ell, \quad P_m = \frac{X}{r_m} - \sum_{\ell < m} d_\ell \eta_\ell. \quad (12.10)$$

The geometric distinction needed for the cover is the following. Call a nonterminal plateau *hard* if its retained zero-output edge does not satisfy the one-end admissibility hypothesis of Proposition 12.1(2). All other plateaux are simple. A hard plateau has one selected zero block and all its other retained edges are earlier nonzero edges. Constants may be chosen so that a hard plateau has $d_j\sqrt{M} \geq 1$; the small-gap variant below also covers the bounded transition range.

Lemma 12.2 (Projected rectangles and two left walls). *A complete $\mathcal{Q}_j$ atom q has actual capacity*

$$\text{Vol}(q) \leq C\pi_G(q)e^{P_j(q)}. \quad (12.11)$$

At a hard plateau, fix a $\mathcal{Q}_{j-1}$ atom α . Read its selected zero block backwards and set

$$C_j(w) = H_B + \sum_{x < w} [\log(r + z_j) - \log z_j \mathbf{1}_{\text{child}}] - rw, \quad Z_j = C_j(M) + Y_A. \quad (12.12)$$

Then $H_B \leq CN_B$ and $Y_A \geq -C(V_{\text{old}} + A)$ wordwise. There are two partial capacities,

$$\text{Vol}(Q_w^{\text{glob}}) \leq C\pi_G(Q_w^{\text{glob}})e^{C_j(w)} \text{Shot}_w, \quad (12.13)$$

$$\text{Vol}(\alpha, Q_w) \leq C\pi_G(\alpha, Q_w)e^{P_{j-1}(\alpha) + C_j(w)} \text{Shot}_w. \quad (12.14)$$

Imposing both left walls

$$C_j(w) \geq -h - \mathcal{L}(w), \quad P_{j-1} + C_j(w) \geq -h - \mathcal{L}(w), \quad \mathcal{L}(w) = A_0 \log(1 + \min(w, M - w)), \quad (12.15)$$

costs at most CSe^{-h} in the union of their failures. Their simultaneous survivors have effective reserve

$$h_{\text{eff}} = h + \min(P_{j-1}, 0) \leq h. \quad (12.16)$$

Here $S \geq 1$ is an uncolored original shot with all fixed moments; it may be chosen jointly for the finitely many plateaux and other local failure covers, without reading the final unread root suffix.

Proof. We first compute the capacity of a completed route. For a hard plateau we then compute two partial capacities, according to whether the ancestor coordinates are left unresolved or fixed. Their failure covers impose the two left walls, whose intersection gives the stated effective reserve.

Completed routes. Let B_j be the sum of the deadlines of the exterior coordinates resolved through plateau j . A resolved divisor contributes width $\log 2$, while an unresolved coordinate retains its full size bound. After normalization their rectangle has capacity Ce^{-B_j} . The coarse information mark is the local log-alphabet for an exterior choice, and that log-alphabet less the log of the compound-child weight for a child choice. Their sum through plateau j gives

$$-\log \pi_G(q) = B_j + P_j(q).$$

Hence $Ce^{-B_j} = C\pi_G(q)e^{P_j(q)}$, which proves (12.11).

The two partial rectangles. For (12.13), leave all ancestor and earlier coordinates in their full rectangles. Reveal the later local word and the crossed points of the current block. The r unresolved exterior widths have common threshold $b - w$, where b is the block's original right endpoint. Their budget deficit is $rw + \sum_{\text{later}} r_g L_g$. The negative observed log mass is the corresponding mark sum, so their difference is $C_j(w)$. Every other original coordinate budget cancels. In particular no earlier private output or selected incoming count occurs in this capacity.

Instead first fix α . Its actual resolved translations give $\pi_G(\alpha)e^{P_{j-1}}$. The same rectangle on its child fiber gives the conditional factor $\pi_G(Q_w | \alpha)e^{C_j(w)} \text{Shot}_w$. Their product is (12.14), with the conditional mass taken under the original law on the ancestor fiber. The shot formed from all original points bounds the selected-list shot by positivity. We use the two resulting covers separately.

The failure covers. At the first failure of either wall, including $w = 0$, substitute the failed inequality into its corresponding capacity. The result is the original class probability times $e^{-h - \mathcal{L}(w)} \text{Shot}_w$. The first-failure classes form an antichain in their own reveal order. For the second cover, first sum conditional Kraft masses and then their original ancestor probabilities. Their total is at most one. Summing the fixed logarithmic shot weights gives CSe^{-h} for each cover. A union bound combines the two failure covers. Their simultaneous survivors have reserve (12.16). The wordwise bounds follow from the positive information marks and the physical rank budgets. $\square$

12.3 Conditional half-line estimates

The capacity exponents split into negative and positive half-lines relative to a chosen threshold. We now bound both half-lines by the same uncolored cost. This uniformity is what permits backward conditioning through the plateau routes.

For a simple plateau the calculation comes from a binomial count middle. A proper local slot is after all ancestor exterior deadlines, so its ancestor category is deterministically child. Conditioning on any $\mathcal{Q}_{j-1}$ atom therefore leaves its local labels with the product law (12.2).

Lemma 12.3 (Simple-plateau kernels). *Choose an original local count middle of n slots, disjoint from the plateau's necessary label banks. Let $E_j(h)$ be their joint event and $p_j(h, G)$ its private probability. For every uncolored threshold ξ_j , set $\zeta_j = \eta_j - \xi_j$. Uniformly in every $\mathcal{Q}_{j-1}$ atom,*

$$\begin{aligned} \mathbb{E}_{\pi_G}[e^{\zeta_j} 1_{\{\zeta_j \leq 0\}} 1_{E_j(h)} \mid \mathcal{Q}_{j-1}] &\leq C b_j(h), \\ \mathbb{E}_{\pi_G}[e^{-d_j \zeta_j} 1_{\{\zeta_j > 0\}} 1_{E_j(h)} \mid \mathcal{Q}_{j-1}] &\leq C b_j(h), \\ b_j(h) &= (1 + d_j \sqrt{1+n})^{-1} p_j(h, G). \end{aligned} \quad (12.17)$$

For a terminal plateau the corresponding conditional mask mass is bounded by $b_m(h) = p_m(h, G)$.

Proof. The threshold may depend on the ancestor word, but the maximum-atom bound is uniform in that dependence. Fix the actual ancestor atom and then the entire current coarse route outside the middle. Its masks are now fixed, and

$$\zeta_j = H_0 - \log z_j K, \quad K \sim \text{Bin}\left(n, \frac{z_j}{r + z_j}\right),$$

where H_0 may depend arbitrarily on the old list and outside word. The maximum binomial atom is $C(1+n)^{-1/2}$. Summing it with the negative half-lattice weights gives $C(1+n)^{-1/2}$; the positive half-lattice gives $C/[d_j \sqrt{1+n}]$, with the probability bound providing the cap one. For $d_j = 0$ we use that probability bound directly. Both half-lines are thus at most $C(1+d_j \sqrt{1+n})^{-1}$, uniformly in H_0 . We may now integrate the outside labels, whose mask mass is $p_j(h, G)$ on each ancestor fiber. For the terminal plateau the same observation gives its mask mass without a count weight. The variation of K has left future child words unfixed. $\square$

For a hard plateau write $P = V_{\text{old}} + A$ and let $E_{<}(h)$ denote its earlier own local masks. Their probability $p_h(G_A)$ reads only original local groups strictly before the selected block. Choose a fixed D_A and put

$$\alpha_h = \min\{1, (2+A)^{-D_A} + p_h(G_A)\}. \quad (12.18)$$

By its polynomial factor floor, D_A can be chosen so that

$$p_h \leq \alpha_h \leq 1, \quad -\log \alpha_h \leq D_A \log(2+A) \leq C(1+P), \quad \mathbb{E} \alpha_h^{\sigma_j} \leq C(1+h)^K \mathcal{P}_{<}. \quad (12.19)$$

An empty earlier collection has $\alpha_h = 1$.

Lemma 12.4 (Hard-plateau kernels and right stops). *On a fixed ancestor atom let*

$$x_j = \xi_j - \frac{\sigma_{j-1}}{\sigma_j} \eta_{j-1} \quad (j > 1), \quad x_1 = \xi_1, \quad \zeta_j = Z_j - x_j. \quad (12.20)$$

In addition to the dual left walls, use the right wall

$$C_j(w) \geq x_j + \log \alpha_h - \log S - A_1 \log(1 + M - w). \quad (12.21)$$

Assume its deterministic gate places right-governed first hits after $M/2$. There is one uncolored coefficient $b_j(h)$, independent of the ancestor private word and of x_j , with all three properties:

1. the total right-stop capacity on the ancestor fiber is at most $C\pi_G(\alpha)e^{P_{j-1}+x_j}b_j(h)$;
2. complete surviving plateau words with $\zeta_j \leq 0$ have conditional weighted mass at most $Cb_j(h)$ under weight e^{ζ_j} ;
3. complete surviving plateau words with $\zeta_j > 0$ have conditional weighted mass at most $Cb_j(h)$ under weight $e^{-d_j\zeta_j}$.

The complete-survivor conclusions themselves do not require the gate. The small-gap variant omits the right wall and internal right stops.

Proof. We use a common cost for three operations: ending a complete route below the threshold, continuing above it, and stopping at the right wall inside the block. For complete routes, signed bands isolate a middle count atom and two end tests. At a right stop, the capacity itself supplies the earlier-mask factor, and the first-hit strip supplies a fixed-width atom.

Complete survivors. The exact numerical identity

$$e^\zeta 1_{\{\zeta \leq 0\}} + e^{-d_j\zeta} 1_{\{\zeta > 0\}} = e^{-x_j} \min\{e^{Z_j}, e^{x_j/u_j - d_j Z_j}\} \quad (12.22)$$

gives the two weights used in the conditional mass estimates. Set $\bar{x} = x_j - \log S$ and divide the full output $Z_j - \bar{x}$ into bands $[n\varrho, (n+1)\varrho)$, $n \in \mathbb{Z}$. The right side of (12.22) is at most $S^{d_j}w_n$, where

$$w_n = e^{-d_j n \varrho} \quad (n \geq 0), \quad w_n = e^{-(|n|-1)\varrho} \quad (n \leq -1). \quad (12.23)$$

For positive bands use the second exponential branch and for negative bands the first. This remains valid if a shifted band crosses $\zeta = 0$, since the minimum is at most either branch everywhere. If $d_j\sqrt{M} \geq 1$, both series of $w_n^{\sigma_j}$ times every fixed polynomial in $|n|$ converge uniformly.

The right wall also turns the output band into a reserve for a reflected end test. Subtracting (12.21) from $Z_j = C_j(M) + Y_A$ gives

$$\begin{aligned} C + Z_j - \bar{x} - Y_A - \log \alpha_h &\leq C_H(1 + V_{\text{old}} + A) + (|n| + 1)\varrho + D_A \log(2 + A) \\ &\leq C_D(1 + V_{\text{old}} + A) + (|n| + 1)\varrho. \end{aligned} \quad (12.24)$$

Both the supplied root level and $\log S$ cancel. The small left reserve is still $h + C(1 + N_B)$: only for this necessary test we enlarge h_{eff} to h .

For each band, condition on the current old and earlier local word, the later local word, and the selected-block labels outside the middle. This fixes the band center. The middle child-count maximum atom is uniform in that center, and the end tests use disjoint original slots. Moreover (12.24) has removed the earlier private word from the right reserve. We can therefore integrate the earlier word with its event $E_<(h)$, obtaining exactly $p_h \leq \alpha_h$, and the conditional incoming word with mass one on its ancestor fiber. Integrate the later word under its original mass as well, using $H_B \leq CN_B$ only for the necessary left reserve. The resulting band mass is at most $C\alpha_h B_M(h, n)$, uniformly in the actual incoming list. With

$$T_{\text{full}}(h) = \sum_{n \in \mathbb{Z}} w_n B_M(h, n),$$

each of the two complete half-lines is at most $CS^{d_j}\alpha_h T_{\text{full}}(h)$.

Internal right stops. At a right first hit, positivity of the jumps and bounded downward speed put the risk in a bounded strip below its wall. This does not require a bound on a tied positive prime batch. Substitution into (12.14), divided by $\pi_G(\alpha)e^{P_{j-1}+x_j}$, gives

$$C \frac{\alpha_h}{S} (1 + M - w)^{-A_1} \text{Shot}_w \quad \text{times the conditional prefix mass.} \quad (12.25)$$

Here α_h comes from the capacity at the stopping boundary. Prior survival and the first-hit strip give the fixed- w end-bank tests and one fixed-width middle atom. Their centers may depend on the fixed ancestor data; the estimates are uniform in those centers. Let $T_{\text{right}}(h)$ sum these bounds with the weights and shots in (12.25). One may therefore take

$$b_j(h) = C\alpha_h[S^{d_j}T_{\text{full}}(h) + S^{-1}T_{\text{right}}(h)]. \quad (12.26)$$

This proves the three separate statements.

The small-gap regime. If $d_j\sqrt{M} < 1$, then $\varrho \asymp \sqrt{M}$ and $R \geq c\sqrt{M}$. Keep only the necessary left bank and omit the right wall. The function in (12.22) is at most one. Whole-word conditioning gives p_h times this bank's probability; its moment is F_j^0 because the count-width factor is comparable to one. This avoids any series with a decay rate tending to zero. $\square$

Lemma 12.5 (Joint moments at all plateau powers). *The simple and hard costs can be chosen so that, for every subset $J \subseteq \{1, \dots, m\}$, every nonnegative integer vector (h_j) , and every fixed p ,*

$$\mathbb{E} \left[S^p \prod_{j \in J} b_j(h_j)^{\sigma_j} \right] \leq C_p \prod_{j \in J} (1 + h_j)^K F_j, \quad (12.27)$$

$$\mathbb{E} \left[(1 + N_{\text{pre}}) S^p \prod_{j \in J} b_j(h_j)^{\sigma_j} \right] \leq C_p (1 + V_{\text{pre}}) \prod_{j \in J} (1 + h_j)^K F_j. \quad (12.28)$$

Moreover $b_j(h) \leq CS^Q$ pointwise, uniformly in h , for a fixed Q . The expectations are taken under the original uncolored Poisson law, with the clocks unconditioned on the stopping route.

Proof. We first average the local bank costs on their disjoint original groups. An opposite-monotonicity inequality handles the one possible overlap with a count middle. We then attach shots by fixed insertions, which preserve the same local estimates and hence every plateau power.

For a hard plateau, α_h reads groups strictly before the selected block. A fixed- w , fixed-band expression reads the selected block and later local counts. Before shots are attached, these are independent original Poisson restrictions. Apply (12.19) and (12.5), both at power σ_j , and then sum the signed bands and weighted positions. The result is $\mathcal{P}_{<F_j^0} = F_j$, multiplied by a fixed polynomial in h .

For a simple plateau, a middle count in an intermediate group is not independent of the corresponding mask reserve. Condition on other groups and use (12.6) once, then integrate the disjoint groups. The same argument covers a zero-output edge with an admissible large left reserve. Every input group belongs to its local plateau, so the functions for different plateaux use disjoint Poisson restrictions. Earlier lengths in a hard reserve are deterministic.

To attach the global shot, make its finite Mecke partition in each fixed bank expression before summing positions or forming reserve envelopes. A fixed insertion has three effects: local inserted points can be deleted with a bounded reserve shift, later counts increase N_B by a bounded amount, and middle insertions decrease the inverse-count bound. The earlier mask obeys $\alpha_h(G + \text{points}) \leq C_k \alpha_{h+C_k}(G)$ with its floor unchanged. Thus the same disjoint-group estimates apply after the insertions, proving (12.27). One additional raw-count insertion costs its mean and proves (12.28). Omitted plateaux have factor one, so the proof holds for every subset. In particular it applies to mixed terms created by deterministic floors. Each plateau retains its own power σ_j .

Finally every private bank mass and truncated maximum-atom upper is at most one, the signed band weights have uniformly bounded sum, and the weighted right-position sum is bounded by S . The exponents d_j are bounded. This pointwise bound controls the positive logarithms in the composition below. $\square$

12.4 Lower cutoffs and composition of the covers

The local costs depend on the cover reserve. To choose one reserve for the whole route, we pool those costs into positive uncolored numbers. Polynomial floors control how small a pool can be; weighted summation over reserves preserves the joint moments. The thresholds built from these pools will make the branch capacities telescope.

For a nonterminal plateau let T_j be its total local length. For the terminal plateau use only its local head before the unread suffix. Choose fixed D_j so that $f_j = (2 + T_j)^{-D_j}$ satisfies $f_j^{\sigma_j} \leq CF_j$. Set

$$a_j = f_j + \sum_{h=0}^{\infty} (1+h)^{-\beta} b_j(h), \quad B_* = \sum_{j=1}^m r_j \log a_j, \theta_j = \frac{1}{r_j} \sum_{\ell > j} r_\ell \log a_\ell, \quad \xi_j = \frac{X}{r_j} + \theta_j. \quad (12.29)$$

The a_j are positive uncolored numbers and need not be at most one. They are fixed before the single cover reserve h is chosen. In particular

$$b_j(h) \leq a_j(1+h)^\beta, \quad -\log a_j \leq D_j \log(2 + T_j). \quad (12.30)$$

The pointwise bound in Lemma 12.5 makes every sum finite when $\beta > 1$.

Lemma 12.6 (Joint moments and the right stopping boundary). *Choose $\beta\sigma_* > K + 2$ and $\beta_* \geq m\beta$, and put*

$$L_S = 1 + \log(2S), \quad D = L_S^{\beta_*} e^{B_*}. \quad (12.31)$$

Then

$$\begin{aligned} \mathbb{E}[S^p D^s] &\leq C_p \prod_j F_j, \\ \mathbb{E}[(1 + N_{\text{pre}} + V_{\text{pre}} + |\log D|) D^s] &\leq C(1 + V_{\text{pre}}) \prod_j F_j. \end{aligned} \quad (12.32)$$

If $w_0 = X + \log D < 0$, choose

$$h = \lceil \log(2S) - w_0 \rceil. \quad (12.33)$$

At every reached hard plateau, whose previous proper outputs continued, the right-hit gate holds with a fixed dimension constant.

Proof. There are two points to check. The pooled costs must retain the local moments, and the resulting thresholds must place each right stop in the latter half of its hard block. The first follows by fractional subadditivity. The second follows from cancellation of all future pool contributions in the stopping level.

Pooled moments. Expand the separate sums and floors using subadditivity at each $\sigma_j < 1$. Apply (12.27) to every resulting plateau subset, and sum $(1+h)^{K-\beta\sigma_j}$. Since $sr_j = \sigma_j$, this gives the first bound in (12.32). Use (12.28) for the raw count. The two signs of the logarithmic term are controlled separately. The floors give $(-\log D)_+ \leq C \log(2 + V_{\text{pre}})$. The pointwise polynomial shot bound gives $\log_+ D \leq C(1 + \log S)$, and an additional fixed shot moment absorbs this term. This proves the second bound.

The right-hit gate. Set $\zeta_\ell = \eta_\ell - X/r_\ell - \theta_\ell$ and

$$c_j = \theta_j - \sum_{\ell < j} d_\ell \theta_\ell \quad (j < m), \quad c_m = - \sum_{\ell < m} d_\ell \theta_\ell. \quad (12.34)$$

Direct telescoping gives, including $j = m$,

$$c_j + \sum_{\ell \leq j} \log a_\ell = B_*. \quad (12.35)$$

The local level at a hard plateau satisfies

$$\begin{aligned} h_{\text{eff}} + x_j &\leq h + P_{j-1} + x_j \\ &= h + X + c_j - \sum_{\ell < j} d_\ell \zeta_\ell \leq h + X + c_j. \end{aligned} \quad (12.36)$$

The first inequality holds for both signs of P_{j-1} , since $\min(P_{j-1}, 0) \leq P_{j-1}$. The last uses the true previous continuations $\zeta_\ell > 0$. The choice (12.33) yields $h + X + B_* \leq \log(2S) + C$. Therefore

$$h_{\text{eff}} + x_j + \log \alpha_h - \log S \leq C - \sum_{\ell \leq j} \log a_\ell \leq C + \sum_{\ell \leq j} D_\ell \log(2 + T_\ell). \quad (12.37)$$

The stopping level now involves only the current and preceding plateau floors. At a hard plateau, (12.3) implies that every preceding plateau's local length is at most $V_{\text{old}} \leq CM$ and the current $T_j \leq CM$. Thus the last expression is at most $D_{\text{gate}} \log(2 + M) + C$. Taking A_1 large relative to D_{gate} and A_0 puts the right logarithmic wall below the left wall until $M/2$. A first hit governed by the right wall must therefore occur later, proving the gate. The argument uses neither a sign restriction on the incoming output nor a bound for a future length. $\square$

Remark 12.7 (Order of constants). First fix dimension compactness, rarity and deterministic bank scales. Choose the prefix floors D_A , plateau floors D_j , and then D_{gate} from those floors. Choose A_1 next, including the condition $A_1 \sigma_* > 2$ for position sums. Determine the scalar reserve exponent K after this choice, and finally choose β and β_* . If a scalar effective-bank threshold depends on A_1 , the bounded effective-length case is treated by the probability bound one and the same middle atom: its persistence ratio is bounded below by a fixed positive constant. One does not reduce the original rarity cutoff in a way which feeds back into D_{gate} . Increasing β_* only improves the gate.

Theorem 12.8 (Profile of the ordered cover). *For the original complete divisor union there is a prefix-uncolored $D > 0$, unread by the final ordinary suffix, such that at its root*

$$I_{\text{actual}} \leq C J_{\beta_*}(X + \log D), \quad J_b(w) = \min\{1, C_b e^w (1 + (-w)_+)^b\}, \quad (12.38)$$

and D satisfies (12.32). All original failure words are included in this upper bound.

Proof. The route has three possible endings: a complete proper stop, an internal hard right stop, or the terminal word. We compute their capacities on the original fibers, integrate the future branch first, and then apply the earlier positive kernels. The pooled offsets make the same product bound apply to each ending.

Use all left-failure covers, keeping their union cost CSe^{-h} . On the survivors route plateaux in order. At a hard plateau allow its internal right stop; otherwise complete its route and stop at the first $\eta_j \leq \xi_j$. If there is no proper stop, use the complete terminal word. These are actual projected branches. A later private filter is never charged at an earlier projection.

Complete proper stops. For a complete stop at plateau $j < m$, the exact capacity exponent is

$$P_j = X + c_j + \zeta_j - \sum_{\ell < j} d_\ell \zeta_\ell. \quad (12.39)$$

Condition on its true $\mathcal{Q}_{j-1}$ fiber and apply the negative kernel, then integrate earlier routes backwards with their positive kernels. Each bound is uniform on every preceding fiber. The result is at most $Ce^{X+c_j} \prod_{\ell \leq j} b_\ell(h)$.

Internal and terminal stops. For an internal hard right stop, its reference exponent is exactly

$$P_{j-1} + x_j = X + c_j - \sum_{\ell < j} d_\ell \zeta_\ell.$$

Use the separate right-stop assertion of Lemma 12.4, followed by the same backward positive kernels. It has the same upper bound. For the terminal branch, (12.10) gives $P_m = X + c_m - \sum_{\ell < m} d_\ell \zeta_\ell$. Integrate its true local mask first and then use the earlier positive kernels.

The conditional tower integrates each future child word by a uniform bound before varying the current local count. The current middle therefore retains its conditional binomial law, and ancestor atoms retain their original probabilities.

The common envelope. By (12.30) and (12.35), each branch is at most $Ce^{X+B_*}(1+h)^{j\beta}$. There are at most C_H branch types, so

$$I_{\text{actual}} \leq C \min\{1, Se^{-h} + e^{X+B_*}(1+h)^{m\beta}\}. \quad (12.40)$$

For $w_0 < 0$, Lemma 12.6 permits the reserve in (12.33). It satisfies $1+h \leq CL_S(1+(-w_0)_+)$, and (12.31) now gives (12.38). For $w_0 \geq 0$, use the global original rectangle and no gated cover. The case $m = 1$ has no proper frontier and follows directly from its terminal local masks. $\square$

12.5 One unread ordinary root and the arithmetic conclusion

It remains to integrate the one root that all these branches share. The prefix coefficient is measurable before the ordinary suffix is read. This allows us to combine the cover envelope with the endpoint and largest-block rectangles of that same suffix, using its root count once.

After the whole-block trimming in Section F.1.6, the unread ordinary suffix has length U , largest block M_* , side lengths A_*, B_* , and deterministic mean surplus Δ_* . It retains the same intrinsic ordinary factor

$$\mathcal{B}_I \asymp (1+U)^{-1/2} \min\left\{1, \frac{(1+B_*)(1+\Delta_*+A_*)}{1+M_*}\right\}, \quad \Delta_* \geq cV_{\text{pre}}. \quad (12.41)$$

The trimming leaves the largest root block unchanged and reads no clock needed by the new nonterminal-plateau banks. On the same original tuple its endpoint and largest-block ordinary rectangles may be combined with (12.38).

Proposition 12.9 (The joint ordinary root estimate). *Let $D > 0$ and κ be arbitrary original-prefix-measurable numbers, and write $X = \kappa + (\sigma_m/s)Z_{\text{suffix}}$. If the same tuple has the profile (12.38), its combination with the ordinary suffix rectangles satisfies*

$$\mathbb{E}_{\text{suffix}}[e^{-sX} I_{\text{actual}} \mid G_{\text{pre}}] \leq CD^s e^{-\sigma_m H} (1+U)^{-1/2} \min\left\{1, \frac{(1+B_*)(1+H_++A_*)}{1+M_*}\right\}, \quad (12.42)$$

Here $H = (-\log D - \kappa)/(\sigma_m/s)$. The estimate is uniform in the root translate and includes all counts, including zero. The largest-block atom and its ballot denominator are estimated jointly using Lemma 11.6, with all $n+1$ gaps retained.

Proof. The same-count layer calculation, including every translate and the polynomial outgoing profile, is given in Section F.1.9. $\square$

Theorem 12.10 (Harmonic upper bound). *At the fixed lifted physical optimizer,*

$$K_k(\ell^+) \leq C_H e^{-J} \mathcal{C}_{\text{count}} \mathcal{B}_I \mathcal{P}_{\text{local}}. \quad (12.43)$$

Together with Theorem 7.1, this gives the two-sided harmonic estimate at the fixed lift. Section 14 passes to arbitrary nonnegative lengths.

Proof. We apply the root estimate conditionally on the prefix, then average its two alternatives by the pooled moments. The natural suffix surplus absorbs the prefix length, identifying the ordinary factor. Finally the exact prime tilt supplies the exponential saddle term.

The profile coefficient of Theorem 12.8 reads no uncolored point of the unread suffix. Apply (12.42) to the root X . The exact prefix normalization gives $|\kappa| \leq C(1 + V_{\text{pre}} + N_{\text{pre}})$. For the first alternative, drop $e^{-\sigma_m H_-} \leq 1$ and use the first pooled moment in (12.32). For the second alternative, use

$$1 + H_+ + A_* \leq C(1 + A_* + V_{\text{pre}} + N_{\text{pre}} + |\log D|)$$

and the second bound. Average these two upper bounds separately and then take their minimum. The inequality in (12.41) compares the result with $C\mathcal{B}_I \prod_j F_j$.

The geometric covers apply pointwise before the prime-series and cell enlargements. The exponential change of measure bounds K_k above by $C_H e^{-J} \mathbb{E}[e^{-sX} I_{\text{actual}}]$; consistently rounded harmonic budgets have only a fixed normalization error. Thus (12.7) proves (12.43) with $\mathcal{P}_{\text{ret}}$. Changing between the finitely many fixed rarity thresholds only changes a dimension constant, so the full $\mathcal{P}_{\text{local}}$ is equivalent. The conditional estimates used in this composition are supplied by Proposition 12.1 and the bank comparisons. $\square$

13 Prime clocks and the upper bound

The cover argument was expressed in physical clocks. We now pass to the harmonic prime clocks without changing the physical optimizer. On deterministic prefix banks a harmonic quantile map changes each clock by a bounded amount, so the necessary tests and their moments remain comparable. On the unread ordinary suffix, complete equal-mass cells give uniform positions and hence the ordered-sum rectangle estimate. A bounded total mass deficit permits this completion by a whole-law likelihood comparison. The exact root is retained by moving the budget discrepancy into its prefix translate.

These steps give the upper bound for every $L_i \geq 1$, including bounded effective banks, and in particular the upper bound at the fixed lift.

Theorem 13.1 (Upper bound for positive lengths). *For each fixed k , the numerical profile of (3.12) satisfies*

$$K_k(\log 3 + L_1, \dots, \log 3 + L_k) \leq C_k \Psi_k(L), \quad L_i \geq 1. \quad (13.1)$$

The optimizer, plateau and endpoint costs are evaluated at the physical vector L and held fixed during prime discretization.

We use the geometric covers and conditional estimates of Section 12. The scalar estimates require only a lower bound on the effective bank lengths; bounded banks are treated below. By Lemma 2.4, it suffices, at a factor of two, to consider the size-truncated tuples. The first prime range includes 2 and 3.

13.1 Prime ranges and harmonic clocks

The prefix tests use deterministic banks, so their atomic prime clocks can be compared separately with homogeneous clocks. The comparison must hold for every mark word, including batches at a single prime, because the positive prime expansion retains every occurrence count.

Put $H = k + 1$. All physical saddle products, Gibbs probabilities and rates are frozen. By (3.3), all parameters needed by the nondegenerate scalar laws belong to dimension-dependent compact sets. For an original block, with either forward or reverse log-log coordinate $t(p)$, the uniform prime harmonic estimate gives

$$\left| \sum_{p: t(p) \in [a, a+v)} \frac{1}{p} - v \right| \leq C_H. \quad (13.2)$$

The constant covers open endpoints and fixed omitted initial primes. An empty terminal piece has bounded length and causes the same error. This estimate is uniform in a, v and the original block.

For clarity about the initial primes, set

$$\eta_* = \min_{2 \leq h \leq H} \frac{\log h}{h-1}, \quad Q_H \geq \max(3, 4/\eta_*).$$

A prime has at most H possible labels in an actual divisor union. Deleting it and summing over whether it was present therefore costs its corresponding insertion factor. For the finitely many primes $p \leq Q_H$, the total cost is at most

$$\prod_{p \leq Q_H} (1 + H/p),$$

a fixed dimension constant. Deletion preserves each original upper size constraint. The remaining calculation still uses the original physical optimizer and budgets: the missing harmonic mass is part of the bounded error in (13.2). This is an upper operation, not an extra roughness assumption on the input.

On a deterministic local bank I , write $H_I = \sum_{p \in I} 1/p$. The generalized inverse T of its harmonic cumulative measure obeys

$$|T(s) - s| \leq C_H \quad (0 \leq s \leq H_I), \quad H_I = |I| + O_H(1). \quad (13.3)$$

Push a homogeneous rate- Z_i Poisson process through T , retaining its iid Gibbs marks. The image is exactly the required prime process with intensity $Z_i \sum_{p \in I} p^{-1} \delta_{t(p)}$. In particular this identity permits arbitrarily many occurrences at a single prime. Disjoint deterministic actual banks are independent.

To compare signed walls, inspect the partial sums inside each atom collapsed by the quantile map. For positive jumps, a homogeneous partial sum is at least the actual pre-jump sum. For negative jumps, it is at least the actual post-jump minimum, since only some of the negative jumps have occurred. The linear clock discrepancy in (13.3), together with the fixed logarithmic allowance, costs $O_H(1)$ in the reserve. An actual necessary wall therefore implies the homogeneous necessary wall with that enlarged reserve, for every multiplicity and count.

The middle-binomial atom is likewise an all-count assertion: conditional on the actual middle count N_I , it is at most $C(1 + N_I)^{-1/2}$, uniformly in the band center and in every outside private word. For fixed $a > 0$,

$$\mathbb{E}(1 + N_I)^{-a} \leq C_{H,a}(1 + |I|)^{-a}. \quad (13.4)$$

For large $|I|$ this follows from the Poisson lower-tail estimate and (13.3); for bounded $|I|$ the right side has a fixed positive lower bound. The translated half-line binomial bound proves the same extension for a proper count window of width $1/d$.

Shots are attached on the actual atomic processes. Condition the marks of a fixed number of inserted occurrences and enlarge each necessary reserve by their number times the largest mark. The original middle occurrences remain free for the maximum-atom bound. A fixed-degree shot polynomial has a finite Mecke partition, hence uses only finitely many insertions of this kind. Its exponentially or summably polynomially weighted intensity moments are bounded by (13.3). The joint subset moment consequently keeps the original power σ_j at each plateau.

13.2 Bounded effective banks

The quantile error is harmless on a long bank. On a bounded bank we use the fixed lower floor of its numerical factor, while retaining any other tests and the count middle. This separates the short-bank case from the sharp scalar survival estimate.

If an effective bank has length exceeding a fixed scalar threshold plus $2C_H$, its harmonic length is comparable to its physical length. Apply the sharp signed one-end bound of Theorem 5.16 after (13.3). The reserve change is harmless because the factor contains $1+\text{reserve}$.

If an effective bank is bounded, its numerical persistence factor has a fixed positive lower bound. Indeed the selected one-end regime has $\bar{T} \asymp_H (1+Y)/\sqrt{\ell}$, and the selected two-end regime has the product of the two normalized reserves. Since each reserve is at least one, bounded ℓ cannot give a vanishing factor. Discard that test in the upper and keep the other banks and the count middle, with the rarity cutoff fixed in the order of Remark 12.7. A bounded total local plateau length also has a count factor bounded below; both weighted half-lines are at most one. These arguments are uniform for all $L_i \geq 1$.

13.3 Comparison with the Poisson law

The unread ordinary root suffix is represented by equal mass cells. Actual cell masses can be slightly deficient. The following likelihood-ratio bound permits them to be completed in one step, with the rectangle minimum, root tilt and private powers all kept in the integrand.

Lemma 13.2 (Poisson mass comparison). *Let $0 \leq \mu_c \leq \mu_c^*$ on a finite collection of cells, and let $\Delta = \sum_c (\mu_c^* - \mu_c)$. Attach the same conditional Gibbs label law to the counts under both product Poisson measures. For every nonnegative measurable functional F of the entire environment and private word,*

$$\mathbb{E}_\mu F \leq e^\Delta \mathbb{E}_{\mu^*} F. \quad (13.5)$$

This remains true conditionally on any unchanged prefix.

Proof. On the support of the first measure its likelihood ratio is

$$\frac{d\mathbb{P}_\mu}{d\mathbb{P}_{\mu^*}}(n) = e^\Delta \prod_c (\mu_c / \mu_c^*)^{n_c} \leq e^\Delta.$$

Use the usual support convention for zero means. The identical Gibbs conditional law cancels from the ratio. Integrating F proves the claim. $\square$

The same bound applies when F contains the rectangle minimum, root tilt, shot sums and conditional masses at different plateau powers. The two laws can be represented by the coupling

$N_c^* = N_c + \text{Pois}(\mu_c^* - \mu_c)$. Inequality (13.5) follows from the likelihood ratio of the laws, so it applies to the entire integrand without a monotonicity hypothesis under added points.

We apply this comparison to the complete nonnegative suffix integrand, after forming the rectangle minimum and fixing the prefix costs and stopping boundaries.

13.4 A bounded total cell deficit and the physical rectangle

To apply the likelihood comparison at constant cost, the deficits must be summable over all cells. Greedy cells have this property because their first primes grow geometrically. The same bound controls the displacement between physical and cell coordinates.

For an ordinary block with h colors put $\eta = \log h/(h-1)$. Greedily group successive primes above Q_H until adding the next prime would exceed mass η . If p_j begins a complete cell and p_{j+1} begins the next one, its harmonic mass H_j satisfies

$$\eta - 1/p_{j+1} \leq H_j \leq \eta, \quad H_j \geq 3\eta/4. \quad (13.6)$$

Since the prime sum is at most the corresponding integer harmonic sum,

$$H_j \leq 1/p_j + \log(p_{j+1}/p_j).$$

Our choice of Q_H gives

$$p_{j+1} \geq e^{\eta/2} p_j, \quad \sum_j (\eta - H_j) \leq \frac{1}{Q_H(e^{\eta/2} - 1)}. \quad (13.7)$$

A final partial cell and an initial clipped piece each add at most η to the deficit. Complete them too. There are at most k blocks and their rates are at most H , so the total tilted Poisson mass added is bounded solely by H . Lemma 13.2 therefore costs a fixed multiplicative constant.

If U'_i is the number of completed cells times η_i , then

$$U'_i = L_i + O_H(1). \quad (13.8)$$

Every partial sum of the cell deficits is bounded by the same constant in (13.7). Combining it with (13.2), a prime in cell j has log-log coordinate within $O_H(1)$ of $j\eta_i$. Consequently replacing actual positions by cell positions multiplies a whole expression $1 + \sum_\ell e^{t_\ell}$ by at most $e^{O_H(1)}$. Its exponent $h-1$ is bounded, so the comparison constant is uniform in the number of occurrences. The actual original rectangle is proved before this replacement and before repeated-prime enlargement.

Under the completed Poisson law, conditional on the total $N = n$, cell indices are iid uniform indices. They can be realized by n independent uniform continuous positions on $[0, U'_i]$ and rounding to cells. For either signed minimum, use the appropriate boundary of the cell; the loss is at most its fixed linear-clock width, irrespective of the number of points in the cell. This realizes the same all- n ordinary ballot bound with all $n+1$ spacings retained. For the actual rectangle the precise ordered-sum estimate is Lemma 11.6; its proof does not impose centrality or exclude count zero.

13.5 The ordinary suffix integral

Cell completion changes the clock budget, but the tilt must still use the original root. We record the compensation explicitly before applying the ordinary root estimate.

Write $w_i = \log h_i$, $c_i = h_i - 1$ on the unread suffix. With its same counts and the original root coefficient r ,

$$\begin{aligned} Z_{\text{phys}} &= \sum_i (w_i N_i - c_i L_i), & Z_{\text{cell}} &= \sum_i (w_i N_i - c_i U'_i), \\ X &= \kappa + r Z_{\text{phys}} = \kappa' + r Z_{\text{cell}}, \\ \kappa' &= \kappa + r \sum_i c_i (U'_i - L_i). \end{aligned} \tag{13.9}$$

This exact identity has $\kappa' - \kappa = O_H(1)$ and preserves the root draw and the suffix's conditional Gibbs law.

Let $D > 0$ be the complete prefix cost after the causal cover. The all-count ordinary rectangle theorem of Proposition 12.9 is valid for every real translate

$$H_* = \frac{-\log D - \kappa'}{r}.$$

On the completed suffix it yields

$$CD^s e^{-t(H_*)} (1 + U')^{-1/2} \min \left\{ 1, \frac{(1 + B')[1 + (H_*)_+ + A']}{1 + M'} \right\}. \tag{13.10}$$

It also permits the established fixed polynomial outgoing envelope. The single largest-block Poisson atom and the ballot denominator were extracted jointly in this bound, and Lemma 13.2 preserves that bound.

All quantities with 1+length in (13.10) are comparable to their physical versions. For a large suffix one may use its physical largest block, whose completed length remains comparable; for a bounded suffix the numerical factors have a fixed positive lower bound. The arbitrary translate in (13.10) absorbs a grid mean negative by $O_H(1)$. The physical KKT inequality $D_{\text{nat}} \geq c_H V_{\text{pre}}$ is used only when averaging the prefix. The bounded change of κ is absorbed by $1 + D_{\text{nat}} + A$. The joint subset moment and the one-count Mecke bound then give the same physical ordinary factor as in Theorem 12.10.

13.6 Proof of the upper bound for positive lengths

Proof of Theorem 13.1. We first assemble the arithmetic comparisons, and then identify their factors with the profile at the frozen physical optimizer.

Apply the covers first to the size-truncated tuples and then enlarge their nonnegative analytic envelope. The KKT comparisons, eligibility rules and stopping boundaries remain valid at positive lengths, and the harmonic comparison gives the all-count prefix estimates. Bounded effective banks have fixed factor floors and can be omitted. The unread suffix is completed by Lemma 13.2, with its true root preserved by (13.9). Thus the complete causal proof, its different plateau powers and its single root integration apply for every $L_i \geq 1$. All constants are independent of L .

It remains to identify the numerical profile. This comparison takes place at the same physical optimizer. Fixed proof reserves and window constants can be replaced by the finite scores in (3.6), because their reference and good-group terms are nonnegative. For paired contexts, the reference increment νL_i is independent of the outside context. The nonnegative reference minima imply $|\nu| L_i \leq X_i + Y_i$, so strongly drifting edges and omitted saturated edges have factors bounded below. On a rare edge, the speed-centered and natural correlations differ by $O_H(|\nu| + 1/L_i) = O_H(T_e)$. The local Lipschitz bound for χ and boundedness of $T_e |\log T_e|$ then replace the exponent at

constant cost. Since only finitely many canonical edges occur, all can be included in the profile. The count widths similarly become $(1 + d\sqrt{1 + T})^{-\sigma}$, and the explicit ordinary suffix has the same factor. This proves (13.1). $\square$

14 Stability and the uniform arithmetic estimates

The sharp estimates have been proved on positive lengths, with the lower construction requiring one fixed dimension-dependent threshold. To finish the uniform theorem, we compare harmonic sums whose cutoffs move by bounded amounts on the logarithmic clock. This comparison is elementary and applies even when a prime range is empty. It lets us evaluate the profile at lifted lengths and then return to the original arithmetic parameters.

14.1 Stability of the original harmonic sum

A change of cutoffs affects only primes whose group memberships change. Removing those primes leaves a common positive sum; reinserting them costs a bounded Euler product. This gives the following explicit comparison.

Write $c = \log 3$ and $m_i = k - i + 2$ as before, and fix an absolute constant C_M for which

$$\left| \sum_{p \leq t} \frac{1}{p} - \log \log t - B_1 \right| \leq C_M \quad (t \geq 3).$$

This form follows, for example, from the explicit estimates of Rosser and Schoenfeld [23, Theorem 5]; enlarging the absolute constant covers the bounded range. Only this uniform coarse bound is used.

Theorem 14.1 (Explicit log-parameter stability). *For $\ell, \ell' \in [c, \infty)^k$ put*

$$\delta_i = \ell'_i - \ell_i, \quad D = \sum_{i=1}^k \left| \sum_{h=1}^i \delta_h \right|, \quad A = \sum_{i=1}^k m_i |\delta_i| + (k+1)(D + 2kC_M).$$

Then

$$e^{-A} K_k(\ell) \leq K_k(\ell') \leq e^A K_k(\ell).$$

In particular, for each fixed $L < \infty$, $K_k(\ell') \asymp_{k,L} K_k(\ell)$ uniformly whenever $\|\ell' - \ell\|_1 \leq L$.

Proof. Let y_i, y'_i be the associated cutoffs. Their definitions give

$$\log \log y'_i - \log \log y_i = \sum_{h \leq i} \delta_h.$$

Every prime that belongs to different blocks in the two configurations, or belongs to only one of their prime ranges, lies in

$$\mathcal{B} = \bigcup_{i=1}^k \{p : \min(y_i, y'_i) < p \leq \max(y_i, y'_i)\}.$$

Indeed, outside this union the truth values of all inequalities $p \leq y_i$ and $p \leq y'_i$ agree. Hence their least valid index, if any, agrees. Mertens's estimate and positivity imply

$$\sum_{p \in \mathcal{B}} \frac{1}{p} \leq D + 2kC_M. \tag{14.1}$$

Retain only the common primes outside $\mathcal{B}$, with their common block labels, and let S_* be their unnormalized harmonic volume sum. This number is positive, since the empty selection contributes $(\log 2)^k$. Write S, S' for the two full unnormalized sums. Inserting a selected block- i prime gives the union of at most m_i translates of the old union, one of them unchanged. Thus it increases the volume by a factor between 1 and $m_i \leq k+1$. Sum first over the common primes, and then over all selections of the remaining primes, retaining their exact reciprocal weights. This gives, for both $S_1 = S$ and $S_1 = S'$,

$$S_* \leq S_1 \leq S_* \prod_{p \in \mathcal{B}} \left(1 + \frac{k+1}{p}\right) \leq S_* \exp \left((k+1) \sum_{p \in \mathcal{B}} \frac{1}{p} \right).$$

Both sums lie between the same lower bound and its exponential multiple. The ratio S'/S is therefore bounded by that multiple and its reciprocal. Finally the exact ratio of the normalizing products is $\exp(-\sum_i m_i \delta_i)$. Combine this identity with (14.1) to obtain the assertion. $\square$

Corollary 14.2 (Contraction of any bounded collection of blocks). *Fix $L \geq c$ and any set $I \subseteq \{1, \dots, k\}$. If $c \leq \ell_i \leq L$ for $i \in I$, replace those coordinates by c and leave the others unchanged, obtaining $\widehat{\ell}$. Then $K_k(\ell) \asymp_{k,L} K_k(\widehat{\ell})$ uniformly over all remaining coordinates. If L is specified in terms of k only, the constants depend on k only.*

Proof. The changed vector remains admissible and its ℓ^1 distance from the original is at most $k(L-c)$. Apply Theorem 14.1. $\square$

14.2 Completion at one fixed lift

We now use stability to remove the lower-bound length restriction. Only the profile is evaluated at the lift; the harmonic representation of the divisor count is applied at the original cutoffs.

The constant Λ_k chosen in Appendix H.2 satisfies the lower-bound thresholds. The upper and lower estimates therefore apply at L^* :

$$c_k \Psi_k(L^*) \leq K_k(\log 3 + L^*) \leq C_k \Psi_k(L^*).$$

Apply Theorem 14.1 to $\delta_i = L_i^* - L_i \in [0, \Lambda_k]$. With $C_M = 100$,

$$A \leq B_k \Lambda_k + 200k(k+1), \quad B_k = \frac{k(k^2 + 3k + 4)}{2}.$$

Absorb this factor into the constants. The empty prime selections give a positive harmonic sum when some lengths vanish. This proves Theorem 2.3. Combining it with (2.4) proves Theorem 2.1 at exactly the original endpoints and stated x range.

14.3 Classical special cases

For $k = 1$, the scalar rule yields $s_1 = \alpha_2 = \log(1/\log 2)/\log 2$. There are no binary canonical edges or nonterminal plateaux. The ordinary suffix has $U = M = L$, $A = B = D = 0$, so

$$\Psi_1(L) = e^{-\delta L} (1+L)^{-3/2}, \quad \delta = 1 + \alpha_2 - 2^{\alpha_2} = 1 - \frac{1 + \log \log 2}{\log 2}.$$

Harmonic stability makes the fixed lift a constant-factor change. Consequently the formula gives Ford's dyadic order $(\log y)^{-\delta} (1 + \log \log y)^{-3/2}$ in the common range $x \geq 2y^2$ [13].

For sufficiently large balanced tables the same calculation has one plateau. The first length grows while the remaining lifted lengths stay bounded, so the saddle determines the logarithmic exponent and the ordinary suffix gives the factor of exponent $3/2$.

Corollary 14.3 (Balanced tables). *For every fixed $k \geq 1$, uniformly for $N \geq 3$,*

$$A_{k+1}(N, \dots, N) \asymp_k \frac{N^{k+1}}{(\log N)^{Q(k/\log(k+1))} (\log \log N)^{3/2}}.$$

Proof. We first verify that the ordered saddle has one plateau, then evaluate the exponential and ordinary factors on that plateau. For equal cutoffs, the lifted vector is $(T, \Lambda_k, \dots, \Lambda_k)$, where $T = \log \log N + O_k(1)$ as $N \rightarrow \infty$. Put $H = k + 1$ and

$$\alpha_H = \frac{\log((H-1)/\log H)}{\log H}.$$

The full equality face has a critical point $\sigma = \alpha_H + O_k(T^{-1})$, by

$$\sum_i L_i \{(h_i - 1) - h_i^\sigma \log h_i\} = 0.$$

The multiplier of its internal constraint at $j < k$ is

$$T \left\{ R - \frac{k(R+1) \log(R+1)}{H \log H} \right\} + O_k(1), \quad R = k - j.$$

It is positive for $1 \leq R < k$, since $t \log t / (t-1)$ is strictly increasing for $t > 1$. The KKT conditions and strict concavity therefore give one plateau for all sufficiently large T . On this plateau,

$$J = \{1 + k\alpha_H - H^{\alpha_H}\}T + O_k(1) = Q(k/\log H)T + O_k(1).$$

There are no strict-separation factors. The first block is the threshold-near ordinary block. Rank separation bounds its proper binary drifts away from zero; the reference-increment inequality then bounds their persistence factors below by a positive constant. All other block lengths are fixed, so their factors have the same property. Full-plateau balance gives $D = 0$, and the ordinary suffix has $U = T + O_k(1)$, $M = T$, $A = 0$ and $B = O_k(1)$. Its factor is therefore comparable to $T^{-3/2}$. Theorem 2.2 proves the assertion for large N . The remaining bounded range is absorbed in the constants, using $1 \leq A_{k+1}(N, \dots, N) \leq N^{k+1}$. $\square$

15 Rectangular multiplication tables

The table theorem follows by transferring the density to the harmonic sum already estimated. For sufficiently large first sides this is Koukoulopoulos's table-to-divisor comparison. Bounded sides require an additional step: delete their axes from the table and compare the resulting harmonic sums in the two dimensions. The proof below gives both comparisons, including the case in which only one unbounded axis remains.

For $r \geq 1$ and $3 \leq t_1 \leq \dots \leq t_r$, put $t_0 = 3$ and $\mathcal{K}_r(\mathbf{t}) = K_r(\ell(\mathbf{t}))$. Thus

$$\mathcal{K}_r(\mathbf{t}) = \prod_{i=1}^r \left(\frac{\log t_i}{\log t_{i-1}} \right)^{-(r-i+2)} \sum_{\mathbf{a} \in \mathcal{A}_r(\mathbf{t})} \frac{L^{(r+1)}(\mathbf{a})}{a_1 \cdots a_r}. \quad (15.1)$$

We also use $A_1(N) = \lfloor N \rfloor$ for $N \geq 1$.

Theorem 15.1 (Harmonic representation of the table density). *Let $k \geq 1$ be fixed, and let $1 \leq N_1 \leq \dots \leq N_{k+1}$ be real. With $P = \prod_{i=1}^{k+1} N_i$ and $Y_i = \max\{3, N_i/2\}$ for $1 \leq i \leq k$, one has*

$$\frac{A_{k+1}(\mathbf{N})}{P} \asymp_k \mathcal{K}_k(\mathbf{Y}). \quad (15.2)$$

Proof. We handle large sides first. We then delete every bounded initial axis and identify the corresponding bounded factors in the harmonic sum.

Koukoulopoulos's Theorem 1.1 [19] states that, for $r \geq 1$ and real $3 \leq M_1 \leq \dots \leq M_{r+1}$,

$$A_{r+1}(\mathbf{M}) \asymp_r H^{(r+1)} \left(\prod_{i=1}^{r+1} M_i, (M_1/2, \dots, M_r/2), (M_1, \dots, M_r) \right). \quad (15.3)$$

Together with the harmonic reduction (2.4), this gives

$$\frac{A_{r+1}(\mathbf{M})}{\prod_i M_i} \asymp_r \mathcal{K}_r(M_1/2, \dots, M_r/2) \quad (M_1 \geq 6). \quad (15.4)$$

Indeed the cutoffs $t_i = M_i/2$ are at least 3, and

$$2^r(t_1 \cdots t_r)t_r = \left(\prod_{i=1}^r M_i \right) \frac{M_r}{2} \leq \frac{1}{2} \prod_{i=1}^{r+1} M_i,$$

so the hypothesis of that reduction is satisfied.

We next remove the initial sides for which this large-side reduction is unavailable. Let

$$b = \#\{i \in \{1, \dots, k\} : N_i < 6\}, \quad m = k - b,$$

and put

$$P_{\text{tail}} = \prod_{i=b+1}^{k+1} N_i, \quad A_{\text{tail}} = A_{m+1}(N_{b+1}, \dots, N_{k+1}).$$

Setting the first b factors equal to 1 embeds the tail table. For each choice of those factors, its distinct products form an integer dilate of the tail table. There are at most $\prod_{i=1}^b \lfloor N_i \rfloor$ such choices. Since every deleted side belongs to $[1, 6)$, division by P gives

$$6^{-b} \frac{A_{\text{tail}}}{P_{\text{tail}}} \leq \frac{A_{k+1}(\mathbf{N})}{P} \leq \frac{A_{\text{tail}}}{P_{\text{tail}}}. \quad (15.5)$$

This argument includes sides in $[1, 2)$, for which the only factor is 1.

It remains to show that inserting the bounded axes changes the harmonic density by only a dimension-dependent factor. Suppose first that $m \geq 1$, and set $z_j = N_{b+j}/2$ for $1 \leq j \leq m$. Then $z_1 \geq 3$ and $\mathbf{Y} = (3, \dots, 3, \mathbf{z})$. Put $h = \log 2$ and $D = \log 12$. We claim that

$$h^b \mathcal{K}_m(\mathbf{z}) \leq \mathcal{K}_k(\underbrace{3, \dots, 3}_b, \mathbf{z}) \leq D^b \mathcal{K}_m(\mathbf{z}). \quad (15.6)$$

For $b = 0$ the claim is an identity. For $b \geq 1$, each tuple in the middle sum has the form $\mathbf{a} = (a_1, 1, \dots, 1, a_{b+1}, \dots, a_k)$ with $a_1 \mid 6$. The correspondence

$$\mathbf{a} \longmapsto \mathbf{a}' = (a_1 a_{b+1}, a_{b+2}, \dots, a_k)$$

is a bijection onto $\mathcal{A}_m(\mathbf{z})$ preserving reciprocal weights: the primes 2, 3 recover a_1 , and all other primes in the first component recover a_{b+1} . This remains true when $z_1 = 3$. Choosing the first b divisor coordinates equal to 1 gives

$$[-h, 0)^b \times \mathcal{L}^{(m+1)}(\mathbf{a}') \subseteq \mathcal{L}^{(k+1)}(\mathbf{a}).$$

Conversely, every first b divisor coordinate is at most 6, since each relevant prefix product divides a_1 . Projecting onto the last m coordinates gives a permissible divisor word for $\mathbf{a}'$. Consequently

$$\mathcal{L}^{(k+1)}(\mathbf{a}) \subseteq [-h, \log 6)^b \times \mathcal{L}^{(m+1)}(\mathbf{a}').$$

The volume ratios are therefore between h^b and D^b . The first b logarithmic ratios in (15.1) are 1, and the remaining exponents are $k - (b + j) + 2 = m - j + 2$. Summing proves (15.6). Now (15.4), (15.5), and (15.6) give the desired comparison.

When every initial axis is bounded, the tail is a one-dimensional table and its density stays between two absolute constants. If $m = 0$, then $\mathbf{Y} = (3, \dots, 3)$ and

$$\frac{A_{\text{tail}}}{P_{\text{tail}}} = \frac{\lfloor N_{k+1} \rfloor}{N_{k+1}} \in [1/2, 1].$$

Indeed, if $q = \lfloor N_{k+1} \rfloor \geq 1$, then $N_{k+1} < q + 1 \leq 2q$. The harmonic sum has the four tuples $(a_1, 1, \dots, 1)$ with $a_1 \in \{1, 2, 3, 6\}$, whose reciprocal weights sum to 2. The divisor word with every coordinate equal to 1 gives volume h^k , and every coordinate is at most 6, giving volume at most D^k . Thus

$$2h^k \leq \mathcal{K}_k(3, \dots, 3) \leq 2D^k.$$

Combined with (15.5), this yields

$$\frac{\mathcal{K}_k(\mathbf{Y})}{4(6D)^k} \leq \frac{A_{k+1}(\mathbf{N})}{P} \leq \frac{\mathcal{K}_k(\mathbf{Y})}{2h^k}.$$

There are only finitely many choices of b and m , depending on k , so the comparison constants depend only on k . $\square$

Proof of Theorem 2.2. Apply Theorem 15.1 and Theorem 2.3 at $\ell(\mathbf{Y}) \in [\log 3, \infty)^k$. $\square$

16 Further questions

Theorem 2.2 permits every rectangular shape, but its comparison constants depend on the fixed number of factors. The next questions ask what happens when the dimension grows, or when only products with a specified arithmetic form are counted. The known special cases indicate two different phenomena that such extensions should capture. For an integer $d \geq 2$ and real $1 \leq N_1 \leq \dots \leq N_d$, put

$$\mathcal{M}_d(\mathbf{N}) = \{n_1 \cdots n_d : n_i \in \mathbb{N}, n_i \leq N_i\}, \quad A_d(\mathbf{N}) = \#\mathcal{M}_d(\mathbf{N}).$$

Variable-dimensional rectangular tables. Can one obtain matching estimates for $A_d(\mathbf{N})$ with absolute comparison constants, uniformly in d and the ordered side-length vector, with the dependence on d explicit in the formula? Can one obtain asymptotic formulae in specified joint growth regimes? For equal integer sides $N \geq 2$, the count is eventually a polynomial in d of degree $\pi(N)$. Sabuncu and Táfula [24, Theorem 1.2] determine the asymptotic size of its leading coefficient as $N \rightarrow \infty$, yielding a formula when d is sufficiently large in terms of N . This addresses a particular equal-side regime. A general rectangular theory should also recover the fixed-dimensional estimate of Theorem 2.2 and describe the transition between these regimes. The question concerns families $(d(t), \mathbf{N}(t))$ with $d(t) \rightarrow \infty$ or $N_{d(t)}(t) \rightarrow \infty$, subject to any further growth hypotheses needed for a precise asymptotic statement. The results cited here do not provide a uniform estimate over all such rectangular families.

Distinct perfect powers. For an integer $r \geq 2$, define

$$A_d^{[r]}(\mathbf{N}) = \#(\mathcal{M}_d(\mathbf{N}) \cap \{m^r : m \in \mathbb{N}\}).$$

The equal-side case $d = r$ has the exact elementary answer $A_r^{[r]}(N, \dots, N) = \lfloor N \rfloor$: every $m \leq \lfloor N \rfloor$ can be used in all r positions, and no larger root is possible. Outside such special cases, what matching estimates hold uniformly in $(d, \mathbf{N}, r)$, with the dependence on d and r explicit? What asymptotic formulae hold along the growing families above, with fixed r or an integer-valued $r = r(t) \geq 2$? The unrestricted-product estimates in this paper and in [24] do not determine this restricted count. We are not aware of a result covering the full simultaneous parameter range. The question concerns that range, with the elementary cases above and other known special cases understood.

A Saddle and endpoint geometry

The endpoint scales in the profile arise from a finite geometric problem: minimize the cost of a lawful history while retaining its actual names and deadlines. This appendix connects that problem to the ordered saddle and constructs count types on which the resulting endpoint constraints can all be imposed.

The argument proceeds from reference costs to actual costs. Clipped rank histories telescope to stationarity multipliers; compression extends their nonnegativity to every lawful reference history. Quantitative secant and curvature estimates then identify the weak transitions that require path constraints. A displaced count box pays the remaining empirical errors simultaneously, and intrinsic normalization removes the incoming exit from the local reserve. Finally, strict-exit bands and terminal completion place these boxes on one joint count source. Throughout, each original prime range keeps its own coefficient and count scale.

A.1 Two reference-cost identities

Two identities organize the reference calculation. The first turns a clipped rank cost into an order-constraint multiplier. The second bounds the coefficient of an incoming mixture by a strict entropy surplus. Both retain the original groups: stationarity is available for the whole incoming mixture, while its individual components can have different intensities.

Lemma A.1 (Clipped margin telescope). *Use the complete chain $p_i = 1$. Let $[a, b]$ be a maximal equal-product plateau, with product σ , child weight z , and original coefficients κ_g as in (3.4). Write $r_g = b - a + 1$ for $g < a$, and $r_g = b - g + 1$ for $a \leq g \leq b$. For integers $0 \leq R \leq b - a + 1$ set*

$$\Lambda_R = \sum_{g \leq b} L_g \{ \min(r_g, R) - \kappa_g E_{\min(r_g, R)}(z) \}.$$

Then

$$\Lambda_R = \lambda_{b-R} \geq 0, \quad \Lambda_0 = \Lambda_{b-a+1} = 0, \quad \lambda_j = - \sum_{i \leq j} \partial_{s_i} J, \quad \lambda_0 = \lambda_k = 0.$$

Proof. Set $1 \leq R \leq b - a + 1$ and $j = b - R + 1$. Raising $s_j, \dots, s_b$ together clips the rank contribution from each earlier group at the suffix rank. We use the derivative in the positive product region; it exists even when the formal direction is not feasible in both directions under the order constraints. The linear budget contributes $\sum_{l=j}^b V_l = \sum_{g \leq b} \min(r_g, R) L_g$.

For $g \geq j$, differentiating $G_g = \sigma \log(r_g + z)$, with the permanent child exponent below b fixed, gives $dz/d\sigma = -z \log z/\sigma$ and

$$\frac{dG_g}{d\sigma} = \frac{E_{r_g}(z)}{r_g + z}, \quad \frac{dZ_g}{d\sigma} = \kappa_g E_{r_g}(z).$$

For an earlier local group the routing probability into the surviving compound alphabet is $(R + z)/(r_g + z)$. For an earlier group it is

$$\left(\prod_{l=g}^{a-1} \beta_l \right) \frac{R + z}{b - a + 1 + z}.$$

The derivative at j is $E_R(z)/(R + z)$; propagation and multiplication by Z_g therefore give $\kappa_g E_R(z)$ in both cases. Subtraction from the linear derivative yields

$$\Lambda_R = \sum_{l=j}^b \partial_{s_l} J = \lambda_{j-1} - \lambda_b = \lambda_{b-R}.$$

Maximality of the plateau gives $\lambda_b = \lambda_{a-1} = 0$, with the stated endpoint conventions. The remaining multipliers are nonnegative. The case $R = 0$ follows from the definition. $\square$

The telescope gives nonnegativity for clipped benchmark paths. For a general legal rank path the capacity varies between groups; shifted compression will compare it with one of these benchmarks while respecting those changing capacities.

Lemma A.2 (Entropy denominator for an old mixture). *Suppose $H \geq 2$, $p > 0$, $S \geq 1$, $0 \leq u \leq 1$, and $p + S^u \leq H$. With*

$$E_p(x) = (p + x) \log(p + x) - x \log x, \quad D(p, x) = E_p(x)/p,$$

one has

$$S^{u-1}(1 + \log S) + \frac{p}{2H} \leq D(p, S^u) \leq 1 + \log H. \quad (\text{A.1.1})$$

Consequently, if $\kappa_{\text{av}} \geq 0$ and $\kappa_{\text{av}} \leq S^{u-1}/D(p, S^u)$, then

$$\kappa_{\text{av}}(1 + \log S) \leq 1 - \frac{p}{2H(1 + \log H)}.$$

Proof. For $x = S^u$, integration of the derivative of $t \log t$ gives

$$D(p, x) = 1 + \log x + \frac{1}{p} \int_0^p \log(1 + t/x) dt.$$

Since $\log(1 + t/x) \geq t/(x + t) \geq t/(x + p)$,

$$D(p, x) \geq 1 + u \log S + \frac{p}{2(x + p)} \geq 1 + u \log S + \frac{p}{2H}.$$

The same expression is the average of $1 + \log(x + t)$, giving the upper bound in (A.1.1). Put $L = \log S$ and $f(u) = 1 + uL - e^{(u-1)L}(1+L)$. Then $f'' \leq 0$, $f(1) = 0$, and $f(0) = 1 - (1+L)e^{-L} \geq 0$. Concavity implies $f(u) \geq 0$, which proves the lower bound. Finally divide it by $D(p, S^u)$ and use its upper bound:

$$\frac{S^{u-1}(1 + \log S)}{D(p, S^u)} \leq 1 - \frac{p}{2HD(p, S^u)} \leq 1 - \frac{p}{2H(1 + \log H)}.$$

Multiplication by the assumed bound for κ_{av} completes the proof. $\square$

In applications κ_{av} is obtained from the whole incoming count balance. The conclusion bounds that average, and does not assign an individual stationarity equation to each earlier physical group.

A.2 Complete-reference geometry and original endpoint boxes

We first keep the fine-word likelihood exact, including the final private names. After deriving its carry cancellation, we build the endpoint geometry in three steps: nonnegative reference costs, a quantitative classification of weak transitions, and a common box of actual counts. The last step supplies compatible local pairs without resampling an already accepted prefix.

A.2.1 Exact carries and the full word density

A routed child contributes to two consecutive levels with opposite coefficients. This cancellation links the uncolored root, the signed node outputs, and the full fine-word probability. It is also the reason that narrow strict-exit bands control likelihood ratios, whereas internal equal-product carries contribute no extra exponential cost.

We first allow a decreasing alphabet chain $H = h_1 > \dots > h_m \geq 2$, $m \geq 2$, with positive lengths L_i . Write $p_j = h_j - h_{j+1}$ for $j < m$, $p_m = h_m - 1$, and $V_j = \sum_{i \leq j} L_i$. At its ordered saddle, set

$$S_m = h_m, \quad S_j = p_j + S_{j+1}^{s_{j+1}/s_j}, \quad Z_i = S_i^{s_i}, \quad J = \sum_i L_i + \sum_j s_j p_j V_j - \sum_i L_i Z_i.$$

The compactness theorem gives $1/2 < s_m \leq \dots \leq s_1 < 1 - \epsilon_H$. Define

$$\begin{aligned} u_j &= s_{j+1}/s_j, & d_j &= u_j^{-1} - 1, & r_j &= s_j/s_1, & a_j &= \log S_j, & \beta_j &= S_{j+1}^{u_j}/S_j, \\ A_j &= a_j - u_j \beta_j a_{j+1} \quad (j < m), & A_m &= a_m, \\ \lambda_j &= - \sum_{i \leq j} \partial_i J, & \lambda_0 &= \lambda_m = 0, & m_j &= \lambda_j - \lambda_{j-1}. \end{aligned}$$

The signed margin m_j is distinct from the number m of levels. Complementarity and summation by parts give

$$\lambda_j \geq 0, \quad \lambda_j(s_j - s_{j+1}) = 0, \quad \sum_j s_j m_j = 0, \quad \sum_j m_j = 0, \quad \sum_{j < m} r_j m_j = r_m \lambda_{m-1}. \quad (\text{A.2.1})$$

Indeed the first sum is $\sum_{j < m} (s_j - s_{j+1}) \lambda_j$. With $\mu_i = L_i Z_i$, differentiation of the perspective recursion yields

$$A_j \sum_{i \leq j} \mu_i \prod_{i \leq q < j} \beta_q = p_j V_j + m_j. \quad (\text{A.2.2})$$

Euler homogeneity of $G_i = s_i \log S_i$ then gives the uncolored root identity

$$T_1 := \sum_j r_j p_j V_j = \sum_i r_i a_i \mu_i. \quad (\text{A.2.3})$$

For the difference, multiplied by s_1 , is $\sum_j s_j \partial_j J = 0$ by (A.2.1).

Use the original recursive color law on a fixed uncolored tuple. At level $j < m$, let M_j count the incoming and local slots and let C_j count those routed to the child. Put

$$Y_j = a_j M_j - u_j a_{j+1} C_j - p_j V_j, \quad X_m = a_m M_m - p_m V_m, \quad X_1 = \sum_i r_i a_i N_i - T_1.$$

Here X_1 denotes the uncolored root residual, not Y_1 . Using $M_{j+1} = C_j + N_{j+1}$ and $r_{j+1} = r_j u_j$, every child count cancels and gives

$$X_1 = \sum_{j < m} r_j Y_j + r_m X_m. \quad (\text{A.2.4})$$

Under the natural marked Poisson law, $\mathbb{E}Y_j = m_j$ and $\mathbb{E}X_m = m_m = -\lambda_{m-1}$. The full word probability $\pi(\omega)$ includes the terminal free names. The negative log probability of an exterior name is a_j , and that of a child route is $a_j - u_j a_{j+1}$; subsequent refinement contributes at its own level. Therefore, term by term,

$$-\log \pi(\omega) = B_{\text{geom}} + \sum_{j < m} Y_j + X_m, \quad B_{\text{geom}} = \sum_j p_j V_j. \quad (\text{A.2.5})$$

If $q_j = s_j/s_m - 1$, $q_m = 0$, elimination of X_m gives

$$1 = A_{\text{eff}} \pi(\omega) \exp \left(- \sum_{j < m} q_j Y_j \right), \quad A_{\text{eff}} = \exp(B_{\text{geom}} + X_1/r_m). \quad (\text{A.2.6})$$

Let y_j be fixed by the uncolored environment before its private refinements, and define carries by

$$Y_j = m_j + y_j + \xi_j - \xi_{j-1}/u_{j-1}, \quad \xi_0 = 0. \quad (\text{A.2.7})$$

The index-zero term is zero. Put $x = \sum_{j < m} r_j y_j$. If $x - C_{\text{root}} < X_1 \leq x$, then

$$X_m = (X_1 - x)/r_m - \lambda_{m-1} - \xi_{m-1}/u_{m-1}. \quad (\text{A.2.8})$$

Thus the terminal margin retains the last cumulative multiplier. For example, solving for the last local count by

$$N_m^* = \left\lfloor \frac{T_1 + x - \sum_{i < m} r_i a_i N_i}{r_m a_m} \right\rfloor$$

would give $C_{\text{root}} = r_m a_m$ when this count is admissible. Centrality and nonnegativity of this count are separate probability requirements; the identity alone supplies neither when L_m is small.

Since $\sum_j q_j m_j = 0$ and $q_j - q_{j+1}/u_j = d_j$, the exact cancellation is

$$\sum_{j < m} q_j Y_j = \sum_{j < m} q_j y_j + \sum_{j < m} d_j \xi_j. \quad (\text{A.2.9})$$

Combining this with the factor $e^{-s_1 X_1}$ gives

$$(r_m^{-1} - s_1)X_1 - \sum_{j < m} q_j Y_j = \sum_{j < m} (1 - s_j)y_j - \sum_{j < m} d_j \xi_j + (r_m^{-1} - s_1)(X_1 - x). \quad (\text{A.2.10})$$

Here $(r_m^{-1} - s_1)r_j - q_j = 1 - s_j$. Thus $y_j = \log(C_j^*/D_j)$ produces the powers $D_j^{s_j-1}$, with no additional multiplier-dependent rate.

A.2.2 Comparison of the original posteriors

An intersection class contains fine words with their original probabilities. To convert its conditional accepted mass to an unnormalized counting mass, we retain the relative likelihood of the anchor and competitor. The full carry identity bounds this likelihood on the retained support; grouping into plateaux then reveals which exit widths are needed.

Fix one complete uncolored environment G , one accepted full-chain weight $0 \leq F_G \leq 1$, and the original prior π_G on all its fine words. The carries in (A.2.7) can be defined for any word; bounds on them are additional support conditions. Write $E(\omega) = \sum_{j < m} d_j \xi_j(\omega)$. For two retained words,

$$\frac{\pi_G(\omega)}{\pi_G(\omega')} = e^{E(\omega) - E(\omega')}.$$

Let $Q_h(\omega)$ be any original comparison class, including classes that mix parent and child cells, and put

$$q_h = \mathbb{E}_\pi[F_G(\omega') \mid Q_h(\omega)], \quad q_h^{\text{rel}} = \mathbb{E}_\pi[F_G(\omega') e^{E(\omega) - E(\omega')} \mid Q_h(\omega)].$$

If the oscillation of E on $\{F_G > 0\}$ is at most B_H , positivity on this same class gives

$$e^{-B_H} q_h \leq q_h^{\text{rel}} \leq e^{B_H} q_h. \quad (\text{A.2.11})$$

For the actual geometric factor Φ_h , define $I_h = \log \Phi_h + \log[\pi_G(Q_h)/\pi_G(\omega)]$. Direct substitution gives

$$e^{I_h} q_h^{\text{rel}} = \Phi_h \sum_{\omega' \in Q_h} F_G(\omega').$$

Consequently (A.2.11) compares the full-history sums for every fixed positive collision power. The same argument applies to an enlarged whole cell with its actual class and original geometric factor. A sufficient oscillation condition is

$$|\xi_j - c_j| \leq w_j, \quad \sum_{j < m} d_j w_j \leq B_H/2,$$

where the centers c_j are common to the environment. If $E_c = \sum d_j c_j$, the nominal mass is

$$\mathbb{E}_\pi[F_G e^{-E}] \asymp e^{-E_c} \mathbb{E}_\pi F_G.$$

The center factor is retained even when the oscillation is bounded.

Now group the levels into maximal plateaux $[a_\nu, b_\nu]$, with products $\sigma_1 > \dots > \sigma_M$. Put

$$u_\nu = \sigma_{\nu+1}/\sigma_\nu, \quad d_\nu = u_\nu^{-1} - 1, \quad \eta_\nu = \xi_{b_\nu}, \quad y^{(\nu)} = \sum_{j=a_\nu}^{b_\nu} y_j, \quad Z_\nu = \sum_{j=a_\nu}^{b_\nu} Y_j \quad (\nu < M).$$

Exterior multipliers vanish, and internal ratios equal one. Hence

$$Z_\nu = y^{(\nu)} + \eta_\nu - \eta_{\nu-1}/u_{\nu-1}, \quad E = \sum_{\nu < M} d_\nu \eta_\nu, \quad (\text{A.2.12})$$

$$Z_M := \sum_{j=a_M}^{m-1} Y_j + X_m = \frac{X_1 - x}{r_m} + \sum_{j=a_M}^{m-1} y_j - \eta_{M-1}/u_{M-1}. \quad (\text{A.2.13})$$

Common plateau sums of the targets suffice in these formulas; individual internal targets may be redistributed while those sums and the grouped bands are maintained. If the strict exits have widths R_ν satisfying

$$0 \leq \eta_\nu \leq C_H R_\nu, \quad R_\nu \leq C_H \sqrt{1 + T_\nu}, \quad d_\nu R_\nu \leq C_H,$$

then $E = O_H(1)$. The lattice-sized floor of $\sqrt{1 + T_\nu}/(1 + d_\nu \sqrt{1 + T_\nu})$ has these properties. Internal equal-product carries have coefficient zero. In the one-plateau case the full word probability is independent of the routing at fixed G , although a proper atomic projection need not be uniform.

Formula (A.2.10) expresses the nominal integrand as

$$\prod_{j < m} D_j^{s_j - 1} \mathbb{E}_\pi[F_G e^{-E}],$$

up to the fixed target constants and bounded root factor. Grouping retains the corresponding plateau powers $\sigma_\nu - 1$. Evaluating this expression requires the conditional first-mass estimate at each stage with its entire type mixture inside the power, retaining the coupled count acceptance in F_G .

For completeness, the distinction between a full word and its atomic projection can be seen from one parent and child. If the parent output is $Z = y + \eta$, then

$$\pi_{\text{full}}(w, v) = \pi_p(w) \pi_c(v \mid w), \quad -\log \pi_p(w) = B_p + Z(w).$$

For two accepted full words,

$$\begin{aligned} \log \frac{\pi_p(w_0)}{\pi_p(w)} &= \eta - \eta_0, \\ \log \frac{\pi_{\text{full}}(w_0, v_0)}{\pi_{\text{full}}(w, v)} &= d(\eta_0 - \eta), \\ \log \frac{\pi_c(v_0 \mid w_0)}{\pi_c(v \mid w)} &= (\eta_0 - \eta)/u. \end{aligned}$$

The child fiber supplies the cancellation even when $d = 0$. Thus full-word posterior comparison is applied after the fine-history estimate, retaining the actual child conditional mass.

A.2.3 A finite shift of the strict-exit targets

Recoloring one immediate-child name as a shallow exterior name raises the current output and lowers the next one. A triangular integer cascade therefore realizes all nonterminal targets while changing the strict carries by bounded amounts. Its weighted output change is zero, so the desired common root must already have been selected.

Let $A_\nu = \log S_{a_\nu}$ denote the mark of the shallowest exterior leaf of plateau ν , and put $b_\nu = u_\nu A_{\nu+1} = \log z_\nu$ for $\nu < M$. In a local bank of plateau ν , change e_ν leaves that are the immediate child's shallowest exterior into the plateau's eligible shallow name. The banks lie in disjoint original local blocks. With $e_0 = 0$,

$$\Delta Z_\nu = b_\nu e_\nu - A_\nu e_{\nu-1} \quad (\nu < M), \quad \Delta Z_M = -A_M e_{M-1}.$$

For desired nonnegative shifts y_ν , choose successively

$$e_\nu = \left\lceil \frac{y_\nu + A_\nu e_{\nu-1}}{b_\nu} \right\rceil, \quad \delta_\nu = b_\nu e_\nu - A_\nu e_{\nu-1} - y_\nu \in [0, b_\nu).$$

Comparison of (A.2.12) before and after the change gives

$$\Delta \eta_\nu = \delta_\nu + \Delta \eta_{\nu-1} / u_{\nu-1} = \sum_{l \leq \nu} \frac{\sigma_l}{\sigma_\nu} \delta_l = O_H(1).$$

Thus an existing exit window is preserved with a fixed enlargement. If $y_\nu = O_H(\log(2 + T_\nu))$, the plateau length hierarchy gives $T_l \leq C_H T_\nu$ for $l < \nu$, and the finite recursion yields $e_\nu = O_H(\log(2 + T_\nu))$. These changes fit a macroscopic local bank with a central required-leaf count once its own total is large. The weighted sum of the output changes is zero because $r_\nu b_\nu = r_{\nu+1} A_{\nu+1}$, so the private changes leave the uncolored root unchanged. Its desired target rises by $\sum r_\nu y_\nu$; the unread reservoir must therefore be selected under that target before the private construction.

A.2.4 Secants, compression, and the reference cost

Throughout this appendix the ambient alphabet has at most H letters; all constants may depend on H . For $z \geq 1$ put

$$E_q(z) = (q + z) \log(q + z) - z \log z, \quad E_0(z) = 0.$$

Compression compares a lawful rank profile with a clipped constant rank. The comparison threshold is a normalized secant of the entropy function. Its quantitative monotonicity both preserves earlier compression comparisons and separates the possible weak transitions.

Lemma A.3 (Normalized secants). *For $1 \leq b < y \leq H$, let*

$$L(b, y) = \frac{y \log y - b \log b}{y - b}, \quad \beta_b(y) = \frac{\log L(b, y)}{\log y}.$$

Then $\beta'_b(y) \leq -1/(24H^2)$.

Proof. Set $F(t) = \log \int_0^1 e^{tu} du$, $0 \leq t \leq \log H$. Its second derivative is the variance under the density $p_t(u) = e^{tu} / \int_0^1 e^{tv} dv \geq H^{-1}$, so

$$F''(t) \geq H^{-1} \int_0^1 (u - \mathbb{E}_t u)^2 du \geq (12H)^{-1}.$$

For $t = \log y$, $\beta_1(y) = 1 - F(t)/t$, whence

$$-\beta'_1(y) = \frac{\int_0^t v F''(v) dv}{yt^2} \geq \frac{1}{24Hy} \geq \frac{1}{24H^2}.$$

For general b , writing $s = b/y$ gives

$$y\partial_y L(b, y) = h(s) = \frac{1 - s + s \log s}{(1 - s)^2}, \quad h'(s) = \frac{(1 + s) \log s + 2(1 - s)}{(1 - s)^3} < 0.$$

Indeed $\log s - 2(s - 1)/(s + 1)$ increases to zero on $(0, 1)$, its derivative being $(s - 1)^2/[s(s + 1)^2]$. Increasing b at fixed y increases L and decreases $y\partial_y L$. It therefore decreases

$$\beta'_b(y) = \frac{(y\partial_y L/L) \log y - \log L}{y(\log y)^2}.$$

Comparison with $b = 1$ proves the assertion. $\square$

Lemma A.4 (Shifted compression). *Let $0 \leq e \leq 1$ and $F_r(q) = q - (r + z)^{-e} E_q(z)$. At finitely many increasing integer capacities r , let integers d_r be nondecreasing and satisfy $0 \leq d_r \leq r$. There is an integer $R \geq 0$ such that*

$$F_r(\min(r, R)) \leq F_r(d_r) \quad \text{at every capacity.}$$

The assertion remains true if only the coefficient at the largest capacity is decreased to an arbitrary positive value. No restriction on the increments $d_{r'} - d_r$ is required.

Proof. Read capacities in increasing order, starting with benchmark $R = 0$. At capacity r , replace $R = A$ by the current count $C = d_r$ only if $F_r(A) > F_r(C)$. Monotonicity of the counts ensures $A < C \leq r$. The failing comparison means

$$L(z + A, z + C) > (z + r)^e.$$

For an earlier capacity $A < h < C$, Lemma A.3 gives $L(z + A, z + h) > (z + h)^e$; for $C \leq h \leq r$ the displayed inequality suffices, and for $h \leq A$ the clipped arguments coincide. Thus every earlier comparison is preserved. If the last coefficient is $a \leq (r + z)^{-e}$, its failing comparison instead gives $L(z + A, z + C) > 1/a \geq (z + r)^e$, which proves the same preservation. For the singleton convention $\phi_m(n) = n - 1 - m^{-e} n \log n$, $1 \leq n \leq m$, the identical argument uses $L(A, C) > m^e$; zero counts may first be replaced by one, since the entropy margin is zero at both zero and one. $\square$

Lemma A.5 (Compression with the original capacities). *Fix $z \geq 1$ and $0 < e < 1$. Read local capacities in increasing order. At capacity r , let the lawful rank be C_r , where $0 \leq C_r \leq r$ and the ranks are nondecreasing in this reading order. Put*

$$E_q(z) = (q + z) \log(q + z) - z \log z, \quad F_r(q) = q - (r + z)^{-e} E_q(z).$$

There is an integer R such that, for every processed local group,

$$F_r(C_r) \geq F_r(\min(r, R)).$$

The assertion remains true after appending original earlier groups of constant maximal capacity. Their ranks must lie between zero and that capacity and must remain nondecreasing through the entire reading order, including the local/old junction. Their coefficients in place of $(r + z)^{-e}$ must be nonincreasing in the appended reading order and no greater than the coefficient at the maximal local capacity. In an earlier group with coefficient κ , use $F(q) = q - \kappa E_q(z)$.

Proof. Start with $R = 0$. At each local capacity r , replace R by $C = C_r$ exactly when $F_r(R) > F_r(C)$. Before a replacement, $R \leq C \leq r$, because the ranks are nondecreasing. The replacement criterion is precisely

$$L(z + R, z + C) > (r + z)^e \geq (z + C)^e.$$

Consider a previously processed capacity $h \leq r$. If $h \leq R$, its clipped benchmark does not change. If $R < h < C$, Lemma A.3 gives

$$L(z + R, z + h) > (z + h)^e,$$

and hence $F_h(h) < F_h(R)$. If $h \geq C$, its coefficient is at least the current coefficient, so $F_h(C) < F_h(R)$. Thus every previous clipped benchmark weakly decreases. All previously established comparisons persist, while the current comparison becomes equality. If no replacement occurs, the current comparison already holds. Induction proves the local assertion.

Continue the induction into the earlier groups. Their rank C is at least the current benchmark R , by the assumed monotonicity. An update at coefficient κ requires $L(z + R, z + C) > 1/\kappa$. Since this threshold is no smaller than the maximal local threshold, the preceding three-case argument preserves every local comparison. The processed old coefficients are also at least κ , so their earlier comparisons are preserved. Thus the same induction covers the entire original list. $\square$

The old-group hypotheses follow from lawfulness in the application. For a plateau $[a, b]$, read original groups in the reverse order $b, b-1, \dots, 1$. The rank function $d(t)$ counts boundaries that have not yet activated; activating a boundary only decreases this count. Thus every endpoint rank list is nondecreasing in the reverse reading order, including the junction between group a and the old groups. Lawfulness places every boundary's activation no later than its name's deadline, so each rank is at most the corresponding capacity. Moreover, for $g < a$, the transport identity gives $\kappa_g = \kappa_{g+1}\beta_g^{1-s_g} \leq \kappa_{g+1}$, including $g = a-1$. Consequently the incoming coefficients are nonincreasing in the reverse reading order and do not exceed the first local coefficient.

Fix a maximal equal-product plateau $[a, b]$ of product σ in the complete natural chain. Write

$$r_i = b - i + 1, \quad S_i = r_i + z, \quad c_i = S_i^{\sigma-1}, \quad V_i = \sum_{l \leq i} L_l, \quad T = V_b.$$

The permanent child has weight $z \geq 1$. Retain every original old physical block, with its natural coefficient

$$\kappa_i = \frac{Z_i}{S_a} \prod_{i \leq l < a} \beta_l \quad (i < a), \quad \kappa_i = c_i \quad (a \leq i \leq b). \quad (\text{A.2.14})$$

Its mean letter intensities are $\kappa_i(1, \dots, 1, z)$. All old blocks have capacity r_a ; local block i has capacity r_i . Put $K_{\text{in}} = \sum_{i < a} L_i \kappa_i$. The plateau stationarity identities give

$$K_{\text{in}} E_{r_a}(z) - r_a V_{a-1} + \sum_{i=a}^b L_i [c_i E_{r_i}(z) - r_i] = 0, \quad (\text{A.2.15})$$

$$\Lambda_R := R V_{a-1} - K_{\text{in}} E_R(z) + \sum_{i=a}^b L_i \{\min(r_i, R) - c_i E_{\min(r_i, R)}(z)\} = \lambda_{b-R} \geq 0. \quad (\text{A.2.16})$$

Here $0 \leq R \leq r_a$ and $\lambda_{a-1} = \lambda_b = 0$. For example, at $R = r_j$ the left side is precisely the incoming buffer minus the remaining local drift, hence the multiplier λ_{j-1} . This also checks the two zero endpoint values. These coefficients describe the unconditioned natural reference; conditional estimates retain each incoming word's actual list.

Proposition A.6 (Reference nonnegativity). *Every lawful original coarse history on this plateau alphabet has nonnegative mean total cost. More precisely, its cost has the form*

$$\bar{C} = D_{\text{partition}} + D_{\text{clock}} + \sum_v \omega_v \left(\lambda_{b-R_v} + \sum_i L'_i \Delta_{i,v} \right), \quad (\text{A.2.17})$$

where all terms on the right are nonnegative, $\sum_v \omega_v = 1$, and the finite list of vertices has cardinality bounded in terms of H alone.

Proof. First retain the chain of cells containing the permanent child, $A_0 \supsetneq \cdots \supsetneq A_q$, with split times $\tau_0 \leq \cdots \leq \tau_q$. Resolve every pure sibling immediately when it leaves this chain. This is lawful, since every name in a live cell is eligible until its split. If $d(t)$ is the number of unresolved boundaries, then d is nonincreasing, $d(t) \leq r_i$ in local block i , and the resulting star-spine cost is exactly

$$\bar{C}_{\text{sp}} = \bar{F}_{\text{full}}(\tau_0) + \sum_{l=1}^q \{ \bar{F}_{A_l}(\tau_l) - \bar{F}_{A_l}(\tau_{l-1}) \} \quad (\text{A.2.18})$$

$$= \int_0^T [d(t) - \kappa(t) E_{d(t)}(z)] dt. \quad (\text{A.2.19})$$

The initial value is zero by (A.2.15); no pure-cell lifetime or original rank delay remains in this comparison history.

The transport identity is $\kappa_i = \kappa_{i+1} \beta_i^{1-s_i}$, including the old/local transition, so κ increases on $[0, V_a]$. Define

$$\bar{c}_a = \frac{K_{\text{in}} + L_a c_a}{V_a} \leq c_a, \quad A(x) = \int_0^x (\bar{c}_a - \kappa(v)) dv.$$

Then $A \geq 0$, $A(0) = A(V_a) = 0$. Since $E_{d(t)}$ is nonincreasing, Stieltjes summation gives the exact correction

$$D_{\text{clock}} = \int_0^{V_a} (\bar{c}_a - \kappa(t)) E_{d(t)} dt = \sum_{0 < x < V_a} \Delta E_d(x) A(x) \geq 0.$$

Replace this first region by $L'_a = V_a$, $c'_a = \bar{c}_a$; for $i > a$ keep $L'_i = L_i$, $c'_i = c_i$. The suffix sums (A.2.16) are unchanged.

We reduce the averaged cost to endpoints. Subdivide the ordered split-time domain at $0, V_a, \dots, V_b$. The cost is affine on each closed chamber. Every vertex has all cuts at endpoints, because a maximal tied group at a nonendpoint could otherwise move in both directions while respecting order, block assignment, and deadlines. Hence the cost is a convex combination with weights ω_v of

$$\sum_i L'_i F_i(d_i(v)), \quad F_i(q) = q - c'_i E_q(z).$$

At a vertex the ranks $d_i(v)$ decrease in physical order. Apply Lemma A.4 with the reduced largest coefficient to choose R_v such that $\Delta_{i,v} = F_i(d_i(v)) - F_i(\min(r_i, R_v)) \geq 0$. The clipped part is a nonnegative multiplier, proving (A.2.17) for the star spine. It also bounds all negative proper-cell increments by the root mean and positive proper-cell increments. The root mean is nonnegative: it starts and ends at zero, and its slopes have the single positive-to-negative crossing proved below.

For a general coarse history let $d(t)$ count all unresolved original boundaries. Its eligible unresolved rank is at most $d(t)$, the remaining rank being original delay. A partition of masses $(1, \dots, 1, z)$ with unresolved rank at most d has entropy at most $E_d(z)$. Indeed merge two non-trivial cells and split off their smallest atom. Convexity of $x \log x$ preserves rank and does not

decrease entropy. Iteration leaves one nontrivial cell, and increasing any chosen mass increases its entropy; it therefore uses the largest $d + 1$ atoms, namely the child and d exterior names. Consequently

$$D_{\text{partition}} = \int_0^T \left\{ \kappa_i E_{d(t)}(z) - \sum_{A \in \Pi(t)} \text{Ent}_{v_i}(A) \right\} dt \geq 0.$$

Apply the preceding chamber argument to the history's actual activation order to obtain (A.2.17). $\square$

For a proper plateau, whose z is bounded away from one, the finite partition inequality has a strict gap unless its unresolved partition is one child-containing cell of full unresolved rank and singletons. It has no gap between different child-containing sets of the same cardinality. At $z = 1$ further equal-weight ties remain. Moreover, Proposition A.6 concerns this plateau alphabet: old cells crossing to a name outside it belong to their larger original context.

A.2.5 Discrete reserves and single crossings

Nonnegativity of the clipped costs is only the starting point. Their discrete curvature supplies a definite reserve on intervening capacities, while the secant thresholds force the drift signs to cross at most once. These two facts let us charge a weak transition to its own block even when some intervening lengths are small. For groups of capacity c_j , length $L_j \geq 0$ and coefficient $0 < \kappa_j \leq 1$, put

$$g_j(q) = q - \kappa_j E_q(z), \quad \Lambda_q = \sum_j L_j g_j(\min(c_j, q)).$$

Assume $c_j + z \leq H$ and all $\Lambda_q \geq 0$.

Lemma A.7 (Discrete curvature reserve). *Let $T_a = \sum_{c_j=a} L_j$ and $W_a = \sum_{c_j>a} \kappa_j L_j$. Then*

$$\Lambda_a + \frac{1}{2} \log H T_a \geq \frac{W_a}{2H}. \quad (\text{A.2.20})$$

If $A < R < B$, set $G_R(a) = (\min(R, a) - A)(B - \max(R, a))/(B - A)$ for $A < a < B$. Then

$$\Lambda_R + \log H \sum_{a=A+1}^{B-1} G_R(a) T_a \geq \frac{(R - A)(B - R)}{2H} \sum_{c_j \geq B} \kappa_j L_j. \quad (\text{A.2.21})$$

In particular a local group of capacity greater than a contributes at least $L_i/(2H^2)$ to the right side of (A.2.20).

Proof. With $Q_a = E_{a+1} - 2E_a + E_{a-1}$,

$$Q_a = \int_{-1}^1 \frac{1 - |t|}{z + a + t} dt \geq H^{-1},$$

and direct subtraction gives

$$D_a := 2\Lambda_a - \Lambda_{a-1} - \Lambda_{a+1} = Q_a W_a + \sum_{c_j=a} L_j [1 - \kappa_j (E_a - E_{a-1})].$$

The last bracket is at least $-\log H$. Since $2\Lambda_a \geq D_a$, this proves (A.2.20). The discrete Green identity expresses Λ_R as the linear interpolation of Λ_A, Λ_B plus $\sum_{a=A+1}^{B-1} G_R(a) D_a$. These endpoint values are nonnegative, $W_a \geq \sum_{c_j \geq B} \kappa_j L_j$, and $\sum_a G_R(a) = (R - A)(B - R)/2$, giving (A.2.21). A local coefficient is at least H^{-1} . $\square$

Lemma A.8 (Weak edges and their order). *On one plateau let $g_r(q) = q - (r + z)^{-e} E_q(z)$, $0 \leq e \leq 1$, and put $\epsilon_* = (\log 2)/(192H^2)$. If $0 \leq q < r$ and $|g_r(r) - g_r(q)| \leq \epsilon_*$, then for $q < h < r$*

$$g_h(q) - g_h(h) \geq \frac{\log 2}{32H^2}. \quad (\text{A.2.22})$$

Two such full-input edges $r_1 \rightarrow q_1$, $r_2 \rightarrow q_2$ with $r_1 > r_2$ necessarily satisfy $q_1 > q_2$.

Proof. Weakness gives $|(r + z)^{-e} L(q + z, r + z) - 1| \leq \epsilon_*/(r - q)$. Its logarithm is at least $-2\epsilon_*$. Integrating Lemma A.3 between $h + z$ and $r + z$, and using $\log(h + z) \geq \log 2$, gives

$$\log[(h + z)^{-e} L(q + z, h + z)] \geq \frac{\log 2}{24H^2} - 2\epsilon_* \geq \frac{\log 2}{32H^2}.$$

Since $g_h(q) - g_h(h) = (h - q)[(h + z)^{-e} L(q + z, h + z) - 1]$, this proves the first assertion. For the second write $\beta(q, r) = \beta_{q+z}(r + z)$. Weakness implies $|\beta(q_j, r_j) - e| \leq 2\epsilon_*/\log 2$. If $q_1 \leq q_2$, increasing the lower secant endpoint and then using Lemma A.3 gives

$$\beta(q_2, r_2) - \beta(q_1, r_1) \geq \beta(q_1, r_2) - \beta(q_1, r_1) \geq \frac{1}{24H^2},$$

whereas the two weak bounds give at most $1/(48H^2)$. $\square$

Thus surviving deepest output sets are strictly nested in physical order within one plateau.

We record explicitly the uniform incoming coefficients used in the estimates. If every natural product lies in $[\varepsilon_H, 1 - \varepsilon_H]$, then (A.2.14) and its transport identity give

$$H^{-H-1} =: \kappa_* \leq \kappa_l \leq 1, \quad \kappa_{l+1} - \kappa_l \geq d_* > 0 \quad (\text{A.2.23})$$

for consecutive earlier groups, where one may take

$$d_* = \min \left\{ \frac{\varepsilon_H}{H^2}, H^{-H-1} [1 - (1 - H^{-1})^{\varepsilon_H}] \right\}.$$

The first term also covers the last old/local transition. This follows by inserting the natural transport ratio $\kappa_l/\kappa_{l+1} = \beta_l^{1-s_l}$ and the bounds on its finite alphabet. In particular all old full-rank slopes decrease. For later quantitative comparisons one may fix

$$\delta_* = \frac{\kappa_*}{4H}, \quad \epsilon_0 = \min\{\delta_*/2, d_*\delta_*/(4H)\}, \quad C_{\text{len}} = H/\delta_*, \quad c_{\text{prefix}} = \frac{d_*}{2(1 + C_{\text{len}})}.$$

Lemma A.9 (Single crossing and length allocation). *Take any equal-product interval $[a, b]$, not necessarily maximal, with common product σ , child size $C = h_{b+1}$ and $z = \exp(G_{b+1}/\sigma)$; at the terminal boundary $C = z = 1$. Put $n_i = h_i - C$, $Z_i = (n_i + z)^\sigma$, and*

$$e_i = (n_i + z)^{\sigma-1} E_{n_i}(z) - n_i, \quad I_{\text{in}} = \sum_{l < a} L_l Z_l \prod_{l \leq v < a} \beta_v, \quad M_H = H(1 + \log H).$$

Then

$$\sum_{i=a}^b L_i e_i = n_a V_{a-1} - I_{\text{in}} \frac{E_{n_a}(z)}{n_a + z} - \lambda_{a-1} + \lambda_b. \quad (\text{A.2.24})$$

Writing $B_0 = M_H V_{a-1} + |\lambda_{a-1}| + |\lambda_b|$, the absolute value of the left side is at most B_0 . The signs of e_i have at most one negative-to-positive change. Except for at most one index, $|e_i| \geq \gamma_H$, where

$$d_H = (24H^2)^{-1}, \quad \gamma_H = 1 - 2^{-d_H/3} > 0.$$

Consequently

$$|e_i|L_i \leq B_0 + M_H \sum_{l \neq i} L_l, \quad (\text{A.2.25})$$

$$\gamma_H \sum_{i \in S} L_i \leq B_0 + M_H \sum_{i \notin S} L_i \quad (\text{A.2.26})$$

whenever S has one sign and avoids the possible small-drift index. If $R_i = V_{a-1} + \sum_{l \neq i} L_l + (|\lambda_{a-1}| + |\lambda_b|)/M_H$, then

$$|e_i| \leq M_H R_i / L_i, \quad |\sigma - \theta_{n_i}(z)| \leq \frac{H M_H R_i}{(\log 2) L_i}, \quad \theta_n(z) = 1 - \frac{\log(E_n(z)/n)}{\log(n+z)}.$$

If the small-drift index is first or last, the other lengths satisfy $\sum_{l \neq i} L_l \leq (B_0 + |e_i|L_i)/\gamma_H$.

Proof. Vary the common product, differentiating the child weight as well. The natural recursion gives $\partial_\sigma G_i = E_{n_i}(z)/(n_i + z)$ on the interval and its transported multiple before the interval. Stationarity, with the two boundary multipliers retained, is exactly (A.2.24). The transport relation $Z_i \beta_i = Z_{i+1} \beta_i^{1-s_i}$ gives $I_{\text{in}} \leq Z_a V_{a-1}$; all displayed coefficients are bounded by M_H , proving its absolute bound. Now

$$\frac{e_i}{n_i} = \exp\{(\sigma - \theta_{n_i}(z)) \log(n_i + z)\} - 1.$$

The normalized-secant estimate implies $0 < \theta_n(z) < 1$ and $\theta_{n'}(z) - \theta_n(z) \geq d_H(n' - n)$ for $n' > n$. (The secant here is $L(z, z+n) = E_n/n$.) The integer capacities decrease in physical order, so the asserted sign pattern follows; at most one threshold is within $d_H/3$ of σ . Outside that interval the last exponential has absolute difference from one at least γ_H , giving the uniform separation. Moving the remaining terms in (A.2.24) proves (A.2.25) and (A.2.26). Finally the derivative of e_i in σ is at least $(\log 2)/H$, yielding the threshold-distance bound. At an extreme small-drift index all the others have one sign, giving the final claim. $\square$

On a maximal plateau the preceding lemma and the increasing old coefficients show that the full-rank slopes $g_i(r_i) = -e_i$ change from positive to negative at most once. Their integral is zero by (A.2.15). This proves the root-mean assertion used in Proposition A.6.

A.2.6 Which weak transitions can remain

A weak slope can require a persistence factor only if both completed endpoint costs are small. We show that this forces the full eligible input and the deepest lawful output. Histories retain their original names and deadlines, and pure siblings are resolved when they leave the star spine. Let $e = 1 - \sigma \geq e_0 > 0$. Compression, deadline charges, and a lawful two-cut exchange exclude the other possibilities.

Proposition A.10 (Local weak transitions). *There are $\epsilon_H, c_H > 0$ such that a synchronous local weak edge $A \rightarrow B$, of ranks $0 \leq q < R \leq r_i$, satisfies*

$$|g_i(R) - g_i(q)| \leq \epsilon_H \implies \overline{C}_i^-, \overline{C}_i^+ \geq c_H L_i,$$

for every lawful endpoint prefix and suffix context, unless A is the full current eligible alphabet and B is its true deepest q -name suffix together with the child. There is at most one exceptional q for a fixed full rank. Across local blocks these exceptional outputs are strictly ordered as in Lemma A.8.

Proof. Start with the compression decomposition on the original groups. Reading an endpoint profile backwards makes its ranks and capacities nondecreasing, so Lemma A.4 applies. The old groups share the largest capacity and have nonincreasing coefficients in this reading order. At an update $A \rightarrow D$, $(E_D - E_A)/(D - A) > 1/\kappa_{\text{current}}$. Every processed old coefficient is at least the current one; its benchmark therefore decreases. The update threshold is also at least the natural largest-capacity threshold, preserving all local comparisons. We obtain

$$\overline{C} = \Lambda_K + \sum_j L_j \Delta_j, \quad \Delta_j = g_j(d_j) - g_j(\min(c_j, K)) \geq 0. \quad (\text{A.2.27})$$

During this algorithm the benchmark *value* at every already processed group only decreases. Splitting a group at a cut preserves this argument, since repeated equal coefficients cause no difficulty.

Set $\eta = 1/(4H^2)$, $\nu = e_0/H$, and initially take $\epsilon \leq 1/(16H^2)$. Since $g_i'' \leq -H^{-2}$, an integer strictly between q and R has value greater than both weak endpoint values by η , whereas an integer outside $[q, R]$ has value less than both by η . To see the latter, compare its secant with the secant on $[q, R]$, whose absolute slope is at most $\epsilon/(R - q)$; strong concavity separates the slopes by at least $(R - k)/(2H^2)$ on the left (and symmetrically on the right). Thus if $\min(r_i, K) \notin \{q, R\}$, either (A.2.27) is impossible or $\Delta_i \geq \eta$, which pays ηL_i .

Suppose now that this clipped value is q or R , with $q \geq 1$. Every future rank is at most q . The reverse benchmark must reach q strictly after block i : if its incoming value were $A < q$, its value on block i would be strictly below both weak endpoint values, so block i would not update it; subsequent updates only decrease this value and cannot finish at either weak endpoint. At the first reverse crossing $A \rightarrow q$, occurring at capacity h , every already processed future group of capacity $c \geq q$ whose actual rank is less than q gains at least

$$g_c(A) - g_c(q) > (q - A) \left[\left(\frac{h + z}{c + z} \right)^e - 1 \right] \geq \frac{e_0}{H} = \nu. \quad (\text{A.2.28})$$

Here $h > c$; a later forward rank cannot rise back to q . The last bound uses $h \geq c + 1$ and $\log(1 + 1/(c + z)) \geq 1/H$. It follows that

$$\overline{C} \geq \min\{\eta L_i, \nu D_B\}, \quad D_B = \sum_{j > i: c_j \geq q, |B \cap E_j| < q} L_j.$$

More strongly, in the accepted benchmark cases the full decomposition retains $\Lambda_K + \nu D_{\text{rank}}$, where D_{rank} sums all future groups of capacity at least q and actual rank below q . This includes both expired names and premature refinements.

The multiplier supplies the remaining current-block charge. If $K = q$ and the output B is wrong, its capacity- q group has rank below q . Hence $\overline{C} \geq \Lambda_q + \nu T_q$, and Lemma A.7 turns this intervening-length bound into

$$c_{\text{curv}} L_i, \quad c_{\text{curv}} = \frac{\min(1, 2\nu/\log H)}{2H^2}.$$

If $K = R < r_i$, the reverse algorithm first reaches R at block i or earlier, of capacity at least $r_i > R$. The same comparison as (A.2.28) pays ν on the already processed capacity- R group, giving the same conclusion. Incoming coefficients are no greater than the corresponding natural one, as required for this comparison.

It remains to consider a wrong output with $R = r_i = r$ and $K \geq r$. Set

$$\epsilon^* = \min\{(16H^2)^{-1}, (\log 2)/(192H^2)\}, \quad \gamma = \frac{\log 2}{48H^2}, \quad \mu = \min(\nu, \gamma), \quad M = 2H(1 + \log H).$$

At every future capacity $q < h < r$, rank q costs at least γ against the full benchmark by (A.2.22); rank below q costs ν by (A.2.28). The capacity- q group has rank below q because B is wrong. Consequently, with $T' = \sum_{a=q}^{r-1} T_a$,

$$\overline{C} \geq \Lambda_K + \mu T'. \quad (\text{A.2.29})$$

Set

$$W = \sum_{c_j \geq r} \kappa_j L_j, \quad A_H = \frac{2H^2 \log H}{\log 2}, \quad \epsilon_H = \min\{\epsilon^*, 1/(16HA_H)\}.$$

With $\ell = L(q+z, r+z)$ and $e' = \log \ell / \log(r+z)$,

$$0 < e' < 1, \quad |e' - e| \leq \frac{2\epsilon_H}{(r-q) \log 2}.$$

For an earlier/current capacity $c \geq r$, put $n = \min(c, K) \geq r$. Since $\kappa_j \leq (c+z)^{-e}$, normalized-secant monotonicity yields

$$\begin{aligned} g_j(q) - g_j(n) &\leq \kappa_j(n-q)[(n+z)^{e'} - (c+z)^e] \\ &\leq A_H \epsilon_H \kappa_j. \end{aligned}$$

The last inequality uses $n-q \leq H$ and $\partial_v(n+z)^v \leq H \log H$ on $[0, 1]$; it is one-sided and remains valid if $e' < e$. Smaller capacities between q and r contribute at most MT' , and capacities at most q contribute zero. Thus

$$\Lambda_q \leq \Lambda_K + A_H \epsilon_H W + MT'.$$

The Green estimate (A.2.21) on $[q-1, r]$, evaluated at q , gives $\Lambda_q + \log H T' \geq (r-q)W/(2H)$. Combining these inequalities and (A.2.29) shows

$$\frac{7W}{16H} \leq \Lambda_K + (M + \log H)T' \leq \max\{1, (M + \log H)/\mu\} \overline{C}.$$

Since $W \geq L_i/H$, this pays $c_{\text{wrong}} L_i$, where $c_{\text{wrong}} = 7 \min\{1, \mu/(M + \log H)\}/(16H^2)$. The case of a deadline completion, with full rank r on the preceding blocks, is included (then $K \leq r$); no lower bound on intervening side lengths has been imposed.

For a proper input $q < R < r_i$, the following time exchange pays the current block even when its output is correct. At the right endpoint b_i choose $c \in A \setminus B$ and an eligible $d \notin A$, already cut at $\tau_d \leq l_i$. Both names have deadlines at least b_i . Move only c to τ_d to form one comparison history, and move only d to b_i to form a second. Both are lawful. Between τ_d and b_i their ranks are $k(t) - 1$ and $k(t) + 1$, surrounding the original $k(t)$, and remain between zero and eligible capacity. Proposition A.6 gives nonnegative comparison costs, while exact subtraction gives

$$2\overline{C} - \overline{C}^- - \overline{C}^+ = \int_{\tau_d}^{b_i} \kappa(t)[E_{k(t)+1} - 2E_{k(t)} + E_{k(t)-1}] dt \geq L_i/H^2.$$

The right endpoint costs at least $L_i/(2H^2)$; the left differs by $L_i[g_i(R) - g_i(q)]$, so both cost at least ηL_i . This also covers $q = 0$. Taking the minimum of the three positive constants proves the stated selection for every context and hence for its finite minimum. Strict concavity makes the weak matching from a fixed full rank unique, and Lemma A.8 gives the ordering across blocks. $\square$

Proposition A.11 (Old weak transitions). *Every weak transition in an original earlier group of a successor plateau has reference endpoint costs at least $c_H L_i$. For a full-input transition both costs are at least $c_H V_{a-1}$. These assertions hold for every lawful star-history context, including $z = 1$ at a terminal plateau.*

Proof. Let the incoming plateau have full exterior rank r , $S = r + z$, and $V = V_{a-1} > 0$. Its averaged incoming coefficient is $\kappa_{\text{av}} = I_{\text{in}}/(SV)$. At the preceding strict node, write $p \geq 1$, $u < 1$, $x = S^u$, $A = E_p(x)/(p+x)$ and $\beta = x/(p+x)$. Stationarity, with $m = -\lambda_{\text{preceding}} \leq 0$, gives

$$I_{\text{in}} = \beta(pV + m)/A \leq \frac{x}{D(p,x)}V, \quad D(p,x) = E_p(x)/p.$$

The entropy integral and concavity give

$$D(p, S^u) \geq S^{u-1}(1 + \log S) + \frac{p}{2H}, \quad D(p, x) \leq 1 + \log H.$$

Indeed integrate $1 + \log(x+t)$ for $0 \leq t \leq p$, using $\log(1+t/x) \geq t/(x+p)$, and then use the tangent bound for S^{1-u} . It follows that

$$\kappa_{\text{av}}(1 + \log S) \leq 1 - \delta_0, \quad \delta_0 = \frac{p}{2H(1 + \log H)}.$$

Put $\beta_H = 1/[4H(1 + \log H)^2]$, and use a weak threshold at most $\min\{\beta_H/4, \kappa_*/(8H)\}$. For a full-input edge $r \rightarrow q$, weakness implies $\kappa_i \geq (1 - \epsilon)/(1 + \log S)$, hence $\kappa_i - \kappa_{\text{av}} \geq \beta_H$. Incoming coefficients increase, so

$$\sum_{l < i} L_l(\kappa_i - \kappa_l) \geq V(\kappa_i - \kappa_{\text{av}}).$$

Move just the names in $A \setminus B$ to time zero, leaving the future B cuts fixed. The comparison is lawful and nonnegative. If the original transition time is τ , exact subtraction is

$$\overline{C}_\tau - \overline{C}_{\text{early}} = (E_r - E_q) \sum_{l < i} L_l(\kappa_i - \kappa_l) + [g_i(r) - g_i(q)]\tau.$$

Since $E_r - E_q \geq 1$, $\tau \leq V$, both endpoints cost at least $\beta_H V/2$. If instead $q < R < r$, use the same legal two-cut exchange as in Proposition A.10. Now its second difference is at least $\kappa_* L_i/H$, giving both endpoint costs at least $\kappa_* L_i/(4H)$.

For the eventual actual-word application, an admitted endpoint row $C_v \geq c_0 \overline{C}_v - h$, together with the original named regularity estimate $E_i(t) \geq -C_H \vartheta L_i - C_H K(1 + L_i^\zeta)$, converts this to $c'_H L_i - C$. Choose ϑ first small, then use Young's inequality on L_i^ζ ($\zeta < 1$). Nonweak edges use their favored endpoint. Multiple coordinates share the same finite deterministic cost budget. $\square$

A.2.7 A terminal suffix without proper weak binary edges

At the terminal plateau the alphabet is uniform. Proper weak binary transitions can occur only in its negative-drift head. The drift balance and separated thresholds therefore leave a deterministic suffix of substantial length with only ordinary constraints. This suffix will supply the unread root reservoir.

Proposition A.12. *In a terminal maximal plateau with positive local lengths there is a deterministic local suffix whose total length is at least $c_H T$, and which has no proper full-input weak binary edge. Its earliest largest original block has length at least $c_H T/H$. The suffix and the tie rule depend only on the natural profile.*

Proof. Let its natural alphabet sizes be $H \geq h_1 > \dots > h_m \geq 2$, $r_i = h_i - 1$, $\kappa_i = h_i^{\sigma-1}$, and $e_i = h_i^\sigma \log h_i - (h_i - 1) = -g_i(r_i)$. The exact terminal balance gives

$$D := \sum_i L_i e_i = nV_{\text{old}} - \frac{E_n(1)}{h_1} I_{\text{in}} \geq b_H V_{\text{old}}, \quad b_H = \frac{1}{2H^2(1 + \log H)^2}.$$

For completeness, when $V_{\text{old}} > 0$ insert $I_{\text{in}} = \beta(pV_{\text{old}} + m)/A$, $m \leq 0$, from the preceding proof. The inequality $E_p(x)/p \geq 1 + \log x + p/[2(x+p)]$ gives a surplus of at least $np/[2H^2(1 + \log H)^2]$ times V_{old} . When there is no incoming block, $D = 0$.

If $r \rightarrow q$ is proper ($1 \leq q < r$) and $|g_i(r) - g_i(q)| \leq \epsilon_w \leq 1/(4H^3)$, then $-g_i'' \geq H^{-2}$ and $g_i(0) = 0$ imply

$$g_i(q) - \frac{q}{r}g_i(r) \geq \frac{q(r-q)}{2H^2}.$$

Writing $v = g_i(r) - g_i(q)$ shows $g_i(r) \geq qr/(2H^2) + rv/(r-q) \geq 1/(4H^2) =: c_w$. Thus every such weak block has $e_i \leq -c_w$. The thresholds α_h solving $e_h(\sigma) = 0$ are strictly increasing in h , by Lemma A.9. Choose disjoint fixed closed neighborhoods $|\sigma - \alpha_h| \leq \tau$ in which $|e_h| \leq c_w/2$; this is possible since $|\partial_\sigma e_h| \leq H(\log H)^2$. Outside these neighborhoods the nonzero drifts have a uniform absolute lower bound $g_0 > 0$.

If a present threshold is near σ , begin the suffix there; otherwise begin at the first positive drift, which exists since $D \geq 0$. The head has negative drift at most $-g_0$, while the suffix has at most one small drift followed by positive drifts and therefore contains no proper weak binary block. Its drift sum obeys

$$D_{\text{tail}} \geq b_H V_{\text{old}} + g_0 L_{\text{head}} \geq 0.$$

Since $|e_i| \leq H(1 + \log H)$, its length is a fixed fraction of $V_{\text{old}} + L_{\text{head}} + L_{\text{tail}} = T$. The largest of its at most H original blocks has length at least $c_H T/H$; choosing the earliest one resolves ties deterministically. $\square$

This suffix supplies an unread reservoir for the one uncolored root count. It does not impose a within-block terminal path event.

A.2.8 Physical prefix guards

The count box can adjust exterior labels, but full-group outputs remain fixed. Before choosing fresh names, we must therefore control the signed remainder at every original endpoint. Clipped positive-prefix costs provide the needed linear reserve, and the two exact expressions for the remainder let us use the shorter of its prefix and tail. Keep the original groups, including earlier groups, indexed by $g = 1, \dots, m$, $m \leq H$, with capacities r_g and transported coefficients κ_g . Zero-length original groups may retain their indices; the coefficients are those of the same fixed complete-chain optimizer. For $\mu_g = \kappa_g(r_g + z)L_g$ and $\beta_g = z/(r_g + z)$, fix actual total and child counts in the own-group windows

$$|n_g - \mu_g| + |k_g - \beta_g \mu_g| \leq B\sqrt{1 + \mu_g}. \quad (\text{A.2.30})$$

The child positions and the incoming word may also be fixed. Define

$$\begin{aligned} w_g &= \log(r_g + z)n_g - \log z k_g - r_g L_g, \\ \zeta_g &= \log(r_g + z)(n_g - \mu_g) - \log z(k_g - \beta_g \mu_g), \\ Y &= \sum_g w_g = \sum_g \zeta_g, \quad b_0 = a_0 - Y, \quad Z_l = \sum_{g>l} \zeta_g. \end{aligned} \quad (\text{A.2.31})$$

The equality for Y is reference balance, not conditional expectation. The entrance a_0 is the actual entrance, with all its dependence on the incoming word retained.

Lemma A.13 (Clipped positive prefixes). *Let m_l be the least reference cost of a lawful endpoint rank profile positive through group l and zero thereafter; set $m_0 = 0$. Then*

$$m_l = \min_{1 \leq R \leq r_a} \sum_{g \leq l} L_g g_g(\min(r_g, R)) \geq c_H \min(V_l, T - V_l), \quad m_m = 0. \quad (\text{A.2.32})$$

If $a_0, b_0 \geq -h_0$, then, for each fixed $\theta > 0$,

$$b_0 + Z_l + \theta m_l \geq -h_{\text{bin}} \quad (0 \leq l \leq m), \quad (\text{A.2.33})$$

where h_{bin} depends only on H, B, h_0, θ .

Proof. Apply original-group reverse compression to the positive prefix, starting the benchmark at one instead of zero. Every update still preserves the already processed comparisons: for earlier groups its threshold is $1/\kappa_{\text{current}}$, with larger coefficients already processed, and for local groups it is the same secant comparison as before. Thus every positive rank profile dominates some clipped profile. Conversely, retain the deepest R eligible names, delete each on expiry, and cut the rest at V_l . This realizes the clipped prefix lawfully and proves the equality in (A.2.32). Each candidate is nonnegative by the full-history comparison; no stationarity for a truncated system is invoked.

For fixed R put $k_g = \min(r_g, R)$ and

$$f_g = g_g(k_g) = k_g(1 - S_g^{t_g}), \quad t_g = \frac{\log(\kappa_g E_{k_g}(z)/k_g)}{\log S_g}, \quad S_g = r_g + z.$$

Across earlier groups and the first local transition, $t_{g+1} - t_g \geq d_*/\log H$. At local capacity h write

$$t_g = \sigma - T_h(R), \quad T_h(R) = 1 - \frac{\log(E_{\min(h, R)}(z)/\min(h, R))}{\log(h + z)}.$$

For $h \leq R$, Lemma A.3 gives $\partial_h T_h(R) \geq 1/(24H^2)$; for $h \geq R$,

$$\partial_h T_h(R) = \frac{\log(E_R/R)}{(h + z) \log^2(h + z)} \geq \frac{\log(2 \log 2)}{H \log^2 H}.$$

The function is continuous at R , and local capacities fall by positive integers. Therefore every original transition raises t_g by at least

$$d_0 = \min \left\{ \frac{d_*}{\log H}, \frac{1}{24H^2}, \frac{\log(2 \log 2)}{H \log^2 H} \right\} > 0.$$

Set $c_H = 1 - 2^{-d_0/3}$. If $f_g \leq c_H$, then $t_g \geq -d_0/3$; hence every later $f_v \leq 1 - 2^{2d_0/3} \leq -c_H$. Thus either all prefix coefficients exceed c_H , or all tail coefficients are at most $-c_H$. In the first case its cost is at least $c_H V_l$; in the second the retained multiplier gives

$$\sum_{g \leq l} L_g f_g = \Lambda_R - \sum_{g > l} L_g f_g \geq c_H (T - V_l).$$

Minimizing proves (A.2.32). At $l = m$, $R = r_a$ gives zero.

Own-group centrality implies for every consecutive group set I

$$\left| \sum_{g \in I} \zeta_g \right| \leq C_B \sqrt{1 + \sum_{g \in I} L_g}.$$

Use both exact expressions for the same remainder,

$$b_0 + Z_l = b_0 + \sum_{g>l} \zeta_g = a_0 - \sum_{g\leq l} \zeta_g.$$

Their lower bounds give $-h_0 - C_B\sqrt{1+D_l}$, where $D_l = \min(V_l, T - V_l)$. The single inequality $C_B\sqrt{D_l} \leq \theta c_H D_l + C_B^2/(4\theta c_H)$ proves (A.2.33), with $h_{\text{bin}} = h_0 + C_B + C_B^2/(4\theta c_H)$. $\square$

Only original physical endpoints occur in this lemma. Splitting a weak block artificially would destroy the strict increase of t_g . Its actual full-stop row is

$$C_{\text{stop}}(l) = b_0 + \sum_{g>l} w_g = a_0 - \sum_{g\leq l} w_g = C_{\text{stop}}^{\text{nat}}(l) + b_0 + Z_l, \quad C_{\text{stop}}^{\text{nat}}(l) = \sum_{g\leq l} L_g g_g(r_g). \quad (\text{A.2.34})$$

In particular the two extreme rows require the actual bounds on a_0, b_0 ; changing exterior names cannot alter them.

It is useful to see why a carry bound alone is insufficient here. At an internal local endpoint j , let n_0 be the incoming count, C_j^{route} the count routed to the next child, and K_j^{past} the permanent-child count in all old and local groups through j . Then

$$\begin{aligned} T_j &= \log S_a n_0 + \sum_{i=a}^j \log S_i n_i - \log S_{j+1} C_j^{\text{route}} - \sum_{l=a}^j V_l, \\ W_j &= W_{\text{in}} + T_j - \sum_{l=a}^j y_l, \\ H_j &= \log S_{j+1} C_j^{\text{route}} - \log z K_j^{\text{past}} - r_{j+1} V_j. \end{aligned}$$

Direct cancellation yields

$$\sum_{g\leq j} w_g = T_j + H_j, \quad C_{\text{stop}}(j) = a_0 + W_{\text{in}} - W_j - \sum_{l=a}^j y_l - H_j. \quad (\text{A.2.35})$$

Fresh naming changes T_j, H_j oppositely. The old routed-child risk in H_j is not a later local wall. Lemma A.13 absorbs this remainder directly, using centrality of each earlier physical group; an aggregate old window would not suffice at an old internal endpoint.

A.2.9 A common count box for every endpoint

We now move from reference costs to actual named counts. Biasing fresh exterior counts toward the shallowest eligible name gives a square-root gain on every proper row that omits it. Rows that retain it pay a reference length charge through a two-cut exchange. A single displaced box thus controls the finite family of endpoint histories simultaneously.

Proposition A.14 (Displaced exterior counts). *Fix all original clocks, total and permanent-child counts, child binary words, and old prerefinement data, with (A.2.30). Assume the natural full balance and nonnegativity of every completed named-star reference history. Conditional on this data, fresh exterior labels in group g are independent uniform labels from its nested eligible set D_g , of size r_g . There is an event of conditional probability at least $c_{\text{box}} > 0$ on which, simultaneously for every completed endpoint history v ,*

$$C_v \geq (1 - \theta)\bar{C}_v + b_0 + Z_{l(v)} + T_{\text{good},v} - C_\theta. \quad (\text{A.2.36})$$

Here $0 < \theta < 1$ is fixed, $l(v)$ is the last positive-rank group, and $T_{\text{good},v}$ is a nonnegative sum of square-root exterior counts defined in the proof. If the guards $b_0 + Z_l + \theta' m_l \geq -h_{\text{bin}}$ hold, the constants can be chosen so that the same event satisfies

$$C_v \geq c_0 \bar{C}_v - h_{\text{bin}} - C \quad \text{for every } v \quad (\text{A.2.37})$$

for any fixed $c_0 < 1$. For every nonnegative prerenement-measurable weight F_0 , $\mathbb{E}(F_0 \mathbf{1}_{\text{box}}) \geq c_{\text{box}} \mathbb{E} F_0$.

Proof. Write $N_g = n_g - k_g$ for the exterior count and n_{gc} for its count of name c . Let c_g be the shallowest eligible name. A proper row of rank $0 < q < r_g$ is called good on this group if it omits c_g , and bad if it retains c_g . A bad row has a missing deeper name d with later deadline. Its cut is no later than the left endpoint of the group, whereas c_g is retained through the right endpoint. Move the cut of c_g to that of d , and, in a second comparison, move that of d to the cut of c_g . Both histories are lawful. The exact second difference is at least $\kappa_g L_g / H$, so reference nonnegativity gives $\bar{C}_v \geq \kappa_g L_g / (2H)$ for each bad group. There are at most H such groups, hence

$$\bar{C}_v \geq \frac{\kappa_*}{2H^2} \sum_{g \text{ bad}} L_g. \quad (\text{A.2.38})$$

For a proper rank- q row retaining A_g , exact expansion of its entropy costs gives

$$C_v = b_0 + \bar{C}_v + Z_{l(v)} + \sum_{g \text{ proper}} e_{vg} - \sum_{g \text{ proper}} \log(q + z) \sum_{c \in A_g} (n_{gc} - N_g / r_g), \quad (\text{A.2.39})$$

where

$$e_{vg} = \left[\log S_g - \frac{q}{r_g} \log(q + z) \right] (n_g - \mu_g) - \left(1 - \frac{q}{r_g} \right) \log(q + z) (k_g - \beta_g \mu_g).$$

There is no fresh-name error for a full or empty row. By centrality, $|e_{vg}| \leq C_B(1 + \sqrt{N_g})$ and $N_g \leq C_B(1 + L_g)$.

Choose a fixed small $\epsilon > 0$ and then a large fixed K . For sufficiently large N_g , require every nonshallow count to lie in the interval

$$n_{gc} = N_g / r_g - K \sqrt{N_g} + [-\epsilon, \epsilon] \sqrt{N_g} \quad (c \neq c_g), \quad (\text{A.2.40})$$

with the shallow count determined by the total. A good proper row then gains $qK\sqrt{N_g} + O_H(\epsilon\sqrt{N_g})$, while a bad row loses at most $C_H K \sqrt{N_g}$. Choosing K after B makes the gain pay the good binary errors and leaves $\sqrt{N_g}$ on every sufficiently large good group. Define their sum to be $T_{\text{good},v}$. For large N_g these boxes are interior, contain $\gg_H N_g^{(r_g-1)/2}$ lattice points, and each point has multinomial mass $\asymp_H N_g^{-(r_g-1)/2}$ by Stirling's formula with all coordinates in fixed central windows. Their probability is therefore bounded below. On bounded groups choose a supported balanced type; its minimum positive probability is a fixed constant. A group with $r_g = 1$ has no exterior choice. The finitely many independent group boxes thus have conditional probability at least c_{box} .

The total remaining bad error is at most $C_K \sum_{g \text{ bad}} \sqrt{1 + L_g}$. Apply Young's inequality once and use (A.2.38) to bound it by $\theta \bar{C}_v + C_\theta$. Substitution in (A.2.39) proves (A.2.36) simultaneously for the finite family. Taking $\theta = \theta' = (1 - c_0)/2$ and using $\bar{C}_v \geq m_{l(v)}$ yields (A.2.37). More economically, the guard can use $\min_{v:l(v)=l} (\theta \bar{C}_v + T_{\text{good},v})$, which is at least θm_l . Finally the probability bound holds for every fixed unread environment; conditioning proves the weighted assertion. It applies to the whole type event before any fractional power. The input likelihood, fixed by the original totals and binary child words, is unchanged. $\square$

A named regularity event whose conditional failure probability is at most $c_{\text{box}}/2$ for every fixed child word may be intersected with this box. The weighted assertion does not permit a weight that reads the fresh exterior labels.

Lemma A.15 (Monotone transport to the box). *Fix the same original clocks and child binary words. Let the source be any subset of exterior count types with $|n_{gc} - N_g/r_g| \leq K_{\text{src}}\sqrt{1 + N_g}$. There is a target distribution of constant conditional mass in a fixed wider central window, satisfying the endpoint box, and a coupling from each source type to each target type that changes only nonshallow names into the shallowest name. For every nonnegative functional increasing under these replacements,*

$$\mathbb{E}[\mathbf{1}_{\text{target}}\Phi] \geq c_{\text{tar}} \mathbb{E}[\mathbf{1}_{\text{source}}\Phi]. \quad (\text{A.2.41})$$

Proof. Choose (A.2.40) with $K > \sqrt{2}K_{\text{src}} + \epsilon$, enlarged as required for the box. Every target nonshallow count m_c is then at most its source count n_c . For bounded $N < N_0$ choose the all-shallow target, of probability at least H^{-N_0} ; this lies in a fixed wider central window. The large-count Stirling estimate proves target mass at least $c_{\text{tar}} > 0$. Given a uniform source word of type n , retain a uniformly selected m_c of the n_c occurrences of each nonshallow c , and change the rest to the shallow name 0. The resulting word is uniform of type m . Indeed a fixed target word has $m_0!/[n_0! \prod_{c \neq 0} (n_c - m_c)!]$ source preimages; multiplying by the source word probability $n_0! \prod_{c \neq 0} n_c!/N!$ and the retention probability $\prod_{c \neq 0} \binom{n_c}{m_c}^{-1}$ gives $m_0! \prod_{c \neq 0} m_c!/N!$, its required uniform probability. All original slots, times and child marks stay fixed. Monotonicity therefore gives $\mathbb{E}(\Phi \mid m) \geq \mathbb{E}(\Phi \mid n)$ for every source and target pair. Average over the normalized source acceptance, then over target types and use source mass at most one. This proves (A.2.41); truncation handles unbounded Φ . The target endpoint support follows from Proposition A.14 and the already fixed two-entrance guards. No within-block path or filtered regularity assertion is part of this transport. $\square$

A.2.10 The strict-exit construction on original counts

Successive output constraints look coupled because a transfer changes two neighboring levels. Their cumulative carries have a simpler law: each designated two-name split changes only its own carry. After coarsening those pairs, the unread split counts are independent binomials, and summing their entire bands gives the product of width factors.

Proposition A.16 (Diagonal strict exits). *Let maximal plateaux be indexed in decreasing products $\sigma_1 > \dots > \sigma_M$, with $u_j = \sigma_{j+1}/\sigma_j$ and $d_j = 1/u_j - 1$. Let T_j be the local length of proper plateau j , and put $V_j = \sum_{a \leq j} T_a$. Fix $C_y \geq 0$ and target numbers $0 \leq y_j \leq C_y \log(2 + T_j)$ for $j < M$, held fixed throughout the count selection and transport. For grouped plateau outputs Z_j , define $\eta_j = \sum_{a \leq j} (\sigma_a/\sigma_j)(Z_a - y_a)$ for $1 \leq j < M$. Suppose $V_j \leq C_H T_j$ and $T_j \geq T_0(H, C_y)$ for the proper plateaux in question. Conditional on admitted original uncolored count windows, there is a count-only acceptance, in fixed full-leaf central windows, of mass*

$$\asymp_{H, C_y} \prod_{j < M} \frac{R_j}{\sqrt{T_j}}, \quad R_j = \max\{R_0, \min(\sqrt{T_j}, d_j^{-1})\}, \quad (\text{A.2.42})$$

on which all strict carries satisfy $aR_j \leq \eta_j \leq bR_j$ for fixed $0 < a < b$. The constants and central windows depend only on H, C_y . This acceptance can be transported through the proper endpoint boxes, with a further factor depending only on H, C_y and without changing the carries.

Proof. Choose in each proper plateau an original block B_j of length at least T_j/H . In it choose its current shallow name t_j and the first shallow fine leaf c_j of the immediate child. Set $A_j = \log S_{a_j}$ and $b_j = u_j A_{j+1} = \log z_j$. These are uniformly bounded above and away from zero. The two original probabilities obey $p_{c_j}/p_{t_j} = S_{\text{child}}^{u_j-1}$ and both lie in a fixed interior probability interval. Coarsen just the pair $\{c_j, t_j\}$, fixing its total count m_j , all coarsened words and all other leaf words. Its unread split count K_j , counting t_j , has the original binomial law $\text{Bin}(m_j, p_j)$, independently for different j .

For $2 \leq j < M$, these carries satisfy $\eta_j = Z_j - y_j + \eta_{j-1}/u_{j-1}$. Changing one c_j to t_j raises Z_j by b_j , lowers Z_{j+1} by A_{j+1} and changes no other grouped output. The expansion

$$\eta_l = \sum_{a \leq l} \frac{\sigma_a}{\sigma_l} (Z_a - y_a)$$

therefore cancels the two effects for $l > j$, giving the simultaneous exact diagonalization

$$\eta_j = h_j + b_j K_j. \quad (\text{A.2.43})$$

There is a fixed central coarsening event of constant probability on which all coarsened categories and all other original-block leaf counts lie in their own-scale central windows. It leaves all K_j unread. On it $m_j \asymp_H T_j$ and $|h_j + b_j m_j p_j| \leq C_{H, C_y} \sqrt{T_j}$. To verify the last estimate, the natural grouped proper output has mean zero at a strict boundary; the sum of finitely many own-group errors through j is $O_H(\sqrt{1 + V_j})$. Since $T_a \leq V_j \leq C_H T_j$ for $a \leq j$, the weighted sum of targets through j is $O_{H, C_y}(\log(2 + T_j))$ and is absorbed by the displayed square-root bound.

Choose R_0 large enough that $(b - a)R_0 > 2 \sup_j b_j$, then $T_0 = T_0(H, C_y) \geq R_0^2$ large enough for all resulting binomial windows to be interior. For each coarsening, the condition $aR_j \leq h_j + b_j K_j \leq bR_j$ is an interval of $\asymp_{H, C_y} R_j$ integers within $C_{H, C_y} \sqrt{m_j}$ of its mean. Each such atom has mass $\asymp_{H, C_y} m_j^{-1/2}$, so independence of the unread split counts proves (A.2.42). The full fine-leaf counts still lie in fixed enlarged central windows. The event is an explicit polytope of original counts with the full band mass just computed.

Its actual entrances are $a_{0j} = y_j + \eta_j \geq 0$ for $j < M$, with $b_{01} = 0$ when $M > 1$ and $b_{0j} = \eta_{j-1}/u_{j-1} \geq 0$ for $2 \leq j < M$. Apply Lemma A.15 successively to the finitely many nonterminal plateaux. Each replacement preserves that plateau's permanent-child word, all preceding coarse words and the finer child word, hence every Z_j and η_j . The source acceptance remains inside the linear integral at each step. The resulting subprobability measure has mass reduced only by a constant depending on H, C_y ; its bounded density can be normalized to an acceptance in $[0, 1]$. This is an unfiltered count transport. A later path condition is imposed only after this count source has been built. Finally $0 \leq d_j \eta_j \leq C_{H, C_y}$ on the bands, so the exact full-word likelihood ratios in (A.2.9) are bounded above and below; the local products themselves have not been replaced or perturbed. $\square$

A weight measurable with respect to the coarsened words can be retained in the conditional diagonal calculation. A weight reading the unread K_j or their paths cannot be inserted at that stage.

A.2.11 Local endpoint representatives and terminal admission

The common count box gives scale bounds for many outside contexts. For a local bridge all those contexts have the same endpoint difference, so one maximum formula yields a smaller compatible pair. Terminal admission requires a further distinction: private names may be averaged

before prefix acceptance, but accepted prefix labels must remain fixed when the unread suffix is completed.

Lemma A.17 (A local representative of a common endpoint pair). *Suppose the actual minima of the two endpoints on original block i satisfy, throughout an admitted whole-count support,*

$$aX_i \leq x_i \leq bX_i, \quad aY_i \leq y_i \leq bY_i, \quad y_i - x_i = \Delta_i(K_i),$$

where $0 < a \leq b$ are fixed, $X_i, Y_i > 0$ are deterministic scores, and K_i is the original local binary count. Then

$$x_i^-(K_i) = \max\{aX_i, aY_i - \Delta_i(K_i)\}, \quad y_i^-(K_i) = x_i^-(K_i) + \Delta_i(K_i) \quad (\text{A.2.44})$$

satisfy the same scale bounds from below and are no greater than the actual minima coordinatewise. They have exactly the original endpoint difference and read only the local count (and its fixed original total).

Proof. The actual x_i is at least both entries in the first maximum; adding its fixed difference proves the right-end comparison. The second maximum is equivalently $\max\{aX_i + \Delta_i(K_i), aY_i\}$, which proves its lower bound. For a local edge the exact difference is

$$\Delta_i(K_i) = (r_i - q_i)L_i - n_i \log S_i + K_i \log B_i.$$

Every outside context uses this same difference. Its centered empirical bridge is also the same original binary bridge $E_i(t)$. Thus the chord between (A.2.44) lies below every actual endpoint chord. If $\theta > 0$ is the fixed allocation fraction, the sufficient wall $\theta \ell_i^-(t) + E_i(t) \geq g d_i(t)^\gamma$ (or its fixed intercept version) implies all corresponding allocated walls on the admitted count support. Choose the fixed shift in the scores so that their lower bounds exceed the scalar reserve threshold. The scalar all-reserve theorem then evaluates this local pair on the same scale, using its required interior type and compact original clock. For bounded blocks use the admitted finite support. A deterministic-mark bridge uses the ordinary scalar theorem rather than a zero-variance substitution in the binary theorem. $\square$

This construction avoids choosing two independent corners of an endpoint rectangle. The path event alone does not assert the count support on which the comparisons hold.

Lemma A.18 (Forgetting private terminal names). *Let π be the full-leaf word law after fixing original uncolored totals. Write C for the complete coarse word retaining all proper names and treating the entire last plateau as one compound symbol, and V for its private refinement. Then*

$$\pi(dC, dV) = \pi_c(dC) \pi_t(dV \mid C).$$

For a count-only acceptance $0 \leq A(C, V) \leq 1$ of mass W , its conditional average $A_c(C) = \mathbb{E}_{\pi_t}[A(C, V) \mid C]$ has the same mass. If A is invariant under within-block permutations, A_c depends only on coarse counts. Every condition depending only on C that holds on the source still holds where $A_c > 0$.

Proof. Conditional terminal labels are uniform among eligible names at each original slot, independently between slots. Applying the same within-block permutation to C and V preserves this conditional law, proving the count assertion. The mass identity is conditional expectation. If a coarse condition fails at C , the original acceptance vanishes for almost every refinement there, so

its average also vanishes. The new measure is $A_c(C)\pi(dC, dV)$, leaving V at its conditional law. Thus proper endpoints, strict exits, proper canonical binary types, own-group coarse centrality and proper coarse paths survive. An old test or arbitrary weight of the private V need not survive. After a count transport this averaging is done after its bounded-density normalization, and introduces no further mass loss or fractional power. $\square$

Proposition A.19 (Prefix source before the unread suffix). *Condition on admitted own-scale original uncolored totals before the terminal suffix of Proposition A.12. There is a count-only prefix acceptance $0 \leq A_{\text{pre}} \leq 1$, independent of the uncolored counts and paths in that suffix, of mass*

$$W \asymp_H \prod_{j < M} w_j, \quad w_j = R_j / \sqrt{T_j}, \quad (\text{A.2.45})$$

with all proper strict-exit bands and endpoint boxes, all admitted proper binary types, and own-group terminal-prefix central and exterior count windows. Its original full-leaf type law is retained. The constants can be fixed uniformly for every suffix count vector in the central lower construction of the root theorem.

Proof. The diagonal count construction and each nonterminal-plateau count transport read only blocks through their nonterminal plateau. In particular they read no terminal local block and can be formed with zero common target before any suffix count. Their mass is (A.2.45). Apply Lemma A.18 to forget the private terminal names; this preserves the mass and the proper coarse conditions. The absence of suffix *uncolored* count dependence follows from the preceding construction, not merely from this marginalization. Refine the terminal lists already present in the prefix with their conditional uniform law. Require permanent-0 counts in fixed sufficiently wide own-group central windows, and then impose the displaced exterior count boxes on these prefix groups. Binomial centrality and Proposition A.14 give a fixed conditional factor on each of at most H groups. These restrictions do not affect any proper coarse condition, and read no newly sampled clock. The resulting acceptance has the asserted mass. All constants are chosen to cover the uniform central windows of the root construction, so this construction works for each of its later admitted root count vectors. Within every exact full-leaf type the bridge words retain their original uniform permutations. $\square$

Proposition A.20 (Terminal counts after the one root atom). *Use the zero common-target prefix source above and the uncolored Euler root band $-r_*W_0 \leq X_1 \leq 0$. After the one suffix reservoir count has been selected by the root theorem, the terminal endpoint box can be imposed with a conditional probability bounded below uniformly for every admitted coarse prefix and root count vector. It changes neither that root count nor any proper coarse word. With $a_0 = 0$ its terminal entrance is*

$$b_0 = \eta_{\text{prev}}/u_{\text{prev}} - X_1/r_* \geq 0, \quad 1 + b_0 \asymp_H 1 + R_{\text{prev}}/u_{\text{prev}}. \quad (\text{A.2.46})$$

For an absent proper predecessor set $R_{\text{prev}} = 0$. The admitted rows satisfy

$$C_v \geq b_0 + \zeta_{\text{tail},v} + \frac{3}{4}\overline{C}_v + G_v - C_{\text{box}}, \quad (\text{A.2.47})$$

where G_v is the good-group square-root gain.

Proof. The exact carry relation for the whole terminal plateau is $Z_{\text{term}} = X_1/r_* - \eta_{\text{prev}}/u_{\text{prev}}$. Thus $a_0 = 0$, $b_0 = -Z_{\text{term}}$ give (A.2.46); the root error is bounded and the proper exit is in its positive band. Condition now on the coarse prefix and every original uncolored count, including

the already chosen root reservoir count. Each earlier group's actual entering terminal total is in its own-scale coarse window. Refine terminal names by the conditional law of Lemma A.18. Require the permanent-name 0 count in a sufficiently wide central interval. Binomial Chebyshev bounds give a fixed simultaneous probability, using the supported finite convention on bounded groups. With these counts and positions fixed, impose Proposition A.14. Its full-stop guards follow deterministically from Lemma A.13 and the two nonnegative entrances. For terminal $z = 1$, each full-group logarithmic sum is $\log(r_g + 1)n_g$, independent of all refinements. Hence neither Z_{term} , the uncolored root, nor b_0 changes. Proper binary words are coarse and remain fixed. The stronger row (A.2.36), with $\theta = 1/4$, is exactly (A.2.47). This is a count admission; terminal within-block paths are imposed subsequently. $\square$

A.2.12 Actual endpoint scores and their intrinsic normalization

The raw endpoint cost includes the exit inherited from the preceding plateau. A local wall uses the intrinsic cost obtained after subtracting that exit. For late endpoints the subtraction cancels in the exact tail formula; for early endpoints the positive prefix reserve pays it. This distinction determines the two-sided scales of the local bridge. For a named endpoint star template v , let $A_{v,g}$ be its retained exterior names in group g , $q_g = |A_{v,g}|$, and $l = l(v)$ its last positive-rank group. Its reference cost is

$$\bar{C}_v = \sum_g L_g g_g(q_g).$$

Let $\mathcal{B}_v$ and $\mathcal{G}_v$ be respectively its bad and good proper groups, as in Proposition A.14, and put

$$L_{\text{bad}} = \sum_{g \in \mathcal{B}_v} L_g, \quad L_{\text{zero}} = T - V_l, \quad S_{\text{good}}(v) = \sum_{g \in \mathcal{G}_v} \sqrt{L_g}.$$

All other groups are full and incur no exposed-name discrepancy. The reference estimates and the displaced box give, simultaneously,

$$\begin{aligned} \bar{C}_v &\geq a_H L_{\text{bad}}, & \bar{C}_v &\geq b_H \min(V_l, T - V_l), \\ C_v &\geq b_0 + Z_l + \frac{3}{4} \bar{C}_v + G_v - C_{\text{box}}, & G_v &\geq c_H S_{\text{good}}(v) - C_H. \end{aligned} \quad (\text{A.2.48})$$

The last comparison follows from own-scale centrality: on a large group $N_g \geq c_H L_g$, whereas bounded N_g forces bounded L_g .

Lemma A.21 (Early and late corner payment). *On this single box, every named endpoint star corner obeys*

$$V_l \leq T/2 : \quad C_v \geq a_0 + c_H V_l - C_H, \quad (\text{A.2.49})$$

$$V_l \geq T/2 : \quad C_v \geq b_0 + c_H L_{\text{zero}} + c_H L_{\text{bad}} + c_H S_{\text{good}}(v) - C_H. \quad (\text{A.2.50})$$

The all-zero and complete-deadline corners have costs exactly a_0 and b_0 , respectively.

Proof. In the early case use $b_0 + Z_l = a_0 - \sum_{g \leq l} \zeta_g$. Its error is $O(\sqrt{1 + V_l})$ and is absorbed once by the positive prefix charge in (A.2.48), retaining a fixed fraction of V_l . In the late case use $b_0 + Z_l = b_0 + \sum_{g > l} \zeta_g$; only the zero-rank tail has a square-root error. Split the $3\bar{C}_v/4$ budget into two equal pieces for L_{bad} and L_{zero} , absorb that error with half the latter piece, and retain the good-group gain. The extreme identities follow directly from (A.2.34). The same proof, with a fixed constant, includes $T = 0$. $\square$

Here is the exact cancellation behind two-sided scores. With $N_g(A)$ the exterior count in A , write

$$H_g = \log S_g n_g - \log z k_g, \quad H_g(A) = \log(q_g + z)[k_g + N_g(A)] - \log z k_g.$$

For an empty group $H_g(A) = 0$; for a full group it is H_g exactly. The notation $\overline{H}_g$ below means the deterministic natural reference potential, including unconditioned old routing. Set

$$\nu_{vg} = [H_g - H_g(A_{v,g})] - [\overline{H}_g - \overline{H}_g(A_{v,g})].$$

Then $\nu_{vg} = 0$ identically on every full group, and direct expansion of $C_v = a_0 + \sum_{g \leq l} [q_g L_g - H_g(A_{v,g})]$ gives

$$C_v = a_0 - \sum_{g \leq l} \zeta_g + \overline{C}_v + \sum_{g \text{ proper}} \nu_{vg} = b_0 + \sum_{g > l} \zeta_g + \overline{C}_v + \sum_{g \text{ proper}} \nu_{vg}. \quad (\text{A.2.51})$$

Every adjusted named count is within $C_H \sqrt{1 + L_g}$ of its own natural group mean. Therefore

$$|\nu_{vg}| \leq C_H \sqrt{1 + L_g}, \quad \left| \sum_{g \in J} \zeta_g \right| \leq C_H \sqrt{1 + \sum_{g \in J} L_g} \quad (\text{A.2.52})$$

for consecutive J . In the late formula the discrepancies of large full groups have canceled, rather than contributing a spurious $\sqrt{T}$ error.

Lemma A.22 (Raw finite scores). *Suppose $a_0, b_0 \geq 0$ lie in fixed comparable intervals of deterministic scales $A_{\text{in}}, B_{\text{in}}$, respectively, allowing scale zero for an identically zero entrance. For a fixed sufficiently large h , set*

$$S_v^{\text{raw}} = \begin{cases} h + A_{\text{in}} + \overline{C}_v, & V_l \leq T/2, \\ h + B_{\text{in}} + \overline{C}_v + S_{\text{good}}(v), & V_l > T/2. \end{cases}$$

Then $c_H S_v^{\text{raw}} \leq C_v + h \leq C_H S_v^{\text{raw}}$ simultaneously on every admitted count type.

Proof. For an early corner all proper groups are in the prefix, and (A.2.51)–(A.2.52) give $C_v \leq a_0 + \overline{C}_v + C_H \sqrt{1 + V_l}$. Since $V_l \ll_H \overline{C}_v$, this is $a_0 + C'_H \overline{C}_v + C'_H$. For the lower bound use the first expression for $b_0 + Z_l$ in (A.2.48), discard $G_v \geq 0$, and absorb its square root with at most $\overline{C}_v/4$ to get

$$C_v \geq a_0 + \frac{1}{2} \overline{C}_v - C_H. \quad (\text{A.2.53})$$

For a late corner, the stronger box row similarly gives

$$C_v \geq b_0 + \frac{1}{2} \overline{C}_v + c_H S_{\text{good}}(v) - C_H. \quad (\text{A.2.54})$$

The tail formula gives an upper bound by $b_0 + \overline{C}_v$ plus the square roots of the zero tail, bad groups and good groups. The first two are bounded by $C_H(1 + \overline{C}_v)$; the last are $C_H S_{\text{good}} + C_H$. A fixed large h completes both comparisons, including the extreme rows. $\square$

These raw scores describe C_v . They are not the local reserve scores, which must account for the normalization in the full cost.

For later use we spell out how the raw payment interacts with exact competitor classes. Let A be a finite nonnegative complete-type weight in the own-group central support, $W = \mathbb{E}_\pi A$, and let $\mathcal{E}_v = \mathcal{B}_v \cup \mathcal{G}_v \cup \{g > l(v)\}$ be its exposed groups. Full groups have one cell containing all

eligible names. The exact count-reveal estimate (C.1.1) therefore gives, in the original class of the retained anchor,

$$W_v := \mathbb{E}_\pi[A \mid \text{Class}_v] \leq C_H W \prod_{g \in \mathcal{E}_v} (1 + n_g)^{(a_g - 1)/2}. \quad (\text{A.2.55})$$

There is no full-group count factor and no class-probability factor. If a predeclared selected-bridge cost satisfies $D_g \geq c_D(1 + n_g)^{-A_D}$ (and $D_g = 1$ on unselected groups), then (A.2.50) pays every polynomial in (A.2.55) on a late corner, giving

$$e^{-C_v} W_v \leq C_H e^{-b_0} W \prod_{g \in \mathcal{E}_v} D_g. \quad (\text{A.2.56})$$

Indeed $n_g \ll_H 1 + L_g$, and a positive fraction of L_g on a bad or zero-tail group, or of $\sqrt{L_g}$ on a good group, dominates any fixed multiple of $\log(1 + L_g)$ up to a fixed constant. The same argument works for any fixed positive fraction of $C_v - b_0$. Under the same-class bank hypotheses of (C.1.9), apply the bank theorem with every $\mathcal{E}_v$ slot exposed and depth zero elsewhere. Its bounds $q_{g,0} \leq C_H D_g$ then give, for an acceptance $0 \leq F \leq A$ with the stated path tests,

$$e^{-C_v} \mathbb{E}_\pi[F \mid \text{Class}_v] \leq C_H e^{-b_0} W \prod_{g \text{ selected}} D_g. \quad (\text{A.2.57})$$

An early corner instead uses the bound $q_v = \mathbb{E}_\pi[F \mid \text{Class}_v] \leq 1$ for $0 \leq F \leq 1$. It yields $e^{-C_v} q_v \leq C_H e^{-a_0} e^{-c_H V_l}$. In the normalization $a_0 = Y, b_0 = 0$, the target condition

$$e^{-Y} \leq C_{\text{target}} D_{\text{global}}, \quad D_{\text{global}} = W \prod_{g \text{ selected}} D_g, \quad (\text{A.2.58})$$

pays all early corners. Combining with (A.2.57), and the dimension-bounded number of endpoint templates, gives for fixed $\delta > 0$

$$\sum_v e^{-\delta C_v} q_v^\delta \leq C_\delta D_{\text{global}}^\delta.$$

Condition (A.2.58) is an additional target input, not a consequence of the count box. In particular a short prefix is never asked to pay count factors from an unrelated long zero-rank tail. The type envelope A may be unbounded as long as the actual early-corner acceptance F is bounded by one.

Theorem A.23 (Intrinsic endpoint reserve). *Let V_{old} be the total original old length and assume*

$$a_0, b_0 \geq 0, \quad b_0 \leq B\sqrt{1 + V_{\text{old}}}. \quad (\text{A.2.59})$$

Define the intrinsic cost

$$K_v = Y - I_v = C_v - b_0 \quad \text{when } C_v = a_0 - I_v, \quad Y = a_0 - b_0.$$

For every lawful endpoint template with $t_v = V_{l(v)} \geq V_{\text{old}}$, set

$$S_v^{\text{intr}} = \begin{cases} h + A_{\text{in}} + \overline{C}_v, & t_v \leq T/2, \\ h + \overline{C}_v + S_{\text{good}}(v), & t_v > T/2. \end{cases} \quad (\text{A.2.60})$$

For a fixed sufficiently large h , uniformly on the same count support,

$$c_H S_v^{\text{intr}} \leq K_v + h \leq C_H S_v^{\text{intr}}, \quad K_v \geq c_0 \overline{C}_v - C_0. \quad (\text{A.2.61})$$

The assertion also holds for terminal $z = 1$ using Proposition A.20; no proper-child branching gap is needed for it.

Proof. For $V_{\text{old}} \leq t_v \leq T/2$, pay the incoming exit with the positive prefix reference cost. Equation (A.2.59) and Young's inequality give $b_0 \leq B\sqrt{1+t_v} \leq \bar{C}_v/4 + C_B$. We can therefore subtract b_0 in (A.2.53) while keeping the full coefficient of a_0 :

$$K_v \geq a_0 + \frac{1}{4}\bar{C}_v - C_B.$$

Its upper bound follows from the exact prefix formula and $b_0 \geq 0$: $K_v \leq a_0 + \bar{C}_v + C_H\sqrt{1+t_v} \leq a_0 + C'_H\bar{C}_v + C'_H$. The case $t_v = V_{\text{old}} = 0$ is covered by the fixed constant. For $t_v > T/2$, subtract b_0 in the exact tail formula itself:

$$K_v = Z_{l(v)} + \bar{C}_v + \sum_{g \text{ proper}} \nu_{vg}.$$

The proof of (A.2.54) now gives directly $K_v \geq \bar{C}_v/2 + c_H S_{\text{good}} - C_H$; its upper proof gives $K_v \leq C_H(\bar{C}_v + S_{\text{good}}) + C_H$. Only the zero-tail and bad-group errors are absorbed, and full-group errors cancel exactly. These inequalities prove the theorem after increasing h once. $\square$

On the zero-target construction, $b_0 = \eta_{\text{prev}}/u_{\text{prev}}$ satisfies (A.2.59): the preceding local length is at most V_{old} , its exit scale is bounded by a fixed multiple of its square root, and u_{prev} is bounded below. Terminal root rounding adds only a fixed constant. The complete-deadline template has $K_v = 0$ and intrinsic score h . The all-zero template has $K_v = a_0 - b_0$ and is excluded when $V_{\text{old}} > 0$; that exclusion is essential. The late score in (A.2.60) contains no B_{in} or preceding exit reserve.

Corollary A.24 (Legal local pairs). *For a local synchronous edge in block $i = [l_i, b_i]$ having at least one exterior cut, retain its lawful outside endpoint contexts. Let v^-, v^+ denote their left and right completions and set*

$$\begin{aligned} X_i^{\text{intr}} &= \min_v S_{v^-}^{\text{intr}}, & Y_i^{\text{intr}} &= \min_v S_{v^+}^{\text{intr}}, \\ x_i^{\text{intr}} &= \min_v (K_{v^-} + h), & y_i^{\text{intr}} &= \min_v (K_{v^+} + h). \end{aligned}$$

These actual minima are in the corresponding common scale boxes. For a canonical full-to-deepest- q edge they satisfy

$$y_i^{\text{intr}} - x_i^{\text{intr}} = \Delta_i = (r_i - q)L_i - \log S_i n_i + \log(q + z)K_{i,B}. \quad (\text{A.2.62})$$

A sufficient local pair, reading only this original local type, is

$$x_i^- = \max\{c_H X_i^{\text{intr}}, c_H Y_i^{\text{intr}} - \Delta_i\}, \quad y_i^- = x_i^- + \Delta_i. \quad (\text{A.2.63})$$

Proof. Every endpoint template retains an actual cut of a name in $A \setminus B$, so $t_v \geq l_i \geq V_{\text{old}}$, also when the cut is at the old/local boundary. Theorem A.23 therefore places every member in the common boxes, and the minima stay in those boxes. Moving the synchronous cut across i changes only its rank and logarithmic count contribution, giving (A.2.62) for each outside context. The intrinsic subtraction of b_0 is common to both endpoints; thus the minimum right endpoint is the minimum left endpoint plus Δ_i . Lemma A.17 gives (A.2.63) on the same types, words, and clocks. Fix h before reserve contraction so that the contracted endpoints exceed the scalar threshold. $\square$

Proposition A.25 (Allocation in a local-containing chamber). *In a lawful activation chamber containing a cut assigned to a local physical block, the simultaneous cost allocation is valid with*

$K = C - b_0$. If E_i is the original centered empirical bridge in coordinate i , L_{intr} its synchronized multilinear endpoint chord, and ℓ_i^{intr} its legal outside-context minimum chord shifted once by C_0 , then, for any $0 < \theta \leq 1/(2H)$ (in our source $\theta = M_1^{-M_1}$),

$$K_{\text{sync}} + C_0 \geq \frac{1}{2}(L_{\text{intr}} + C_0) + \sum_i [\theta \ell_i^{\text{intr}} + E_i]. \quad (\text{A.2.64})$$

For a surviving local canonical edge its sufficient support is

$$\theta \ell_i^{\text{intr}}(t) + E_i(t) \geq g d_i(t)^\gamma - h_W, \quad (\text{A.2.65})$$

and the local pair (A.2.63) suffices for it.

Proof. Fix one local cut of the chamber. Every endpoint vertex retains that cut at a legal local boundary, hence has $t_v \geq V_{\text{old}}$. By Theorem A.23, every such vertex obeys $K_v \geq c_0 \bar{C}_v - C_0$ with $\bar{C}_v \geq 0$. The synchronization identity is unchanged on subtracting the constant b_0 throughout this same-word chamber:

$$K_{\text{sync}} = L_{\text{intr}} + \sum_i E_i, \quad L_{\text{intr}} = L_{\text{multi}} - b_0.$$

The endpoint minima, shifted once, satisfy $0 \leq \ell_i^{\text{intr}} \leq L_{\text{intr}} + C_0$. There are at most H coordinates, so $\theta \sum_i \ell_i^{\text{intr}} \leq (L_{\text{intr}} + C_0)/2$; substitution proves (A.2.64). There is no leftover $-\theta b_0$ term because the cost was normalized before allocation.

Strong edges and reference-paid weak edges use the same named regularity bounds and $K_v \geq c_0 \bar{C}_v - C_0$. For the remaining local canonical edges retain (A.2.65). An old coordinate uses its old charge (Proposition A.11) and the same regularity estimate. For a proper atomic child, the branching normal form and occupancy estimate then apply with this normalized cost: synchronization differences, spread losses and pure-cell lifetimes are unchanged by subtracting the chamber constant. Divide the available reference, spread and curve budgets by the same fixed fractions as in that allocation; every original cut and its multiplicity remains. At $z = 1$ this argument first concerns star histories, and extension to branching uses the separate floating-spine argument with this adjusted endpoint input. Entirely old chambers are outside the stated domain. $\square$

A final two-color computation makes the normalization concrete. On its last local block $[l, T]$, with every other fine name cut at its own deadline, moving only the final boundary to t gives the exact global fine cost

$$K_{\text{fullstop}}(t) = \log 2 N((t, T]) - (T - t). \quad (\text{A.2.66})$$

Put $D = \log 2 N((l, T]) - (T - l)$. After a fixed shift its forward pair is $(h + D, h)$ and its reverse pair $(h, h + D)$. This context is among the legal outside contexts of the same $q = 0$ coordinate. Its shifted chord therefore dominates the adjusted minimum chord at both ends, and hence throughout the block. Since $\ell_{\text{min}}^{\text{intr}} \geq 0$, the sufficient allocated wall implies

$$\theta \ell_{\text{min}}^{\text{intr}} + E \geq g d^\gamma - h_W \implies K_{\text{fullstop}} + h \geq g d^\gamma - h_W.$$

More generally two lawful pairs at one exact count have the same increment and differ only by a constant translation; their lower pair is still lawful. The ordinary all-reserve theorem evaluates the pair in (A.2.66) only after verifying positivity, its exact root count and unread-clock conditioning. Up to fixed buffers it has the form $(1, 1 + D)$, and contains no incoming b_0 term.

A.3 The optimizer and endpoint costs

We collect the preceding inputs in the notation of the profile. This identifies the deterministic score minima with common boxes for the actual local counts, and records how the terminal construction changes when the common root target becomes positive. The identification uses the complete-chain optimizer throughout.

A.3.1 Geometric inputs to the endpoint construction

The endpoint construction uses the ordered optimizer in (3.2), with $H = k + 1$ and $V_j = \sum_{i \leq j} L_i$. Theorems 4.1 and 4.8 give its uniqueness and dimension-only compact bounds. For $k = 1$ the sole product is α_2 ; for $k \geq 2$ the products exceed α_2 and are bounded away from one. These statements include equal-product plateaux and do not require a positive gap between distinct products.

For a maximal plateau $[a, b]$ of product σ , write $r_i = b - i + 1$ and let z be its permanent-child weight, with $z = 1$ on the final plateau. Its local coefficients are $c_i = (r_i + z)^{\sigma-1}$; original earlier groups have capacity r_a and coefficients κ_i from (A.2.14). The exact plateau balance is (A.2.15), and the clipped suffix balances are (A.2.16). In particular their exterior endpoint multipliers vanish. The length hierarchy in Theorem 4.17 gives

$$V_{a-1} \leq 2H^3(1 + \log H)^3 \sum_{i=a}^b L_i$$

without a factor involving a strict product gap.

Proposition A.6 supplies reference nonnegativity for every lawful original history. Its proof retains the number of unactivated original boundaries, rather than replacing it by the eligible unresolved rank: a pure cell resolved after expiry can leave a delay in the former count. Each unactivated boundary is assigned to a still-eligible name, so that its count is at most the original capacity. The partition comparison, old-coefficient Stieltjes averaging, and endpoint compression therefore apply with these original ranks. Lemmas A.3 and A.4 justify the compression, including the decreased coefficient after averaging earlier groups. Equal split times and the equal-weight terminal case are included.

Keep all original groups, indexed by g , and put $g_g(q) = q - \kappa_g E_q(z)$. Lemma A.13 gives the exact least positive-prefix cost and its charge in (A.2.32), including the empty and complete prefixes. Together with own-group centrality (A.2.30) and the two expressions for the signed remainder in (A.2.31), this yields the endpoint guard (A.2.33). Zero-length original groups contribute zero and retain their indices. All these comparisons use the complete-chain optimizer, including when an endpoint history ends before the final group.

A.3.2 Actual displaced counts and the common intrinsic box

Proposition A.14 supplies own-scale central fine counts and a fixed positive displaced gain on every good proper group. A bad group (retaining shallow and omitting a deeper eligible name) has $\overline{C} \geq \kappa_* L / (2H)$ by the lawful two-cut exchange and the second difference of E_q . There are finitely many groups, so $\overline{C}$ controls their total bad length. The charge in (A.2.32) controls the zero tail and early prefix.

For an actual endpoint template let l be its last positive-rank group and ν_{vg} its proper-group potential discrepancy. Full groups have $\nu_{vg} = 0$ exactly. At every admitted complete type,

$$C_v = a_0 - \sum_{g \leq l} \zeta_g + \bar{C}_v + \sum_{g \text{ proper}} \nu_{vg} = b_0 + \sum_{g > l} \zeta_g + \bar{C}_v + \sum_{g \text{ proper}} \nu_{vg}. \quad (\text{A.3.1})$$

All discrepancies on a proper group are $O(\sqrt{1+L_g})$, uniformly in that count support. In an early corner use the prefix formula; its prefix square root is absorbed by (A.2.32), retaining coefficient one on a_0 . In a late corner use the tail formula; only zero-tail, bad, and good groups have remaining errors. The first two are absorbed by their reference lengths, and the displaced good counts contribute a positive multiple of $S_{\text{good}} = \sum_{\text{good}} \sqrt{L_g}$. The same formulas give the corresponding uppers. Thus after one fixed shift,

$$C_v + h \asymp \begin{cases} h + a_0 + \bar{C}_v, & t_v \leq V_b/2, \\ h + b_0 + \bar{C}_v + S_{\text{good}}(v), & t_v > V_b/2. \end{cases} \quad (\text{A.3.2})$$

For a local-containing endpoint template, $t_v \geq V_{\text{old}}$. The incoming strict exit satisfies $b_0 \leq B\sqrt{1+V_{\text{old}}}$; its preceding local length is bounded by V_{old} , its carry has its band size at most a fixed square root, and u is bounded below. In an early corner (A.2.32) absorbs this incoming subtraction into a fraction of $\bar{C}_v$, retaining coefficient one on a_0 . In a late corner subtract it in the exact tail formula (A.3.1), where it cancels. Consequently $K_v = C_v - b_0$ satisfies

$$K_v + h \asymp \begin{cases} h + a_0 + \bar{C}_v, & t_v \leq V_b/2, \\ h + \bar{C}_v + S_{\text{good}}(v), & t_v > V_b/2. \end{cases} \quad (\text{A.3.3})$$

At zero proper targets, a_0 lies in its strict band $aR_j \leq a_0 \leq bR_j$. These are precisely the construction's deterministic intrinsic scores, up to fixed shifts and constants. Taking minima over the same finite paired contexts proves their simultaneous common scale box at every admitted complete type. The moving local bundle's cost difference is exactly

$$\Delta_i = (r_i - q)L_i - n_i \log(r_i + z) + K_{i,B} \log(q + z),$$

independent of outside context. Thus the minimum right endpoint equals the minimum left endpoint plus this same increment; the max formula produces the common legal pair.

These are the source instances of Theorem A.23 and Corollary A.24. The all-zero template when $V_{\text{old}} > 0$ is deliberately excluded from a local pair: it has $K = a_0 - b_0$ and cannot satisfy an incoming-free positive box. The local coordinate's actual cut at its left local boundary ensures $t_v \geq V_{\text{old}}$, so that exclusion is automatic and lawful.

A.3.3 Terminal completion at zero and positive root targets

The terminal prefix pair is constructed at zero common root target, before the suffix spatial clock is read. This numerical count completion leaves that clock unread. Prefix terminal fine types and their exterior boxes remain fixed after their paths have been accepted.

Fix a retained full prefix type and an admitted central uncolored suffix vector. Fill the unread suffix terminal names, retaining all prefix labels. Permanent-name centrality and displaced exterior boxes have fixed positive conditional mass on each new group; the prefix groups already satisfy these restrictions. Since the endpoint-box proof is groupwise, it applies to the combined list without further prefix refinement. This is the form of Proposition A.20 needed after a full prefix word has been accepted.

At the zero target, root rounding $\epsilon \leq 0$ gives $a_0 = 0$, $b_0 = \eta_{\text{prev}}/u - \epsilon/r_m \geq 0$; (A.3.3) supplies the zero-root pair and its upper and lower boxes. Any such numerical completion has the same

local prefix increment and therefore gives the same local pair in (A.2.63). The suffix clock remains unread.

For a later target $x \geq 0$, the actually selected reservoir count gives root $X_1 = x + \epsilon$. Its correct entrances are

$$a_0 = x/r_m, \quad b_0 = \eta_{\text{prev}}/u - \epsilon/r_m \geq 0. \quad (\text{A.3.4})$$

The new suffix count stays in a fixed central window under the allowed logarithmic target cap. Fill only its private suffix names as just described. Applying (A.2.51) and the intrinsic comparison (A.2.61) now shows that the terminal early lower bounds gain the nonnegative common entrance, while terminal late intrinsic lowers retain neither entrance nor incoming carry. Every previously chosen zero-root prefix pair is still below the actual minima. Its private path law, its local count, its fixed sufficient curve, and its envelope are unchanged.

An upper box for the positive-root early minimum is neither needed nor implied by the zero-root box; that minimum may grow by x/r_m . Likewise $b_0 = \eta_{\text{prev}}/u - X_1/r_m$ is the zero-target formula only when $a_0 = 0$; using it at positive target would be wrong. Equation (A.3.4) is the normalization used in the comparison histories of Section D.1.

The ordinary weak reservoir, if present, needs its separate actual count max-formula and ordinary bridge estimate. Section B.2 establishes the proper binary prefix inputs and does not replace this one root/suffix construction by a fictitious zero-root clock.

A.4 Selection of binary constraints and unrestricted intervals

The lower construction retains every canonical weak wall. The upper construction needs only genuinely rare factors, whose rank intervals must also be disjoint. We verify that both selections recover the same profile up to fixed constants, with all scalar parameter ranges chosen before the later reserve constants. We then identify free intervals inside the actual comparison classes.

A.4.1 A reserve-independent lower selection

A saturated numerical factor can still correspond to a wall needed by the geometric lower argument. We therefore select by weak drift rather than by the final factor size. Strong concavity leaves at most one selected binary bundle per original block, and the reference increment bounds all factors omitted by this rule.

Let

$$\epsilon_w = M_1^{-100}, \quad \mathcal{I}_{\text{low}} = \{e : |\nu_e| \leq \epsilon_w\}. \quad (\text{A.4.2})$$

Strict concavity and the normalized-secant argument in Section A.2.4 imply that for a fixed full rank there is at most one such output, and outputs decrease strictly in physical order. First consider two outputs at the same input rank r . On $0 \leq a \leq r$ the function $g_r(a) = a - \kappa_r E_a(z)$ satisfies

$$-g_r''(a) = \frac{\kappa_r}{a+z} \geq H^{-2},$$

since $r+z \leq H$ and $\kappa_r = (r+z)^{\sigma-1} \geq H^{-1}$. Choose the weakness threshold also to satisfy $\epsilon_w < 1/(4H^2)$. Suppose that $q_1 < q_2 < r$ are two integer outputs and that both corresponding

edges are weak. Strong concavity, applied to the chord with endpoints q_1, r , gives

$$\begin{aligned} g_r(q_2) - g_r(r) &\geq \frac{r - q_2}{r - q_1} \{g_r(q_1) - g_r(r)\} + \frac{(q_2 - q_1)(r - q_2)}{2H^2} \\ &\geq -\epsilon_w + \frac{1}{2H^2} > \epsilon_w. \end{aligned}$$

This contradicts weakness of the second edge. Thus each input rank has at most one weak output. In particular, after the fixed rarity threshold has been chosen to imply weakness, each original local block has at most one retained rare edge.

The two completed reference minima m_-, m_+ have the exact common increment $m_+ - m_- = \nu_e L_i$. This holds because every paired outside context has that same increment, so taking its minimum commutes with adding the constant. Reference nonnegativity gives

$$|\nu_e| L_i \leq m_- + m_+ \leq X_e + Y_e,$$

and hence

$$T_e \geq \frac{|\nu_e| L_i}{1 + H L_i} \geq \frac{|\nu_e|}{2H}. \quad (\text{A.4.3})$$

Every omitted nonweak edge factor is consequently at least $\min(1, \epsilon_w/(2H))$, since $2\chi \leq 1$. There are at most $k(k+1)/2 \leq H^2$ factors. Keeping all weak edges, including weak edges whose persistence factor is saturated, changes the profile by at most a fixed multiplicative constant.

The lower construction may use this larger selection. At a final complete type the original block clocks and uniform multiset words are independent, and there is one binary bundle per block. Its local scalar theorem is stated for both the rare branch and the saturated branch. On the saturated branch the conditional curved-wall plus binary/named regularity mass has a fixed positive fractional moment; its free-bank envelope may be one. No rank-interval disjointness is needed for this lower multiplication: that condition belongs to the upper current-plus-intermediate-bank construction. The joint count source already fills all original fine names, so it needs no extra count adjustment when a saturated weak edge is retained.

The source-star comparison therefore has every weak local wall it uses, including walls whose numerical factor is saturated. The same selection is used in Section 7 and Section C.1.

A.4.2 Intrinsic scores and added reserves

The source bridge theorem requires a fixed positive reserve. Adding it to the finite endpoint scores changes their comparison with the profile only by a dimension-dependent constant. The comparison is made after the scalar parameter ranges have been fixed; it does not alter the underlying clock or mark law.

The common endpoint box uses

$$S_v^I = \begin{cases} h + R_j + \overline{C}_v, & \text{early proper,} \\ h + \overline{C}_v, & \text{early terminal,} \\ h + \overline{C}_v + S_{\text{good}}(v), & \text{late,} \end{cases} \quad (\text{A.4.4})$$

not the physical profile's score with leading constant one. The intrinsic endpoint theorem gives, with constants independent of the chosen later $h \geq M_0^{20}$,

$$M_0^{-20} S_v^I \leq K_v + h \leq M_0^{20} S_v^I. \quad (\text{A.4.5})$$

Theorem A.23, in particular (A.2.61), gives (5): all additive count errors have already been absorbed by the reference charge or the fixed shift. Increasing h adds the same positive quantity to both sides' scales. The complete-deadline intrinsic cost is zero exactly. The global all-zero raw endpoint cost is its entrance a_0 exactly; when an incoming old length is positive it is excluded from a local intrinsic pair, whose current local cut ensures $t_v \geq V_{\text{old}}$. If there is no incoming old length and the terminal root is zero, its intrinsic cost is zero as well.

Since the fixed width floor is $R_0 = 100H^2$, independent of h , its comparison with the profile's $1 + \widehat{R}$ has constants at most $1000H^2$. Taking finite minima yields

$$(1000H^2)^{-1} X_e \leq X_e^I \leq 1000H^2(1+h)X_e, \quad (1000H^2)^{-1} Y_e \leq Y_e^I \leq 1000H^2(1+h)Y_e. \quad (\text{A.4.6})$$

Multiplying the scales and applying the clipped power changes each edge factor by at most $[10^6 H^4(1+h)^2]$, since $2\chi \leq 1$. This is a final dimension-only comparison constant. It is not an input to the compact family of scalar laws.

A.4.3 Actual-type compactness before reserve selection

The scalar theorem is applied at actual admitted counts. Own-scale centrality and weakness keep their centered binary fraction, signed speed, and correlation in a common compact range. The same remains true after contracting the endpoint chord. These bounds keep the later choice of a large reserve from changing the scalar hypotheses.

For every selected final complete type, centrality gives

$$|n_i - \lambda L_i| \leq B\sqrt{1 + L_i}, \quad |k_i - pn_i| \leq 2HB\sqrt{1 + L_i}.$$

Consequently the actual legal endpoint increment is

$$\Delta_i = (r - q)L_i - an_i + bk_i = \nu_e L_i + E_i, \quad |E_i| \leq 4H^2 B\sqrt{1 + L_i}. \quad (\text{A.4.7})$$

For $L_i \geq M_0^{100}$, these bounds imply n_i/L_i lies in $[1/2, 2H]$, k_i/n_i lies in $[1/(2H), 1 - 1/(2H)]$, and

$$p_0 = \frac{a - (r - q)L_i/n_i}{b}, \quad |p_0 - p| \leq 4H^3 \epsilon_w + 20H^4 B/\sqrt{L_i}. \quad (\text{A.4.8})$$

The right side is below $1/(4H)$ at the threshold in (H.2.1). Thus the centered fraction p_0 is interior with a bound chosen before h . The original and contracted speeds remain between fixed positive constants: the contraction mixes $(r - q)$ and the actual total jump rate $(an_i - bk_i)/L_i$; the latter is positive and bounded away from zero by the displayed count support. The split weights, gap means and channel variances are controlled by the same compact construction. No endpoint upper coefficient proportional to h is used.

For the scalar correlation replacement, smoothness of its explicit rational formula on that compact parameter set gives

$$|\rho_i^0 - \rho_e| \leq C_H(|\nu_e| + B/\sqrt{L_i}). \quad (\text{A.4.9})$$

On $T_e < 1$, (3) gives $|\nu_e| \leq 2HT_e$, while $X_e, Y_e \geq 1$ gives $T_e \geq 4/(1 + HL_i)$. Therefore

$$(|\nu_e| + B/\sqrt{L_i})|\log T_e| \leq 2H/e + B \frac{\log(1 + HL_i)}{\sqrt{L_i}}. \quad (\text{A.4.10})$$

The last quotient is at most $2\sqrt{H}$ by $\log(1+t) \leq \sqrt{t}$ (using the weaker constant 2 also covers the displayed 1+HL convention). Local Lipschitz continuity of χ then bounds the exponent replacement by $\exp(C_H(1+B))$ uniformly in all lengths, including the weak-drift limit. Replacing T_e by the intrinsic score ratio invokes (6) only as a final constant.

The intrinsic rare test $T_e^I < t_0 = M_1^{-100}$ is used solely to choose the scalar expression's rare or saturated branch. Compactness (8) holds on both branches. The physical selection threshold need not be shrunk by $(1+h)^{-2}$, and $1/t_{\text{rare}}$ does not enter new probability estimates after h is chosen.

For the contracted source wall this statement is applied to the contracted scales, not to the same unscaled scores with a new tiny box coefficient. Put $\theta = \theta_{\text{intr}}$ and $\tilde{X} = \theta X_e^I$, $\tilde{Y} = \theta Y_e^I$. If the actual intrinsic chord is $\ell(t) = x + \Delta t/P$ and its centered bridge is $E(t)$, then the contracted path $\theta\ell(t) + E(t)$ has speed

$$c_\theta = \theta c + (1-\theta)(an - bk)/P, \quad p_{0,\theta} = \theta p_0 + (1-\theta)k/n. \quad (\text{A.4.10a})$$

Both remain in the same compact parameter set by (8) and the actual central type, uniformly for $\theta \in [0, 1]$. Its legal endpoints are $\theta x, \theta y$, so the common lower/upper box coefficients $M_0^{\pm 20}$ are unchanged when the reference scales are $\tilde{X}, \tilde{Y}$. The source allowance h_W can be added to both endpoints: at the final reserve $h_W \leq \theta M_0^{-20} h/100$, so that addition changes their box coefficients by at most a factor two. In particular the scalar initial reserve is at least $\theta M_0^{-20} h$, which exceeds the already fixed scalar threshold; it does not require choosing a new compact parameter set after h .

The centered correlation from (10a) differs from ρ_e by $C_H(\theta|\nu_e| + B/\sqrt{L_i})$. Equation (10) therefore applies without worsening its constant. The common scale multiplication by θ changes the persistence factor by at most θ^{-2} , since its power is $2\chi \leq 1$. This and (6) are final dimension-only comparison constants. An asymmetric continuation estimate applied at $\tilde{X}, \tilde{Y}$ consequently retains its original a_*, a_0, κ constants; it never receives θ^{-1} through a box coefficient.

A.4.4 The upper rare selection and an explicit disjointness margin

The upper proof reads intermediate groups as well as the current block. Two retained weak rank intervals that overlap would force a positive endpoint cost on one of them, contradicting rarity. The following explicit margin makes this exclusion uniform and also separates the treatment of bounded path banks from the choice of rarity threshold.

The upper uses only edges with $T_e < \tau_0$, where τ_0 is fixed before h . Set

$$c_s = 1/(200H^4), \quad A_s = c_s/H, \quad \tau_0 = \min\{\epsilon_w/(4H), A_s/4\}. \quad (\text{A.4.11})$$

By (3), these edges are weak. The normalized-secant estimates in Section F.1 give the following strict comparisons with coefficient at least c_s : an earlier natural or transported rank a in $[r, h]$ costs at least c_s against the weak output q ; a rank below a positive weak output costs at least c_s on the current block; and an intervening capacity $q < h < r$ costs at least c_s against its full rank. The first coefficient follows from $1 - \exp(-\log 2/(48H^2)) \geq \log 2/(96H^2)$ and $1 - \exp(-(1-\sigma)/H)(1+\epsilon_w) \geq 1/(40H^4)$; the other two coefficients are at least $1/(4H^2)$ and $\log 2/(32H^2)$.

Suppose two upper-retained weak edges have $r_1 > r_2$ but $q_1 < r_2$. Output order gives $q_2 < q_1$. Lowering all earlier ranks in a left completion for the second edge to q_2 is lawful

and proves $X_2 \geq 1 + c_s L_1$. In a right completion for the first, all surviving q_1 names remain eligible until the second block. The ordered cut-time cost on that interval is concave because the original coefficients increase with physical time. Minimize over its simplex by a vertex, leaving one constant rank $a \leq q_1$. If $a < q_1$, the current-block strict comparison pays $c_s L_1$; if $a = q_1$, filling the second block pays $c_s L_2$. Complete-reference nonnegativity of the lawful comparison yields $Y_1 \geq 1 + c_s \min(L_1, L_2)$ in every context.

Since the remaining scores are at least one and rates at most H ,

$$T_1 \geq A_s \frac{1 + \min(L_1, L_2)}{1 + L_1}, \quad T_2 \geq A_s \frac{1 + L_1}{1 + L_2}. \quad (\text{A.4.12})$$

If $L_2 \geq L_1$, the first ratio is at least A_s . Otherwise their product is at least A_s^2 . Both edges cannot satisfy (11). Thus upper-retained rank intervals $(q_i, r_i]$ are disjoint, and a proper zero-output edge is last. Every discarded upper factor is at least τ_0 ; at most H^2 such factors occur.

The cutoff (11) is not decreased after the scalar estimates's fixed minimum length n_0 is chosen. If a deterministic upper path bank has length $\ell < n_0$, the necessary hard endpoint inequality gives $T_e \geq c_H / \sqrt{1 + \ell}$. Its persistence factor is therefore bounded below by a dimension-only multiple of $n_0^{-1/2}$, since $2\chi \leq 1$. On this fixed bounded experiment one may discard its local path mask and pay $n_0^{O_H(1)}$, retaining its proper middle count test. In the effective construction $\log n_0$ is at most $\exp(M_2^{4000})$; this loss is below the final geometry intercept. This treatment changes a fixed comparison constant; it does not change τ_0 or introduce an h -dependent scalar compact bound.

The lower selection (2) and upper selection (11) differ, for good reason: only the upper proof reads intermediate local groups. Both recover exactly the same canonical factor product up to dimension-only constants. Neither selection depends on h .

A.4.5 Free intervals in the comparison classes

History reduction is useful for probabilities only if its geometric charges describe the original class. We show that each late block either pays a positive length or depth cost, or is unrestricted away from short endpoint strips. This yields genuinely free count banks and the two free intervals required by the conditional path estimate.

The full source/target geometry gives the actual nonnegative charge

$$\Lambda = J(v) + B(h^*) + \Pi(h^*) + D_{\text{gap}} + D_* + S_{\text{pure}} + S_{\text{off}}. \quad (\text{A.4.13})$$

Here v is the maximum-weight endpoint representative of the same original history. The token proof in Section D.2 (5)–(6) controls every original cut, with multiplicity. For each physical block choose an endpoint radius r_g at least one and large enough to contain all original cuts and every movement interval from h^* to v in that block, unless that radius reaches $L_g/2$, when use the entire block. This is a purely geometric choice. The number of tokens, spreads and movements is at most a fixed power of H , so

$$\sum_g r_g^\gamma \leq C_H(1 + \Lambda). \quad (\text{A.4.14})$$

If $r_g < L_g/4$, the original partition is constant on the middle half: its boundaries are outside that half. The comparison star agrees there with v , because all h^* -to- v movement intervals have also been put in the strips. Each proper projection and its same-cut star preserve the eligible

unresolved rank. At the terminal floating-chain step, the only possible increase is from expired first names of off-spine nonsingleton cells. The number of these increases is at most the original off-spine internal-cell count. Hence, pointwise on that middle interval,

$$0 \leq r(h^*) - r(h) \leq m_{\text{off}}. \quad (\text{A.4.15})$$

If v is full but the original partition is nonfull there, the left side is at least one on length $L_g/2$, so $S_{\text{off}} \geq L_g/2$. This pays any fixed polynomial loss at that block. If v is nonfull in a late-plateau local block, its endpoint has a zero-tail, a bad proper group, or a displaced good proper group. The first two have the reference linear charge; the last has its positive good-name square-root charge. The endpoint comparison therefore gives $J(v) \geq c_H \sqrt{L_g} - C_H$. This also pays any fixed polynomial loss. If $r_g \geq L_g/4$, (14) pays L_g^γ directly. These cases exhaust every late local block. Thus each unpaid late local class is literally unrestricted outside its endpoint strips. This is a statement about its per-slot signatures, before conditioning labels.

Every nonterminal plateau's designated count bank is in a local block of length at least T_j/H and stays a fixed positive fraction from both ends. If its strips reach the bank, (14) pays a positive power of T_j ; if its middle partition is nonfull, one of the preceding endpoint charges pays a positive power of T_j . Otherwise the original class is unrestricted on that complete bank. Its unread two-name split is therefore free under the original product word law. For a selected path block the same cases leave either a paid block or the two actual free end intervals just inside its strips.

Finally, the inverse polynomial floors are dominated by these positive powers: for all $d \geq 0$ and $0 < \alpha \leq 1$,

$$A \log(2 + d) \leq \varepsilon d^\alpha + C(A, \alpha, \varepsilon),$$

with an explicit constant obtained from $\log(2 + d) \leq \log 2 + 2d^{\alpha/2}/\alpha$ and Young's inequality. The fixed block/cell count supplies only a bounded number of such losses. This proves the free/paid classification in Section D.1 from (13), without using the posterior probability being bounded. The exact original-law disintegration in Section D.5 then applies on these target classes. It averages all pair totals, including zero; its bank-size estimate uses the complete permanent-slot signatures on each coarsened fiber.

B Conditional probability and bridge estimates

This appendix supplies the probability estimates used in Sections 5 and 6. The first subsection proves finite Gaussian association, records the joint density for selected exact gaps and marks, and establishes small-tube and positive-drift continuation estimates. The next subsection combines these ingredients at a final complete type and constructs the uncolored envelope required in the arithmetic lower bound.

The remaining subsections make the constants explicit. Their order reflects the proof dependencies: the rank coupling and its wall action; the exact middle tube; the Gaussian endpoint and wall bounds; the flat preparation and relative coupling errors; and the separate ordinary bridge construction. Large numerical constants serve only to give a uniform fixed length threshold.

B.1 Auxiliary probability estimates

We begin with the association inequality used both to join adjacent Gaussian segments and to compare environmental tilts. All conditioning in its applications is specified separately; the

inequality itself concerns the unconditioned finite Gaussian law.

Lemma B.1 (Finite Gaussian association). *Let Z be a centered nondegenerate Gaussian vector whose covariance matrix Σ has nonnegative entries. For bounded coordinatewise nondecreasing Borel functions f, g ,*

$$\mathbb{E}[f(Z)g(Z)] \geq \mathbb{E}f(Z) \mathbb{E}g(Z).$$

The assertion also holds whenever truncation makes both sides converge.

Proof. For smooth bounded functions with bounded derivatives, let (X_t, Y_t) have diagonal covariance blocks Σ and cross block $t\Sigma$, $0 \leq t \leq 1$. This is a positive semidefinite covariance: one may take $X_t = Z$ and $Y_t = tZ + \sqrt{1-t^2}Z'$ with Z' an independent copy. Gaussian integration by parts gives

$$\frac{d}{dt} \mathbb{E}[f(X_t)g(Y_t)] = \sum_{i,j} \Sigma_{ij} \mathbb{E}[\partial_i f(X_t) \partial_j g(Y_t)] \geq 0.$$

The two off-diagonal covariance blocks account for the absence of a factor $1/2$. The covariance derivative identity follows by differentiating $\exp(-\langle \xi, C(t)\xi \rangle/2)$ and integrating twice by parts, first for smooth compactly supported test functions; cutoff approximation and covariance regularization give the displayed form. At $t = 0$ the two vectors are independent; at $t = 1$ they coincide. Integration proves the claim for smooth functions.

Convolution with a positive smooth product kernel preserves monotonicity. Bounded coordinatewise monotone functions have a Lebesgue-null discontinuity set, so the convolutions converge almost everywhere for the nonsingular Gaussian law. Dominated convergence proves the Borel assertion. At distinct finite Ornstein–Uhlenbeck times the covariance is positive definite; coincident times can simply be deleted. Applying this finite result on nested dense time grids and using path continuity proves its survival applications. The continuous survival applications therefore follow from the unconditioned finite-grid inequalities. Any conditioned association used elsewhere is established through the stated MTP₂ structure. $\square$

B.1.1 Joint exact-total comparisons and small tubes

Exposing two ends of an exact bridge leaves a common residual total and a common residual type. A single likelihood ratio handles those constraints jointly and compares the exposed variables with independent reference coordinates. On central exposed totals it also gives the lower comparison needed for the middle-tube construction.

Let $f_{r,\tau}$ denote the density of a sum of r independent exponentials of mean τ . If n marks have exact low count k , $\pi = k/n$, and their $n + 1$ gaps have total $P = (n + 1)\tau$, exposing m specified marks and m distinct specified gaps has joint density

$$\frac{f_{n+1-m,\tau}(P - T)}{f_{n+1,\tau}(P)} \frac{\mathbb{P}\{\text{Bin}(n - m, \pi) = k - K\}}{\mathbb{P}\{\text{Bin}(n, \pi) = k\}} \quad (\text{B.1.1})$$

relative to independent reference gaps and marks. Here T, K are the exposed totals, and the ratio is zero outside its support. Independence of the two unobserved sums proves the identity. The same formula with m replaced by $2m$ applies jointly to the first m and last m marks and to the distinct gap sets

$$E_1, \dots, E_m; \quad E_{n-m+2}, \dots, E_{n+1}.$$

There are $n - 2m$ residual marks and $n + 1 - 2m$ residual gaps.

For $m \leq n/4$, or for $m = \lfloor n/8 \rfloor$ at each end, this ratio is bounded above on its entire support, uniformly on compact positive τ and interior π . On each fixed central box

$$|T - m\tau| + |K - m\pi| \leq C\sqrt{n}$$

it is bounded below positively for sufficiently large n ; in the two-end formula replace m by $2m$. Indeed the Gamma maximum density is $O((1+r)^{-1/2})$, its density at the reference mean is $\asymp (1+r)^{-1/2}$, and the corresponding binomial bounds have the same order. Central Stirling bounds give the lowers after the remaining totals are inside their support. These denominators are at their exact reference means. Near-central variants require the large-total hypotheses of Lemma 5.17.

Lemma B.2 (Exact multicolor small tube). *Fix $r, \epsilon > 0$, $0 < \tau_- \leq \tau_+ < \infty$, and $a > 0$. Let a word of length n be uniform of exact type $v = (v_1, \dots, v_r)$, $\pi_c = v_c/n \geq \epsilon$. Independently let n exponential gaps be conditioned to have total $n\tau$, $\tau \in [\tau_-, \tau_+]$. For all sufficiently large n ,*

$$\mathbb{P} \left\{ \max_{j \leq n} |T_j - j\tau| \leq a\sqrt{n}, \quad \max_{c,j} |N_c(j) - j\pi_c| \leq a\sqrt{n} \right\} \geq c_a > 0.$$

The threshold depends on the fixed data and a .

Proof. We first obtain a maximal tail estimate by comparison on each half with independent coordinates. We then retain central subblock totals and use that estimate inside shorter bridges to obtain a small tube of any prescribed fixed scaled radius.

For both exact bridges,

$$\mathbb{P}\{\max_j |T_j - j\tau| > u\} + \sum_c \mathbb{P}\{\max_j |N_c(j) - j\pi_c| > u\} \leq C \exp\{-c \min(u^2/n, u)\}. \quad (\text{B.1.2})$$

To see this, expose at most $3n/4$ coordinates. Their likelihood relative to independent reference coordinates is the product of

$$\frac{f_{n-m,\tau}(n\tau - \sum_I E_j)}{f_{n,\tau}(n\tau)}, \quad \frac{\mathbb{P}\{\text{Mult}(n-m, \pi) = v - N_I\}}{\mathbb{P}\{\text{Mult}(n, \pi) = v\}}.$$

Gamma and multinomial maximum-density and central-density Stirling bounds make both ratios bounded above. The bounded ratios compare the entire selected set with its independent reference law; the selected coordinates remain dependent under the exact bridge law. Independent centered gaps and centered color indicators have moment generating functions bounded by $\exp(C\theta^2)$ for $|\theta| \leq \theta_0$. The exponential supermartingale maximal inequality gives the right side of (B.1.2) on the first half. Reverse the second half, using exchangeability and the zero centered complete total. This proves (B.1.2); $n = 1$ is immediate.

Choose a fixed integer $J \geq 2$, and split the coordinates into J blocks of lengths between $n/(2J)$ and $2n/J$. In the first $J-1$ blocks, let the first $r-1$ color counts vary in integer intervals of width $\delta\sqrt{n}$ about their means. Determine the remaining color count by the block length and the last block by the full type. All deviations are $O_r(J\delta\sqrt{n})$ and all counts are nonnegative for sufficiently large n . The exact conditional atom is

$$\frac{\prod_{b=1}^J \mathbb{P}\{\text{Mult}(n_b, \pi) = v_b\}}{\mathbb{P}\{\text{Mult}(n, \pi) = v\}}.$$

Central Stirling gives a lower bound $c_{J,\delta} n^{-(J-1)(r-1)/2}$ for each choice. There are at least $c_{J,\delta} n^{(J-1)(r-1)/2}$ choices. Thus this event has positive constant probability, including $r = 1$.

Likewise prescribe the first $J - 1$ block gap totals in real intervals of width $\delta\sqrt{n}$ about $n_b\tau$, with the last determined by the full total. Their density is $\prod_b f_{n_b,\tau}(A_b)/f_{n,\tau}(n\tau)$, bounded below by $c_{J,\delta} n^{-(J-1)/2}$ on a box of volume $\asymp n^{(J-1)/2}$. Its probability is positive, and the gap and color channels remain independent.

After J is fixed, choose δ small enough that all cumulative boundary deviations are at most $a\sqrt{n}/4$. Conditional on these block totals, the blocks are independent exact bridges with compact new parameters. By (B.1.2), their combined failure probability for tubes of radius $a\sqrt{n}/2$ about their own linear interpolants is at most $CJ \exp(-ca^2J)$ for sufficiently large n . Choose J to make this at most $1/2$, and then choose δ and the length threshold. Boundary and internal deviations sum to less than $a\sqrt{n}$. All choices precede n . $\square$

Apply the gap and color conclusions separately at $n + 1$ and n , respectively, to obtain the small tube for a bridge with n marks and all $n + 1$ gaps. The prescribed tube radius determines its large-length threshold.

B.1.2 Positive drift under exact conditioning

In the asymmetric endpoint regime, the effective prefix ends at a height comparable to the inverse remaining drift. The following continuation estimate converts that height into a constant success probability on the entire remaining exact bridge, however long it is.

Lemma B.3 (Ordinary positive-drift continuation). *Fix compact positive ranges for two marks w_0, w_1 (which may be equal), a speed c , and $\tau = P/(n+1)$, with k/n in a fixed compact subinterval of $(0, 1)$. On $[0, P]$ take n uniform points and, independently, a uniform k -subset bearing mark w_1 ; the others bear w_0 . For $\varepsilon \in \{-1, 1\}$ put*

$$Z(t) = x + \varepsilon\{w_0 N_0(t) + w_1 N_1(t) - ct\}, \quad v = \varepsilon \left(\frac{(n-k)w_0 + kw_1}{P} - c \right).$$

Suppose $0 < v \leq v_{\max}$. For every fixed $a > 0$ there are $R(a) < \infty$ and $b(a) > 0$ such that

$$x \geq a/v, \quad Pv^2 \geq R(a) \quad \implies \quad \mathbb{P}\{Z(t) \geq 0 \ (0 \leq t \leq P)\} \geq b(a).$$

This probability averages both the exact clock and exact type.

Proof. Section B.2.9 proves the two-color continuation directly. Its first half compares the exact clock and exact type jointly with two independent Poisson processes. An exponential supermartingale gives a positive first-half survival probability; a central count window retains a fixed fraction of it. The reversed half-density has a uniform upper bound, and its maximal inequality bounds late failure by $C_{\text{den}} e^{-\kappa Pv^2/4}$. Choosing $R(a)$ as specified there leaves a positive fraction of the first-half mass. This argument uses the given exact type and both marks, including the case of equal marks. $\square$

The reference rank drift is $\varepsilon(\text{EW} - c\tau) = (y - x + \varepsilon c\tau)/n$. The terminal unmarked gap accounts for its difference from the physical drift $(y - x)/P$. It is retained in both the density and the path calculations.

B.2 Uniform scalar estimates at final complete types

We assemble the binary bridge estimate in the form used after the arithmetic count adjustment. The complete type is fixed before the spatial clock is read. We choose a legal common endpoint pair, obtain its strong fractional survival moment, and then bound every admitted type by one uncolored envelope. This separates the lower family at a specified type from the upper envelope that must work uniformly over all types and reveal assignments.

B.2.1 Count types and endpoint pairs

Condition on one original uncolored total n . The physical block has length P , with n/P in a fixed positive compact interval. Its fine word at any final feasible complete type is a uniform multiset permutation, as proved in Section C.2. A fixed binary bundle has exact size k , so its mark word is a uniform k -subset of the n original slots, independently of the uniform uncolored clock. All $n + 1$ gaps are retained.

Write its local risk as

$$R(t) = ct - aN(t) + bN_B(t), \quad a > b > 0, \quad \Delta = cP - an + bk, \quad (\text{B.2.1})$$

with $c, b, a - b, n/P$ in fixed positive compact intervals. Its final binary fraction $\pi = k/n$ is in a fixed interior interval; adjusting exterior counts changes it by a fixed multiple of $n^{-1/2}$, not by an order-one fraction. For the selected edge assume the actual intrinsic endpoint minima, at every final admitted type, satisfy

$$a_*X \leq x_{\text{act}} \leq b_*X, \quad a_*Y \leq y_{\text{act}} \leq b_*Y, \quad y_{\text{act}} - x_{\text{act}} = \Delta, \quad X, Y \geq h_*. \quad (\text{B.2.2})$$

Here the scores X, Y are fixed before the unread spatial clock. The common local pair is

$$x^- = \max\{a_*X, a_*Y - \Delta\}, \quad y^- = x^- + \Delta. \quad (\text{B.2.3})$$

It obeys the same positive lower bounds and the upper bounds in (2), and is below every actual endpoint minimum. Its chord is therefore below every actual intrinsic chord with the same local increment. Keeping the increment fixed is what makes this common pair legal; its endpoints are chosen together.

Set

$$T = \frac{(1 + X)(1 + Y)}{n}, \quad p_0 = \frac{a - cP/n}{b}, \quad \rho_0 = \frac{(cP/(n + 1))^2}{(cP/(n + 1))^2 + b^2p_0(1 - p_0)}. \quad (\text{B.2.4})$$

On a fixed rare branch $T < t_0$, (1)–(3) give exactly

$$|\pi - p_0| = |\Delta|/(bn) \leq C(X + Y)/n \leq CT. \quad (\text{B.2.5})$$

Take t_0 small so p_0 is uniformly interior. The resulting interior bound verifies feasibility at the actual binary type as well as at the uncolored total.

B.2.2 Gaussian estimates

The Gaussian input consists of three comparisons: endpoint tails under the survival tilt, constant-factor multiplicativity, and bounded-action walls after Brownian time change. We collect them here so that the relative discrete transfer below has a fixed reference lower constant.

We apply the estimates of Section 5 with η in place of λ . For the private endpoint distribution μ , with mode m , Lemma 5.3 gives

$$|W - m| \preceq_{\text{st}} |N(0, 1)|, \quad |\mathbb{E}W - m| \leq \sqrt{2/\pi}. \quad (\text{B.2.6})$$

Its proof also gives the decreasing environmental majorant

$$m_t(E) \leq M_t(E) = C \left[1 + \sum_{l \geq 0} e^{-l/2} \sup_{s \in I_l} (-\sqrt{\rho/(1-\rho)} E_s)_+ \right], \quad (\text{B.2.7})$$

where each interval I_l has length at most two, and $\sup_t \mathbb{E} \exp(cM_t^2) < \infty$ for some $c > 0$. In particular,

$$\mathbb{E}_E \left[e^{aE_t^2} (\mathbb{E}_W [1_{\text{survive}} e^{aW_t^2} \mid E])^\eta \right] \leq CF(t) \quad (\text{B.2.8})$$

for a sufficiently small fixed $a > 0$, also simultaneously at both endpoints. Theorem 5.4 supplies

$$F(s)F(t) \leq F(s+t) \leq CF(s)F(t), \quad C^{-1}e^{-\chi t} \leq F(t) \leq e^{-\chi t}. \quad (\text{B.2.9})$$

Together with the bounded-action wall comparison, this yields

$$\mathbb{E} q_{\text{Brownian}}^\eta \asymp \min\{1, ((1+h)/\sqrt{n})^{2\chi(\eta, \rho)}\} \quad (\text{B.2.10})$$

when $|\delta|\sqrt{n} \leq D$ and the two channel variances lie in fixed positive compact intervals; see Corollary 5.12. The endpoint boxes and strong curves in that corollary can be retained. Proposition 5.5 controls changes in the parameters. Quantitative constants for these estimates are given in Section B.5.

B.2.3 The relative discrete-to-Gaussian comparison

For iid exponential gaps and iid binary marks, the reference one-end risk is a sum of two independent centered channels plus drift. Couple the channels separately on independent successive dyadic blocks. Section B.3 derives the required numerical coupling from [5, Theorem 2.2(ii), equation (2.4)]. On a block of length L , its error is at most $1000M^2(A+4)^2 \log^3(L \vee e)$ and its failure probability is at most $5L^{-A}$, for every fixed $A \geq 1$. The environmental and private channels use independent copies. Summing successive dyadic errors gives a fourth-degree logarithmic majorant; its normalized Cameron–Martin action and variation are bounded in Section B.3.4.

The coupling error must be paid at the scale of survival. We obtain that relative estimate from flat preparation and reserve comparisons, before using the Gaussian exponent. For either risk sign let $F_n(h)$ first mean its flat-wall fractional one-end survival moment. Optional stopping at a fixed lower reserve or upper height B , followed by a scaled positive tube, prepares height u in time $O((1+u)^2)$ with ordinary probability $\geq c/(1+u)$. The walk and both channel sums can be kept central: their stopped second moments are $O(B)$, so exclusion beyond RB costs $O(1/(R^2B))$, a small fraction of the B^{-1} preparation. Exponential overshoots and bounded opposite overshoots justify optional stopping by truncation. Bounded scaled drift is paid by an exponential-mean

likelihood ratio on that central preparation. Concatenate independent survival from $h + u$. Time monotonicity, $\mathbb{E}p^\eta \geq \mathbb{E}p$, and the same tube prove

$$F_n(h) \geq cn^{-1/2}, \quad F_n(h+u) \leq C(1+u)^C F_n(h), \quad F_{2n}(h) \geq cF_n(h). \quad (\text{B.2.11})$$

For doubling, first remove environments whose unfavorable centered gap sum exceeds $R\sqrt{n}$. The unfavorable event has monotonicity opposite to survival, so association pays its weighted loss by its ordinary loss. On the retained environments, a favorable private centered endpoint of size $C\sqrt{n}$ has constant probability. Its monotonicity agrees with survival, so their intersection retains that fraction of the conditional mass.

The resulting reserve permits an independent future interval with constant ordinary success probability. For its completed private mass, $0 \leq U \leq q$ and $\mathbb{E}U \geq cq$ imply $\mathbb{E}U^\eta \geq cq^\eta$. Averaging the future in this order proves the last part of (11).

The flat preparation also proves the reserve comparison for a negative soft wall $-gj^\gamma$. After preparation at time m , the future condition at $-gj^\gamma$ is stronger than the shifted global condition at $-g(m+j)^\gamma$. For a positive strong wall we instead use flat-survival domination below. The flat and negative-wall comparisons hold for Gaussian increments as well.

If a bad event D reads the first $b \leq n/2$ slots and their coupling auxiliaries, discard initial survival and bin the positive prefix height in unit bins. It has Gaussian/exponential tails, and each bin's probability is at most both $\mathbb{P}(D)$ and its height-tail probability. Apply (11) to the remaining reserve, use the geometric mean of those two probability bounds, and sum the bins. This gives

$$\mathbb{E}\mathbb{P}_{\text{private}}(\text{survival}, D \mid \text{extended environment})^\eta \leq C(1+b)^2 \mathbb{P}(D)^{\eta/4} F_n(h). \quad (\text{B.2.12})$$

If D begins after $n/4$, retain necessary survival on the first $n/8$ slots, independent of D , and use Jensen and finitely many doublings in (11). Its relative cost is at most $C\mathbb{P}(D)^\eta F_n(h)$. The argument applies on each marginal of the coupling; the environmental coupling stays outside the private conditional probability and the private coupling stays inside. All independent reference increments belong to the original coupling space; survival remains inside the conditional private integral.

By (12), all dyadic coupling failures after a fixed scale L_0 have total relative fractional cost at most

$$CF_n(h) \sum_{L \geq L_0, L \text{ dyadic}} L^{2-A\eta/4}. \quad (\text{B.2.13})$$

Choose A , then L_0 , so this coefficient is below the fixed Gaussian lower constant from (10). The first finite block has an exponentially decaying maximum-path tail. Choose its reserve threshold last; (12) then also makes its bad relative cost small. On the good coupling event, the accumulated discrepancy at rank time $t \geq L_0$, including the within-block interpolation, is at most $C(1+\log t)^4$. Indeed the error on a block of length L is $O(\log^3(L \vee e))$, and there are $O(1+\log(t/L_0))$ completed or partly read dyadic blocks before t . In Lamperti time $u = \log(t/L_0)$ a smooth majorant for the normalized discrepancy is

$$w(u) = \frac{C}{\sqrt{L_0}} (1 + \log L_0 + u)^4 e^{-u/2}.$$

Lemma B.4, with $m = 4$, bounds its Cameron–Martin action and the square of its variation uniformly in the final horizon. The explicit calculation in Section B.3.4 gives $A(w) + V(w)^2 < M^{-100}$

at the stated starting scale. The first finite coupling block is treated separately and consumes a fixed fraction, for example $h/8$, of the reserve. Path inclusions and fractional subadditivity therefore first give the discrete flat-wall upper and lower comparisons from (10), with each failure absorbed against the corresponding flat marginal moment in (13). This proves flat discrete/Gaussian comparability before a positive strong curve is imposed. Positive strong-wall survival is a subset of flat survival, so its bad-event moment is bounded by (12) for the flat experiment. The Gaussian strong-wall lower with compatible central separate endpoint boxes is at least a fixed positive multiple of the Gaussian flat moment by the bounded-action comparison above. The already proved flat comparability now absorbs that error. Choose the failure coefficient smaller than this fixed multiple divided by the flat comparison constant. Thus strong curves and their compatible boxes survive, without any linear reserve comparison for a positive wall. A negative weak wall, when needed for an upper path inclusion, uses its valid negative-wall version of (12); its bounded-action Gaussian upper and an error coefficient below $1/2$ absorb its own error first. These choices use the fixed Gaussian comparison and the already proved flat relative bounds, so they precede the strong discrete normalization.

Retaining regularity. Before exact conditioning, interval tails and the same relative early/late decomposition permit local REG to be added at a fixed fraction of the survival moment. A failure on a rank interval of length l and endpoint depths d_1, d_2 has probability at most $C \exp[-cl - cK(1 + d_1^\zeta + d_2^\zeta)]$. For early intervals apply (12); for late intervals retain independent early necessary survival; intervals joining the two regions have length a fixed fraction of the horizon and their length tail pays the reciprocal of the crude polynomial lower (11). Summing the fixed-grid pairs and using fractional subadditivity deletes at most $C \exp(-cK)F_n(h)$ of the moment. Gap REG converts rank depths to physical depths; the discrepancy in a strong γ curve is a small multiple of rank depth to γ plus a fixed intercept, because $\zeta < \gamma$. Choose curve amplitude before the intercept and final minimum reserve. Thus the discrete one-end theorem holds with physical strong curve, binary REG, and compatible central boxes of the separate channels.

B.2.4 Joint conditioning on the total and type

The uniform clock is $n + 1$ independent exponential gaps conditioned on their sum $P = (n + 1)\tau$; the marks are iid Bernoulli $\pi = k/n$ conditioned on their exact count. Exposing m marks and m distinct gaps has joint reference density

$$\frac{f_{n+1-m,\tau}(P - T_m) \mathbb{P}\{\text{Bin}(n - m, \pi) = k - K_m\}}{f_{n+1,\tau}(P) \mathbb{P}\{\text{Bin}(n, \pi) = k\}}. \quad (\text{B.2.14})$$

Its upper is constant on the whole support for $m \leq n/4$; its lower is constant on fixed central exposed totals, above a fixed length threshold. This follows directly from Gamma/binomial maximum atoms and central Stirling bounds. The two-end formula uses $2m$ jointly, including the final unmarked gap at the reversed end. Thus both exposed ends are compared in one density, retaining their common residual total and type.

Suppose the legal reserves x, y are both of order at most $\sqrt{n}$, and expose $m = \lfloor n/8 \rfloor$ slots at each end. Their reference drift is $\epsilon(w_+ - b\pi - c\tau) = (y - x + \epsilon c\tau)/n$, with bounded scaled magnitude. The density upper in (14) bounds the moment by the product of independent reference one-end moments.

For the lower, retain the compatible positive central inner heights and the separate central totals from the relative coupling. The remaining exact clock and type have compact parameters.

Their mean path joins heights at least $a\sqrt{n}$, with an $O(1)$ pre-jump correction. A tube of radius $a\sqrt{n}/2$ therefore gives constant ordinary success probability.

To obtain this tube, split each exact bridge into a fixed number J of subblocks. Central Stirling bounds give constant probability that their totals lie in intervals of width $\delta\sqrt{n}$. Conditional on these totals, exponential maximal tails control the internal fluctuations. Choose J large enough to make the combined failure at most $1/2$, and then choose δ small. The gap and binary channels remain independent. Finally, interval Bernstein bounds and the depth summation give ordinary middle REG failure at most $C\exp(-cK)$, which can be paid against this fixed tube probability.

For each fixed outer clock let $q = q_L q_R$. The full private mass U , as a function of the remaining clock, obeys $0 \leq U \leq Cq$ and $\mathbb{E}_{\text{middle}} U \geq cq$. Hence

$$E_{\text{middle}} U^\eta \geq (Cq)^{\eta-1} E_{\text{middle}} U \geq c'q^\eta. \quad (\text{B.2.15})$$

Use the central lower in the single joint outer density (14), then average the independent reference ends. Thus the exact-type strong-wall/REG fractional moment is comparable to

$$Q_n(x, y) = \min\{1, [(1+x)(1+y)/n]^{2\chi(\eta, \rho)}\}, \quad \rho = (c\tau)^2 / [(c\tau)^2 + b^2\pi(1-\pi)]. \quad (\text{B.2.16})$$

For the remaining endpoint regimes, two large reserves are treated by a whole-bridge tube and unequal reserves by an effective prefix. If both exceed $\sqrt{n}$, a whole-bridge small tube has constant ordinary probability. If one is smaller and the other much larger, expose only an effective prefix of length m comparable to $n^2/(1+y)^2$. Its bounded scaled drift gives the one-end factor in (16). For the lower its retained height is H comparable to $\sqrt{m}$ and remaining drift v comparable to y/n , so $Hv \geq a_0 > 0$ and the remaining horizon times v^2 is as large as needed after fixing the asymmetry cutoff. For the remaining exact bridge, compare its first half to two independent color Poisson clocks. Their joint density is bounded above everywhere and below on central half totals. The exponential supermartingale $\exp(-\kappa v Z)$ gives ordinary survival from a_0/v with positive constant probability on that first half. On the reversed half, a centered maximal inequality bounds failure by $C\exp(-cPv^2)$. Choose its fixed threshold to make this smaller than half the retained first-half mass. A half-chord version stays above $(H + vt)/2$, which dominates the sub-square-root curve uniformly in the remaining horizon. The same REG deletion applies. Integrate this remaining clock by (15). When the effective m is bounded, the physical drift is bounded below; one fixed minimum reserve pays the curve and the same continuation gives the constant factor.

This proves Theorem 6.1 with the relative marked fractional moment, the feasible remaining exact type, and all $n+1$ gaps. The middle and tail clocks are integrated before the fractional power. One fixed large reserve and regularity intercept cover the bounded supported experiments. Deterministic marks use the separate ordinary bridge argument.

B.2.5 Contracted actual-type chord and the required persistence factor

We now return to the endpoint scores fixed by the arithmetic type. Contracting their legal chord spends only a constant fraction of the reserve budget. The same-word identity below changes the speed by a small amount on the rare branch, and parameter stability gives the required canonical persistence factor.

Write the original exact bridge as $\ell(t) + E(t)$. For a fixed $\theta \geq \theta_0 > 0$, its contracted chord has the same word identity

$$\theta\ell(t) + E(t) = \theta x + \varepsilon\{S(t) - c_\theta t\}, \quad c_\theta = \theta c + (1-\theta)M/P. \quad (\text{B.2.17})$$

The speed remains in a fixed positive compact interval. On the rare branch, $|c_\theta - c| \leq C(X+Y)/n \leq CT$. Together with (5) and the $n/(n+1)$ gap correction, local Lipschitz continuity gives

$$|\chi(\eta, \rho_{\text{contracted, type}}) - \chi(\eta, \rho_0)| \leq CT.$$

The contracted endpoint product is comparable to T . Changing the exponent therefore costs at most $\exp[CT|\log T|]$, bounded uniformly. Equation (16), with (3), proves that the contracted curved binary path and binary REG have fractional moment at least

$$c\Psi, \quad \Psi = T^{2\chi(\eta, \rho_0)} \quad (T < t_0), \quad \Psi = 1 \quad (T \geq t_0). \quad (\text{B.2.18})$$

The constants are uniform over every final complete type in (2). We can now impose fine-name regularity pointwise on each retained binary word. At a fixed fine refinement type, the two parent lists have uniform multiset refinements. An interval containing m parent positions has hypergeometric named count with mean $(k_c/n_{\text{parent}})m$ and tail $2\exp[-cu^2/(m+u)]$.

Parent binary regularity bounds m and the difference between this mean and the physical empirical rate. Choose the parent slope below the prescribed named slope, then choose the named intercept large. A named failure has bound $C\exp[-cl - cK(1 + d_1^\zeta + d_2^\zeta)]$, whose grid-pair sum is uniformly finite. Thus at every retained binary word and complete fine type the refinement succeeds with probability at least $1 - \epsilon$. The named restriction retains $(1 - \epsilon)^\eta$ of the binary fractional moment before the rare path is averaged. Only the uniform multiset laws inside the parent lists are needed.

Macroscopic recoloring banks can also be admitted. At a central complete fine type and fixed interior interval of length comparable to the block, its uncolored occupancy is binomial and its fine counts are hypergeometric. Failure of the broad bank window has ordinary probability at most $C\exp[-cn/\log^2(2+n)]$, uniformly in the type. If q_{bad} is the already restricted conditional mass lost to this failure, Jensen gives

$$Eq_{\text{bad}}^\eta \leq (Eq_{\text{bad}})^\eta \leq Ce^{-c\eta n/\log^2(2+n)}.$$

Since $\Psi \geq c(2+n)^{-1}$ from $\chi \leq 1/2$ and the endpoint floor, this is smaller than a fixed fraction of (18), above one fixed length threshold. Fractional subadditivity $(q - q_{\text{bad}})^\eta \geq q^\eta - q_{\text{bad}}^\eta$ proves the broad bank retention.

B.2.6 The type-uniform uncolored envelope and its matching moment

The lower family just constructed is type-dependent. For the reveal argument we also need an upper envelope depending only on the uncolored clock. At each reveal depth, two unread intervals give necessary one-end tests. Their lengths adapt to the larger reserve, and a summable depth weight combines their private masses into a single envelope with the same fractional scale.

On the saturated branch $T \geq t_0$, set $q_d = D = 1$ at every reveal depth. Then $\Psi = 1$ and the envelope upper bounds are immediate. The source retains the separate fixed positive lower bound established above. The construction below therefore concerns $T < t_0$, as in Appendix C.

Binary REG implies the endpoint occupancy bound $N([0, t]) + N([P - t, P]) \leq C_{\text{occ}}(1 + t)$. At a prescribed reveal depth d , leave strips of width $C_s(1 + d)$ and use two disjoint original free intervals of length

$$L_d = \epsilon_0 \min\{P, (n/M_d)^2\}, \quad U_d = 1 + X + d, \quad V_d = 1 + Y + d, \quad M_d = \max(U_d, V_d). \quad (\text{B.2.19})$$

On active depths $d \leq [C_{\text{cut}} \log(2+n)]^{1/\gamma}$, a fixed large length threshold and small ϵ_0 ensure their combined occupancy is at most half of the unread slots. Fix the complete type and every actually revealed label. If their counts are m, e , the remaining binary word is an exact $(k-e)$ -subset of $n-m$ slots. Its fraction differs from p_0 by at most $C(X+Y+1+d)/n$. The joint exact subset density on both free intervals, relative to iid Bernoulli of that remaining fraction, is bounded by a constant: its numerator is a maximum binomial atom on at least half the slots and its denominator is a central atom on all unread slots.

Use Bernoulli probabilities $p_+ = p_0 + C_1(X+Y+1+d)/n$ for the increasing left necessary test, and $p_- = p_0 - C_1(X+Y+1+d)/n$ for the decreasing reversed right test. Coupling uniforms coordinatewise gives the necessary path upper $Cq_d(G) = Cq_L(G)q_R(G)$ at every supported type and reveal assignment. These coordinatewise couplings bound necessary tests under the original exact type; their reference Bernoulli laws define the upper experiment only.

The uniform exact-total clock on both fixed intervals has joint density relative to a rate n/P Poisson clock

$$\frac{\mathbb{P}\{\text{Pois}(n(1-2L_d/P)) = n - N_J\}}{\mathbb{P}\{\text{Pois}(n) = n\}} \leq C. \quad (\text{B.2.20})$$

Only after this one density comparison are the reference intervals independent. Their physical drifts times $\sqrt{L_d}$ are bounded. The physical one-end upper (proved by a rank prefix and the same Gamma conditioning/relative comparison above) gives

$$Eq_d^\eta \leq C \min\{1, (U_d/\sqrt{1+L_d})^{2\chi_+}\} \min\{1, (V_d/\sqrt{1+L_d})^{2\chi_-}\} \leq C(1+d)^C \Psi. \quad (\text{B.2.21})$$

To compare the right side with the canonical factor, separate the two length regimes. If $M_d \leq \sqrt{n}$, then L_d is comparable to n . Otherwise the larger-reserve factor is saturated, and the smaller reserve ratio is comparable to $U_d V_d/n$. The exponent changes by $O(T + (1+d)/n)$. Every nontrivial reserve ratio is at least a fixed multiple of $\min(1, T)$, so the logarithmic exponent-change costs are uniformly bounded. The remaining depth loss follows from $U_d V_d \leq (1+d)^2(1+X)(1+Y)$.

If a free interval lies below the scalar threshold, take $q_d = 1$. Then a reserve is of order n , and Ψ is bounded below. Beyond the active depth cutoff, take $q_d = 1$ as well; the coefficient tail in the envelope pays for these depths.

Choose fixed $B_0, b_0 > 0$, and D_0 with $D_0\eta_- > 2\sup \chi + 1$. Set

$$D(G) = \min \left\{ 1, (2+n)^{-D_0} + \sum_{d \geq 0} (1+d)^{B_0} e^{-b_0 d^\gamma} q_d(G) \right\}. \quad (\text{B.2.22})$$

Choose C_{cut} large after D_0 so the coefficient tail raised to any allowed plateau power is $O((2+n)^{-D_0\eta})$. Fractional subadditivity and (21) then prove

$$(2+n)^{-D_0} \leq D \leq 1, \quad ED^\eta \leq C\Psi. \quad (\text{B.2.23})$$

At depth zero the necessary original path bound gives $q \leq CD$ for every complete type. Additional curved walls, named regularity and broad banks only reduce q . The uncolored envelope is fixed before any target and before any complete type is read. If several blocks share a history-depth charge, their finite number permits choosing b_0 below its allocated positive coefficient; the floor exponent is unchanged.

B.2.7 Result at actual complete count types

The strong lower family and its named refinement give the lower moment, while the necessary tests give the pointwise envelope and its upper moment. Combining (18), the relative broad-bank retention, and (23) yields

$$\boxed{q \leq CD, \quad Eq^\eta \geq c\Psi, \quad ED^\eta \leq C\Psi, \quad (2+n)^{-D_0} \leq D \leq 1}. \quad (\text{B.2.24})$$

The expectations are in the original uncolored clock at its fixed actual total. The private probability is in the original uniform mark/permutation law at the fixed final complete type. Each plateau uses its own power in its compact interval. All selected spatial tests are imposed after the count adjustment, under the final complete type; the lower family is constructed at that type rather than transported from a previously accepted path.

Ordinary deterministic marks have their separate exact bridge factor $\min(1, (1+X)(1+Y)/(1+n))$. The same two-end preparation, middle tube, asymmetric continuation and REG deletion establish that factor. Its deterministic private marks are treated by the ordinary bridge construction, with its own ballot factor.

B.2.8 The full relative bad-event estimate

We give the full calculation behind the relative deletion used above. A bad prefix can improve later survival through a favorable ending height, so we bin that height and pay it by the reserve comparison. A late bad block instead factors from a necessary early survival prefix. Both arguments include coupling auxiliaries in their sigma-fields.

Take $b \leq n/2$. Given the extended prefix environment, let p_r be the private probability of D and positive prefix increment in $[r, r+1)$, using $r = 0$ for all nonpositive increments as well. Dropping prefix survival and using independence of the remaining original channels gives

$$Eq_{\text{survive},D}^\eta \leq \sum_{r \geq 0} Ep_r^\eta F_{n-b}(h+r+1).$$

Here F is the flat-wall experiment. Its preparation proof gives the linear reserve comparison $F_m(h+u) \leq C(1+u)F_m(h)$. In detail, stop a zero-mean walk on leaving the interval between a fixed positive lower reserve and a height of order $1+u$. Optional stopping, bounded mean overshoot and a fixed central tube give a preparation event of ordinary probability at least $c/(1+u)$ which ends above $h+u$ by time $C(1+u)^2$. For $u \leq C_0\sqrt{m}$ the bounded scaled drift is paid by its exponential likelihood on the central preparation totals. For $u > C_0\sqrt{m}$, the crude lower $F_m(h) \geq cm^{-1/2}$ and $F_m(h+u) \leq 1$ prove the reserve comparison directly, with no long preparation at nonzero drift. Attach an independent future of length m . The preparation may lengthen the full survival event, which only strengthens its implication of survival for the first m steps. Given its successful clock/mark preparation, use the future conditional private mass from reserve $h+u$. Averaging the preparation before raising to η retains at least its ordinary success probability, since $p^\eta \geq p$. Independence then gives $F_m(h) \geq c(1+u)^{-1}F_m(h+u)$.

This argument is also valid for a negative soft power wall $-gj^\gamma$: the flat preparation is stronger, and the attached future wall $-gj^\gamma$ is above the global wall $-g(m+j)^\gamma$. It is not the proof for a positive wall $+gj^\gamma$, whose shifted global wall is higher. For positive strong walls only the flat bad-event majorant is used. For a negative soft wall, dropping its first b steps changes the remaining wall by at most gb^γ , because $(b+j)^\gamma \leq b^\gamma + j^\gamma$. Include that amount in the remaining reserve;

the factor is at most $C(1 + r + b^\gamma)$ and is absorbed in the quadratic b loss below. Doubling and time monotonicity give $F_{n-b}(h) \leq CF_n(h)$ since $n \leq 2(n - b)$. Jensen then implies

$$Ep_r^\eta \leq \mathbb{P}(D, \text{bin } r)^\eta \leq \mathbb{P}(D)^{\eta/2} \mathbb{P}(H_b^+ \geq r)^{\eta/2} \quad (r \geq 1).$$

Bounded scaled drift implies $|\mathbb{E}H_b| \leq C_D\sqrt{b}$; exponential moments of the centered gap or Gaussian channel give $\mathbb{P}(H_b^+ \geq r) \leq C_D \exp[-c \min(r^2/(1+b), r)]$. The fixed drift displacement is absorbed in the constant for $r \leq 2C_D\sqrt{b}$ and in a smaller tail exponent otherwise. For $r = 0$, use $Ep_0^\eta \leq \mathbb{P}(D)^\eta$. Summing the Gaussian part on $r \leq 1 + b$ and the exponential part afterwards gives

$$\sum_{r \geq 0} (2 + r) Ep_r^\eta \leq C(1 + b) \mathbb{P}(D)^{\eta/2} \leq C(1 + b)^2 \mathbb{P}(D)^{\eta/4}.$$

This proves (12) for the flat experiment and its negative-wall variant. In the bounded-positive-jump orientation use $H_b \leq lb$ directly to get the stronger $C(1 + b) \mathbb{P}(D)^\eta F_n(h)$. The tails used here are unconditional intersection and height tails on a marginal of the product coupling. Coupling auxiliaries only enlarge the prefix σ field; the unconditional intersection and tail bounds above remain valid on either marginal of the coupling.

For a late D , retain necessary survival on the first $\lfloor n/16 \rfloor$ steps. Every coupling block read by D , including all auxiliaries, is disjoint from every whole coupling block used by that prefix; these complete sigma-fields are independent on the product coupling space. The bounded range in which the initial block prevents this separation is treated by Jensen and the crude survival lower bound. Conditional private masses factor for this relaxed event. Thus

$$Eq_{\text{survive}, D}^\eta \leq F_{\lfloor n/16 \rfloor}(h) E \mathbb{P}_{\text{private}}(D \mid E_{\text{future}})^\eta \leq CF_n(h) \mathbb{P}(D)^\eta,$$

by Jensen and a fixed number of doublings. A negative soft wall has the same necessary initial survival on its original prefix and the same doubling argument. This supplies the two relative failure cases needed for every dyadic block in (13). Positive strong-wall failure deletion is performed only after flat comparability has been established by the relative coupling argument above.

B.2.9 Uniform exact-type asymmetric continuation

The continuation is needed separately for every retained complete prefix. We compare the first half with independent color Poisson clocks, retain a central count window within its survival mass, and use a reversed maximal estimate to pay for late failure.

For two exact color counts n_i on duration P , write $\lambda_i = n_i/P$, let all marks be at most w_+ , and let $\tau = P/(n + 1) \geq \tau_-$. The exact type/clock law is equivalent to independent uniform positions for the two colors. Relative to their independent Poisson clocks, the density on the first half is

$$\prod_{i=0}^1 \frac{\mathbb{P}\{\text{Pois}(n_i/2) = n_i - N_i(P/2)\}}{\mathbb{P}\{\text{Pois}(n_i) = n_i\}}.$$

It is bounded above by C_{den} everywhere and below by $c_{\text{den}}(K) > 0$ on $|N_i(P/2) - n_i/2| \leq K\sqrt{n_i}$ once $n_i \geq N_0(K)$. The same statements hold on the reversed half. Set

$$C_{\text{exp}} = w_+^2 e^{w_+} / (2\tau_-), \quad \kappa = \min\{v_{\text{max}}^{-1}, (2C_{\text{exp}})^{-1}\}.$$

For positive signed drift v and either orientation, Taylor's bound gives exponential compensator at parameter $\theta \leq 1$ at most $-v\theta + C_{\text{exp}}\theta^2$. At $\theta = \kappa v$, $\exp(-\theta Z)$ is a nonnegative supermartingale. Starting from $x \geq a/v$, the Poisson first-half exit probability is at most $\exp(-\kappa a)$, so survival has probability at least $p_a = 1 - \exp(-\kappa a) > 0$. Choose $K \geq \max(1, \sqrt{2/p_a})$. Chebyshev on the two Poisson counts has total failure at most $K^{-2} \leq p_a/2$. Consequently exact first-half survival has probability at least

$$b_{\text{left}}(a) = c_{\text{den}}(K)p_a/2 > 0.$$

At a time $P - u$ on the late half the exact identity is $Z(P - u) = x + v(P - u) - \epsilon M_{\text{rev}}(u)$, where M_{rev} is the centered reversed color-weight sum. Failure forces $\sup_{u \leq P/2} \epsilon M_{\text{rev}}(u) \geq vP/2$. The Poisson exponential maximal inequality, and then the exact half-density upper, give

$$P_{\text{exact}}(\text{late-half failure}) \leq C_{\text{den}}e^{-\kappa P v^2/4}.$$

Choose $R(a)$ so that

$$\begin{aligned} R(a) &\geq (4/\kappa) \log\{2C_{\text{den}}/b_{\text{left}}(a)\}, \\ R(a)/v_{\text{max}}^2 &\geq \tau_+[N_0(K)/\epsilon + 1], \end{aligned}$$

where both color fractions are at least ϵ . Then $Pv^2 \geq R(a)$ ensures the central half-density thresholds and a late failure at most $b_{\text{left}}(a)/2$. The full exact bridge survives with probability at least $b_{\text{left}}(a)/2$, uniformly in its individual exact type.

In the asymmetric binary bridge's retained effective prefix, m is comparable to δ^{-2} , its height H is comparable to $\sqrt{m}$, and its remaining physical drift v' is comparable to δ . Increasing the fixed asymmetry cutoff makes $Hv' \geq a_0$ and $P'(v')^2 \geq 4R(a_0/4)$ for every retained prefix. Use the same actual remaining marks with speed $c_{\text{half}} = (c + M'/P')/2$, entrance $H/2$, and drift $v'/2$. The preceding continuation then has constant probability and implies the original path stays above $(H + v's)/2$. Because H and v' are comparable to $\sqrt{m}$ and $1/\sqrt{m}$, respectively,

$$(H + v's)/2 \geq c\sqrt{m}(1 + s/m), \quad d_P(T_{\text{pre}} + s)^\gamma \leq Cm^\gamma(1 + s/m)^\gamma.$$

One fixed effective-time threshold pays any prescribed global sub-square-root curve for all remaining horizons. Empirical remaining color rates differ from the full ones by $O(\sqrt{m}/n) \leq O(n^{-1/2})$; choose ordinary regularity intercept after the above continuation lower constant. The central prefix density (14) has a uniform lower for all retained complete prefix totals. Thus, at each fixed outer clock, the full mass satisfies $0 \leq U \leq Cq_{\text{pre}}$ and $\mathbb{E}_{\text{remaining}}U \geq cq_{\text{pre}}$. Equation (15) retains its fractional moment. The central density and continuation estimates apply to each retained prefix, with the remaining total and type determined by that prefix.

B.3 Coupling of the rank channels

We now derive the two scalar rank couplings and their constants. The exponential channel is obtained by integrating an empirical process coupling; truncation near the singular endpoint produces the additional logarithmic factor. Independent copies then give a product coupling for the environmental and private channels.

The primary source is Castelle and Laurent-Bonvalot, *Strong approximations of bivariate uniform empirical processes*, Ann. IHP 34 (1998), 425–480, Theorem 2.2(ii), equation (2.4), printed page 429, together with the construction in Section 3.2, pp. 438–439 [5]. The latter writes

$K(m, u) = \sum_{i \leq m} B_i(u)$ and proves the bound with this index. We follow that construction, correcting the printed $K(n, u)$ in (2.4) to $K(m, u)$. The theorem is available at Numdam.

In their convention $\ell_n = \log(n \vee e)$. For iid uniform U_i and their centered counting process $R_m(u) = \sum_{i \leq m} (\mathbf{1}_{U_i \leq u} - u)$, a Kiefer process can be coupled so that

$$\mathbb{P} \left(\max_{1 \leq m \leq n} \sup_{0 \leq u \leq 1} |R_m(u) - K(m, u)| > (x + 60\ell_n)\ell_n \right) \leq 0.67e^{-x/30}. \quad (\text{B.3.1})$$

The Kiefer covariance is $\min(s, t)[\min(u, v) - uv]$. This empirical-process estimate is the sole coupling input. The Bernoulli specialization, exponential integral, interpolation error, and dyadic accumulation are calculated below.

B.3.1 Uniform Bernoulli marks

Fix any $p \in [0, 1]$. Let $B_i = \mathbf{1}_{U_i \leq p}$ and $W_p(m) = K(m, p)$. The increments of W_p are independent centered Gaussians of variance $p(1 - p)$. At $x = 30(A + 2)\ell_n$, (1) gives

$$\max_{m \leq n} \left| \sum_{i \leq m} (B_i - p) - W_p(m) \right| \leq 30(A + 4)\ell_n^2 \quad (\text{B.3.2})$$

with failure probability at most $0.67n^{-A-2}$, for every $A \geq 1$, $n \geq 2$. This holds uniformly in p ; positivity of the variance is only needed later for a scalar persistence theorem.

B.3.2 Centered exponential gaps

Use an independent copy of (1), and set $E_i = -\log(1 - U_i)$. These are iid mean-one exponentials. Define the Gaussian process

$$W_E(m) = - \int_0^1 \frac{K(m, u)}{1 - u} du \quad (\text{B.3.3})$$

as a mean-square integral. It is Brownian motion at integer times: its Gaussian increments are independent, and its variance per increment is

$$\int_0^1 \int_0^1 \frac{\min(u, v) - uv}{(1 - u)(1 - v)} du dv = 1.$$

The empirical identity is

$$\sum_{i \leq m} (E_i - 1) = - \int_0^1 \frac{R_m(u)}{1 - u} du. \quad (\text{B.3.4})$$

Choose $\epsilon = \exp[-(A + 2)\ell_n]$. On the event $\max_{i \leq n} U_i \leq 1 - \epsilon$, the omitted empirical integral over $(1 - \epsilon, 1)$ has absolute value exactly $m\epsilon \leq n\epsilon$. The exceptional probability is at most $n\epsilon \leq n^{-A-1}$.

The omitted Gaussian integral is a Brownian motion in the first time coordinate of variance per unit time

$$v_\epsilon = \int_0^\epsilon \int_0^\epsilon \frac{\min(a, b) - ab}{ab} da db = 2\epsilon - \epsilon^2 \leq 2\epsilon.$$

By reflection, for $h > 0$,

$$\mathbb{P}\left(\max_{m \leq n} |W_{E,\text{tail}}(m)| > h\right) \leq 4 \exp[-h^2/(4n\epsilon)].$$

At $h = 4(A+2)\ell_n$ this is at most n^{-A-2} . Indeed $n\epsilon \leq n^{-A-1}$ and $4(A+2)^2\ell_n^2 n^{A+1} \geq (A+2)\log n + \log 4$ for $n \geq 2$, $A \geq 1$. On the good event in (1), the retained integral error is at most

$$30(A+4)\ell_n^2 \log(1/\epsilon) = 30(A+4)(A+2)\ell_n^3.$$

Adding the two tails gives the fully numerical estimate

$$\mathbb{P}\left(\max_{m \leq n} \left| \sum_{i \leq m} (E_i - 1) - W_E(m) \right| > 100(A+4)^2\ell_n^3\right) \leq 3n^{-A}. \quad (\text{B.3.5})$$

The last constants include all preceding three failure terms. A mean- τ exponential channel is obtained simply by multiplying (3)–(5) by τ .

B.3.3 The marked risk and independent dyadic blocks

Let $M \geq 1$, let c, τ, b be positive and bounded above by M , and take the centered rank risk

$$S_m = c\tau \sum_{i \leq m} (E_i - 1) + b \sum_{i \leq m} (B_i - p).$$

Couple the exponential and Bernoulli channels by independent copies of (1). The resulting two Brownian motions are independent, with the exact environmental/private variances $(c\tau)^2$ and $b^2p(1-p)$. From (2), (5),

$$\mathbb{P}\left(\max_{m \leq n} |S_m - W(m)| > 200M^2(A+4)^2\ell_n^3\right) \leq 4n^{-A}. \quad (\text{B.3.6})$$

The environmental coupling is outside the private conditional probability, and the private coupling is inside. Survival is evaluated under the original channel laws on this product space. Independent copies of the entire construction on successive deterministic dyadic blocks supply the block independence used in the relative bad-event proof.

For continuous Gaussian interpolation, reflection on each unit interval and a union bound add at most $20M^2(A+4)\ell_n$ to (6), with probability at most n^{-A} . Thus the enlarged constant $1000M^2(A+4)^2\ell_n^3$ has total failure at most $5n^{-A}$.

On dyadic blocks of lengths $2^j L_0$, up to physical rank s , the accumulated endpoint errors have deterministic good-event bound

$$1000M^2(A+4)^2 \sum_{j \leq \log_2(s/L_0)+1} [\log(2^j L_0 \vee e)]^3. \quad (\text{B.3.7})$$

This is a fixed polynomial of degree four in $\log(2+s)$ with explicit coefficients depending on M, A, L_0 . After Lamperti normalization it has bounded Cameron–Martin action. This accumulated wall therefore fits the scalar bridge comparison after the explicit wall-action and minimum-reserve inequalities are adjusted. The power-wall exponent and the parameter regimes remain the same.

For $n = 1$, a direct quantile coupling or the fixed initial-block treatment handles the bounded experiment; no asymptotic statement about that case is used in (2)–(7).

B.3.4 Action of the accumulated coupling-error wall

The good-event discrepancy becomes a polynomial times a decaying exponential in OU time. To insert it as a wall, we must control its derivatives as well as its amplitude. The next lemma bounds both the action and the total variation of its Cameron–Martin functional, uniformly in the horizon.

Lemma B.4 (A logarithmic coupling wall). *Fix an integer $1 \leq m \leq 4$, $a \geq 4m$, and $c \geq 0$. On any finite OU time interval $[0, T]$, put*

$$w(s) = c(a + s)^m e^{-s/2}.$$

With the action and variation functionals

$$A(w) = w(0)^2 + \int_0^T (w' + w/2)^2 ds,$$

$$V(w) = |w'(T) + w(T)/2| + |w(0)/2 - w'(0)| + \int_0^T |w/4 - w''| ds,$$

one has, uniformly in T ,

$$A(w) \leq 3m^2 c^2 a^{2m}, \quad V(w) \leq 10m^2 c a^m.$$

The same bounds hold for the reversed wall $w(T - s)$. For a sum of one wall at each end, the action is at most twice the sum of the individual actions and the variation is at most their sum.

Proof. For every $0 \leq j \leq 2m - 2$,

$$(a + s)^j \leq a^j e^{js/a} \leq a^j e^{s/2},$$

and for $0 \leq j \leq m - 1$ the last exponential can be replaced by $e^{s/4}$. Also

$$w' + w/2 = cm(a + s)^{m-1} e^{-s/2},$$

$$w/4 - w'' = ce^{-s/2} \{m(a + s)^{m-1} - m(m-1)(a + s)^{m-2}\},$$

where the last term is absent for $m = 1$. The corresponding integrals are bounded using $\int_0^\infty e^{-s/2} ds = 2$ and $\int_0^\infty e^{-s/4} ds = 4$. The endpoint term $(a + T)^{m-1} e^{-T/2}$ is at most a^{m-1} because $a \geq 2(m-1)$. Thus

$$A(w) \leq c^2 a^{2m} + 2m^2 c^2 a^{2m-2} \leq 3m^2 c^2 a^{2m},$$

$$V(w) \leq c\{a^m + 2ma^{m-1}\} + 4cm\{a^{m-1} + (m-1)a^{m-2}\} \leq 10m^2 c a^m.$$

Reversal preserves the action: expanding the two squares makes their difference an endpoint term that cancels the difference of the initial squares. It also interchanges the two boundary variation terms and preserves the interior one. Finally, the assertions for sums follow from $(u + v)^2 \leq 2u^2 + 2v^2$ and the triangle inequality. $\square$

Take the numerical compact parameter set $M \geq 10$, with the two centered channel standard deviations at least M^{-2} , the coefficients and mean gaps at most M^2 , and $\rho \geq M^{-1}$. In the

numerical rank coupling choose the tail power $A = 400M$. The continuous-interpolation bound on a block of length L is

$$1000M^4(A+4)^2 \log^3(L \vee e) \leq 2 \cdot 10^8 M^6 \log^3(L \vee e).$$

A block crossing rank time t has length at most $2t$. There are at most $2 + \log_2(t/L_0)$ blocks up to that time, and $\log(2t) \leq 2(1 + \log t)$. Summing these bounds gives

$$2 \cdot 10^8 M^6 \{2 + \log_2(t/L_0)\} \log^3(2t) \leq 10^{10} M^6 (1 + \log t)^4 \quad (t \geq L_0).$$

The total Gaussian standard deviation is at least M^{-2} . Dividing by it and then by $\sqrt{\rho}$ to place the wall in the environmental channel is therefore covered by the larger coefficient

$$C = 10^{10} M^{12}.$$

In particular the separate gap factors c and τ enter through the product $c\tau$ already present in the rank coupling. The coefficient bound is uniform in the time horizon and reserve.

On the good event, the accumulated stepwise error, including every jump and within-block interpolation, is pointwise dominated by this smooth majorant. In Lamperti time $s = \log(t/L_0)$, its wall is

$$w(s) = \frac{C}{\sqrt{L_0}} (1 + \log L_0 + s)^4 e^{-s/2}.$$

Apply the lemma with $m = 4$, $a = 1 + \log L_0$, and $c = C/\sqrt{L_0}$. Both its action and the square of its variation are bounded by $26000C^2 a^8/L_0$, uniformly in the final horizon. For

$$L_0 = \left\lceil \exp\{\exp(M^{400})\} \right\rceil,$$

one has $a \leq 2 \exp(M^{400})$. The logarithm of the last bound is at most

$$70 + 24 \log M + 8M^{400} - \exp(M^{400}) < -100 \log M.$$

For example, the last inequality follows from $e^u \geq u^2/2$, $u = M^{400} > 300$, and $\log M \leq u$. Thus $A(w) + V(w)^2 < M^{-100}$. The two-end version follows from the last assertion of the lemma. This calculation establishes the claimed bounded action uniformly over all later lengths.

A physical power wall of exponent $\gamma = 1/8$ has Lamperti decay exponent $1/2 - \gamma = 3/8$, which belongs to the numerical wall construction's interval $[1/3, 1/2]$. Reserve terms have exponent $1/2$. Therefore the logarithmic wall and all the reserve and power walls used here have the stated finite action and variation bounds.

B.4 Explicit exact-clock and exact-type middle continuation

Here we quantify the middle tube used when both inner endpoint heights are central and positive. We retain nearly central totals in a fixed number of subblocks, then control fluctuations inside each exact subblock. A final chord calculation converts the rank bounds to the whole physical path, including both sides of every jump and the terminal gap. A common added reserve does not affect this calculation.

Let $M \geq 2$, $0 < a \leq 1$. Consider n exact marks with a uniform K -subset, $p = K/n \in [1/M, 1 - 1/M]$, and independent exact uniform clock gaps: $n + 1$ exponentials conditioned on their sum $P = (n + 1)\tau$, $\tau \in [1/M, M]$. Let a risk have coefficients $0 < c, b, w_+ \leq M$, jumps

$w_+, w_+ - b$ positive, and endpoint reserves $x, y \geq a\sqrt{n+1}$. Their legality means that for one fixed sign $\epsilon \in \{-1, 1\}$ they satisfy the exact identity

$$y - x = \epsilon(cP - nw_+ + bK).$$

At time t , with $r(t)$ the number of marks already read and $s(t)$ their selected-subset count, the risk and its physical chord are

$$R(t) = x + \epsilon\{ct - w_+r(t) + bs(t)\}, \quad C(t) = x + (y - x)t/P.$$

The endpoint difference is fixed by the exact count and clock totals. Adding a common reserve raises the chord without changing that difference or the fluctuation law.

Set

$$J = \lceil (10^8 M^4 / a^2)^2 \rceil, \quad \delta = \frac{a}{100M^2 J}, \quad N_0 = \lceil 10^8 M^8 J^4 / a^4 \rceil, \\ c_{\text{mid}} = \frac{1}{2} e^{-100MJ} \left(\frac{\delta}{100M} \right)^{2J}. \quad (\text{B.4.1})$$

For $n+1 \geq N_0$, the actual exact bridge stays within $a\sqrt{n+1}/2$ of its positive chord with probability at least c_{mid} . In particular it stays above $a\sqrt{n+1}/2$. The constants do not depend on n, K, P, x, y . Every one of the $n+1$ gaps, including the final unmarked gap, is retained.

B.4.1 Subblock totals have explicit positive mass

Divide the n mark slots into J consecutive groups of sizes as equal as possible. Allocate their corresponding gaps to these same groups and allocate the one final gap to the last group. If q_j is a gap shape, then, at (1), $(n+1)/(2J) \leq q_j \leq 2(n+1)/J$. Mark group sizes satisfy the same bounds with n in place of $n+1$.

For the first $J-1$ groups retain their gap sums within $\delta\sqrt{n+1}$ of τq_j , and their subset counts within $\delta\sqrt{n}$ of pm_j . The last totals are fixed by the two global exact totals and differ from their means by at most $J\delta\sqrt{n+1}$ and $J\delta\sqrt{n}$. All these totals are feasible: (1) makes their displacements smaller than half the smallest gap mean and half each of the two smallest mark means. Each first mark interval contains at least $\delta\sqrt{n}$ integers.

The joint density of the first gap sums is exactly

$$\frac{\prod_{j=1}^J f_{q_j, \tau}(t_j)}{f_{n+1, \tau}(P)}, \quad t_J = P - \sum_{j < J} t_j. \quad (\text{B.4.2})$$

Elementary Stirling bounds imply, for $m \geq 2$, $|u| \leq m\tau/2$,

$$f_{m, \tau}(m\tau + u) \geq \frac{1}{4\tau\sqrt{m}} \exp[-2 - 2u^2/(m\tau^2)], \quad f_{N, \tau}(N\tau) \leq \frac{1}{\tau\sqrt{N}}.$$

In (2), summing squared standardized displacements gives at most $4J^3\delta^2M^2$. Integrating the $J-1$ intervals yields a mass at least

$$e^{-10J-8J^3\delta^2M^2} \left(\frac{\delta}{10M} \right)^{J-1}. \quad (\text{B.4.3})$$

For marks the exact joint probability of a specified count vector is $\prod_j \mathbb{P}(\text{Bin}(m_j, p) = k_j) / \mathbb{P}(\text{Bin}(n, p) = K)$. On these central displacements, Stirling and the elementary relative-entropy upper $D(p + u/m || p) \leq 4u^2/(m^2 p_{\min})$ give

$$\mathbb{P}(\text{Bin}(m, p) = mp + u) \geq (4\sqrt{m})^{-1} e^{-4M - 4Mu^2/m}.$$

The denominator at its mean is at most $2\sqrt{M/n}$. Summing the $J-1$ integer intervals therefore gives a mass at least

$$e^{-10MJ - 8MJ^3\delta^2} \left(\frac{\delta}{10}\right)^{J-1}. \quad (\text{B.4.4})$$

The numerical lower bound in (1) is smaller than one half the product of (3)–(4); use $J^3\delta^2 M^2 \leq J/10^4$ and $M \geq 2$. Gap and subset total vectors are independent before and after their own respective exact conditioning.

B.4.2 Conditional subblock fluctuations have constant success

After these two complete total vectors are fixed, different subblocks are independent. Inside a gap subblock the exact-total law is a Dirichlet spacing law. On its first half its density relative to iid exponentials of its exact mean is at most 100: divide a Gamma maximum density by its exact-mean density and use Stirling. The same statement holds on its reversed half.

For iid centered mean- τ exponentials, the elementary exponential martingale gives

$$\mathbb{P}\{\max_{r \leq m} |\sum_{i \leq r} (E_i - \tau)| > u\} \leq 2 \exp[-\frac{1}{4} \min\{u^2/(m\tau^2), u/\tau\}].$$

Applying this on the two halves after the density comparison bounds the exact subblock gap failure by $400 \exp[-c \min(u^2/(m\tau^2), u/\tau)]$, with $c = 1/16$ sufficient.

For an exact binary subset of size k in m positions, write $B_r = \sum_{i \leq r} B_i - rk/m$. The process $B_r/(m-r)$ is a martingale for $r < m$. On the first half its increments have absolute value at most $4/m$. The maximal Azuma bound, then reversal for the second half, gives

$$\mathbb{P}\{\max_{r \leq m} |B_r| > u\} \leq 4e^{-u^2/(16m)}.$$

Take $u = a\sqrt{n+1}/(100M^2)$. The true conditional gap means stay in $[1/(2M), 2M]$; at (1) both subblock failure exponents dominate $a^2 J/(2 \cdot 10^6 M^6)$. Consequently the total failure over all subblocks and both channels is at most

$$800J \exp[-a^2 J/(2 \cdot 10^6 M^6)] < \frac{1}{2}. \quad (\text{B.4.5})$$

The last strict inequality follows directly from the explicit choice of J : its exponent is at least $5 \cdot 10^9 M^2/a^2$, while $\log(1600J) \leq 100M^2/a^2$.

Here is the physical-chord calculation, including the final gap. Put $N = n+1$, $A = nw_+ - bK$, and let T_r be the time of the r th mark, S_r its cumulative number of selected marks. At all physical times, including the final unmarked gap,

$$R(t) - C(t) = -\epsilon [w_+ \{r(t) - nt/P\} - b \{s(t) - Kt/P\}].$$

At time T_r , the bracket is

$$rw_+ - bS_r - (A/P)T_r. \quad (\text{B.4.6})$$

The coefficient A/P is in $[0, M^2]$, since all jumps are positive. Writing $T_r = r\tau + e_r$ and $S_r = pr + v_r$, (6) becomes $r(w_+ - bp)/N - bv_r - (A/P)e_r$; its first term is at most M . The retained total displacements and the within-block events give $|e_r| \leq J\delta\sqrt{N} + u$ and $|v_r| \leq J\delta\sqrt{n} + u$. Thus the absolute value in (6) is at most $M + 2M^2(J\delta\sqrt{N} + u)$. Immediately before a mark there is one further jump of size at most M . Between marks the chord error is linear, so its maximum occurs at one of these endpoints. On the final unmarked gap it is again linear and ends at zero. Consequently the complete physical-path error is at most

$$2M + 2M^2(J\delta\sqrt{N} + u) \leq 2M + \frac{a\sqrt{N}}{25} < \frac{a\sqrt{N}}{2},$$

where the last inequality follows from $N \geq N_0$. Both risk signs use this same absolute deviation. Combining (3)–(5) proves (1).

B.4.3 Completing the conditional private mass

This is a joint ordinary continuation for each retained prefix word with compatible inner endpoint heights at least $a\sqrt{n}$. For a fixed prefix environment let its already accepted private mass be q , and let the completed private mass be U as a function of the remaining environment. Then $0 \leq U \leq q$ and $\mathbb{E}U \geq c_{\text{mid}}q$; hence $\mathbb{E}U^\eta \geq c_{\text{mid}}q^\eta$ before averaging the prefix. The prefix remains fixed in this inequality; its rare mass is averaged only after the remaining-clock fractional moment has been bounded.

If a physical strong curve has amplitude g and exponent $\gamma < 1/2$, additionally require

$$n + 1 \geq (4gM^\gamma/a)^{1/(1/2-\gamma)}.$$

Then the retained tube dominates that curve. Named REG is completed by its pointwise uniform conditional refinement bound; if ordinary REG is imposed here, choose its explicit error smaller than $c_{\text{mid}}/4$ before subtracting it. That is a finite elementary tail condition once the desired slopes/intercepts are specified.

This gives explicit middle-continuation constants. The next subsection supplies the endpoint-box and wall constants for the OU comparison.

B.5 Numerical OU endpoint, wall and multiplicativity constants

Fix $M \geq 2$ and

$$M^{-1} \leq \eta \leq 1, \quad M^{-1} \leq \rho \leq 1 - M^{-1}.$$

Let E, W be independent stationary OU processes with covariance $\exp(-|s - t|/2)$, and put

$$q_t(E) = \mathbb{P}_W\{\sqrt{\rho}E_s + \sqrt{1-\rho}W_s \geq 0, 0 \leq s \leq t \mid E\}, \quad F(t) = \mathbb{E}q_t(E)^\eta.$$

At $\eta = 1$, $F(t) = \pi^{-1} \arcsin(e^{-t/2})$; in this case set $\chi(1, \rho) = 1/2$.

The following constants cover the endpoint tails, multiplicativity, and wall comparisons. We choose them with ample room for the later relative-error estimates:

$$a_* = 2^{-48}M^{-2}, \quad C_* = 2^{10}, \quad B_* = 2^{28}M^2, \quad C_Q = 2^{80}M^2. \quad (\text{B.5.1})$$

Every assertion below first holds on a finite time grid; monotone refinement and continuity give the continuous assertion. No constant in (1) depends on the time or on the grid cardinality.

B.5.1 The conditional private endpoint

As in Lemma 5.3, fluctuations around the endpoint mode have a Gaussian bound. The mode contracts over each fixed-length segment and is driven by the negative part of the environmental wall. We make both estimates numerical here.

For a fixed environment, the endpoint density of W conditioned on survival is $\phi(z)g(z)$, where g is increasing and log-concave. The monotonicity follows by running the OU bridge backwards from its endpoint: its conditional mean at every earlier time is an increasing multiple of z . Log-concavity is the Gaussian Prékopa's theorem applied to the convex set of path inequalities. Consequently its log density has curvature at most -1 , and its mode m is nonnegative. The usual density-ratio proof gives

$$|W - m| \preceq_{\text{st}} |N(0, 1)|, \quad |\mathbb{E}W - m| \leq 1, \quad \mu \leq_{\text{lr}} m + |N(0, 1)|. \quad (\text{B.5.2})$$

For the last comparison, the ratio of $\mu(m + u)$ to the half-normal density is decreasing for $u \geq 0$: differentiate and use the curvature bound and the zero derivative at the mode. Below m the half-normal density is zero. OU transition and killed transition kernels are TP2, so this likelihood-ratio order propagates through a future segment.

Consider a segment of length $v \in [1, 2]$, and let $B = \sup(-\sqrt{\rho/(1-\rho)}E)_+$ there. Raise its wall to B and raise its incoming law to $m + |N|$. Let $r = \exp(-v/2)$. An OU noise segment is $\exp(-s/2)B_{\exp(s)-1}$, and $\exp(2) - 1 < 7$. The explicit Brownian tube bound

$$\mathbb{P}\{\sup_{u \leq 7} |B_u| \leq 1\} \geq e^{-100} \quad (\text{B.5.3})$$

is enough. For example, divide $[0, 7]$ into 28 intervals of length $1/4$, retain each interval endpoint in $[-1/4, 1/4]$, and require its bridge fluctuation to have absolute value at most $1/2$. The Gaussian transition density on that endpoint interval is at least $e^{-1/2}/\sqrt{2\pi/4}$ and the Brownian bridge reflection bound gives bridge success at least $1 - 2e^{-2}$. The product is larger than e^{-100} .

If $rm \geq B + 2$, (3) gives survival probability at least e^{-100} . The unrestricted endpoint relative to rm is bounded by $r|N_1| + \sqrt{1 - r^2}N_2$; its positive tail is at most $4 \exp(-u^2/8)$. Tail integration after dividing by e^{-100} gives conditional mean at most $rm + 100$.

If $rm < B + 2$, retain the starting half-normal in $[e(B+3), e(B+3)+1]$. Since $0 \leq m \leq e(B+2)$, this costs at most $\exp[-101(B+1)^2]$ in the exponent. With (3) the survival probability is at least $\exp[-201(B+1)^2]$. The same tail integration with cutoff $100(B+1)$ gives conditional mean at most $rm + 101(B+1)$. By (2), in either case the new mode satisfies

$$m_{\text{new}} \leq rm + 1024(B+1). \quad (\text{B.5.4})$$

For an initial segment of duration at most one the same proof starts at the unfiltered stationary normal and gives the same $1024(B+1)$ bound. Decompose backwards into intervals of lengths in $[1, 2]$, with at most one initial short interval. Since $\sqrt{\rho/(1-\rho)} \leq \sqrt{M} \leq M$, (4) gives the pointwise bound

$$m_t(E) \leq \mathcal{M}_t(E) := 8192M \left(1 + \sum_{j \geq 0} e^{-j/2} \sup_{s \in I_j} (-E_s)_+ \right). \quad (\text{B.5.5})$$

The functional on the right is decreasing in every environment coordinate. For a stationary OU interval of length at most two, its supremum Z in absolute value satisfies $\mathbb{P}(Z > u) \leq 6 \exp(-u^2/56)$ by the representation above and reflection. Thus $\mathbb{E} \exp(Z^2/112) \leq 7$. Put $S = \sum \exp(-j/2) < 3$. Weighted Cauchy–Schwarz and Jensen, without independence of the intervals, give

$$\mathbb{E} \exp(2^{-40} M^{-2} \mathcal{M}_t^2) \leq 8. \quad (\text{B.5.6})$$

B.5.2 Relative moments of the tilted environment

The environment law tilted by q_t^η is MTP2 and lies in stochastic order between its original law and the law tilted by q_t . This follows from monotonicity of q_t , MTP2 of q_t , and association under each intermediate tilt. Applying these orders separately to the positive and negative parts suffices; the square moment is assembled from these two monotone functions.

Under the q_t tilt, rotate into the positive composite channel $Z = \sqrt{\rho}E + \sqrt{1-\rho}W$ and its independent orthogonal OU channel. At a time $u \in [0, t]$ its conditional density is

$$\frac{\phi(z)p_u(z)p_{t-u}(z)}{F_1(t)}, \quad z \geq 0, \quad p_v(z) = 2\Phi\left(\frac{z}{\sqrt{e^v-1}}\right) - 1, \quad F_1(t) = \pi^{-1} \arcsin(e^{-t/2}).$$

For $v = 0$, take $p_0 = 1$. The inequalities $p_v(z) \leq \min(1, \sqrt{2/\pi}z/\sqrt{e^v-1})$ and $F_1(t) \geq \pi^{-1} \exp(-t/2)$ show, by splitting according as both durations exceed one, that this density is at most $10(1+z)^2\phi(z)$. Therefore the environment at every time under the q_t^η tilt satisfies

$$\mathbb{E}_{\text{tilt}} e^{uE_s^2} \leq 2^8, \quad 0 \leq u \leq 1/8. \quad (\text{B.5.7})$$

For the negative part use the original normal law; for the positive part use the rotated density just displayed. Inequality (6), under the tilt, follows from its decreasing functional and stochastic order. Together with (2), Cauchy–Schwarz, and time reversal, (6)–(7) give the two-end bound

$$\mathbb{E}_E \left[e^{a_*(E_0^2+E_t^2)} \left(\mathbb{E}_W[1_{\text{survive}} e^{a_*(W_0^2+W_t^2)} \mid E] \right)^\eta \right] \leq C_* F(t). \quad (\text{B.5.8})$$

All exponents left after the four Cauchy–Schwarz steps are smaller than the exponents in (6) and (7); the numerical factors are smaller than 2^{10} .

B.5.3 Quasi-multiplicativity and endpoint boxes

The mixed endpoint moment controls the Mehler density across a fixed gap. It also lets us discard large endpoints relative to the surviving mass and then translate the retained endpoints into positive boxes. Both operations have costs independent of the horizon.

Separate two surviving segments by $g = 2 \log(1/a_*) + 4$. Their Mehler density relative to independent endpoints is at most $(1-r^2)^{-1/2} \exp(r(x^2+y^2)/(2(1+r)))$, $r = \exp(-g/2)$. Use it first inside the private integral and then outside. Its quadratic coefficient is at most a_* , so (8) gives

$$F(s+g+t) \leq 2^{22} F(s) F(t).$$

Association gives $F(s+g+t) \geq F(s+t)F(g)$, and $F(g) \geq F_1(g) \geq \exp(-g/2)/\pi$. Hence

$$F(s)F(t) \leq F(s+t) \leq C_Q F(s)F(t), \quad C_Q^{-1} e^{-\chi t} \leq F(t) \leq e^{-\chi t}. \quad (\text{B.5.9})$$

The exponential rate exists by superadditivity, and $\eta/2 \leq \chi \leq 1/2$, as in the fractional persistence theorem.

Remove every endpoint outside $[-B_*, B_*]$. The environmental tail uses (8) directly. For a private tail, fractional subadditivity and (8) give cost at most $C_* \exp(-a_* \eta B_*^2) F(t)$. Since $a_* \eta B_*^2 \geq 256M$, the four deletions cost less than $F(t)/4$. Translate both channels by

$$h(s) = 4B_* \{e^{-s/2} + e^{-(t-s)/2}\}.$$

This raises the whole path, and takes both endpoints into $[3B_*, 9B_*]$. For one channel its Cameron–Martin norm squared is at most $4(4B_*)^2$, while its linear functional on the retained box has absolute value at most $8B_*^2$. The two translations therefore cost at most $\exp(-80B_*^2)$, including the fractional private density factor. Thus the moment with these four positive endpoint restrictions is at least

$$\frac{1}{2} e^{-80B_*^2} F(t). \quad (\text{B.5.10})$$

Wider compatible boxes only increase this lower bound.

B.5.4 Explicit bounded-action walls

An environmental translation imposes the desired composite wall. After integration by parts, its likelihood uses a signed measure of path values. The tilted pointwise Gaussian moment therefore controls the likelihood through the total variation of that measure.

For a deterministic twice differentiable OU shift h , write

$$A(h) = h(0)^2 + \int_0^t (h' + h/2)^2 ds,$$

$$V(h) = |h'(t) + h(t)/2| + |h(0)/2 - h'(0)| + \int_0^t |h/4 - h''| ds.$$

Integration by parts expresses its Cameron–Martin functional as a signed path-value measure of total variation at most $V(h)$. From (7), Jensen with its normalized variation measure, and $u|z| \leq z^2/8 + 2u^2$, one obtains

$$\mathbb{E}_{\text{tilt}} e^{\pm L_h} \leq 2^8 e^{2V(h)^2}, \quad |\mathbb{E}_{\text{tilt}} L_h| \leq 8V(h).$$

Consequently the fractional moment for the shifted environmental path satisfies the explicit comparison

$$e^{-8V(h)-A(h)/2} F(t) \leq \mathbb{E}_{q_t}(E+h)^\eta \leq 2^8 e^{2V(h)^2} F(t). \quad (\text{B.5.11})$$

To impose a composite wall $b(s)$, take $h = b/\sqrt{\rho}$ and the appropriate sign. For a sum of d terms $C_l \exp(-\alpha_l s)$ or $C_l \exp(-\alpha_l(t-s))$, with $\alpha_l \in [1/3, 1/2]$, let $W = \sum |C_l|$. Direct integration gives

$$A(h) \leq 12MW^2, \quad V(h) \leq 16\sqrt{M}W.$$

Thus all Lamperti reserve, $\gamma \leq 1/6$ power-wall, and bounded scaled-drift terms of total amplitude at most W have an upper comparison constant $2^8 \exp(512MW^2)$ and lower constant $\exp(-128\sqrt{MW} - 6MW^2)$. Their endpoint displacement is at most $\sqrt{MW}$; enlarge the four endpoint boxes by this fixed amount. These bounds specify the widths of the Gaussian endpoint boxes and the separation needed for the Mehler comparison. The dimension-dependent values of M, W , together with the hierarchy, regularity, count, and grid bounds, are chosen in Section H.2.

B.6 Numerical flat preparation, relative failures and whole-block independence

This subsection makes the relative coupling argument effective. First we prepare a larger reserve at a deterministic time, so that a fresh future is independent of the preparation. We then prove doubling, sum the favorable-prefix-height bins, and verify that late failures read blocks disjoint from the retained early prefix. These comparisons are established for the flat walk before its persistence exponent is used.

Let $M \geq 10$. All positive coefficients, gap means, nonzero jump sizes, and centered channel standard deviations are in $[M^{-2}, M^2]$; both binary fractions and their complements are at least M^{-2} ; $M^{-1} \leq \eta \leq 1$; the associated Lamperti correlation satisfies $M^{-1} \leq \rho \leq 1 - M^{-1}$; and the signed mean drift obeys $|\delta|\sqrt{n} \leq M$. Choose M large enough to satisfy all these bounds. Let $F_n(h)$ be the conditional fractional survival moment for the flat walk, with the continuous pre/post-jump convention retained.

One possible construction is

$$h \geq M^{100}, \quad C_{\text{prep}} = \exp\{\exp(M^{200})\}, \quad C_{\text{double}} = \exp(M^{5000}), \quad C_{\text{bad}} = \exp\{\exp(M^{300})\}. \quad (\text{B.6.1})$$

For every $n \geq 1$, every $u \geq 0$, and every event D reading a prefix of $b \leq n/2$ original channels and their independent coupling auxiliaries, the following inequalities hold after a fixed initial finite convention with the same bounds:

$$F_n(h) \geq C_{\text{prep}}^{-1} \min\{1, (1+h)/\sqrt{1+n}\},$$

$$F_n(h+u) \leq C_{\text{prep}}(1+u)F_n(h), \quad F_n(h) \leq C_{\text{double}}F_{2n}(h), \quad (\text{B.6.2})$$

$$\mathbb{E}q_{\text{survive}, D}^\eta \leq C_{\text{bad}}(1+b)^2\mathbb{P}(D)^{\eta/4}F_n(h). \quad (\text{B.6.3})$$

For a bad event whose whole coupling-block sigma-field is independent of every whole block used by the retained first $\lfloor n/16 \rfloor$ slots, the right side can instead be $C_{\text{bad}}\mathbb{P}(D)^\eta F_n(h)$. At small n , retain the actual event and use the constant in (1). The finite convention is thus fixed by the original walk and the displayed compact input, before the Gaussian exponent or the conditional survival normalization enters.

B.6.1 Numerical ordinary preparation at a deterministic time

The stopped walk reaches an upper height with a controlled probability, but its hitting time is random. The tube construction extends each retained entrance to one deterministic terminal time while keeping both channel totals central. This is the point at which the later concatenation obtains independent future channels.

First center the walk by changing the exponential mean by its signed mean drift. For $n \geq M^{100}$, this changed mean remains in $[M^{-3}, M^3]$, since $|\delta| \leq M/\sqrt{n}$. For smaller n , the reserve $h \geq M^{100}$ and the exponential tail give ordinary survival of probability at least one half; all comparisons below are then covered by (1). Continuous survival in the orientation with negative clock motion costs at most one extra bounded jump: start its discrete argument from reserve $h - M^2 \geq h/2$.

For the centered walk stopped on leaving $[-h/2, u]$, put $l = u + h + 2M^3$. Every exponential-side overshoot has conditional mean at most $2M^3$ and second moment at most $8M^6$, by memorylessness; the opposite overshoot is bounded by M^2 . Optional stopping of the walk and of its square minus variance times time gives

$$\mathbb{P}(\text{upper exit}) \geq \frac{h}{4l}, \quad \mathbb{E}\tau_{\text{exit}} \leq M^{20}(h+1)l. \quad (\text{B.6.4})$$

For the upper inequality on the mean exit time, first bound the upper-exit probability above by $(h + 2M^3)/u$ when $u \geq h$, then its square contribution by $M^6(h+1)l$. If $u < h$, the crude bound $M^6l^2 \leq 4M^6(h+1)l$ suffices. Divide by variance at least M^{-4} . Truncation justifies the stopped identities.

Take $R = \exp(M^{40})$. Markov's inequality on (4) removes less than one tenth of the upper exits when the stopping time exceeds $Rl^2/4$. The individual centered clock and mark martingales have stopped second moments at most $M^{30}(h+1)l$. Their Doob bound removes less than another tenth if either absolute value exceeds $M^{30}l$ before the retained exit.

Extend each retained exit to the deterministic time $m = \lceil Rl^2 \rceil$, staying positive and ending above the desired enlarged reserve. For a fresh segment of length between l^2/M^{30} and twice that length, retain binary counts in a central interval of width $\sqrt{\text{length}}/M^{20}$ and the Gamma total in an interval giving signed risk increment between zero and $\sqrt{\text{length}}/M^{20}$. Central Stirling bounds give this joint total event probability at least $\exp(-M^{100})$.

Conditional on the retained exact totals, apply the middle tube in Section B.4 with compact parameter $M' = M^{10}$ and endpoint fraction $a = M^{-20}$. Its explicit constants J, N_0, c_{mid} satisfy

$$J \leq M^{300}, \quad N_0 \leq M^{2000}, \quad c_{\text{mid}} \geq e^{-M^{1000}}.$$

For a segment of length below M^{2000} , retain each increment in a fixed favorable interval. Such an interval has probability at least $\exp(-M^{20})$ and positive increment at least M^{-10} . In the exponential-positive orientation choose its gap in a unit interval far enough above its mean and choose the favorable binary mark; in the opposite orientation choose its gap below half its mean and the favorable mark. Thus every short segment has a tube probability at least $\exp(-M^{3000})$. If $l < M^{2000}$, this same direct event handles the entire fixed preparation time; it costs at most $\exp[-M^{20}RM^{4002}] \geq \exp[-\exp(M^{100})]$.

For $l \geq M^{2000}$, split the remaining time after the stopped entrance into the displayed segment sizes, with a possibly short last segment. There are at most $2RM^{30} + 1$ segments. All starting and ending reserves are at least a fixed fraction of l , so the exact middle tube applies. Every segment has nonnegative endpoint increment, and its within-segment error is less than $l/100$. Its two individual channel totals remain in central windows of width at most $M^{30}l$ per segment. The product lower bound is at least $\exp[-\exp(M^{100})]$, uniformly in the retained stopping time.

Changing the exponential mean back to its original value on these retained total windows costs at most $\exp[-M^{100}R^2]$; expand its exact Gamma likelihood and use $|\delta|l \leq M$ when $l \leq \sqrt{n}$. This is again larger than $\exp[-\exp(M^{150})]$. The final preparation has probability

$$\mathbb{P}(\text{fixed-time preparation}) \geq C_{\text{prep}}^{-1} \frac{h}{u + h + 2M^3}. \quad (\text{B.6.5})$$

The final time is deterministic, so its two fresh future channels are independent of the preparation. If the private preparation probability at a fixed environment is p , then $\mathbb{E}p^\eta \geq \mathbb{E}p$. Concatenation therefore proves the reserve inequality when $l \leq \sqrt{n}$, since the full concatenated event implies the original first n steps. The preparation may be longer than n ; its likelihood bound above only used $|\delta|l \leq M$, so that causes no difficulty. Independently, ordinary survival follows by preparation at height $\sqrt{n/(10R)}$ in at most $n/2$ steps and its positive tube to time n . For this continuation use the actual smaller height fraction $a_R = (100M^{10}\sqrt{R})^{-1}$ and compact parameter set $M' = M^{10}$, rather than $a = M^{-20}$. Its explicit formulas give

$$J_R \leq e^{M^{41}}, \quad N_R \leq e^{M^{42}}, \quad c_R \geq \exp[-\exp(M^{43})].$$

For example $\log J_R \leq 24 \log 10 + 120 \log M + 2M^{40} \leq M^{41}$; the displayed N_R and c_R bounds follow by substitution. Sum the unconditioned count and gap totals in compatible central intervals of width $a_R\sqrt{n}/(100M^{10})$ to obtain legal endpoints both at least $a_R\sqrt{n}$; their total mass is at least $\exp(-M^{100})a_R^2$. Then apply the exact middle tube. If $n < N_R$, a fixed favorable increment event costs at most $\exp[-M^{20}N_R] \geq \exp[-\exp(M^{43})]$. Thus a uniform ordinary continuation constant $\exp[-\exp(M^{50})]$ suffices. It is much larger than the inverse C_{prep} used in (2). This gives the first line of (2), including saturation when the original reserve already exceeds that height. If $l > \sqrt{n}$, then either $h \geq \sqrt{n}/4$ and this ordinary lower is a fixed constant, or $u \geq \sqrt{n}/4$ (the fixed $2M^3$ allowance is harmless as $h \geq M^{100}$). In the latter case $(1+u)F_n(h)$ has a fixed lower bound. Use $F_n(h+u) \leq 1$. All lost powers of R are smaller than C_{prep} . This proves the crude lower and reserve comparison using only stopped moments, central total probabilities, and the explicit middle tube.

B.6.2 Numerical doubling for flat survival

Remove prefix environments whose unfavorable centered clock sum exceeds $100M^4\sqrt{n}$. Its ordinary probability is at most $M^{-4}/100$; its mass under the survival tilt is at most the same fraction, since survival and the unfavorable event have opposite coordinate monotonicities. Product association bounds the weighted loss directly by the unweighted unfavorable-clock probability.

At every remaining environment, force the *one-sided favorable binary-count tail* whose displacement is at least $M^{12}\sqrt{n}$. It is a lower count tail when fewer selected marks are favorable, and an upper count tail in the opposite orientation. This event is monotone in the same direction as survival. For $n \geq M^{100}$, obtain its probability lower bound $\exp(-M^{50})$ by a central subinterval contained in that tail. The subinterval is used only to estimate the tail probability; it is not the event used in association or in the continuation. If feasibility fails, n is below a fixed power of M and the favorable supported word has probability at least $\exp(-M^{200})$. Its intersection with private survival retains at least this fraction by binary product association. We retain the whole favorable count tail, since continuation uses only the lower bound $M^{-10}\sqrt{n}$ on the ending reserve.

An independent second half from this reserve has ordinary survival probability at least $\exp(-M^{4000})$. Use the same central total windows and explicit exact middle tube as above; when the sample size is below its displayed threshold, the fixed favorable word/gap event costs at most $\exp(-M^{4000})$. If a prefix private mass is f and its completion at the future environment has mass Z , then $0 \leq Z \leq f$ and $\mathbb{E}Z \geq cf$. Thus $\mathbb{E}Z^\eta \geq cf^\eta$. Combining the two retentions proves the numerical doubling constant in (1).

B.6.3 Summing the relative bad-event costs

Bin the positive prefix risk increment in intervals $[r, r + 1)$, putting every nonpositive value in bin zero. If p_r is its conditional private mass intersected with D , independence of the remaining original channels and the reserve comparison give

$$Eq_{\text{survive}, D}^\eta \leq C_{\text{prep}} C_{\text{double}} F_n(h) \sum_{r \geq 0} (2 + r) E p_r^\eta.$$

The centered exponential and bounded binary Chernoff bounds give

$$\mathbb{P}(H_b^+ \geq r) \leq 10 \exp \left[-\frac{1}{1000M^{10}} \min \left(\frac{r^2}{1+b}, r \right) \right]$$

after shifting the mean by at most $M\sqrt{b}$, which only enlarges the constant to $\exp(M^{30})$. Jensen and Cauchy–Schwarz give $\mathbb{E} p_r^\eta \leq \mathbb{P}(D)^{\eta/2} \mathbb{P}(H_b^+ \geq r)^{\eta/2}$ for $r \geq 1$, and $\mathbb{E} p_0^\eta \leq \mathbb{P}(D)^\eta$. Sum the Gaussian part through $r \leq 1 + b$ and then the exponential part. The sum is at most $\exp(M^{50})(1+b)\mathbb{P}(D)^{\eta/2}$. The constants from (1), and one doubling to compare $n - b$ with n , are smaller than C_{bad} ; weakening the power and the polynomial proves (3).

For a late bad event the separation is on the *whole product block-coupling space*, including every auxiliary. In the dyadic construction, a block of length $2^j L_0$ has left endpoint $a = (2^j - 1)L_0$ and right endpoint $b = (2^{j+1} - 1)L_0 = 2a + L_0$. Thus $b > n/4$ and $n \geq 16L_0$ imply $a > 3n/32 > n/16$. Retain necessary survival only on the first $\lfloor n/16 \rfloor$ original channels. Every whole coupling block used by this prefix has a different block index from every late block just identified. Consequently the complete σ fields, including all auxiliaries, are independent. Jensen bounds the other factor by $\mathbb{P}(D)^\eta$; a fixed number of doublings gives the displayed late inequality. If $n < 16L_0$, handle the initial block by the early estimate or the final reserve chosen larger than $100ML_0^2$; the early estimate controls that overlapping-block range. For a general event the late variant requires its entire coupling blocks to be disjoint from the retained prefix. Every early event can read auxiliaries because the intersection estimate and the height tail are unconditional.

B.6.4 An explicit coupling starting scale

Apply Section B.3 with $A = 400M$ on dyadic blocks. The block error is at most $1000M^4(A + 4)^2 \log^3(L \vee e)$ after a harmless coefficient enlargement, and its failure is at most $5L^{-A}$. An explicit starting scale is

$$L_0 = \left\lceil \exp\{\exp(M^{400})\} \right\rceil. \tag{B.6.6}$$

The relative bad-block series is bounded by $10C_{\text{bad}}L_0^{2-A\eta/4} < \exp(-M^{100})$. Section B.3.4 gives a smooth pointwise majorant for the accumulated good errors, including jumps and interpolation. Its action plus squared variation is less than M^{-100} , uniformly in the final horizon. The quantitative OU wall estimates absorb the accumulated good errors, while the relative series pays the bad errors at the same survival scale.

A negative soft power wall uses the same preparation because its attached future wall is stronger than the shifted global wall. Choose $h \geq (100g)^{1/(1-2\gamma)}$ first; its normalized Lamperti amplitude is at most $1/100$. A positive strong wall is treated by flat bad-event domination and the Gaussian lower comparison. For that wall the error is paid by the established flat relative bound.

These probability estimates depend only on the stated physical compact input M . Sections H.1 and H.2 explicitly bound the deterministic, count, and regularity constants in terms of k and imply the choice of Λ_k .

B.7 Numerical ordinary bridge estimates

The constant-mark bridge requires its own effective construction. There is one gap channel and no private count variance. A stopped quadratic potential gives the drifted upper bound; one-channel Gaussian survival and relative coupling give the strong lower bound. The exact bridge is then completed by a middle tube or an asymmetric continuation. We keep the physical coefficient bound separate from the larger probability constants, so that the final reserve does not enter the definition of an earlier threshold.

B.7.1 Parameters and explicit bounds

Two constants must be kept separate. Let $A \geq 10$ bound the physical coefficients, and choose

$$M \geq \exp(A^{1000}). \quad (\text{B.7.1})$$

Assume $w, c, \tau \in [A^{-1}, A]$, $P = (n+1)\tau$, $n \geq 1$, and $\epsilon \in \{-1, 1\}$. The clock is the uniform clock of n marks in $[0, P]$; equivalently all $n+1$ gaps are exponentials of mean τ conditioned on their total P . Put

$$R(t) = x + \epsilon\{wN(t) - ct\}, \quad y - x = \epsilon(nw - cP), \quad C(t) = x + (y - x)t/P. \quad (\text{B.7.2})$$

The exact chord identity is

$$R(t) - C(t) = \epsilon w\{N(t) - nt/P\}. \quad (\text{B.7.3})$$

The exact signed increment links the two endpoints. Adding a common reserve changes the chord but leaves the centered fluctuation and the coefficient in (3) unchanged.

Fix $\gamma = 1/8$, $\zeta = 1/16$, $0 \leq g \leq A$, a REG slope $\nu \geq M^{-1}$, and a grid with at most A boundaries per physical unit interval. Rates within $A/\sqrt{1+n}$ of n/P are permitted. The following construction suffices:

$$L_0 = \lceil e^{e^{M^{2300}}} \rceil, \quad h_{\text{pre}} = \lceil e^{e^{M^{2400}}} \rceil, \quad N_{\text{core}} = \lceil e^{e^{M^{2600}}} \rceil, \\ \boxed{h_{\text{ord}} = \lceil e^{e^{M^{3000}}} \rceil, \quad K_{\text{ord}} = \lceil e^{e^{M^{3500}}} \rceil, \quad C_{\text{ord}} = e^{e^{M^{4000}}}.} \quad (\text{B.7.4})$$

For every legal $x, y \geq h_{\text{ord}}$, with regularity intercept at least K_{ord} ,

$$C_{\text{ord}}^{-1}\beta_n(x, y) \leq \mathbb{P}\{R(t) \geq gd_P(t)^{1/8} \ (0 \leq t \leq P), \text{ REG}\} \leq C_{\text{ord}}\beta_n(x, y), \\ \beta_n(x, y) = \min\{1, (1+x)(1+y)/(1+n)\}. \quad (\text{B.7.5})$$

Here $d_P(t) = \min(t, P-t)$ and REG means

$$|N(t) - N(s) - \lambda_0(t-s)| \leq \nu(t-s) + K[1 + d_P(s)^{1/16} + d_P(t)^{1/16}] \quad (\text{B.7.6})$$

for grid pairs, or all physical pairs after reducing the initial slope by a factor 100 and enlarging K by a factor $100A^4$. These fixed factors are smaller than (4). For a contracted chord $\theta C + (R - C)$, $0 < \theta \leq 1$, use (5) at endpoints θx , θy whenever both exceed h_{ord} . Its speed is

$$c_\theta = \theta c + (1 - \theta)nw/P, \quad (\text{B.7.7})$$

which lies in $[1/(2A), 2A^2]$; replacing A by $2A^2$ in (1) is an explicit compact-input enlargement. The factor changes by at most θ^{-2} . The estimate (5) is an ordinary probability estimate for this one-channel clock.

The powers in (4) are chosen to fit the existing bounds for the fixed length threshold: physical and asymmetry constants can be bounded by $A = M_1$, probability constants by $M = M_2$, after harmless enlargement of A in (1). Every threshold in (4) is smaller than h_W , and K_{ord} is smaller than K_{REG} . These choices precede the final geometry reserve h .

B.7.2 Elementary drifted one-end upper bound

Let E_j be independent mean-one exponentials and consider the rank walk

$$Z_j = h + \epsilon \sum_{l \leq j} (w - c\tau E_l), \quad \delta = \epsilon(w - c\tau).$$

Its variance per step lies in $[A^{-4}, A^4]$. Both exit overshoots from any interval have conditional first moment at most A^2 and second moment at most $2A^4$: on the unbounded side this is memorylessness, and the other side is bounded by w . Put

$$q = 10A^4, \quad u = \min \left\{ \frac{\sqrt{n}}{100A^6}, \frac{1}{100A^6|\delta|} \right\}, \quad (\text{B.7.8})$$

where the second entry is infinity when δ is zero. For $h < u$ stop at the first exit from $(0, u)$ and use $V(z) = (z + q)(u + q - z)$. While inside,

$$E[V(Z_{j+1}) - V(Z_j) \mid Z_j = z] = (u - 2z)\delta - E(Z_{j+1} - Z_j)^2 \leq -A^{-4}/2. \quad (\text{B.7.9})$$

At exit the negative contribution to V has expectation at most $3A^2(u + 3q)$. If $u \geq 10q$, truncation of (9), followed by the overshoot bounds, gives

$$E\tau \leq 20A^4(h + q)u, \quad P(Z_\tau \geq u) \leq 3(h + q)/u. \quad (\text{B.7.10})$$

For the probability bound use $\mathbb{E}Z_\tau = h + \delta\mathbb{E}\tau$ and lower overshoot mean at most A^2 ; $20A^4|\delta|u \leq 1/5$ absorbs the drift term. If $u < 10q$, the trivial probability bound already has the claimed form. The event of survival to n is contained in upper exit or $\tau > n$. Markov's inequality and (8)–(10) prove

$$P(\min_{j \leq n} Z_j \geq 0) \leq 1000A^{10} \min\{1, (h + q)(n^{-1/2} + |\delta|)\}. \quad (\text{B.7.11})$$

The same bound with a larger factor $10A^2$ handles the physical pre/post-jump convention. In particular, when $|\delta|\sqrt{n} \leq M$ it is bounded by $M^{100} \min(1, (1+h)/\sqrt{1+n})$. The stopped-potential argument thus gives an explicit upper bound directly from the overshoot moments.

For a Gaussian rank walk with increment mean δ and standard deviation σ , the normal Mills-ratio inequality bounds either exit overshoot by $|\delta| + \sigma$ in first moment and $2\delta^2 + 2\sigma^2$ in second

moment. Here $|\delta|, \sigma \leq A^2$, so the bounds are $2A^2$ and $4A^4$. The same potential argument, with its intermediate constants enlarged by a factor at most four, remains within the prefactor $1000A^{10}$ in (11).

B.7.3 Flat preparation and relative coupling failures, with one channel

The following is the ordinary specialization of the proof of the flat-preparation argument in Section B.6, rather than a specialization of its binary statement. Delete the Bernoulli coordinate, its central count constraint, and its favorable-count event. All remaining events are in the exponential channel. Use the internal compact parameter $M' = M^{10}$. Its variance, mean, speed and bounded scaled drift cover $[A^{-1}, A]$ and $|\delta|\sqrt{n} \leq M$. The numerical exit, deterministic preparation, Gamma-total and exact-total tube calculations in that section are then unchanged, or improved by deleting a constraint. The middle-tube proof in Section B.4 needs only its Gamma formulas (2)–(3), spacing formula (5), and physical identity (6); its binary factor can be deleted. One may retain the original smaller c_{mid} .

For completeness, a short favorable gap event is explicit. For positive jumps choose $E \leq w/(2c\tau)$; it has probability at least $A^{-3}/4$ and positive increment at least $w/2$. For the opposite orientation retain E in a unit interval starting beyond $2w/(c\tau) + 1$; its probability is at least $\exp(-10A^4)$ and its positive increment is at least A^{-2} . The fixed-time preparation uses at most $2\exp((M')^{40})(M')^{30} + 1$ tubes, exactly as in that proof. The actual smaller saturation-tube fraction is $(100(M')^{10}\sqrt{\exp((M')^{40})})^{-1}$; its J , N_0 and inverse probability logarithm are bounded there by $\exp((M')^{41})$, $\exp((M')^{42})$, $\exp((M')^{43})$. All these costs are smaller than

$$C_{\text{flat}} = e^{M^{2100}}. \quad (\text{B.7.12})$$

Combining this explicit preparation lower with (11), for $h \geq M^{1000}$ and both orientations, gives

$$C_{\text{flat}}^{-1} \min(1, (1+h)/\sqrt{1+n}) \leq F_n(h) \leq C_{\text{flat}} \min(1, (1+h)/\sqrt{1+n}). \quad (\text{B.7.13})$$

The preparation includes saturation for the reserve h and gives the linear reserve comparison by the same deterministic concatenation. All preparation events belong to the exponential channel, as required for deterministic marks.

An even simpler bad-event calculation is now available. If D reads a prefix of b original gaps, including its coupling auxiliaries, drop prefix survival and apply (11) to the remaining independent walk. Prefix positive increments have second moment at most $M^{100}(1+b)^2$. Cauchy–Schwarz, (13) and one time comparison therefore give

$$P(\text{flat survival}, D) \leq C_{\text{bad}}(1+b)^2 P(D)^{1/2} F_n(h), \quad C_{\text{bad}} = e^{M^{2200}}, \quad b \leq n/2. \quad (\text{B.7.14})$$

For an event whose entire coupling blocks start after $n/4$, retain only the necessary prefix of length $\lfloor n/16 \rfloor$. Its *whole product σ field*, auxiliaries included, is disjoint from D . Equations (11)–(13) give instead $C_{\text{bad}}\mathbb{P}(D)F_n(h)$. For $n < 16L_0$ use the early estimate; the first finite block is covered by $h_{\text{pre}} > 100M^2L_0^2$. This is the same separation adjustment already checked for the binary construction. The Gaussian walk has the same proof, with its normal second moment and overshoot bounds. The constants use the explicit flat brackets and the original gap-tail estimates.

B.7.4 The one-channel Gaussian estimates and strong one-end boxes

For one stationary standard OU process of covariance $\exp(-|s - t|/2)$, reflection gives exactly

$$F_1(t) = P(Z_s \geq 0, 0 \leq s \leq t) = \pi^{-1} \arcsin(e^{-t/2}), \quad \pi^{-1}e^{-t/2} \leq F_1(t) \leq \frac{1}{2}e^{-t/2}. \quad (\text{B.7.15})$$

At any time s under this survival tilt its density is

$$\frac{\phi(z)p_s(z)p_{t-s}(z)}{F_1(t)}, \quad z \geq 0, \quad p_v(z) = 2\Phi(z/\sqrt{e^v - 1}) - 1,$$

with $p_0 = 1$. Splitting according as both durations exceed one and using $p_v(z) \leq \min(1, \sqrt{2/\pi}z/\sqrt{e^v - 1})$ gives the uniform majorant $10(1 + z)^2\phi(z)$. Direct Gaussian integration therefore gives

$$E_{\text{survive}}e^{Z_s^2/8} \leq 256, \quad E_{\text{survive}}e^{(Z_0^2 + Z_t^2)/16} \leq 256. \quad (\text{B.7.16})$$

Delete endpoints outside $[0, B]$, $B = 2^{20}$. The deleted mass is at most $512 \exp(-B^2/16)F_1(t) < F_1(t)/4$. Translate by $4B(e^{-s/2} + e^{-(t-s)/2})$. This is a sum of covariance representers; its norm squared is at most $64B^2$ and its linear functional has magnitude at most $8B^2$ on the retained box. Hence both endpoints can be confined to $[4B, 9B]$ with mass at least $e^{-100B^2}F_1(t)/2$.

For a twice differentiable wall shift h define $A(h), V(h)$ exactly as in the bounded-action wall argument in Section B.5. Equation (16), normalized-variation Jensen and $u|z| \leq z^2/8 + 2u^2$ yield, directly for this single law,

$$e^{-8V(h) - A(h)/2}F_1(t) \leq P(Z + h \geq 0) \leq 256e^{2V(h)^2}F_1(t). \quad (\text{B.7.17})$$

For d decaying exponentials with exponents in $[1/3, 1/2]$ and total absolute amplitude W , $A(h) \leq 12W^2$, $V(h) \leq 16W$. Reserve, bounded scaled drift and the $\gamma = 1/8$ physical power wall are just these terms under Brownian Lamperti time. This proves the one-channel Brownian one-end factor $\min(1, (1 + h)/\sqrt{1 + n})$, with compatible positive end boxes, with comparison loss at most $\exp(M^{1000})$. The initial Brownian tube of duration of order h^2 is the explicit fixed increment/bridge tube from the scalar estimates; if h is already a fixed fraction of $\sqrt{n}$, use that same tube for saturation. For compact variance and drift bounds, all normalized endpoint boxes have endpoints below M^{100} and positive inner endpoint at least M^{-100} . For an effective asymmetric prefix with $|\delta|\sqrt{m} \leq 1$ and initial cap $A^{20}\sqrt{m}$, these bounds can instead be taken $[A^{-100}, A^{100}]$; the strong wall's normalized amplitude is below one once m exceeds h_{pre}^2 . The large M controls the coupling cutoff, not these latter physical box coefficients.

The conversion to a physical curve is also literal. Use half the initial reserve for the rank comparison and rank-wall amplitude $10A^4(1 + j)^{1/8}$. In the positive-jump orientation flat survival implies $t \leq (h + Aj)/c \leq Ah + A^2j$ before the j th jump. Hence $gt^{1/8} \leq g(Ah)^{1/8} + g(A^2j)^{1/8}$; its first term is less than $h/4$ and its second is paid by the rank wall. In the opposite orientation the same argument applies when $t \leq Ah + 2A^2j$. Beyond that value $ct - wj \geq ct/2$, which pays $gt^{1/8}$ once t exceeds $(2g/c)^{8/7}$; the half reserve pays the remaining fixed interval. Pre-jump/post-jump differences are at most w , already included in h_{pre} . These two time ranges verify that the rank-wall event enforces the stated physical curve in either orientation.

Use the exponential coupling (5) of Section B.3, without its Bernoulli part, independently on dyadic blocks. Take coupling tail power 200. The accumulated good-event error is at most $10^8 M^2 \log(2 + s)^4$; include continuous interpolation. Its Lamperti amplitude and action are below

M^{100} after the cutoff L_0 . Indeed $\log(2+s)^4/s^{3/8}$ is decreasing for $\log s \geq 100$, and substitution of (4) gives this bound at L_0 . The good-event wall can be inserted in (17).

For early bad blocks, (14) bounds the loss by a constant times $C_{\text{bad}}L_0^{-90}$; for late blocks use whole-block separation. The geometric series is less than $\exp(-\exp(M^{2250}))$, which is smaller than the Brownian strong-box retention and all flat comparison costs above. The initial block's exponential tail is even smaller, because h_{pre} exceeds $100M^2L_0^2$. Thus, for $h \geq h_{\text{pre}}$ and $n \geq h_{\text{pre}}^2$, the one-ended exponential risk has the strong curve, a central total and the compatible capped-reserve positive end box defined below, with probability between

$$e^{-e^{M^{2500}}} \min(1, (1+h)/\sqrt{1+n}) \quad \text{and} \quad e^{e^{M^{2500}}} \min(1, (1+h)/\sqrt{1+n}). \quad (\text{B.7.18})$$

The upper follows already from (11). REG is deleted relatively by (14) and the Gamma Chernoff bound. After reducing its slope by $100A^{10}$, this bound has coefficient at least M^{-12} . The elementary grid-pair series of polynomial degree at most ten is bounded by $\exp(-K/(10^6M^{20}))$ once $K > \exp(M^{100})$; use the change of variable $u = t^{1/16}$ in its two integrals. With K_{ord} this is smaller than half the retained lower in (18). Crossing long intervals are paid by their length tail; late short intervals use the whole-block separation. Inversion of rank-gap REG gives physical REG; its factors $100A^{10}$ and the two unit boundary adjustments are paid by (4).

Endpoint boxes with a large common reserve. In (18), the auxiliary box is imposed on a capped-height surrogate, whose centered fluctuation is the original one. Retain the original gap mean, drift and centered fluctuation S_j , and replace only the initial height by $h_{\text{cap}} = \min(h, M^{20}\sqrt{n})$. The lower event reads $h_{\text{cap}} + S_j$, with its ending height in $[M^{-100}\sqrt{n}, M^{100}\sqrt{n}]$. Its actual ending height is $h - h_{\text{cap}} + H_{\text{cap}}$; the actual risk is the capped risk plus the same constant $h - h_{\text{cap}}$ at every physical time. The retained Gamma-total window is central because $H_{\text{cap}} - h_{\text{cap}} - \delta n = O(M^{100}\sqrt{n})$, independent of that large common reserve. Thus when h is large, the actual endpoint adds the uncapped common reserve to this auxiliary box and can be arbitrarily larger than $\sqrt{n}$. Since $n \geq h_{\text{pre}}^2$, h_{cap} is still at least h_{pre} . Shorter experiments are handled by the final reserve/whole-chord branches of the exact central and asymmetric bridge constructions below and do not need this auxiliary box assertion. For the asymmetric prefix use the physical cap $A^{20}\sqrt{m}$ and its physical box $[A^{-100}, A^{100}]\sqrt{m}$.

B.7.5 Exact central bridge: one joint density and one middle tube

Put $K_{\text{asym}} = \exp(A^{700})$; it is smaller than M by (1). First suppose $x, y \leq K_{\text{asym}}\sqrt{n}$. The reference independent exponential gap mean is τ , and its signed drift satisfies $|\delta|\sqrt{n} \leq 2A^2K_{\text{asym}} < M$. If both endpoints exceed $\sqrt{n}$, replace the chord by the smaller constant chord $\sqrt{n}$ and the speed by nw/P ; its event implies the original event by (3). Otherwise take disjoint outer rank segments with at most $n/4$ marks each. At the left and reversed right use (18), its strong curve and positive endpoint box. Their ordinary probabilities multiply, because their entire reference gap sets are disjoint. The pre/post-mark bounded adjustments are paid by h_{pre} .

There is one exact-clock density for both outer segments:

$$\frac{f_{n+1-m_L-m_R, \tau}(P - T_L - T_R)}{f_{n+1, \tau}(P)}. \quad (\text{B.7.19})$$

Stirling gives its all-support upper at most 100, and its lower on the retained central outer totals at least $\exp(-M^{300})$; with wider boxes the exponent can be bounded by M^{1000} . The joint density retains the common residual total of both exposed ends. If an outer box is saturated,

truncate its initial reserve to a fixed multiple of $\sqrt{n}$ for the lower; the omitted common height can only help the actual chord.

At each retained pair of outer paths, the unread middle has an exact-total spacing law. Its legal endpoints are at least $M^{-100}\sqrt{n}$. Delete the subset coordinate in the proved Section B.4 tube and apply its displayed $J, \delta, N_0, c_{\text{mid}}$ with enlarged coefficient compact bound M^{100} and fraction M^{-100} . The loss has logarithm less than M^{2000} , and its length threshold is below N_{core} . Its chord identity is (3), so huge outer common reserves do not change its fluctuation coefficient. Since the remaining horizon is a fixed fraction of n , the same tube pays the global power curve for $n \geq N_{\text{core}}$; join its REG with the outer REG after reserving a fixed fraction of the slope. The long crossing pairs are deleted by their unconditional exact Binomial/Gamma tail, whose exponent dominates the reciprocal polynomial ballot factor. Thus (19) and the middle continuation give (5) in the central range with loss much smaller than C_{ord} .

For the upper, use (11) on both reference ends and the all-support upper in (19). Their product is at most a fixed multiple of β . This gives the central flat upper with an explicit constant from the one-end potential estimate and the joint Gamma density.

B.7.6 All reserves: explicit asymmetric constants

Reverse the whole exact clock when necessary. Outside the central box it suffices to handle $x < \sqrt{n}$, $y > K_{\text{asym}}\sqrt{n}$. The reference signed drift is positive and satisfies exactly

$$\delta = \frac{y - x + \epsilon\tau}{n}.$$

Take $m = \lfloor A^{-100}\delta^{-2} \rfloor$, $x' = \min(x, A^{20}\sqrt{m})$ and $y' = y - x + x'$. Increasing K_{asym} ensures $m \leq n/(16A^2)$; the remaining reference clock mean stays in $[1/(2A), 2A]$. Above the fixed threshold $m_* = h_{\text{pre}}^8 M^{1000}$, (18) gives a retained prefix of probability at least $\exp(-\exp(M^{2500}))$ times

$$\min(1, (1 + x')/\sqrt{m}) \asymp_A \beta_n(x, y),$$

with central total and capped ending height $H \in [A^{-100}\sqrt{m}, A^{100}\sqrt{m}]$. The original risk and endpoint have the additional common shift $x - x'$. Its density in the full exact clock is the ratio $f_{n+1-m, \tau}(P - T_m)/f_{n+1, \tau}(P)$, with uniform central lower. For each retained prefix the remaining physical drift v' obeys

$$\delta/(4A^2) \leq v' \leq 4A^2\delta, \quad Hv' \geq A^{-200}, \quad P'(v')^2 \geq 16\exp(A^{600}). \quad (\text{B.7.20})$$

These follow by inserting the retained $O_A(\sqrt{m})$ endpoint and duration errors: $\sqrt{m}\delta \geq A^{-51}$, whereas $n\delta^2 \geq K_{\text{asym}}^2/(4A^4)$. Thus every error is at most $A^{200}/K_{\text{asym}}^2$ of the main increment. The estimates hold separately for every retained prefix path and its actual duration and endpoint.

Here are numerical constants for the residual exact continuation:

$$\kappa = A^{-20}e^{-A}, \quad a = A^{-250}, \quad K_c = 4A^{200}e^A, \quad R_c = e^{A^{600}}. \quad (\text{B.7.21})$$

A rate- λ Poisson clock, $\lambda = n'/P' \in [1/(2A), 2A]$, with deterministic jump w has compensator at $\theta \leq 1/A$ bounded above by $-\nu\theta + 2A^4e^A\theta^2$. At $\theta = \kappa v$ it is at most $-\nu\theta/2$. Optional stopping from entrance a/v therefore gives first-half survival probability $p_a = 1 - \exp(-\kappa a) \geq \kappa a/2$. Retain its count within $K_c\sqrt{n'}$ of its mean; Chebyshev removes at most $p_a/2$. The exact first-half density relative to that Poisson clock is at most 25 everywhere and at least $(1/16)\exp(-4 - 8K_c^2)$ on this count event, whenever $n' \geq 16K_c^2$. The resulting first-half lower is

$$b_L \geq (p_a/32)e^{-4-8K_c^2}.$$

On the reversed late half, failure of the positive drift path requires a centered count fluctuation at least $vP'/2$. The same exponential maximal inequality and density upper give $25 \exp(-\kappa P'v^2/4)$. Since

$$(4/\kappa) \log(50/b_L) \leq 1000A^{420}e^{3A} < R_c,$$

and $R_c/(100A^6) > 16K_c^2$, (21) guarantees continuation probability at least $b_L/2$. This is the one-channel Poisson comparison for the constant mark law. Apply it to the half-chord process $Q(s) = R_{\text{rem}}(s) - (H + v's)/2$, whose entrance is $H/2$, drift is $v = v'/2$, and speed is $(c + n'w/P')/2$. Its reversed entrance is $(H + v'P')/2$ and its reversed drift is $-v'/2$. Consequently late-half failure means

$$\epsilon M_{\text{rev}}(u) > H/2 + v'(P' - u)/2 \geq v'P'/4, \quad 0 \leq u \leq P'/2.$$

The maximal bound is therefore $25 \exp(-\kappa P'(v')^2/16)$, exactly the bound above with $v = v'/2$. Equation (20) has the extra factor 16, so no margin is lost by this factor-four change. Moreover the entrance/drift product is $Hv'/4 \geq A^{-200}/4 \geq a$ in (21). Its event is the half-chord bound

$$R_{\text{rem}}(s) \geq (H + v's)/2 \geq A^{-210}\sqrt{m}(1 + s/m). \quad (\text{B.7.22})$$

The global power curve is at most $A^2m^{1/8}(1 + s/m)^{1/8}$ because $T_m \leq 2Am$. Already $m \geq m_*$ makes (22) dominate it. Unconditional ordinary REG failure is less than $b_L/4$ at K_{ord} . Rate discrepancies at the junction are $O_A(m^{-1/2})$ and $O_A(\sqrt{m}/n)$, smaller than the reserved slope above m_* . Thus each retained prefix has the required completion with this fixed lower, and integration in its single exact-prefix density proves the asymmetric lower. The logarithm of every continuation inverse loss above is below $\exp(A^{700}) < M$; it fits (4) with a large margin.

If $m < m_*$, use the whole half-chord construction with the final reserve h_{ord} . The positive physical drift is at least $v_0 = 1/(10A^{102}\sqrt{m_*} + 1)$. This bound and m_* are fixed before h_{ord} . The elementary maximum

$$\sup_{t \geq 0} \{gt^{1/8} - v_0t/4\} \leq 2A^2v_0^{-1/7}$$

is smaller than $h_{\text{ord}}/4$, and $h_{\text{ord}}v_0/4 > 1$. The first-half lower therefore follows from (21), even more strongly with the larger entrance. If $Pv_0^2 < R_c$, the horizon is at most $R_c/v_0^2 < N_{\text{core}}$; on that bounded range the final reserve is larger than $100A^2(P + n + 1)$ and the strong event is automatic. K_{ord} makes REG automatic there as well. This treatment does not feed the larger output reserve back into m_* .

The asymmetric upper uses only (11) and the all-support exact prefix density upper. Since $\delta \leq 4A^2y/n$ and $y \geq \sqrt{n}$, it is at most $C_A \min(1, (1 + x)y/n) \leq C_A\beta$. This completes all-reserve upper without an asymptotic ballot constant.

Finally, on $n < N_{\text{core}}$ with legal $x, y \geq h_{\text{ord}}$, the full path is at least $\min(x, y) - An$, whereas its curve is at most $A[A(n + 1)]^{1/8}$. The chosen h_{ord} makes this deterministic strong inequality hold. Every count displacement is at most $2n + A^2\sqrt{n}$, below K_{ord} , so REG is automatic. The whole bounded-horizon range is therefore covered deterministically at these final reserve and intercept thresholds, within C_{ord} .

B.7.7 Application to the terminal suffix and fixed lift

Equations (2)–(22) establish an effective ordinary strong/REG bracket with its exact clock and signed endpoint identity. Uniform fine-name refinement at a fixed complete type is then the same pointwise hypergeometric calculation as in the history estimates: the path reads no fine name, so the refinement retains a fixed fraction before averaging this ordinary path. Broad banks are retained by their exponentially small conditional failure relative to $\beta \geq c/(1+n)$. These add at most $\exp(M^{2000})$ to the inverse-loss logarithm and are covered by C_{ord} . The ordinary source is linear in its one reservoir count atom; this construction changes only its conditional path bracket and does not introduce a second root atom.

Specifically, after the accepted terminal prefix is frozen, the arithmetic construction reads the remaining suffix count and its conditional uniform clock. On a weak ordinary original block the imposed local event is $\theta C_{\text{actual}}(t) + E_{\text{actual}}(t) \geq gdp(t)^{1/8}$, together with uncolored regularity, original fine-name REG, and the broad-bank conditions when designated. The actual endpoint pair is $\theta x_{\text{actual}}, \theta y_{\text{actual}}$, with its exact signed increment; the suffix/root count is unchanged. Apply (5)–(7) to that pair. The Gaussian and Gamma end boxes in (18)–(19) are auxiliary proof events on the independent reference end segments. They are not additional upper restrictions on the actual source suffix endpoint or on its prescribed root count. If both legal endpoints exceed $\sqrt{n}$, the whole constant-chord surrogate in the exact central bridge construction has the same centered fluctuation and lies below the actual chord. If an asymmetric initial endpoint is capped, both legal endpoints are shifted by its one common constant, leaving the increment and clock law unchanged. At each retained prefix the actual middle pair is still legal and positive; no already sampled terminal label or accepted prefix path is changed. Complete-cell lower events follow from this continuous event; the reverse comparison adds only $c\Delta_* + g\Delta_*^{1/8}$ to reserves as in Section E.3, paid by h_{ord} .

For the specified fixed lift, take the already fixed physical compact bound $A = 2M_1^2$ and $M = M_2$. Requirement (1) follows from $(2M_1^2)^{1000} < M_1^{100000}$. Every threshold in (4), and the continuation inequalities (20)–(22) are then dominated by the existing $K_{\text{REG}} = e^{e^{M_2^{5000}}}$ and $h_W = e^{e^{M_2^{5500}}}$. The actual contracted endpoints exceed h_{ord} because $\theta M_0^{-20}h > h_{\text{ord}}$ for the existing $\theta = M_1^{-M_1}$ and $h = e^{e^{M_2^{6000}}}$. Hence the ordinary case $\rho = 1$ uses the same Λ_k , scalar compact parameter, or the h -independent weak selection.

The ordinary bridge estimate and its application to the actual terminal suffix therefore fit the same fixed lift as the binary construction.

C Count types and the selected measure

This appendix constructs the bounded word weight used in the lower bound and proves its first-mass estimate. Count admission and spatial acceptance play different roles: counts may couple every block, whereas at a fixed complete type the clocks and uniform multiset words remain independent. We exploit this distinction before introducing the adaptive targets by recoloring.

The construction has four stages. Partial-revelation and free-interval estimates produce one uncolored dominating cost for each selected block. A mixed-moment identity combines these costs over the whole count source. An ordinary suffix supplies one root atom and its path factor. Finally, a bounded-density transport realizes the targets, and an exact likelihood calculation evaluates the nominal first mass. The complete-cell construction and finite disintegration at the end verify that these stages retain their stated conditioning.

C.1 The joint lower measure and its transport

This appendix supplies the probability and measure constructions used in Section 7. We keep the original physical blocks throughout. Conditional on its uncolored total n_i , a block has n_i uniform points on its interval and all $n_i + 1$ spacings. Its fine labels have their original categorical law. Conditional on a complete named count vector, they are a uniform multiset permutation. Constants in this appendix depend on the fixed dimension and the fixed compact scalar, count, and wall parameters.

C.1.1 Revealing part of an original word

A comparison class reveals some labels but leaves the rest under their original law. The cost of retaining a central count band is then a ratio of multinomial atoms. Summing this ratio over the whole coupled count weight, rather than choosing one type, gives a bound that charges only the blocks and strips actually revealed.

Lemma C.1 (Central counts after partial revelation). *In block i , let N_i labels be independent with alphabet size $a_i \leq H$ and probabilities $p_{i,c} \geq p_* > 0$. The blocks are independent under their original law π . Let $A(K) \geq 0$ be a finite weight on their complete count vectors, supported on*

$$|K_{i,c} - N_i p_{i,c}| \leq B\sqrt{1 + N_i}, \quad W = \mathbb{E}_\pi A.$$

Fix m_i slots in block i . For any positive-probability event Q measurable in just these slots, including coupled or partial signatures,

$$\mathbb{E}_\pi[A \mid Q] \leq CW \prod_{i:m_i>0} \left(\frac{1 + N_i}{1 + N_i - m_i} \right)^{(a_i-1)/2} \leq CW \prod_{i:m_i>0} (1 + m_i)^{(a_i-1)/2}. \quad (\text{C.1.1})$$

If $m_i \leq N_i/2$ in every affected block, the ratio is bounded by a fixed constant. A block with no revealed slots incurs no factor.

Proof. For a fixed a -letter law in this compact simplex, its multinomial mass $P_{n,p}(k)$, with $d = a - 1$, satisfies

$$\sup_k P_{n,p}(k) \leq C(1 + n)^{-d/2}, \quad P_{N,p}(k) \geq c(1 + N)^{-d/2} \quad \text{on the stated central window.} \quad (\text{C.1.2})$$

For the first bound, Fourier inversion, with the last coordinate set to zero, bounds a coefficient by the integral of $|p_a + \sum_{j \leq d} p_j e^{it_j}|^n$. The square of this modulus is at most

$$1 - 2p_*^2 \sum_{j \leq d} (1 - \cos t_j) \leq 1 - c \sum_{j \leq d} t_j^2, \quad |t_j| \leq \pi.$$

Its integral is $O((1 + n)^{-d/2})$. For the lower bound, Stirling's inequalities give $P_{N,p}(k) \geq cN^{-d/2} \exp\{-ND(k/N\|p)\}$ when N is large. The entropy Hessian is bounded on the segment between p and k/N , so its exponent is $O(B^2)$. Bounded N is covered by compactness and feasibility. The case $a = 1$ is deterministic.

Expose the specified labels exactly, and let b_i be their count vector. Independence under the original law gives the remaining type mass explicitly:

$$\pi(K = k \mid \text{revealed labels}) = \prod_i P_{N_i - m_i, p_i}(k_i - b_i).$$

When $m_i = 0$, the factor is exactly its original type mass and cancels in the ratio. On each affected block, (C.1.2) bounds the ratio by the corresponding factor in (C.1.1). Multiply by $A(k)$ and sum the entire coupled type vector. Conditioning on Q then averages these uniform bounds over its exact revealed assignments. The last inequality follows from $(1+N)/(1+N-m) \leq 1+m$. If $W = 0$, all the relevant feasible type weights vanish, and the conclusion is immediate. $\square$

For a fixed original history and anchor, a signature class is an event Q of this kind: it reads only the slots with nontrivial partitions. This applies to that fixed class under the original law. It does not change that law to a law conditioned on an adaptively chosen history. In particular, if a class fully exposes blocks in I and reads m_j additional slots elsewhere, its count cost is at most

$$C \prod_{i \in I} (1 + N_i)^{(a_i-1)/2} \prod_{j \notin I} (1 + m_j)^{(a_j-1)/2}.$$

Only counts in actually revealed blocks or strips appear.

C.1.2 Two free intervals in the original class

The local path upper bound uses two disjoint intervals just inside the exposed endpoint strips. At a fixed complete type the remaining binary labels form one uniform subset, so both intervals must first be compared jointly with an independent Bernoulli law. Opposite monotonicities of the forward and reverse walls then give one envelope uniform over all retained types.

We next bound the path part after the complete type and exposed labels have been fixed. Consider one selected block of length P , total $n \asymp P$, and binary risk

$$R(t) = ct - aN(t) + bN_B(t), \quad c, b, a - b > 0,$$

with these parameters in fixed compact intervals. Every accepted word has an actual endpoint pair satisfying

$$0 \leq x \leq C_{\text{box}}X, \quad 0 \leq y \leq C_{\text{box}}Y, \quad y - x = cP - an + bk, \quad x + R(t) \geq -h, \quad (\text{C.1.3})$$

where $X, Y \geq h_* > 0$. All additional walls and refinements can remain in the acceptance. Assume the uncolored endpoint bound

$$N([0, t]) + N([P - t, P]) \leq C_{\text{occ}}(1 + t), \quad 0 \leq t \leq P/2. \quad (\text{C.1.4})$$

The original class may read prescribed slots in endpoint strips of width $C_s(1 + d)$, but is unrestricted between them. This last condition is a statement about its signatures.

Put

$$\mathcal{T} = \frac{(1 + X)(1 + Y)}{n}, \quad p_0 = \frac{a - cP/n}{b}, \quad \rho_0 = \frac{(cP/(n + 1))^2}{(cP/(n + 1))^2 + b^2 p_0(1 - p_0)}. \quad (\text{C.1.5})$$

On the rare branch $\mathcal{T} < t_0$, the accepted interior type and (C.1.3) imply that p_0 belongs to a fixed compact subinterval of $(0, 1)$. Define $\Psi = \mathcal{T}^{2\chi(\eta, \rho_0)}$ on this branch and $\Psi = 1$ otherwise, where η lies in a fixed compact subinterval of $(0, 1)$.

Fix $0 < \gamma < 1/2$ and

$$D_n = [C_{\text{cut}} \log(2 + n)]^{1/\gamma}.$$

Set $q_d = 1$ if the branch is saturated, if $n < n_0$, or if $d > D_n$. Here n_0 is chosen so that, in the remaining cases, the endpoint strips contain at most a fixed small fraction of n slots and the

following two intervals fit in the free middle. Choose their starting and ending boundaries s_d, t_d just inside those strips. Write

$$U_d = 1 + X + d, \quad V_d = 1 + Y + d, \quad M_d = \max(U_d, V_d), \quad L_d = \varepsilon_0 \min\{P, (n/M_d)^2\}. \quad (\text{C.1.6})$$

Take intervals $[s_d, s_d + L_d]$, $[t_d - L_d, t_d]$. The constant ε_0 is small enough that they are disjoint and their combined occupancy is at most half the unread slots, by (C.1.4). Grid rounding changes only fixed allowances. If the resulting length is below a fixed scalar threshold, again set $q_d = 1$.

In the remaining case set

$$p_{\pm} = p_0 \pm C_1(X + Y + 1 + d)/n.$$

Reduce t_0 and increase n_0 to keep these probabilities uniformly interior. Let q_L be the iid Bernoulli p_+ probability on the actual slots of the left interval that

$$cv - aN([s_d, s_d + v]) + bN_B([s_d, s_d + v]) \geq -C_2 U_d, \quad 0 \leq v \leq L_d.$$

Let q_R be the iid Bernoulli p_- probability on the right interval, read backwards, that

$$-cv + aN([t_d - v, t_d]) - bN_B([t_d - v, t_d]) \geq -C_2 V_d.$$

Define $q_d = \mathbf{1}_{(\text{C.1.4})} q_L q_R$. These iid probabilities are upper bounds for the original conditional law, not the law of the accepted word.

To prove this, fix all revealed labels and a complete named type. If m slots and e binary successes have been revealed, the remaining binary word is a uniform $(k - e)$ -subset of $N = n - m$ slots. Its fraction $p' = (k - e)/(n - m)$ satisfies

$$|k - np_0| \leq C(X + Y), \quad |p' - p_0| \leq C(X + Y + 1 + d)/n,$$

so $p_- \leq p' \leq p_+$. Let $M_J \leq N/2$ be the combined number of slots in both free intervals. Relative to iid Bernoulli p' , their joint density under the exact subset law is

$$\frac{\mathbb{P}\{\text{Bin}(N - M_J, p') = Np' - j\}}{\mathbb{P}\{\text{Bin}(N, p') = Np'\}} \leq C. \quad (\text{C.1.7})$$

The numerator is bounded by a maximum atom and the denominator is a central atom. This is one comparison for both intervals and holds for every assignment on them. The endpoint identity and occupancy bound show that (C.1.3) requires both displayed tests, with the stated allowances. Under the iid law the intervals are independent. The left event increases and the right event decreases with binary successes, so coordinatewise uniform coupling gives

$$\mathbb{P}\{\text{accepted path} \mid \text{complete type, revealed labels}\} \leq Cq_d(G). \quad (\text{C.1.8})$$

Additional named restrictions within either binary list do not change this uniform binary-subset marginal.

For several blocks the multiplication is made at a fixed whole type and fixed exposed labels. Suppose $0 \leq F \leq A(K)$ times the necessary path indicators, with A supported on the original central windows and $W = \mathbb{E}_{\pi} A$. Multiply (C.1.8) under the independent residual block laws, sum the whole type, and then apply Lemma C.1. This gives

$$\mathbb{E}_{\pi}[F \mid Q] \leq CW R_{\text{strip}} \prod_{j \in I} (1 + n_j)^{(a_j - 1)/2} \prod_{\substack{i \notin I \\ i \text{ selected}}} q_{i, d_i}(G_i), \quad (\text{C.1.9})$$

where I denotes fully exposed blocks and

$$R_{\text{strip}} = \prod_{i \notin I: m_i > n_i/2} (1 + m_i)^{(a_i-1)/2}.$$

At active depths, $m_i \leq \varepsilon n_i$, so this extra factor only concerns the discarded large-depth regime. Conditioning on a partial-signature event Q last averages the uniform bound over its exact revealed assignments. There is no factor $1/\pi(Q)$, and the type acceptance need not factor across blocks.

C.1.3 Moments of the free-interval costs

The preceding envelope is pointwise in the clock. Its fractional moment is evaluated by a second joint comparison, now between the exact-total clock and a Poisson clock on the two free intervals. Summing the envelopes over depths with decaying coefficients produces a positive uncolored cost that dominates every needed local test and has the correct persistence moment.

For $d \leq D_n$, the functions just constructed satisfy

$$\mathbb{E}_{G|n} q_d(G)^\eta \leq C(1+d)^C \Psi. \quad (\text{C.1.10})$$

Here and below $A(K)$ and deterministic parameters must be fixed before these unread clocks for a clock moment assertion. The class bound itself is pointwise after fixing the clocks.

The exact-total uniform clock restricted jointly to the two free intervals has density, relative to a rate n/P Poisson clock,

$$\frac{\mathbb{P}\{\text{Pois}(n(1-\kappa)) = n - N_J\}}{\mathbb{P}\{\text{Pois}(n) = n\}} \leq C, \quad \kappa = 2L_d/P < 1/2. \quad (\text{C.1.11})$$

Conditional point locations also agree. Only after this single joint comparison are the two reference restrictions independent. Drop the occupancy indicator for the upper bound. The auxiliary risk drifts have magnitude $O((X+Y+1+d)/n)$, whose product with $\sqrt{L_d}$ is bounded. Theorem 5.23, for either sign, therefore gives

$$\mathbb{E} q_d^\eta \leq C \min \left\{ 1, \left(\frac{U_d}{\sqrt{1+L_d}} \right)^{2\chi_+} \right\} \min \left\{ 1, \left(\frac{V_d}{\sqrt{1+L_d}} \right)^{2\chi_-} \right\}. \quad (\text{C.1.12})$$

The one-end theorem is applied to the original positive mark pair in both orientations, with compact drift changes; all original spacings were retained before (C.1.11).

If $M_d \leq \sqrt{n}$, then $1+L_d \asymp n$ and the product of the reserve ratios is $\asymp U_d V_d/n$. If $M_d > \sqrt{n}$, then $1+L_d \asymp (n/M_d)^2$; the larger reserve factor is bounded by one and the smaller ratio is $\asymp U_d V_d/n$. The exponent changes satisfy

$$|\chi_+ - \chi(\eta, \rho_0)| + |\chi_- - \chi(\eta, \rho_0)| \leq C\{\mathcal{T} + (1+d)/n\}.$$

Every nontrivial reserve ratio in (C.1.12) is at least $c \min(1, \mathcal{T})$. Multiplying its negative logarithm by the last bound is uniformly bounded, because $\mathcal{T}|\log \mathcal{T}|$ is bounded and $d \leq D_n$. Replace both exponents by $\chi(\eta, \rho_0)$ and use $U_d V_d \leq (1+d)^2(1+X)(1+Y)$ to prove (C.1.10). If the free interval was too short for the one-end theorem, then $M_d \gg n$, and $d = o(\sqrt{n})$ forces one of X, Y to have order n . Thus $\mathcal{T}$ has a fixed positive lower bound and the convention $q_d = 1$ is valid. Bounded $n < n_0$ is treated in the same upper estimate; this does not enlarge the support convention of the source lower.

Fix $b_0 > 0$ and an integer B_0 large enough to sum the later pivot multiplicities. Choose D_0 with $D_0\eta_- > 2\sup\chi + 1$, and set

$$D^{\text{raw}}(G) = (2+n)^{-D_0} + \sum_{d \geq 0} (1+d)^{B_0} e^{-b_0 d^\gamma} q_d(G), \quad D(G) = \min(1, D^{\text{raw}}(G)). \quad (\text{C.1.13})$$

Choose C_{cut} after D_0 so that the η -powers of the numerical coefficient tail beyond D_n sum to at most $C(2+n)^{-D_0\eta}$, uniformly in η . Subadditivity and (C.1.10) then give

$$(2+n)^{-D_0} \leq D \leq 1, \quad \mathbb{E}D^\eta \leq C\Psi. \quad (\text{C.1.14})$$

At depth zero (C.1.8) gives $q \leq CD$ for each supported complete type. Clipping at one preserves this bound because $q \leq 1$.

Beyond the active cutoff the convention $q_d = 1$ provides no small moment. The geometric depth charge pays this regime directly. When it contains cd^γ , enlarge C_{cut} until

$$e^{-cd^\gamma/2}(1+n)^{H/2} \leq C(2+n)^{-D_0}, \quad d > D_n. \quad (\text{C.1.15})$$

This pays both the selected bridge floor and any large-strip reveal ratio in that block, leaving half the penalty to sum pivots. An unselected block with $m > n/2$ instead has $n \leq 2m \leq C(1+d)$, so its reveal ratio is a polynomial in its actual depth. For active depths use $q_d \leq CD(1+d)^{-B_0} e^{b_0 d^\gamma}$, choosing b_0 below the available geometric coefficient. When several blocks share a depth parameter, divide its fixed penalty among the dimension-bounded number of factors before choosing these constants.

A related count extraction will sometimes be useful in a completely free interior interval. Conditional on every label outside it, a band of at most $C(1+R)$ possible counts there has probability at most $\min(1, C(1+R)/\sqrt{1+N_I})$, irrespective of its center. This is the binomial maximum-atom bound in (C.1.2). The whole band is summed before taking a power. If this interval and the two path intervals are disjoint with total length less than half the original block, the single density comparison (C.1.11) applies to all three at once. The count factor has moment $O(\min(1, (1+R)/\sqrt{1+P})^\eta)$; atypically small occupancy has exponentially small probability. This explains the ordinary count-window factor without replacing the original class by three independently conditioned experiments.

C.1.4 Named regularity on each retained word

Binary acceptance leaves two actual lists to be refined into fine names. Regularity must survive this refinement conditionally on each retained parent word, however rare that word may be. A uniform hypergeometric estimate gives this per-word statement before any fractional moment is averaged.

Lemma C.2 (Uniform named refinement). *Let a fixed parent list of $n \leq C(1+L)$ points in an original block satisfy, at the fixed grid boundaries,*

$$|N((a, b]) - (n/L)(b-a)| \leq \theta_{\text{in}}(b-a) + K_{\text{in}}\{1 + d_L(a)^\zeta + d_L(b)^\zeta\}, \quad (\text{C.1.16})$$

where $0 < \zeta < 1/2$ and the grid has bounded boundary density. For every feasible full refinement type (k_c) , its original uniform multiset refinement satisfies, simultaneously in the finitely many names,

$$|N_c((a, b]) - (k_c/L)(b-a)| \leq \theta_{\text{out}}(b-a) + K_{\text{out}}\{1 + d_L(a)^\zeta + d_L(b)^\zeta\} \quad (\text{C.1.17})$$

with conditional probability at least $1 - \varepsilon$. Here θ_{in} is sufficiently small in terms of the prescribed $\theta_{\text{out}} > 0$, and K_{out} is sufficiently large in terms of the fixed data and ε . The assertion holds for every fixed parent word, however it was previously selected.

Proof. Condition on the parent positions. On an interval containing m positions, the name- c count is hypergeometric with mean αm , where $\alpha = k_c/n$. Its exponential moments are bounded by those of independent Bernoulli sampling: apply the elementary-symmetric-mean inequality to the n values e^t or 1, and use complementation for the lower tail. Chernoff's argument therefore gives

$$\mathbb{P}\{|N_c - \alpha m| > u\} \leq 2 \exp\{-cu^2/(m + u)\}.$$

The displacement of the mean from $(k_c/L)(b - a)$ is bounded by (C.1.16), and $m \leq C(b - a) + K_{\text{in}}(1 + d_L(a)^\zeta + d_L(b)^\zeta)$. After the stated choices, the probability of violating (C.1.17) is at most

$$C \exp\{-c_1(b - a) - c_2 K_{\text{out}}(1 + d_L(a)^\zeta + d_L(b)^\zeta)\}.$$

The sum over all grid endpoint pairs is uniformly finite. Pairs near an original endpoint are summed by the convergent series $\sum_{j \geq 0} e^{-cj^\zeta}$; pairs in opposite halves also pay $e^{-c_1(b-a)}$. Increasing K_{out} makes the finite union over names and blocks have probability at most ε . Bounded blocks have bounded counts, absorbed by the fixed intercept; empty lists have their unique empty refinement. $\square$

Apply this lemma to both actual lists of a retained binary word, conditioning on that word and its complete type. It supplies named REG with constant conditional success before any rare bridge is averaged. If natural rather than empirical rates are used, a central type changes the rate by $O(L^{-1/2})$, absorbed by a spare linear slope above a fixed length threshold. The lemma concerns this per-word refinement only; any later function of the selected spatial lists remains in its own conditional measure.

C.1.5 Admission of the macroscopic recoloring intervals

The later transport needs many occurrences of two specified names in an interior interval. Its admission failure is exponentially small in the interval length, whereas the bridge factor has a polynomial lower bound. We use this separation inside the local fractional moment, so bank admission retains a fixed proportion even of a rare bridge mass.

Consider a prescribed original block with $P, n \asymp T$, complete central fine type $|K_c - np_c| \leq B\sqrt{1 + n}$, and a fixed interior interval I of length at least ρP , a fixed positive distance from both endpoints. For two specified actual fine names s, t , require

$$m = N(I) \geq c_I n, \quad |K_{I,s} - p_s m| + |K_{I,t} - p_t m| \leq \frac{m}{\log(2 + T)}. \quad (\text{C.1.18})$$

The same requirement may be imposed in finitely many prescribed intervals. Conditional on n , the uncolored occupancy failure has probability $O(e^{-cn})$. Conditional on the complete type and a clock with $m \geq c_I n$, the named occupancy is hypergeometric, with mean mK_c/n . Its distance from $p_c m$ is $O(m/\sqrt{n})$, which is less than $m/(8 \log(2 + T))$ above a fixed threshold. The same Chernoff bound as in Lemma C.2 gives, at each full central type,

$$\mathbb{P}_{G,\pi}\{(\text{C.1.18}) \text{ fails} \mid K\} \leq C \exp\{-cT/\log^2(2 + T)\}. \quad (\text{C.1.19})$$

Let $q(K, G)$ already include the allocated curved path, binary REG, and full named regularity, with

$$q \leq CD, \quad \mathbb{E}q^\eta \geq c\Psi, \quad \mathbb{E}D^\eta \leq C\Psi, \quad \Psi \geq c_0(2+n)^{-B_1}. \quad (\text{C.1.20})$$

The all-reserve and contracted-chord bridge theorems 6.1 and 6.5, followed by Lemma C.2, supply the lower assertion. The free-interval construction supplies the other two assertions. Let q_{good} also impose (C.1.18), and put $q_{\text{bad}} = q - q_{\text{good}}$. Subadditivity, then Jensen, give

$$\begin{aligned} \mathbb{E}q_{\text{good}}^\eta &\geq \mathbb{E}q^\eta - \mathbb{E}q_{\text{bad}}^\eta \\ &\geq c\Psi - (\mathbb{E}q_{\text{bad}})^\eta \geq c\Psi - C \exp\{-c\eta T / \log^2(2+T)\} \geq c'\Psi. \end{aligned} \quad (\text{C.1.21})$$

The final inequality follows after one fixed threshold enlargement, uniformly in the compact range of η , from the polynomial lower bound. All walls and REG restrictions were already inside q , and the dominating cost D does not change. The same argument applies to an unselected regular block with $D = \Psi = 1$. This local fractional estimate is taken before any later scale or root probability is integrated. For large T , (C.1.18) also guarantees at least cn occurrences of each of its two names, enough for every prescribed $O(\log T)$ recoloring.

C.1.6 Mixed moments of the coupled count measure

The count source is a single measure on complete types. For each fixed type, Hölder turns the local fractional survival estimate into a mixed first moment with the dominating cost. Only these mixed integrands are multiplied across independent clocks; Tonelli then integrates the coupled type measure. This order retains every plateau's own exponent.

The count construction in Section A.2 precedes the unread positions. More precisely, the diagonal exit bands and their endpoint count transports (A.16, A.14, and A.15) use only the nonterminal plateaux. Projecting out terminal private names and then filling their prefix lists under the original conditional law gives the whole prefix count measure of A.19. Conditional on every original prefix total in its own central window, this is one positive measure μ on full fine types and count-only state, independent of newly unread prefix positions, with

$$W = \mu(1) \asymp \prod_{j < m} w_j. \quad (\text{C.1.22})$$

It retains the displaced good-name box. The conditional terminal completion in A.20 has constant mass for each retained prefix type and each suffix root vector in the specified central construction. Independence from suffix uncolored counts follows from the preceding support of the nonterminal-plateau construction.

For each fixed complete type, take the legal pair of A.17, namely (7.5). All its outside contexts have the same actual increment, so it is below both actual intrinsic endpoint minima and belongs to their common positive boxes. Write its path as $\ell_i^- + E_i$, with ℓ_i^- the joining chord. Impose on that original clock

$$\theta \ell_i^-(t) + E_i(t) \geq 2g \min(t, L_i - t)^\gamma - h_W, \quad 0 < \zeta < \gamma < 1/2, \quad 0 < \theta < 1, \quad (\text{C.1.23})$$

together with binary REG. Conditional on this word, refine both lists by Lemma C.2. On unselected blocks require only the appropriate ordinary and named regularity. Each local mass $q_i(\omega, G_i)$ then satisfies (C.1.20) at $\eta_i = \sigma_{\text{plateau}(i)}$, with the factor Ψ_i^f of (7.6). The necessary flat path follows from (C.1.23) because its uncontracted chord is larger and nonnegative, so the

depth-zero bank bound applies at these intrinsic scales. Bounded supported experiments retain their fixed positive conditional mass and have $D_i = \Psi_i^I = 1$; a scalar parameter is not assigned to a zero count. Applying (C.1.21) incorporates all prescribed macroscopic recoloring intervals without changing these estimates.

The following measure identity retains the individual plateau powers.

Lemma C.3 (Whole-type mixed moment). *Let μ be any finite positive measure on complete types, independent of the original independent environments G_i , and suppose (C.1.20) holds uniformly in its support, with possibly different $\eta_i \in [\eta_-, \eta_+] \subset (0, 1)$. Then, for $f(G) = \int \prod_i q_i(\omega, G_i) d\mu(\omega)$,*

$$\mathbb{E} \left[f(G) \prod_i D_i(G_i)^{\eta_i-1} \right] \asymp \mu(1) \prod_i \Psi_i. \quad (\text{C.1.24})$$

Proof. For each fixed ω and i , Hölder applied to

$$q_i^{\eta_i} = (q_i D_i^{\eta_i-1})^{\eta_i} (D_i^{\eta_i})^{1-\eta_i}$$

gives

$$\mathbb{E}[q_i D_i^{\eta_i-1}] \geq \frac{(\mathbb{E} q_i^{\eta_i})^{1/\eta_i}}{(\mathbb{E} D_i^{\eta_i})^{(1-\eta_i)/\eta_i}} \gg \Psi_i.$$

The pointwise domination $q_i \leq C D_i$ gives both the upper estimate and finiteness. At each complete type, independence of the original environments lets us multiply the local mixed integrands. Their two-sided estimates are uniform in the type, so Tonelli integrates the product over the full measure μ . $\square$

For the present prefix, $f_{\text{pre}}(G) = \int \prod_i q_i(\omega, G_i) d\mu(\omega)$. The plateau costs in (7.10) give the deterministic width multiplier $\prod_{j < m} w_j^{\sigma_j-1}$. Multiplying (C.1.22) by this multiplier yields exactly $\prod_{j < m} w_j^{\sigma_j}$, while every local path retains its own plateau power.

C.1.7 The ordinary suffix and its one count atom

The suffix estimate must hold pointwise in the complete retained prefix and uniformly over its later uncolored root target. We first identify the ordinary weak block's two small endpoint scales, then select the actual reservoir count and impose the path event at that same count. This gives one suffix kernel containing both the ordinary factor and the root atom. Suppose first that the suffix contains the designated near-threshold group in the convention of (3.11). We call this its ordinary weak block; throughout this construction that term refers to the fixed profile neighborhood, with no additional drift cutoff. It is the first suffix block. Let M_0 be its length, P the entire preceding original length, and B the following positive-drift length. All later suffix drifts have a fixed positive lower bound, by the separation of the finitely many thresholds. The clipped-reference and KKT surplus estimates give

$$A_{\text{mean}} = \sum_{g < i} L_g g_g(r_g) \asymp P, \quad B_{\text{mean}} = - \sum_{g > i} L_g g_g(r_g) \asymp B, \quad A_{\text{mean}} + M_0 g_i(r_i) = B_{\text{mean}}. \quad (\text{C.1.25})$$

Thus $D = A_{\text{mean}} \asymp P$, $P \leq C(M_0 + B)$, and $B \leq C(P + M_0)$, including zero-side cases. This sums all original earlier groups and does not assign a common drift sign to each of them.

At the weak block's left endpoint, reference clipping gives cost at least $c \min(P, M_0 + B) \gg P$. The full-prefix context has cost exactly A_{mean} and no proper good-group term. At the right

endpoint the same argument gives a lower bound $c \min(P + M_0, B) \gg B$, attained up to constants by the full-prefix context of cost B_{mean} . Therefore the intrinsic endpoint minima are

$$X_0^I \asymp 1 + P, \quad Y_0^I \asymp 1 + B. \quad (\text{C.1.26})$$

In both comparisons the common terminal target is initially zero; there is no incoming-exit reserve in the small endpoint. The constant-mark theorem 6.7, applied to the legal representative with actual increment $r_i M_0 - n \log(r_i + 1)$, gives the local scale

$$\beta_{\min} \asymp \min\{1, (1 + P)(1 + B)/(1 + M_0)\}. \quad (\text{C.1.27})$$

This uses the deterministic-mark theorem directly.

If the weak block is a largest suffix block, this is the bracket in (7.11), by (C.1.25). If a later positive block is largest, then $B \geq M_0$, so (C.1.27) is bounded below. Write M for that largest length and A_*, B_* for the preceding and following lengths within the suffix; these are the lengths A, B in (7.11). There are fixed $g_H, C_H > 0$ such that each later drift is at least g_H and the absolute first-block drift is at most C_H . Thus

$$D \geq 0, \quad D \geq g_H M - C_H M_0, \quad A_* \geq M_0.$$

If $M \geq 2C_H M_0/g_H$, then $D \geq g_H M/2$; otherwise $A_* \geq g_H M/(2C_H)$. In either case $D + A_* \geq c_H M$, so the bracket in (7.11) is also bounded below by a fixed positive constant. If no designated near-threshold group is present, all suffix drifts are uniformly positive, giving $D \gg U$ and the same conclusion. These comparisons preserve the suffix convention and its fixed threshold neighborhoods.

Condition on a complete retained prefix. Its uncolored discrepancy Δ_{pre} satisfies $|\Delta_{\text{pre}}| \leq C\sqrt{1 + V_{\text{pre}}}$, and the Euler identity gives the suffix target

$$E_x = D + (x - \Delta_{\text{pre}})/r_m, \quad |E_x - D| \leq C_A \sqrt{1 + D} \quad (\text{C.1.28})$$

for every prefix-uncolored $0 \leq x \leq A \log(2 + V_{\text{pre}})$, using $D \gg V_{\text{pre}}$. Fix the nonreservoir suffix counts in their own central windows. In a predetermined largest block use the actual new count

$$n_*^x = \left\lfloor \frac{\sum_{\text{suffix}} c_i L_i + E_x - \sum_{i \neq *} w_i n_i}{w_*} \right\rfloor, \quad w_i = \log(r_i + 1), \quad c_i = r_i. \quad (\text{C.1.29})$$

The support threshold and central root construction of Theorem 6.9 make it nonnegative and central, with Poisson mass $\asymp (1 + U)^{-1/2}$. All its original $n_*^x + 1$ gaps are then sampled. In the unconditional Poisson experiment this means retaining the event that the sampled total equals (C.1.29); it does not alter a previously fixed tuple. The bounded floor error is $-C \leq \epsilon \leq 0$.

The count n_*^0 with $x = 0$ is a numerical device for verifying the prefix boxes by A.20; the experiment samples a clock only at its actual selected count. As x increases, early terminal lower bounds gain the common entrance, while late intrinsic lower bounds are independent of both that entrance and the incoming carry. The prefix paths, costs, and local pairs can therefore be fixed before evaluating x .

If the weak block is not the reservoir, its local count is unchanged. If it is, put $k = n_*^x - n_*^0 \geq 0$. Then $k \leq C(1 + x)$, and the max formula for the legal pair gives

$$\Delta_x = \Delta_0 - w_* k, \quad x_0^- \leq x_x^- \leq x_0^- + w_* k, \quad a_* Y_0^I \leq y_x^- \leq y_0^-. \quad (\text{C.1.30})$$

Equation (C.1.26) absorbs $w_*k \leq C_A \log(2 + P)$ in the left box, while the right endpoint keeps its positive lower box. Its scale is still (C.1.27); an incoming exit or x has not been added to its small right endpoint.

Keep every terminal fine label already sampled in the accepted prefix fixed. Refine only the newly unread suffix labels under their conditional categorical law. The zero-target prefix completion supplies the deterministic guards (A.2.30), (A.2.36) and the clipped-prefix lower bound; the actual positive-root completion retains these guards by the exact intrinsic cancellation. Section C.2 gives the complete finite disintegration, and Section B.7 supplies the actual ordinary strong-wall event with capped auxiliary end boxes. Impose that event and uncolored regularity on the actual $n_*^x + 1$ -gap clock. No accepted prefix label is resampled and no clock at the virtual zero-root count is reused. The other suffix blocks have separated drift. Reading a positive slope from the left, or a negative slope from the right, their REG implies the allocated inward wall: absorb $K(1 + t^\zeta) + gt^\gamma$ into a fixed intercept and a fraction of the linear drift. Long central blocks preserve the drift gap; bounded supported blocks use their fixed finite convention. The same argument pays the strong and reference-paid local comparisons. After counts are fixed, the original suffix clocks are independent; there is at most one weak ordinary probability. Fill named regularity on each retained clock by Lemma C.2. This constructs a joint completion of order $\mathcal{B}_I = (1 + U)^{-1/2}\beta_0$ on the same word.

For the matching upper and the support needed later, retain the lawful full-stop context for every suffix $q = 0$ comparison: all eligible names remain until its moving cut, then disappear. Its intrinsic cost is the original reverse suffix risk, with no incoming reserve. At a fixed count its centered fluctuation agrees with that of the minimum representative; the latter's positive chord, including its θ -contraction, lies below the full-stop chord. Thus the imposed weak and paid strong walls imply on their actual joint support

$$h_{\text{base}} + R_{\text{tail}}(v) \geq 0 \quad (0 \leq v \leq U) \quad (\text{C.1.31})$$

for one fixed reserve, large enough for floor and fixed intercept errors. Drop other filters and apply the all-count upper in Theorem 6.9 at this reserve and target (C.1.28). It gives $O(\mathcal{B}_I)$, with the reservoir denominator retained before its Poisson supremum. Consequently, for every retained complete prefix word and every permitted x , its actual suffix completion kernel is comparable to $\mathcal{B}_I$. Integrating the private prefix gives the pointwise relation

$$\mathbb{E}_{\text{suffix}} f_0^x(G_{\text{pre}}, G_{\text{suffix}}) \asymp \mathcal{B}_I f_{\text{pre}}(G_{\text{pre}}). \quad (\text{C.1.32})$$

This expectation includes the one root atom, all suffix positions, and the original named refinements. It may be multiplied by any nonnegative prefix-uncolored cost, but it does not scalarize a future private-list weight.

Apply (C.1.32) first, then Lemma C.3 at each full type, and finally (C.1.22). This proves the source mixed moment

$$\mathbb{E} \left[f_0^x \prod_j (D_j^{\text{own}})^{\sigma_j - 1} \right] \asymp \mathcal{B}_I \prod_{j < m} w_j^{\sigma_j} \prod_{i \in \mathcal{I}} \Psi_i^I(\sigma_{\text{plateau}(i)}). \quad (\text{C.1.33})$$

The ordinary factor is linear and occurs once.

C.1.8 Targets fixed by the costs and a fine-word density

The cost floors bound the entire logarithmic target range in advance. Within that range, one triangular cascade chooses the required numbers of recolorings. The exact preimage count of the

uniform subset kernel then gives a bounded output density, pointwise in the full environment and its actual root indicator.

We specify the order of constants before defining the adaptive targets. First fix compact scalar ranges, θ , $0 < \zeta < \gamma < 1/2$, and the finite count and displacement support. Choose a polynomial floor exponent D_0 as in (C.1.14); it depends on the compact exponent range and number of factors, not on later fixed wall constants. Use the widths $w_j \leq 1$, with $-\log w_j \leq \frac{1}{2} \log T_j$. The floors imply predetermined bounds

$$y_j := -\log D_j^{\text{own}} \leq C_0 + C_1 \log(2 + T_j), \quad y_j \geq 0. \quad (\text{C.1.34})$$

Only selected head blocks enter the terminal cost, so its target is bounded in terms of V_{pre} . The hierarchy $T_l \leq C_H T_j$ for $l < j$ and bounded reciprocal child entropies now fix all logarithmic transfer ranges described below. Construct the common upper costs simultaneously for these full ranges, then choose the source with spare strong-curve and REG margins. Finally enlarge the macro-block and root support thresholds. These later constants do not alter the floor exponent or logarithmic target bounds.

Set

$$x = \sum_{j=1}^m r_j y_j, \quad r_j = \sigma_j / \sigma_1. \quad (\text{C.1.35})$$

This is prefix-uncolored and lies in the preassigned range of (C.1.28). Choose the actual root by (C.1.29), obtaining $X_1 = x + \epsilon$, $-C \leq \epsilon \leq 0$. Let $F_0^x \in [0, 1]$ be the joint source acceptance just constructed, with private mass f_0^x . Proper targets are still zero at this stage. The suffix is sampled only once, and (C.1.33) applies to this x .

Let A_j be the logarithm of the first shallow effective alphabet of nonterminal plateau j , and put $b_j = u_j A_{j+1}$. In its prescribed macro block take the immediate child's first shallow fine label c_j , and that block's eligible shallow exterior fine label t_j . Both are actual original labels with probabilities bounded below, and their ratio is

$$p(c_j)/p(t_j) = \exp\{(u_j - 1)A_{j+1}\}.$$

The destination need not be the first name of the entire plateau. Define, using only the uncolored prefix,

$$e_0 = 0, \quad e_j = \left\lceil \frac{y_j + A_j e_{j-1}}{b_j} \right\rceil, \quad \delta_j = b_j e_j - A_j e_{j-1} - y_j \in [0, b_j), \quad j < m. \quad (\text{C.1.36})$$

Equation (C.1.34) and the hierarchy give the predetermined bound $e_j \leq A'_H \log(2 + T_j)$. Condition (C.1.18) and the final threshold ensure that all these occurrences exist. Choose a uniform e_j -subset of the c_j -slots in its bank and recolor it t_j . The banks lie in disjoint original blocks. This defines a single probability kernel on the full fine word, preserving every clock and the root.

Here is its exact density calculation. In one bank, write s, t for source and destination names, and k', l' for their output counts. Every preimage is obtained by choosing e of the output t -slots, and has $k' + e$ available source slots. Its original word probability is larger by $(p_s/p_t)^e$. Thus the unrestricted push-forward density is

$$R_e(k', l') = (p_s/p_t)^e \frac{\binom{l'}{e}}{\binom{k'+e}{e}}. \quad (\text{C.1.37})$$

On an output with a retained preimage, expand the e factors. The two bank counts differ from $p_s m, p_t m$ by at most $m/\log(2+T)$, and the shift is at most e . Since both probabilities are uniformly positive,

$$|\log R_e| \leq C\{e/\log(2+T) + e^2/m\} \leq C_A. \quad (\text{C.1.38})$$

Restricting to accepted preimages decreases this density. Multiply the bounds over the disjoint banks to obtain C_R , and write ν_y for the push-forward of πF_0^x . The normalized target acceptance is

$$F_y^I = \frac{1}{C_R} \frac{d\nu_y}{d\pi}, \quad 0 \leq F_y^I \leq 1, \quad \mathbb{E}_\pi F_y^I(G, \cdot) = f_0^x(G)/C_R. \quad (\text{C.1.39})$$

The mass identity holds for every fixed full uncolored environment, including its root indicator and the adaptive vector $e(G)$. Thus (C.1.39) preserves the integrated mass.

C.1.9 The reconstructed support and its common upper costs

A bounded output density controls mass, but the intersection argument also needs path support for every positive target word. We recover its source endpoint pair from the inverse complete type and compare the source and target paths on the actual changed slots. A spare source curve pays their logarithmic recoloring discrepancy. The common upper costs are constructed over the full transfer range before the adaptive target is evaluated. Fix a selected block of length P that is recolored. Its canonical retained binary set contains s and excludes t . The inverse type is determined by the full target type K : it is $K - \sum_j e_j v_j$, with v_j the source-to-destination count shift. Hence the source pair x_0, y_0 is known even before the flipped slots are identified. Writing $L_{\text{flip}}(t)$ for the number of recolorings before t , we reconstruct the path by the exact identity

$$\begin{aligned} \tilde{Z}(t) &= x_0 + (c + be/P)t - aN(t) + bN_{B,\text{target}}(t) \\ &= Z_{\text{source}}(t) - b\{L_{\text{flip}}(t) - et/P\}, \quad \tilde{Z}(0) = x_0, \quad \tilde{Z}(P) = y_0. \end{aligned} \quad (\text{C.1.40})$$

This follows from $N_{B,\text{target}} = N_{B,\text{source}} - L_{\text{flip}}$.

For every possible chosen subset in the fixed interior bank,

$$|L_{\text{flip}}(t) - et/P| \leq CeP^{-\gamma} \min(t, P-t)^\gamma. \quad (\text{C.1.41})$$

Before the bank the left side is et/P , after it it is $e(P-t)/P$, and inside it the depth is $\gg P$ and the left side is at most e . The spare $2g$ curve in (C.1.23) and $e = O(\log P)$ therefore imply

$$\theta \ell_0(t) + E_{\text{target}}(t) \geq g \min(t, P-t)^\gamma - h_W \quad (\text{C.1.42})$$

above the final threshold. Its endpoints are exactly the original source pair. On an unflipped local block the word and local type are unchanged, and the inverse complete type recovers the same pair. A previous plateau's recoloring preserves every point of the later local clock, so the estimate uses that block's own length.

Named REG is preserved with the predetermined spare constants. Any interval meeting the macro bank either has length $\gg P$ or has an endpoint of depth $\gg P$. Its $O(\log P)$ count change is paid respectively by a spare linear slope or a spare depth-to-the- ζ intercept. The empirical-center shift $O(\log P/P)$ is paid by the same margins. The terminal suffix has no recoloring bank; its root, clocks, and fixed-reserve ordinary wall remain unchanged. The resulting support is the source-reconstructed support (C.1.42). The separate same-word information and carry comparison in Section D.1 transfers full fine histories.

To construct one bank envelope in advance for all the permitted transfers, retain U_d, V_d, L_d of (C.1.6) and let $E_{max} = \lfloor A \log(2 + T) \rfloor$. Use

$$p_{\pm} = p_0 \pm C_1(X + Y + 1 + d + E_{max})/n, \quad c_{max} = c + bE_{max}/P. \quad (\text{C.1.43})$$

The left free interval uses speed c_{max} and probability p_+ ; the reversed right interval uses speed c and probability p_- . These opposite choices both enlarge their necessary events. The allowances remain $C_2 U_d, C_2 V_d$. After fixing the true revealed assignments and full target type, (C.1.40) gives $|K_B - np_0| \leq C(X + Y) + e$; thus its remaining binary fraction lies between p_- and p_+ . The joint exact-subset comparison (C.1.7) then applies to both intervals on the original target word. This argument uses the direct support (C.1.42).

The joint clock comparison (C.1.11) is unchanged. The new drift term times $\sqrt{L_d}$ is at most $C \log(2 + T)/\sqrt{n}$, uniformly bounded because the recolored block is macroscopic. The additional exponent error is $O(E_{max}/n)$, and its product with the logarithm of any active reserve ratio is $O(\log^2(2 + n)/n)$. The proof of (C.1.10) therefore gives

$$\mathbb{E}(q_d^*)^\eta \leq C(1 + d)^C \Psi_{\text{source}}, \quad \mathbb{E}(D^*)^\eta \leq C \Psi_{\text{source}}, \quad (\text{C.1.44})$$

using the same floor and coefficient tail. All costs here are fixed from the entire range E_{max} , before any target is evaluated, with the original source reserves.

For the class upper, use the count-only prefix acceptance A_{pre} , extended by one on suffix names, and drop the source's suffix root and terminal filters. The full F_0^x itself is not a clock-independent count weight. Every preimage of a target type has the same inverse type, so positivity and (C.1.38) bound F_y^I by the shifted weight

$$A_e(K) = A_{\text{pre}}(K - \sum_j e_j v_j).$$

Its whole original type mass W_e satisfies

$$cW \leq W_e \leq CW, \quad W = \mathbb{E}_\pi A_{\text{pre}}. \quad (\text{C.1.45})$$

Indeed, in one changed block the adjacent multinomial ratio is

$$\frac{\pi(k + ev)}{\pi(k)} = (p_t/p_s)^e \prod_{l=0}^{e-1} \frac{k_s - l}{k_t + l + 1}, \quad \left| \log \frac{\pi(k + ev)}{\pi(k)} \right| \leq C(e/\sqrt{n} + e^2/n).$$

All shifted types are feasible above the fixed threshold, and unchanged block factors cancel exactly. Shift the entire coupled type sum once to prove (C.1.45). Uniformity over every allowed e makes the bound pointwise also at $e(G)$. The weight A_e can depend on G . At a fixed whole type, apply the original free-interval estimate; then sum the shifted acceptance and bound $W_e(G)$ by CW before averaging the clocks. This proves (C.1.9) for the transported density using the all-transfer envelopes. The count factor belongs to this conditional class calculation; the root atom is counted only in the separate source first mass.

C.1.10 Exact exits, likelihood, and the first mass

The cascade changes the carries by bounded amounts, while the root already supplies the terminal target. Substituting these exact outputs into the full fine-word likelihood exposes the plateau cost powers. A direct expansion of the natural Poisson law then gives the current saddle exponent and the complete nominal first mass.

The grouped residuals change under (C.1.36) by

$$Z_j^y - Z_j^0 = b_j e_j - A_j e_{j-1} = y_j + \delta_j \quad (j < m), \quad Z_m^y - Z_m^0 = -A_m e_{m-1}. \quad (\text{C.1.46})$$

Their r_j -weighted sum is zero because $r_j b_j = r_{j+1} A_{j+1}$. Define the new exits by the identity $Z_j^y = y_j + \eta_j^y - \eta_{j-1}^y / u_{j-1}$. Subtract the source identity, solve its triangular recursion, and get

$$\eta_j^y - \eta_j^0 = \sum_{l \leq j} (\sigma_l / \sigma_j) \delta_l = O_H(1). \quad (\text{C.1.47})$$

All these changes are nonnegative. The positive source exit bands remain positive and, since $d_j R_j \ll 1$,

$$0 \leq E_y := \sum_{j < m} d_j \eta_j^y \leq C_H, \quad Z_m^y = y_m + \epsilon / r_m - \eta_{m-1}^y / u_{m-1}. \quad (\text{C.1.48})$$

The latter follows from the unchanged root $X_1 = \sum_j r_j y_j + \epsilon$. In particular, its y_m comes from the actual root; no terminal private transfer is made.

Here is the exact likelihood calculation, including its deterministic normalization. Put $B_{\text{geom}} = \sum_i (h_i - 1) L_i = \sum_{j=1}^k V_j$. The full recursive fine-word law, including the terminal free labels, satisfies, term by term,

$$-\log \pi(\omega) = B_{\text{geom}} + \sum_{j=1}^m Z_j(\omega), \quad X_1 = \sum_{j=1}^m r_j Z_j. \quad (\text{C.1.49})$$

At an exterior choice the negative log probability is its local log-alphabet; at a child route it is that log-alphabet minus the child's weighted log-alphabet. The child contributions cancel against the next stage, proving both identities. Equal-product internal margins cancel in grouping, by complementarity in (4.20).

Let $q_j = \sigma_j / \sigma_m - 1$, so $q_m = 0$. Eliminating the terminal output in (C.1.49) gives

$$1 = e^{B_{\text{geom}} + X_1 / r_m} \pi(\omega) e^{-\sum_{j < m} q_j Z_j}.$$

Since $q_j - q_{j+1} / u_j = d_j$, the exit identities give $\sum_{j < m} q_j Z_j = \sum_{j < m} q_j y_j + E_y$. Therefore

$$(1/r_m - \sigma_1) X_1 - \sum_{j < m} q_j Z_j = \sum_{j=1}^m (1 - \sigma_j) y_j - E_y + (1/r_m - \sigma_1) \epsilon. \quad (\text{C.1.50})$$

This uses $(1/r_m - \sigma_1) r_j - q_j = 1 - \sigma_j$, including $j = m$. No exponential in the logarithmic target is discarded. Also, for two retained words in the same uncolored environment, (C.1.49) and the sum of their exit identities give

$$\frac{\pi(\omega)}{\pi(\omega')} = \exp\{E_y(\omega) - E_y(\omega')\}. \quad (\text{C.1.51})$$

It is bounded by (C.1.48), while its actual class mass still requires the separate class estimates.

For a complete cell occupancy g , put $W_g = \prod_{i,c} |c|^{g_{i,c}} / g_{i,c}!$ and $Z_F(g) = \sum_{\omega} F_y^I(g, \omega)$. Under the natural Poisson law the cell intensity is $Z_i = e^{G_i}$. Expanding that law in $e^{-\sum_i h_i L_i} \sum_g W_g Z_F(g)$ gives deterministic factor $e^{-\sum_i h_i L_i + \sum_i Z_i L_i}$, random factor $e^{-\sum_i G_i n_i}$, and $\pi(\omega)^{-1}$. The definitions of the root and the saddle coordinates give $\sum_i G_i n_i = \sigma_1 X_1 + \sum_j s_j V_j$.

Combining with (C.1.49), the deterministic exponent is exactly $-J(L; s)$. Equation (C.1.50) proves

$$e^{-\sum_i h_i L_i} \sum_g W_g Z_F(g) = e^{-J(L; s)} \mathbb{E}_{\text{nat}} \mathbb{E}_\pi \left[F_y^I e^{-E_y} \prod_j (D_j^{\text{own}})^{\sigma_j - 1} e^{(1/r_m - \sigma_1)\epsilon} \right]. \quad (\text{C.1.52})$$

This is algebra on the actual count/root support. The factors involving E_y and ϵ are bounded above and below. Using the pointwise mass identity (C.1.39) inside this same expectation, followed by (C.1.33), evaluates it as

$$e^{-J(L; s)} \mathcal{B}_I \prod_{j < m} w_j^{\sigma_j} \prod_{i \in \mathcal{I}} \Psi_i^I(\sigma_{\text{plateau}(i)}). \quad (\text{C.1.53})$$

The root indicator occurs once, linearly.

To integrate the remaining original prefix totals, let $\mu_i = Z_i L_i$. On their own central windows, $|n_i - \mu_i| \leq B\sqrt{1 + \mu_i}$. The deterministic intrinsic score minima do not depend on n_i . Replacing n_i by μ_i in (7.6), with products frozen, changes p_i^0, ρ_i^0 and the logarithm of its base by $O(\mu_i^{-1/2})$ on the compact rare branch. The base is at least $c/(1 + \mu_i)$, so local Lipschitz continuity of χ gives

$$|\log \Psi_i^I - \log \bar{\Psi}_i^I| \leq C \log(2 + \mu_i) / \sqrt{\mu_i} \leq C. \quad (\text{C.1.54})$$

If a branch threshold is crossed, both bases are comparable to that fixed threshold and the same conclusion holds. The dimension-bounded number of original Poisson central windows has constant joint mass; the source vanishes off them. Integrating (C.1.53) gives (8.10). Fixed positive allowances can then be changed to the profile's normalization because all scores are finite minima of sums of nonnegative terms, and all exponents remain in compact ranges.

C.1.11 Construction on complete cells

Arithmetic realization requires a weight on cell words. We construct it directly from boundary walls, regularity, and count admission. Uniform extension to every positioning of the slots inside their cells gives both the required lower event and the necessary upper walls. Since the resulting costs are constant on each cell fiber, their fractional and mixed moments are preserved exactly when the within-cell positions are integrated.

Before counts, positions, or labels are sampled, fix a partition of each original block into cells with endpoints including its physical endpoints, bounded mesh Δ_* , and bounded boundary density. A cell word records the occupancy $g_{i,c}$ and its exchangeable labeled slots. Every final event is invariant under permutations within a cell. Only the declared bounded experiments are used for short blocks, with bounded total, feasible types, and fixed positive retained mass; these blocks are not recoloring banks. A later numerical cutoff for a bank upper estimate does not expand this bounded support convention.

First consider any lift of a fixed cell word to distinct positions inside its cells. For positive marks and either orientation, $Z(t) = x + \varepsilon(\sum_{s \leq t} W_s - ct)$, monotonicity of jumps gives, on $[l, r]$,

$$Z(t) \geq Z(l) - c\Delta_* \quad (\varepsilon = 1), \quad Z(t) \geq Z(r) - c\Delta_* \quad (\varepsilon = -1). \quad (\text{C.1.55})$$

Use the same half-open convention for all boundary counts. There is no dependence here on the number of points in the cell. Since $g \min(t, L - t)^\gamma$ oscillates by at most $g\Delta_*^\gamma$, a boundary strong wall implies the continuous strong wall on every lift after increasing its allowance by

$c_{max}\Delta_* + g\Delta_*^\gamma$. The reverse inclusion requires no loss. Logarithmic soft walls are handled by their fixed local modulus of continuity.

For an allocated bridge $\theta\ell + E$, its endpoints are $\theta x, \theta y$, and its signed-jump representation has speed

$$c_\theta = \theta c + (1 - \theta)S_{\text{total}}/L, \quad (\text{C.1.56})$$

where S_{total} is its positive mark sum. On the large central experiments this speed stays in a fixed positive compact interval, independent of the sizes of the smaller intrinsic endpoints. The same is true after the compensating change $c + be/L$. Thus (C.1.55) also applies to the allocated walls at these smaller pairs.

Boundary REG implies continuous regularity on every lift. To bound the count on $(s, t]$, expand it to its neighboring outer boundaries for the upper bound and shrink it to the inner boundaries for the lower. Length and mean change by at most $2\Delta_*$ and $2\lambda_{max}\Delta_*$. Also

$$|d_L(s)^\zeta - d_L(s')^\zeta| \leq \Delta_*^\zeta$$

for a neighboring boundary s' . If the inner interval is empty, its length is at most $2\Delta_*$, and nonnegativity gives the required lower bound. This proves the assertion with fixed additional intercept, even though an interior cell's permitted occupancy can grow with its depth.

Define the cell source using boundary allocated walls, boundary binary REG, and its count box from the outset. The continuous relative strong/REG event at the same exact pair is contained in this boundary event. The reverse inclusion just proved provides the continuous necessary upper wall with a fixed larger allowance. Changing such a fixed allowance changes the all-reserve factors by fixed constants only. For named filling, condition on a complete binary word and refine both actual lists. Lemma C.2 applies to each lift with the enlarged fixed regularity constants. A boundary binary word is a mixture of these lifts; integrating their uniform success bound gives the same relative retention for the named boundary event.

Construct fresh costs as cell functions. At each active depth choose all strips and free-interval endpoints at original boundaries and round L_d down by at most two meshes. A length below the fixed threshold uses the already justified $q_d = 1$ convention. Use the all-transfer parameters (C.1.43) and test its two necessary walls at boundaries, with only the fixed mesh allowance. Denote the resulting product by $q_{i,d}^{\text{cell}}(g_i)$. Both the exact-subset comparison (C.1.7) and the exact-clock comparison (C.1.11) push to cells with the same density bound. The either-sign pointwise implication (C.1.55) then gives

$$\mathbb{E}_{g_i|n_i}(q_{i,d}^{\text{cell}})^\eta \leq C(1+d)^C \Psi_i^I(\eta), \quad d \leq D_{n_i}.$$

Use these functions in (C.1.13), including its floor, numerical tail, and clipping, to define $D_i^{I,\text{cell}}(g_i)$. The proofs of (C.1.14) and (C.1.44) give

$$q_i^{I,\text{cell}}(\omega, g_i) \leq C D_i^{I,\text{cell}}(g_i), \quad \mathbb{E}(q_i^{I,\text{cell}})^\eta \geq c \Psi_i^I(\eta), \quad \mathbb{E}(D_i^{I,\text{cell}})^\eta \leq C \Psi_i^I(\eta). \quad (\text{C.1.57})$$

The middle lower bound follows from the contained continuous event, with its proved finite scalar support in bounded-count cases.

The macroscopic recoloring intervals are unions of complete cells, still a fixed positive distance from both block endpoints. Their occupancy is binomial conditional on n_i , and the two-name counts are hypergeometric conditional on the full type and occupancy. Thus (C.1.19) and the local subtraction (C.1.21) apply directly to these cell events. They preserve (C.1.57) before the coupled type or root is integrated.

The exact uncolored cell law at fixed total is

$$\mathbb{P}(g_i \mid n_i) = n_i! \prod_c \frac{(|c|/L_i)^{g_{i,c}}}{g_{i,c}!}. \quad (\text{C.1.58})$$

It is the push-forward of the original uniform-point law, equivalently of its full $(n_i + 1)$ -spacing simplex law. Conditional on the occupancy, positions are uniform inside cells, labels have their original categorical law, and conditional on full type the labeled slots have their uniform multiset law. By construction,

$$q_i^{I,\text{cell}}(\omega, G_i) = q_i^{I,\text{cell}}(\omega, g_i), \quad D_i^{I,\text{cell}}(G_i) = D_i^{I,\text{cell}}(g_i).$$

Thus integration over a cell fiber preserves q_i , q_i^η , and $q_i D_i^{\eta-1}$ exactly. At a fixed complete type the block experiments remain independent; Lemma C.3 therefore applies before the full measure μ is integrated.

The plateau costs, y_j , and x are now functions of prefix uncolored cell counts alone. For every retained prefix cell word, boundary-to-continuous REG supplies the hypotheses of the suffix argument on each lift. The actual new reservoir total is still (C.1.29). Sample its cell occupancy directly by (C.1.58). Counts, endpoint identities, and the bounded root error are unchanged. Impose the weak ordinary representative's boundary strong wall and boundary regularity, with the same constant-mass named completion. The continuous ordinary lower is contained in it; strong blocks are paid by their separated drifts. Conversely the intrinsic full-stop contexts imply the global ordinary wall at boundaries, and (C.1.55) extends it with one fixed allowance. The all-count root upper still applies. Consequently

$$\mathbb{E}_{\text{suffix}} f_0^{I,\text{cell},x}(g_{\text{pre}}, g_{\text{suffix}}) \asymp \mathcal{B}_I f_{\text{pre}}^{I,\text{cell}}(g_{\text{pre}}) \quad (\text{C.1.59})$$

pointwise on the retained prefix. Its root is a single count atom, and the endpoint change of a weak reservoir is still exactly (C.1.30).

Apply the recoloring kernel to these cell words. Its uniform subset choice commutes with every within-cell permutation, as do the source count, wall, REG, bank, and root events. Formula (C.1.37) and its bound remain exact, so

$$F_y^{I,\text{cell}} = C_R^{-1} d\nu_y / d\pi \in [0, 1], \quad \mathbb{E}_\pi F_y^{I,\text{cell}}(g, \cdot) = f_0^{I,\text{cell},x}(g) / C_R \quad (\text{C.1.60})$$

is cell-invariant. At every boundary the same slot identity (C.1.40) holds. Macro interiority and the spare strong wall prove (C.1.42) at these boundaries, then (C.1.55) extends it to every lift with a fixed allowance uniform in the batch size and target. Named boundary regularity is retained by the same macro-length or macro-depth argument, and the terminal suffix is unchanged.

The output and exit identities (C.1.46)–(C.1.48) are count identities and incur no cell-rounding error. Apply (C.1.59), then the local mixed moments from (C.1.57), and finally the whole count mass (C.1.22). Equation (C.1.60) transfers that mass with its fixed normalization. The exact likelihood calculation (C.1.52) and the central-count comparison (C.1.54) consequently give the same e^{-J} times the factors in (8.10).

All constants can be fixed in the order stated before (C.1.34), with the mesh bound included among the initial data: floor exponent and widths first, the logarithmic transfer ranges next, then free-interval envelopes and spare strong/REG/refinement and mesh allowances, and the final macro, total, and root thresholds last. The subsequent arithmetic realization may therefore use this bounded cell-invariant density without reading actual prime values inside its slots.

C.2 Exact count-source disintegration and accepted-prefix retention

We verify the conditional laws used above on a finite count space. Diagonalization computes the full band mass; count transport preserves uniform final permutations; terminal marginalization builds a prefix independent of the unread suffix. Once prefix paths have been accepted, completion fills only new suffix names. These facts justify the spatial multiplication at each complete type.

C.2.1 The finite count space

All count operations take place under the original product label law on fixed physical blocks. The inputs listed below summarize the geometric and categorical properties already proved. They specify exactly which coarse data a count adjustment preserves and which private names remain available for later completion.

There are at most H original physical blocks. Conditional on their uncolored totals, block i contains n_i original slots with a fixed eligible fine alphabet and fixed fine probabilities. All slots are independent under the original law π . Every physical block's alphabet and probabilities are constant throughout that block.

Let nonterminal plateaux be $j = 1, \dots, M-1$, in decreasing products σ_j . Write $u_j = \sigma_{j+1}/\sigma_j$, $d_j = u_j^{-1} - 1$, and T_j for the sum of plateau j 's local physical lengths. Fix

$$R_j = \max\{R_0, \min(\sqrt{T_j}, d_j^{-1})\}, \quad w_j = R_j/\sqrt{T_j}. \quad (\text{C.2.1})$$

The construction uses the following actual inputs, not extra probabilistic assumptions.

1. All effective fine probabilities are bounded away from zero, and all two-name split probabilities are in $[p_*, 1 - p_*]$. In a selected proper local block of length $P_j \geq T_j/H$, its original slot total is comparable to T_j .
2. The local hierarchy satisfies $V_j = \sum_{a \leq j} T_a \leq C_H T_j$. The natural proper grouped output has mean zero. The exact atomic fine-output identities are $\Delta Z_j = b_j$, $\Delta Z_{j+1} = -A_{j+1}$, with $b_j = u_j A_{j+1}$; the coefficients have fixed positive upper and lower bounds.
3. Every original/routed group's admitted totals and permanent-child counts are in own-scale central windows. The natural full balance and every lawful completed named-star reference are nonnegative. The two actual proper entrances are nonnegative on the construction.
4. The original fine probabilities give equal probabilities to exterior names of a fixed plateau. Proper exterior refinements do not change that plateau's permanent-child classification or any descendant word. Ancestors see all those names as the same child.
5. Terminal private names can be refined under their conditional law; their endpoint count restrictions change neither any proper coarse word nor the uncolored root output.

Items 1–3 follow from the complete-reference and intrinsic-box estimates in Sections A.2 and A.3. The following construction uses these inputs. Item 4 is the nested categorical law. Item 5 follows from that law and terminal child weight $z = 1$; it is justified below.

C.2.2 The diagonal band mass uses only original counts

The cumulative exit coordinates diagonalize the designated two-name splits. Coarsen each pair before reading its split, compute each full binomial band mass, and average the central coarsening only afterwards. This produces one count acceptance with the product width mass before any spatial path is imposed.

In each selected original local block coarsen just the two fine names c_j, t_j . Fix their total m_j , every other category's count, and all other original-block counts. Leave their split $K_j = N(t_j)$ unread. Because selected physical blocks are distinct, these splits have the independent original laws

$$K_j \sim \text{Bin}(m_j, p_j).$$

Define the exact strict carry

$$\eta_l = \sum_{a \leq l} (\sigma_a / \sigma_l) (Z_a - y_a).$$

Flipping c_j to t_j changes η_j by b_j and every other η_l by zero. For $l > j$, the two changed summands cancel since $\sigma_j b_j = \sigma_{j+1} A_{j+1}$. Thus simultaneously

$$\eta_j = h_j + b_j K_j, \tag{C.2.2}$$

where h_j reads no unread split. This exact diagonalization is (A.2.43).

A coarsened own-scale central count event has constant probability. On it $c_m T_j \leq m_j \leq C_m T_j$. Natural mean-zero output, the central errors from at most H earlier blocks, and $V_j \leq C_H T_j$ imply

$$|h_j + b_j m_j p_j| \leq B_h \sqrt{T_j}. \tag{C.2.3}$$

Predetermined targets $y_a \leq C_y \log(2 + T_a)$ add only a fixed multiple of $\sqrt{T_j}$ to this bound. No unread split or its path is used in selecting the coarsened event.

Fix $0 < a < b$ and choose $R_0 \geq 2b^*/(b - a)$, where b^* is an upper bound for all b_j . Then the lattice interval

$$\{k : aR_j \leq h_j + b_j k \leq bR_j\}$$

contains at least $(b - a)R_j/(2b^*)$ integers and at most $[(b - a)/b_* + 2/R_0]R_j$ integers. By (3) every such integer is within a fixed multiple of $\sqrt{m_j}$ of $m_j p_j$. Increase the fixed length threshold until this whole interval lies in $[p_* m_j/2, (1 - p_*/2)m_j]$. Stirling's upper and lower factorial bounds then give a uniform two-sided atom estimate

$$c/\sqrt{m_j} \leq \mathbb{P}(K_j = k) \leq C/\sqrt{m_j}$$

there. Consequently its original probability is comparable to $R_j/\sqrt{T_j} = w_j$. Multiply only the independent unread split probabilities at this stage. Averaging over the entire central coarsening gives a count-only acceptance $A_0 \leq 1$ of mass

$$c_0 \prod_{j < M} w_j \leq W_0 \leq C_0 \prod_{j < M} w_j. \tag{C.2.4}$$

All resulting fine counts are in fixed enlarged central windows. There is no conditioning of a current split on a later carry; equation (2) diagonalizes them under the original law first.

C.2.3 Count transport: constant density and uniform final permutations

The endpoint box is reached by moving nonshallow exterior names to the shallow name. A direct preimage count verifies uniformity within each target type and bounds the transported density by a fixed constant. Permutation equivariance then preserves the independent uniform block words at the final complete type, including their routed old lists.

Fix one plateau's coarse child word and all its original/routed groups. In group g , the N_g exterior slots have uniform labels from $r_g \leq H$ names. Let 0 be its shallowest name. Suppose a source type obeys

$$|n_c - N_g/r_g| \leq K_s \sqrt{1 + N_g}.$$

Choose the endpoint box parameter $K > \sqrt{2}K_s + \epsilon$, enlarged to pay all displayed good-row errors. For every nonshallow name choose target counts

$$m_c \in N_g/r_g - K\sqrt{N_g} + [-\epsilon, \epsilon]\sqrt{N_g}. \quad (\text{C.2.5})$$

For sufficiently large N_g , every such target satisfies $m_c \leq n_c$ and is positive. The remaining count belongs to the shallow name. The target box's original multinomial probability is at least a fixed $c_b > 0$: it has a fixed positive multiple of $N_g^{(r_g-1)/2}$ lattice points, each with probability comparable to $N_g^{-(r_g-1)/2}$. For the fixed bounded range of N_g , use the all-shallow target, of probability at least H^{-N_0} . All counts remain in a wider fixed central window.

For each source and target type pair, retain a uniformly chosen m_c of the source occurrences of every nonshallow name c and recolor the rest shallow. This is an explicit coupling. For any fixed target word, its number of preimages and transition weight give exactly

$$\frac{m_0!}{n_0! \prod_{c \neq 0} (n_c - m_c)!} \frac{n_0! \prod_{c \neq 0} n_c!}{N_g!} \prod_{c \neq 0} \binom{n_c}{m_c}^{-1} = \frac{m_0! \prod_{c \neq 0} m_c!}{N_g!}. \quad (\text{C.2.6})$$

Thus it maps a uniform source type word to a uniform target type word. It fixes every original position and the child/exterior classification. Changes among own exterior names leave all grouped outputs and carries unchanged by input 4. Different original fine types can map to the same target type; no injectivity is asserted.

The density bound needed later is elementary. Conditional on the fixed coarse context, let the source subprobability have mass $s \leq 1$. For each source type draw a target type from the original multinomial law restricted to (5), normalized by its box probability $p_b \geq c_b$. Equation (6) shows that the resulting target measure has density exactly $s/p_b \leq 1/c_b$ on that box, relative to the original conditional exterior law. Hence dividing the transported measure by $1/c_b$ restores a density in $[0, 1]$ and loses at most the fixed factor c_b . Apply this separately to the finite number of original groups and nonterminal plateaux. The resulting original-law acceptance $A_1 \leq 1$ still satisfies (4), with altered fixed constants, and has all proper endpoint boxes and strict carry bands.

Every step is equivariant under slot permutations within the original physical blocks: this holds for count admission, uniform subset selection, child routing, and the target boxes. A count-only density therefore remains invariant under the product permutation group, whose orbits are the complete fine block types. Conditional on a final complete type, *the original uniform permutations in all physical blocks remain independent*. This full blockwise statement also controls the old routed lists.

If a binary bundle contains only some of the plateau’s exterior names, this count adjustment can change its binary count. At this stage no spatial path has been accepted. The new count stays in a fixed enlarged central window and becomes fixed on disintegration of the final complete type. This is why exterior count adjustment precedes all binary path acceptance.

C.2.4 Terminal marginalization and the prefix source

Averaging terminal private names preserves every proper coarse condition and the full count mass. We next fill only the terminal names already present in the prefix. This yields a complete count source before any unread suffix environment is sampled; the subsequent root completion keeps all of its accepted labels fixed.

Let C retain all proper fine names and replace every private terminal name by one compound terminal symbol. Write V for their private refinement. Under the original law,

$$\pi(dC, dV) = \pi_c(dC)\pi_t(dV | C).$$

For the bounded count acceptance just constructed set $A_c(C) = \mathbb{E}_\pi[A_1(C, V)|C]$. This has the same mass. Every condition which reads only C survives on its positive support. Permutation invariance implies that A_c reads only coarse block counts. Replacing A_1 by A_c leaves terminal labels at their conditional law and introduces no density inflation or mass loss.

The proper count construction is supported on blocks through the nonterminal plateaux, all before the terminal local blocks. It can therefore be performed at zero common target before any suffix uncolored count is read. This support property proves independence from the unread suffix environment; terminal marginalization then preserves the mass and coarse restrictions.

Now fill terminal labels already present in the prefix under $\pi_t(\cdot|C)$. On a terminal earlier group each entering compound count is central in its own physical scale by the retained coarse windows. Require its permanent terminal name count in a wider central binomial window; this has constant probability uniformly in that coarse input. Conditional on that total, its exterior names have the uniform law. Their displaced count box (5) has a further constant probability. There are at most H groups. Thus the prefix source has bounded original-law density and mass

$$W \asymp_H \prod_{j < M} w_j. \quad (\text{C.2.7})$$

All these restrictions concern counts only, so the complete-type permutation statement proved above survives. They depend on no newly unread prefix position or suffix environment.

This proves the complete prefix count assertion. It does not authorize another terminal refinement after prefix paths have been accepted. The later ordinary-root completion is proved in the transport and arithmetic estimates and is used with the following essential restriction: *every terminal label already sampled in the accepted prefix stays fixed*. Conditional on that full retained prefix and the admitted central uncolored suffix count vector, refine only newly unread suffix terminal names under their conditional laws. Their permanent-name central windows and displaced exterior boxes have a fixed conditional mass; all prefix groups already satisfy those windows and boxes. This statement is justified in Section A.3.3 and must not be inferred from permutation invariance of a coarse-prefix completion.

The deterministic prefix guard inputs are own-group centrality (A.2.30), the retained displaced good-name gain (A.2.36), and the original clipped-prefix costs $m_l \geq c_H \min(V_l, T - V_l)$ of A.13. Write ζ_g for the actual full-group residual and $Z_l = \sum_{g > l} \zeta_g$. Once the root has been selected, the exact two formulas

$$b_0 + Z_l = b_0 + \sum_{g>l} \zeta_g = a_0 - \sum_{g\leq l} \zeta_g$$

and own-group centrality give, for each fixed $\theta' > 0$,

$$b_0 + Z_l + \theta' m_l \geq -h_{\text{bin}}, \quad h_{\text{bin}} = C_B + C_B^2/(4\theta' c_H), \quad (\text{C.2.7a})$$

when the two actual entrances are nonnegative. These are the guards (A.2.33); no new random prefix condition is imposed. At terminal child weight $z = 1$ the full-group output is $\log(r_g + 1)n_g$, independent of private refinement. Thus the unread suffix filling changes neither the root count nor any full-group residual or proper coarse datum.

For the zero-target numerical comparison, $X_1 = \epsilon$ with $-r_* W_0 \leq \epsilon \leq 0$ and $a_0 = 0$, $b_0 = \eta_{M-1}/u_{M-1} - \epsilon/r_* \geq 0$. For a later nonnegative common target x the actual root is $X_1 = x + \epsilon$, and the correct entrances instead are

$$a_0 = x/r_*, \quad b_0 = \eta_{M-1}/u_{M-1} - \epsilon/r_* \geq 0. \quad (\text{C.2.7b})$$

The zero-root early endpoint box is solely a numerical comparison; no spatial clock at that virtual reservoir count is sampled. The positive-root early lower gains x/r_* , while the late intrinsic lower cancels both entrance and incoming carry. It is unnecessary, and in general false, to claim a positive-root early upper box at the old zero-root scale. The fixed prefix's binary path, named word and law remain untouched throughout this later completion.

C.2.5 Complete types and independent spatial clocks

We can now use the finite construction in the spatial estimates. Its count measure may be coupled, but it reads no uncolored position. At a fixed complete type each local path mass depends only on its own clock, with the rest of the type treated as a fixed parameter. The following mixed-moment calculation uses exactly this independence.

Let μ be the push-forward of the prefix subprobability in (7) to its complete fine types, conditional on the original uncolored totals. It is one coupled measure: count acceptance can couple every block. The count construction above reads no uncolored spatial position. Consequently μ is independent of the remaining original independent uncolored block environments G_i . At a fixed complete type ω , uniform fine permutations are independent between original physical blocks.

Each selected block's path restriction at its legal local pair therefore defines its own mass $q_i(\omega, G_i)$. That function may read the entire complete type as a parameter; uniform local bounds must hold on the whole support. It reads only the spatial environment G_i . No spatial law is taken under accepting-type independence.

Suppose the displayed scalar inputs have been proved uniformly:

$$q_i \leq C D_i, \quad E q_i^{\eta_i} \geq c \Psi_i, \quad E D_i^{\eta_i} \leq C \Psi_i, \quad 0 < \eta_- \leq \eta_i \leq \eta_+ < 1. \quad (\text{C.2.8})$$

For positive D_i , Holder gives

$$E(q_i D_i^{\eta_i-1}) \geq \frac{(E q_i^{\eta_i})^{1/\eta_i}}{(E D_i^{\eta_i})^{(1-\eta_i)/\eta_i}} \geq c' \Psi_i.$$

The upper bound is $q_i D_i^{\eta_i-1} \leq C D_i^{\eta_i}$. If a zero-cost convention is needed, define the product zero wherever $D_i = q_i = 0$ and obtain the assertion by positive regularization; the envelope has

the positive polynomial floor. Constants are uniform in the compact range of plateau powers. At a fixed ω the environments are independent, so these local integrands multiply. Tonelli, with no signed interchange, yields

$$E \left[\left(\int \prod_i q_i(\omega, G_i) d\mu(\omega) \right) \prod_i D_i(G_i)^{\eta_i-1} \right] \asymp \mu(1) \prod_i \Psi_i. \quad (\text{C.2.9})$$

This proves (C.1.24). Multiplying the nonterminal plateau width factors gives exactly

$$\mu(1) \prod_{j < M} w_j^{\sigma_j-1} \asymp \prod_{j < M} w_j^{\sigma_j}.$$

No power has been replaced by a common final plateau power.

D Comparison histories and overlap sums

An intersection history records when the fine labels of two divisor boxes are allowed to separate. Its cost combines a geometric multiplier with the probability of its original label class. To bound the overlap row, we reduce that cost to a finite endpoint history and retain positive charges for every cut moved or delayed.

The proof has two components. The deterministic component projects histories through the alphabet chain, resolves branching, and interpolates the surviving cuts to endpoint stars. It keeps the root coefficient one and tracks every original pivot. The probabilistic component estimates labels under the actual target class, first at complete count types and then through diagonal exit bands. Early endpoint targets and late free-bank costs are complementary; together they pay every plateau cost. The forest estimates sum the positive movement and depth penalties, including the full multiplicity of pivots that share a cell boundary.

D.1 Comparison histories and their geometric costs

We begin with the exact information cost under the original product label law. Three reductions organize the argument: lawful parent-child projection separates the alphabets, star comparison resolves branching, and synchronization exposes the endpoint chords. Each reduction is a comparison on one fixed word. Estimates for the probability of its class are deferred until the resulting geometry has identified the free and paid blocks.

All comparisons in this appendix retain one original word and its original physical groups. A history assigns an activation time $0 \leq t_c \leq d_c$ to each boundary of an anchored permutation of the fine names. At a physical position one first splits this line at its activated boundaries and then intersects its cells with the eligible alphabet. A competitor belongs to the class $Q_h(\omega)$ when its label is in the anchor's eligible cell at every original slot. For a split $e = (A; B, C)$ define the logarithmic loss

$$\ell_e(a) = \mathbf{1}_{a \in A} \log \frac{\pi(A \cap E)}{\pi(\text{side}_e(a) \cap E)}.$$

It is zero when only one side is eligible. If $B_e(t)$ is its sum on tail slots, the split cost is $C_e(t) = B_e(t) - (d_e - t)$. Product-law telescoping gives $-\log \pi(Q_h) = \sum_e B_e(t_e)$. Use complete-cell boundary tests, processing the cell on the left before its right boundary activates. Equivalent one-sided conventions give the same assertions when used consistently. All original slot representatives, including their multiplicities within a cell, remain in the history sums. Thus, for a

slot-indexed arithmetic history, the class in this appendix is the complete-cell enlargement $\bar{Q}_h$ defined in Section 9.1. Its cuts are the right boundaries of the pivot cells, with repeated rounded pivots permitted and indexed by the original history. The arithmetic multiplier is unchanged. The arithmetic class is contained in this class, so its unnormalized row is bounded by the row established here.

D.1.1 Legal interpolation on the original time cells

Within a fixed activation chamber, ordered cuts have a finite endpoint-simplex representation. Reference costs interpolate exactly. For the actual word, the difference from this interpolation is an empirical bridge on the same block; future groups cancel. This isolates the path constraint that must remain when endpoint costs are small.

Fix the original permutation, the weak activation order, and a physical block for every variable cut. In a block $[a, b]$ of length $L > 0$ its assigned cuts satisfy $a \leq t_1 \leq \dots \leq t_m \leq b$ and have deadlines at least b . Put $t_0 = a, t_{m+1} = b$ and

$$w_q = (t_{q+1} - t_q)/L, \quad 0 \leq q \leq m.$$

The vertex v_q puts the first q cuts at a and the remaining cuts at b . The weights are nonnegative, sum to one, and $t_r = \sum_q w_q(v_q)_r$, since $\sum_{q \leq r} w_q = (t_r - a)/L$. Over the physical blocks take the product weights w_v . There are at most 2^n vertices for $n \leq H - 1$ cuts, and only a bounded number of permutations, activation orders and block assignments. Every vertex is deadline-legal, although it need not be an available arithmetic pivot.

The reference calculation now reduces to finitely many endpoints. Each block contribution is affine, so with $\bar{M}_h = \sum_e \bar{C}_e$ we have

$$\bar{M}_h(t) = \sum_v w_v \bar{M}_h(v), \quad \min_t \bar{M}_h(t) = \min_v \bar{M}_h(v). \quad (\text{D.1.1})$$

Subtract a common offset once from both sides. Evaluate vertex ties in the chamber's weak order. A different consistent tie resolution gives the same total, because the intermediate entropy terms telescope. Reference nonnegativity is a separate geometric input.

For the actual word and a split assigned to $[a, b]$, define its empirical bridge, centered at the actual block total, by

$$\mathcal{E}_e(u) = \sum_{s \in [a, u]} \ell_e(\omega_s) - \frac{u - a}{L} \sum_{s \in [a, b]} \ell_e(\omega_s).$$

Direct subtraction cancels all future groups and the deadline, giving

$$C_e(u) = \frac{b - u}{L} C_e(a) + \frac{u - a}{L} C_e(b) - \mathcal{E}_e(u), \quad \sum_e C_e(t_e) = \sum_v w_v \sum_e C_e(v_e) - \sum_e \mathcal{E}_e(t_e). \quad (\text{D.1.2})$$

The cancellation is on the same anchor and retains all of its old lists. When several cuts occur simultaneously, their losses combine by the pointwise chain rule:

$$\sum_{e \text{ simultaneous}} \ell_e(c) = \log \frac{\pi(\Pi_-(c))}{\pi(\Pi_+(c))}.$$

They therefore form one empirical bridge of the bundled partition loss. For a synchronous full-to-singleton split in a uniform alphabet this loss is the constant $\log H$; its bridge is purely uncolored.

Suppose the original named counts on a block satisfy

$$|N_c([s, t]) - \nu_c(t - s)| \leq \vartheta(t - s) + K[1 + d(s)^\zeta + d(t)^\zeta], \quad d(u) = \min(u - a, b - u), \quad 0 < \zeta < 1. \quad (\text{D.1.3})$$

Using the shorter end interval in the empirical bridge yields

$$|\mathcal{E}_e(u)| \leq C_H \vartheta d(u) + C_H K[1 + d(u)^\zeta]. \quad (\text{D.1.4})$$

For $u - a \leq L/2$, subtract the reference mean in its prefix and whole-block terms; their linear errors sum to at most $2\vartheta(u - a)$, and their endpoint errors to $K[1 + d(u)^\zeta] + 2K(u - a)/L$. The other half uses the negative suffix. Sum over boundedly many colors and bounded loss coefficients. Consequently (D.1.2) has the corresponding lower error with $\sum_e d(t_e)$ and $\sum_e d(t_e)^\zeta$. A small vertex cost cannot automatically pay these errors; its bundled bridge must then retain an actual path constraint.

D.1.2 Original lifetimes, projection, and delay

Projection separates parent and child entropy, but it must also preserve the geometric rank. An internal cell can outlive its first name, creating a nonnegative delay between original boundaries and eligible cells. The exact entropy chain rule supplies a further gain when child cells are mixed with an outside name. We keep both corrections before comparing projected histories.

A current original internal cell A is born at s_A and split at t_A . If $a(A)$ is its first original name, every other name $c \in A$ has its unactivated original boundary inside A , so $t_c \geq t_A$ and $d_c \geq t_c \geq t_A$. Only $a(A)$ can expire during its life. An anchor-containing cell loses no names; a pure n_A -cell has eligible size n_A or $n_A - 1$. Singleton leaves contribute no delay.

Let q_h be the number of nonempty eligible cells minus one, and $q_{\text{geom}} = \#\{c \neq * : t_c \leq t < d_c\}$. Every active original boundary starts exactly one nonanchor cell. The second rank counts those whose first name is eligible, and the first those with any eligible name. Hence

$$D_h := \int (q_h - q_{\text{geom}}) = \sum_{A \text{ properpureinternal}} [t_A - \max(s_A, d_{a(A)})]_+ \geq 0. \quad (\text{D.1.5})$$

This holds for all original permutations, zeros and ties. For the tail-entropy potential

$$F_A(t) = \sum_{s \geq t, \omega_s \in A} \log \frac{\pi_s(A \cap E(s))}{\pi_s(\omega_s)} - \int_t^T (|A \cap E(u)| - 1)_+ du,$$

put

$$\hat{F}_A(t) = F_A(t) + \int_0^t [(n_A - 1) - (|A \cap E(u)| - 1)_+] du.$$

The extra increment on a pure cell's life is exactly its term in (D.1.5), and is zero for an anchor-containing cell. The exact split telescope can thus be written

$$\sum_e C_e(t_e) = F_{A_0}(t_{\text{root}}) + \sum_{A \text{ properinternal}} [\hat{F}_A(t_A) - \hat{F}_A(s_A)]. \quad (\text{D.1.6})$$

All original names are eligible before the first root split, so its normalization stays unchanged. The modified potentials already include the delay. Equivalently one can use the unmodified potentials together with the explicit delay, counting it once.

For a parent alphabet $O \sqcup C$, merge all child-touching cells of a fine partition Π into one parent star cell, retaining pure-outside cells, and intersect its cells with C for the child partition. For a law ν , write $H_\nu(\Pi) = \sum_a \nu(a) \log(\nu(\Pi(a))/\nu(a))$ for residual entropy. If $P(a)$ is the parent name, $Z(a)$ the fine cell and $\bar{Z}(a)$ its parent cell, the conditional entropy chain rule is

$$H_\pi(\Pi) = H_{\pi_p}(\Pi_p) + \pi(C)H_{\pi(\cdot|C)}(\Pi_c) - M(\Pi), \quad M(\Pi) = I(P; Z | \bar{Z}) \geq 0. \quad (\text{D.1.7})$$

If the child occupies one cell, or no child-touching cell contains an outside name, the contrast is identically zero. Otherwise choose an outside name c in a mixed cell and a different child-touching cell A' . In their merged parent cell Q , the joint atom (c, A') is zero, while its product marginal is $\pi(c)\pi(A')/\pi(Q)^2$. If every fine probability is at least p_* , the elementary relative-entropy inequality $D(P\|Q) \geq 2\|P - Q\|_{\text{TV}}^2$ gives

$$M(\Pi) \geq 2\pi(Q)[\pi(c)\pi(A')/\pi(Q)^2]^2 \geq 2p_*^4. \quad (\text{D.1.8})$$

The actual per-label contrast is

$$D(a) = \log \frac{\pi_p(\Pi_p(P(a)))}{\pi_p(P(a))} + \mathbf{1}_{a \in C} \log \frac{\pi(\Pi_c(a))}{\pi(a)} - \log \frac{\pi(\Pi(a))}{\pi(a)}.$$

Its mean is M , and it is bounded. It can be negative on child names: in a mixed cell A it is $\log[\pi(Q)\pi(A \cap C)/(\pi(C)\pi(A))]$ there, and $\log(\pi(Q)/\pi(A))$ on outside names. Original named regularity, applied on the bounded number of constant-partition intervals, gives

$$\sum_{\text{slots}} D(\omega_s) \geq cS_{\text{mix}} - C[1 + \sum_{t \text{ mixed endpoints}} d_g(t)^\zeta]. \quad (\text{D.1.9})$$

Here S_{mix} counts only times with at least two child-touching cells and an outside name in one of them. The original intensities have a positive lower bound. This is a fixed-word regularity estimate, including when the history is chosen adaptively.

Eligible unresolved ranks add exactly. If there are u pure outside cells and v child-touching cells, then

$$|O| + |C| - u - v = (|O| + 1 - (u + 1)) + (|C| - v).$$

Let d_h be the original unactivated-boundary rank and r_h the eligible unresolved rank. Then $D_{\text{orig}} = \int (d_h - r_h) \geq 0$: the number of empty original cells is at most the number of ineligible names. This is the same expired-first delay as above. If effective costs use eligible ranks and the two actual information terms in (D.1.7), their exact same-word decomposition is

$$-\tilde{I}_h = C_p^{\text{eff}} + C_c^{\text{eff}} + D_{\text{orig}} + \sum_s D(\omega_s).$$

Any entrance/root offset is assigned once.

To realize the projections lawfully, assume all outside deadlines are at most F and all child deadlines at least F . Assign a fine split to the child when both children touch the child alphabet, and to the formal parent when one child is pure outside or the split is entirely outside. This partitions the original event multiset. Delete outside names from the original line to obtain the child line, assigning each new boundary the minimum of the intervening original times. Those segments are disjoint, and each resulting child boundary meets its deadline. Augment the formal parent by isolating each outside name at its deadline if necessary. Its eligible partition is unchanged: before F no child has died; afterwards only the parent star is eligible. Every augmented nonsingleton

contains only unexpired names. Represent it as a lawful anchored line by putting the star first in star cells, and a maximal-deadline name first in pure cells. At a split put the child containing that anchor first. Every right-child anchor is then unexpired. Simultaneous refinements have only a bounded number of resolutions.

For these lawful projections $D_p = 0$, $D_c \geq 0$, and $D_{\text{gap}} = D_{\text{orig}} - D_c \geq 0$. Before F the child delay is zero; after F the full and child eligible ranks agree, whereas each unactivated child boundary needs an entire disjoint unactivated original segment. Thus $d_h \geq d_c$ pointwise. Rank additivity gives

$$\sum t_h = \sum t_p + \sum t_c + D_{\text{gap}}, \quad \tilde{\Phi}_h = \Phi_p \Phi_c e^{-D_{\text{gap}}}, \quad \tilde{I}_h = I_p + I_c - D_{\text{gap}} - \sum_s D(\omega_s). \quad (\text{D.1.10})$$

The child is evaluated on the selected list itself. We next sum the projection fibers. Cancel common formal-parent and augmented-parent times with their multiplicities. If the remaining erased times are $e_1 \leq \dots \leq e_j$ and the added deadlines are $a_1 \leq \dots \leq a_j$, cumulative refinement gives $a_l \leq e_l$ and $D_{\text{gap}} = \sum_l (e_l - a_l)$. Unchanged projected cuts retain their original pivot labels. Endpoint regularity bounds the original representatives in $[a + l, a + l + 1)$ by $C(1 + l)^K$, so for a fixed projected history

$$\sum_{\text{erased fiber}} e^{-\delta D_{\text{gap}}} \leq C_H \left(C \sum_{l \geq 0} (1 + l)^K e^{-\delta l} \right)^H < \infty. \quad (\text{D.1.11})$$

Crossing physical endpoints costs only the fixed number of groups. No retained child pivot is renumbered as a child-only representative.

D.1.3 Realizing added deadlines on the actual grid

Lawful projection may introduce physical deadline cuts that are not arithmetic pivots. Uncolored regularity places the last occupied cell within a fixed distance of each deadline. Moving only these added cuts to that boundary preserves every slot signature and changes the multiplier by a bounded factor. Original pivots retain their full indexing and multiplicity.

Assume original uncolored occupancy obeys, at complete-cell endpoints in each block,

$$N([s, t)) \geq \lambda_0(t - s) - K[1 + d(s)^\zeta + d(t)^\zeta], \quad \lambda_0 > 0.$$

For a physical deadline d , let $r(d)$ be the right boundary of its last occupied cell, or zero if there is none. The empty suffix from $r(d)$ to d has length at most

$$B = \max\{1, (2K/\lambda_0)^{1/(1-\zeta)}\} + HK/\lambda_0.$$

Indeed every whole empty block has length at most K/λ_0 ; the one possible initial partial block of length u satisfies $\lambda_0 u \leq K(1 + u^\zeta)$, giving the first term. Replacing at most H added deadline cuts by $r(d)$ changes no activation status at any original slot. Thus the comparison classes and all their weighted masses are identical for every acceptance F , even if it couples old labels, types or future words. The multipliers change by the exact factor

$$\Phi_h / \Phi_{h_r} = e^{-\sum (d - r(d))} \in [e^{-HB}, 1].$$

Apply this only to added parent deadlines. The projection becomes

$$\Phi_h = \Phi_{p_r} \Phi_c e^{-D_{\text{gap}} - E_{\text{round}}}, \quad 0 \leq E_{\text{round}} \leq HB.$$

Retained original and child pivots are unchanged. The finite assignment choices preserve (D.1.11). Projected parent rows must allow repeated representatives: different deadlines can share the same last occupied cell. The interpolation, signature and geometric bounds below hold in this enlarged positive row because they use named cuts with ties and count tuples by a product of representative counts. No distinct-pivot theorem is invoked on that enlarged parent row.

D.1.4 Branching and the terminal floating chain

For a proper plateau the heavier permanent child gives a strict entropy gain when pure branching cells are absorbed into its star. The terminal alphabet has equal weights, so this gain needs a different comparison: a surviving name carries the last nonsingleton chain, which may float away from the permanent anchor. Its off-spine gain and clipping delay are retained, and the complete reference reserve pays the eventual return to the anchor.

Consider a nonterminal plateau with exterior names of relative weight one and permanent child of weight z , where $1 < z_- \leq z \leq z_+$ and $r + z \leq H$. In each original group the reference intensities are κ_g per exterior name and $z\kappa_g$ for the child, with $0 < \kappa_- \leq \kappa_g \leq \kappa_+$. Assume named regularity on all original groups used, including old ones. On the natural complete chain $z \geq \sqrt{2}$ follows from $s_m > 1/2$ and the terminal weight. These are reference intensities even after an incoming word has been fixed.

For an original history, let $B_h(t) = \{*\} \cup \{c : t_c > t\}$. All its names are eligible. Ordering the names by decreasing activation time gives a lawful star history h^* with the same named cuts and multiplier; its only nonsingleton cell is $B_h(t)$. On the same word,

$$C_h - C_{h^*} = I_{h^*} - I_h = \sum_s L_h(\omega_s), \quad (\text{D.1.12})$$

where

$$L_h(a) = \mathbf{1}_{a \in B_h} \log \frac{\pi_g(B_h)}{\pi_g(a)} - \log \frac{\pi_g(A_h(a))}{\pi_g(a)}.$$

Suppose the current star cell has u exterior names and the pure original internal cells have original sizes $n_l \geq 2$ and eligible sizes $q_l = n_l - \delta_l$, $\delta_l \in \{0, 1\}$. Then $d_h = u + \sum_l (n_l - 1)$, and the reference contrast per unit time equals

$$\kappa_g \{E_{d_h}(z) - E_u(z) - \sum_l q_l \log q_l\}.$$

Move the $n_l - 1$ nonfirst names into the star one cell at a time. For $k = n_l - 1$ and current mass $w \geq z$, the entropy gain is $E_k(w) - q_l \log q_l$. If the first name is alive, it is at least

$$E_k(z) - E_k(1) \geq (z_- - 1) \log(1 + 1/z_+) =: \gamma_z > 0.$$

If expired, it is at least $(k + 1) \log(k + 1) - k \log k \geq 2 \log 2 > \gamma_z$. Thus the contrast rate is at least c_0 times the number of pure original internal cells. When that number is zero the partitions agree, so the contrast vanishes pointwise.

Let S_{pure} be the sum of these cell lifetimes, and β_l the births of the positive-length components where they are present. Subdivide at activations and physical endpoints and apply (D.1.3) to the bounded coefficients. Since $d_E(t)^\zeta \leq d_E(\beta_l)^\zeta + |J_l|^\zeta$ inside a component, choosing the slope small and absorbing $K|J_l|^\zeta$ into its linear cost gives

$$C_h \geq C_{h^*} + cS_{\text{pure}} - C[1 + \sum_l d_E(\beta_l)^\zeta]. \quad (\text{D.1.13})$$

The original rank already includes expired-first delay. To pay a birth at large depth with a short lifetime, we retain the growing wall used below; lifetime alone does not control that error. Shifting to a neighboring fixed cell adds its regularity depth error and a bounded geometric term, so the estimate remains valid for original representatives. The comparison concerns information costs; class posteriors will be estimated under their own label laws.

For a terminal plateau all eligible names have equal reference intensity, and the preceding strict-child gap is unavailable. Select instead the right endpoint τ of the last positive-length interval with a nonsingleton eligible cell. If none exists, its entropy is zero and $I_h = -\sum t_e$. Otherwise choose two names in that last cell that remain eligible up to τ from the left and retain one, c , as a spine name. Its ancestral cell contains both for every $t < \tau$. It therefore has eligible size $q_{\text{sp}}(t) \geq 2$ and $d_c \geq \tau$. Clip activations after τ to τ ; this changes no eligible entropy and pays $\sum_e (t_e - \tau)_+$ in geometric delay.

For $t < \tau$, let $B(t)$ consist of the eligible names in the spine cell and the nonfirst names of every other original internal cell. This set is nested. At an off-spine split it loses the right child's first name; at a spine split keeping the left child it does the same; at one keeping the right child it loses the old first name if alive; a spine first-name expiration removes that name at its deadline. Off-spine first-name expirations change nothing. Remove all but c at τ . Equivalently label the refinement tree by representatives, retaining c on its ancestral branch and the original left child on off-spine subtrees. Assign each removed representative its split time, clipped to its own deadline if it has expired and then to τ . The time multiset before clipping is the original one. If $\delta_{\text{sp}}(t)$ indicates an expired spine first name, then

$$|B(t)| - 1 = d_{\text{orig}}(t) - \delta_{\text{sp}}(t), \quad D^* = \int_0^\tau \delta_{\text{sp}}(t) dt + \int_\tau^T d_{\text{orig}}(t) dt \geq 0$$

is exactly the loss of the sum of activation times.

This chain has a lawful representation anchored at the original name 0. If $c = 0$, use decreasing removal times. Otherwise let s_0 be 0's removal time and use the line $0, c$, followed by all other names in decreasing removal order; give the boundary before c time s_0 . It is legal since $s_0 \leq \tau \leq d_c$. Call this comparator h^{sp} ; after s_0 its nonsingleton cell floats away from 0. The exact multiplier discrepancy remains D^* .

If off-spine internal cells have original sizes n_l and eligible sizes $q_l = n_l$ or $n_l - 1$, their $k_l = n_l - 1$ nonfirst names give $|B| = q_{\text{sp}} + \sum_l k_l$. The mean contrast is

$$\kappa_g \{ |B| \log |B| - q_{\text{sp}} \log q_{\text{sp}} - \sum_l q_l \log q_l \}.$$

Absorbing a cell into a mass $w \geq 2$ gains, when its first name is alive, $E_k(w) - (k+1) \log(k+1) = \int_1^w \log(1 + k/u) du \geq \log(1 + 1/H)$; expiration only increases the gain. Let S_{off} be the off-spine lifetime. The same regularity argument as for (D.1.13) gives

$$C_h \geq C_{h^{\text{sp}}} + D^* + c S_{\text{off}} - C[1 + \sum_l d_E(\beta_l)^\zeta]. \quad (\text{D.1.14})$$

A birth erased because its first name expired has depth at most its distance from that physical deadline, which is paid by D^* . Absorb these errors into a fraction of D^* and a constant. Other births remain comparison cut times.

Fiber summation distinguishes physical endpoints from interior cuts. A physical endpoint has a bounded exponentially weighted fiber. At a retained interior τ , named regularity instead gives

$$\sum_{\text{original } t \geq \tau} e^{-a(t-\tau)} \leq C_a [1 + d_E(\tau)^\zeta]. \quad (\text{D.1.15})$$

There are at most $H - 1$ such factors. A retained $d_E(\tau)^\gamma$ wall with $\gamma > \zeta$ will pay their polynomial. No bounded occupancy at an arbitrary interior point is presumed.

The reference cost itself pays the floating interval. This uses nonnegativity of every complete lawful nonincreasing reference rank history with the same entrance normalization. In a group with $r + 1$ unit names its integrand is $g_g(d) = d - \kappa_g(d + 1) \log(d + 1)$. On $[s_0, \tau]$ the floating rank lies in $1 \leq d \leq r - 1$. Increasing or decreasing it by one on that whole interval gives two lawful reference competitors: the rank drop at s_0 admits the increase, the capacity bound holds, and both end at zero at τ . Discrete concavity gives

$$2g_g(d) - g_g(d - 1) - g_g(d + 1) = \kappa_g \Delta^2[(d + 1) \log(d + 1)] \geq \kappa_*/H.$$

The second difference is the integral of $1/x$ against the unit triangular kernel on $[d, d + 2]$. Integrating and using nonnegative competitor costs yields

$$\overline{C}_{h^{\text{sp}}} \geq a_H(\tau - s_0), \quad a_H = \kappa_*/(2H). \quad (\text{D.1.16})$$

Replace c by 0 in the retained set on $[s_0, \tau]$, removing c at s_0 instead of 0 and keeping all other times. This gives a lawful permanent-0 star h^0 with the same time multiset and reference cost. On the actual word their difference is

$$I_{h^{\text{sp}}} - I_{h^0} = \sum_{[s,t) \subset [s_0, \tau)} \log |B|[N_c([s, t)) - N_0([s, t))].$$

The finite subdivision and regularity therefore give

$$|C_{h^{\text{sp}}} - C_{h^0}| \leq C_H \vartheta(\tau - s_0) + C_K[1 + d_E(s_0)^\zeta + (\tau - s_0)^\zeta]. \quad (\text{D.1.17})$$

We reserve part of the global reference and depth budgets below to pay this same-word reanchoring difference.

Mixed time is preserved sufficiently well by these comparisons. The augmented proper parent has no delay, so its same-cut star has exactly the same eligible unresolved rank. The terminal comparator has rank at least that of the original terminal partition: before τ they are respectively $d_{\text{orig}} - \delta_{\text{sp}}$ and $d_{\text{orig}} - \delta_{\text{all}}$, with $\delta_{\text{sp}} \leq \delta_{\text{all}}$; afterwards both are zero. Iterating eligible-rank additivity implies that every comparison child has unresolved rank at least its original child. Thus if it has at least two cells, so did the original. If a proper parent has no pure internal cell, its original and star partitions agree pointwise. Writing M_j, M_j^* for the original and comparison mixed indicators,

$$M_j^*(t) \leq M_j(t) + \mathbf{1}_{\{A_j(t) > 0\}}, \quad \sum_j S_{\text{mix},j}(h^*) \leq \sum_j S_{\text{mix},j}(h) + \sum_{j < m} S_{\text{pure},j}. \quad (\text{D.1.18})$$

Indeed mixedness means precisely that the merged parent star has an outside name and the child has at least two cells. Consequently a small fixed fraction of original mixed charge can be reserved for the global comparator while retaining half each proper pure-lifetime charge. This is a deterministic comparison of indicators, not of the signed per-label contrast.

D.1.5 Synchronizing cuts and sharing one endpoint budget

The spread of cuts within a block has a positive curvature cost. After synchronizing them at their mean, the remaining cost is an endpoint chord plus the actual empirical bridge. We divide one endpoint budget among the finitely many coordinates before imposing walls. Strong drifts pay distance from the cheaper end; the remaining weak coordinates use their allocated intrinsic walls.

Here first consider a proper coarse plateau, with $g_i(d) = d - \kappa_i E_d(z)$ and $\kappa_i \geq \kappa_* > 0$. Suppose all lawful endpoint stars on the supported actual word satisfy $C_v \geq c_0 \bar{C}_v - h_E$, with $\bar{C}_v \geq 0$ and one common normalization. In a fixed chamber, synchronize its m star cuts in a block at their average $\tau = (t_1 + \dots + t_m)/m$, and let $D_i = t_m - t_1$. The actual before/after named sets have ranks $R, q = R - m$ and remain legal. The rank integral is unchanged because $\int_a^b (d(t) - q)/m dt = \tau - a$. Uniform curvature, $g_i''(d) = -\kappa_i/(d + z) \leq -\kappa_*/H$, quantifies the gain from collapsing the spread to its mean. The reference excess over the synchronous chord is at least

$$\frac{\kappa_*}{2H} \int_a^b (d - q)(R - d) dt \geq \frac{\kappa_*}{2H} D_i.$$

For one cut both sides vanish. The actual contrast is supported in $[t_1, t_m]$; named regularity on its finitely many pieces, including the adjacent cell at a nongrid average, bounds the error by a small linear multiple of D_i and $C[1 + d_i(\tau)^\zeta + D_i^\zeta]$. Absorb the spread power into its linear charge. Combining with (D.1.13), whose births occur at these cuts, gives

$$C_h \geq C_{\text{sync}} + c \sum_i D_i + c_p S_{\text{pure}} - C[1 + \sum_i d_i(\tau_i)^\zeta]. \quad (\text{D.1.19})$$

The mean τ_i is an auxiliary comparison time; original arithmetic pivots keep their labels. Place each synchronized bundle at the left or right endpoint of its block, and write L_{multi} for the interpolation of these endpoint costs. The same-word identity is

$$C_{\text{sync}}(\tau) = L_{\text{multi}}(\tau) + \sum_i \mathcal{R}_i(\tau_i),$$

where $\mathcal{R}_i$ is minus the empirical bridge of its actual bundled partition loss, with zero endpoints. Its difference Δ_i between endpoint costs and its remainder are independent of the outside endpoint choices. Thus L_{multi} is an affine sum of one-coordinate functions. Shift every cost by h_E once and put

$$a_i^* = \min_{\text{outside}} (C_{i,L,\text{outside}} + h_E), \quad b_i^* = a_i^* + \Delta_i, \quad \ell_i(t) = (1 - u_i)a_i^* + u_i b_i^*, \quad u_i = (t - a_i)/L_i.$$

Both endpoints are nonnegative, and the same outside choice minimizes both. Positive interpolation gives $L_{\text{multi}} + h_E \geq \ell_i(\tau_i)$ for every i . For $\theta_{\text{alloc}} = 1/(2H)$, set $W_i = \theta_{\text{alloc}} \ell_i + \mathcal{R}_i$. Since there are at most H coordinates,

$$C_{\text{sync}} + h_E \geq \frac{1}{2}(L_{\text{multi}} + h_E) + \sum_i W_i(\tau_i). \quad (\text{D.1.20})$$

Thus one common endpoint budget supports all the local wall terms, with a positive half retained for the remaining geometric charges.

If $|g_i(R) - g_i(q)| \geq \epsilon > 0$, every reference endpoint at the more expensive side exceeds ϵL_i , and $\ell_i(t) \geq c_0 \epsilon$ times the distance from the cheaper endpoint. Equation (D.1.4) gives $W_i(t) \geq c \text{dist}(t, \text{cheaper end}) - C$. If both reference endpoints in every legal outside context exceed ηL_i , it gives $W_i(t) \geq c L_i - C$ instead. All remaining weak coordinates retain the actual allocated wall

$$\theta_{\text{alloc}} \ell_i(t) + \mathcal{R}_i(t) \geq g d_i(t)^\gamma - h_W, \quad \gamma > \zeta. \quad (\text{D.1.21})$$

The reference classification leaves precisely the full-current-input to deepest-suffix local edges for these walls. Old weak blocks use their separately proved charges or support. At a nongrid

synchronized time, adjacent-cell regularity costs $C[1 + d_i(t)^\zeta]$; half the strong curve absorbs it. The chord changes by a bounded amount across one cell, since its endpoint difference is $O(1 + L_i)$.

Combining (D.1.19)–(D.1.21) gives positive spread and pure-lifetime penalties, linear cheaper-end distance penalties for strong coordinates, full-length penalties for paid weak ones, and $(g/2)d_i(\tau_i)^\gamma$ for unpaid weak ones, minus a fixed constant. Their exponential pivot sum is bounded. To verify multiplicity, bin spreads and endpoint distances in unit intervals. Every cut is within its spread of the average, so original endpoint occupancy bounds the number of tuples by a fixed polynomial in these quantities. Exponential or stretched-exponential penalties sum that polynomial, and only a bounded number of chambers occur. Repeated representatives in the enlarged parent space obey the same product upper bound. Throughout, the comparison class remains the original one.

D.1.6 Global endpoint costs and the uncontracted root

The local intrinsic costs must now be summed along the full fine chain. Early and late endpoint formulas keep their entrance coefficients exactly. Adverse changes of endpoint flag are paid at their own length scale by a prefix or mixed-time charge. This leaves the terminal root outside the bridge allocation, with coefficient one, and prepares the exact target cascade.

Use the actual common-root source of Section 7. Fix throughout the global source and target arguments $0 < \zeta < \gamma < 1/2$, with the regularity slope and the intrinsic wall contractions chosen before constructing the source. Owners have products σ_j , ratios $u_j = \sigma_{j+1}/\sigma_j$, $d_j = u_j^{-1} - 1$ and $r_j = \sigma_j/\sigma_1$. Let T_j be their local lengths and $V_j = \sum_{a \leq j} T_a$. The hierarchy gives $V_j \ll T_j$. Original groups remain separate, including short or zero ones. The actual tuple has root $X_1 = x + \epsilon$, with

$$x = \sum_j r_j y_j, \quad y_j \geq 0, \quad -C \leq \epsilon \leq 0.$$

On its source word the proper targets are zero, the strict exits obey $0 \leq \eta_j \ll \sqrt{1 + T_j}$ and $\sum_{j < m} d_j \eta_j \ll 1$. Set

$$a_j = \eta_j, \quad b_j = \eta_{j-1}/u_{j-1} \quad (j < m), \quad a_m = x/r_m, \quad b_m = \eta_{m-1}/u_{m-1} - \epsilon/r_m,$$

with initial incoming term zero. Then $Z_j^0 = a_j - b_j$ is the exact signed plateau output, and $R_0 = \sum_j Z_j^0$ is the full fine residual. In particular the bounded root error remains in a nonnegative b_m . Write $K_j(v) = C_j(v) - b_j = Z_j^0 - I_j(v)$.

A fine endpoint star isolates each nonanchor fine name at a lawful original physical endpoint, keeping all unresolved names in the permanent anchor's cell. Its atomic projections retain the same named cuts. Let t_j be plateau j 's last positive cut and $s_j = \mathbf{1}_{\{t_j \leq V_j/2\}}$. The endpoint count estimates for the displaced source boxes give the coefficient-one inequalities

$$K_j(v_j) \geq s_j Z_j^0 + cH_j - C, \tag{D.1.22}$$

$$K_j(v_j) \geq s_j Z_j^0 + cJ_j - C, \tag{D.1.23}$$

where

$$H_j = s_j t_j + (1 - s_j)[V_j - t_j + L_{\text{bad},j} + S_{\text{good},j}], \quad J_j = \overline{C}_{v_j} + S_{\text{good},j}.$$

The second retains the positive reference terms of the finite-score proof: at early endpoints use the exact prefix expression and absorb its $O(\sqrt{1 + t_j})$ centered error into a fraction of reference cost; at late endpoints use the exact tail and pay only its zero-tail square root. The good-group

displacement remains positive. Reference cost controls the early prefix and late bad/zero lengths, so $H_j \ll J_j + 1$. The exact prefix and tail formulas retain a_m at an early terminal endpoint and b_m at a late one. Full-group discrepancies cancel, so unrelated full groups contribute no $\sqrt{V_j}$ error.

There are no added cuts in a fine star's projections. The exact entropy chain rule gives

$$C_0(h) := R_0 - I_h(\omega_0) = \sum_j K_j(v_j) + \sum_{j,s} D_j(\omega_s).$$

On each mixed physical group, full named central counts and (D.1.8) give $cL_g - C\sqrt{1+L_g}$; absorb half its mean. Nonmixed contrasts vanish identically. With

$$J(h) = \sum_j J_j + \sum_j S_{\text{mix},j}, \quad H(h) = \sum_j H_j + \sum_j S_{\text{mix},j},$$

one therefore obtains $C_0(h) \geq \sum_j s_j Z_j^0 + cJ(h) - C$, and also its version with H .

Every adverse flag change $s_j = 0, s_{j+1} = 1$ has a charge at its own scale. If $t_{j+1} \geq t_j/2$, the child's early prefix contributes at least $V_j/4$. Otherwise, on (t_{j+1}, t_j) an eligible parent exterior remains in the fine anchor cell while an immediate-child exterior has become a separate singleton. All those child names are eligible up to $V_j \geq t_j$. This is a mixed interval of length greater than $V_j/4$. Thus $H \geq V_j/4$, and $J \gg V_j - C$, for every adverse change. Expanding the outputs exactly,

$$\sum_j s_j Z_j^0 = s_m(x + \epsilon)/r_m + \sum_{j < m} \eta_j(s_j - s_{j+1}/u_j).$$

When both flags are one, their carry terms total $-\sum d_j \eta_j = O(1)$. Retain the positive η_j terms at 1-to-0 changes. At each 0-to-1 change, the negative term is $O(\sqrt{1+V_j})$ and the preceding case analysis provides a linear charge at that same scale. A small share of this charge absorbs it, leaving

$$C_0(h) \geq s_m a_m + \sum_{j < m: s_j=1, s_{j+1}=0} \eta_j + cJ(h) - C. \quad (\text{D.1.24})$$

The same proof holds when the actual mixed contrast is replaced by any fixed sufficiently small positive multiple of mixed time.

We record the relevant same-word transport calculation. In plateau j 's local interior bank of length $\gg T_j$, recolor e_j copies of the immediate child's first shallow leaf to the local shallow exterior. Put $A_j = \log S_j(\text{first})$, $b_j^{\text{flip}} = u_j A_{j+1}$. The construction gives

$$b_j^{\text{flip}} e_j - A_j e_{j-1} = y_j + \delta_j, \quad e_0 = 0, \quad 0 \leq \delta_j < b_j^{\text{flip}}, \quad e_j \ll \log(2 + T_j), \quad (\text{D.1.25})$$

and the original fine prior ratio is $\log[p(c_j)/p(d_j)] = b_j^{\text{flip}} - A_{j+1}$. Thus the full residual changes by $\Delta R = \sum_{j < m} e_j (b_j^{\text{flip}} - A_{j+1})$. At a regular endpoint bank the possible signatures and per-flip increments of the fine cost are

plateau full	0
plateau zero, child full	b_j^{flip}
plateau zero, child zero	$b_j^{\text{flip}} - A_{j+1}$.

Here “full” means the entire relevant eligible cell, with no mixed state at those two levels. The missing signature, plateau full with a split child, is mixed. The increments follow directly by subtracting ΔI from the stated ΔR ; for instance in the middle case $\Delta I = -A_{j+1}$. This keeps the prior change.

Any discrepancy from the ideal coefficient $b_j^{\text{flip}} s_j - A_{j+1} s_{j+1}$ is paid by $J(h) \gg \sqrt{T_j} - C$. The exhaustive cases are: a proper parent or child state contributes an early/bad length or good-group square root in its local or earlier group; a mixed state contributes the whole group; an early full plateau or child has an early prefix reaching that group; a late zero plateau has a zero tail containing it; a late zero child has a tail containing its entire local length, which is $\gg T_j$ by the hierarchy. Every per-flip coefficient is bounded. Since $\log(2 + T_j) \leq \epsilon' \sqrt{T_j} + C_{\epsilon'}$, these exceptional banks consume an arbitrarily small fixed part of J . Regular banks keep their exact increments.

The main coefficients telescope without any monotonicity of the flags:

$$\sum_{j < m} e_j (b_j^{\text{flip}} s_j - A_{j+1} s_{j+1}) = \sum_{j < m} s_j (y_j + \delta_j) - s_m A_m e_{m-1}.$$

Since $r_j b_j^{\text{flip}} = r_{j+1} A_{j+1}$, the actual root gives

$$a_m - A_m e_{m-1} = y_m - \sum_{j < m} (r_j / r_m) \delta_j = y_m + O_H(1). \quad (\text{D.1.26})$$

Thus every fine endpoint star on the target satisfies

$$C_y(h) \geq \sum_{j: s_j=1} y_j + cH(h) - C. \quad (\text{D.1.27})$$

This bound retains coefficient one on every early target, including the terminal target. The telescope applies to every pattern of early and late plateaux.

For later use, if a late plateau's endpoint template is not full on an original local block g , its charge is at least $c\sqrt{L_g} - C$. Indeed a zero parent pays its zero tail; a proper parent pays its bad length or good square root; a full parent with a nonfull fine class is mixed on the entire group. For the terminal plateau the atomic and fine alphabets agree. These charges add over the finite set of affected groups after reducing the common coefficient.

D.1.7 Intrinsic chords within a global fine history

A local atomic wall remains useful inside a global history only if its chord is dominated by the global endpoint budget. We prove this for a proper coordinate even when its child is partly split, using either the intrinsic chord or a linear charge on the current block. Terminal coordinates use a root-separated budget and their actual same-count source representatives.

Fix a local canonical full-current-input to deepest-suffix edge in plateau j and physical block $i = [l_i, b_i]$. First consider a global endpoint context in which the whole immediate child remains intact on the open block. Its varying bundle is exactly the indicated plateau exteriors, with deadlines at least b_i ; all other cuts are fixed lawful endpoints. The global endpoint difference and centered bridge are precisely the atomic ones:

$$C_0(h^+) - C_0(h^-) = K_j(v_j^+) - K_j(v_j^-) = \Delta_i, \quad \Delta_i = (r_i - q)L_i - \log S_i n_i + \log(q + z)K_{i,B}.$$

Let ℓ_i^{intr} be the adjusted chord from the two actual all-context intrinsic minima, shifted by their fixed reserve h_{intr} . If the child is early, $t_{j+1} \geq b_i \geq L_i$, so both global endpoints have a charge $cL_i + h_*$ after one fixed shift. If the child is late, the intrinsic score upper is

$$K_j(v_j) + h_{\text{intr}} \leq C(h_{\text{intr}} + s_j a_j + J_j).$$

For an early nonterminal plateau followed by a late child, (D.1.24) retains exactly $a_j = \eta_j$; for a late plateau J_j suffices. At the terminal plateau it retains $s_m a_m$. Hence in these cases both global endpoints dominate a fixed positive multiple of the intrinsic endpoints. Interpolation gives the same chord domination.

Choose a global budget share $\theta_{\text{glob}} > 0$. The linearly paid case plus (D.1.4) pays a strong inward wall. In the chord case choose in advance $0 < \theta_{\text{intr}} \leq c_* \theta_{\text{glob}}$. The source's actual contracted wall $\theta_{\text{intr}} \ell_i^{\text{intr}} + \mathcal{R}_i \geq g d_i^\gamma - h_W$ then implies the global allocated wall. This is why the contractions are fixed before the source probability construction.

Terminal coordinates require the root to remain outside this allocation. The source supplies same-root representatives with actual local increment and $x_i^- \leq C_A X_i^0$, $y_i^- \leq C_A Y_i^0$, where the deterministic scores X_i^0, Y_i^0 are computed at zero terminal target. For a terminal prefix these follow by a virtual zero-root count completion of its actual type; for the ordinary reservoir the actual root count stays fixed and the extra left reserve is $O_A(\log(2 + V_{\text{pre}})) \ll X_i^0$, while the right stays in its fixed box. No virtual clock is used. After subtracting $a_m s_m$ from a completed terminal endpoint, (D.1.24) dominates its zero-target score. Consequently its root-separated endpoint chord dominates a fixed multiple of the actual source representative chord, even if the separated endpoint difference itself is no longer Δ_i .

Now allow an arbitrary fine star. Synchronize by plateau within each original physical block: $\alpha = (j, i)$ has mean time τ_α , spread D_α , and endpoint depth d_α . Atomic curvature pays the within-block spread, also when the terminal child has weight $z = 1$. Thus

$$\sum_j K_j(h_j) \geq \sum_j K_j(h_j^{\text{sync}}) + c_s \sum_\alpha D_\alpha - C[1 + \sum_\alpha d_\alpha^\zeta].$$

For a fine star its mixed time has the exact formula $S_{\text{mix},j} = (t_j - \theta_j)_+$, where t_j is the plateau's largest exterior cut and θ_j is the smallest cut of any nonanchor fine name in its entire child. All child names are eligible until $V_j \geq t_j$. Maxima, minima and positive parts are Lipschitz, so synchronization changes total mixed time by at most $C_H \sum D_\alpha$. The actual mixed contrast is at least $c_M \sum S_{\text{mix},j}$ minus $C[1 + \sum(d_\alpha^\zeta + D_\alpha^\zeta)]$: every nontrivial error endpoint is an original cut, within its spread of its mean. Choose a small fixed $\epsilon_{\text{mix}} > 0$ relative to c_M, c_s . The fine cost therefore dominates

$$F(\tau) = \sum_j K_j(h_j^{\text{sync}}) + \epsilon_{\text{mix}} \sum_j S_{\text{mix},j}(h^{\text{sync}})$$

plus a positive spread charge and the stated depth error.

In each physical block order all its synchronized plateau-group times. On this ordered chamber the mixed term is affine. Use ordered-simplex endpoint weights within that block, and products only across distinct blocks. If F_v is the endpoint value and $a = a_m$, then exactly

$$F(\tau) = a\mathcal{S} + \widehat{L} + \sum_\alpha \mathcal{R}_\alpha, \quad \mathcal{S} = \sum_v \omega_v s_m(v), \quad \widehat{L} = \sum_v \omega_v (F_v - a s_m(v)). \quad (\text{D.1.28})$$

The enhanced endpoint argument applies with $\epsilon_{\text{mix}} S_{\text{mix}}$ and yields

$$F_v - a s_m(v) \geq \sum_{j: s_j=1, s_{j+1}=0} \eta_j + cJ(v) - C, \quad \widehat{L} + C \geq c \sum_v \omega_v J(v) \geq 0.$$

The coordinate marginal remains $\mathbb{P}(v_\alpha = b_i) = (\tau_\alpha - l_i)/L_i$. We keep the root term $a\mathcal{S}$ separate from $\widehat{L}$ throughout the wall allocation.

A proper local weak coordinate can have a partially split child; its $\mathcal{R}_\alpha$ is still the atomic projection bridge. Fix only interpolation choices in other physical blocks. Let t_{out} be the largest

cut of the immediate child plateau there, including fixed boundary cuts. It is at most l_i or at least b_i . If $t_{\text{out}} \geq b_i$, the child's flag is fixed. An early child pays cL_i at every current vertex; a late child leaves either the positive plateau exit for an early plateau or its J_j for a late one, so the intrinsic chord comparison applies. If $t_{\text{out}} \leq l_i$ and there are child cuts in this block, a right child cut is entirely old and its reference cost is at least $cb_i \geq cL_i$. If all child cuts are left but the parent bundle is right, the whole block is mixed. Thus every parent-right vertex has charge cL_i , and the coordinate marginal pays $c(\tau_\alpha - l_i)$. With no child cut in this block, its flag is fixed: an early child gives the same mixed payment and a late one allows the intrinsic comparison. Averaging the outside choices gives

$$\widehat{L} + C \geq c \min\{\tau_\alpha - l_i, \ell_\alpha^{\text{intr}}(\tau_\alpha)\}. \quad (\text{D.1.29})$$

The first branch plus regularity pays the inward curve; the second plus the same actual $\mathcal{R}_\alpha$ is supplied by the contracted source wall. The identity $\min(A, B) + E = \min(A + E, B + E)$ completes this case.

Strong coordinates have a fixed reference drift gap and are paid from their cheaper endpoint. Old weak and local proper-input or wrong-name weak coordinates have two-ended reference charge cL_i . These statements apply vertex by vertex using the actual marginals. Remaining terminal coordinates use the zero-target representative comparison above, including the actual ordinary root count. Choose $\theta_{\text{glob}} \leq 1/(4H^2)$ and all source contractions small enough in advance. For every coordinate,

$$\theta_{\text{glob}}(\widehat{L} + C) + \mathcal{R}_\alpha \geq g_0 d_\alpha^\gamma - C_0.$$

There are at most H^2 shares. Their sum leaves at least three quarters of $\widehat{L} + C$. After absorbing the earlier d^ζ errors,

$$C_0(h) \geq a\mathcal{S} + c \sum_v \omega_v J(v) + c \sum_\alpha (d_\alpha^\gamma + D_\alpha) - C. \quad (\text{D.1.30})$$

Every empirical bridge has been paid once.

D.1.8 Selecting an endpoint without losing the root

Endpoint interpolation gives an average of costs, while history summability needs one representative. A maximum-weight lawful vertex has uniformly positive weight and controlled distance from every cut. We transfer the terminal early flag to this vertex with a small part of the retained reference and depth budgets, preserving the root coefficient.

Choose in each physical block an ordered-simplex vertex of maximum weight, with a fixed tie rule. Their product $v^\#$ is lawful and has weight at least $(H + 1)^{-H}$. A chosen endpoint side has marginal probability at least $1/(H + 1)$. If it is the nearer side, its movement is d_α ; if farther, that marginal is d_α/L_i , so its movement is at most $(H + 1)d_\alpha$. Original unsynchronized cuts add their spread. In particular

$$\sum_v \omega_v J(v) \geq (H + 1)^{-H} J(v^\#), \quad |t_c - v_c^\#| \leq C_H(d_\alpha + D_\alpha). \quad (\text{D.1.31})$$

The maximum-weight rule supplies the uniform marginal lower bound needed to control movement. Coordinatewise nearest choices need not have such a bound.

Suppose $0 \leq a \leq A \log(2 + T)$, where $T = V_m$. Then for every $\epsilon > 0$,

$$a|\mathcal{S} - s_m(v^\#)| \leq \epsilon \sum_v \omega_v \overline{C}_m(v) + \epsilon \sum_\alpha d_\alpha^\gamma + C_\epsilon. \quad (\text{D.1.32})$$

Only the unique physical block $[l, b]$ crossing $T/2$ can vary the flag, unless another fixed-late cut forces all flags to zero. The coordinate marginals and controlled movement give $|\mathcal{S} - s_m(v^\#)| \leq C_H \sum d_\alpha/L$ in this block, as follows: for an early representative use the union bound for a right choice; for a late one use one chosen-right coordinate to bound the all-left event. If $L \geq T/4$, then $A \log(2+T)d/L \leq A \log(2+4L)L^{-\gamma}d^\gamma$, whose coefficient is arbitrarily small at large L ; bounded L costs a constant. If $L < T/4$, both endpoints lie inside $(T/4, 3T/4)$, and every contributing endpoint has last terminal cut there. The clipped-prefix reference lower $\bar{C}_m(v) \geq c_H \min(t_v, T-t_v)$ therefore pays $c_H T$. It absorbs $A \log(2+T)$ with an arbitrarily small share and a constant. This proves (D.1.32) and keeps the full root coefficient. Applying it to (D.1.30) gives

$$C_0(h) \geq a s_m(v^\#) + c J(v^\#) + c \sum_\alpha (d_\alpha^\gamma + D_\alpha) - C.$$

We need a slightly stronger weighted form. Put

$$F_\epsilon^0(h^*) = \sum_j K_j^0(h_j^*) + \epsilon \sum_{j < m} S_{\text{mix},j}(h^*), \quad B(h^*) = \sum_j \bar{C}_j(h_j^*), \quad \Pi(h^*) = \sum_\alpha (D_\alpha + d_\alpha^\gamma).$$

For fixed sufficiently small $\epsilon > 0$, the same proof yields

$$F_\epsilon^0(h^*) \geq a s_m(v^\#) + c [J(v^\#) + B(h^*) + \Pi(h^*)] - C. \quad (\text{D.1.33})$$

Indeed mixed-time Lipschitz and atomic synchronization apply directly to this deterministic mixed-length term. At endpoints any fixed positive mixed coefficient pays adverse flag changes. Reserve positive shares of reference and spread, in addition to all bridge allocations:

$$0 \leq B(h^*) - B(h^{\text{sync}}) \leq C_H \sum_\alpha D_\alpha.$$

The lower is concavity; the upper follows because the rank profiles differ only between each group's first and last cut, with uniformly bounded integrands. These shares retain $B(h^*)$, while further shares pay J , curves, spread and the flag interpolation. All allocations are fixed before constructing the source.

D.1.9 From a general fine history to its source and target costs

We can now assemble the geometric reductions. Projection and branching supply delay, lifetime, and mixed-time gains. Synchronization and endpoint selection supply reference and depth gains. Their positive shares absorb all named-regularity errors and still leave a summable charge on every original history. Recoloring transfers the same structure to the target word.

Proposition D.1 (Original-history source geometry). *On the source support just constructed, every lawful original fine history h has a lawful comparison star h^* and endpoint representative $v = v^\#$ such that*

$$C_0(h) \geq a s_m(v) + c \Lambda(h) - C, \quad \Lambda = J(v) + B(h^*) + \Pi(h^*) + D_{\text{gap}} + D_* + S_{\text{pure}} + S_{\text{off}} \geq 0. \quad (\text{D.1.34})$$

For every fixed $\delta > 0$, the associated geometric envelope has a uniformly bounded sum over all original histories and original pivot representatives. All constants are independent of length ratios and strict product gaps.

Proof. Recursively apply the lawful projection (D.1.10), keeping original time labels and selected lists. The exact identity is

$$C_0(h) = \sum_j K_j^0(h_j) + D_{\text{gap}} + M_{\text{orig}},$$

with $M_{\text{orig}} \geq c_M S_{\text{orig}} - C[1 + \sum_{\text{original cuts } t} d_E(t)^\zeta]$ from (D.1.9). Apply (D.1.13) to nonterminal plateaux and (D.1.14), (D.1.17) to the terminal one. Their signed output normalizations cancel in each same-word difference. They retain positive proper pure lifetimes, terminal off-spine lifetimes and true clipping delay D_* . The remaining terminal loss is

$$C_H \vartheta T_{\text{float}} + C_K [1 + T_{\text{float}}^\zeta + \sum_{\text{births}} d_E(\beta)^\zeta], \quad B(h^*) \geq a_H T_{\text{float}}.$$

If there is no terminal nonsingleton interval, choose its comparison cuts zero, $D_* =$ the sum of its original times, and $S_{\text{off}} = T_{\text{float}} = 0$; this case is exact. By (D.1.18), a sufficiently small $\epsilon < \min(c_M/2, c_p/4)$ retains $\epsilon S_{\text{mix}}(h^*)$ while leaving positive shares of original mixed and pure time. The weighted quantity in (D.1.33) is therefore available.

We pay the depth errors by tracking their original events. An erased projection event is paired to a physical deadline, with distance charged to D_{gap} . A terminal event is retained, clipped to its deadline, or clipped to τ . In the last case, $d_E(e) \leq d_E(\tau) + (e - \tau)$, and $e - \tau$ belongs to D_* ; τ is a retained terminal cut when it is not a physical endpoint. The same alternatives treat expired-first births. Retained cuts lie within their group spread of its synchronized time. Thus all errors are bounded by

$$C[1 + D_{\text{gap}}^\zeta + D_*^\zeta + \sum_\alpha (d_\alpha^\zeta + D_\alpha^\zeta)]. \quad (\text{D.1.35})$$

First choose the regularity slope so that a reserved part of $B(h^*) \geq a_H T_{\text{float}}$ pays the linear floating error. Young's inequality pays the sublinear floating error from another part. Delay and spread pay their ζ -powers; the inward curves with $\gamma > \zeta$ pay d_α^ζ . Each allocation leaves a fixed positive share of its budget, proving (D.1.34).

The projection fibers sum by (D.1.11). Terminal fibers at a physical endpoint sum by the same bound; at interior τ they incur the finite polynomial in (D.1.15), paid by the retained inward and spread budgets. Remaining cuts lie in the endpoint strips (D.1.31); their polynomial multiplicity is summed by exponential spread and stretched-exponential depth weights. Finite assignment chambers, names, tie resolutions and permutations only change the constant. Generated repeated deadline representatives remain in the enlarged positive row. This proves summability without replacing an original class by a comparison class. $\square$

Proposition D.2 (Original-history target geometry). *For the transported bounded density F_y on the same actual root tuple, every word of positive weight and every original fine history satisfy, with the constructions above,*

$$C_y(h) := R_y - I_h(\omega_y) \geq \sum_{j:s_j(v)=1} y_j + c\Lambda(h) - C. \quad (\text{D.1.36})$$

Its geometric envelope has the same original-pivot summability.

Proof. Fix any retained source preimage of the target word. Such a preimage exists for every positive value of the transported density. For a comparison fine star define $F_\epsilon^q = \sum_j K_j^q + \epsilon S_{\text{mix}}$ for $q = 0, y$. Mixed time depends only on names and cuts, so exactly

$$\Delta F_\epsilon(h^*) = \Delta R - \sum_j \Delta I_j(h_j^*).$$

Each atomic information is a sum over original slots, zero when a slot does not route to that plateau; the formula remains exact when recoloring changes child membership. Compare this change with the change at the endpoint representative v on the same source/image pair. The residual ΔR cancels. The bounded atomic mark vector changes only if a flipped slot meets a movement strip. Bank j has depth $\gg T_j$ in its original block; such an intersection forces $\Pi(h^*) \geq c_H T_j^\gamma - C_H$. Since it contains $O(\log(2 + T_j))$ flips, for every fixed $\alpha > 0$

$$|\Delta F_\epsilon(h^*) - \Delta F_\epsilon(v)| \leq \alpha \Pi(h^*) + C_\alpha. \quad (\text{D.1.37})$$

At a regular endpoint bank, the three per-flip increments computed above apply to $\sum_j K_j$ itself: the relevant mixed contrasts vanish identically, ancestor exteriors have expired, and deeper plateaux read neither changed shallow name. Exceptional signatures have $J(v) \gg \sqrt{T_j} - C$, by the exhaustive endpoint cases already proved. Their logarithmic number consumes any small fixed fraction of $J(v)$. Hence

$$\Delta F_\epsilon(v) \geq \sum_{j < m} e_j(b_j^{\text{flip}} s_j - A_{j+1} s_{j+1}) - \alpha J(v) - C_\alpha.$$

Apply (D.1.33), (D.1.37), and the exact telescope (D.1.26); nonnegative δ_j only help. For small α ,

$$F_\epsilon^y(h^*) \geq \sum_{j: s_j(v)=1} y_j + c[J(v) + B(h^*) + \Pi(h^*)] - C. \quad (\text{D.1.38})$$

The telescope retains coefficient one on the terminal root of the fixed source-image pair. It uses the exact output changes and endpoint charges, keeping the same root tuple throughout.

The target retains the original named regularity with prescribed spare margins. An interval meeting a flipped interior bank either has length $\gg T_j$ or an endpoint of depth $\gg T_j$; otherwise it cannot reach that bank. A spare linear slope in the first case, or a spare ζ -depth coefficient in the second, pays $O(\log(2 + T_j))$ flips after fixing a sufficiently large plateau threshold. Routed atomic counts are sums of finitely many fine counts. Thus choose target regularity constants first and construct the source inside them; no clock or reference intensity changes.

Project the original history on this target word. Its exact identity is $C_y(h) = \sum_j K_j^y(h_j) + D_{\text{gap}} + M_y$. Use the proper and terminal same-word comparisons and mixed-time domination exactly as in the preceding proof. The available weighted budget is now (D.1.38). The floating reference cost pays the linear and sublinear floating errors; (D.1.35) is paid by the same true delays, spread and inward curves. Positive shares remain, proving (D.1.36). Fiber counting is unchanged and keeps every original representative. This argument holds simultaneously for all histories on the chosen source/image pair; it does not compare class posteriors. $\square$

D.1.10 The actual comparison class and the complete row

The geometric representative estimates cost, but the conditional probability is taken in the original complete-cell class. Endpoint strips identify which labels this class reveals. On a late block, nonfull signatures are geometrically paid; the remaining blocks have genuinely free intervals. We then sum complete types and diagonal count bands under the original target law before taking a fractional power.

Throughout this geometric row calculation, Q_h is the complete-cell class $\bar{Q}_h$ of Section 9.1; its $I_h, C_y(h)$ and relative posterior are computed with that same class and the unchanged original multiplier. The original slot history remains its index.

Set, for the predeclared environmental costs and count widths,

$$D_j^{\text{own}} = w_j \prod_{i:\text{plateau}(i)=j} D_i \quad (j < m), \quad D_m^{\text{own}} = \prod_{i:\text{plateau}(i)=m} D_i, \\ y_j = -\log D_j^{\text{own}}, \quad D_{\text{all}} = \prod_j D_j^{\text{own}}.$$

These targets are fixed before reading the private word. Fix its actual class $Q_h(\omega)$ and the bounded transported acceptance $F = F_y$. It has the actual root, named count support, strict exits, reconstructed source tests, and bounded full relative likelihood established in the transport construction.

In a physical block $g = [l_g, b_g]$, choose an integer endpoint-strip radius $r_g \geq 1$ dominating, by a dimension factor, the depths of all original cuts and all movement strips from h^* to v ; cap it at $L_g/2$ when necessary. From the exact depth bookkeeping,

$$\sum_g r_g^\gamma \leq C_H(1 + \Lambda). \quad (\text{D.1.39})$$

If $r_g < L_g/4$, the original partition is constant on the middle interval and h^* there agrees with v . These are partition facts, independent of label probabilities. Proper projection and star comparison preserve eligible unresolved rank. The only possible increase for h^* is from expired terminal off-spine first names, whose number is at most the original terminal off-spine internal-cell count. Thus, pointwise,

$$0 \leq r(h^*) - r(h) \leq m_{\text{off}}.$$

If v is full but the original middle is not, this rank difference is at least one over an interval of length $L_g/2$, so $S_{\text{off}} \geq L_g/2$. If v is nonfull on a late plateau's local block, its endpoint charge is $J(v) \geq c\sqrt{L_g} - C$, as proved above. If the strips occupy a quarter block, their own L_g^γ charge pays. Every late block is therefore geometrically paid or the original class is full outside its endpoint strips. Any missing polynomial floor on a paid block is absorbed by an arbitrarily small fixed fraction of Λ at its own length scale.

For a proper late plateau's count bank, either those strips reach its interior or the original class is full on the bank. In the first case Λ has a positive power of T_j and pays the width floor $w_j \gg (2 + T_j)^{-1/2}$. Let J_{free} be the remaining free banks. The original class reads no label in them. Condition first on complete types and exactly read strip labels. Unread permutations in different blocks are independent uniform residual multisets, and the type-uniform path envelopes give

$$\mathbb{P}(\text{necessary path tests} \mid \text{complete types, read strips}) \leq C \prod_{i \in J_{\text{free}}} q_{i, r_i}(G_i). \quad (\text{D.1.40})$$

Sum the coupled type law inside this bound before taking any fractional power. To estimate its exit probability, next reveal the pair-coarsened bank labels and the labels outside those pairs. The diagonal identities $\eta_j = h_j + b_j^{\text{flip}} K_j$ leave every h_j independent of the unread splits. Maximum split atoms bound a band by $C(1 + R_j)/\sqrt{1 + M_j}$ at the actual pair total M_j . Averaging all original pair totals, including zero, gives

$$\mathbb{P}(\text{exit restrictions} \mid Q_h(\omega)) \leq C \prod_{j \in J_{\text{free}}} \min\{1, R_j/\sqrt{1 + T_j}\}. \quad (\text{D.1.41})$$

Indeed M_j is binomial under the fully free bank's original law; Chernoff gives $\mathbb{E}(1 + M_j)^{-1/2} \leq C(1 + T_j)^{-1/2}$. The original bank total is supplied by named regularity; if it fails, the accepting

density is zero. This proof does not infer an exit-only bound from empty accepting support on a small-pair fiber. Combining the two finite disintegrations proves

$$\mathbb{E}_\pi[F \mid Q_h(\omega)] \leq C \prod_{j \in J_{\text{free}}} w_j \prod_{i \in I_{\text{free}}} q_{i,r_i}(G_i). \quad (\text{D.1.42})$$

The fully stated lemma, its elementary proof, the actual diagonal cancellation and the joint free-bank density construction appear in Section D.5. The class in that proof is the target class with its changed old lists.

The same environmental envelopes can be chosen for any sufficiently small fixed $b_0 > 0$ so that $q_{i,r} \leq CD_i e^{b_0 r^\gamma}$. Their stretched-exponential coefficient sum is finite, so clipping and normalization into $[0, 1]$ changes only a fixed constant. Their prescribed polynomial floors and source moments persist. Choose b_0 small relative to the coefficient in (D.1.36) and (D.1.39). Pay all missing late widths and path floors with the local payments above. Then

$$\mathbb{E}_\pi[F \mid Q_h(\omega)] \leq C e^{(c/2)\Lambda} \prod_{j: s_j(v)=0} D_j^{\text{own}}.$$

The floor fixes the logarithmic target range independently of b_0 ; that range fixes the finite geometry; then choose b_0 , and only afterwards construct envelopes and source. Combining with (D.1.36) gives

$$e^{-C_y(h)} \mathbb{E}_\pi[F \mid Q_h(\omega)] \leq CD_{\text{all}} e^{-c'\Lambda}.$$

For retained compatible competitors the full likelihood ratio is $e^{E_y(\omega) - E_y(\omega')}$, whose amplitude is bounded by the strict-exit weighted likelihood identity. Therefore the true relative posterior

$$q_h^{\text{rel}} = \mathbb{E}_\pi[F(\omega')\pi(\omega)/\pi(\omega') \mid Q_h(\omega)]$$

satisfies the same bound with a fixed extra factor. Raising to a fixed $\delta > 0$ and using original-pivot summability proves

$$\sum_{h \text{ original}} e^{-\delta C_y(h)} (q_h^{\text{rel}})^\delta \leq C_\delta D_{\text{all}}^\delta. \quad (\text{D.1.43})$$

D.1.11 Bounded family weights and the arithmetic first mass

The relative row becomes an unnormalized counting row through an exact finite class identity. Combined with the prime-cell kernel, it bounds a higher moment of the weighted box count by its first mass. Hölder then retains that first mass to the first power in the volume lower bound. The final linear projection removes auxiliary nominal data without moving any fractional power through it.

For the arithmetic quantities in this paragraph, write Q_h^{arith} for the class (E.1.4), and Q_h^{cell} for its complete-cell enlargement. The row just obtained for Q_h^{cell} bounds the row by $\sum_{Q_h^{\text{arith}}} F \leq \sum_{Q_h^{\text{cell}}} F$, with the same $\Phi_g(h)$ and the same original index set.

Fix an original cell occupancy g , its finite legal label space Ω , and $0 < \delta \leq 1$. An acceptance $F : \Omega \rightarrow [0, 1]$ may couple all slots, but is independent of actual prime values and invariant under within-cell slot permutations. Set

$$Z_F = \sum_Y F(Y), \quad M_h(F; Y) = \sum_{Z \in Q_h^{\text{arith}}(Y)} F(Z), \quad R_F(Y) = \sum_h \Phi_g(h)^\delta M_h(F; Y)^\delta.$$

The finite weighted class identity is exactly

$$\sum_h \Phi_g(h)^\delta \Psi_{h,1+\delta}(F) = \sum_Y F(Y) R_F(Y). \quad (\text{D.1.44})$$

It is simply the expansion of the class power sum, so a pointwise bound $R_F \leq C$ on $F > 0$ gives a total at most CZ_F . For any positive product law π ,

$$M_h(F; Y) = \frac{\pi(Q_h^{\text{arith}}(Y))}{\pi(Y)} q_{h,\text{arith}}^{\text{rel}}(F; Y),$$

where the relative posterior on the right is conditioned on $Q_h^{\text{arith}}(Y)$ itself. This follows by inserting $\pi(Z)/\pi(Z)$ in the finite class sum; bounded F alone does not remove its relative likelihood. A pointwise row for a support indicator also holds for every smaller bounded weight, by positivity; a merely averaged row does not imply this.

Suppose the correlated prime-cell kernel applies and the retained occupancy has squarefree supply $cW_g \leq S_g \leq W_g$, with all tuples inside the required size cutoff. On harmonic tuple measure times Lebesgue space define $f(\mathbf{a}, t) = \sum_Y F(Y) \mathbf{1}_{B_Y(\mathbf{a})}(t)$, where the original boxes have side $b = \log 2$. Then $\int f = b^k S_g Z_F \gg W_g Z_F$. At a point of B_Y , every other containing box overlaps B_Y , so its divisor log vector is within b in each coordinate. The correlated kernel and (D.1.44) consequently give $\int f^{1+\delta} \ll W_g Z_F$. Hölder on this same unnormalized measure yields

$$\text{meas}\{f > 0\} \geq \frac{(\int f)^{(1+\delta)/\delta}}{(\int f^{1+\delta})^{1/\delta}} \gg W_g Z_F.$$

Its support is in the original divisor union. Occupancies partition the tuple space and can be summed, retaining the linear dependence on the bounded acceptance.

Finally, extra nominal data may be removed in a linear first-mass identity when their support is already cell-measurable. Suppose that, conditional on g , the joint law of (Y, Ξ) and an acceptance $a(Y, \Xi) \in [0, 1]$ are equivariant under within-cell permutations, and accepted realizations have an invariant cell-label support $S(Y)$. Let $\pi_0(Y) = \mathbb{P}_{\text{joint}}(Y \mid g)$ be the actual marginal, and set $F(Y) = \mathbb{E}_{\text{joint}}[a(Y, \Xi) \mid Y, g]$ where $\pi_0(Y) > 0$, with $F(Y) = 0$ otherwise. Then F is invariant and $0 \leq F \leq \mathbf{1}_S$. For every nonnegative Y -measurable weight w ,

$$\sum_Y \pi_0(Y) w(Y) F(Y) = \mathbb{E}_{\text{joint}}[w(Y) a(Y, \Xi) \mid g].$$

For a desired nonnegative counting weight v , take $w = v/\pi_0$ where $\pi_0 > 0$ and $w = 0$ otherwise; the left side is then $\sum_Y v(Y) F(Y)$. Use Tonelli for an outer occupancy integral. The conditional law of Ξ may depend on Y . This preserves linear likelihood factors; it does not commute conditional expectation with a fractional power. Uniform deletion of specified-color slot subsets and count-dependent target laws are permutation equivariant, so the transport has this property whenever its source and boundary support do. The complete-cell construction in the main proof retains its nested fractional quantities before this linear arithmetic projection.

D.2 Endpoint forests and summability of original pivots

The reductions above change names and sometimes replace a cut by a deadline. A cut-token construction tracks these changes monotonically and identifies their total backward motion with the delay budget. It then bounds every original pivot depth by final depths, spreads, and delay. A single stretched-exponential estimate sums all original histories and their within-cell multiplicities.

D.2.1 Original data and notation

Fix a dimension bound $H \geq 2$. There are $B \leq H$ original physical blocks $[a_g, b_g]$, with $L_g = b_g - a_g > 0$, and at most $n \leq H - 1$ original cuts. The physical endpoint set is

$$E = \{a_g, b_g : 1 \leq g \leq B\}, \quad d_E(t) = \min_{e \in E} |t - e|. \quad (\text{D.2.1})$$

Within a block this is its ordinary endpoint depth. The original pivot grid includes its original representatives with their slot multiplicities. Its uncolored regularity implies, for fixed constants C_{occ}, κ ,

$$\#\{\text{original representatives } t : d_E(t) \in [q, q+1)\} \leq BC_{\text{occ}}(1+q)^\kappa, \quad q = 0, 1, \dots \quad (\text{D.2.2})$$

The zero pivot contributes one extra representative. Original permutations, zero pivots and ties remain included.

For example, the named regularity inequality at Section D.1, gives (2) by summing over the at most H names and applying the resulting uncolored upper bound to a unit interval at depth q near either end. Its linear reference intensity is bounded in terms of H . Its endpoint error is at most $C(1 + (q+1)^\zeta)$. A bounded complete-cell mesh only enlarges that interval by a fixed amount and therefore changes the constant. Thus one may take $\kappa = \zeta$; any larger fixed power also suffices. This argument includes every original representative in an occupied cell, rather than assuming bounded interior occupancy. In particular, when several distinct original slot pivots round to the same cell boundary, each is counted separately in (2). The summability theorem below is therefore a sum indexed by the original histories, with the rounded geometric times and their full original multiplicities.

The original history is a refinement tree of an anchored permutation, with activation times legal for each original boundary's deadline. Owners form the nested chain of exterior alphabets and a compound child. Their proper child weights satisfy $z \geq z_- > 1$; the terminal alphabet has equal weights. Throughout,

$$0 < \zeta < \gamma < 1/2. \quad (\text{D.2.3})$$

After lawful plateau projections, proper star comparisons, terminal floating-chain comparison and reanchoring, let h^* be the resulting fine comparison star. Its cuts in a given plateau/physical-block group α are synchronized at their mean τ_α . Write

$$D_\alpha = \max_{r \in \alpha} t_r^* - \min_{r \in \alpha} t_r^*, \quad d_\alpha = d_E(\tau_\alpha), \quad \Pi(h^*) = \sum_{\alpha} (D_\alpha + d_\alpha^\gamma). \quad (\text{D.2.4})$$

There are at most $H - 1$ nonempty groups. Let D_{gap} be the accumulated projection delay, D_* the terminal clipping delay, and $D = D_{\text{gap}} + D_*$. All are nonnegative. Let S_{pure} and S_{off} be the proper pure-cell and terminal off-spine lifetimes. These are the same quantities defined above.

D.2.2 The forest-token lemma

Each internal refinement event carries one token. Projection may rename the token or move it to a deadline, but does not create an untracked event. The cumulative rank comparison matches erased times to earlier added deadlines. The terminal spine uses the same tokens, so all backward displacements add to the two delay quantities.

Lemma D.3 (Monotone cut tracking). *The canonical constructions can be implemented by retaining n cut tokens. Their original times are $t_1, \dots, t_n$; their final comparison-star times, with multiplicity, are $t_1^*, \dots, t_n^*$. Token names can change, but*

$$t_r^* \leq t_r \quad (1 \leq r \leq n), \quad \sum_r (t_r - t_r^*) = D_{\text{gap}} + D_*. \quad (\text{D.2.5})$$

Physical endpoint cuts added in projections are included as the new positions of old tokens, not as extra untracked cuts.

Proof. At one exterior/child projection, the formal parent events and child events partition the original event multiset. A child boundary is the minimum-time event in its intervening original segment; distinct child boundaries use disjoint segments. The other events belong to the formal parent. This retains all the times before augmentation.

The augmented parent isolates an expired exterior name at its physical deadline when necessary. Here is a rank proof that its activated-event count is at least the formal parent's at every time. Let F separate the exterior deadlines from all child deadlines. Before F the child has no expired first name, so its unactivated rank equals its eligible unresolved rank. The augmented parent also has zero expired-first delay. Eligible unresolved ranks add under projection. Thus, before F , original unactivated rank minus the two projected unactivated ranks is exactly the non-negative original expired-first delay density. After F the parent rank is zero. Every unactivated child boundary uses an entire disjoint intervening segment of unactivated original boundaries, so original unactivated rank is at least the child's. Since the formal-parent and child event multisets partition the original one, this proves, on both sides of F , that formal-parent unactivated rank is at least augmented-parent unactivated rank. Their activated-event counts have the reverse inequality. Cancel the common event times with multiplicity. If $e_1 \leq \dots \leq e_j$ are the remaining formal-parent times and $a_1 \leq \dots \leq a_j$ the remaining added deadlines, the cumulative count inequality gives $a_l \leq e_l$ for every l . Move token l from e_l to a_l . The loss of total activation time is $\sum_l (e_l - a_l) = D_{\text{gap}}$ for this projection: integrating the difference of unactivated-boundary ranks gives exactly the same quantity. This is the explicit rank identity. Repeat recursively. Each stage moves existing tokens backwards; all displacements add.

A proper same-cut star changes the tree and its name order but retains the event-time multiset. In the terminal plateau, choose the surviving spine representative in the last nonsingleton interval as above. At each binary refinement retain the spine representative on its ancestral branch and retain the left-child representative on every off-spine subtree. The discarded representative receives that event's original time. There is exactly one discarded representative per internal tree vertex, so before clipping the times are the original multiset. Clip a discarded name's time to its own physical deadline if necessary and then to the final nonsingleton time τ . Both operations move times backwards.

For $t < \tau$, the retained set consists of the eligible names in the spine cell and the nonfirst names of off-spine internal cells. Consequently its unresolved rank is the original unactivated rank minus the indicator of an expired spine first name. After τ its rank is zero. Integrating the rank difference gives precisely

$$D_* = \int_0^\tau \delta_{\text{sp}}(t) dt + \int_\tau^T d_{\text{orig}}(t) dt = \sum_r (t_r^{\text{before}} - t_r^{\text{after}}).$$

Reanchoring the floating chain at the permanent original anchor only exchanges the spine and anchor names; its removal-time multiset is unchanged. If no nonsingleton interval exists, all

terminal cut tokens are moved to zero and the displacement is their original time sum, as specified above. This proves (5). $\square$

Lemma D.4 (Original pivot depths). *The tracked construction satisfies the dimension-only bound*

$$\sum_{r=1}^n d_E(t_r)^\gamma \leq H^2(1 + D_{\text{gap}} + D_* + \Pi(h^*)). \quad (\text{D.2.6})$$

Proof. Distance to a fixed set is 1-Lipschitz. If token r belongs to final group α , then

$$d_E(t_r) \leq (t_r - t_r^*) + D_\alpha + d_\alpha.$$

Because $0 < \gamma < 1$, the γ power of a sum is at most the sum of its γ powers. Hölder's inequality and concavity give $\sum_r (t_r - t_r^*)^\gamma \leq n^{1-\gamma} D^\gamma$. There are at most n tokens in a group, and $v^\gamma \leq 1 + v$. Hence

$$\sum_r d_E(t_r)^\gamma \leq n^{1-\gamma}(1 + D) + n \sum_\alpha (D_\alpha^\gamma + d_\alpha^\gamma) \leq n^{1-\gamma}(1 + D) + n^2 + n\Pi.$$

For $n \leq H - 1$, the last expression is at most the right side of (6). The case $n = 0$ is immediate. $\square$

The depth bound also gives (D.1.39). Up to a dimension factor, the largest original depth or movement in a block bounds its required strip radius. The sum above, together with the corresponding final spread and mean-depth charges, therefore bounds the γ th power of that radius.

D.2.3 Summability of all original pivots

The token depth bound converts the geometric penalty into a product of penalties in the original pivot depths. Since original regularity bounds the number of representatives at each depth by a polynomial, these one-dimensional sums converge uniformly. Relaxing order, deadline, and distinctness restrictions enlarges the sum and makes its factorization immediate.

Theorem D.5 (Summability of original pivots). *Suppose the geometry assigns*

$$\Lambda(h) \geq D_{\text{gap}} + D_* + \Pi(h^*) \quad (\text{D.2.7})$$

and uses the tracked constructions of Lemma D.3. Then, for every fixed $\delta > 0$,

$$\sum_{h \text{ original}} e^{-\delta \Lambda(h)} \leq e^\delta H! \left[1 + BC_{\text{occ}} \sum_{q=0}^{\infty} (1+q)^\kappa e^{-\delta q^\gamma / H^2} \right]^{H-1} < \infty. \quad (\text{D.2.8})$$

The bound is independent of block lengths, length ratios, strict product gaps, original grid size and the number of representatives. If the row convention duplicates a history by finitely many named tie resolutions, multiply the right side by that fixed dimension-only number.

Proof. Equations (6), (7) imply

$$e^{-\delta \Lambda(h)} \leq e^\delta \prod_{r=1}^n e^{-\delta d_E(t_r)^\gamma / H^2}.$$

Sum this product over every original time tuple, ignoring deadline constraints, ordering constraints and distinctness constraints. All terms are nonnegative, so this only enlarges the sum. There are at most $H!$ original name permutations. Bound each one-dimensional pivot sum using (2) and the zero pivot. This proves (8). The numerical series converges because a stretched exponential with any positive exponent dominates every fixed polynomial. $\square$

In particular, the full original-pivot summability assertion in D.1, follows from its specified nonnegative Λ , once its cost estimate is established. No separate bounds on individual projection fibers are needed.

D.2.4 The arbitrary-history cost reduction

To use the token summability theorem, we need a cost estimate retaining all of its positive charges. The following deterministic reduction separates pure-cell, off-spine, and mixed-time gains from their named regularity errors. Section D.3 supplies the weighted source-star inequality that pays those errors. Let $q = 0$ or $q = y$ denote one source or target word. Define

$$F_\epsilon^q(h^*) = \sum_j K_j^q(h_j^*) + \epsilon \sum_{j < m} S_{\text{mix},j}(h^*), \quad B(h^*) = \sum_j \bar{C}_j(h_j^*). \quad (\text{D.2.9})$$

Assume the actual fine names obey the original named regularity, with fixed depth-intercept coefficient K and a sufficiently small linear slope ϑ . There exist positive constants c_p, c_o and finite C_H, C_K such that the canonical reduction gives

$$C_q(h) \geq F_\epsilon^q(h^*) + D_{\text{gap}} + D_* + c_p S_{\text{pure}} + c_o S_{\text{off}} - C_H \vartheta T_{\text{float}} - C_K \left[1 + T_{\text{float}}^\zeta + \sum_{r=1}^n d_E(t_r)^\zeta \right], \quad (\text{D.2.10})$$

for sufficiently small fixed $\epsilon > 0$. Here T_{float} is the terminal interval away from the permanent anchor, and

$$B(h^*) \geq a_H T_{\text{float}}, \quad a_H > 0. \quad (\text{D.2.11})$$

These facts are deterministic same-word comparisons. Their proofs are as follows; they do not use the posterior row being proved.

The lawful projection has the exact cost decomposition into atomic costs, nonnegative projection delay and per-slot mixed contrasts. On a nonterminal plateau a current pure original internal cell has $k + 1$ original names and either $k + 1$ or k eligible names. Moving its k nonfirst names into the star increases reference residual entropy by at least

$$\gamma_z = (z_- - 1) \log(1 + 1/z_+) > 0$$

per unit reference coefficient when its first name remains alive; expiration increases this gain. Sum over pure-cell lifetimes. Named regularity on the fixed, bounded number of constant-partition intervals converts this reference gain to $c_p S_{\text{pure}}$, with a small linear error and errors at the original birth/split depths. The output normalization cancels in the same-word difference. This proves the claimed comparison.

In the terminal floating chain, the spine cell has at least two eligible names. Absorbing an off-spine internal cell into its mass gains at least $\log(1 + 1/H)$ per reference coefficient, giving $c_o S_{\text{off}}$. The loss of activation time is exactly D_* . Reanchoring exchanges the two equal-probability fine names only on the floating interval, so its remaining actual-word error is bounded by $C_H \vartheta T_{\text{float}} + C_K [1 + T_{\text{float}}^\zeta + \sum_{\alpha \in \mathcal{D}_{\text{orig}}} d_\alpha^\zeta]$. Here $\mathcal{D}_{\text{orig}}$ indexes the original birth and split depths in this reanchoring, and d_α denotes the corresponding depth. These are the required floating-chain error bounds, with every original depth error retained.

For (11), while the chain floats its rank d lies between 1 and eligible capacity minus one. Retaining the permanent anchor on the whole floating interval increases that rank by one; discarding the other last-cell survivor at the interval's left end decreases it by one. Both comparison histories are lawful. They share the original entrance normalization and have nonnegative complete reference costs. For $g_g(d) = d - \kappa_g(d+1) \log(d+1)$,

$$2g_g(d) - g_g(d-1) - g_g(d+1) \geq \kappa_*/H.$$

Integrate over the floating interval and use the two nonnegative comparison costs. Its reference cost is at least $a_H T_{\text{float}}$ with $a_H = \kappa_*/(2H)$; reanchoring preserves that reference cost. Other atomic reference costs are nonnegative. This proves (11), using the complete-reference nonnegativity theorem proved in Section A.2.4.

Finally, a comparison child has unresolved rank at least that of the original child, so it has no more eligible cells. If a comparison mixed indicator is one and its proper original parent has no pure internal cell, that parent already agrees with its same-cut star and the original child has at least two eligible cells. The original indicator is therefore one. If the parent has a pure internal cell, charge its lifetime. Thus, pointwise and then integrated,

$$\sum_j S_{\text{mix},j}(h^*) \leq \sum_j S_{\text{mix},j}(h) + S_{\text{pure}}.$$

The original mixed contrast has positive reference gap on every mixed interval; REG supplies its linear gain minus the same depth errors. Choose ϵ smaller than half this mixed gain and half the proper pure gain. This retains the mixed term in (9) and positive shares of both original gains, proving (10).

D.3 Partial-child, root-separated and weighted source-star comparisons

This subsection proves the two chord comparisons needed by the global wall allocation. A partially split proper child is paid either by a linear old-prefix charge or by the parent's intrinsic chord. A terminal coordinate is paid by its source chord after the root has been separated. We then combine these comparisons with curvature and a maximum-weight endpoint to retain the full reference, spread, and inward-depth budgets.

D.3.1 Actual source quantities

Use the fixed ordered KKT profile and the source constructed in Section 7. Owners have products $\sigma_1 > \dots > \sigma_m$,

$$u_j = \sigma_{j+1}/\sigma_j, \quad d_j = u_j^{-1} - 1, \quad r_j = \sigma_j/\sigma_1.$$

All these products are in a dimension-only compact subinterval of $(1/2, 1)$. Write T_j for local plateau length and V_j for its total old plus local length. The independently derived plateau balance gives

$$V_{j-1} \leq C_{\text{hier}} T_j, \quad C_{\text{hier}} = 2H^3(1 + \log H)^3. \quad (\text{D.3.1})$$

This is Section 4. It is a bounded-ratio inequality for a complete maximal plateau, not a claim that every local block is longer than its predecessors.

The actual source word has proper targets zero, root $X_1 = x + \epsilon$, $-C \leq \epsilon \leq 0$, and strict exits satisfying

$$\eta_j \geq 0, \quad \eta_j \leq C\sqrt{1 + T_j}, \quad \sum_{j < m} d_j \eta_j \leq C. \quad (\text{D.3.2})$$

Set

$$a_j = \eta_j, \quad b_j = \eta_{j-1}/u_{j-1} \quad (j < m), \quad a_m = x/r_m, \quad b_m = \eta_{m-1}/u_{m-1} - \epsilon/r_m.$$

The atomic outputs are $Z_j = a_j - b_j$; an absent incoming term is zero. The actual intrinsic endpoint cost is $K_j(v_j) = Z_j - I_j(v_j)$. For an endpoint fine star put

$$s_j(v) = \mathbf{1}_{\{t_j(v) \leq V_j/2\}}, \quad J_j(v) = \overline{C}_j(v_j) + S_{\text{good},j}(v_j), \quad J(v) = \sum_j J_j(v) + \sum_{j < m} S_{\text{mix},j}(v). \quad (\text{D.3.3})$$

Here t_j is the last positive cut of plateau j itself. Cuts in deeper plateaux do not enter this maximum.

For fixed sufficiently small $\epsilon_{\text{mix}} > 0$, define the endpoint value

$$F(v) = \sum_j K_j(v_j) + \epsilon_{\text{mix}} \sum_{j < m} S_{\text{mix},j}(v). \quad (\text{D.3.4})$$

All these terms use the same source word; histories change cuts and partitions, not the word's labels or its old lists.

D.3.2 Two endpoint estimates

The displaced count box gives a lower cost with coefficient one on the early signed output and an upper cost in terms of the intrinsic score. Both follow from exact prefix and tail expressions. Keeping these formulas separate prevents a full-group square-root error from entering the late endpoint scale.

The displaced own-scale source count box gives, simultaneously on its actual support,

$$K_j(v_j) \geq s_j(v)Z_j + c_0 J_j(v) - C_0. \quad (\text{D.3.5})$$

For a named endpoint context containing a local cut, one also has

$$K_j(v_j) + h_{\text{intr}} \leq C'_0(h_{\text{intr}} + s_j(v)a_j + J_j(v)). \quad (\text{D.3.6})$$

Here all constants are dimension-only once the fixed count-box and reserve constants are chosen. These are not assertions about a conditioned mean. They follow from the exact prefix/tail formulas for the actual named counts, as follows.

The box row is $C_v \geq b_j + Z_{\text{tail}} + (3/4)\overline{C}_v + G_v - C_{\text{box}}$, with $G_v \geq cS_{\text{good}} - C$. Subtract b_j . At an early endpoint the exact prefix formula replaces Z_{tail} by Z_j minus centered prefix discrepancies of size $O(\sqrt{1+t_j})$. Its positive reference prefix charge pays this square root, retaining a fixed part of $\overline{C}_v$ and the good-group gain. At a late endpoint use the exact tail expression; only the zero-rank tail has a centered square-root error. Reference cost controls its length and the bad-group lengths, so it pays those errors while retaining the good-group gain. This proves (5).

For the upper bound at an early local-containing endpoint, $K_j \leq a_j - b_j + \overline{C}_v + O(\sqrt{1+t_j})$. The local cut ensures $t_j \geq V_{\text{old}}$. Reference cost controls t_j when the endpoint is early, and $b_j \geq 0$ can be discarded, giving (6). At a late endpoint the exact intrinsic tail formula cancels the incoming b_j ; zero-tail and bad-group square roots are paid by reference cost, and good-group square roots are S_{good} . This again gives (6), without any incoming-exit reserve in the late score. These exact formulas and cancellations are proved in Section A.2.12.

Endpoint root removal. There are constants $c_E > 0, C_E$ such that

$$E(v) := F(v) - a_m s_m(v) + C_E \geq c_E \left[J(v) + \sum_{j < m: s_j(v)=1, s_{j+1}(v)=0} \eta_j \right] \geq 0. \quad (\text{D.3.7})$$

Proof. Sum (5), retaining a fixed positive coefficient of the mixed lengths in (4). The output identity is exactly

$$\sum_j s_j Z_j = s_m a_m + s_m \epsilon / r_m + \sum_{j < m} \eta_j (s_j - s_{j+1} / u_j).$$

Equal-one terms sum to $-\sum d_j \eta_j$, bounded by (2). Keep every positive η_j at a $1 \rightarrow 0$ transition. A $0 \rightarrow 1$ loss is at most $C\sqrt{1+V_j}$. It has a charge at its own scale: if $t_{j+1} \geq t_j/2$, the child's early prefix has length at least $V_j/4$; otherwise on (t_{j+1}, t_j) an eligible parent exterior is retained in the fine anchor cell and an immediate-child exterior is already a singleton. This is a mixed interval of length at least $V_j/4$. In either case $J(v) \geq cV_j - C$ for that transition. Spend at most a $1/(2H)$ fraction of the retained J budget for each such loss, using $C\sqrt{V} \leq \epsilon V + C_\epsilon$. There are at most $H - 1$ losses. A positive coefficient of J and every positive $1 \rightarrow 0$ exit remain; bounded floor errors enter C_E . This proves (7). $\square$

D.3.3 A child cut in this block pays its own old prefix

A cut in a parent's local block is an old cut for its child. If that child reaches the right endpoint, its complete reference prefix cost pays the current block even when its own local lengths are uneven. We recover the clipped-prefix estimate on the original coefficient chain and use the plateau hierarchy to compare the child's remaining length with this old interval.

We need the reference fact that a plateau endpoint with last positive cut t has

$$\overline{C}(v) \geq c_{\text{ref}} \min(t, V - t), \quad c_{\text{ref}} > 0. \quad (\text{D.3.8})$$

This is the clipped-positive-prefix statement in Section A.2.8. Its proof can be checked entirely on the original reference profile. Reverse compression makes any positive endpoint rank profile dominate a clipped rank profile $\min(r_g, R)$, $R \geq 1$. The compression update $A \rightarrow D$ is legal because the normalized secant $\log((f(z+D) - f(z+A))/(D-A))/\log(z+D)$, $f(v) = v \log v$, decreases in its upper endpoint; hence an update lowering the current benchmark also lowers every already processed local benchmark. For earlier groups, already processed coefficients are

larger, giving the same assertion directly. This is the original secant/compression proof, not a stationarity assertion for a truncated vector.

For each fixed R , write its coefficient as

$$f_g = g_g(\min(r_g, R)) = \min(r_g, R)(1 - S_g^{t_g}).$$

The original transported coefficients and decreasing integer capacities make consecutive t_g increase by at least

$$d_0 = \min \left\{ \frac{d_*}{\log H}, \frac{1}{24H^2}, \frac{\log(2 \log 2)}{H \log^2 H} \right\} > 0,$$

where the old coefficient gap has the explicit lower bound

$$d_* = \min \left\{ \frac{\varepsilon_H}{H^2}, H^{-H-1} \left[1 - (1 - H^{-1})^{\varepsilon_H} \right] \right\}, \quad \varepsilon_H = \frac{2 \log 2 - 1}{2H(\log H)^2}.$$

Put $c_{\text{ref}} = 1 - 2^{-d_0/3}$. If any prefix coefficient is at most c_{ref} , its $t_g \geq -d_0/3$ and every later coefficient is at most $-c_{\text{ref}}$. Otherwise every prefix coefficient exceeds c_{ref} . The first alternative gives prefix cost at least c_{ref} times tail length, because the complete clipped reference cost is nonnegative; the second gives at least c_{ref} times prefix length. Minimizing over R proves (8). This uses the complete reference comparison, not a lower bound on the lengths of intervening groups.

Now take a parent plateau j and one of its local physical blocks $[a, b]$, $L = b - a$. All cuts of the immediate child plateau $j + 1$ in this block are old cuts. If a child endpoint has last positive cut b , then (1) and (8) give

$$J_{j+1}(v) \geq \overline{C}_{j+1}(v) \geq \frac{c_{\text{ref}}}{\max(1, C_{\text{hier}})} b \geq \frac{c_{\text{ref}}}{\max(1, C_{\text{hier}})} L. \quad (\text{D.3.9})$$

Indeed $b \leq V_j$, and the child's length after b is at least its local total $T_{j+1} \geq V_j/C_{\text{hier}} \geq b/C_{\text{hier}}$. This is precisely the old-interval payment needed in the partial-child argument.

D.3.4 Partial-child comparison on the endpoint forest

Ordered-simplex interpolation couples parent and child endpoint choices within the same physical block. We therefore fix only the choices in other blocks. According to the last child cut outside the current block, every parent-right vertex is paid either by an old-prefix or mixed-time charge, or by the parent's intrinsic endpoint. Its exact coordinate marginal then gives the required chord bound.

Fix a proper local canonical weak coordinate α of plateau j in $[a, b]$. Its synchronized cut is τ , and $u = (\tau - a)/L$. Its atomic projection is full current input to the true deepest suffix. All contexts below are original lawful named contexts.

Let $x_{\text{intr}}, y_{\text{intr}}$ be the actual intrinsic minima over paired outside contexts, shifted by the fixed reserve. Their difference is the actual local increment Δ , because moving only the current atomic bundle across this original block changes no outside term. Write

$$\ell_{\text{intr}}(\tau) = (1 - u)x_{\text{intr}} + uy_{\text{intr}}. \quad (\text{D.3.10})$$

If the source uses the adjusted sufficient pair instead, its two endpoints are no larger than these minima: both actual minima lie above the lower common box, their difference is Δ , and the

adjustment is $\max(a_*X, a_*Y - \Delta)$. Thus proving the inequality with (10) also proves it for the actual sufficient source chord.

Synchronize every plateau in the current block and use its lawful ordered-simplex vertices. Take products of their interpolation weights only across different physical blocks. Fix an outside-block vertex choice. Define t_{out} to be the largest cut of plateau $j + 1$ in those outside blocks, including its fixed boundary cuts. Therefore $t_{\text{out}} \leq a$ or $t_{\text{out}} \geq b$. The child may have deeper plateaux; their cuts remain present but do not enter this definition of t_{out} .

There are two possible bounds for the conditional current-block average of $E(v)$:

$$\mathbb{E}_{\text{current}} E(v) \geq c(\tau - a) \quad \text{or} \quad \mathbb{E}_{\text{current}} E(v) \geq c\ell_{\text{intr}}(\tau), \quad (\text{D.3.11})$$

with the same dimension-only positive c . Here and below these expectations are finite geometric interpolation averages, not probabilities of random words.

If $t_{\text{out}} \geq b$, the immediate child's plateau flag is fixed. If it is early, its prefix has length at least b , so (7) pays cL at every current vertex and yields the first bound in (11). If it is late, (7) retains η_j when the parent is early, while J_j pays when the parent is late. Equation (6) therefore gives, at every current vertex,

$$E(v) + C \geq c(K_j(v_j) + h_{\text{intr}}).$$

When the parent coordinate is left, its right-hand side is at least cx_{intr} ; when it is right, at least cy_{intr} . Atomic parent endpoints do not depend on how the fine child is partitioned: the parent projection merges all child-touching cells into its compound child. The actual incoming word and local counts remain the same. The parent's right marginal is exactly u , even with dependent ordered choices for every child coordinate in this block. Interpolation gives the second bound in (11).

If $t_{\text{out}} \leq a$ and plateau $j + 1$ has cuts in this block, consider every vertex at which the parent bundle is right. A child bundle at the right endpoint has last child cut b , and (9), (7) pay cL . If all child bundles are left, an immediate-child exterior is already a singleton throughout the block; a parent exterior in the varying bundle remains in the fine anchor cell until b . Both are eligible on the entire block. Thus the block is mixed and (7) again pays cL . Other current vertices have $E(v) \geq 0$ by (7). The parent-right marginal u yields $cLu = c(\tau - a)$.

If plateau $j + 1$ has no cut in this block, its flag is fixed. A late child gives the intrinsic bound as above. An early child has all its exterior cuts at or before a , so a parent-right vertex is again mixed throughout the block and gives the linear bound. These cases exhaust every lawful current vertex, regardless of cuts of deeper plateaux in the same block.

In the intrinsic cases a fixed additive constant was displayed; increase C_E once by that dimension-only amount. Each outside choice therefore gives at least the smaller of the two quantities in (11). Average the outside choices. Since $\sum_v \omega_v E(v) = \widehat{L} + C_E$, the result is exactly

$$\boxed{\widehat{L} + C \geq c \min\{\tau - a, \ell_{\text{intr}}(\tau)\}}. \quad (\text{D.3.12})$$

This proves (D.1.29) for the actual partially split fine child. No independence within one physical block, no replacement of an actual class, and no assumed comparison of global and local endpoint differences has been used.

D.3.5 Terminal root removal preserves the actual source chord

Subtracting the terminal root flag changes the global endpoint budget, but leaves a positive budget for the zero-target intrinsic score. We compare that score with the actual source representative: a terminal prefix keeps its original type, and an ordinary reservoir uses the newly selected count. Endpoint domination then transfers the source wall without requiring equality of global and local endpoint differences.

For a terminal local coordinate, let X_0, Y_0 be the minima of the zero-common-target intrinsic scores on its original lawful left and right contexts. The zero-target endpoint score is

$$S_0(v_m) = \begin{cases} h + \bar{C}_m(v_m), & s_m = 1, \\ h + \bar{C}_m(v_m) + S_{\text{good},m}(v_m), & s_m = 0. \end{cases}$$

Thus $S_0(v_m) \leq h + J_m(v)$ at every endpoint, including endpoints whose actual source root is positive. Equation (7), with its fixed shift increased once, gives

$$E(v) \geq cS_0(v_m). \quad (\text{D.3.13})$$

The actual source representative endpoints satisfy

$$x^- \leq C_P X_0, \quad y^- \leq C_P Y_0 \quad (\text{D.3.14})$$

with a uniform dimension-only C_P . Here is why (14) holds for the actual source construction, rather than being a new assumption.

For a selected terminal binary prefix edge, the source pair and its local full type are fixed before the prefix-uncolored root target is read. The reservoir lies in the unread ordinary suffix. At a numerical zero-root count completion, the same prefix type has actual endpoint minima in the common X_0, Y_0 boxes, with the same local increment Δ . The source's max-formula pair is no larger than these minima. Evaluating the later root target changes no point, label or local increment in this prefix block. This proves (14) for prefix edges. The completion only verifies a finite count identity; no clock at a virtual count is sampled or reused. These are exactly the admission and sampling rules in Section C.1.

For an ordinary suffix edge, X_0 is comparable to $1 + P$, where P is the entire preceding physical length, and Y_0 to $1 + B$, where B is the following positive-drift length. Let its positive reference prefix and negative reference tail be $A_{\text{mean}}, B_{\text{mean}}$, respectively. They satisfy

$$cP \leq A_{\text{mean}} \leq CP, \quad cB \leq B_{\text{mean}} \leq CB.$$

The exact actual terminal output and own-scale central side counts give for the weak block's actual increment

$$\Delta = B_{\text{mean}} - A_{\text{mean}} - x/r_m + O(\sqrt{1+P} + \sqrt{1+B}). \quad (\text{D.3.15})$$

The incoming strict exit is included in the $O(\sqrt{1+P})$ term. The actual root has $x \leq A \log(2 + V_{\text{pre}}) \leq A \log(2 + P)$. Take the fixed max-formula constant a_* sufficiently small compared with the positive constants in these two reference comparisons. Then

$$x^- = \max\{a_* X_0, a_* Y_0 - \Delta\} \leq C(1 + P), \quad y^- = \max\{a_* Y_0, a_* X_0 + \Delta\} \leq C(1 + B).$$

Indeed the negative linear $-cB$ pays the $\sqrt{B}$ error in the first maximum, and the negative linear $-cP$ pays the $\sqrt{P}$ error in the second. The remaining $\log(2+P)$ term is at most a fixed multiple of $1+P$; its sign in the second maximum is favorable. This proves (14) directly at the actual root count, without changing that count.

Equivalently, if the weak block is the reservoir, its actual count increase $k = n_*^x - n_*^0$ is nonnegative and $O(1+x)$. The max formula gives exactly $x_0^- \leq x_x^- \leq x_0^- + w_*k$ and $a_*Y_0 \leq y_x^- \leq y_0^-$. Since X_0 is comparable to $1+P$, the logarithmic root increase is absorbed in its left box. The direct argument (15) also covers the case where the reservoir is another suffix block. No incoming-exit or root reserve is added to Y_0 .

At a current endpoint with the terminal coordinate left, $S_0(v_m) \geq X_0$; at a right endpoint, $S_0(v_m) \geq Y_0$. Combine (13), (14), then use the exact coordinate marginal u . The root-separated global chord therefore satisfies

$$\boxed{\hat{L} + C \geq (c/C_P)[(1-u)x^- + uy^-]}. \quad (\text{D.3.16})$$

This is the actual source representative chord, including the true ordinary reservoir count. Its endpoint difference need not equal the root-separated global endpoint difference. That equality is not needed: (16) follows from endpoint domination and the same atomic centered remainder. This proves the claimed comparison.

D.3.6 Closing the weighted source-star estimate

We assemble the local comparisons while reserving positive shares for later use. Curvature pays synchronization spread; the allocated source walls pay inward depth; endpoint interpolation retains reference and good-count costs. A maximum-weight representative converts the average to a single endpoint, and the terminal flag transfer keeps the root coefficient one.

Let h^* be any of the actual fine comparison stars in Section D.2. Synchronize plateau cuts inside each original physical block, retaining all names. Let D_α, d_α be its spreads and mean depths, and $\Pi = \sum_\alpha (D_\alpha + d'_\alpha)$.

Atomic reference curvature is uniform: $g''_g(d) = -\kappa_g/(d+z) \leq -\kappa_*/H$. Between the first and last of a bundle's original cuts, its integer intermediate rank lies strictly between its before and after ranks. The curvature excess over the synchronous chord is therefore at least $\kappa_*D_\alpha/(2H)$. The actual same-word excess differs from that reference excess only on this spread interval. Named REG gives error at most

$$C_H \vartheta D_\alpha + C_K[1 + d_\alpha^\zeta + D_\alpha^\zeta].$$

Choose the regularity slope below a fixed fraction of the curvature constant and pay each D_α^ζ by another fraction of D_α , plus a constant. Mixed length is Lipschitz under synchronization, with loss at most $C_H \sum D_\alpha$; choose ϵ_{mix} below the remaining curvature fraction. This proves

$$F(h^*) \geq F(h^{\text{sync}}) + c_D \sum_\alpha D_\alpha - C_K[1 + \sum_\alpha d_\alpha^\zeta]. \quad (\text{D.3.17})$$

There is no reference error on a whole unaltered group. All constants are fixed before source construction.

In each physical block use the ordered-simplex endpoint weights. The deterministic mixed term is affine in that chamber, while each atomic same-word cost has its exact centered bridge. Consequently

$$F(h^{\text{sync}}) = a_m \mathcal{S} + \widehat{L} + \sum_{\alpha} \mathcal{R}_{\alpha}, \quad \mathcal{S} = \sum_v \omega_v s_m(v). \quad (\text{D.3.18})$$

Equation (7) gives, with $\widehat{E} = \widehat{L} + C_E$,

$$\widehat{E} \geq c_E \sum_v \omega_v J(v) \geq 0. \quad (\text{D.3.19})$$

Choose $\theta_{\text{glob}} \leq 1/(4H^2)$. For every weak proper coordinate, (12) and its actual contracted source wall give $\theta_{\text{glob}} \widehat{E} + R_{\alpha} \geq g_0 d_{\alpha}^{\gamma} - C$. On the linear branch of (12), REG pays the same inequality; its distance $\tau - a$ is at least d_{α} . Terminal weak coordinates use (16). Strong coordinates have a fixed reference slope gap: the expensive endpoint has reference charge at least cL , so (19) pays its distance from the cheaper endpoint, which is again at least d_{α} . Reference-paid weak coordinates have charge at least cL at both endpoints and are paid in the same way. In all linear cases, first choose ϑ small, then absorb d_{α}^{ζ} and the desired γ curve into a fixed fraction of the linear charge plus a constant. Thus simultaneously,

$$\theta_{\text{glob}} \widehat{E} + \mathcal{R}_{\alpha} \geq g_0 d_{\alpha}^{\gamma} - C_0. \quad (\text{D.3.20})$$

For proper and terminal weak coordinates choose the intrinsic source contraction no larger than their proved chord-domination constant times θ_{glob} . These choices are made before sampling the source. There are at most H^2 coordinates, so summing (20) in (18) leaves at least three quarters of $\widehat{E}$. Combine with (17), and pay its ζ powers by a small fraction of the retained γ powers:

$$F(h^*) \geq a_m \mathcal{S} + c_J \sum_v \omega_v J(v) + c'_D \sum_{\alpha} D_{\alpha} + c_{\gamma} \sum_{\alpha} d_{\alpha}^{\gamma} - C. \quad (\text{D.3.21})$$

The reference cost $B(h^*)$ may be retained as well. Concavity and the rank-integral identity for synchronization give

$$0 \leq B(h^*) - B(h^{\text{sync}}) \leq C_B \sum_{\alpha} D_{\alpha}, \quad B(h^{\text{sync}}) = \sum_v \omega_v B(v) \leq \sum_v \omega_v J(v). \quad (\text{D.3.22})$$

Both inequalities are pointwise: rank profiles change only on their spread intervals, and all reference integrands are dimensionally bounded. Splitting the positive budgets in (21) therefore retains $cB(h^*)$ together with positive J , spread and inward-depth charges.

Choose the maximum-weight vertex in each physical block. Its product $v^{\#}$ is lawful and has weight at least $\omega_0 = (H+1)^{-H}$. Hence $\sum_v \omega_v J(v) \geq \omega_0 J(v^{\#})$. It remains to move the root coefficient from $\mathcal{S}$ to $s_m(v^{\#})$. This costs arbitrarily small fixed fractions of the two existing budgets, plus a constant:

$$a_m |\mathcal{S} - s_m(v^{\#})| \leq \epsilon \sum_v \omega_v \overline{C}_m(v_m) + \epsilon \sum_{\alpha} d_{\alpha}^{\gamma} + C_{\epsilon}. \quad (\text{D.3.23})$$

Here is a direct proof. Only the physical block crossing $V_m/2$ can vary the terminal flag, unless an outside later terminal cut fixes every flag at zero. In that crossing block $[a, b]$, put $L = b - a$. The selected endpoint marginal is at least $1/(H+1)$, because the whole block vertex has at least that weight. If it selects the farther endpoint, its distance is at most H times the nearer distance. Thus $|S - s_m(v^{\#})| \leq Hd_{\alpha}/L$. If $L \geq V_m/4$, use $a_m \leq A \log(2 + V_m)$ and $d_{\alpha}/L \leq L^{-\gamma} d_{\alpha}^{\gamma}$;

the coefficient $AH \log(2 + 4L)L^{-\gamma}$ tends to zero. Bounded L costs a constant. If $L < V_m/4$, both endpoints of the crossing block lie in $(V_m/4, 3V_m/4)$. Every contributing terminal endpoint has last cut there, so (8) pays at least $c_{\text{ref}}V_m/4$ in each reference endpoint cost. This absorbs $A \log(2 + V_m)$ with any fixed positive fraction and a constant. These two cases prove (23).

Apply (23) to (21), with ϵ below the retained positive coefficients. Use (22) and the maximum-vertex weight. The conclusion is

$$\boxed{F_{\epsilon_{\text{mix}}}^0(h^*) \geq a_m s_m(v^\#) + c_*[J(v^\#) + B(h^*) + \Pi(h^*)] - C_*}. \quad (\text{D.3.24})$$

All coefficients remain dimension-only positive constants. The root has coefficient one. Equation (24) is (D.1.33), proved from the actual endpoint count support and the actual source walls, with the partial-child and terminal comparisons supplied by (12), (16).

Together with the endpoint-forest and original-pivot proof, this establishes the source cost used by the target recoloring and posterior sections.

D.4 Recoloring and weighted history costs

The source geometry must hold for every positive target word produced by recoloring. We compare one accepted source with one of its images on the same uncolored environment and root count. Exact atomic increments telescope at regular endpoint banks; exceptional signatures and movement strips pay their logarithmic discrepancies from already retained geometric charges. Spare regularity margins then extend the estimate to arbitrary target histories.

D.4.1 Actual source/image pair and deterministic geometry

Fix one full uncolored environment G , the physical ordered KKT profile, and the predetermined environmental targets $y_j \geq 0$. There are at most $H - 1$ nonterminal plateaux. Retain one actual accepted source word ω_0 ; its actual root count is the source construction's count at

$$x = \sum_j r_j y_j, \quad r_j = \sigma_j / \sigma_1, \quad a_m = x / r_m. \quad (\text{D.4.1})$$

Let ω_y be any permitted image of this word under the actual uniform-subset recoloring kernel. Each nonterminal plateau j has one macroscopic original local bank in a block of length P_j with

$$P_j \geq T_j / H, \quad \min_{t \text{ in bank}} d_E(t) \geq \rho P_j \geq \rho T_j / H, \quad \rho > 0. \quad (\text{D.4.2})$$

Every bank is a union of complete cells. A fixed cell mesh changes the right side by only a dimension-only additive constant, absorbed by the fixed plateau threshold. Banks for distinct plateaux belong to distinct original physical blocks. In bank j , recolor e_j copies of the immediate child's first shallow fine name c_j to the current block's eligible shallow exterior name t_j . Set

$$\begin{aligned} A_j &= \log S_j(\text{first}), & b_j &= u_j A_{j+1}, & e_0 &= 0, \\ b_j e_j - A_j e_{j-1} &= y_j + \delta_j, & 0 &\leq \delta_j < b_j, \\ e_j &\leq A_e \log(2 + T_j). \end{aligned} \quad (\text{D.4.3})$$

These are the actual integer transfers; the root and every uncolored position remain fixed. All effective alphabets have size at most H , so $A_j, b_j \leq \log H$.

Fix any original history h . Its lawful projections, star comparisons, terminal spine and clipping, synchronization, and maximum-weight endpoint representative depend on names, cuts and

deadlines only. They therefore produce the same h^* , v , D_{gap} , D_* , Π , reference B , and geometric $J(v)$ on this source/image pair. The labels used to evaluate their information costs do change. This distinction is essential.

Write

$$F_\epsilon^q(h^*) = \sum_l K_l^q(h_l^*) + \epsilon \sum_l S_{\text{mix},l}(h^*), \quad q = 0, y. \quad (\text{D.4.4})$$

The mixed lengths in (4) are geometric and do not change with the word. The weighted source-star inequality proved in Section D.3, equation (24), is

$$F_\epsilon^0(h^*) \geq a_m s_m(v) + c_*[J(v) + B(h^*) + \Pi(h^*)] - C_*. \quad (\text{D.4.5})$$

D.4.2 Exact atomic changes, including the old target lists

A transfer changes only the current plateau and its immediate child. The child's incoming list loses the recolored original slot, so its information is evaluated with the changed membership indicator. Subtracting these exact information increments from the output increments gives the three regular endpoint coefficients used in the target telescope.

Changing one c_j to t_j changes the atomic outputs exactly by

$$\Delta Z_j = b_j, \quad \Delta Z_{j+1} = -A_{j+1}, \quad \Delta Z_l = 0 \quad (l \neq j, j+1). \quad (\text{D.4.6})$$

At an ancestor plateau all its exterior names have expired on this local bank; both changed labels belong to its only eligible compound child. Its atomic information increment is zero. At a deeper plateau $l > j+1$, both labels are outside its incoming fine alphabet, so neither contributes to its atomic information. The immediate child's *old list loses this actual original slot*. It is not identified with its source old list. Every atomic information is evaluated as a sum over original slots with its fine-membership indicator; the indicator is zero when that slot does not route to that plateau. Thus (6) is valid on the changed lists.

For the fixed endpoint history v , atomic state full means all eligible exterior names remain in the compound-child cell; zero means none remains; other states are proper. All states are constant on the original bank block. Directly subtracting ΔI_l from (6) gives the following per-flip increments in $\sum_l K_l$:

$$\begin{array}{l|l} \text{parent full, immediate child full} & 0 \\ \text{parent zero, immediate child full} & b_j \\ \text{parent zero, immediate child zero} & b_j - A_{j+1}. \end{array} \quad (\text{D.4.7})$$

For example, in a full parent cell replacing its compound-child mark by its exterior mark increases residual information by $\log z = b_j$, canceling ΔZ_j . In a full immediate-child cell, removing its first shallow name decreases information by A_{j+1} , canceling ΔZ_{j+1} . In a zero cell its information is identically zero. This proves every entry in (7). Deeper fine splits do not invalidate these atomic calculations; the changed shallow name lies outside every deeper plateau. The deterministic mixed term in (4) has zero increment.

Every atomic residual-information coefficient is between zero and $\log H$. Only plateaux $j, j+1$ can change at this bank. Consequently any endpoint per-flip increment is bounded in absolute value by $4 \log H$, and its discrepancy from the ideal $b_j s_j - A_{j+1} s_{j+1}$ by $6 \log H$.

D.4.3 Geometric charges for exceptional endpoint intervals

The ideal transfer coefficient is determined by the two early flags. When an endpoint bank's signatures disagree with that coefficient, a proper, mixed, early-full, or late-zero charge pays at least its square-root length scale. Since only logarithmically many slots are recolored, a small fixed share of that charge pays the entire exceptional bank.

Here s_j is the plateau's actual early/late endpoint flag, exactly as in Section D.3. No monotonicity of the sequence of flags is imposed. Unless the states and flags match one of

$$\begin{array}{l|l} \text{parent full, child full} & s_j = s_{j+1} = 0 \\ \text{parent zero, child full} & s_j = 1, s_{j+1} = 0 \\ \text{parent zero, child zero} & s_j = s_{j+1} = 1, \end{array} \quad (\text{D.4.8})$$

the endpoint has

$$J(v) \geq c_{\text{ex}} \sqrt{T_j} - C_{\text{ex}}, \quad c_{\text{ex}} > 0. \quad (\text{D.4.9})$$

For completeness, these are all the cases.

- If either relevant atomic state is proper, it is either bad (retaining its eligible shallow name) or good (omitting it). A bad row has reference cost at least $\kappa_* P_j / (2H^2)$ by exchanging the shallow retained name and a deeper missing name. The two comparison histories are lawful; their discrete second difference is at least $\kappa_* P_j / H$, and reference nonnegativity pays half that amount. A good row contributes $\sqrt{P_j}$ to S_{good} . This includes the immediate child's earlier group; its eligible shallow name is precisely c_j .
- A full parent with a nonfull immediate child is mixed on the entire original bank block: a retained parent exterior lies in the fine anchor cell while an immediate-child exterior is a singleton. The length charge is P_j .
- If a full parent is early, its last cut reaches the bank block; its early prefix has length at least P_j . The clipped-prefix reference lower bound pays that length. The same argument applies to an early full immediate child, whose last cut also reaches this old bank block.
- If a zero parent is late, every later eligible parent name is already cut. Its zero tail contains the original bank block and pays P_j .
- If a zero immediate child is late, all its exterior names have been cut by this old bank block. Its zero tail contains its entire local total T_{j+1} . The plateau hierarchy gives $T_{j+1} \geq V_j / C_{\text{hier}} \geq T_j / C_{\text{hier}}$, so this tail also pays at the current bank's scale.

The clipped-prefix constant c_{ref} and hierarchy constant are explicit in Section D.3. For large supported banks one may, for example, take a smaller constant than

$$\min\{\kappa_*/(2H^3), H^{-1/2}, H^{-1}, c_{\text{ref}}/H, c_{\text{ref}}/C_{\text{hier}}\}$$

in (9); bounded supported conventions only change C_{ex} . This establishes (9) on original source reference geometry, independently of which recoloring subset was chosen.

On banks satisfying (8), (7) is exactly the ideal coefficient. On all other banks, use the uniform coefficient bound and (9). Because there are at most $H - 1$ banks and $\log(2 + T) \leq \epsilon\sqrt{T} + C_\epsilon$, for every fixed $\alpha > 0$,

$$\Delta F_\epsilon(v) \geq \sum_{j < m} e_j(b_j s_j - A_{j+1} s_{j+1}) - \alpha J(v) - C_\alpha. \quad (\text{D.4.10})$$

Sharing the single J budget among the finite exceptional banks only changes the choice of ϵ ; it creates no length-ratio or product-gap condition.

D.4.4 Moving from the endpoint to the actual comparison star

The change in residual output is common to the comparison star and its endpoint, so it cancels in their difference. Only flipped slots inside a movement strip contribute. Reaching an interior bank forces a macroscopic movement or depth, whose positive-power penalty absorbs the logarithmic number of flips.

The residual output increment cancels when comparing the changes at h^* and at its representative v :

$$\Delta F_\epsilon(h^*) - \Delta F_\epsilon(v) = - \sum_l [\Delta I_l(h_l^*) - \Delta I_l(v_l)]. \quad (\text{D.4.11})$$

A flipped slot contributes to the right side only if it belongs to a movement strip between an original comparison-star cut and its chosen endpoint. Outside those strips the two relevant eligible partitions are identical, so their mark coefficients agree exactly. Every per-flip contribution to (11) has absolute value at most $4 \log H$.

For a synchronized group α , maximum-weight endpoint selection gives

$$|t_r^* - v_r| \leq D_\alpha + (H + 1)d_\alpha. \quad (\text{D.4.12})$$

Indeed its chosen side has marginal at least $1/(H + 1)$; when it is the farther side its distance is at most H times the nearer one. If a movement strip meets bank j , (2), (12) force either $D_\alpha \geq cT_j$ or $d_\alpha \geq cT_j$ for one of the bounded number of groups in that physical block. Therefore

$$\Pi(h^*) \geq c'T_j^\gamma - C'.$$

The corresponding $O(\log(2 + T_j))$ flips can be paid by any fixed positive fraction of this charge, plus a constant. Sum over the finite banks, as in the previous section. For every $\alpha > 0$,

$$|\Delta F_\epsilon(h^*) - \Delta F_\epsilon(v)| \leq \alpha \Pi(h^*) + C_\alpha. \quad (\text{D.4.13})$$

This proves (D.1.37) using the original source and image's atomic membership indicators. It remains valid when a flip removes an original slot from a later old list. Nothing asserts equality or monotonicity of those old lists.

D.4.5 Exact cascade and the weighted target-star inequality

After the exceptional-bank and movement-strip errors have been paid, the regular transfer coefficients telescope. The terminal contribution combines with the already selected root to produce the terminal target. Thus every early target remains with coefficient one while positive shares of all geometric budgets survive.

The finite coefficients telescope exactly:

$$\sum_{j < m} e_j(b_j s_j - A_{j+1} s_{j+1}) = \sum_{j < m} s_j(y_j + \delta_j) - s_m A_m e_{m-1}. \quad (\text{D.4.14})$$

Since $r_j b_j = r_{j+1} A_{j+1}$, multiplying (3) by r_j and summing gives

$$A_m e_{m-1} = \sum_{j < m} (r_j / r_m) (y_j + \delta_j), \quad a_m - A_m e_{m-1} = y_m - \sum_{j < m} (r_j / r_m) \delta_j. \quad (\text{D.4.15})$$

The last sum is bounded by a dimension-only constant: products lie in $(1/2, 1)$, so $r_j / r_m \leq 2$, and $\delta_j < \log H$. No gap or length enters this bound. Combine (5), (10), (13), (14), and (15). The proper terms $s_j \delta_j$ are nonnegative. Choose $\alpha < c_*/4$ for each of the two error allocations. A fixed part of every geometric budget remains, and

$$\boxed{F_\epsilon^y(h^*) \geq \sum_{j: s_j(v)=1} y_j + c_y [J(v) + B(h^*) + \Pi(h^*)] - C_y}, \quad c_y > 0. \quad (\text{D.4.16})$$

This is the target weighted-star estimate on the actual source/image pair. Each target has coefficient one. The terminal root is the actual fixed root of that pair; (15) does not compare different root tuples or different local minima.

D.4.6 Image regularity and spare margins

The information estimates for the image require named regularity about the same original reference intensities. Interior recoloring changes an interval count by at most a logarithmic amount. Every affected interval has either macroscopic length or a deep endpoint, so fixed spare slope and depth margins pay this change uniformly. The order of constants below makes this choice independent of the sampled word.

Let an original bank block have length P and its bank have depth at least ρP . Every interval $[s, t]$ in that block which meets the bank either has length at least $\rho P/2$, or an endpoint with depth at least $\rho P/2$. Otherwise an interval shorter than $\rho P/2$ with both endpoints within $\rho P/2$ of physical endpoints cannot meet that interior bank.

For each individual fine name the count change on such an interval is at most $e \leq A_e \log(2+T)$. Choose fixed target REG parameters ϑ_t, K_t with spare margins over source parameters ϑ_s, K_s . Because $P \geq T/H$, one fixed length threshold ensures

$$A_e \log(2+T) \leq (\vartheta_t - \vartheta_s) \rho P/2, \quad A_e \log(2+T) \leq (K_t - K_s) (\rho P/2)^\zeta. \quad (\text{D.4.17})$$

Use the first inequality in the long-interval case, the second in the deep-endpoint case. Intervals missing the bank have no change. Thus the image retains the target named regularity about the *same unconditional original fine intensities*, with fixed margins. If REG is centered at an empirical type, its rate changes by at most e/P ; the resulting interval error is also at most e and is paid by doubling the fixed allowance in (17).

Every routed atomic named count, including a count on a changed old list, is a sum of at most H original fine-name indicators. Summing their REG inequalities supplies atomic REG about the corresponding original routed reference intensities. The same calculation applies to each original physical group; it does not regard the new old list as a clock sampled from the old source conditional law.

These constants can be fixed without a circular dependence. First fix the compact profile, source count-box parameters, width floors and the envelope floor exponent. The floors determine the entire logarithmic target/transfer cap A_e . Fix the positive reference, mixed and curvature coefficients, global wall allocation and small source contractions. Choose the target linear REG slope below the geometric mean-gain thresholds and the source slope smaller again. Choose

source wall and refinement constants, including a finite K_s ; then choose K_t larger by a fixed margin, allowing the factor H for routed names. The larger depth-intercept constant only enlarges additive constants in the geometric absorption inequalities: it does not change their positive linear or inward coefficients. Finally enlarge the fixed plateau-length threshold to satisfy (17) and all count/bridge thresholds. The envelope's small depth coefficient is chosen below the already fixed positive history charge; its floor stays fixed. Construct the source and the image only after all these constants have been fixed.

Hence requiring image REG does not ask the source to meet a smaller intercept after it has been sampled, and does not change any physical clock, fine prior, local point or root count.

D.4.7 Arbitrary original target histories and classes

The weighted target-star estimate now enters the canonical reduction for every original history. Image regularity supplies the same branching and mixed gains as on the source. Token summability handles all original pivots. Since every positive image density has a retained preimage, the estimate holds throughout the transported support; the probability of its changed class is evaluated separately.

Apply the deterministic canonical reduction in Section D.2 to the target word. The same original history gives the same projection/spine geometry. Its actual cost is the exact sum of target atomic costs plus projection delay and target per-slot mixed contrasts. Image REG, proved above, supplies the proper pure-cell gain, terminal off-spine gain and mixed gain about the original reference profile. The same terminal floating reference budget pays its anchor-comparison error.

Insert (16) in that reduction and pay the ζ -depth errors by positive γ inward charges and true delays. One obtains

$$C_y(h) = R_y - I_h(\omega_y) \geq \sum_{j:s_j(v)=1} y_j + c\Lambda(h) - C, \quad \Lambda = J(v) + B(h^*) + \Pi(h^*) + D_{\text{gap}} + D_* + S_{\text{pure}} + S_{\text{off}}. \quad (\text{D.4.18})$$

Theorem D.5 gives $\sum_h \exp(-\delta\Lambda(h)) \leq C_\delta$ for every fixed $\delta > 0$. This proves (D.1.36), including original-pivot summability, on every retained source/image pair.

The target density is the scaled push-forward of the accepted source measure. Positive target density therefore has at least one retained source preimage. The preceding argument works for each such preimage and every original history simultaneously; all geometry and targets are fixed by G and the history. Thus it holds for every target word of positive density.

The target class $Q_h(\omega_y)$ uses the original eligible per-slot signatures containing the target label. Recoloring can change those signatures, so its conditional mass is evaluated anew under the original label law. Section D.5 provides that disintegration; equation (18) supplies its arithmetic cost input.

D.5 Conditional label laws and comparison sums

We finish with the finite probability argument used by the geometric row. Complete-type conditioning controls the path events under uniform residual multisets. A second disintegration coarsens the free count banks and uses the diagonal binomial exit variables. These are successive estimates under one original law, so a count bank and a path block may belong to the same physical block.

D.5.1 Finite statement

The hypotheses distinguish three kinds of data: the fixed uncolored environment, the labels revealed by the comparison class, and the unread labels in free banks. Path envelopes are uniform after full types and exact revealed assignments have been fixed. Exit bounds instead use independent pair splits after coarsening; their centers may depend on all the coarsened data.

Fix an uncolored environment G : it specifies the original physical blocks, every slot's position and eligible alphabet, and all deterministic targets. In original block i , let its n_i labels be independent categorical variables with probabilities $p_{i,c} \geq p_* > 0$; labels in different blocks are independent. Write π_G for this finite product law and K_i for the complete named count vector of block i .

A comparison class Q is a positive-probability product of per-slot signature restrictions. In each free path block i , these restrictions read only an endpoint-strip slot set R_i ; every other slot in that block has the full eligible alphabet. Paid blocks may be restricted arbitrarily.

There are finitely many disjoint count banks B_j , belonging to different original blocks, such that Q imposes no restriction on any slot of B_j . Each has two designated fine names s_j, t_j , whose original probabilities are bounded below by p_* . Its uncolored number of slots n_{B_j} satisfies

$$n_{B_j} \geq c_* T_j, \quad T_j \geq T_* > 0. \quad (\text{D.5.1})$$

Let η_j be a function of the complete named count vectors only. Condition on every label outside all designated pairs in their banks, and on the pair-coarsened labels inside those banks. If M_j is the number of slots coarsened to $\{s_j, t_j\}$ and L_j the number of their t_j labels, suppose the exact identity is

$$\eta_j = h_j + \beta_j L_j, \quad 0 < \beta_- \leq \beta_j \leq \beta_+, \quad (\text{D.5.2})$$

where every h_j is measurable in the coarsened σ -field and depends on none of the unread split variables L_l . Allowed exit bands are $\eta_j \in I_j$, where every I_j has length at most $C_R R_j$, $R_j \geq 1$. Its location may depend arbitrarily on G and the coarsened data, provided its defining restriction is measurable in the complete type K and the fixed environment. These intervals are the actual fixed-target carry bands, so this last condition holds.

An arbitrary additional type-only admission set T_0 may be imposed; in particular it may contain the original central fine-count windows. In each free path block i , let E_i be a local necessary path event. Its deterministic parameters may depend on the whole complete type K and all exactly revealed strip labels. Suppose that for every feasible such type and strip assignment in T_0 and the exit bands,

$$\mathbb{P}_{\pi_G}(E_i \mid K, \text{all exactly revealed labels}) \leq q_i(G), \quad (\text{D.5.3})$$

where $q_i(G)$ is the same nonnegative envelope for every type and revealed assignment. Local path events concern only the remaining labels of their own block after these data are fixed. No factorization of any accepting density is assumed.

Theorem. For any acceptance $0 \leq F \leq 1$ supported on T_0 , all the exit bands and all the necessary events E_i ,

$$\mathbb{E}_{\pi_G}[F \mid Q] \leq C \prod_j \min \left(1, \frac{R_j}{\sqrt{1 + T_j}} \right) \prod_i q_i(G). \quad (\text{D.5.4})$$

The constant depends only on the fixed number of banks and blocks, $p_*, c_*, \beta_-, \beta_+, C_R$. Free count banks and free path blocks may belong to the same original block. The proof does not assume that their accepted events are independent.

D.5.2 Proof

First bound the path probability at each complete type and sum that type law inside the original class. This leaves the conditional probability of the count admission. Then reveal the data outside the free pairs, use maximum binomial atoms for the exit bands, and average all pair totals. Neither stage factors the accepting density.

First condition on all exact labels in the strip sets read by Q and on all exact labels in paid blocks. Denote this revealed data by A . The class restriction Q becomes fixed. Next fix the whole type $K = (K_i)_i$. By the original product categorical law, the unrevealed labels in each block are a uniform permutation of the residual multiset of that block's type, and these residual permutations are independent across blocks. This follows directly by writing the joint probability of an assignment: its product probability is constant over permutations of a fixed type; the type and revealed constraints factor over blocks once their values are fixed.

The type-only event C comprising the exit bands and T_0 is fixed by K and G . Thus (3), dropping all extra accepting restrictions, gives

$$\mathbb{E}_{\pi_G}[F \mid K, A] \leq \mathbf{1}_C(K) \prod_i q_i(G). \quad (\text{D.5.5})$$

Summing the original conditional type law and then the revealed assignments compatible with Q gives exactly

$$\mathbb{E}_{\pi_G}[F \mid Q] \leq \prod_i q_i(G) \mathbb{P}_{\pi_G}(C \mid Q). \quad (\text{D.5.6})$$

This finite disintegration uses independence only after the complete type and exact revealed labels have been fixed. Summing them afterwards retains the original coupled posterior law.

For the remaining count probability, drop T_0 , reveal the labels outside the count banks, and coarsen each designated pair. Let S be this σ -field. Since Q reads no bank slot, it is S -measurable. Conditional on S , the unread splits have independent laws

$$L_j \mid S \sim \text{Bin}(M_j, \theta_j), \quad \theta_j = \frac{p_{j,t_j}}{p_{j,s_j} + p_{j,t_j}}. \quad (\text{D.5.7})$$

Both θ_j and its complement are bounded below. By (2), each exit band admits at most $C(1 + R_j)$ integer split counts, uniformly in every band center and every other coarsened datum. The elementary maximum-atom bound for a binomial with compact probability parameter gives

$$\mathbb{P}(C \mid S) \leq \prod_j \min \left(1, \frac{C(1 + R_j)}{\sqrt{1 + M_j}} \right). \quad (\text{D.5.8})$$

Conditional on Q , the labels in the fully free banks still have their original product laws, independently for different banks. Consequently the variables M_j remain independent binomials with parameters

$$(n_{B_j}, p_{j,s_j} + p_{j,t_j}).$$

The maximum-atom bound (8) is uniform in the centers h_j from (2), so their dependence on outside labels no longer enters. A Chernoff bound below half the pair-total mean gives

$$\mathbb{E}(1 + M_j)^{-1/2} \leq \frac{1}{\sqrt{1 + \mathbb{E}M_j/2}} + \mathbb{P}(M_j < \mathbb{E}M_j/2) \leq \frac{C}{\sqrt{1 + n_{B_j}}}. \quad (\text{D.5.9})$$

Since $R_j \geq 1$, (1), (8), (9), and independence of the pair totals imply

$$\mathbb{P}(C \mid Q) \leq C \prod_j \min \left(1, \frac{R_j}{\sqrt{1 + T_j}} \right). \quad (\text{D.5.10})$$

Combining (6) and (10) proves (4). The maximum-atom bound itself can be proved without a local limit theorem: Fourier inversion bounds its mass by $(2\pi)^{-1} \int_{-\pi}^{\pi} |1 - \theta + \theta \exp(iu)|^m du$, whose integrand is at most $\exp(-cmu^2)$ for compact θ ; the integral is $O((1 + m)^{-1/2})$. $\square$

D.5.3 Diagonal exit variables

The finite theorem requires a simultaneous diagonal identity, not just a recursive description of the carries. The atomic transfer coefficients provide it: a flip changes its own carry, while its two neighboring output contributions cancel in every later carry. Disjoint original banks then leave all split variables unread.

For the plateau products $\sigma_1 > \dots > \sigma_m$, let $u_j = \sigma_{j+1}/\sigma_j$,

$$\eta_l = \sum_{a \leq l} \frac{\sigma_a}{\sigma_l} (Z_a - y_a).$$

In the bank of plateau j , changing its immediate child's first shallow fine label c_j into its current shallow exterior label t_j changes the grouped outputs by

$$\Delta Z_j = b_j = u_j A_{j+1}, \quad \Delta Z_{j+1} = -A_{j+1}, \quad \Delta Z_a = 0 \quad (a \neq j, j+1).$$

For $l < j$, this changes no term of η_l . For $l = j$, it changes η_j by b_j . For $l > j$, the change is exactly

$$\frac{\sigma_j}{\sigma_l} b_j - \frac{\sigma_{j+1}}{\sigma_l} A_{j+1} = 0.$$

Hence every unread bank split changes exactly its own exit. Because the original banks are disjoint, repeated single-slot changes prove (2) simultaneously, with all h_j independent of every unread split. This is the actual cancellation at Section A.2.8, proved by the exact identity. It is not merely triangular independence or a change to a selected-word law.

D.5.4 Direct construction of the path envelope

The envelope must bound every feasible remaining type after the class has exposed its endpoint strips. One joint exact-subset density comparison handles both free intervals. Forward and reverse wall monotonicities allow opposite Bernoulli parameters, chosen to cover all types and permitted inverse transfers before the target is read.

The uniformity required in (3) follows from the two-free-interval argument (Section C.1), as follows.

Fix the complete type and exact revealed strip labels in one binary projection. With n total labels, k binary successes, m revealed labels and e revealed successes, the unread word is a uniform $k - e$ subset of $N = n - m$ slots. Its success proportion is

$$p' = (k - e)/(n - m).$$

The actual endpoint identity $y - x = cP - an + bk$, with the allowed endpoint box and strip occupancy bound, gives a deterministic interval $p_- \leq p' \leq p_+$ whose endpoints depend only on the uncolored environment, the allowed endpoint-box scales and strip width. If $M \leq N/2$ slots are retained jointly in two disjoint free end intervals, the exact subset law has the following density relative to iid Bernoulli p' on those slots:

$$\frac{\mathbb{P}\{\text{Bin}(N - M, p') = Np' - j\}}{\mathbb{P}\{\text{Bin}(N, p') = Np'\}} \leq C. \quad (\text{D.5.11})$$

The denominator is the atom at the integer mean of a compact binomial law and is bounded below by $c(1 + N)^{-1/2}$ by central Stirling; the numerator is at most $C(1 + N - M)^{-1/2}$. This is one joint density for the two intervals, including every allowed type and revealed assignment.

The necessary left signed wall is increasing in binary successes, and the necessary reversed right wall is decreasing in them. Under the iid reference the two interval restrictions are independent. Increasing the first success parameter to p_+ and decreasing the second to p_- therefore bounds the joint event by the product of the two corresponding iid wall probabilities. The density constant in (11) then proves (3) with one envelope valid for every type in the subsequent sum. Dropping further refinements, source regularity, and accepting restrictions only enlarges this necessary-event probability.

For transported source words, the permitted inverse recolorings are bounded by the predeclared uncolored range $E_{\max} = O(\log(2 + T))$. Enlarge $p_+ - p_-$ and the signed speed to cover that entire fixed range before defining the envelopes. Thus they remain uniform in the inverse type and do not need to be independent of the final accepting density. This proves the finite conditional inequality; the separate moment evaluation of these envelopes still uses scalar persistence estimates.

D.5.5 The uncolored bank-size condition

The maximum-atom factors above are averaged over all original pair totals, including zero. The bank-size condition concerns the uncolored bank alone and is supplied by named regularity.

The required uncolored bank-size bound (1) follows on any nonempty accepting support from its named regularity inequality: summing over the finite alphabet gives an uncolored lower count on a fixed interior interval of length comparable to T_j . Choose the REG linear slope smaller than one quarter of the compact lower uncolored intensity. The endpoint-depth error is $O(1 + T_j^\zeta)$ with $\zeta < 1$; a fixed sufficiently large length threshold absorbs it into half that interval's linear expected count. If this deterministic uncolored condition fails, the accepting density vanishes on that environment, so its posterior row is vacuous.

D.5.6 Normalization of the relative comparison sum

The endpoint cost pays early plateaux, and the conditional class estimate pays late plateaux. Multiplying the two inequalities includes every uncolored plateau cost once. A bounded retained-word likelihood then gives the relative row; the exact carry identity supplies the last normalization needed for the arithmetic counting row.

In the geometric application of the finite lemma below, the classes Q_h are the complete-cell enlargements of Section 9.1. Every information I_h , cost $C_y(h)$, class probability and relative posterior uses that same class; the original arithmetic multiplier and original index h are retained. After (16), the arithmetic row follows from nonnegative class inclusion. No separate comparison of the two information costs or the two conditional posteriors is required.

The source and target geometric estimates and the finite label bound now combine by multiplication. The following finite lemma records the resulting row estimate for Section D.1.

Fix an accepted anchor ω in the same finite environment. For each original history h , let its geometric representative assign early/late flags $s_j(h)$ and a nonnegative charge $\Lambda(h)$. Set $D_j = \exp(-y_j)$, $y_j \geq 0$, and $D_{\text{all}} = \prod_j D_j$. Suppose, with constants independent of the histories,

$$C_y(h) \geq \sum_{j:s_j(h)=1} y_j + c\Lambda(h) - C_0, \quad \mathbb{E}_\pi[F \mid Q_h(\omega)] \leq C_1 e^{c\Lambda(h)/2} \prod_{j:s_j(h)=0} D_j, \quad (\text{D.5.12})$$

and suppose the likelihood ratio between any two accepted words in this environment is at most C_R . Then the relative posterior

$$q_h^{\text{rel}} = \mathbb{E}_\pi \left[F(\omega') \frac{\pi(\omega)}{\pi(\omega')} \mid Q_h(\omega) \right] \quad (\text{D.5.13})$$

satisfies, pointwise in the anchor and every original history,

$$e^{-C_y(h)} q_h^{\text{rel}} \leq e^{C_0} C_1 C_R D_{\text{all}} e^{-c\Lambda(h)/2}. \quad (\text{D.5.14})$$

Indeed, the likelihood ratio is only needed where $F(\omega') > 0$, so $q_h^{\text{rel}} \leq C_R \mathbb{E}[F \mid Q_h]$. Multiplying the two inequalities in (12) then includes each plateau cost exactly once, since its flag is either zero or one. This proves (14). For any fixed $\delta > 0$, if the original history sum satisfies

$$\sum_{h \text{ original}} e^{-\delta c\Lambda(h)/2} \leq A_\delta, \quad (\text{D.5.15})$$

raising (14) to the power δ and summing gives the asserted full relative row, with constant $A_\delta (\exp(C_0) C_1 C_R)^\delta$. All type summations have already been performed in the proof of (4); no fractional power is moved through a conditional expectation here.

For completeness, the subsequent passage to the unnormalized finite counting row has one additional exact factor that should be kept visible. With the actual arithmetic history multiplier Φ_h and its information

$$I_h = \log \Phi_h + \log \frac{\pi(Q_h(\omega))}{\pi(\omega)}, \quad C_y(h) = R_y - I_h,$$

the finite class identity gives

$$\Phi_h M_h(F; \omega) = e^{I_h} q_h^{\text{rel}} = e^{R_y} e^{-C_y(h)} q_h^{\text{rel}}. \quad (\text{D.5.16})$$

Consequently the counting row is bounded by a constant times $(\exp(R_y) D_{\text{all}})^\delta$, rather than directly by D_{all}^δ . The exact carry identity is

$$R_y = \sum_j y_j + \epsilon/r_m - \sum_{j < m} (u_j^{-1} - 1) \eta_j^y, \quad -C \leq \epsilon \leq 0, \quad \eta_j^y \geq 0. \quad (\text{D.5.17})$$

Thus $\exp(R_y)D_{\text{all}} \leq 1$; the extra factor in (16) is harmless. The same identity bounds the full likelihood ratio: its only word-dependent contribution in a fixed environment is the exit sum, which lies in a fixed compact interval by strict exit support. This gives the constant C_R in (13). The inputs (12) and (15) follow from the source/target comparisons in Section D.2 and the original-pivot sum in Theorem D.5. For the arithmetic classes, replace M_h on the left of (16) by the smaller sum over $Q_h^{\text{arith}}(\omega)$. The same unnormalized row bound follows, with every original positive pivot, zero pivot and within-cell multiplicity intact.

E Prime grids and harmonic mass

The probabilistic lower construction produces weighted label words on complete cells. To convert their mass into divisor-box volume, we need two arithmetic facts: enough squarefree prime selections at every accepted occupancy, and a bound for the overlap of their boxes. The first is a cellwise distinct-draw estimate. The second reduces collisions to a finite family of triangular pivot constraints and keeps the correlations of the label weight intact.

We prove these estimates and then carry the accepted family to the prime grid at the fixed physical optimizer. The later subsections assemble the transfer in the notation of the original harmonic sum and give numerical choices for the grid constants.

E.1 Prime supply, collisions, and the arithmetic grid

Actual prime values are assigned only after the accepting density is defined on complete cell words. This order lets us first prove an arithmetic collision bound for any admissible weight. We then verify squarefree supply on its supported occupancies and use Hölder’s inequality to recover one first power of the selected mass. The remaining work is to compare the grid construction with the physical profile, including its count bands and ordinary root.

E.1.1 A fractional collision kernel for correlated label weights

Two divisor boxes can meet only if their coordinate logarithms are close. For a fixed pair of labels, successive maximal pivots turn these closeness conditions into a triangular system in at most one prime per coordinate. Integrating those primes produces a history factor. The remaining label contribution is a power sum over history classes.

The accepting weight may couple different slots, so this class power sum must be evaluated for the whole weight. The triangularization is related to the unweighted argument of [19, Section 7.1]; the finite history construction below gives the weighted form needed here.

Cells, slots, and the arithmetic moment We separate the choice of prime cells from the label weight. Named slots record the occupancy of each cell; permutation invariance will later recover the exact factorial multiplicity of an arithmetic tuple.

Fix $k \geq 1$, put $b = \log 2$, and set

$$\eta_i = \frac{\log(k - i + 2)}{k - i + 1}, \quad \rho_i = e^{\eta_i}, \quad y_0 = 3 \leq y_1 \leq \cdots \leq y_k.$$

Use pairwise disjoint prime cells $D_{i,t} = (\lambda_{i,t-1}, \lambda_{i,t}] \cap \mathbb{P}$ for $1 \leq t \leq v_i$, satisfying

$$\lambda_{i,t-1} \geq 3, \quad \sum_{p \in D_{i,t}} p^{-1} \leq \eta_i,$$

$$\log \lambda_{i,t-1} \geq c_* \log y_{i-1} \rho_i^t, \quad \left| \log \frac{\log y_i}{\log y_{i-1}} - \eta_i v_i \right| \leq A_*. \quad (\text{E.1.1})$$

The constants $c_* > 0, A_* < \infty$ are fixed. The elementary greedy cells described below have $c_*^{-1}, A_* \ll_k 1$. Empty blocks with $v_i = 0$ are allowed; the last inequality still applies to them.

Fix integers $g_{i,t} \geq 0$, and define

$$r_i = \sum_t g_{i,t}, \quad R_i = \sum_{h \leq i} r_h, \quad R_0 = 0, \quad G_g = \prod_{i,t} g_{i,t}!, \quad M_g = \prod_i \eta_i^{r_i}.$$

The slots in block i are $\mathcal{B}_i = (R_{i-1}, R_i] \cap \mathbb{Z}$. Assign its first $g_{i,1}$ slots to cell 1, its next $g_{i,2}$ to cell 2, and so on. Write $E_g(q)$ for the assigned cell of a positive slot q , and set $E_g(0) = 0$. No ordering of prime values inside one cell is imposed.

A label is a map Y assigning to a slot $q \in \mathcal{B}_i$ one color in $C_q = \{0, i, i+1, \dots, k\}$. Equivalently,

$$\mathcal{P}_{\mathbf{r}} = \{(Y_1, \dots, Y_k) : Y_j \subseteq \{1, \dots, R_j\}, \quad Y_j \cap Y_h = \emptyset \ (j \neq h)\},$$

where unused slots have color zero. Let $F : \mathcal{P}_{\mathbf{r}} \rightarrow [0, \infty)$ be finite, independent of all prime values, and invariant under every permutation of slots within each assigned cell. No product structure is assumed for F . It may depend on the entire occupancy vector g .

Let $\mathcal{A}(g)$ consist of squarefree tuples $\mathbf{a} = (a_1, \dots, a_k)$ having $g_{i,t}$ prime factors from each $D_{i,t}$, with all prime factors coming from these cells. Choose any named-slot representation $\mathbf{p}$ of such a tuple. For labels Y, Z , put

$$Y \sim_{\mathbf{p}} Z \iff \left| \sum_{q \in Y_j} \log p_q - \sum_{q \in Z_j} \log p_q \right| < b \quad (1 \leq j \leq k).$$

For $1 < P \leq 2$, define the arithmetic moment and its harmonic sum by

$$\begin{aligned} W_F^P(\mathbf{a}) &= \sum_{Y \in \mathcal{P}_{\mathbf{r}}} F(Y) \left(\sum_{Z: Y \sim_{\mathbf{p}} Z} F(Z) \right)^{P-1}, \\ \mathcal{W}_F^P(g) &= \sum_{\mathbf{a} \in \mathcal{A}(g)} \frac{W_F^P(\mathbf{a})}{a_1 \cdots a_k}. \end{aligned} \quad (\text{E.1.2})$$

Within-cell invariance makes these definitions independent of the chosen representation. Distinct labels correspond to distinct divisor tuples because all the primes are distinct.

Lemma E.1 (Elementary cell supply). *For every ordered cutoff vector above there are cells containing exactly the primes $\max\{k, y_{i-1}\} < p \leq y_i$ and satisfying (E.1.1) with constants depending only on k .*

Proof. For one block put $u = \max\{k, y_{i-1}\}$ and $\eta = \eta_i$. First greedily partition all primes above u into successive maximal finite cells whose reciprocal mass is at most η . This is possible because every such prime satisfies $p^{-1} < \eta$ and the prime reciprocal sum diverges. Let λ_t be the last prime in complete cell t , with $\lambda_0 = u$, and let q_{t+1} be the first prime in the next cell. The mass H_t of complete cell t satisfies

$$\eta - q_{t+1}^{-1} < H_t \leq \eta.$$

Apart from at most $Q_k = \lceil 2/\min_i \eta_i \rceil$ initial cells, we have $q_{t+1} > Q_k$ and hence $H_t \geq \eta/2$. The uniform Mertens estimate for $3 \leq u \leq v$ gives

$$\sum_{u < p \leq v} \frac{1}{p} = \log \frac{\log v}{\log u} + O(1).$$

It follows that $\log \lambda_t \geq c_k \log u e^{\eta t/2}$. Consequently

$$\sum_{t \geq 1} q_{t+1}^{-1} \leq \sum_{t \geq 1} \exp(-c_k \log 3 e^{\eta t/2}) \ll_k 1.$$

Summing the deficits in H_t now gives $\sum_{s \leq t} H_s = t\eta + O_k(1)$, whence

$$\log \lambda_t = \log u e^{t\eta + O_k(1)}.$$

Now truncate at y_i : keep the nonempty cells intersecting $(u, y_i]$ and give the last retained cell endpoint y_i . Only the last cell can be incomplete, and its extra deficit is at most η . The cumulative mass comparison therefore yields $\log(\log y_i / \log u) = \eta v_i + O_k(1)$ for $y_i > u$, also when there are no retained primes. If $y_i \leq u$, use no cells and note that $\log y_i / \log y_{i-1}$ is bounded above and below in terms of k . Finally $\log u / \log y_{i-1} \asymp_k 1$; the estimates at preceding endpoints, including $t = 1$, give (E.1.1). $\square$

All histories and their explicit partitions A history records the coordinate removal order and its pivot slots. It also specifies which competitors can remain after each removal. Writing this as a partition of each slot's alphabet makes the compatibility relation explicit, including zero pivots and empty blocks.

A history $h = (\sigma, \mathbf{I})$ consists of a permutation $\sigma = (\sigma_1, \dots, \sigma_k)$ of $\{1, \dots, k\}$ and integers satisfying

$$0 \leq I_j \leq R_{\sigma_j}, \quad \text{all positive } I_j \text{ are distinct.} \quad (\text{E.1.3})$$

Assign a positive pivot I_j to its slot block and every zero pivot to block 1. Define

$$\mathcal{J}_i(h) = \{j : I_j \text{ is assigned to block } i\}, \quad n_i(h) = |\mathcal{J}_i(h)|, \quad J_i(h) = \sum_{s \leq i} n_s(h).$$

Let $\mathfrak{H}_{\mathbf{r}}$ be the histories in (E.1.3) with $J_i(h) \geq i$ for every i . Necessarily $J_k(h) = k$. No monotonicity of $I_1, \dots, I_k$ is imposed. Zero pivots can precede positive pivots.

Put $A_j(h) = \{\sigma_j, \dots, \sigma_k\}$. For a slot q , partition its color set C_q by the following explicit equivalence:

$$c \equiv_{h,q} c' \iff 1_{c \in A_j(h)} = 1_{c' \in A_j(h)} \text{ for every } j \text{ with } I_j < q. \quad (\text{E.1.4})$$

Color 0 belongs to none of the $A_j(h)$. Write $\Pi_{h,q}$ for the nonempty equivalence cells. Declare Z compatible with Y , written $Z \equiv_h Y$, if their colors lie in the same cell of $\Pi_{h,q}$ for every slot q . Equivalently,

$$\bigcup_{s=j}^k (Z_{\sigma_s} \cap (I_j, R_k]) = \bigcup_{s=j}^k (Y_{\sigma_s} \cap (I_j, R_k]) \quad (1 \leq j \leq k). \quad (\text{E.1.5})$$

This equivalence follows simply by checking the membership of each individual slot in the two unions. It includes repeated zero pivots, empty blocks, and arbitrary ties among cell indices.

Define

$$\Phi_g(h) = \prod_{i=1}^k \rho_i^{-(k-J_i(h))v_i - \sum_{j \in \mathcal{J}_i(h)} E_g(I_j)}, \quad (\text{E.1.6})$$

$$M_F(Y; h) = \sum_{Z: Z \equiv_h Y} F(Z), \quad \Psi_{h,P}(F) = \sum_Y F(Y) M_F(Y; h)^{P-1}. \quad (\text{E.1.7})$$

These are finite quantities determined by the occupancy and the label weight. We sum over the stated history set, including any history that is not generated by a pair; its nonnegative contribution only enlarges the majorant.

Theorem E.2 (Correlated fractional collision majorant). *Under (E.1.1), uniformly for $1 < P \leq 2$ and every weight F specified above,*

$$\boxed{\mathcal{W}_F^P(g) \leq C(k, c_*, A_*) \frac{\prod_i \eta_i^{r_i}}{\prod_{i,t} g_{i,t}!} \sum_{h \in \mathfrak{H}_r} \Phi_g(h)^{P-1} \Psi_{h,P}(F)}. \quad (\text{E.1.8})$$

For the cells of Lemma E.1 the constant is C_k only. It is independent of $P, \mathbf{y}, \mathbf{r}, g, F$.

Proof. The proof has three steps. We express the moment on the original named-prime measure, derive a linear collision bound for each pair by triangular pivots, and interpolate on the whole measure before summing the history classes. This order keeps both the original factorials and the correlations of the weight.

The named-prime measure. Every arithmetic tuple in $\mathcal{A}(g)$ has exactly G_g named representations by distinct primes in the prescribed cells. Hence, writing $d\mu(\mathbf{p}) = (\prod_q p_q)^{-1}$ for this finite named-prime measure and $\delta = P - 1 \in (0, 1]$, we have the exact identity

$$\mathcal{W}_F^P(g) = \frac{1}{G_g} \sum_Y F(Y) \sum_{\mathbf{p} \text{ distinct}} d\mu(\mathbf{p}) \left(\sum_Z F(Z) 1_{Y \sim_{\mathbf{p}} Z} \right)^\delta. \quad (\text{E.1.9})$$

The factorial therefore belongs to the original measure throughout the argument. Its total mass is at most M_g by (E.1.1). For any nonnegative function f , finite-measure Hölder (or concavity) gives

$$\sum d\mu f^\delta \leq M_g^{1-\delta} \left(\sum d\mu f \right)^\delta. \quad (\text{E.1.10})$$

This includes $\delta = 1$; if the measure has zero mass the desired theorem is immediate.

Canonical pivots and prime integration. Fix a pair Y, Z . Its canonical history depends on the labels alone, before the prime values are chosen. Starting with all coordinate colors remaining, at step j let U_j be the set of slots belonging to exactly one of $Y_c \triangle Z_c$, as c runs through the remaining coordinates. If $U_j \neq \emptyset$, choose $I_j = \max U_j$, let σ_j be the unique remaining coordinate containing that slot in its symmetric difference, and remove σ_j . If $U_j = \emptyset$, choose $I_j = 0$, let σ_j be the largest remaining coordinate, and remove it. Continue for exactly k steps.

A positive pivot belongs to $Y_{\sigma_j} \triangle Z_{\sigma_j}$, giving $I_j \leq R_{\sigma_j}$. After that coordinate is removed, the pivot is in no remaining difference and cannot be chosen again. Removing a coordinate at a zero step can create exclusive slots, so later pivots may still be positive. Every removed coordinate $c \leq i$ has its pivot in the first i blocks, with a zero counted in block 1. Hence $J_i \geq i$ and the resulting history is in $\mathfrak{H}_r$.

The exclusive slots are exactly the symmetric difference of the unions of remaining coordinate sets. To check this, use the fact that each slot has at most one color in either label. If both of

its different colors remain, it occurs in two coordinate differences; if exactly one color remains, it occurs in one. Maximality of I_j therefore gives (E.1.5), also at $I_j = 0$.

Fix the nonpivot primes, and keep the collision row at every positive pivot step. Order both rows and pivot columns by those steps. A pivot already removed has coefficient zero in each later row, while a pivot's own coefficient is 1 or -1 . Thus the coefficient matrix is upper triangular, with entries in $\{0, 1, -1\}$ and dimension at most k . Back substitution, or the finite unit-triangular inverse expansion, bounds its inverse row sums by 2^k . Discarding the zero-step rows enlarges the feasible set. The remaining constraints place every positive pivot logarithm in a product of intervals of length at most $B_k = 2^{k+1}b$, whose centers depend only on the nonpivot primes and labels. We can therefore integrate over this containing rectangle; the collision constraints themselves may still be correlated.

Chebyshev's prime-counting upper bound implies, for a fixed positive B_k , that

$$\sum_{p \in D_{i,t} \cap [X, e^{B_k} X]} \frac{1}{p} \leq \frac{C_k}{\log \max\{3, \lambda_{i,t-1}, X\}} \leq \frac{C_k c_*^{-1} \rho_i^{-t}}{\log y_{i-1}}. \quad (\text{E.1.11})$$

For the first inequality, if the intersection is nonempty take $L = \max\{\lambda_{i,t-1}, X\} \geq 3$ and contain it in $[L, e^{B_k} L]$; then bound its number of primes by $\pi(e^{B_k} L)$ and each reciprocal by L^{-1} . Integrate first over this rectangle. We may then drop distinctness and use η_i as the mass of each nonpivot slot. Removing the pivot masses from the nominal product loses at most k factors η_i^{-1} , a constant depending only on k . For the canonical history $h(Y, Z)$, this gives

$$\sum_{\mathbf{p} \text{ distinct}} d\mu(\mathbf{p}) 1_{Y \sim_{\mathbf{p}} Z} \leq C(k, c_*) M_g \prod_i (\log y_{i-1})^{-n_i(h)} \rho_i^{-\sum_{j \in \mathcal{J}_i(h)} E_g(I_j)}. \quad (\text{E.1.12})$$

Here a zero pivot deletes no prime. We inserted a factor $(\log y_0)^{-1}$ for each such pivot; their number is at most k , so this only changes the displayed constant.

The physical cutoff factors equal the factors in (E.1.6) up to the stated constant. More explicitly, summation by parts gives

$$\begin{aligned} \sum_i n_i(h) \log \log y_{i-1} &= k \log \log 3 + \sum_{i < k} (k - J_i(h)) \log \frac{\log y_i}{\log y_{i-1}} \\ &= \sum_i (k - J_i(h)) \eta_i v_i + O(k + k^2 A_*). \end{aligned}$$

Therefore the left side of (E.1.12) is at most $C(k, c_*, A_*) M_g \Phi_g(h(Y, Z))$.

Interpolation and the history sum. For fixed Y , partition the competitors Z by their canonical histories, and enlarge each generating class to all $Z \equiv_h Y$. Weights are nonnegative, so

$$\sum_Z F(Z) \sum_{\mathbf{p} \text{ distinct}} d\mu(\mathbf{p}) 1_{Y \sim_{\mathbf{p}} Z} \leq C(k, c_*, A_*) M_g \sum_{h \in \mathfrak{H}_r} \Phi_g(h) M_F(Y; h).$$

Use (E.1.10) in the named-measure identity (E.1.9), followed by $(\sum_h a_h)^\delta \leq \sum_h a_h^\delta$ for $a_h \geq 0$. The total nominal factor is exactly M_g , because the two powers of M_g add to one. Taking the fixed constant at least one makes its δ power bounded by itself, uniformly for $0 < \delta \leq 1$. Summing over anchors with weight $F(Y)$ now proves (E.1.8). Pivot choices are already recorded in the history sum. Removing the corresponding primes consequently introduces no factor $g_{i,t}$ or falling factorial, and G_g remains the original multiplicity. $\square$

The correlated partition term The theorem's label term has a useful exact interpretation: it is the sum of powers of the total weights of the history classes. This is also the form to which the relative-class estimates will be applied.

Let $\mathcal{Q}_h$ be the partition of the whole label space into the equivalence classes of $\equiv_h$. This is the Cartesian partition induced by the slot partitions (E.1.4), but the weights of its classes need not factor. For every nonnegative label weight, the exact identity is

$$\Psi_{h,P}(F) = \sum_{Q \in \mathcal{Q}_h} \left(\sum_{Y \in Q} F(Y) \right)^P. \quad (\text{E.1.13})$$

Indeed $M_F(Y; h)$ is constant on Q , with value $\sum_{Z \in Q} F(Z)$, so the anchors in Q supply the remaining first power. If $F(Y) = \prod_q w_{q,Y(q)}$, and only in that additional case for this factorization, the right side becomes

$$\prod_q \sum_{B \in \Pi_{h,q}} \left(\sum_{c \in B} w_{q,c} \right)^P.$$

For a Gibbs weight depending on aggregate counts from several slots, we use (E.1.13) with those correlations intact. The product formula applies when the weight has the stated product structure.

The theorem is pointwise in g . Thus one may multiply both sides by any nonnegative occupancy test $T(g)$ and sum over g , allowing $F = F_g$ separately at every occupancy. This preserves exactly the nominal harmonic factorial mass M_g/G_g and the same constant. To obtain the lower bound for K_k , we next establish actual prime supply at each accepted occupancy and bound the weighted history sum on the right of (E.1.8), with its required polynomial factors. The occupancy measure in that sum remains the nominal harmonic measure.

E.1.2 Squarefree mass at every supported occupancy

The collision theorem is an upper estimate on a nominal cell measure. The lower construction also needs actual squarefree mass. Because acceptance can be rare, we compare the two masses at each supported occupancy. A two-sided occupancy tube bounds the loss from repeated prime draws, while the rapid growth of the cells makes the total loss summable.

For a nonempty finite prime cell D , put

$$H_D = \sum_{p \in D} p^{-1}, \quad p_D = \min D, \quad e_g(D) = \sum_{\substack{p_1 < \dots < p_g \\ p_j \in D}} (p_1 \cdots p_g)^{-1}, \quad e_0(D) = 1.$$

Drawing g independent primes with probabilities $1/(pH_D)$ gives the exact identity

$$g!e_g(D) = H_D^g \mathbb{P}\{\text{all draws distinct}\}.$$

Conditional on s distinct previous draws, their total probability mass is at most $s/(H_D p_D)$. If $(g-1)/(H_D p_D) \leq 1/2$, then

$$\mathbb{P}\{\text{all distinct}\} \geq \prod_{s=0}^{g-1} \left(1 - \frac{s}{H_D p_D} \right) \geq \exp \left(-\frac{g(g-1)}{H_D p_D} \right).$$

For a complete greedy cell with target mass η , $0 \leq \eta - H_D < 1/p_{\text{next}} \leq 1/p_D$. If $p_D \geq 2/\eta$ and $(g-1)/(H_D p_D) \leq 1/2$, then $H_D \geq \eta/2$ and

$$1 \geq \frac{g!e_g(D)}{\eta^g} \geq \exp \left(-\frac{2g^2}{\eta p_D} \right). \quad (\text{E.1.14})$$

This includes $g = 0$.

Use only complete retained cells $D_{i,j}$, omitting the final incomplete cell, and set $p_{i,j} = \min D_{i,j}$, $\eta_* = \min_i \eta_i$, and

$$W_g = \prod_{i,j} \frac{\eta_i^{g_{i,j}}}{g_{i,j}!}, \quad S_g^{\text{all}} = \prod_{i,j} e_{g_{i,j}}(D_{i,j}).$$

The second expression is exactly the squarefree harmonic tuple mass, since the cells are disjoint. Provided the single-cell capacity conditions above hold, multiplication of (E.1.14) gives

$$1 \geq S_g^{\text{all}}/W_g \geq \exp \left(-\frac{2}{\eta_*} \sum_{i,j} \frac{g_{i,j}^2}{p_{i,j}} \right). \quad (\text{E.1.15})$$

Occupancy bounds on the accepting support. We verify the cell capacity conditions uniformly on the accepting support. Let $L_i = \eta_i v_i$, suppose the total count obeys $r_i/L_i \leq D_0$, and each of its h_i fine colors satisfies at cell boundaries

$$\left| N_{i,c}(t) - \frac{t}{L_i} q_{i,c} \right| \leq H_0 + \theta \min(t, L_i - t).$$

An empty block has count zero. Summing over colors gives, for the uncolored cumulative count $G_i(j) = \sum_{s \leq j} g_{i,s}$,

$$\left| G_i(j) - \frac{j}{v_i} r_i \right| \leq h_i H_0 + h_i \theta \eta_i \min(j, v_i - j).$$

The left tube bounds $g_{i,j} \leq G_i(j)$, and the right tube bounds $g_{i,j} \leq r_i - G_i(j-1)$. With $A_0 = (k+1)H_0$, $A_1 = (\max_i \eta_i)(D_0 + (k+1)\theta)$, these two bounds give

$$g_{i,j} \leq A_0 + A_1 \min(j, v_i - j + 1). \quad (\text{E.1.16})$$

An accepted labeling supplies the tube for its uncolored occupancy. That occupancy is shared by all labelings and is preserved by recoloring. The named regularity bounds in the construction imply the displayed linear tubes since $t^\zeta \leq 1 + t$ for $0 < \zeta < 1$.

A uniform distinctness ratio. Choose all retained primes greater than a fixed $Q \geq 2/\eta_*$. Each complete cell has mass at least $\eta_i/2$. The previous $j-1$ retained cells and the reciprocal-integer bound imply

$$p_{i,j} \geq \frac{Q}{2} e^{(j-1)\eta_i/2}.$$

Let $q = e^{-\eta_*/2}$ and define

$$B_1 = \frac{A_0}{1-q} + \frac{A_1}{(1-q)^2},$$

$$B_2 = \frac{A_0^2}{1-q} + \frac{2A_0A_1}{(1-q)^2} + \frac{A_1^2(1+q)}{(1-q)^3}.$$

These are the sums of $(A_0 + A_1j)q^{j-1}$ and its squared-polynomial counterpart. Thus

$$\frac{g_{i,j}}{H_{i,j}p_{i,j}} \leq \frac{4B_1}{\eta_*Q}, \quad \sum_{i,j} \frac{g_{i,j}^2}{p_{i,j}} \leq \frac{2kB_2}{Q}.$$

Taking $Q \geq \max(3, 2/\eta_*, 8B_1/\eta_*)$ makes every capacity condition valid, and yields pointwise

$$1 \geq S_g^{\text{all}}/W_g \geq \exp(-4kB_2/(\eta_*Q)) > 0. \quad (\text{E.1.17})$$

Thus actual squarefree mass retains a fixed fraction of nominal mass at every supported occupancy, including occupancies supporting a rare accepting event.

The arithmetic size constraint. The right-hand tube also gives a fixed size bound. Write $P_{i,j} = \max D_{i,j}$. The $v_i - j$ later cells and the uniform Mertens difference estimate give

$$\log P_{i,j} \leq e^{C_M} \log y_i e^{-\eta_i(v_i-j)/2}.$$

The right-hand tube in (E.1.16) therefore implies

$$\log a_i \leq \sum_j g_{i,j} \log P_{i,j} \leq e^{C_M} B_1 \log y_i.$$

Choose the allowed size exponent at least $A_{\text{tube}} = \max(1, e^{C_M} B_1)$. Then the restricted actual mass equals S_g^{all} on every supported occupancy, proving (9.8). The support constants are fixed before this roughness and grid choice.

E.1.3 Cancellation and one arithmetic first mass

We now combine the collision theorem with the relative-class bound. The likelihood normalization cancels the selected costs, leaving a uniform collision row for every accepted anchor. Squarefree supply then bounds the first integral of the weighted box multiplicity, and the row bounds its fractional higher integral. Hölder's inequality converts these two integrals into the desired volume lower bound.

Apply the relative-class estimate of Section D.1 to the complete-cell enlargement $\bar{Q}_h$ of each original arithmetic history. The multiplier $\Phi_g(h)$ and all original pivot multiplicities are retained. Class inclusion and the exact identity in Section 9.1 then give

$$\sum_h e^{-\delta C_y(h)} (q_h^{\text{rel}})^\delta \leq C \left(\prod_j D_j^{\text{own}} \right)^\delta.$$

The exact outputs give $R_y = \sum_j y_j + \epsilon/r_m - E_y$, with $E_y = \sum_{j < m} d_j \eta_j^y \geq 0$ and $-C \leq \epsilon \leq 0$. Since $\prod_j D_j^{\text{own}} = e^{-\sum_j y_j}$, $e^{R_y} \prod_j D_j^{\text{own}} \leq 1$. Also, with the actual class $Q_h(Y)$,

$$\Phi_g(h) \sum_{Z \in Q_h(Y)} F(Z) = e^{I_h} q_h^{\text{rel}}, \quad C_y(h) = R_y - I_h.$$

Multiplication proves the unnormalized row bound

$$\sum_h \Phi_g(h)^\delta \left(\sum_{Z \in Q_h(Y)} F(Z) \right)^\delta \leq C \quad (F(Y) > 0).$$

Every original pivot is retained. A bounded clock discrepancy changes the constant once per original cut.

With the collision row established, work on harmonic tuple measure times Lebesgue measure and let $f(\mathbf{a}, t) = \sum_Y F(Y) 1_{B_Y(\mathbf{a})}(t)$ and $Z_F = \sum_Y F(Y)$. The weight is bounded, invariant under permutations inside each cell and independent of the actual prime values there. By (E.1.17), $\int f = (\log 2)^k S_g Z_F \geq c W_g Z_F$. If $t \in B_Y$, any other box through t has its center within $\log 2$ of that of B_Y in every coordinate. Thus, pointwise,

$$f^{1+\delta} \leq \sum_Y F(Y) 1_{B_Y} \left(\sum_{Z: Y \sim_{\mathbf{p}} Z} F(Z) \right)^\delta.$$

Integrate and apply Theorem E.2 and the row bound to obtain $\int f^{1+\delta} \leq CW_g Z_F$. Hölder on its support gives

$$\text{meas}\{f > 0\} \geq \frac{(\int f)^{(1+\delta)/\delta}}{(\int f^{1+\delta})^{1/\delta}} \geq cW_g Z_F.$$

Its support is contained in the original divisor union. Summing over the disjoint occupancies retains exactly the first power of the selected nominal mass.

E.1.4 Bounded perturbations with the physical optimizer fixed

The preceding arithmetic estimates concern complete cells. We compare that construction with the physical profile in four stages: bound the reference defects, handle the two exceptional endpoints exactly, compare the finite endpoint scores, and compare their scalar exponents. Count selection and root completion are addressed in the next subsubsection.

Fix the geometric, scalar, count, regularity and occupancy constants, then choose the roughness and cell grids. The initial roughness deletion has bounded harmonic mass, and omitting the final incomplete cell loses less than one fixed cell mass in each group. The coarse prime sum comparison therefore bounds the total resulting length loss by a constant $R = O_k(1)$. Choose the lift above all source thresholds and above $C_H(1 + R)^2$. At the lifted physical vector L optimize once and keep its products, plateau partition, fine probabilities, selected edges and ordinary suffix fixed. The complete-cell lengths L' satisfy $\sum_i |L'_i - L_i| \leq R$. Every original group remains present.

Reference defects. Map each grid block affinely onto its physical block. This preserves the order of activations, deadlines and eligible partitions of every lawful history. All reference integrands are bounded on the compact parameter ranges. Thus, uniformly over the full history space,

$$\begin{aligned} |\overline{C}_{L',s}(h') - \overline{C}_{L,s}(h)| &\leq C_H R, & \overline{C}_{L',s}(h') &\geq -C_H R, \\ |\partial_{s_i} J(L', s) - \partial_{s_i} J(L, s)| &\leq C_H R, & |J(L', s) - J(L, s)| &\leq C_H R. \end{aligned} \quad (\text{E.1.18})$$

Every fixed-coefficient reference charge inherits this additive defect, including those for wrong names, old weak directions, mixed times, pure lifetimes and terminal rank curvature. For instance, the same two lawful competitors give $\overline{C}_{\text{float}} \geq c_H T_{\text{float}} - C_H R$. The old-mixture node equation has residual $O_H(R)$. Dividing it by the old length gives the entropy error; multiplying by that length in the old-group charge leaves an additive $O_H(R)$ error. After the lift, a nonempty old collection is large, so $V_{\text{old}} \leq C_H T_{\text{plateau}} + C_H R$ still has fixed-factor strength. Compression and interpolation use L' itself, with the intercept only in their KKT lower bounds. The exact likelihood, projection and carry identities do not use stationarity.

Exceptional endpoint rows. For one plateau let $m_{\text{ref}} = \sum_g (\mathbb{E}H_g - r_g L'_g) = O_H(R)$ and $\zeta_g = H_g - \mathbb{E}H_g$. Its accepted exact output is $Y = a_0 - b_0 = m_{\text{ref}} + \sum_g \zeta_g$, so

$$\begin{aligned} C_v &= a_0 - \sum_{g \leq l} \zeta_g + \overline{C}'_v + \sum_{\text{proper}} \nu_{v,g}, \\ &= b_0 + m_{\text{ref}} + \sum_{g > l} \zeta_g + \overline{C}'_v + \sum_{\text{proper}} \nu_{v,g}. \end{aligned} \quad (\text{E.1.19})$$

For the all-zero template $C_v = a_0$ exactly. For the complete-deadline template $\nu = 0$ and $\overline{C}'_v = -m_{\text{ref}}$, so $C_v = b_0$ exactly and its intrinsic cost is zero. Neither small endpoint is enlarged by R . For every other endpoint, an intrinsic charge absorbs the defect. If its last positive time is strictly internal, clipped charging contributes at least $c_H \min_g L_g$. If it reaches the end without

being everywhere full, a proper group contributes either a linear bad-group charge or a displaced good-group square-root charge. Hence

$$\overline{C}_v + S_{\text{good}}(v) \geq c_H \sqrt{\min_g L_g} - C_H. \quad (\text{E.1.20})$$

The chosen lift absorbs all bounded defects within the already reserved fractions of these charges. The displaced named box is retained; central uncolored counts alone would not imply this step.

Finite score comparison. The finite context sets at L, L' have identical indexing. Formal perturbed scores may use $\max(0, \overline{C}'_v)$, with the two exceptional templates given their exact scores. Reference errors are $O_H(R)$ and $|\sqrt{L'_g} - \sqrt{L_g}| \leq \sqrt{|L'_g - L_g|}$. Local exit scales are comparable. A change of early/late flag can occur only when $|t_v - V_b/2| \leq C_H R$, where V_b is the total earlier-plus-local clock. The reference charge is then $\geq c_H V_b - C_H R$. For large V_b , the plateau hierarchy makes this dominate either addition to the score; for bounded V_b , use the positive score floor. Thus the paired scores, and their separate minima, remain within fixed factors. The actual grid pair still uses $\Delta'_i = c_i L'_i - a_i n_i + b_i k_i$ in the max construction.

Scalar and ordinary factors. At natural mean $\mu'_i = Z_i L'_i$, the centered parameter $p_i^0 = (a_i - c_i/Z_i)/b_i$ is unchanged; the gap-clock parameter $L'_i/(\mu'_i + 1)$ changes by $O_H(R)/(1 + L_i)^2$. The local Lipschitz estimate for χ and $|\log T_i| \leq C_H \log(2 + L_i)$ on the rare branch therefore give a bounded exponent-times-log error. Score comparability gives comparability of all binary factors, including threshold crossings. Similarly $w'_j = R'_j/\sqrt{T'_j} \asymp w_j$. The frozen ordinary suffix lengths change by $O_H(R)$; its numerical factor is stable with $1 + \max(D', 0)$ in this comparison. If two largest blocks are comparable, choosing either places the other on a side and gives a fixed positive ballot factor. The numerical factor comparison covers this choice of largest block. When the grid mean is negative by a bounded amount, the exact root argument below supplies the legal endpoint pair required by the probability theorem.

E.1.5 Count bands and the exact root on the final grid

The grid comparison must use counts that are realizable under the original coupled count law. We translate the proper split bands and multinomial boxes stage by stage, comparing their factorial weights before sampling paths. We then solve the root equation at the grid budget and check the ordinary endpoint pair separately according to whether its prefix is empty.

For the same complete integer type, all grouped outputs and all recursive exit coordinates change by deterministic $O_H(R)$ because the count coefficients are fixed. Condition on the common pair-coarsened field in the strict-exit construction. Each proper macro split remains an independent $\text{Bin}(m_j, p_j)$, with $m_j \asymp T_j$, compact p_j , and the same coefficient in its exit; its effect on every deeper exit cancels exactly. Band centers move by $O_H(R)$ without reading any remaining split.

Begin in a fixed interior subband and translate each split by a deterministic integer of size $O_H(R)$. The width floor exceeds the finite lattice steps, so rounding leaves a fixed fractional margin; that margin also absorbs the small width change. On the central range, factorial ratios give logarithmic binomial weight change $O_H(R/\sqrt{m_j} + R^2/m_j)$. The windows retain spare width because $\sqrt{m_j} \gg R$. The same factorial comparison applies to a finite multinomial thick box: its conditional center and incoming count move by $O_H(R)$, and its fixed $\sqrt{L_g}$ width stays unchanged. Make these comparisons stage by stage, before sampling any path.

The natural uncolored means also change by $O_H(R)$. For $|n - \mu| \leq C\sqrt{\mu}$, the logarithm of their Poisson ratio is $n \log(\mu'/\mu) - (\mu' - \mu) = O_H(1)$. Hence integration of the complete count

measure retains its true weights and mass $W' \asymp \prod_{j < m} w_j$. Equations (E.1.19) and (E.1.20) supply the same strong-box fractions and legal physical-score endpoint boxes. Terminal completion is checked by the same thick-box calculation at every admitted root count, retaining the original root clock.

The grid root. The actual root count is the floor solving the exact grid equation. At the same other counts it changes by at most $C_H(1+R)$ from the physical-budget floor. Since $V_{\text{pre}} \leq C_H U$ and $U \gg (1+R)^2$, the Euler identity with its bounded residual gives $|n'_* - \mu'_*| \leq C_H \sqrt{U}$. Its Poisson atom is comparable to $U^{-1/2}$, uniformly over retained prefix environments and the allowed logarithmic targets. All $n'_* + 1$ gaps are then sampled at this count.

The ordinary endpoint pair. Let P be the entire original length before the ordinary weak block and B its later positive-drift length. If $P > 0$, the lift makes it large; the physical relation $D \asymp P$ gives $D_{\text{grid}} \geq c_H P - O_H(R) > 0$. The actual ordinary pair remains in the boxes $X \asymp 1+P$, $Y \asymp 1+B$, so the ordinary exact-type theorem applies. When $P = 0$, there are no earlier plateaux and no terminal binary head. The common targets vanish and $x = 0$. The root equation still fixes total risk to its bounded floor error, while the natural grid mean may have either sign $O_H(R)$. For $B > 0$, central counts in the later blocks give surplus at least $c_H B - C_H \sqrt{B} - O_H(R) \geq c'_H B$. Those counts and the root floor determine the weak first-block increment exactly. One legal endpoint is fixed and the other is comparable to $1+B$. For $B = 0$, the increment is just the bounded root floor, and both reserves are fixed.

These legal pairs satisfy the all-reserve ordinary exact-type strong-wall and REG theorem: their endpoint relation is exact, their reserves are positive, and their marks, speed and gap parameter stay compact. The theorem requires these conditions even when the natural mean is negative. At central counts the other suffix blocks keep their separated signed drifts. Conditional on all counts, the original clocks and their constant-retention named refinements give one joint kernel

$$\asymp U^{-1/2} \min \left\{ 1, \frac{(1+P)(1+B)}{1+M_{\text{weak}}} \right\}$$

when a weak block exists, and the saturated path factor otherwise. The physical positive-tail identities identify this with the frozen ordinary factor. Both sides use the same central root atom and the same full-gap ordinary bridge; a later largest positive block makes the path factor saturated.

Completion of the transfer. All binary blocks now have supported central types, their actual grid increments and positive common boxes. The curved-wall, regularity and all-transfer estimates apply at the fixed physical products. Integrate the mixed moments at a fixed full type and then sum the complete type measure. The exact plateau powers and tilt identity give

$$K_k(\log 3 + L) \geq c_H e^{-J(L'; s)} \mathcal{P}_{\text{fixed}}(L'; s) \geq c_H e^{-J(L; s)} \mathcal{P}_I(L).$$

The first inequality is the first-mass transfer already proved; the second uses (E.1.18) and the factor comparisons above. Products have remained fixed at the physical optimizer throughout. All regularity and occupancy constants remain those used to choose the grid; only the final length threshold and additive geometric constants depend on its loss bound. Finally Theorem 14.1 compares the original vector with its one fixed lift, as used in Theorem 7.1.

E.2 Rectangles, root counts, and arithmetic multiplicities

For later reference, we assemble the preceding estimates in the notation of the original divisor-box sum. The upper transfer starts with projected rectangles, makes the positive prime expansion, and

integrates one ordinary root. The lower transfer uses the correlated collision row and pointwise squarefree supply to recover one first mass. The formulas below keep their arithmetic normalizations and counting ranges explicit.

E.2.1 The arithmetic quantity and its exact counting range

Let $k \geq 1$, $H = k + 1$, $3 \leq y_1 \leq \dots \leq y_k$, $y_0 = 3$,

$$L_i = \log \frac{\log y_i}{\log y_{i-1}}, \quad h_i = k - i + 2, \quad B_{\text{geo}} = \sum_i (h_i - 1) L_i.$$

The first prime block is all primes at most y_1 , including 2 and 3; later blocks are $(y_{i-1}, y_i]$. A tuple has squarefree, pairwise coprime components supported on those blocks. Put

$$K_k = e^{-\sum_i h_i L_i} \sum_{\mathbf{a}} \frac{V(\mathbf{a})}{a_1 \cdots a_k}, \quad V(\mathbf{a}) = \text{vol} \bigcup_{d_1 \cdots d_j | a_1 \cdots a_j} \prod_{j=1}^k [\log(d_j/2), \log d_j]. \quad (\text{E.2.1})$$

Koukoulopoulos [19, Theorem 1.7] proves

$$\frac{H^{(k+1)}(x, \mathbf{y}, 2\mathbf{y})}{x} \asymp_k K_k, \quad x \geq 2^k \left(\prod_i y_i \right) y_k. \quad (\text{E.2.2})$$

The convention $y_0 = 3$ agrees with the published logarithmic normalization. Formula (2) applies throughout its displayed range. It compares the full integer count with the squarefree harmonic representation, so prime powers in n , repeated factors in a possible divisor, and the integer-count tail are all included in (2).

The positive size truncation in Section 2 applies to the full volume-weighted sum before any probabilistic argument. Lemma 2.4 shows that for one fixed τ_k , this sum is at most twice its restriction to $a_i \leq y_i^{\tau_k}$. The inserted-prime inequality is used on the original volume, and the weighted logarithmic first moment is bounded by $C_k \log y_i$. Markov's inequality and a union over the k groups give the factor two. The resulting actual prefix size bound $\log(a_1 \dots a_i) \leq k\tau_k \log y_i$ supplies the full rectangles below. The truncation thus supplies the rectangles before the profile is estimated.

E.2.2 Actual projected rectangles

With the normalized volume defined in (12.8), the full divisor-box union has volume

$$I_0(\mathbf{a}) = e^{-B_{\text{geo}}} V(\mathbf{a}). \quad (\text{E.2.3})$$

The cylinder estimates below use this same normalization and the absolute probabilities of the original label classes, as in Section F.1.

Fix a size-truncated actual squarefree tuple and its uncolored prime positions. For an active exterior set $\mathcal{C}$, let E_c be the original deadline and choose $0 \leq \tau_c \leq E_c$ for each $c \in \mathcal{C}$. Reveal that name at prime positions right of τ_c , leaving the actual descendant names compounded. Let Q_τ be a cylinder under the original fine categorical law, and let its probability be $\pi_G(Q_\tau)$. Write

$$\Xi_\tau = \sum_{c \in \mathcal{C}} (\tau_c - E_c) - \log \pi_G(Q_\tau), \quad Z(t) = 1 + \sum_{x \leq t} e^{x-t}.$$

Each unresolved log-divisor displacement is at most $C_H e^{\tau_c} Z(\tau_c)$. Earlier full coordinates use their original size rectangles; compounded children use their original full-size rectangles, without shots at their future deadlines. Their geometric budgets cancel in the normalization. Lemma F.1 therefore gives the cylinder cap

$$\text{Vol}_{\text{norm}}(Q_\tau) \leq C_H \pi_G(Q_\tau) e^{\Xi_\tau} \prod_{c \in \mathcal{C}} Z(\tau_c). \quad (\text{E.2.4})$$

The original class probability in (4) cancels with $-\log \pi$ in Ξ , expressing the deterministic rectangle width as a likelihood-weighted capacity. Fixing a true ancestor route multiplies by its probability and its fixed projected sizes; the same absolute cap remains valid.

A fixed mesh of bounded density changes each shot by at most $e^{\Delta_{\max}}$. On a first mesh failure of a wall lowered by $A \log(1 + \text{depth})$, (4) gives e^{-h} times a summable depth weight and fixed shot powers. The first-failure cylinders form an antichain in the real reveal filtration: their original categorical probabilities sum to at most one. Therefore their union has normalized volume at most $C_H S(G) e^{-h}$. This is the cover in Section F.1, and is the basis of both dual left walls and of the right first-hit cover. Its shots are then integrated by finite occurrence partitions/Mecke, before summing locations, as proved in Section F.2.

E.2.3 The exact positive prime expansion retains every tail

Let $H_i = \sum_{p \in \text{block } i} 1/p$. Freeze a physical optimizer and its rates Z_i , products s_i , and top product s . Set

$$J = \sum_i L_i + \sum_i s_i V_i - \sum_i Z_i L_i, \quad X = \frac{\sum_i N_i \log Z_i - \sum_i s_i V_i}{s}.$$

Under P_Z , prime p in block i has an independent count $\text{Pois}(Z_i/p)$. For every nonnegative symmetric function A of the named lists, expansion of these finitely many Poisson atoms gives the identity

$$e^{-\sum_i L_i} \sum_{n \geq 0} \prod_i \frac{1}{n_i!} \sum_{\text{ordered prime lists}} A \prod_q p_q^{-1} = e^{-J + \sum_i Z_i (H_i - L_i)} \mathbb{E}_Z[e^{-sX} A]. \quad (\text{E.2.5})$$

Section F.1 proves the same identity. On distinct lists the factorials count each squarefree tuple exactly. Extend (3) by zero off the actual squarefree size-truncated support; (5) and the positive truncation give

$$K_k \leq 2e^{-J + \sum_i Z_i (H_i - L_i)} \mathbb{E}_Z[e^{-sX} I_0]. \quad (\text{E.2.6})$$

The geometric caps hold on the support of I_0 . Their nonnegative numerical minimum majorizes the zero-extended integrand, and can therefore be summed over unrestricted lists. The resulting Poisson expectation includes repeated occurrences at a prime and every count, while still bounding the original squarefree contribution. Mertens' uniform difference gives $|H_i - L_i| \leq C_H$; hence the extra factor in (6) is bounded above and below by constants depending only on H . Finitely many initial primes can alternatively be deleted at the explicit fixed Euler-factor cost $\prod_{p \leq Q_H} (1 + H/p)$ (Section 13).

For a deterministic actual prime bank, its cumulative harmonic quantile map obeys $|T(u) - u| \leq C_H$. Push forward a homogeneous Poisson process through this map, retaining its iid Gibbs marks.

This produces exactly the repeated atomic prime process. For a positive jump test use pre-jump values; for a negative jump test use post-jump values. A partial sum within a collapsed atom is then on the favorable side. Only a bounded clock error changes the reserve; there is no bound on multiplicity or total count. Section 13 also keeps count-middle atoms and shot insertion powers at their original plateau powers.

The unread root suffix uses completed equal-mass cells. Greedy cells of target $\eta = \log h/(h-1)$ have mass at least $3\eta/4$, and

$$p_{j+1} \geq e^{\eta/2} p_j, \quad \sum_j (\eta - H_j) \leq [Q_H(e^{\eta/2} - 1)]^{-1}.$$

Final/initial partial cells add at most two target masses per original group. For all cells together the added tilted Poisson mean is bounded. If $\mu_c \leq \mu_c^*$ and $\Delta = \sum (\mu_c^* - \mu_c)$, the exact whole-law ratio is $e^\Delta \prod (\mu_c/\mu_c^*)^{n_c} \leq e^\Delta$. It bounds one entire nonnegative integrand, including its minimum, root tilt and private powers. It needs no monotonicity under inserted points. The original root is retained exactly by

$$X = \kappa + r \sum_i (w_i N_i - c_i L_i) = \kappa' + r \sum_i (w_i N_i - c_i U_i'), \quad \kappa' = \kappa + r \sum_i c_i (U_i' - L_i). \quad (\text{E.2.7})$$

Here $U_i' = L_i + O_H(1)$. Section 13 gives this comparison for the complete Poisson law, including count zero.

E.2.4 The ordinary root atom and ballot denominator are joint

The ordinary factor comes from the root atom and the ordered-sum ballot denominator at the same reservoir count. Section F.1 proves the needed upper estimate for all counts. On an unread suffix put

$$Z = \sum_i (w_i N_i - c_i L_i), \quad X = \kappa + rZ, \quad r = t/s, \quad 0 < t \leq s < 1.$$

Let M be a largest original block, U the total suffix length, and A, B its earlier/later lengths. Condition on the outside counts. Three actual rectangle caps and the prefix cap give

$$I \leq C_H \min\{1, e^Z, e^{B_{\text{act}}} W_{*, N_*}, D e^X\}. \quad (\text{E.2.8})$$

The capped ordered-sum rectangle $W_{*, n}$ has the all- n tail

$$\mathbb{P}(W_{*, n} > e^{-\ell} \mid n) \leq \min \left\{ 1, \frac{C_H(1+\ell)(C_1 + Z_* + \ell)_+}{1+n} \right\}. \quad (\text{E.2.9})$$

Section 11 derives (9) from the $A = 0$ part of [19, Lemma 5.5], with exact integer rounding and with $n = 0$ handled directly. The proof only needs the published flat and crowding order-statistics estimates stated in Section 11. There is no central-count assumption in (9).

At the same reservoir count, Stirling and the lower Poisson tail give $\sup p_\mu(n) \leq C(1+\mu)^{-1/2}$ and $\sup p_\mu(n)/(1+n) \leq C(1+\mu)^{-3/2}$. Outside a fixed-width root strip there are at most $1 + W/\log 2$ reservoir integers. The actual two reserves in (9) become $1+\ell+B_{\text{act}}$ and $C_1+Z+\ell+A_{\text{act}}$. Their positive parts may be enlarged to independent outside positive parts. Since $\mathbb{E}(B_{\text{act}})_+ \leq CB$ and $\mathbb{E}(A_{\text{act}})_+ \leq CA$, averaging the two mode bounds separately gives their minimum, with the ballot denominator still present in the second bound.

Set $H_0 = (-\log D - \kappa)/r$, $w = Z - H_0$. Layer cake in (8) gives the exact integration weight $D^s \exp[-(1-s)\ell]$. At fixed outside counts $a = w + \ell/r$ is a translated lattice of spacing at least $\log 2$; $a > 0$ and the endpoint cap imply $a \geq -H_0 - C$. Thus

$$\sum e^{-ta} \leq C e^{-t(H_0)_-}, \quad \sum e^{-ta} (C + H_0 + a + b)_+ \leq C e^{-t(H_0)_-} [1 + (H_0)_+ + b].$$

Integration yields

$$\mathbb{E}[e^{-sX} I] \leq C_H D^s e^{-t(H_0)_-} (1+U)^{-1/2} \min \left\{ 1, \frac{(1+B)[1 + (H_0)_+ + A]}{1+M} \right\}. \quad (\text{E.2.10})$$

There is no polynomial in a negative root deficit. A fixed polynomial outgoing cap $\min(1, C e^v (1 + (-v)_+)^{\beta})$ changes the layer threshold only to $\ell + 2\beta \log(2 + \ell) + C$; its layer integral is finite uniformly since $1 - s$ has a fixed positive lower bound. Formula (10), then the causal tower in Section F.2, proves the root portion of the upper cover. Completing the suffix changes (10) only by the bounded translate in (7).

E.2.5 Canonical rare banks and the unread suffix

The bank selection is determined by the endpoint geometry. Complete-reference nonnegativity in Proposition A.6 and the normalized secants in Lemma A.3 give the weakness and disjointness assertions of Section F.1.

Output ordering is supplied by the numerical comparison in Lemma A.8 of Section A.2.4. If two weak full-input edges have $r_1 > r_2$ but $q_1 \leq q_2$, monotonicity in the lower secant endpoint and the strict derivative bound give $\beta(q_2, r_2) - \beta(q_1, r_1) \geq 1/(24H^2)$, while weakness gives at most $1/(48H^2)$. Hence $q_1 > q_2$. This applies also to proper zero output, because the secant lemma's hypothesis is $b \geq 1$, not $b \geq 2$.

If the rank intervals overlap, the lawful completion changes in Section F.1 gives

$$X_2 \geq 1 + \delta_H L_1, \quad Y_1 \geq 1 + c_H \min(L_1, L_2).$$

Their persistence ratios are consequently bounded below on at least one edge: if $L_2 \geq L_1$, the first edge ratio is bounded below; if $L_2 < L_1$, the product of the displayed ratio bounds is bounded below. Choose the fixed rarity cutoff below that bound. The intervals $(q_i, r_i]$ of the retained rare edges are then disjoint. In particular there is at most one retained edge per block, and a proper zero-output edge is last. All discarded canonical factors have a fixed positive lower bound; there are only $O_H(1)$ of them.

The deterministic one-end bank in Section F.1 has length $\ell = c_{\text{bank}} L^2 / (1 + X)^2$, belongs to the original current block, and has $\ell \leq L/16$. The endpoint geometry gives $1 + X \geq c\sqrt{1 + L}$, $1 + Y \leq C\tau\sqrt{1 + L}$ and intermediate local length at most CY . Its exact projected rectangle yields a necessary bank test with reserve $h + CN_{\text{intermediate}}$. Integrate the intermediate word with its original mass; only its uncolored count remains in the test. Its moments are at most $C_p(1 + Y)^p$. The scalar one-end upper then gives precisely the retained edge factor at the original plateau power. In a proper zero-output edge the same statement requires the explicit large-left-score hypothesis; the remaining case is the hard two-wall cover proved in Section F.2.

The current-plus-intermediate sets are disjoint by rank separation. Shared count middles are separated by the proved covariance inequality for a decreasing count weight and an increasing reserve, conditioning on the other groups. No independent future acceptance law is used.

Terminal binary rare edges belong to the negative ordinary-drift head. If a test reads counts in the selected suffix, trim its initial intermediate groups and start at the first capacity at most its output. The removed length is at most $C_H \tau \sqrt{1 + L}$; the current negative head pays the resulting mean change. With τ fixed sufficiently small, the largest original suffix block is retained, $U' \asymp U$, and $1 + D' + A' \asymp 1 + D + A$. Thus every mask is prefix measurable for the final unread suffix. Section F.1 contains this check; its terminal geometry inputs are part of the independent deterministic estimate.

E.2.6 A weighted collision theorem with exact original factorials

The collision theorem of Section E.1 applies to any finite nonnegative label weight F , invariant under permutations inside each assigned prime cell and independent of prime values. It may couple all slots.

For occupancy g , denote the original multiplicity by $G_g = \prod_{i,j} g_{i,j}!$ and nominal harmonic mass by $M_g = \prod_i \eta_i^{r_i}$. The exact named-prime identity is

$$\mathcal{W}_F^{1+\delta}(g) = G_g^{-1} \sum_Y F(Y) \sum_{\mathbf{p} \text{ distinct}} \left(\prod_q p_q^{-1} \right) \left(\sum_Z F(Z) 1_{Y \sim_{\mathbf{p}} Z} \right)^\delta, \quad 0 < \delta \leq 1. \quad (\text{E.2.11})$$

For each fixed Y, Z , choose a canonical original history by repeatedly taking the largest slot belonging to exactly one remaining coordinate difference, or a zero pivot when none exists. Positive pivot rows and columns in their removal order have an upper triangular matrix with diagonal $+/-1$. Its inverse row sums are at most 2^k . Conditional on all nonpivot primes, the positive pivot logarithms therefore lie in a product of intervals of length at most $2^{k+1} \log 2$. Chebyshev gives each pivot harmonic mass at most $C_k / [\log y_{i-1} \rho_i^{\text{cell}(\text{pivot})}]$. Only after this rectangle containment are distinctness conditions discarded for an upper bound. Zero pivots delete no prime and only insert a bounded first-block normalizing factor.

Summation by parts identifies these factors with the construction's history factor $\Phi_g(h)$, with error bounded by $O_H(1)$ since each block has a fixed cell-budget defect. Finite-measure concavity on the whole original prime measure in (11), followed by fractional subadditivity over histories, proves

$$\mathcal{W}_F^{1+\delta}(g) \leq C_H \frac{M_g}{G_g} \sum_h \Phi_g(h)^\delta \sum_Y F(Y) \left(\sum_{Z \equiv_h Y} F(Z) \right)^\delta. \quad (\text{E.2.12})$$

The original history sum already records each pivot choice, so integration of a pivot introduces no falling factorial or extra $g_{i,j}$ factor. Uniformity as δ decreases to zero follows because the fixed majorant constant to power δ is bounded by that same constant.

E.2.7 Pointwise squarefree supply and one linear first mass

We use the squarefree supply estimate from Section E.1. In a complete cell D , let $H_D = \sum 1/p$, $p_D = \min D$. Then

$$g! e_g(D) = H_D^g \mathbb{P}(\text{all } g \text{ iid weighted draws distinct}).$$

If $p_D \geq 2/\eta$ and $(g-1)/(H_D p_D) \leq 1/2$, successively excluding the at most $s/(H_D p_D)$ mass of previous draws gives

$$1 \geq \frac{g!e_g(D)}{\eta^g} \geq \exp[-2g^2/(\eta p_D)]. \quad (\text{E.2.13})$$

Accepted named regularity and a fixed central total bound imply $g_{i,j} \leq A_0 + A_1 \min(j, v_i - j + 1)$, with A_0, A_1 fixed before the roughness cutoff. Define $q = \exp(-\eta_*/2)$ and

$$B_1 = \frac{A_0}{1-q} + \frac{A_1}{(1-q)^2}, \quad B_2 = \frac{A_0^2}{1-q} + \frac{2A_0A_1}{(1-q)^2} + \frac{A_1^2(1+q)}{(1-q)^3}.$$

For retained primes greater than $Q \geq \max(3, 2/\eta_*, 8B_1/\eta_*)$, greedy-cell growth gives $p_{i,j} \geq (Q/2) \exp[(j-1)\eta_i/2]$. Hence all capacity conditions in (13) hold and, at every supported occupancy,

$$1 \geq \frac{S_g^{\text{all}}}{W_g} \geq \exp[-4kB_2/(\eta_*Q)] > 0, \quad W_g = \prod_{i,j} \frac{\eta_i^{g_{i,j}}}{g_{i,j}!}. \quad (\text{E.2.14})$$

The right-end tube and Mertens give the actual size bound $\log a_i \leq e^{C_M} B_1 \log y_i$. One fixed allowed size exponent covers it. Thus (14) remains true in the actual size-restricted mass. The squarefree comparison is therefore multiplicative on the entire accepting support.

The source and target history estimates first give the complete-cell row. Enlarge each arithmetic class by rounding its positive pivots to the right boundaries of their cells as in Section 9.1. The multiplier and original summation index are unchanged. Nonnegative class inclusion therefore proves, on every accepted target word, the following row for the classes Q_h^{arith} :

$$\sum_h \Phi_g(h)^\delta \left(\sum_{Z \in Q_h^{\text{arith}}(Y)} F(Z) \right)^\delta \leq C_H. \quad (\text{E.2.15})$$

Its exact cancellation is retained: if $R_y = \sum y_j + \epsilon/r_m - E_y$, $E_y \geq 0$, $\epsilon \leq 0$, and $\prod D_j^{\text{own}} = \exp(-\sum y_j)$, then $\exp(R_y) \prod D_j^{\text{own}} \leq 1$. The complete-cell class probabilities and relative likelihoods are all computed under the original conditional word law, as proved in Section D.5 and Section D.4; the original-pivot sum is finite uniformly by Section D.2 and Section D.3.

On harmonic tuple measure times Lebesgue measure define $f(a, t) = \sum_Y F(Y) \mathbf{1}_{B_Y(a)}(t)$. Because each box has volume $(\log 2)^k$, (14) gives $\int f \geq c_H W_g \sum_Y F(Y)$. If two boxes meet, their centers are within $\log 2$ in every coordinate. Consequently (12) and (15) give $\int f^{1+\delta} \leq C_H W_g \sum_Y F(Y)$. Hölder's inequality on the support then proves

$$\text{meas}\{f > 0\} \geq \frac{(\int f)^{(1+\delta)/\delta}}{(\int f^{1+\delta})^{1/\delta}} \geq c_H W_g \sum_Y F(Y). \quad (\text{E.2.16})$$

This is one first power of the selected nominal mass. Its support is contained in the original divisor union. Summation over disjoint occupancies proves the lower arithmetic transfer; it does not raise the entire rare probability to one terminal power. The chosen δ is a fixed dimension constant for this last step.

E.2.8 Completion of the ordinary suffix

Section 6 supplies the ordinary suffix and root kernel. Its count atom and path factor are constructed on one complete type. Conditional on all nonreservoir central counts, the floor solving the actual root picks a nonnegative central reservoir count. Its atom is of order $(1+U)^{-1/2}$;

all its original $n + 1$ gaps are then sampled. There is at most one nearly centered alphabet because the finite thresholds α_h strictly increase. Strong side blocks have outward path probability bounded below. At the nearly centered block the actual reserves are comparable to $1 + h + B$ and $1 + h + D + A$, by its exact junction values and the central count errors. The ordinary ballot lower gives precisely the bracket in (10). If another block is largest, the separated drift makes both brackets saturated. The all-count upper uses the same reservoir count atom and ballot denominator, before its Poisson supremum.

For the lower source, the required stronger suffix kernel is Section C.1. Its weak block is first; complete balance gives $D \asymp P$, where P is the full original prefix before it. Its common endpoint scales are $1 + P, 1 + B$, from the actual complete reference. The logarithmic target changes only the reservoir count by $O(1 + x)$; if it is the weak block, the max formula raises the left reserve by at most this amount and lowers the right reserve while keeping its positive lower floor. Since $x \leq A \log(2 + P)$, its scale is absorbed by $1 + P$. The ordinary exact strong-wall/REG theorem of Section B.2, applied to these legal actual endpoint pairs, gives the same path bracket. All suffix clocks are independent after this complete count vector is fixed. Genuine named refinement retains a fixed fraction on every retained clock; it does not resample an accepted prefix.

The terminal completion of Section A.3.3 retains the accepted prefix terminal word and fills *only the previously unread suffix names* at the actual positive-root count. A zero-root count is solely a numerical endpoint-box comparison. No clock at that virtual count is sampled, and no positive-root early upper box at zero-root scale is asserted. The early actual lower gains the root entrance; the late intrinsic lower cancels it. The root atom then appears once, linearly, before all prefix mixed moments.

E.3 Complete cells, frozen physical optimizer and actual root counts

The grid transfer can be read as a comparison of four objects: the accepted cell-word density, reference costs, coupled count boxes, and the ordinary root pair. We summarize each comparison here. Their common feature is that the physical optimizer and the original conditional word laws stay fixed.

E.3.1 The complete-cell construction

The construction in Section C.1 defines walls, REG conditions, banks and costs directly on cell occupancies and labeled slots. This is the appropriate discretization for fractional moments: conditional averaging of an already accepted continuous private mass would not preserve those moments. We instead compare cell events with continuous events on every lift of the same word.

Cells have fixed maximum width Δ_* and bounded boundary density. For a positive-jump risk on a cell $[l, r]$, $Z(t) \geq Z(l) - c\Delta_*$; for the negative-jump risk, $Z(t) \geq Z(r) - c\Delta_*$. These inequalities hold for every lift of the cell word to positions, regardless of cell occupancy. A strong $gd(t)^\gamma$ wall changes by at most $g\Delta_*^\gamma$, and its contracted chord has a fixed compact positive speed. Boundary walls therefore imply continuous walls with one fixed reserve allowance. The continuous strong event is contained in the corresponding boundary event. Boundary REG likewise implies continuous REG: expand or shrink to adjacent boundaries, losing at most two meshes in length and a fixed intercept. This uses no bound on interior-cell occupancy.

Free-bank envelopes are rebuilt as cell functions with endpoints rounded to original boundaries. Exact subset and clock densities push forward with the same upper bound; apply the preceding every-lift wall implication. Their fractional moments have the same bounds as the continuous envelopes. Fine named filling is uniform at each fixed complete binary word on each lift; averaging

a mixture of lifts retains the pointwise fixed success bound. It does not average a rare path probability before taking its power.

Conditional on total n_i , the exact cell law is

$$\mathbb{P}(g_i \mid n_i) = n_i! \prod_c \frac{(|c|/L_i)^{g_{i,c}}}{g_{i,c}!}. \quad (\text{E.3.1})$$

All cell costs and conditional private source masses are constant on that fiber, so integration preserves q , q^η and $qD^{\eta-1}$ as identities. The original complete-type laws and independent original block clocks are retained. Macroscopic banks are unions of complete cells and retain their binomial/hypergeometric estimates. The actual root floor is sampled once under (1), and its ordinary suffix kernel is proved both ways by continuous comparison.

The uniform recoloring subset choice commutes with every within-cell permutation. Its exact fine-word density calculation is unaffected. Thus the final density F_y^{cell} is bounded, cell invariant and independent of the actual prime values assigned later. It is precisely the weight allowed in the correlated arithmetic collision theorem. The entire original-history proof applies on these boundary events: the slot/output identities are count identities, and every boundary wall and named regularity extends to every lift with only the fixed allowance.

E.3.2 The physical optimizer is frozen through the grid change

Section E.1 chooses the roughness/grid after fixing the geometric, count, regularity and occupancy constants. It then chooses one sufficiently large lift. Optimize the physical vector L once; retain its products, plateaux, probabilities, selected edges and ordinary suffix on the grid vector L' . Omitting rough initial primes and incomplete final cells produces

$$\sum_i |L'_i - L_i| \leq R = O_H(1). \quad (\text{E.3.2})$$

Map each grid block affinely onto its physical block. This preserves the original activation/deadline order and every lawful named context. Each reference integrand is bounded uniformly in the compact products and alphabet. At a fixed mapped history, its integral changes by at most $C_H R$. The same bound applies to J and to each stationarity derivative, because those expressions are linear in the block lengths at the frozen products. Hence

$$|\overline{C}'(h') - \overline{C}(h)| \leq C_H R, \quad \overline{C}'(h') \geq -C_H R, \quad |J(L'; s) - J(L; s)| \leq C_H R. \quad (\text{E.3.3})$$

All deterministic source/history comparisons acquire an additive defect only. For an old entropy identity, divide its node residual by the old length to compute the entropy error, and multiply back by that same length in the charge; the total error is $O_H(R)$. Every nonempty original group is large after the lift, so hierarchy and the positive charge fractions remain fixed-factor inequalities. Exact categorical likelihood, slot, output and carry identities do not invoke stationarity and acquire no defect.

E.3.3 The zero and full-deadline endpoints

The two exceptional templates can have small endpoint costs, so they are compared by exact identities rather than by an undifferentiated $O(R)$ reference allowance. If $m_{\text{ref}} = \sum_g (\mathbb{E}H_g - r_g L'_g)$, the exact endpoint rows are

$$C_v = a_0 - \sum_{g \leq l} \zeta_g + \bar{C}'_v + \sum_{\text{proper}} \nu_{v,g} = b_0 + m_{\text{ref}} + \sum_{g > l} \zeta_g + \bar{C}'_v + \sum_{\text{proper}} \nu_{v,g}. \quad (\text{E.3.4})$$

For the all-zero template, $C_v = a_0$ exactly. For the complete-deadline template, $\nu = 0$ and $\bar{C}'_v = -m_{\text{ref}}$, so $C_v = b_0$ exactly and its intrinsic cost is zero. No grid defect is added to either small endpoint.

Every other endpoint has its own charge. If its last positive time is an internal deadline, reference clipping gives at least $c_H \min_i L_i$. If it reaches the end but is not everywhere full, an original group is proper: a bad group gives a linear charge, and a displaced good group gives a square-root charge. Thus

$$\bar{C}_v + S_{\text{good}}(v) \geq c_H \sqrt{\min_i L_i} - C_H. \quad (\text{E.3.5})$$

Choose the final lift so that the unused fraction of (5) pays $C_H R$ from (3). The displaced named count box is retained. It is this named displacement, in addition to central uncolored totals, that supplies the square-root charge in (5).

The finite context sets on L and L' have identical indexing. Square-root terms change by at most $\sqrt{|L'_g - L_g|}$. If an early/late flag changes, its last positive time is within $C_H R$ of $V_b/2$, the midpoint of the total earlier-plus-local clock. The clipped reference is then at least $c_H V_b - C_H R$ and, by the plateau hierarchy, dominates both possible additions to the score. The exceptional templates do not change flags. Therefore every paired score, and its separate minimum, is comparable to the physical one. The actual grid pair still uses its exact increment $\Delta'_i = c_i L'_i - a_i n_i + b_i k_i$ in the max construction.

E.3.4 The grid count boxes and the root are actual counts

Condition on the common pair-coarsened field. The original proper macro splits remain independent $\text{Bin}(m_j, p_j)$ with $m_j \asymp T_j$ and compact p_j ; their effect on every deeper exit cancels by the same count identity. Grid band centers shift by $O_H(R)$ without reading any unread split. Translate a central interior subband by the nearest deterministic integer. The fixed width floor only needs to exceed the finite lattice step, not the magnitude of the translate. The factorial ratio on a central range is

$$\log \frac{p_{m,p}(n+u)}{p_{m,p}(n)} = O_H(R/\sqrt{m} + R^2/m), \quad |u| = O_H(R). \quad (\text{E.3.6})$$

After the lift $\sqrt{m} \gg R$, so these are feasible central bands with unchanged fixed margins and comparable mass. The same elementary factorial calculation treats every finite multinomial thick box, before any path is sampled. Means change by $O_H(R)$, and the central Poisson ratio $n \log(\mu'/\mu) - (\mu' - \mu)$ is bounded. Thus the whole coupled count measure retains mass comparable to $\prod_j w_j$, and the earlier complete-type path proofs apply at the frozen physical products.

The reservoir floor solves the exact grid root equation. Relative to the physical floor it changes by at most $C_H(1+R)$. Its distance from its own natural mean is at most $C_H \sqrt{U}$, using the Euler identity with bounded residual and $V_{\text{pre}} \leq C_H U$. Choosing $U \gg (1+R)^2$ makes it nonnegative and central; its true Poisson atom is comparable to $U^{-1/2}$. All its original gaps are sampled at this actual count.

For the ordinary weak block let P be the whole preceding length and B its later positive-drift length. If $P > 0$, it is large after the lift and $D_{\text{grid}} \geq c_H P - C_H R > 0$. Its actual legal pair still

has scales $1 + P, 1 + B$. If $P = 0$, there is no prefix and the common target is exactly zero. The root sets total risk to its bounded floor error, even when the nominal grid drift is negative by $O_H(R)$. If $B > 0$, its separated positive drift is at least $c_H B - C_H \sqrt{B} - C_H R \geq c'_H B$. The weak block's actual endpoint increment is determined by these actual counts and the root floor: one endpoint is fixed and the other is comparable to $1 + B$. If $B = 0$, both endpoints are fixed. The exact-type strong ordinary bridge theorem applies directly to these pairs; it needs positive legal endpoints and compact speed/gap parameters, not a nonnegative nominal mean.

Consequently the grid source kernel is $U^{-1/2} \min(1, (1 + P)(1 + B)/(1 + M_{\text{weak}}))$ when a weak block exists, and the bracket is saturated otherwise. A later largest positive block also saturates it. This is the frozen physical ordinary factor. Combined with the per-type mixed moments and the linear first-mass transfer of Section E.2, it gives the fixed-lift lower at the physical profile, with constants depending on H only. The rounded cell vector uses the original physical optimizer.

E.3.5 Stability of the original harmonic sum, including zero axes

The stability argument of Section 14 compares the original harmonic sums directly by removing primes whose block membership changes and then reinserting them. For two vectors ℓ, ℓ' , put

$$\delta_i = \ell'_i - \ell_i, \quad D = \sum_{i=1}^k \left| \sum_{h \leq i} \delta_h \right|, \quad A = \sum_i h_i |\delta_i| + H(D + 2kC_M).$$

Every prime whose block membership changes is in the union of the k intervals between the old and new cutoffs. Its harmonic mass is at most $D + 2kC_M$. Remove these primes in both configurations and keep the common unnormalized volume sum S_* . It is positive because the empty tuple contributes $(\log 2)^k$. Inserting a prime in block i makes at most h_i translates of the old union, one unchanged. Therefore, for each of the two full sums,

$$S_* \leq S_j \leq S_* \prod_{p \in \mathcal{B}} (1 + H/p) \leq S_* e^{H(D + 2kC_M)}.$$

The ratio S'/S lies between the last exponential and its reciprocal; its exponent is not doubled. Multiplying by the exact normalizing ratio gives

$$e^{-A} K_k(\ell) \leq K_k(\ell') \leq e^A K_k(\ell). \quad (\text{E.3.7})$$

This proof allows coincident cutoffs and zero lengths. It changes no integer-count x range, since it compares only the arithmetic K .

For $L_i^\lambda = \max(L_i, \lambda)$, (7) has exponent at most $B_k \lambda + C_0(k)$, where $B_k = k(k^2 + 3k + 4)/2$ and $C_0 = 2kHC_M$. Fix the single numerical constant Λ_k specified in the full quantitative estimates, put $L_i^* = \max(L_i, \Lambda_k)$ and $\ell_i^* = L_i^* + \log 3$. The positive-core upper and the explicit fixed-lift lower, both proved at this same vector, give

$$c_k \Psi_k(\mathbf{L}^*) \leq K_k(\ell^*) \leq C_k \Psi_k(\mathbf{L}^*).$$

Combining these bounds with (7) gives

$$c_k e^{-B_k \Lambda_k - C_0(k)} \Psi_k(\mathbf{L}^*) \leq K_k(\ell) \leq C_k e^{B_k \Lambda_k + C_0(k)} \Psi_k(\mathbf{L}^*). \quad (\text{E.3.8})$$

Both displayed exponential factors depend only on k . The final rule therefore evaluates the raw profile at this one fixed lift, including vectors on equality faces. The complete scalar-inverse evaluator is proved in Appendix I.

E.4 Numerical harmonic estimates, complete cells and squarefree supply

We finish with numerical versions of the harmonic and squarefree estimates. The elementary Mertens difference controls the clock defect; geometric cell growth makes that defect and the distinctness loss summable. Once the occupancy tube constants are fixed, the roughness cutoff below determines every grid constant used in the quantitative lift.

Put $H = k + 1$,

$$\eta_i = \frac{\log(k - i + 2)}{k - i + 1}, \quad \eta_* = \min_i \eta_i, \quad q = e^{-\eta_*/2}.$$

For $H \geq 2$, $1/(2H) \leq \eta_* \leq \eta_i < 1$.

E.4.1 An elementary numerical Mertens difference

For every real $x \geq 3$,

$$-1 \leq \sum_{p \leq x} p^{-1} - \log \log x \leq 17. \quad (\text{E.4.1})$$

Thus $C_M = 40$ is safe in every two-end Mertens difference used in the harmonic estimate. Here is a proof that requires no numerical asymptotic prime theorem. If $N \geq 2$ is an integer, the primes in $(\lceil N/2 \rceil, N]$ divide $\binom{N}{\lceil N/2 \rceil} \leq 2^N$. Iteration gives $\theta(N) \leq 3N \log 2 < 4N$, hence $\theta(x) \leq 4x$. Partial summation implies

$$\sum_{p \leq x} \frac{\log p}{p} \leq 4 + 4 \log x, \quad \sum_{p \leq x} \frac{\log p}{p-1} \leq 8 + 8 \log x.$$

For $s = 1 + 1/\log x$, integrate the Euler-factor logarithmic derivative from 1 to s . Its difference is at most 16, so

$$\prod_{p \leq x} (1 - p^{-1})^{-1} \leq e^{16} \zeta(s) \leq e^{16} (1 + \log x) \leq e^{17} \log x.$$

This proves the upper side of (1), since the product logarithm dominates the prime reciprocal sum. Conversely this product contains the whole harmonic sum up to $\lfloor x \rfloor$, which is at least $\log x$. The terms of order at least two in its logarithm have sum at most $\sum_{m \geq 2} 1/[m(m-1)] = 1$. This proves the lower side. In particular prime reciprocal sums diverge, so the greedy complete cells used below always exist. The weighted-prime constant in Section 2 can also be set to $C_W = 8$, since $4 + 4 \log x \leq 8 \log x$ for $x \geq 3$.

E.4.2 Explicit cells and their clock defect

Given the two occupancy constants $A_0, A_1 \geq 0$, define

$$B_1 = \frac{A_0}{1-q} + \frac{A_1}{(1-q)^2}, \quad B_2 = \frac{A_0^2}{1-q} + \frac{2A_0A_1}{(1-q)^2} + \frac{A_1^2(1+q)}{(1-q)^3},$$

$$Q = \left\lceil \max \left(3, 16H, \frac{2}{\eta_*}, \frac{8B_1}{\eta_*}, \frac{8kB_2}{\eta_*} \right) \right\rceil. \quad (\text{E.4.2})$$

Delete primes up to Q . In every original prime block start at $u = \max(Q, y_{i-1})$, greedily take the maximal successive prime cell of reciprocal mass at most η_i , and omit the final incomplete cell at y_i . Each complete cell has mass between $\eta_i - 1/p_{\text{next}}$ and η_i ; (2) makes it at least $\eta_i/2$.

The reciprocal-integer bound gives for the first prime in the j th retained cell

$$p_{i,j} \geq (Q/2)e^{(j-1)\eta_i/2}. \quad (\text{E.4.3})$$

Indeed the preceding cells have total mass at least $(j-1)\eta_i/2$, whereas the harmonic mass of all integers in (Q, p) is at most $\log(2p/Q)$. Therefore the sum of the greedy mass deficits is at most

$$\sum_{j \geq 1} p_{\text{next},j}^{-1} \leq \frac{2}{Q} \frac{1}{1 - e^{-\eta_i/2}} \leq \frac{16H}{Q} \leq 1. \quad (\text{E.4.4})$$

Here $1 - e^{-\eta_i/2} \geq \eta_i/4 \geq 1/(8H)$ was used. Combined with (1), (4), and the omitted incomplete cell, the ideal complete clock length $L'_i = \eta_i v_i$ differs from $\log(\log y_i / \log u)$ by at most 82. If the entire block lies below Q , its lost physical length is at most $\log \log Q$. Hence a numerical total defect bound is

$$\sum_i |L'_i - L_i| \leq R_k(Q) := k(\log \log Q + 100). \quad (\text{E.4.5})$$

The cell clocks have mesh at most one and boundary rate at most $2H$; an inclusive interval of length one contains at most $2H + 1 \leq 3H$ boundaries. If an affine endpoint map is used, its constant defect in the quantitative fixed-lift construction can consequently be taken from (5); no unspecified Mertens or cell-supply constant is needed.

For completeness, the collision-cell geometric constants can be set to $c_* = \exp(-600H^2)$ and $A_* = 600H^2 + \log \log Q + 100$. To see this before any roughness deletion, at most $5H$ greedy cells have the next prime below $\lceil 2/\eta_* \rceil$; all later cells have mass at least $\eta_i/2$. Applying (1) yields $\log \lambda_t \geq \log u \exp(-100H) \exp(\eta_i t/2)$. For the numerical series detail, put $a = \exp(-100H) \log 3$, $b = \eta_i/2 \geq 1/(4H)$. Monotonicity gives

$$\sum_{t \geq 1} e^{-ae^{bt}} \leq \frac{1}{b} \int_a^\infty e^{-v} \frac{dv}{v} \leq 4H \{\log(1/a) + e^{-1}\} < 401H^2 < 500H^2.$$

The first inequality is the decreasing integral test. In the second, bound the integral on $[a, 1]$ by $\log(1/a)$ and on $[1, \infty)$ by e^{-1} . Summing the deficits once more yields $\log \lambda_t \geq \log u \exp(\eta_i t - 600H^2)$. The upper cumulative comparison and final incomplete cell give the displayed A_* . After deletion, (4) improves the deficit bound, so these choices remain safe.

E.4.3 Pointwise squarefree supply

On the actual accepted support suppose

$$g_{i,j} \leq A_0 + A_1 \min(j, v_i - j + 1).$$

The exact elementary-symmetric-sum calculation in the pointwise squarefree supply argument in Section E.2, with (E.4.2)–(E.4.3), gives

$$1 \geq \frac{S_g^{\text{all}}}{W_g} \geq \exp\left(-\frac{4kB_2}{\eta_* Q}\right) \geq e^{-1/2} \quad (\text{E.4.6})$$

at every supported occupancy. The capacity condition follows from $4B_1/(\eta_*Q) \leq 1/2$; complete cell means are at least $\eta_i/2$. This gives a fixed relative squarefree supply bound even for a rare accepting mass. Formula (2) is a fixed roughness rule once A_0, A_1 are fixed, even for a huge number of cells.

The size exponent used for the lower harmonic source is allowed to be at least $\exp(80)B_1$; its lower sum is a subset of the unrestricted K_k sum. It need not equal the upper-truncation exponent $2kHC_W + 1$. Consequently this source does not require deleting an additional number of last cells merely to match that upper-truncation exponent.

Section H.1 bounds the deterministic coefficient and both occupancy constants used in these grid inequalities.

F Covers and conditional upper estimates

The upper bound rests on a geometric capacity estimate and two kinds of conditional integration. A cylinder of partial label information lies in a projected rectangle; its width can be written as its original class probability times a risk exponent and a deterministic shot sum. Necessary local tests control those exponents. Finally the ordinary suffix is integrated with the root atom and its ballot denominator at the same count.

We prove these ingredients and explain how their conditional bounds compose in the original routing filtration. Keeping the geometric capacities separate from the private masses makes the reveal order and the dependence of each estimate explicit.

F.1 Geometric covers and conditional estimates for the upper bound

We begin with the projected rectangles and positive prime expansion. The rank-separation argument then selects disjoint rare banks, whose one-end and whole-band estimates give the local assertions of Proposition 12.1. After checking that the terminal masks leave a suffix unread, we prove the ordinary root estimate of Proposition 12.9. Clocks and eligibility sets retain their original block labels throughout.

F.1.1 Rectangles and deterministic sums at stopping locations

A cut fixes the large-prime part of a divisor coordinate and leaves its smaller-prime part free. The free displacement is bounded by an exponential sum at that cut. The following lemma packages the product of these width bounds as a capacity under the original fine-label law.

Lemma F.1 (Cylinder rectangles). *Fix a size-truncated squarefree tuple as in Lemma 2.4 and its uncolored prime positions. Let $\mathcal{C}$ be a set of exterior coordinate names. For $c \in \mathcal{C}$, let E_c be its original logarithmic deadline and choose $0 \leq \tau_c \leq E_c$. A cylinder Q_τ specifies the assignments to these names at positions to the right of their cuts; all remaining eligible names stay compounded. Endpoint batches are placed on a fixed prescribed side. Under the original fine categorical law π , put*

$$B_{\mathcal{C}} = \sum_{c \in \mathcal{C}} E_c, \quad \Xi_\tau = \sum_{c \in \mathcal{C}} \tau_c - B_{\mathcal{C}} - \log \pi(Q_\tau), \quad Z(t) = 1 + \sum_{x \leq t} e^{x-t}.$$

Then

$$\text{Vol}_{\text{norm}}(Q_\tau) \leq C_H \pi(Q_\tau) e^{\Xi_\tau} \prod_{c \in \mathcal{C}} Z(\tau_c). \quad (\text{F.1.1})$$

Coordinates outside $\mathcal{C}$ retain their deterministic full-size rectangles; they contribute no shot at a future deadline.

Proof. The revealed factors determine translations, so only unresolved widths need to be bounded. The sum in Z uses every original occurrence and therefore bounds the contribution of any selected descendant list. The unresolved displacement in coordinate c is at most

$$\log 2 + (\log 3) \sum_{x \leq \tau_c} e^x \leq C_H e^{\tau_c} Z(\tau_c).$$

For $c \notin \mathcal{C}$, the size truncation gives an interval of width at most $C_H e^{E_c}$. Multiplying these widths and dividing by the full geometric normalization gives $C_H \exp(\sum_{c \in \mathcal{C}} (\tau_c - E_c)) \prod_{c \in \mathcal{C}} Z(\tau_c)$. Multiplication and division by the absolute original probability $\pi(Q_\tau)$ proves the formula. For a fixed ancestor atom α , use its resolved coordinate translations and inherited geometric multiplier. Temporary conditioning replaces the joint mass by $\pi(Q_\tau \mid \alpha)$; summing against $\pi(\alpha)$ restores the absolute joint mass. $\square$

To use this capacity at first failures, we sum over a deterministic mesh. This allows the moment estimates to be made at fixed locations before the occurrence process is drawn. Partition a physical interval $[a, b]$ into cells of length at most Δ , with a bounded number of endpoints per unit interval. If v is the right endpoint of a cell and t belongs to that cell, positivity gives

$$Z(t) \leq e^\Delta Z(v). \quad (\text{F.1.2})$$

Every term present at t remains present at v , and its kernel changes by e^{v-t} . At successive mesh points the reveal partitions refine. The cylinders stopped at a first failure are disjoint because each word has one first failure; their original probabilities hence sum to at most one. We use this argument for each cover separately. On an ancestor fiber, we then sum against the original ancestor mass. Suppose each first-failure cylinder has capacity at most

$$C\pi(q)e^{-h}\omega(t)Z(t)^d \prod_c Z(\tau_c), \quad d \leq H,$$

where $\omega(t) = (1 + \text{dist}(t, \{a, b\}))^{-A}$, or its one-ended version, and the outside cuts τ_c are deterministic. Within one mesh cell $\omega(t) \leq C_{\Delta, A}\omega(v)$ and the sum of the atom probabilities is at most one. Thus the total failure capacity is bounded by $Ce^{-h}S$, with

$$S = \sum_v \omega(v)Z(v)^d \prod_c Z(\tau_c).$$

The stopped atom probabilities have already been summed in this expression.

The moment bound follows from the exponential decay in each shot. For a Poisson process with bounded intensity per unit interval, $Z(v)$ has all fixed moments uniformly in deterministic v : split into unit intervals and sum their exponentially decreasing coefficients, or use the Laplace functional. Taking fixed A large enough, Minkowski's inequality and $\sum_v \omega(v) < \infty$ then give $\mathbb{E}S^p \leq C_{H,p}$ for every required fixed p . Finite sums over the endpoint contexts have the same property. A cut within bounded distance of an endpoint is covered by finitely many extra mesh cells. An incoming selected-list sum is bounded pointwise by the original-list sum at the same deterministic threshold.

The probability tests themselves may be made on the complete deterministic mesh, including empty cells. The risk has one-sided jumps and bounded linear speed; survival on that mesh implies

the corresponding continuous necessary wall after a fixed reserve increase, using the appropriate side for the jump convention. The rising wall has bounded derivative on the fixed interval, so the same conclusion holds for its first-hit strip. Consequently the fixed-time bank estimates below are only applied at deterministic mesh locations.

Under insertion of q fixed original occurrences, every deterministic $Z(v)$ increases by at most q . Keeping the mesh and contexts fixed therefore gives

$$S(G + \text{insertions}) \leq C_{H,q} S(G). \quad (\text{F.1.3})$$

The categorical log-likelihood changes at each fixed test by at most $C_H q$, which is paid by the reserve. A Poisson–Mecke expansion of a fixed shot monomial uses finitely many partitions of its occurrences; apply this insertion bound in each term before summing the deterministic locations. This proves the fixed-moment and insertion assertions for the stopping covers. The stopped locations have already been summed on the deterministic mesh, so these assertions concern fixed-location shots.

F.1.2 Positive prime expansion with the original root

We next express the harmonic sum under a tilted occurrence law. The identity is exact on the original prime ranges. Geometric caps are proved first on actual squarefree tuples and then extended by positivity to all occurrence lists. This order allows the analytic expectation to retain repeated occurrences and every count.

Let $\mathcal{P}_i$ be the original prime ranges, including 2, 3 in $\mathcal{P}_1$, and put

$$H_i = \sum_{p \in \mathcal{P}_i} \frac{1}{p}, \quad B_{\text{geo}} = \sum_i V_i = \sum_i (h_i - 1) L_i.$$

Take fixed positive Z_i, s_i , put $s = s_1 > 0$, and define

$$J = \sum_i L_i + \sum_i s_i V_i - \sum_i Z_i L_i, \quad X(G) = \frac{\sum_i N_i(G) \log Z_i - \sum_i s_i V_i}{s}.$$

Let $\mathbb{P}_Z$ give independent occurrence counts $\text{Pois}(Z_i/p)$ at each $p \in \mathcal{P}_i$. For every nonnegative symmetric functional A on occurrence lists, Tonelli and the Poisson exponential formula give the exact identity

$$\begin{aligned} e^{-\sum_i L_i} \sum_{\mathbf{n} \geq 0} \prod_i \frac{1}{n_i!} \sum_{p_{i,a} \in \mathcal{P}_i} A(G) \prod_{i,a} \frac{1}{p_{i,a}} \\ = e^{-J + \sum_i Z_i (H_i - L_i)} \mathbb{E}_Z[e^{-sX(G)} A(G)]. \end{aligned} \quad (\text{F.1.4})$$

To verify the exponent, the unnormalized list sum equals $e^{\sum_i Z_i H_i} \mathbb{E}_Z[A \prod_i Z_i^{-N_i}]$, and $\prod_i Z_i^{-N_i} = e^{-sX - \sum_i s_i V_i}$.

For the saddle, $\log Z_i = G_i$ and X is exactly the uncolored routing root. The routing root agrees with the prime tilt by telescoping. At node j , minus the local log-probability, multiplied by s_j , is G_j for an exit and $G_j - G_{j+1}$ for a child route. Summing along the route leaves G_i at the occurrence's original block. Subtracting the physical budgets proves the identity at the fixed coefficients; stationarity is not needed and every child list is the original one.

Define $I_0(G) = e^{-B_{\text{geo}} L^{(k+1)}(\mathbf{a}(G))}$ on the squarefree size-truncated support, and $I_0 = 0$ elsewhere. Each squarefree selection occurs $n_i!$ times among ordered lists, so Lemma 2.4 and (F.1.4) imply

$$K_k(\ell) \leq 2e^{-J + \sum_i Z_i(H_i - L_i)} \mathbb{E}_Z[e^{-sX} I_0(G)]. \quad (\text{F.1.5})$$

The geometric caps hold first on that actual support. If they give $I_0 \leq CC_l(G)$ there, with each C_l nonnegative on every list, then $I_0 \leq C \min_l C_l$ on all lists, because of the zero extension. Their numerical cap minimum is therefore evaluated on unrestricted repeated lists under the same atomic Poisson law and root.

The uniform prime harmonic estimate gives $|H_i - L_i| \leq C_H$; the discrepancy in (F.1.5) is bounded since $Z_i \leq H$. Deleting a fixed initial prime set pays its finite Euler product and changes each H_i by only a fixed amount. For later cell completion, if $\mu_c^* \geq \mu_c$ and $\sum_c (\mu_c^* - \mu_c) \leq C_H$, the product Poisson density ratio is at most e^{C_H} . It is applied once to the whole nonnegative integrand, retaining its minimum, powers and root. A change of the physical cell budget is compensated in the external root translate.

F.1.3 Separation of the rank intervals of rare comparisons

We arrange disjoint inputs before multiplying the local probability estimates. We prove that two sufficiently rare comparisons in one plateau have disjoint rank intervals. Weakness places their secants near the same level. If their intervals overlap, lawful completion changes force a large endpoint score for at least one comparison, contradicting rarity.

In one maximal plateau, write $\varepsilon = 1 - \sigma > 0$ and

$$\kappa_r = (r + z)^{-\varepsilon}, \quad g_r(a) = a - \kappa_r E_a(z).$$

The transported incoming coefficients are retained individually. They are no larger than the natural coefficient at the first local capacity. Every lawful complete reference history has nonnegative cost by the KKT and reference-compression results of Section A.2.

Choose a fixed sufficiently small $\epsilon_0(H) > 0$. An edge $r \rightarrow q$ is weak when $|g_r(r) - g_r(q)| \leq \epsilon_0$. If m_-, m_+ are the two minimum reference costs, their common increment is $m_+ - m_- = L_i \nu_i$. Both are nonnegative and the endpoint scores dominate $1 + m_-$ and $1 + m_+$. Hence

$$|\nu_i| L_i \leq m_- + m_+ \leq X_i + Y_i.$$

Since the natural mean is comparable to L_i , sufficiently small $(1 + X_i)(1 + Y_i)/(1 + \mu_i)$ implies this fixed weakness condition.

We need three strict comparisons: a weak edge viewed from a larger capacity, a rank below its output, and a capacity between its output and input. These supply the charges in the overlap argument. The normalized secant estimate of Section A.2 is

$$\beta_b(y) = \frac{\log((y \log y - b \log b)/(y - b))}{\log y}, \quad \beta'_b(y) \leq -\frac{1}{24H^2}.$$

For a weak edge $r \rightarrow q$ and an earlier capacity $h > r$, put $T_a = (a + z)^{-\varepsilon}(E_a - E_q)/(a - q)$. Weakness gives $|T_r - 1| \leq \epsilon_0/(r - q)$, whence $\beta_{q+z}(r + z) - \varepsilon \leq 2\epsilon_0/\log 2$. If $\epsilon_0 \leq \log 2/(192H^2)$, then $\log T_h \leq -\log 2/(48H^2)$. At the other endpoint,

$$g_h(r) - g_h(q) = (r - q)\{1 - (\kappa_h/\kappa_r)T_r\}, \quad \kappa_h/\kappa_r \leq e^{-\varepsilon_{\min}/H}.$$

Taking also $\epsilon_0 \leq (e^{\epsilon_{\min}/H} - 1)/2$ gives a positive lower bound at both r and h . Concavity then proves

$$g_h(a) - g_h(q) \geq \delta_H > 0 \quad (r \leq a \leq h). \quad (\text{F.1.6})$$

Reducing the coefficient increases this difference, so it also holds in the relevant earlier groups. If $q \geq 1$ and $0 \leq a < q$, strict concavity $-g_r'' \geq H^{-2}$ gives

$$g_r(r) - g_r(a) \geq -H\epsilon_0 + \frac{(q-a)(r-a)}{2H^2} \geq \frac{3}{4H^2}$$

after $\epsilon_0 \leq 1/(4H^3)$. Finally the forward secant comparison gives, for $q < h < r$,

$$g_h(q) - g_h(h) \geq \gamma_H > 0. \quad (\text{F.1.7})$$

All thresholds are chosen before any lengths.

Suppose the rank intervals of two weak edges overlap, with $r_1 > r_2$ but $q_1 < r_2$. Output ordering gives $q_2 < q_1$, hence $q_1 \geq 1$. Consider an arbitrary left completion for the second edge. Its earlier retained ranks are at least r_2 ; lowering all of them to q_2 and keeping the later history gives a legal complete history. The changed history has nonnegative cost. By (F.1.6), the original completion therefore pays $X_2 \geq 1 + \delta_H L_1$.

For the right score of the first edge, keep fixed the prefix through its block and the suffix after the second block. All capacities between their right endpoints b_1, b_2 exceed q_1 , so the remaining names have no deadline there. The reference cost, as a function of their ordered cut times, is a constant plus

$$\sum_{a=k_{\text{end}}+1}^{q_1} \int_{b_1}^{\tau_a} \{g_t(a) - g_t(a-1)\} dt.$$

Since $\kappa(t)$ is nondecreasing, each summand is concave. A minimum on the ordered simplex is attained at a vertex, where every cut is at b_1 or b_2 . We may consequently replace the ranks on that interval by one constant rank $a \leq q_1$ without increasing cost. There are two cases. If $a < q_1$, lower the first-block rank to a ; strict concavity gives an original cost of at least $c_H L_1$. If $a = q_1$, fill the intervening groups to their eligible capacities; (F.1.7) gives an original cost of at least $c_H L_2$. In both cases the complete outside history is preserved. Consequently

$$Y_1 \geq 1 + c_H \min(L_1, L_2), \quad \bar{T}_1 \geq c_1 \frac{1 + \min(L_1, L_2)}{1 + L_1}, \quad \bar{T}_2 \geq c_2 \frac{1 + L_1}{1 + L_2}.$$

If $L_2 \geq L_1$ the first ratio is bounded below; otherwise their product is. Hence two sufficiently rare edges must have $q_1 \geq r_2$. Applying this to every pair proves disjointness of the rank intervals $(q_i, r_i]$. The argument also applies to terminal nonzero edges: $z = 1$, but their secant bases are at least two and their marks remain nondegenerate. The terminal zero-output edge remains ordinary.

F.1.4 The two scales of a proper zero-output comparison

A rare full-to-zero comparison has two different reserves. One contains the old and earlier local lengths together with the exit width; the other contains only the later local length. The following endpoint calculation identifies these scales and shows that the selected block dominates the other lengths. This is the geometry needed for the hard-plateau two-wall estimate.

Consider a proper zero-output edge with sufficiently small persistence ratio $\bar{T}$. Let M be the selected local block, A, B the preceding and following local lengths, and V_{old} the complete old length. Put

$$P = V_{\text{old}} + A, \quad T = P + M + B, \quad R_{\text{exit}} = \max\{R_0, \min(\sqrt{1 + A + M + B}, d^{-1})\}.$$

Full input and zero output force the two completion times to be P and $P + M$ in every outside context. The clipped-prefix estimate from Section A.2 gives $\bar{C}_v \geq c_H \min(t_v, T - t_v)$. If the block fails to cross $T/2$ strictly, one of its endpoint costs is at least $c_H M$ and the persistence ratio is bounded below. Thus sufficiently small ratio forces $P < T/2 < P + M$.

For the converse score bounds, use the two explicit contexts that retain each earlier name to its deadline and remove the current names at the selected endpoint. Neither context has a good group. Complete plateau balance gives

$$\bar{C}_{\text{full}}^- = \sum_{\text{old}} L_g g_g(r_{\text{top}}) + \sum_{\text{local before}} L_g g_g(r_g), \quad \bar{C}_{\text{full}}^+ = - \sum_{\text{local after}} L_g g_g(r_g).$$

Their costs are nonnegative and at most $C_H P$ and $C_H B$, respectively. Together with the clipped lower bound and the early/late score definition, this proves

$$X \asymp_H 1 + V_{\text{old}} + A + R_{\text{exit}}, \quad Y \asymp_H 1 + B. \quad (\text{F.1.8})$$

In particular rarity implies $V_{\text{old}} + A + B \leq C_H \tau M$; for a small fixed τ , M dominates the entire old and local length. Uniformly even as $d \downarrow 0$,

$$R_{\text{exit}} \asymp_H \varrho := \max \left\{ \varrho_0, \frac{c_0 \sqrt{1+M}}{1+d\sqrt{1+M}} \right\}.$$

Conversely a sufficiently small $(1+B)(1+V_{\text{old}}+A+\varrho)/(1+M)$ forces that dominance and midpoint crossing, so the scales characterize the same rare region. For the original mark $W = \log(r+z) - \log z 1_{\text{child}}$, complete balance also gives

$$eM = \sum_{\text{old}} L_g g_g(r_{\text{top}}) + \sum_{\text{local} \neq i} L_g g_g(r_g), \quad e = (r+z)^\sigma \mathbb{E}W - r,$$

and therefore $|e|M \leq C_H(V_{\text{old}} + A + B)$. The two reserves in (F.1.8) remain separate when applying a one-end or two-end probability estimate.

F.1.5 Necessary one-end tests on original local intervals

For a nonzero output, the left endpoint score is at least diffusive. A short deterministic bank near the other end then sees the smaller reserve at the correct scale. Its necessary test may read counts in intermediate local groups, but those groups are disjoint from the bank and have length controlled by the smaller score. We first prove this geometry, then derive the test from a projected rectangle, and finally average its conditional mass.

Consider a weak edge $r \rightarrow q$ with $q \geq 1$ on $[a, b]$, of length L , in the rare region $\bar{T} < \tau$, where $\tau > 0$ is sufficiently small in terms of H . Write $S = r + z$, $B = q + z$, $\nu = g_r(r) - g_r(q)$, and let $\mathcal{J}$ consist of the later local groups with $q < h < r$. Their total length is $M_{\mathcal{J}}$. We first prove

$$1 + X \geq c_H \sqrt{1+L}, \quad 1 + Y \leq C_H \tau \sqrt{1+L}, \quad M_{\mathcal{J}} \leq C_H Y. \quad (\text{F.1.9})$$

A left completion retains a proper deepest subset in the current block. For a late completion, that block is good and contributes $\sqrt{L}$. For an early completion, the last positive time is at least b , so clipped-prefix charging contributes at least $c_H L$. These alternatives prove the first inequality; rarity then gives the second. For any right completion, compress the cuts between b and the end of $\mathcal{J}$ to the two endpoints, by the same concavity argument as in Section F.1.3. The compressed rank is some $a' \leq q$. If $a' < q$, lowering the current block costs at least $\delta_H L$; if $a' = q$,

filling the intermediate groups costs at least $\gamma_H M_{\mathcal{J}}$. Complete reference nonnegativity thus gives $Y \geq \min(\delta_H L, \gamma_H M_{\mathcal{J}})$. For sufficiently small τ , rarity excludes the first alternative, proving the last claim. These arguments also hold for $z = 1$ and positive q .

Selecting the bank. The endpoint reference minima differ by νL , so $|\nu|L \leq X + Y$. Choose the deterministic interval length

$$\ell = c_{\text{bank}} \frac{L^2}{(1 + X)^2}. \quad (\text{F.1.10})$$

By choosing c_{bank} sufficiently small and then τ sufficiently small, (F.1.9) and rarity imply

$$\ell_{\text{scalar}} \leq \ell \leq L/16, \quad |\nu|\sqrt{\ell} \leq C_H, \quad \frac{1 + Y}{\sqrt{\ell}} \asymp_H \bar{T} := \frac{(1 + X)(1 + Y)}{1 + S^\sigma L}.$$

Indeed rarity gives $1 + X \leq C_H \tau L / (1 + Y)$ and hence $\ell \geq c_H \tau^{-2}$. Neither ℓ nor its location is selected from the sampled word.

The necessary test from the rectangle. Group the deepest q names and the true compound child into the category B , keeping the other names separate. Reveal the complete coarse word in $\mathcal{J}$, and put

$$H_{\mathcal{J}} = \sum_{q < h < r} \{N_h \log(h + z) - K_{B,h} \log B - (h - q)L_h\}.$$

Read the current interval backwards and set

$$W = \log S - \log B \mathbf{1}_B, \quad C(t) = H_{\mathcal{J}} + \sum_{x < t} W_x - (r - q)t.$$

The original rate is S^σ and its mean drift is $-\nu$. Use full original sizes for the earlier exterior coordinates, the shorter deadlines for the intermediate coordinates, and the original full rectangle for the B coordinates. The unresolved exterior budget is the full budget less $(r - q)t + \sum_{\mathcal{J}} (h - q)L_h$. The observed negative log-probability is the mark sum in the risk expression. Formula (F.1.1) therefore gives the actual partial capacity $C_H \pi_t e^{C(t)}$ Shot $_t$. Stop at the first mesh failure of $C(t) \geq -h - A \log(1 + t)$, including $t = 0$. Section F.1.1 bounds the union of failures by $CS e^{-h}$ with all required moments.

Because $H_{\mathcal{J}} \leq (\log H)N_{\mathcal{J}}$ pointwise, every survivor satisfies the necessary test

$$\sum_{x < t} W_x - (r - q)t \geq -h - C_H N_{\mathcal{J}} - A \log(1 + t), \quad 0 \leq t \leq \ell. \quad (\text{F.1.11})$$

First make this implication with the entire intermediate word fixed, then integrate that word with its original mass. The right side now reads only $N_{\mathcal{J}}$; any future private test remains in the joint surviving-word operator.

The fractional moment. Let Q_h be the conditional private probability of (F.1.11). The signed logarithmic-wall one-end estimate, on the original clock with bounded scaled drift, gives

$$\mathbb{E}_{G_i} Q_h^\sigma \leq C_H \min \left\{ 1, \left(\frac{1 + h + C_H N_{\mathcal{J}}}{\sqrt{\ell}} \right)^{2\chi(\sigma, \rho)} \right\}, \quad \rho = \frac{(\mathbb{E}W)^2}{\mathbb{E}W^2}.$$

The intermediate count is independent of this original local clock, with mean at most $C_H M_{\mathcal{J}} \leq C_H Y$. Its fixed moments are at most $C_p(1 + Y)^p$. Integrating over all its counts proves

$$\mathbb{E} Q_h^\sigma \leq C_H (1 + h)^K \bar{T}^{2\chi(\sigma, \rho)}. \quad (\text{F.1.12})$$

Matching the profile parameter. To compare with the exact-type parameter, write $a_0 = \log S$, $b_0 = \log B$, $c = r - q$, $\lambda = S^\sigma$, $\mu = \lambda L$, and $p = B/S$. Its centered type parameter and correlation are

$$\bar{p} = (a_0 - c/\lambda)/b_0, \quad \bar{\rho} = \frac{(cL/(\mu + 1))^2}{(cL/(\mu + 1))^2 + b_0^2 \bar{p}(1 - \bar{p})}.$$

The identity $\bar{p} - p = -\nu/(b_0\lambda)$ gives $|\bar{p} - p| \leq C_H(|\nu| + 1/L)$. As $\bar{T} \geq c_H/\sqrt{\ell}$, $|\log \bar{T}| \leq C_H + \frac{1}{2} \log \ell$. The scaled-drift bound and $\ell \leq L/16$ bound the product of this logarithm and the correlation error. Compact local Lipschitz continuity of χ therefore permits either parameter in (F.1.12) at fixed comparison cost. This includes every count and all $n + 1$ conditional spacings. Fixed insertions change the reserve by at most a constant per point; the Mecke argument of Section F.1.1 attaches the required shot powers at the same exponent σ .

Zero output with a large left score. For a proper $r \rightarrow 0$ edge, assume explicitly $1 + X \geq c_X \sqrt{1 + L}$. The child weight satisfies $z \geq 2^{u_{\min}} > 1$, so the two marks remain nondegenerate. Here $\mathcal{J}$ consists of all later local groups. At a right completion, postpone the current cuts to their individual deadlines. The complete earlier history is unchanged, and the difference of reference costs is $\sum_{0 < h < r} [-g_h(h)] L_h \geq \gamma_H M_{\mathcal{J}}$. Thus $M_{\mathcal{J}} \leq C_H Y$ directly. The preceding proof applies with $B = z$, $c = r$, and the child category in place of the deepest suffix. In particular it supplies precisely the same moment (F.1.12). The left-score hypothesis is needed: there is no current proper-good group when the output is zero.

Disjoint inputs and the shared count. Rank separation makes the current-plus-intermediate group sets disjoint for distinct rare edges of a plateau; they are also disjoint across plateaux. Their label intervals have length at most $L/16$ and avoid a count middle in the same block. If the count middle lies in an intermediate group, its weight is decreasing in that count and the necessary reserve is increasing. For independent copies N, N' ,

$$2 \operatorname{Cov}(f(N), g(N)) = \mathbb{E}[(f(N) - f(N'))(g(N) - g(N'))] \leq 0$$

when f is decreasing and g increasing. Applying this with all other groups fixed supplies the required separation without an independence assertion for the shared count.

F.1.6 Leaving a terminal suffix unread

The final root integral requires a suffix independent of all prefix mask inputs. Some terminal binary tests initially read counts just inside the chosen suffix. We remove those whole intermediate groups from its beginning. Their short total length leaves the largest block and the intrinsic ordinary factor comparable.

All terminal binary weak blocks lie in the negative ordinary-drift head. The ordinary-suffix geometry gives, before trimming,

$$D_0 \geq b_H V_{\text{old}} + g_0 L_{\text{head}}, \quad U \geq c_H T_{\text{total}}, \quad M_* \geq c_H T_{\text{total}}/H.$$

A terminal necessary test may read intermediate counts within this suffix. If so, start the suffix at the first capacity at most its output q . This is an original group boundary, so the removed segment consists of whole original blocks. Its length satisfies

$$\delta \leq M_{\mathcal{J}} \leq C_H Y \leq C_H \tau \sqrt{1 + L} \leq C'_H \tau L.$$

For several tests use the latest one; rank separation puts all earlier read sets before it. The old head contains its current length L . Choosing the fixed τ small enough, bounded ordinary drifts

give $E_H \delta \leq D_0/4$, $\delta \leq D_0/4$ and $\delta < M_*/2$. Since a whole block of length M_* cannot lie in a segment of length less than $M_*/2$, the largest block and the earliest choice in a tie are retained. Thus

$$U' = U - \delta \asymp U, \quad M'_* = M_*, \quad A'_* = A_* - \delta, \quad B'_* = B_*, \quad D' = D_0 - \sum_{\text{removed}} e_h L_h \asymp D_0.$$

The new suffix still has $D' \geq c_H V'_{\text{pre}}$ and $U' \geq c_H T_{\text{total}}$. Moreover $1 + D' + A'_*$ is comparable to $1 + D_0 + A_*$, so its ordinary factor is unchanged up to constants. Every clock read by the necessary masks now belongs to the actual prefix of this suffix. When no terminal binary rare edge is present, no trimming is needed.

F.1.7 Whole-band estimates with two variable reserves

The hard-plateau kernels need a count band together with two signed end tests. The band is summed before its fractional power is taken. A middle maximum atom supplies its width factor; end banks supply the persistence factor. When one reserve is larger than diffusive, only the opposite, smaller-reserve bank is needed.

Fix a compact nondegenerate positive binary law W , a compact positive rate ν and speed c , and a power σ in a compact subinterval of $(0, 1)$. Suppose $M, L, R \geq 1$ and the band width r satisfies

$$1 \leq r \leq C\sqrt{M}, \quad r \leq CR, \quad |\nu \mathbb{E}W - c|M \leq C(L + R). \quad (\text{F.1.13})$$

All interval locations are deterministic. Use a middle interval of length comparable to M , disjoint from the retained end intervals. An additional later-side count N_B is independent, with Poisson mean at most CL . Let the positive left test have reserve at most $C(1 + h + N_B)$ and the reflected negative right test have reserve at most $C(1 + h + R + (1 + |j|)r)$. Both allow a fixed logarithmically lowered wall. After fixing the entire outside private word, assume the whole band permits at most Cr values of the middle binary count.

Put $T = LR/M$ and $\rho_{\text{nat}} = (\mathbb{E}W)^2/\mathbb{E}W^2$. There is a numerical majorant $B_{h,j}(G)$ for the conditional probability of the band and its retained tests such that

$$\mathbb{E}B_{h,j}^\sigma \leq C(1 + h)^K(1 + |j|)^K(r/\sqrt{M})^\sigma \min\{1, T^{2\chi(\sigma, \rho_{\text{nat}})}\}. \quad (\text{F.1.14})$$

Here the entire permitted count band is summed before the power. *The band and end-bank estimate.* If $\max(L, R) \leq C_0\sqrt{M}$, retain end intervals of length $M/16$. Otherwise retain only the smaller-reserve end, of length $\ell = \min(M/16, c_*M^2/\max(L, R)^2)$. Conditioning on all outside labels, the middle count is binomial, so its maximum atom gives the pointwise majorant

$$B_{h,j} = \min\{1, Cr/\sqrt{1 + N_{\text{mid}}}\} q_{\text{left}} q_{\text{right}},$$

with an omitted end factor equal to one. The bound is uniform in the outside-dependent center. Integrate those outside labels afterwards. The Poisson negative moment supplies $C(r/\sqrt{M})^\sigma$, including small middle counts. On every retained end the scaled drift is bounded by (F.1.13). Thus the signed one-end estimate gives

$$\mathbb{E}q_H^\sigma \leq C \min\{1, [(1 + H)/\sqrt{1 + \ell}]^{2\chi_c}\}, \quad \chi_c = \chi\left(\sigma, \frac{(c/\nu)^2}{(c/\nu)^2 + \text{Var } W}\right).$$

When both ends are retained, their independence and the moments of N_B give $\min(1, (LR/M)^{2\chi_c})$ with the displayed reserve polynomials. The inequality $r \leq CR$ controls every translated right-band reserve. When one end is retained, divide its reserve by $\sqrt{\ell}$; the result is bounded by the

same polynomials times LR/M , so the same path factor follows. If ℓ is bounded, $\max(L, R) \geq cM$ and $T \geq c$; discard the path test at fixed cost while retaining the middle atom. On $T < 1$ the drift hypothesis implies $|\chi_c - \chi(\sigma, \rho_{\text{nat}})| \leq C(L + R)/M \leq 2CT$. The replacement costs at most $e^{C|T \log T|}$. This proves (F.1.14); for $T \geq 1$ both numerical path factors equal one.

Atomic prime clocks. The same bounds hold on actual prime clocks by a pointwise quantile comparison. On a deterministic physical interval of length ℓ , its harmonic cumulative mass is $t + O_H(1)$ for every initial length t , with the same statement for half-open intervals and their reflections. The harmonic-to-physical quantile map therefore satisfies $|T(s) - s| \leq C_H$. Its image of a homogeneous Poisson process is the required atomic prime process, with repeated occurrences allowed. A positive-jump wall is tested at its pre-jump minimum; a negative wall at its post-jump minimum. During the collapse of any atom the homogeneous partial sums dominate these minima up to the bounded clock error. An actual necessary wall is consequently contained in the homogeneous wall with a fixed reserve enlargement, for every mark word and every atom multiplicity. Logarithmic allowances change by only a fixed amount.

The homogeneous estimate applies to the full count law, as follows from Theorem 5.16. Let $m = \lfloor \nu\ell/2 \rfloor$. The probability of fewer than m arrivals by ℓ is exponentially small. Otherwise survival implies a rank wall through the first m independent exponential gaps. For the positive sign, survival at the j th arrival implies $cT_j \leq w_+j + h + A \log(1 + T_j)$ and hence $T_j \leq C(j + h + 1)$. For the negative sign, when this latter bound fails the signed risk is already nonnegative; in the complementary case the same logarithmic comparison applies. In either direction the rank-wall reserve is $C(1 + h)$ with a fixed logarithmic allowance. Theorem 5.16 gives $C(1 + h)^{2\chi}(1 + \ell)^{-\chi}$, and the exponentially small exception is absorbed. This proof includes all original counts.

Fixed-width bands and insertions. At a first hit we use the following fixed-width version. Split a deterministic interval $[0, z)$ into its left quarter, middle half, and right quarter. Given both end words, a fixed terminal band admits at most a fixed number of middle low counts. Hence, uniformly over every band center b ,

$$\sup_b q_b(G) \leq C(1 + N_M)^{-1/2} \ell_h(G_L) r_{r_0}(G_R),$$

and independence of these three deterministic restrictions gives

$$\mathbb{E}(\sup_b q_b)^\sigma \leq C(1 + h)^{2\chi}(1 + r_0)^{2\chi}(1 + z)^{-\sigma/2 - 2\chi}. \quad (\text{F.1.15})$$

The terminal band contributes just its middle atom, without an additional height factor. The bound is uniform for z in a prescribed comparable range. In a stopping cover we sum over those deterministic z locations with their original weights.

Insertion of q fixed occurrences decreases the middle negative count factor, relaxes the positive left reserve by at most qw_+ , and makes the negative right wall stronger. The same middle-binomial argument can leave the original middle marks free after conditioning on the inserted marks. Thus both displayed bounds remain valid with that left reserve shift. To attach shots, for $\beta = 4/\sigma_{\min}$ form

$$C_z(G) = \sum_{h, r_0 \geq 0} (1 + h)^{-\beta} (1 + r_0)^{-\beta} (1 + N_M)^{-1/2} \ell_h(G_L) r_{r_0}(G_R).$$

Fractional subadditivity and (F.1.15) show $\mathbb{E}C_z^\sigma \leq C(1 + z)^{-\sigma/2 - 2\chi}$, since each reserve exponent is at most -3 . Reindexing after insertion gives $C_z(G + q \text{ points}) \leq C_q C_z(G)$. Expanding an integrably weighted fixed-degree shot polynomial into Poisson occurrence partitions and applying Mecke now preserves this power of $1 + z$. The expansion is valid for atomic intensities as well.

For the proper zero-output application, take the separate scales $L = 1 + B$ and $R = 1 + V_{\text{old}} + A + \varrho$ from (F.1.8) and band width ϱ . They satisfy all of (F.1.13), including $\varrho \leq CR$. If $d\sqrt{M} < 1$, then $\varrho \asymp \sqrt{M}$ and $R \geq c\sqrt{M}$; the left-only calculation gives the same bound with a width factor comparable to one, uniformly as $d \downarrow 0$.

F.1.8 Joint count weights and the conditional routing tower

We now combine the simple-bank estimates with the proper count weights. There are two distinct operations. Averaging over the uncolored clocks gives the product of the local profile factors; backward conditioning on private routes converts those costs into geometric capacities. The latter operation uses bounds uniform in every preceding atom, so future child masks are integrated before a current count middle is varied.

For each nonterminal plateau choose a fixed middle in a largest local block, write n_j for its count, and set $M_j = (1 + d_j\sqrt{1 + n_j})^{-1}$. For each simple edge let $Q_i(h)$ be its necessary-test probability. The only possible overlap between a count middle and these tests is when the middle lies in an intermediate set $\mathcal{J}_i$. With all other clocks fixed, $M_j^{\sigma_j}$ decreases in n_j and $Q_i^{\sigma_{o(i)}}$ increases. The covariance inequality proved above extracts $\mathbb{E}M_j^{\sigma_j}$. Each middle lies in at most one such set by rank separation. The remaining test inputs are disjoint original groups, so

$$\mathbb{E} \left[\prod_{j < m} M_j^{\sigma_j} \prod_i Q_i(h)^{\sigma_{o(i)}} \right] \leq \prod_{j < m} \mathbb{E}M_j^{\sigma_j} \prod_i \mathbb{E}Q_i(h)^{\sigma_{o(i)}}.$$

Splitting a Poisson count below and above half its mean gives

$$\mathbb{E}M_j^{\sigma_j} \leq C_H(1 + d_j\sqrt{1 + \mu_{C_j}})^{-\sigma_j},$$

uniformly even as $d_j \downarrow 0$. Together with (F.1.12) this gives all proper count and simple local factors at their respective plateau powers. Fixed insertions only decrease M_j and shift each reserve by a fixed amount; the same Mecke argument attaches the permitted shots. One unweighted count insertion pays its original mean.

We next turn these environmental costs into a conditional capacity bound. Fix the complete original uncolored word G and let $\mathcal{Q}_j$ reveal its first j coarse plateau routes. In a local block of plateau j , every earlier plateau has already expired; its earlier routes are deterministically child. Thus on every $\mathcal{Q}_{j-1}$ atom its current labels still have probabilities $1/(r + z_j)$ for each exterior name and $z_j/(r + z_j)$ for the child, independently over original slots. Let E_j avoid the count middle and satisfy $\pi_G(E_j \mid \mathcal{Q}_{j-1}) \leq p_j(G)$ uniformly on those atoms; for the terminal plateau assume the corresponding conditional bound on E_m . Local-only masks have this property.

For any uncolored-measurable thresholds ξ_j , put $\zeta_j = \eta_j - \xi_j$. Condition on the entire current coarse route outside its middle. Then E_j is fixed and

$$\zeta_j = H_0 - \log z_j K_j, \quad K_j \sim \text{Bin}(n_j, z_j/(r + z_j)),$$

where H_0 may depend arbitrarily on the actual incoming list. The binomial maximum atom and the two exponential half-lattice sums give, uniformly in H_0 and also for $n_j = 0$,

$$\begin{aligned} \mathbb{E}[e^{\zeta_j} 1_{\{\zeta_j \leq 0\}} 1_{E_j} \mid \mathcal{Q}_{j-1}] &\leq CM_j p_j, \\ \mathbb{E}[e^{-d_j \zeta_j} 1_{\{\zeta_j > 0\}} 1_{E_j} \mid \mathcal{Q}_{j-1}] &\leq CM_j p_j. \end{aligned}$$

For the negative side use the geometric sum with ratio $e^{-\log z_j}$; for the positive side use $e^{-d_j \log z_j}$ and cap the bound by one. At $d_j = 0$ the probability bound suffices. The bound is uniform in the arbitrary center, so the outside route can now be integrated. Future child words remain to be integrated by the backward tower.

Backward conditioning now bounds every product of positive half-line factors followed by one negative half-line factor by $C^j \prod_{l \leq j} M_l p_l$, and the fully surviving branch by $C^{m-1} p_m \prod_{l < m} M_l p_l$. For completeness set $a_j = M_j p_j$ for $j < m$, $a_m = p_m$, $r_j = \sigma_j / \sigma_1$, and

$$B_0 = \sum_{j \leq m} r_j \log a_j, \quad \theta_j = r_j^{-1} \sum_{l > j} r_l \log a_l, \quad \xi_j = X/r_j + \theta_j.$$

Stop the actual projected route at the first $\eta_j \leq \xi_j$. The original capacity is $C_H \pi_G(q) e^{P_j}$, and the carry identity gives on this branch

$$e^{P_j} = e^{X+c_j} e^{\xi_j} \prod_{l < j} e^{-d_l \xi_l}, \quad c_j = \theta_j - \sum_{l < j} d_l \theta_l = B_0 - \sum_{l \leq j} \log a_l.$$

The terminal branch has $c_m = -\sum_{l < m} d_l \theta_l$ and the same last expression. Summing stopped capacities and using the backward estimates proves for a surviving set $\mathcal{S} \subseteq \bigcap_j E_j$ that

$$I(\mathcal{S}) \leq C_H \min\{1, e^X \prod_j a_j^{r_j}\}, \quad (\prod_j a_j^{r_j})^{\sigma_1} = \prod_{j < m} M_j^{\sigma_j} \prod_j p_j^{\sigma_j}.$$

If a p_j vanishes, the conditional bound makes the finite surviving support empty, and no logarithm is required. For $m = 1$ the direct full-word capacity proves the assertion. This calculation preserves the actual descendant lists and is the simple-plateau tower used in the causal proof; hard plateaux supply their own three conditional kernels in Lemma 12.4.

F.1.9 The ordinary rectangle and the same-count root integral

We prove Proposition 12.9. The ordinary factor has two sources: a local root atom and a ballot denominator from the largest-block rectangle. Both depend on the reservoir count, so we estimate them jointly before averaging outside counts. Positive layer integration then combines the rectangle with an arbitrary prefix translate and with the polynomial outgoing cover profile.

The proof proceeds through three operations: construct the rectangle minimum, establish two Poisson mode bounds on its largest-block count, and sum the translated root lattice in each layer. Write the unread original suffix as blocks with $h_1 > \dots > h_m \geq 2$, $w_i = \log h_i$, $c_i = h_i - 1$, lengths $L_i \geq 0$, and $U = \sum_i L_i$. Let $0 < t \leq s < 1$, $r = t/s$, in the fixed compact ranges. With the entire prefix fixed, take arbitrary $D > 0$ and $\kappa \in \mathbb{R}$, and set

$$Z = \sum_i (w_i N_i - c_i L_i), \quad X = \kappa + rZ.$$

After the positive enlargement, the N_i are independent Poisson counts with means comparable to L_i . The natural means are $h_i^t L_i$. Conditional on each count there are its original uniform positions, with all $n_i + 1$ gaps. Completed harmonic-cell budgets are used consistently in Z , and their bounded difference from physical budgets is included in κ , keeping X unchanged.

Choose a largest original block $M = L_{i_*}$ and put

$$A = \sum_{i < i_*} L_i, \quad B = \sum_{i > i_*} L_i, \quad B_{\text{act}} = \sum_{i > i_*} (w_i N_i - c_i L_i), \quad A_{\text{act}} = - \sum_{i < i_*} (w_i N_i - c_i L_i).$$

Then $Z = B_{\text{act}} + Z_* - A_{\text{act}}$ and $M \leq U \leq mM$. The two outside sums are independent and retain all their counts.

The rectangle minimum. Three actual rectangles are used before enlargement. A threshold at the beginning of the suffix assigns its primes freely and bounds earlier contributions by their full original rectangles, giving $I \leq Ce^Z$. A threshold within the largest block, with its j smallest prime logarithms retained, gives $I \leq Ce^{B_{\text{act}}} W_{*,N_*}$, where $W_{*,n}$ is the capped ordered-sum rectangle of (11.11). To verify this, the $h_* - 1$ unresolved widths are bounded by the prefix scale times $1 + \sum_{l \leq j} e^{(w_*/c_*) \text{cell}_l}$, whereas the free later labels contribute h_*^{-j} after normalization. Minimize over j , and use the right endpoint rectangle for the cap one. The remaining rectangle is $I \leq C$. Thus, together with a prefix cap $I \leq CDe^X$, the same tuple obeys

$$I \leq C\hat{V}, \quad \hat{V} = \min\{1, e^Z, e^{B_{\text{act}}} W_{*,N_*}, De^X\}. \quad (\text{F.1.16})$$

For every $n \geq 0$ and $\ell \geq 0$, Lemma 11.6 gives

$$\mathbb{P}(W_{*,n} > e^{-\ell} \mid n) \leq \min \left\{ 1, \frac{C(1+\ell)(C_1 + Z_* + \ell)_+}{1+n} \right\}.$$

For negative ℓ the event is empty.

The root atom with the ballot denominator. Fix all outside counts and restrict Z to an interval $[E, E+W]$ of fixed width. There are at most $1+W/\log 2$ possible N_* . The two full Poisson mode bounds

$$\sup_n p_\mu(n) \leq C(1+\mu)^{-1/2}, \quad \sup_n \frac{p_\mu(n)}{1+n} \leq C(1+\mu)^{-3/2} \quad (\text{F.1.17})$$

follow from Stirling near the mode and the elementary Poisson tails; they include $n = 0$ and all noncentral counts. In the event $e^{B_{\text{act}}} W_{*,N_*} > e^{-\ell}$, the left reserve is $1 + \ell + B_{\text{act}}$ and the right reserve is $C_1 + Z + \ell + A_{\text{act}}$. The endpoint rectangle also requires $Z + \ell \geq 0$. Apply the two bounds in (F.1.17) separately, then average the independent sides. Since $\mathbb{E}(B_{\text{act}})_+ \leq CB$ and $\mathbb{E}(A_{\text{act}})_+ \leq CA$, this gives

$$\begin{aligned} & \mathbb{P}\{Z \in [E, E+W], e^Z > e^{-\ell}, e^{B_{\text{act}}} W_{*,N_*} > e^{-\ell}\} \\ & \leq C(1+U)^{-1/2} \min \left\{ 1, \frac{(1+\ell+B)(1+(E+\ell)_+ + A)}{1+M} \right\}. \end{aligned}$$

The root atom and ballot denominator have already been obtained from the same N_* .

Layer integration at the translated root. Put $H_0 = (-\log D - \kappa)/r$ and $w = Z - H_0$. Then $e^{-sX} = D^s e^{-tw}$ and $De^X = e^{rw}$. Positive layer cake in (F.1.16) gives

$$D^{-s} \mathbb{E}[e^{-sX} \hat{V}] = \int_0^\infty e^{-\ell} \mathbb{E}[e^{-tw} 1_{\{w > -\ell/r\}} 1_{\{Z+\ell \geq 0\}} 1_{\{W_* > e^{-\ell-B_{\text{act}}}\}}] d\ell.$$

At fixed outside counts, $a = w + \ell/r > 0$ runs along a translated lattice with spacing w_* . Moreover $Z + \ell = H_0 + a - (1/r - 1)\ell$, so its nonnegativity forces $a \geq -H_0$ up to a fixed geometric translation. Uniformly over the lattice translate and $b \geq 0$,

$$\begin{aligned} & \sum_{a>0, a \geq -H_0-C} e^{-ta} \leq Ce^{-t(H_0)-}, \\ & \sum_{a>0, a \geq -H_0-C} e^{-ta} (C + H_0 + a + b)_+ \leq Ce^{-t(H_0)-} (1 + (H_0)_+ + b). \end{aligned}$$

For $H_0 < 0$, shifting the lattice by $(H_0)_-$ gives the displayed exponential factor. For $H_0 \geq 0$, the sum begins at zero. The negative deficit thus appears only in that exponential factor. Using first the first mode bound and then the second with the ballot tail bounds the conditional integrand before $e^{-\ell}$ by the minimum of

$$Ce^{s\ell-t(H_0)-}(1+M)^{-1/2}, \quad Ce^{s\ell-t(H_0)-}(1+M)^{-3/2}[1+\ell+(B_{\text{act}})_+][1+(H_0)_++(A_{\text{act}})_+].$$

Average the independent sides separately in these two bounds, then take their minimum. Since $1-s$ has a fixed positive lower bound, integration against $e^{-(1-s)\ell}$ yields

$$\mathbb{E}[e^{-sX}\hat{V}] \leq CD^s e^{-t(H_0)-}(1+U)^{-1/2} \min \left\{ 1, \frac{(1+B)(1+(H_0)_++A)}{1+M} \right\}. \quad (\text{F.1.18})$$

Missing sides contribute exactly one; an empty suffix is treated by the deterministic minimum with the same 1+length conventions.

The polynomial outgoing profile. The outgoing profile needed in the causal theorem is $J_\beta(X + \log D)$ instead of De^X , where $J_\beta(v) = \min\{1, C_\beta e^v(1+(-v)_+)^{\beta}\}$ and $\beta \geq 0$ is fixed. If $J_\beta(v) > e^{-\ell}$, then

$$v > -L_\beta(\ell), \quad L_\beta(\ell) = \ell + 2\beta \log(2 + \ell) + C'_\beta.$$

Indeed $(1+u)^\beta \leq Ce^{u/2}$ first gives $u = (-v)_+ \leq 2\ell + C$, which substituted back gives the stated bound. Now set $q = r(Z - H_0) + L_\beta(\ell) > 0$. The tilt becomes $D^s e^{sL_\beta(\ell)} e^{-sq}$ and $Z + \ell \leq H_0 + q/r$. The endpoint cap still forces $q \geq r(H_0)_-$ up to a fixed translation. The same lattice sums give $e^{-t(H_0)-}$ and the same right reserve. The only change in the layer integration is the weight

$$e^{-\ell+sL_\beta(\ell)} \leq Ce^{-(1-s)\ell}(2+\ell)^{2s\beta},$$

whose integral, also multiplied by $1+\ell$, is uniformly finite. This proves (F.1.18) with the polynomial profile and hence Proposition 12.9.

Averaging the prefix deviation. For the natural suffix put $D_{\text{mean}} = \sum_i L_i(h_i^t \log h_i - (h_i - 1)) \geq 0$ and

$$\mathcal{B}_I = (1+U)^{-1/2} \min\{1, (1+B)(1+D_{\text{mean}}+A)/(1+M)\}.$$

The preceding bound also gives

$$\mathbb{E}[e^{-sX}\hat{V}] \leq CD^s \mathcal{B}_I e^{-t(H_0)-} \left[1 + \frac{(H_0 - D_{\text{mean}})_+}{1 + D_{\text{mean}} + A} \right].$$

The root identity is $\kappa = -rD_{\text{mean}} + \Delta_{\text{pre}}$, so $H_0 - D_{\text{mean}} = -(\Delta_{\text{pre}} + \log D)/r$. This is the explicit prefix deviation weight used in the final upper expectation. It remains together with D^s under that expectation; the conditional estimate allows arbitrary correlation among prefix costs, counts and geometric sums.

F.2 Conditional plateau tower, normalized capacities and shot moments

For clarity, we isolate the finite conditional mechanism behind the ordered cover. Half-line kernels compose by backward conditioning; the chosen offsets make every stopping capacity have the same pooled cost; and polynomial floors place a hard right stop after the middle of its block. A relative Poisson–Mecke argument then attaches the shared shots while preserving each plateau’s power.

F.2.1 The conditional plateau filtration

Fix the whole uncolored environment G . Let π_G be its product fine-label law, $Q_0 \subset \dots \subset Q_m$ its plateau-route filtration, and

$$\sigma_1 > \dots > \sigma_m > 0, \quad r_j = \sigma_j / \sigma_1, \quad d_j = r_j / r_{j+1} - 1 \quad (j < m).$$

Let the grouped outputs Z_j be Q_j -measurable and define

$$\eta_j = \sum_{l \leq j} \frac{r_l}{r_j} Z_l, \quad P_j = \sum_{l \leq j} Z_l, \quad X = \sum_{l \leq m} r_l Z_l.$$

Assume X is uncolored and thus fixed under π_G . Directly expanding the sums gives the identities

$$P_j = \eta_j - \sum_{l < j} d_l \eta_l, \quad P_m = X / r_m - \sum_{l < m} d_l \eta_l. \quad (\text{F.2.1})$$

For thresholds ξ_j fixed by G , put $\zeta_j = \eta_j - \xi_j$. Let E_j be the current plateau mask. Suppose the following bounds hold on every preceding plateau atom:

$$\begin{aligned} \mathbb{E}_{\pi_G}[e^{\zeta_j} \mathbf{1}_{\zeta_j \leq 0} \mathbf{1}_{E_j} \mid Q_{j-1}] &\leq C b_j, \\ \mathbb{E}_{\pi_G}[e^{-d_j \zeta_j} \mathbf{1}_{\zeta_j > 0} \mathbf{1}_{E_j} \mid Q_{j-1}] &\leq C b_j, \end{aligned} \quad j < m, \quad (\text{F.2.2})$$

and $\mathbb{P}_\pi(E_m \mid Q_{m-1}) \leq C b_m$. All $b_j = b_j(G)$ are fixed uncolored numbers. Then any product of positive half-line factors followed by one negative half-line factor has expectation at most $C^j \prod_{l \leq j} b_l$. The fully continuing terminal branch is at most $C^m \prod_{l \leq m} b_l$.

Proof. Write the terminal negative factor as g_j and the earlier positive factors as f_l . Since $\prod_{l < j} f_l$ is Q_{j-1} -measurable, the tower property gives

$$\mathbb{E} \left[g_j \prod_{l < j} f_l \right] = \mathbb{E} \left[\mathbb{E}(g_j \mid Q_{j-1}) \prod_{l < j} f_l \right] \leq C b_j \mathbb{E} \prod_{l < j} f_l.$$

Apply the second bound of (2) successively in reverse plateau order. The same calculation starts from $\mathbf{1}_{E_m}$ for the terminal branch. Throughout this calculation the uncolored environment stays fixed under its original law. A future mask is bounded uniformly before the current middle is integrated, leaving that middle with its original binomial law on each preceding atom. $\square$

The local product law on preceding atoms, required in the simple-plateau kernel, is also exact. Original block i can assign its primes only to names with indices $c \geq i$ or to the unused name. If it is local to plateau j , all preceding exterior names have smaller indices and are ineligible. Therefore all their routes are deterministically child on those slots. Conditioning any preceding atom does not observe a random label in a current local block; it leaves its original categorical product law. This supplies the local-law assertion used in the simple kernels of Section 12.

F.2.2 Offsets and stopping capacities

The offsets distribute the future pool costs among the current thresholds. Their telescoping identity makes each stopping branch carry exactly the pool factors already encountered.

Choose arbitrary positive uncolored pools a_j and set

$$B = \sum_j r_j \log a_j, \quad \theta_j = r_j^{-1} \sum_{l>j} r_l \log a_l, \quad \xi_j = X/r_j + \theta_j.$$

Define

$$c_j = \theta_j - \sum_{l<j} d_l \theta_l \quad (j < m), \quad c_m = - \sum_{l<m} d_l \theta_l.$$

Then an exact coefficient calculation gives

$$c_j + \sum_{l \leq j} \log a_l = B \quad (1 \leq j \leq m). \quad (\text{F.2.3})$$

Indeed $d_l/r_l = 1/r_{l+1} - 1/r_l$; summing this identity determines the coefficient of every $\log a_l$ on the left to be exactly r_l . Using (1), a complete first stop at plateau $j < m$ has exponent

$$P_j = X + c_j + \zeta_j - \sum_{l<j} d_l \zeta_l. \quad (\text{F.2.4})$$

If each complete projected atom has geometric capacity at most $C\pi_G(\text{atom}) \exp(P_j)$, (2) and (4) bound the total capacity of that branch by

$$C e^{X+c_j} \prod_{l \leq j} b_l. \quad (\text{F.2.5})$$

For the fully continuing terminal branch the same bound follows from $P_m = X + c_m - \sum_{l<m} d_l \zeta_l$ and its terminal mask estimate.

A hard internal stop has the same conclusion whenever its established conditional capacity bound on the preceding atom is

$$C\pi_G(\alpha) e^{P_{j-1}+x_j} b_j, \quad x_j = \xi_j - \eta_{j-1}/u_{j-1}.$$

This is because

$$P_{j-1} + x_j = X + c_j - \sum_{l<j} d_l \zeta_l.$$

The preceding positive half-line kernels are then applied on the original ancestor fibers, exactly as for a complete stop.

Suppose all left-failure covers have total capacity at most $CS \exp(-h)$, and $b_j(h) \leq a_j(1+h)^\beta$. There are only C_H complete or internal branch types. Equations (3)–(5) then prove

$$I_{\text{actual}} \leq C \min\{1, S e^{-h} + e^{X+B} (1+h)^{m\beta}\}. \quad (\text{F.2.6})$$

This is the passage to (12.40). The geometric capacities provide the branch exponents; the conditional tower integrates their private masses on the original fibers.

F.2.3 Choice of the gate from preceding data

A right stop must occur late enough to leave deterministic end banks and a count middle. The offset identity removes all future pool scales from the stopping level, so the gate can be chosen from the current and preceding lengths alone.

Let $L_S = 1 + \log(2S)$, $\beta_* \geq m\beta$, $D = L_S^{\beta_*} \exp(B)$, and $w_0 = X + \log D$. When $w_0 < 0$, take

$$h = \lceil \log(2S) - w_0 \rceil.$$

At a hard plateau whose earlier routes continued, $\zeta_l > 0$. Its dual left reserve is $h_{\text{eff}} = h + \min(P_{j-1}, 0)$. Hence

$$h_{\text{eff}} + x_j \leq h + P_{j-1} + x_j = h + X + c_j - \sum_{l < j} d_l \zeta_l \leq h + X + c_j. \quad (\text{F.2.7})$$

Since $h + X + B \leq \log(2S) + 1$, (3) and $\log \alpha_h \leq 0$ imply

$$h_{\text{eff}} + x_j + \log \alpha_h - \log S \leq C - \sum_{l \leq j} \log a_l. \quad (\text{F.2.8})$$

A floor $a_l \geq (2 + T_l)^{-D_l}$ gives a bound involving *only prior and current lengths*. If the hard-edge geometric dominance gives $T_l \leq CM$ for every $l \leq j$, the right side of (8) is at most $C + D_{\text{gate}} \log(2 + M)$.

For $0 \leq w \leq M/2$, the right wall lies below the left wall whenever

$$h_{\text{eff}} + x_j + \log \alpha_h - \log S \leq A_1 \log(1 + M - w) - A_0 \log(1 + w).$$

The right side is at least $(A_1 - A_0) \log(1 + M/2)$. Since $\log(2 + M)/\log(1 + M/2)$ is bounded for $M \geq 1$, choosing A_1 once larger than $A_0 + CD_{\text{gate}} + C'$ proves that a right-governed first hit cannot occur before $M/2$. The scalar polynomial exponent K and pool exponent β may then be chosen after A_1 ; they do not occur in (8) except through the already eliminated nonnegative $\beta_* \log L_S$. The floors come from the numerical profile's polynomial lower bounds and can be chosen before A_1 . This gives the stated order of the constants.

Using this admissible h in (6),

$$Se^{-h} \leq \frac{1}{2}e^{w_0}, \quad e^{X+B}(1+h)^{m\beta} \leq Ce^{w_0}(1+(-w_0)_+)^{\beta_*}.$$

For $w_0 \geq 0$, the cap $I_{\text{actual}} \leq C$ suffices. This proves the claimed polynomial rising profile once its local geometric capacities and half-line kernels are available.

F.2.4 Relative Poisson shot attachment preserves plateau powers

The shared shot may read the same clocks as the rare local events. To preserve their profile factors, we need insertion bounds relative to those events. Expanding a fixed-degree shot into occurrence partitions and applying Mecke supplies exactly such bounds.

Let independent Poisson restrictions N_j represent disjoint local plateau banks, with deterministic intensity measures. Suppose a nonnegative functional $f_j(h, N_j)$ satisfies, for every fixed number q of inserted occurrences and all their deterministic positions,

$$f_j(h, N_j + \text{insertions})^{\sigma_j} \leq C_q f_j(h + C_q, N_j)^{\sigma_j}, \quad \mathbb{E} f_j(h, N_j)^{\sigma_j} \leq C(1+h)^K F_j. \quad (\text{F.2.9})$$

Finite sums of deterministic bank-position expressions are permitted: apply fractional subadditivity before their deterministic position sums. Assume a nonnegative shot S is a finite-degree positive polynomial in integrable Poisson shot sums, with the integrals of every occurrence partition of the required degree bounded uniformly in the physical lengths. Then for every fixed $p \geq 0$ and every plateau subset J ,

$$\mathbb{E} \left[S^p \prod_{j \in J} f_j(h_j, N_j)^{\sigma_j} \right] \leq C_p \prod_{j \in J} (1 + h_j)^K F_j. \quad (\text{F.2.10})$$

Proof. Put $r = \lceil p \rceil$. By $S^p \leq 1 + S^r$, it is enough to estimate the zeroth moment and one integer moment. Expand the positive polynomial for that moment. In each monomial, partition the occurrence indices according to equality of Poisson points. The number of partitions is finite, and each uses a bounded number q of distinct points. The multivariate Mecke identity changes its expectation into an integral over these q deterministic inserted positions of the original expectation of the product of functionals with those insertions. Apply (9) to each plateau, then independence of the original disjoint local restrictions. The product expectation is bounded by $C_q \prod (1 + h_j)^K F_j$; the fixed reserve enlargement does not change this order. Every remaining partition integral is bounded by the shot hypothesis. Summing the finitely many positive terms proves the bound at the original plateau powers. $\square$

The actual shots in Section F.1 have the form

$$Z(v) = 1 + \sum_{x \leq v} e^{x-v}, \quad S = \sum_v \omega(v) Z(v)^d \prod_c Z(\tau_c).$$

The local Poisson intensity is bounded on unit intervals, including atomic prime intensities. Every occurrence-partition integral of a product of the exponential kernels is bounded: integrate one kernel against its bounded-unit-mass intensity and sum its exponentially decreasing unit-interval tails. The outside endpoint locations are deterministic; $\sum_v \omega(v)$ is bounded for the stated sufficiently large fixed decay exponent. Products and finite moments are handled by positive expansion and Minkowski before the position sums. This verifies the occurrence-partition integrability required in (10), in addition to the ordinary bound $\mathbb{E} S^p \leq C$.

A raw prefix count N_{pre} is attached by one additional Mecke point. Its intensity integral is its mean, bounded by $C V_{\text{pre}}$. This proves the corresponding bound with $1 + N_{\text{pre}}$ and factor $C(1 + V_{\text{pre}})$.

For a count middle overlapping an intermediate reserve, use a decreasing weight $f(N)$ and increasing reserve bound $g(N)$. Their covariance inequality holds for every distribution of N : with an independent copy N' ,

$$2 \text{Cov}(f(N), g(N)) = \mathbb{E}[(f(N) - f(N'))(g(N) - g(N'))] \leq 0.$$

After this extraction the other deterministic bank inputs are disjoint. This justifies the product-moment step because the proved rank-separation geometry gives that each middle lies in at most one intermediate reserve group. The required insertion estimates are obtained by conditioning the finitely many inserted private marks and deleting those occurrences at a bounded reserve enlargement; middle inverse count factors decrease under insertion. In a fixed-width band the original middle marks may remain unobserved, so its maximum atom is unchanged. Arbitrary band centers have already disappeared from the max-atom bound.

G Scalar large deviations of conditional survival

The positive persistence spectrum describes Laplace transforms of the conditional survival cost. To describe the cost itself, we also need inverse survival moments. Their finite domain has a boundary determined by the environmental-to-private variance ratio. We prove existence and regularity of the limiting pressure throughout that domain, show that its derivative reaches every positive cost, and then derive the full scalar large-deviation principle by a direct change of measure.

There are two new ingredients beyond the positive-order theory. Reverse hypercontractivity identifies the inverse-moment domain, and a covariance interpolation supplies the product inequality needed for existence at those orders. Convex Gaussian curvature controls the tilted variance. These bounds also identify the almost-sure conditional cost and the scalar value selected by each environmental tilt.

G.1 Pressure and the rate function

Throughout this appendix, $0 < \rho < 1$ is fixed and X, Y are the independent stationary standard OU processes of (5.1). Put

$$\beta^2 = \frac{\rho}{1-\rho}, \quad S_t = -\log q_t(X), \quad \theta_c = \beta^{-2}.$$

Stationarity at time zero and equality of the two covariance functions are part of the model throughout this appendix. They permit the common whitening and the endpoint comparison used below. Every continuous environmental path is bounded on $[0, t]$, and a private OU path in a sufficiently high fixed tube has positive probability. Consequently $q_t > 0$ almost surely. The initial constraint gives $q_t \leq \Phi(\beta X_0) < 1$, so $0 < S_t < \infty$.

Theorem G.1 (Pressure and full scalar large deviations). *For every $\theta < \theta_c$, the finite limit*

$$\Lambda_\rho(\theta) = \lim_{t \rightarrow \infty} \frac{1}{t} \log \mathbb{E} e^{\theta S_t} \quad (\text{G.1.1})$$

exists. It is nondecreasing, convex and belongs to $C_{\text{loc}}^{1,1}((-\infty, \theta_c))$. For every $t > 0$, $\mathbb{E} e^{\theta S_t} = \infty$ precisely when $\theta \geq \theta_c$. Moreover,

$$\Lambda_\rho(\theta) \rightarrow \infty, \quad \Lambda'_\rho(\theta) \rightarrow \infty \quad (\theta \uparrow \theta_c), \quad \Lambda'_\rho(\theta) \rightarrow 0 \quad (\theta \rightarrow -\infty). \quad (\text{G.1.2})$$

The variables S_t/t satisfy a full large-deviation principle at speed t , with good convex rate function

$$I_\rho(y) = \sup_{\theta < \theta_c} \{\theta y - \Lambda_\rho(\theta)\}. \quad (\text{G.1.3})$$

Thus, for every Borel set $B \subset \mathbb{R}$,

$$\begin{aligned} -\inf_{B^\circ} I_\rho &\leq \liminf_{t \rightarrow \infty} t^{-1} \log \mathbb{P}(S_t/t \in B) \\ &\leq \limsup_{t \rightarrow \infty} t^{-1} \log \mathbb{P}(S_t/t \in B) \leq -\inf_B I_\rho. \end{aligned} \quad (\text{G.1.4})$$

The rate is finite for every $y > 0$ and infinite for $y \leq 0$. Its unique zero is $\gamma_\rho = \Lambda'_\rho(0)$, and

$$\lim_{y \rightarrow \infty} \frac{I_\rho(y)}{y} = \frac{1-\rho}{\rho}. \quad (\text{G.1.5})$$

G.2 The inverse-moment domain

Whitening turns the finite-grid survival constraint into a convex cone. Reverse hypercontractivity then bounds its negative moments in terms of ordinary survival. The initial-time constraint gives the matching obstruction at the domain boundary. We keep this finite-time domain calculation separate from the later extensive lower bound for pressure.

On a finite ordered grid let Σ be its OU covariance matrix. Simultaneously whitening the environmental and private vectors gives the convex cone $C = \Sigma^{-1/2}\mathbb{R}_+^n$ and

$$q(x) = \mathbb{P}\{Z + \beta x \in C\} = T_{\sqrt{\rho}} \mathbf{1}_C(x), \quad x, Z \sim N(0, I_n),$$

where $T_r f(x) = \mathbb{E}f(rx + \sqrt{1-r^2}Z)$. The second equality uses the cone property. The Gaussian reverse-hypercontractive inequality [6, Section 3.3, equation (3.14)] states

$$\|T_r f\|_s \geq \|f\|_p, \quad s < p < 1, \quad r^2 \leq \frac{1-p}{1-s}.$$

For $s < 0$ the convention is $\|f\|_s = (\mathbb{E}f^s)^{1/s}$, with zero values handled by positive approximation. For $0 < \theta < \theta_c$ choose $s = -\theta$ and $p = 1 - \rho(1 + \theta) > 0$. Raising the norm inequality to the negative power gives

$$\mathbb{E}q_{\text{grid}}^{-\theta} \leq a_{\text{grid}}^{-\theta/[1-\rho(1+\theta)]}, \quad a_{\text{grid}} = \mathbb{E}q_{\text{grid}}.$$

Nested dense grids give $q_{\text{grid}} \downarrow q_t$ and $a_{\text{grid}} \downarrow a(t) > 0$ by continuity. Monotone convergence and (5.4) therefore prove

$$\mathbb{E}e^{\theta S_t} \leq a(t)^{-\theta/[1-\rho(1+\theta)]}, \quad \limsup_{t \rightarrow \infty} t^{-1} \log \mathbb{E}e^{\theta S_t} \leq \frac{\theta}{2[1-\rho(1+\theta)]}. \quad (\text{G.2.1})$$

Nonpositive orders are finite since $S_t \geq 0$.

Conversely, $q_t \leq \Phi(\beta X_0)$. The elementary Gaussian bound $\Phi(-z) \leq (2\pi)^{-1/2} z^{-1} e^{-z^2/2}$ for $z > 0$ shows that $\mathbb{E}\Phi(\beta X_0)^{-\theta}$ dominates a positive constant times

$$\int_1^\infty x^\theta \exp\{(\theta\beta^2 - 1)x^2/2\} dx.$$

This is infinite for $\theta \geq \theta_c$, including equality. This identifies the finite-time moment domain. Divergence of the limiting pressure requires a lower bound proportional to the horizon, which we prove below using a fixed-density time grid.

G.3 A covariance interpolation for existence

For inverse moments we need a product comparison in the direction opposite to the positive-order association bound. We interpolate the single OU correlation joining two blocks. The MTP₂ property of the Gaussian distribution function controls the sign of the resulting derivative throughout the full moment domain.

Divide an ordered finite OU grid into two consecutive nonempty blocks. Its Markov representation is

$$U_{j+1} = r_j U_j + \sqrt{1-r_j^2} G_{j+1}, \quad 0 < r_j < 1,$$

with independent standard Gaussian innovations and stationary U_1 . Replace only the correlation joining the blocks by vr_j , $0 \leq v \leq 1$, and denote the resulting covariance by Σ_v . Within-block covariances are unchanged, cross-block covariances are multiplied by v , and hence

$$\dot{\Sigma}_{ii} = 0, \quad \dot{\Sigma}_{ij} \geq 0 \quad (i \neq j).$$

The precision matrix remains tridiagonal with nonpositive off-diagonal entries. Its Gaussian density is therefore MTP₂; see [11, Section 4.1].

Let $u_v(z) = \mathbb{P}\{N(0, \Sigma_v) \leq z\}$. Its derivatives satisfy

$$u_i \geq 0, \quad \partial_{ij} \log u \geq 0 \quad (i \neq j). \quad (\text{G.3.1})$$

For the second assertion, the relation $w_i \leq z_i$ for all i is closed under coordinatewise minimum and maximum, so its indicator times the Gaussian density in w is MTP₂ in (w, z) . Insert any positive product Gaussian density in z to make the joint integral finite, apply MTP₂ marginalization [11, Proposition 3.4(ii)], and cancel that product density. Thus u_v is MTP₂. Since it is positive and smooth, its two-coordinate lattice inequality gives the second derivative inequality in (G.3.1).

For $\eta > -\theta_c$ put $H_\eta(v) = \mathbb{E}_{U \sim N(0, \Sigma_v)} u_v(\beta U)^\eta$. Differentiation of the environmental Gaussian integral and the private Gaussian heat equation give

$$H'_\eta(v) = \frac{\eta}{2} \sum_{i,j} \dot{\Sigma}_{ij} \mathbb{E}[u^{\eta-2} \{(1 + \beta^2) u u_{ij} + \beta^2(\eta - 1) u_i u_j\}]. \quad (\text{G.3.2})$$

Here all derivatives of $u = u_v$ are evaluated at βU . The expression in braces is exactly

$$(1 + \beta^2) u^2 \partial_{ij} \log u + (1 + \eta \beta^2) u_i u_j.$$

It is nonnegative for $i \neq j$, by (G.3.1) and $\eta > -\beta^{-2}$. Consequently H_η increases for $\eta > 0$ and decreases for $-\theta_c < \eta < 0$.

To justify differentiation at negative orders, we use a slightly larger inverse moment to dominate the derivative integrands. On this fixed grid the covariance path is uniformly positive definite. Lemma 5.7, after whitening, bounds $D^2 \log u_v$ uniformly; its gradient is therefore bounded by a fixed multiple of $1 + |z|$, uniformly in v . Values and gradients at zero are bounded by compactness of the covariance path. Choose $-\eta < \theta' < \theta_c$. The reverse-hypercontractive bound at order θ' is uniform in v : the corresponding orthant probabilities are positive and have a positive minimum on this compact path. Hölder then bounds all polynomial-weighted integrands in (G.3.2). This justifies Gaussian integration by parts and removal of spatial cutoffs. For $\eta > 0$ the same derivative bounds and $u^\eta \leq 1$ suffice.

We now pass from the finite interpolation to the continuous product inequalities. At $v = 0$ the blocks are independent. In two adjacent continuous intervals choose dense grids with no repeated common endpoint; points approaching the endpoint from either side impose it in each marginal limit. For negative powers use monotone convergence and (G.2.1); positive powers use bounded convergence. Writing $M_\theta(t) = \mathbb{E} e^{\theta S_t}$ gives

$$M_\theta(s+t) \leq M_\theta(s) M_\theta(t) \quad (0 < \theta < \theta_c), \quad M_\theta(s+t) \geq M_\theta(s) M_\theta(t) \quad (\theta < 0). \quad (\text{G.3.3})$$

For positive θ , (G.2.1) bounds $0 \leq \log M_\theta(t) \leq C_\theta(t+1)$. For negative $\theta = -p$, Jensen and $q^p \geq q$ when $p \leq 1$ give

$$-\max(1, p)[t/2 + \log \pi] \leq \log M_{-p}(t) \leq 0.$$

The continuous-parameter subadditivity and superadditivity arguments applied to (G.3.3) now prove existence and finiteness in (G.1.1); specifically,

$$\Lambda_\rho(\theta) = \begin{cases} \inf_{t>0} t^{-1} \log M_\theta(t), & 0 < \theta < \theta_c, \\ 0, & \theta = 0, \\ \sup_{t>0} t^{-1} \log M_\theta(t), & \theta < 0. \end{cases}$$

The displayed linear bounds control the remainder in $T = nt + r$, $0 \leq r < t$, giving the continuous-parameter limiting slope. Convexity and monotonicity pass from the finite-time log moment generating functions to their limit.

G.4 Pressure regularity

Throughout the inverse-moment domain, the negative-order tilt is strongly log-concave. The curvature bound from Lemma 5.7 therefore gives a dimension-free variance estimate under every interior tilt. This supplies uniform control of the pressure derivatives as both the time grid and horizon vary.

On a whitened grid write $A = -\log q$ and $d\nu_\theta = e^{\theta A} d\gamma / \mathbb{E} e^{\theta A}$. The potential $V_\theta = |x|^2/2 - \theta A(x)$ has

$$D^2 V_\theta \succeq \kappa_\theta I, \quad \kappa_\theta = \begin{cases} 1, & \theta \leq 0, \\ 1 - \theta\beta^2, & 0 < \theta < \theta_c. \end{cases}$$

The Brascamp–Lieb variance inequality used in Lemma 5.7 and its gradient estimate yield

$$\text{Var}_{\nu_\theta}(A) \leq \kappa_\theta^{-1} \mathbb{E}_{\nu_\theta} |\nabla A|^2 \leq \frac{2\beta^2}{\kappa_\theta} \mathbb{E}_{\nu_\theta} A.$$

Thus, with $L_{t,\text{grid}}(\theta) = t^{-1} \log \mathbb{E} e^{\theta A}$,

$$0 \leq L''_{t,\text{grid}}(\theta) \leq \frac{2\beta^2}{\kappa_\theta} L'_{t,\text{grid}}(\theta). \quad (\text{G.4.1})$$

For $t \geq 1$, the preceding moment bounds control $L_{t,\text{grid}}$ uniformly on every compact subinterval of $(-\infty, \theta_c)$, independently of the grid: $a_{\text{grid}} \geq a(t)$ in the inverse-moment bound, and the same inequality applies to the positive-order bound. Convex secants on a slightly larger compact interval bound $L'_{t,\text{grid}}$ uniformly. Therefore (G.4.1) bounds its second derivative uniformly.

For any fixed t and compact order interval, an interior neighboring order θ' larger than all its orders dominates $A^j e^{\theta A}$, $j = 0, 1, 2$. If $\theta' > 0$, $\mathbb{E} e^{\theta' S_t} < \infty$ by (G.2.1); if $\theta' \leq 0$, the polynomial factors are bounded using the strict order margin. Dominated convergence passes the first two derivatives to the continuous-time function $L_t(\theta) = t^{-1} \log M_\theta(t)$. Its first derivatives are thus locally uniformly bounded and equicontinuous for $t \geq 1$. Arzelà–Ascoli, pointwise convergence of L_t , and $L_t(b) - L_t(a) = \int_a^b L'_t(\theta) d\theta$ prove

$$\Lambda_\rho \in C_{\text{loc}}^{1,1}((-\infty, \theta_c)), \quad L'_t \longrightarrow \Lambda'_\rho \quad \text{locally uniformly.} \quad (\text{G.4.2})$$

In particular the tilted variance of S_t/t is $O(t^{-1})$ on each compact order interval. The conclusion controls the first derivative by a local Lipschitz bound; it does not require a limiting second derivative.

G.5 Both pressure boundaries

To obtain a full scalar LDP, the derivative of the pressure must cover all positive costs. At the finite right boundary we prove an extensive lower bound on inverse moments. At the left boundary we force a high environmental tube, where private survival is arbitrarily likely. These two constructions give the derivative limits needed for the change-of-measure argument.

Take $n = \lfloor t \rfloor + 1$ unit-spaced grid points in $[0, t]$. Their covariance is $\Sigma_{ij} = r^{|i-j|}$, $r = e^{-1/2}$, and write $Q = \Sigma^{-1}$. The eigenvalues of both matrices lie in $[K^{-1}, K]$ for $K = (1+r)/(1-r)$. This follows directly from the tridiagonal OU precision matrix and its row sums. For an environmental vector x with $Qx \leq 0$, put $h = -\beta Qx \geq 0$. Markov's inequality for $e^{h^T Z}$ gives

$$q_{\text{grid}}(x) = \mathbb{P}\{Z \geq -\beta x\} \leq \exp\{\beta h^T x + h^T \Sigma h/2\} = \exp\{-\beta^2 x^T Q x/2\}.$$

Since $q_t \leq q_{\text{grid}}$, for $\delta = 1 - \theta\beta^2 > 0$ we obtain

$$M_\theta(t) \geq \int_{\{Qx \leq 0\}} e^{\theta\beta^2 x^T Qx/2} dN(0, \Sigma)(x) = \delta^{-n/2} \mathbb{P}\{QX \leq 0\}.$$

The last equality uses homogeneity of the cone and Gaussian scaling. Now $QX \sim N(0, Q)$. Its density on $[-1, 0]^n$ is at least $(2\pi K)^{-n/2} e^{-Kn/2}$. Hence

$$\Lambda_\rho(\theta) \geq \frac{1}{2} \log \frac{1}{1 - \theta\beta^2} - C, \quad 0 < \theta < \theta_c, \quad (\text{G.5.1})$$

where C is an absolute constant for this fixed unit grid. This proves pressure divergence at the finite boundary, and convexity then forces $\Lambda'_\rho(\theta) \rightarrow \infty$ there.

At the left boundary, a high environmental tube supplies moments whose logarithmic cost per order can be made arbitrarily small. Fix $a > 0$ and set

$$p_a = \mathbb{P}\{X_s \geq a : 0 \leq s \leq 1\}, \quad r_a = \mathbb{P}\{Y_s \geq -\beta a : 0 \leq s \leq 1\}.$$

These numbers are positive, and $r_a \uparrow 1$ as $a \rightarrow \infty$. Association on consecutive unit intervals implies

$$M_{-p}(t) \geq p_a^{[t]} r_a^{p[t]}, \quad p > 0.$$

Consequently $0 \leq \Lambda_\rho(-p)/(-p) \leq -\log r_a - (\log p_a)/p$. Letting $p \rightarrow \infty$ and then $a \rightarrow \infty$ gives $\Lambda_\rho(-p)/(-p) \rightarrow 0$. For a nondecreasing convex function finite on a left half-line, this sublinear asymptotic implies $\Lambda'_\rho(-\infty) = 0$.

G.6 The full LDP bounds

We prove the lower bound by tilting to each prescribed positive cost. The tilted variance estimate gives concentration there. For the upper bound, local Chernoff estimates cover compact sets and exponential tightness removes compactness. We then verify the rate at the nonpositive endpoint and its large-cost slope.

Choose $0 < \theta_0 < \theta_c$. Since $S_t/t \geq 0$, Chernoff's bound

$$\limsup_{t \rightarrow \infty} t^{-1} \log \mathbb{P}(S_t/t > R) \leq \Lambda_\rho(\theta_0) - \theta_0 R$$

proves exponential tightness. The continuous derivative and its limits in (G.1.2) show that every $y > 0$ equals $\Lambda'_\rho(\theta)$ for an interior θ . Convexity gives $I_\rho(y) = \theta y - \Lambda_\rho(\theta) < \infty$. Under the tilted environmental law

$$\frac{d\mathbb{P}_{\theta,t}}{d\mathbb{P}} = e^{\theta S_t - tL_t(\theta)},$$

the mean of S_t/t tends to y , by (G.4.2), and its variance is $L''_t(\theta)/t = O(t^{-1})$. It follows that $\mathbb{P}_{\theta,t}(|S_t/t - y| < \epsilon) \rightarrow 1$. Changing measure back gives

$$\liminf_{t \rightarrow \infty} t^{-1} \log \mathbb{P}(|S_t/t - y| < \epsilon) \geq -I_\rho(y) - |\theta|\epsilon.$$

For any open set containing y , let $\epsilon \downarrow 0$ inside that set and then optimize over its positive points.

At $y < 0$, the sublinear left asymptotic of Λ_ρ makes $I_\rho(y) = \infty$. Also $I_\rho(0) = \sup_{p>0} \chi(p, \rho) = \infty$, by the unboundedness of the positive spectrum proved in Theorem 5.8. Thus the rate is infinite at every nonpositive point, and the open-set lower bound is completely determined by the positive points.

For the upper bound, fix a compact set K and a number $a < \inf_K I_\rho$. At each $y \in K$ choose an interior parameter θ_y with $\theta_y y - \Lambda_\rho(\theta_y) > a$. The strict inequality persists on a sufficiently small neighborhood of y . Chernoff's bound on that neighborhood has upper rate at most $-a$. A finite subcover proves the compact-set upper bound. Exponential tightness extends it to every closed set, including those with infinite rate infimum. The function I_ρ is convex and lower semicontinuous; for $y \geq 0$ it dominates $\theta_0 y - \Lambda_\rho(\theta_0)$, while it is infinite on $y \leq 0$. Its sublevel sets are therefore compact. This proves the full LDP in (G.1.4).

Finally, differentiability at zero shows that I_ρ has its unique zero at $\gamma_\rho = \Lambda'_\rho(0)$. For $y \geq \gamma_\rho$, convexity gives $\Lambda_\rho(\theta) \geq \gamma_\rho \theta$, so negative θ cannot improve the dual value at zero. For $0 \leq \theta < \theta_c$, $\Lambda_\rho(\theta) \geq 0$; therefore $I_\rho(y) \leq \theta_c y$. Every fixed $\theta < \theta_c$ gives $I_\rho(y)/y \geq \theta - \Lambda_\rho(\theta)/y$. Let $y \rightarrow \infty$ and then $\theta \uparrow \theta_c$ to prove (G.1.5).

G.7 Quenched slope, scalar duality and entropy

Differentiability at zero identifies the ordinary conditional cost, and the exponential concentration bounds strengthen convergence in probability to almost-sure convergence. More generally, each interior order tilt selects the corresponding pressure derivative. Convex duality describes this selection and expresses its relative entropy as a scalar large-deviation cost.

Corollary G.2. *The quenched exponent satisfies*

$$\frac{S_t}{t} \longrightarrow \gamma_\rho \quad \text{almost surely}, \quad \frac{\mathbb{E} S_t}{t} \longrightarrow \gamma_\rho, \quad \frac{1}{2} + \frac{\rho}{16\pi} \leq \gamma_\rho \leq \frac{1}{2(1-\rho)}. \quad (\text{G.7.1})$$

Define the extended spectrum by $\tilde{\chi}(p, \rho) = -\Lambda_\rho(-p)$ for $p > -\theta_c$; it agrees with χ for $p > 0$ and vanishes at $p = 0$. Then

$$\tilde{\chi}(p, \rho) = \inf_{y \geq 0} \{I_\rho(y) + py\}, \quad I_\rho(y) = \sup_{p > -\theta_c} \{\tilde{\chi}(p, \rho) - py\}. \quad (\text{G.7.2})$$

The infimum has the unique minimizer $y_p = \Lambda'_\rho(-p) = \partial_p \tilde{\chi}(p, \rho) > 0$. Under the pure environmental tilt $d\nu_{p,t} = q_t^p d\mathbb{P}/\mathbb{E} q_t^p$, the variables S_t/t have the full LDP with good rate

$$I_{p,\rho}(y) = I_\rho(y) + py - \tilde{\chi}(p, \rho). \quad (\text{G.7.3})$$

Moreover,

$$\lim_{t \rightarrow \infty} \frac{1}{t} \text{D}_{\text{KL}}(\nu_{p,t} \| \mathbb{P}) = I_\rho(y_p) = \tilde{\chi}(p, \rho) - p \partial_p \tilde{\chi}(p, \rho). \quad (\text{G.7.4})$$

Proof. We first identify the quenched slope and its bounds, then establish the dual variational formulas. The tilted LDP follows by shifting the pressure, and the entropy identity follows from its order derivative.

The mean convergence is (G.4.2) at zero. Differentiability there and Chernoff's bound with small positive and negative orders give, for each $\epsilon > 0$, a constant $c_\epsilon > 0$ such that $\mathbb{P}(|S_t/t - \gamma_\rho| > \epsilon) \leq e^{-c_\epsilon t}$ for all sufficiently large t . Borel–Cantelli proves convergence along integers. Monotonicity of S_t supplies the continuous-time conclusion. Jensen and (5.4) give $\gamma_\rho \geq 1/2$; the upper bound follows by taking the right derivative at zero in (G.2.1). For $0 < p \leq 1/2$, Theorem 5.9 gives $\chi(p, \rho)/p \geq 1/2 + \rho/(16\pi)$; let $p \downarrow 0$.

For any interior $\theta = -p$, the supporting-line inequality for Λ_ρ gives $I_\rho(\Lambda'_\rho(\theta)) = \theta \Lambda'_\rho(\theta) - \Lambda_\rho(\theta)$. It also proves that this value is the unique minimizer in the first formula of (G.7.2): equality forces y into the singleton subgradient of Λ_ρ at θ . The second formula is just (G.1.3) with $\theta = -p$. For

$p > 0$, positivity of y_p follows from strict increase of the positive spectrum; for $p \leq 0$, it follows from $\Lambda'_\rho(-p) \geq \gamma_\rho > 0$.

The tilted limiting cumulant is

$$\Lambda_{p,\rho}(\theta) = \Lambda_\rho(\theta - p) - \Lambda_\rho(-p), \quad \theta < \theta_c + p.$$

It has the same regularity, variance bounds and derivative boundary limits as the pressure already treated, and its finite right boundary is positive. The identical exponential-tightness and tilt argument therefore gives a full LDP. Taking its convex dual yields (G.7.3). Finally the exact finite-time identity is

$$D_{\text{KL}}(\nu_{p,t} \| \mathbb{P}) = -p \mathbb{E}_{\nu_{p,t}} S_t - \log \mathbb{E} q_t^p.$$

Derivative convergence proves (G.7.4). This entropy rate belongs to the pure environmental OU tilt. The arithmetic accepting measures are constructed separately and are outside the scope of this identity. $\square$

Remark G.3 (Brownian normalization). The Lamperti representation $X_u = e^{-u/2} B_{e^u}^X$, $Y_u = e^{-u/2} B_{e^u}^Y$ transfers these statements to Brownian random-wall survival on $[1, R]$, with speed $\log R$. The law at the left endpoint is the one induced by Brownian time 1; the appendix does not assert the same finite-time moment domain for prescribed deterministic initial values.

H Constants and the length threshold

The profile uses a single length threshold depending only on the dimension. This appendix makes that choice explicit and checks that all earlier geometric, probabilistic, and arithmetic requirements can be met in a consistent order. The constants are deliberately generous; their role is to establish uniform dependence on the dimension. Harmonic stability then extends the estimate to vanishing lengths.

The verification has three stages. First we bound the primitive coefficients, history sums, and exponent sensitivity. Next we choose the proof parameters, followed by the regularity intercept, endpoint reserve, arithmetic grid, and final length. Finally we explain why the ordered-sum constant enters only the output comparison and prove the equivalence of the lifted and original harmonic profiles.

H.1 Coefficient bounds, scalar sensitivity and pivot sums

The following majorants organize the deterministic coefficients. They also distinguish polynomial degrees from multiplicative probability constants: a large comparison constant need not enlarge the degree of a history weight.

Put $H = k + 1$,

$$A_H = 10^6 H^{3H+30}, \quad B_0 = 10H, \quad B_{j+1} = A_H(B_j + 1) \quad (j < H^2),$$

$$U_H = \max\{(10H)^{10H}, (2A_H)^{H^2+1}\}, \quad Q_H = (100U_H)^{16^{100H^4}}, \quad M_0 = 2^{2^{10000H^5}}, \quad M_1 = M_0^{1000}. \quad (\text{H.1.1})$$

For $H \geq 2$, elementary logarithms give $Q_H < M_0$. Indeed $\log U_H \leq 100H^3 \log(10H)$ and $\log_2(16^{100H^4}) = 400H^4$, while $10000H^5 > 400H^4 + \log_2[100H^3 \log(10H) + \log 100]$. The double exponential in (1) dominates the finite sequence of powers arising from repeated applications of Young's inequality.

H.1.1 Primitive constants

The table records uniform lower bounds for positive coefficients and upper bounds for their sizes. Its denominators involve only the dimension, so the bounds remain valid when lengths or strict plateau gaps have very different scales.

Calculation and location	Explicit bound
Fine probabilities and transported coefficients, Section 3, Section 4	$p_* \geq H^{-H}$, $\kappa_* \geq H^{-H-1}$, rates at most H
Product compactness, Section 4	$\sigma \geq \alpha_2 > 1/2$, $1 - \sigma \geq 1/(10H^3)$
Nondegenerate proper child, Section 3	$z \geq \sqrt{2}$, $z \leq H - 1$, $\log z > 1/3$
Split/loss coefficients, Section D.1	absolute loss at most $H \log H$; absolute contrast at most $3H \log H$
Complete reference, Section A.2.4	nonnegative complete comparison costs; local/old rank capacities are integers in $[1, H]$
Clipped-prefix derivative and charge, Section A.2.8	$d_0 \geq 1/(100H^{H+6})$, $c_{\text{ref}} \geq 1/(1000H^{H+6})$
Local hierarchy, Section 4	$V_{\text{old}} \leq 16H^6 T_{\text{local}}$
Proper nontrivial mixed entropy, Section D.1	per-slot gap at least $2H^{-4H}$, rate gap at least $2H^{-5H}$
Pure child lifetime, Section D.1	$\gamma_z > 1/(4H)$, resulting linear gap at least $H^{-H-1}/(4H)$
Terminal float, Section D.1	rank second difference at least H^{-H-2} ; retained slope at least $H^{-H-1}/(2H)$
Atomic spread, Section D.1	curvature charge at least $H^{-H-1}/(2H)$ times the spread
Original pivot reassignment, Section D.1; independent token proof	at most $H - 1$ tokens, depth inflation coefficient at most H^2
Positive endpoint representative, Section D.1	representative weight at least $(H + 1)^{-H}$, marginal at least $1/(H + 1)$

Every positive entry and its reciprocal is bounded by U_H . Within an eligible cell, a retained fine letter has conditional probability $p_c/\mathbb{P}(\text{cell}) \geq p_c$, so the same lower bound H^{-H} applies. None of these bounds uses a strict adjacent-product gap.

The pure lifetime row uses $(z - 1) \log(1 + 1/z)$ with $z \geq \sqrt{2}$, $z \leq H$; since $\sqrt{2} - 1 > 2/5$, $\log(1 + 1/H) \geq 1/(H + 1)$, and $H/(H + 1) \geq 2/3$, its lower bound exceeds $4/(15H) > 1/(4H)$. The U_H bound therefore covers the quarter coefficient without relying on a limiting estimate.

H.1.2 Selected local and old weak edges

The endpoint formulas in Section A.2.4 give the following explicit weak-edge bounds. In its notation use $e_0 = \epsilon_H \geq 1/(10H^3)$ and

$$\begin{aligned} \nu &= e_0/H \geq 1/(10H^4), & \eta &= 1/(4H^2), \\ \gamma_w &= \log 2/(48H^2) \geq 1/(100H^2), \end{aligned}$$

$$\mu = \min(\nu, \gamma_w) \geq 1/(100H^4).$$

The wrong-output constants obey

$$\begin{aligned} c_{\text{curv}} &= \frac{\min(1, 2\nu/\log H)}{2H^2} \geq 1/(10H^7), \\ c_{\text{wrong}} &= \frac{7 \min(1, \mu/[2H(1 + \log H) + \log H])}{16H^2} \geq 1/(2000H^8). \end{aligned} \quad (\text{H.1.2})$$

The proper-input two-cut exchange pays $L_i/(4H^2)$. The local weak threshold can be at least $1/(1000H^4)$ by the explicit entropy formulas above. Choose the even smaller threshold $\epsilon_w = M_1^{-100}$. Then every nonexceptional local weak edge has both reference endpoints at least $L_i/(2000H^8)$. The possible local exception is exactly the full-input, deepest-output edge; its actual source wall is retained, so no inverse of its vanishing drift is taken.

For old edges, the same comparison gives $\beta_H = 1/[4H(1 + \log H)^2] \geq 1/(16H^3)$. Its full-input charge is at least $V_{\text{old}}/(32H^3)$; its proper-input charge is at least $H^{-H-1}L_i/(4H)$. The same ϵ_w is below both required thresholds $\beta_H/4$ and $\kappa_*/(8H)$. Nonweak edges are charged from the favorable endpoint with gap at least ϵ_w . These bounds keep the selected-edge constants uniform as a local critical drift, output drift, or adjacent saddle-product difference tends to zero.

The local weak-order comparison is an integer-capacity/secant comparison, with normalized-secant derivative at least $1/(24H^2)$; zero-output proper edges use $z \geq \sqrt{2}$. The sole terminal zero-output edge is ordinary and is omitted.

H.1.3 Endpoint/count coefficients and root-separated chords

The count recurrence bounds the errors at every exposed group. Reference charges absorb those errors and leave a definite fraction of each intrinsic chord. The root keeps coefficient one throughout this comparison.

On the coupled source the own-group count errors are bounded by the recurrence in (1). Each count step has at most H eligible labels and at most H earlier groups. Its smallest unread split variance is at least $H^{-(H+1)}T/16$, and the hierarchy bounds the old error by $4H^4B_j\sqrt{T}$. The split coefficient is at least $(\log 2)/2$; therefore the factor A_H covers each diagonal selection and each displaced-count adjustment. At most H^2 such steps occur. The binomial/multinomial probabilities are evaluated separately by Stirling; they are not denominators in the deterministic endpoint coefficient bounds.

The shifted intrinsic common-pair coefficient can be taken at least M_0^{-20} , with upper coefficient at most M_0^{20} , after the fixed reserve exceeds M_0^{20} . This follows from the exact early/tail cancellation, the good-group displacement, and one Young inequality against c_{ref} per exposed group; there are at most H groups. The all-zero and full-deadline rows are exact. These inputs are proved in Section A.2.12, including the terminal unread-suffix refinement. Its scales are the h -leading S_v^I , not the physical profile's leading-one scales. The two selections and their final comparison are made explicit in Section A.4: the lower imposes every weak canonical wall, including saturated ones; the upper's rare cutoff is fixed before h . All physical count compactness and correlation replacement bounds are proved there without a reciprocal h -dependent rarity threshold.

The root is retained with coefficient one in Section D.1. Its bounded rounding is divided by $r_j \geq 1/2$, never by a strict product gap. For allowed targets $y_j \leq D_0 \log(2 + T_j)$ the root-prefix representative comparison may use an extra polynomial in D_0 , because its reserve difference is

at most $2HD_0 \log(2 + V_{\text{pre}})$. The old-input reference charges just listed pay that difference. The coefficient therefore includes this dependence on D_0 .

With $D_0 = M_1^{10}$, its additional coefficients are below M_1^{100} . Choose

$$\theta_{\text{intr}} = \epsilon_{\text{mix}} = M_1^{-M_1}, \quad \theta_{\text{glob}} = 1/(4H^2). \quad (\text{H.1.3})$$

These choices are below the partial-child/root-separated chord fraction times θ_{glob} , including the D_0 factor. The partial-child alternatives in Section D.3 use either a linear cL_i vertex charge or this actual intrinsic chord; the same empirical bridge is used in each branch. The maximum-weight vertex step costs only $(H + 1)^H$. After all finite shares, the retained positive coefficient can safely be bounded below by

$$\delta_H = M_1^{-M_1^2}. \quad (\text{H.1.4})$$

Its actual primitive positive factors involve at most H^3 entries from the primitive-constant table and at most H^3 source/mixing/bank shares, each with reciprocal at most $M_1^{10M_1}$. Their product inverse is smaller than $M_1^{20H^3M_1}$, which is below $M_1^{M_1^2}$. Thus (4) is conservative even without assuming that only a few tiny shares are multiplied. A parameter approaching a merge face causes no denominator in this calculation.

H.1.4 Explicit control of all REG/interpolation errors

We next absorb the regularity and interpolation errors into the available linear and inward-curve charges. The important point is that the number of coefficient operations along a dependency branch is bounded in terms of the dimension, even though pivot locations range over a long interval.

The shorter-end REG calculation bounds one empirical bridge by

$$|\mathcal{E}_e(t)| \leq 4H^2 \log H \{\vartheta d(t) + K[1 + d(t)^\zeta]\}. \quad (\text{H.1.5})$$

For the mixed contrast, at most $2H$ constant-partition intervals per plateau occur, with each per-label coefficient at most $3H \log H$; its error coefficient is at most $12H^3 \log H$. Projection moves at most $H - 1$ original boundaries and token depth gains have coefficient at most H^2 ; concavity of the powers propagates these factors without any endpoint-sized count. Synchronization changes maxima/minima by the sum of at most H^2 spreads. Thus every coefficient of ϑ -length or $K \cdot \text{depth}^\zeta$ before finite budget shares is below U_H^4 .

Take $\zeta = 1/16$, $\gamma = 1/8$ and $\vartheta = M_1^{-M_1^3}$. It is less than one hundredth of every retained linear charge divided by its regularity coefficient. The concrete absorption formulas are

$$av^{1/16} \leq \frac{1}{2}bv^{1/8} + a^2/(2b), \quad av^{1/16} \leq \frac{1}{2}bv + 4b(1 + a/b)^2, \quad a\sqrt{v} \leq \frac{1}{4}bv + a^2/b. \quad (\text{H.1.6})$$

The middle bound follows from $v^{1/16} \leq 1 + \sqrt{v}$ and $a\sqrt{v} \leq bv/4 + a^2/b$: its remainder $a + a^2/b$ is bounded by the displayed constant for all $a, b > 0$. Source and target inequalities use at most H^3 group/boundary charges; applying these absorptions successively never creates a new depth exponent. The original pure-lifetime ζ -power versus a linear charge uses exponent $16/15 < 2$; original depth transfer into inward γ uses exponent 2. Even allowing power 16 at every step is therefore safe for the bound Q_H in (1).

Here the finite-operation count is per coefficient formula, not the number of lawful endpoint contexts. Each original history has at most $H - 1$ cuts; projection moves at most $H - 1$ tokens; each of at most H plateaux has at most H physical groups. Synchronization, partial-child alternatives, endpoint budget shares and original depth transfer therefore require at most $100H^4$ coefficient operations along any dependency branch. Different lawful contexts are bounded by the same branchwise maximum. Taking their maximum does not raise the bound to another power. A sum over the at most $H!H^H2^H$ endpoint templates contributes only that dimension-only multiplier, already below U_H . Original pivot position summation is the separate convergent one-dimensional series, not an iteration of coefficient definitions indexed by its number of terms.

After the small proof parameters are installed, their reciprocals are at most $M_1^{M_1^4}$. If a branch starts from $100M_1^{M_1^4}(1+K)(1+h)$ and each of its at most $100H^4$ steps raises the running majorant to at most power 16, its final bound is at most

$$[100M_1^{M_1^4}(1+K)(1+h)]^{16^{100H^4}} < (M_1Kh)^{M_1^{10}}, \quad (\text{H.1.6a})$$

for $K, h \geq 2$. The inequality follows by comparing exponents: $(M_1^4 + 10)16^{100H^4} < M_1^{10}$. It also includes the inverse retained charge δ_H^{-1} and the inverse upper rare cutoff $\tau_0^{-1} \leq 4HM_1^{100} + 800H^5$; no h -dependent rare cutoff is used. Applying (6a) without the final h gives the REG-only bound below. Thus the final intercept is bounded by the displayed finite arithmetic recurrence.

The terminal flag error is $A \log(2+T)d/L$. At $L \geq T/4$ its coefficient of d^γ is at most $A \log(2+4L)L^{-\gamma}$. At $L < T/4$, the clipped-reference charge is at least $c_{\text{ref}}T/4$ and pays $A \log(2+T)$. For $t \geq 1$, $\log(2+4t) \leq 100t^{1/16}$. The first coefficient is therefore below $\delta_H/100$ whenever

$$L \geq (10^4 A / \delta_H)^{16}.$$

The second reference share is available whenever

$$T \geq (4 \cdot 10^4 A / (\delta_H c_{\text{ref}}))^{16/15}.$$

With $A \leq 2HD_0$, $D_0 = M_1^{10}$, and $\delta_H = M_1^{-M_1^2}$, the logarithms of both cutoffs are below M_1^{100000} . Thus both cutoffs are at most M_2 .

H.1.5 Target recoloring and the final intercept bound

Recoloring contributes a second kind of error: the target displacement can depend on the preceding data. The deterministic cascade below bounds that displacement for every admissible target, so it remains valid for adaptive choices.

The exact target cascade in Section A.2.3 has $b_j e_j - A_j e_{j-1} = y_j + \delta_j$, $0 \leq \delta_j < b_j$, with $b_j > 1/3$, $A_j \leq H$. Hence

$$e_j \leq (4H)^H \{H + D_0 H \log(2 + T_j)\} \leq M_1^{100} \log(2 + T_j). \quad (\text{H.1.7})$$

For $i < j$, the hierarchy gives $T_i \leq C_H T_j$, so

$$\log(2 + T_i) \leq \log(2 + T_j) + \log C_H.$$

At the specified lift, $\log C_H \leq \frac{1}{4} \log(2 + T_j)$. The recurrence with $b_j^{-1} < 3$ and $A_j \leq H$ then gives the factor $(4H)^H$ in (7). It is a deterministic bound for every admissible target vector, including one chosen from the prefix data. Each flip changes any loss by at most $4H \log H$. The deep-bank

margin may be chosen as the smaller fraction $\rho_{\text{bank}} = M_1^{-10M_1}$. Image regularity only uses (7), the linear macro-length margin or ρ_{bank}^ζ endpoint-depth margin; all have explicit coefficients at least powers of $M_1^{-10M_1}$. The scalar cost cutoff/penalty can use $b_0 = M_1^{-M_1^3}$, $C_{\text{cut}} = M_1^{M_1^4}$. All these requirements are paid at the final lift in the fixed-lift rule in Section H.2.

Set $M_2 = \exp(M_1^{100000})$, $K = \exp(\exp(M_2^{5000}))$ and scalar reserve $h = \exp(\exp(M_2^{6000}))$, as in that rule. Combining the preceding coefficient estimates and (6), every deterministic error coefficient or intercept excluding the added reserve is bounded by $(M_1 K)^{M_1^{10}}$. Thus it is below $h_W = \exp(\exp(M_2^{5000}))$. Terms that read the added reserve, including root-separated endpoint shifts, are instead bounded by $(M_1 K h)^{M_1^{10}}$, which is below the explicit

$$C_{\text{geom}}^{\text{final}} = \exp\{\exp(M_2^{6500})\}. \quad (\text{H.1.8})$$

The final source intercept includes terms depending on h , whereas h_W bounds only the regularity terms chosen earlier. The geometric argument carries the logarithmic density cost of the larger intercept. At the fixed lift $\Lambda = \lceil \exp(\exp(M_2^{10000})) \rceil$, (8) is smaller than $\delta_H \sqrt{\Lambda}/100$. The numerical grid defect is at most M_2^{7000} , and the same inequality pays it.

The probability choices precede this geometric intercept. Path and regularity retention are fixed at each exact type before target transport and the arithmetic first-mass argument, using the constants of the scalar estimates. Thus these choices do not depend on the rare arithmetic acceptance probability that they help to estimate.

The displayed tables and inequalities bound the coefficients in the source, classification, and history comparisons proved above. They supply the finite deterministic inputs to the explicit fixed-lift construction.

H.1.6 Scalar exponent sensitivity uses the physical margin

The exponent sensitivity is controlled by the physical variance margin. This calculation keeps the degree of the depth polynomial separate from the large multiplicative probability constant M_2 . For a finite whitened Gaussian grid write $q = T_s f$, $\exp(-s) = \sqrt{\rho}$, $0 \leq f \leq 1$. The one-dimensional Gaussian rearrangement bound gives

$$|\nabla q| \leq \sqrt{\rho/(1-\rho)} \phi(\Phi^{-1}(q)).$$

For every $0 < q < 1$,

$$\phi(\Phi^{-1}q)^2 \leq 2q^2(1 - \log q). \quad (\text{H.1.9})$$

One direct proof puts $q = Q(t)$, $m(t) = \phi(t)/Q(t)$. The truncated normal variance gives $m'(t) = m(t)(m(t) - t) \leq 1$. Hence the derivative of $2(1 - \log Q(t)) - m(t)^2$ is $2m(t)(1 - m'(t)) \geq 0$; its limit at minus infinity is 2. This proves (9) without an unspecified Mills constant.

For the plateau range $0 < \eta < 1$, let $F = \mathbb{E}q^\eta$. Noise differentiation, Gaussian integration by parts, and convexity of $\log \mathbb{E}q^\eta$ in η give

$$0 \leq \partial_s \log F \leq \frac{2(1-\eta)\rho}{1-\rho}(\eta - \log F), \quad -\partial_\eta \log F \leq -\log F/\eta.$$

Since $ds/d\rho = -1/(2\rho)$, divide by the horizon, use the exact ordinary OU survival comparison, and pass to the infinite-horizon rate. The resulting local Lipschitz inequalities (derivatives when they exist, otherwise their corresponding increment bounds) are

$$|\partial_\rho \chi| \leq \frac{(1-\eta)\chi}{1-\rho}, \quad |\partial_\eta \chi| \leq \chi/\eta. \quad (\text{H.1.10})$$

For plateau powers $1/2 \leq \eta < 1$, these bounds are at most $1/[2(1-\rho)]$ and 1. Finite-grid approximation is monotone; the bounds are uniform in the grid dimension. The additive η /horizon term tends to zero.

At a selected actual local type the count compactness gives $p \in [1/(2H), 1 - 1/(2H)]$, $\tau \in [1/(2H), 2]$, $b \geq \log(2)/2$, $b \leq \log H$. Both original and contracted speeds are in $[1/(4H), 2H \log H]$. Thus the environmental variance is at least $1/(64H^4)$, the private variance is at least $1/(100H)$, and their sum is at most $20H^2(1 + \log H)^2$. Consequently both ρ and its complement are at least

$$\epsilon_{\text{phys}} = 1/(10^5 H^{10}). \quad (\text{H.1.11})$$

This conservative floor covers the contracted case as well as the canonical centered model. Equations (10)–(11) bound the physical exponent sensitivity by $10^5 H^{10} < U_H$; the type-error constants come from the deterministic count recurrence and are bounded by U_H . In particular the replacement $O(T + (1+d)/n)$ has an explicit coefficient below U_H^4 . The log-ratio contribution is a fixed constant, using $T|\log T| \leq 1/e$ and $\log(2+n)/\sqrt{1+n} \leq 2$. Its dependence on d comes only from $(1+X+d)(1+Y+d) \leq (1+d)^2(1+X)(1+Y)$ and the finite history polynomial weights. Their degree is covered by $Q_H < M_0$, hence by M_1 . The very large one-end probability comparison constant is not a power of $(1+d)$ and is not assigned to D_0 .

H.1.7 Original-pivot series and the active-depth cutoff

The token reduction leaves one convergent series for each original pivot. We bound these series explicitly, then verify that the active depth cutoff lies inside the permitted bank. For $0 < b \leq 1$, $B \geq 0$, $\gamma = 1/8$, monotone interval comparison and $u = t^{1/8}$ give

$$\sum_{m \geq 0} (1+m)^B e^{-bm^{1/8}} \leq 2^{B+10} (1+b^{-8(B+1)}) \Gamma(8(B+1)+1). \quad (\text{H.1.12})$$

For noninteger B use the larger integer ceiling before this bound. The elementary Gamma inequality $\Gamma(v+1) \leq 2(v+1)^{v+1}$ for $v \geq 1$ follows by writing its integrand as $(t^v \exp(-t/2)) \exp(-t/2)$: the first factor is maximized at $2v$ and is $(2v/e)^v$, and the integral of the second factor is 2. Since $2/e < 1$, this proves the stated bound. One can also use the integer factorial inequality after taking another ceiling. Thus the logarithm of (12) is at most

$$100(B+1)\{1 + \log(B+2) + \log(1/b)\}. \quad (\text{H.1.13})$$

Original unit-strip occupancies have an extra finite prefactor, bounded by a fixed power of $(M_1 Kh)$ by (5). They do not change the degree B in (12). There are at most $H-1$ original tokens and at most H plateaux, so at most C_H products of (12) exhaust original pivot multiplicities, including repeated pivots. For $B \leq M_1^2$ and $b \geq b_0/100 = M_1^{-M_1^3}/100$, the entire logarithmic core-series cost is below M_1^{100} . The additional occupancy prefactor is covered by (6a), including its dependence on h . The original-pivot series is therefore bounded by a dimension-dependent constant, independently of the endpoint size and rare acceptance probability.

For the envelope tail let $\eta \geq 1/2$, $B \leq M_1^2$ and $r_0 = C_{\text{cut}} \log(2+n)$. Substitution $r = d^{1/8}$ in its integral gives a polynomial of degree at most $8B+8$ times $\exp(-b_0 \eta r)$. Since $\log r \leq \sqrt{r}$ for $r \geq 1$, the polynomial is at most $\exp(b_0 r/4)$ as soon as

$$r \geq [100(B+1)/b_0]^2.$$

The explicit $C_{\text{cut}} = M_1^{M_1^4}$ exceeds that threshold and also $100D_0/b_0$, where $D_0 = M_1^{10}$. Hence the summed tail, including its fixed integral prefactor, is bounded by $(2+n)^{-10D_0}$ for $n \geq \Lambda_k/(4H)$. The nontrivial plateau power can therefore be retained; a common final power is not substituted. Also $D_0\eta > 1$ makes the $(2+n)^{-D_0}$ floor smaller than the least possible persistence factor, which is of order $(2+n)^{-1}$. The active depth is precisely $[C_{\text{cut}} \log(2+n)]^8$, already checked to be inside $\rho_{\text{bank}}n/100$ at the specified lift.

The sensitivity bounds and the convergent original-pivot series give the stated coefficient bounds. The ordinary bridge, causal cover, and correlated arithmetic estimates use these same constants.

H.2 An explicit length threshold

For a fixed integer $k \geq 1$, put $H = k + 1$. Here is one explicit rule containing only arithmetic, exponentials and a ceiling:

$$M_0 = 2^{2^{10000H^5}}, \quad M_1 = M_0^{1000}, \quad M_2 = \exp(M_1^{100000}),$$

$$\boxed{\Lambda_k = \left\lceil \exp\{\exp(M_2^{10000})\} \right\rceil.} \quad (\text{H.2.1})$$

The rule has fixed depth and depends only on the dimension. The subsections below verify the geometric margins, count boxes, probability and regularity parameters, and arithmetic grid, in that order. Their final inequalities show that this one threshold meets all the requirements simultaneously.

H.2.1 Numerical complete-chain physical bounds

We begin with the compact ranges for saddle coordinates, fine-label probabilities, and binary variances. These bounds are the common inputs to the endpoint and scalar estimates.

At the actual ordered optimizer, every product is in $[\alpha_2, 1 - \epsilon_H]$, where

$$\epsilon_H = \frac{2 \log 2 - 1}{2H(\log H)^2} \geq \frac{1}{10H^3}.$$

Every original fine-leaf probability is at least H^{-H} ; the conservative transported reference-coefficient floor is H^{-H-1} . Indeed each route has at most $H - 2$ factors $\beta \geq 1/2$, followed by a factor at least $1/H$; the all-child final route is no smaller. Local coefficients $(r+z)^{\sigma-1}$ are at least $H^{-1/2}$. Fine probabilities, reference intensities, and original Poisson rates are at most H . Every split log coefficient has absolute value at most $H \log H$, including a routed original fine split.

For a canonical nondegenerate local edge, $a = \log(r+z)$, $b = \log(q+z)$, $p = (q+z)/(r+z)$. A proper child has $z \geq \sqrt{2}$ because $G_{b+1} \geq \alpha_2 \log 2$ and $\sigma \leq 1$. The terminal nondegenerate edge has $q \geq 1$. Hence

$$b \geq \frac{1}{2} \log 2, \quad a - b \geq \log \frac{H}{H-1} \geq H^{-1}, \quad H^{-1} \leq p \leq 1 - H^{-1},$$

$$m = a - pb \geq \frac{\log 2}{H}, \quad \frac{1}{100H^6} \leq \rho_e \leq 1 - \frac{1}{100H^6}. \quad (\text{H.2.2})$$

Here $\rho_e = m^2/(m^2 + b^2p(1-p))$; the upper denominator is at most $5(\log H)^2/4$. The natural local risk speed is the integer $r - q$, in $[1, H]$. On own-scale central counts its clock mean is in $[1/(2H), 2]$. The contracted speed $\theta c + (1 - \theta)A/P$ also lies in $[1/(4H), 2H \log H]$ for every $0 \leq \theta \leq 1$: the total jump rate A/P has this lower bound on central original counts. Thus making the reserve allocation small does not make the clock speed degenerate. The rare branch uses a threshold $t_0 = M_1^{-100}$. Its actual type and centered type remain in the compact bounds (2), with $100H^6$ replaced by M_1 , once the own-scale count and score bounds below hold. This follows from $\max(X, Y)/n \leq t_0/\min(X, Y)$ and the exact type increment. These bounds remain uniform when a strict product gap tends to zero.

H.2.2 Numerical strict reference charges and suffix thresholds

The next bounds control the charges used to absorb count and path errors. The terminal suffix also needs a definite separation between successive ordinary thresholds.

The hierarchy bound in Theorem 4.17 is

$$V_{\text{old}} \leq C_H T_{\text{local}}, \quad C_H = 2H^3(1 + \log H)^3 \leq 16H^6.$$

The explicit old transition constant in Section A.2.8 is

$$d_* = \min \left\{ \frac{\epsilon_H}{H^2}, H^{-H-1}[1 - (1 - H^{-1})^{\epsilon_H}] \right\} \geq \frac{1}{20H^{H+5}}.$$

Use $1 - e^{-v} \geq v/2$ for $0 \leq v \leq 1$. The local secant derivative is at least $1/(24H^2)$ and the alternative branch at least $\log(2 \log 2)/(H \log^2 H)$. Thus the clipped-reference gap and charge can be set to

$$d_0 \geq \frac{1}{100H^{H+6}}, \quad c_{\text{ref}} = 1 - 2^{-d_0/3} \geq \frac{1}{1000H^{H+6}}. \quad (\text{H.2.3})$$

Local and terminal rank curvature is at least H^{-H-2} . The bad-group charge is at least $H^{-H-1}/(2H^2)$, and the nontrivial mixed entropy gap is at least $2H^{-4H}$. Each bound exceeds M_0^{-1} . Reference integrands and their first two derivatives are bounded in absolute value by M_0 : their denominators involve at most H positive rates or probabilities bounded below by H^{-H} , or logarithms at least $(\log 2)/2$. There are at most $H!H^H 2^H$ physical endpoint histories, fewer than M_0 . These estimates also cover ties and zero first-root templates; they use no inverse of a strict product difference.

For the ordinary suffix one also needs an explicit separation of the thresholds α_h . Put $z = (\log h)/2$ and $R(z) = z \coth z - 1 - \log(\sinh z/z)$. Since $\sinh^2 u - u^2 \geq u^4/3$ and $\sinh u \leq ue^u$,

$$R'(u) \geq ue^{-2u}/3, \quad R(z) \geq z^2 e^{-2z}/8, \quad \alpha'_h \geq 1/(32h^2).$$

Integrating gives $\alpha_{h+1} - \alpha_h \geq 1/(32H^2)$. Consequently the suffix threshold in Section 3 can be retained with

$$\tau \geq 1/(128H^5),$$

and every positive tail drift separated from its weak window is at least $1/(400H^5)$. The first-plateau transport gap $1 - q_+$ is at least $\epsilon_H/(2H)$, so every effective-time comparison coefficient lies between M_0^{-1} and M_0 .

H.2.3 Explicit count-box estimates

We now quantify the count boxes from Section A.2.8 and the fresh-name displacement from Section A.2. The recurrence bounds the centrality constants before transport; a final estimate includes all transport displacements. Start the uncolored Poisson windows at width $10H\sqrt{\mu}$. On the pair-coarsened event give every coarse category an own-scale window of width $10H\sqrt{n}$. Chebyshev and a union bound over at most H^2 categories retain probability at least 9/10. Require original counts at least M_0^{100} so these windows also make every selected two-name bank total at least $H^{-H}T/(4H)$. These conditions concern unread splits only.

For a bound on successive centrality constants set

$$A = 10^6 H^{3H+30}, \quad B_0 = 10H, \quad B_{j+1} = A(B_j + 1), \quad 0 \leq j < H^2. \quad (\text{H.2.4})$$

The H^2 upper bound covers all proper split selections, their successive count adjustments, and the terminal-prefix completion. Every such operation uses at most H earlier groups; the hierarchy turns their summed square-root errors into at most $4H^4 B_j \sqrt{T}$. Its entropy coefficients are at most $H \log H$, its exit coefficient is at least $(\log 2)/2$, and its bank standard deviation is at least $H^{-(H+1)/2} \sqrt{T}/4$. Their product and quotient are bounded by the factor A in (4), including integer rounding and the width floor $R_j \leq \sqrt{T}$. A count transport changes at most H exterior categories, with displacement chosen below; its new own-scale centrality is covered by the same recurrence. Since $(2A)^{H^2+1} \leq M_0$, all these centrality constants can therefore be taken below M_0 .

Choose the fresh displaced box width $\epsilon_{\text{box}} = 1/(100H)$ and displacement $K_{\text{box}} = 100H^3(B_{H^2} + 1) \leq M_0^2$. When $N \geq M_0^{100}$, every nonshallow interval $N/r - K_{\text{box}}\sqrt{N} + [-\epsilon_{\text{box}}, \epsilon_{\text{box}}]\sqrt{N}$ is interior, contains at least $\epsilon_{\text{box}}\sqrt{N}$ integers, and its determined shallow count is also interior. Elementary Stirling at these central points gives its simultaneous mass at least $\exp(-M_0^{10})$. For smaller N , the all-shallow target mass is at least $H^{-M_0^{100}}$; it has wider central constant at most M_0^{51} . Such small original groups are absent in the fixed-lift source, whose original means and retained counts exceed M_0^{100} . The finite bounded-support convention therefore need not be used in (1).

These count transports preserve the preceding coarse words and the finer child word. The binary errors used in the next plateau selection therefore retain the pretransport bound B_{H^2} . Each of at most H transports changes a final fine-leaf count by at most $2HK_{\text{box}}\sqrt{N_g}$, where $N_g \leq 2HL_g$ at the fixed lift. Consequently

$$B_{\text{final}} \leq B_{H^2} + 4H^3 K_{\text{box}} \leq 500H^6(U_H + 1) < M_0.$$

The last inequality follows from $\log U_H \leq 100H^3 \log(10H)$ and $\log M_0 = (\log 2)2^{10000H^5}$. Multiplication by the finitely many split coefficients bounds the endpoint errors by $M_0^5 \sqrt{L_g}$. Since $c_{\text{ref}}^{-1} < M_0$, their Young remainders are at most M_0^{12} .

The full proper/terminal count source uses at most H^2 such steps. Its fixed normalizing density bound is at most $\exp(M_0^{12})$, its own-group count centrality at most M_0^{10} , and its remaining box mass at least $\exp(-M_0^{12})$ times the proper-band mass. The coefficient-one endpoint cancellation is exact. The square-root errors are at most $M_0^5 \sqrt{L}$; Young's inequality against (3) costs at most M_0^{12} . Thus intrinsic common-pair comparisons may use

$$M_0^{-20} S_v \leq K_v + h \leq M_0^{20} S_v, \quad S_v = S_v^I, \quad h \geq M_0^{20}. \quad (\text{H.2.5})$$

The terminal-prefix count box is retained; only unread suffix terminal names are filled at the later positive root. Its type increments use that actual root count. In particular the exceptional complete-deadline and all-zero rows acquire no nominal reference defect.

Here S_v^I is the shifted intrinsic score, with leading term h , as defined in Section A.4 (4). It is not the profile's leading-one score. Comparing the two costs only a final factor at most $1000H^2(1+h)$ per endpoint. The lower selects every canonical weak edge, including saturated ones, so all source weak walls are imposed. The upper retains only $T_e < \tau_0$, with the explicit h -independent τ_0 in Section A.4, (11). All omitted factors are bounded below by a fixed constant. Actual-type compactness is proved directly from weak drift and own-scale central counts in that section, (7)–(10), before h is chosen; no inverse of an h -dependent rarity threshold enters the scalar estimates.

H.2.4 Fixed proof parameters and their finite inequalities

With (2)–(5) fixed, choose the curve exponents, budget fractions, regularity slope, bank fraction, and depth cutoff as follows:

$$\begin{aligned} \gamma = 1/8, \quad \zeta = 1/16, \quad \theta_{\text{glob}} = 1/(4H^2), \quad \theta_{\text{intr}} = M_1^{-M_1}, \quad \epsilon_{\text{mix}} = M_1^{-M_1}, \\ \vartheta = M_1^{-M_1^3}, \quad b_0 = M_1^{-M_1^3}, \quad \rho_{\text{bank}} = M_1^{-10M_1}, \quad D_0 = M_1^{10}, \quad C_{\text{cut}} = M_1^{M_1^4}. \end{aligned} \quad (\text{H.2.6})$$

Actual banks are much larger than ρ_{bank} ; using a smaller bank merely fixes a still positive allowed fraction. The scalar floor power D_0 exceeds every compact plateau exponent and every finite history polynomial. C_{cut} exceeds its required quotient by the stretched-exponential penalty b_0 .

For named refinement slopes start below $\vartheta M_1^{-10H-20}$ and enlarge each successive level by at most M_1^{10} . All final original fine-name slopes remain below $\vartheta/4$; allow target slopes below $\vartheta/2$. There are at most H levels. Every Chernoff coefficient on these fixed original laws is then at least M_2^{-1} .

The primitive geometric coefficients before the weak cutoff, small proof parameters and regularity intercept are installed can be bounded by M_1 , with primitive positive fractions at least M_1^{-1} . The cutoff $\epsilon_w = M_1^{-100}$ is then included among the explicit small proof parameters; the primitive lower bound M_1^{-1} is applied before that cutoff is introduced. The finite number of reference, rank, projection, and carry operations is at most H^3 , each uses the explicit primitive rates and the charges (3), and its Young exponents are 2 or fixed powers at most 16. Even repeated raising to power 16 over H^3 algebra steps has logarithm below 2^{10000H^5} ; this is why M_0 uses the extra exponential. After (6), use the safer positive-fraction floor $M_1^{-M_1^2}$ and bound every finite error intercept that does not read the added reserve by $(M_1 K_{\text{REG}})^{M_1^{10}}$. An endpoint shift that reads the added reserve can instead cost $(M_1 K_{\text{REG}} h)^{M_1^{10}}$; the larger final intercept below pays this separately. These estimates bound the deterministic coefficients in the selected-edge and source-history inequalities.

The asymmetric ordinary continuation uses compact parameter M_1 and height times drift at least M_1^{-20} . The half-Poisson upper density is at most 25; at deviations $K\sqrt{n_i}$, its lower density is at least $(1/16)\exp(-4 - 8K^2)$ for $n_i \geq 16K^2$. With $\kappa = M_1^{-3}\exp(-M_1)$, choose $K = 4M_1^{12}\exp(M_1)$ and $R_{\text{asy}} = \exp(1000M_1^2)$. The first-half and late-half inequalities of Section B.2.9 then hold. The fixed asymmetry cutoff and the scaled drifts in the one-end Gaussian experiment are below M_2 .

Use the scalar estimates with this M_2 . Its proved OU constant is at most $2^{80}M_2^2$, its positive-box loss is at least $\exp(-M_2^{100})$, and its exact middle constants are the displayed formulas in

Section B.4. The relative bad construction for the binary construction is $C_{\text{bad}} = \exp(\exp(M_2^{300}))$, $L_0^{\text{bin}} = \lceil \exp(\exp(M_2^{400})) \rceil$. The ordinary one-channel estimates have $L_0^{\text{ord}} = \lceil \exp(\exp(M_2^{2300})) \rceil$, $h_{\text{ord}} = \lceil \exp(\exp(M_2^{3000})) \rceil$, $K_{\text{ord}} = \lceil \exp(\exp(M_2^{3500})) \rceil$, and $C_{\text{ord}} = \exp(\exp(M_2^{4000}))$; see Section B.7. These constants do not use the binary theorem at $\rho = 1$. Define

$$\begin{aligned} K_{\text{REG}} &= \exp\{\exp(M_2^{5000})\}, & h_W &= \exp\{\exp(M_2^{5500})\}, \\ h &= \exp\{\exp(M_2^{6000})\}, & C_{\text{geom}}^{\text{final}} &= \exp\{\exp(M_2^{6500})\}, \\ R_0 &= 100H^2. \end{aligned} \tag{H.2.7}$$

These choices make the Chernoff and refinement tails smaller than the preceding relative retention losses: their first exponent exceeds K_{REG}/M_2 , whereas the logarithm of every preceding inverse retention is at most $\exp(M_2^{4500})$, including the ordinary $\log C_{\text{ord}} = \exp(M_2^{4000})$. Fixed grid-pair series have powers at most M_1^{1000} ; their integral bounds are also swallowed by that exponent. This fixes REG before lengths are inspected.

Take the strong curve amplitude to be one. Its history errors are paid by the explicit inequality

$$C_K d^{1/16} \leq \frac{1}{2} g_0 d^{1/8} + \frac{C_K^2}{2g_0},$$

where $g_0 \geq M_1^{-M_1^2}$ and $C_K \leq (M_1 K_{\text{REG}})^{M_1^{10}}$. The last constant is below h_W : its logarithm is at most $4M_1^{10} \exp(M_2^{5000}) < \exp(M_2^{5500})$. This also pays the finite repeated Young absorptions. The strong-wall amplitude can therefore be fixed before the regularity intercept is enlarged. The minimum reserve h dominates h_W , all OU endpoint boxes divided by their allocation fractions, $100M_2 \max(L_0^{\text{bin}}, L_0^{\text{ord}})^2$, and every strong-wall/relative-coupling initial reserve.

H.2.5 Arithmetic, grid, and the final length inequalities

The last choice is the prime grid. Its total clock defect is bounded in terms of the parameters already fixed. The final length then makes this defect and every remaining intercept small relative to the available count and reference margins.

Set $A_0 = A_1 = hM_1^{100000}$ and use the exact numerical Q from Section E.4, equation (2). This dominates every actual named regularity linear occupancy tube and gives pointwise squarefree supply at least $e^{-1/2}$. Its elementary bound is $Q \leq h^4 M_1^{300000}$. Consequently

$$R = k(\log \log Q + 100) \leq M_2^{7000}. \tag{H.2.8}$$

At the value of Λ_k in (1), the following inequalities hold:

$$\Lambda_k > 100M_1^{100000} h^2, \quad \Lambda_k > 100M_1^{100000} R_0^2, \quad \Lambda_k > 100M_1^{100000} (1+R)^2, \quad \Lambda_k > M_0^{1000},$$

$$M_1 \{\log(2+T)\}^{M_1^{10}} < M_1^{-M_1^3} T^{1/16} \quad (T \geq \Lambda_k),$$

$$[C_{\text{cut}} \log(2+n)]^8 < \rho_{\text{bank}} n/100 \quad (n \geq \Lambda_k/4H),$$

$$\max(M_1 R, C_{\text{geom}}^{\text{final}}) < M_1^{-M_1^2} \sqrt{\Lambda_k}/100. \tag{H.2.9}$$

For the logarithm inequalities, the relevant functions $\log(2+t)^a/t^b$ are decreasing once $\log(2+t) > 2a/b$; substitute $t = \Lambda_k/(4H)$ and use the strict power gap $10000 > 7000 > 6500 > 6000 > 5500 > 5000 > 400$. This verifies them for every later length, without endpoint enumeration. The first line of (9) makes every actual width band interior and pays the central root/grid count translates. The last line pays every nonexceptional square-root reference charge; the two exceptional rows are exact. The middle lines pay all logarithmic recoloring/image-REG margins and keep every active bank cutoff inside its allowed local block. Affine grid maps lose at most R by (8), so all original groups remain present.

H.3 Harmonic stability and equivalent profiles

Two points complete the dependence check. The ordered-sum estimate changes only the final upper comparison constant. Harmonic stability then compares the given vector with its lift, including zero lengths.

H.3.1 The ordered-sum constant enters only the final upper bound

Let

$$C_H^{\text{os}} = \max_{2 \leq h \leq H} C_{h^{1/(h-1)}}.$$

This is finite by Corollary 11.4. Its value enters only the multiplicative upper constant. To see this directly, fix the geometric constant C_0 before applying the ordered-sum bound and put

$$C_1 = 1 + \log C_0 + (H-1) \log 2 + \log H, \quad C_{\text{tail}} = 1 + \frac{2C_H^{\text{os}} C_1}{(\log 2)^2}.$$

In the notation of Section 11,

$$\begin{aligned} \gamma + 1 &= \frac{Z + \ell + \log C_0 + (h-1) \log 2 + \log h}{\log h} \leq \frac{C_1 + Z + \ell}{\log 2}, \\ \gamma - n + v + 1 &\leq \frac{C_1(1 + \ell)}{\log 2}. \end{aligned}$$

On a nonempty event with $n = 0$ one has $C_1 + Z + \ell > 1$. Thus the all-count conclusion, including $n = 0$, is

$$\mathbb{P}(W_n > e^{-\ell} \mid n) \leq \min \left\{ 1, \frac{C_{\text{tail}}(1 + \ell)(C_1 + Z + \ell)_+}{1 + n} \right\}.$$

The constant C_H^{os} occurs only in C_{tail} . It changes neither the event nor its root translate, count reservoir, geometric prefix cap, plateau powers, or all-count hypotheses. The two Poisson mode bounds in Section F.1 multiply this estimate by fixed factors and retain the same-count minimum. Subsequent layer integrals have already fixed $1 - s$, the root span, and the pool exponent. Consequently enlarging C_{tail} only enlarges the output upper constant.

The pool gate in Section 12 is chosen earlier from past inverse floors and physical linear information. It does not use the ordered-sum estimate. The root estimate concludes the backward upper tower, so its multiplicative constant does not feed back into a gate, acceptance probability, bank length, or scalar cutoff. The positive-core upper holds at every vector with $\min_i L_i \geq 1$. The lower at the explicit lift uses the direct one-channel theorem in Section B.7; its reserve and cutoff, and all collision, history, count, and grid choices, are independent of C_H^{os} . This establishes the claimed separation between the explicit profile and the final comparison constants.

H.3.2 Harmonic stability, including zero lengths

For $L_i \geq 0$, write $\ell_i = L_i + \log 3$ and, for any $\lambda \geq 1$,

$$L_i^\lambda = \max(L_i, \lambda), \quad \ell_i^\lambda = L_i^\lambda + \log 3, \quad \delta_i = L_i^\lambda - L_i \in [0, \lambda].$$

The unconditional harmonic stability proved in Section E.3 gives

$$e^{-A} K_k(\ell) \leq K_k(\ell^\lambda) \leq e^A K_k(\ell), \tag{H.3.1}$$

$$A = \sum_{i=1}^k h_i |\delta_i| + H \left(\sum_{i=1}^k \left| \sum_{j \leq i} \delta_j \right| + 2kC_M \right), \quad h_i = k - i + 2.$$

The harmonic-prime discrepancy estimate permits $C_M = 100$. Since the increments are nonnegative,

$$\sum_{i=1}^k \left| \sum_{j \leq i} \delta_j \right| = \sum_{j=1}^k (k - j + 1) \delta_j \leq \frac{k(k+1)}{2} \lambda,$$

$$\sum_i h_i \delta_i \leq \frac{k(k+3)}{2} \lambda.$$

Therefore

$$A \leq B_k \lambda + C_0(k), \quad B_k = \frac{k(k^2 + 3k + 4)}{2}, \quad C_0(k) = 200k(k+1). \tag{H.3.2}$$

No positive separation of adjacent cutoffs is assumed. Removing the primes whose block memberships change leaves one common unnormalized sum, positive because the empty tuple contributes $(\log 2)^k$. Insertion bounds compare both full sums with that same sum, so the exponent is not doubled. Equal cutoffs, empty prime blocks, and $L_1 = \dots = L_k = 0$ are included.

At $\lambda = \Lambda_k$, the positive-core upper and the explicit-lift lower give

$$c_k \Psi_k(L^{\Lambda_k}) \leq K_k(\ell^{\Lambda_k}) \leq C_k \Psi_k(L^{\Lambda_k}).$$

The lower threshold inequalities hold at Λ_k and at every larger original block length; all plateaux and contexts are computed from the same lifted vector. Combining these bounds with (H.3.1)–(H.3.2) proves

$$K_k(\ell) \asymp_k \Psi_k(L^{\Lambda_k}) = \Phi_k(\ell). \tag{H.3.3}$$

The factor $\exp\{B_k \Lambda_k + C_0(k)\}$ depends only on k .

To return to the divisor count, apply the arithmetic reduction at the original cutoffs and its original domain $x \geq 2^k(y_1 \dots y_k)y_k$. Substituting (H.3.3) into that reduction proves the dyadic theorem without imposing a condition on fictitious lifted cutoffs. The same original-vector harmonic formula enters the multiplication-table transfer.

I Evaluation of the ordered saddle

The profile requires the ordered saddle at every length vector, including vectors on which plateaux merge. This appendix evaluates that saddle by eliminating coordinates one at a time. Each elimination is a monotone scalar equation with endpoint clipping. The gradient formula retains the contribution of a moving constraint, so the same recursion applies on every equality face.

We first define the scalar operation and the elementary value and gradient data. We then state the recursion. A profiling lemma proves its continuity and monotonicity, and the final reconstruction shows that it returns the ordered maximum.

I.1 The scalar operation

For a continuous strictly decreasing function D on $[a, b]$, define the clipped scalar inverse

$$\mathfrak{R}(a, b; D) = \begin{cases} a, & D(a) \leq 0, \\ b, & D(b) \geq 0, \\ \text{the unique } z \in (a, b) \text{ with } D(z) = 0, & D(a) > 0 > D(b). \end{cases} \quad (\text{I.1})$$

For $a = b$, the interval has only one possible value. On a nondegenerate interval, the endpoint tests select a boundary value, or the intermediate value theorem gives the unique interior root. Bisection evaluates the interior branch to any specified accuracy. This is the scalar operation used at every stage of the recursion.

The recursion below gives a continuous strictly decreasing function D at each step, so the clipped inverse is uniquely defined.

I.2 Initial elementary data

The recursion needs both the objective and its gradient. We compute them on a full positive box so that profiling remains valid before the remaining coordinates are ordered.

Fix $k \geq 1$, positive lengths L_i , and $V_j = \sum_{i \leq j} L_i$. Set $a = 1/4$, $b = 1$. On the whole box $[a, b]^k$, not merely its ordered part, compute

$$G_k = s_k \log 2, \quad G_i = s_i \log(1 + e^{G_{i+1}/s_i}), \quad Z_i = e^{G_i},$$

and

$$J(s) = \sum_i L_i + \sum_j s_j V_j - \sum_i L_i Z_i. \quad (\text{I.2})$$

Its gradient has a direct scalar recursion. For $i < k$ put

$$\beta_i = \frac{e^{G_{i+1}/s_i}}{1 + e^{G_{i+1}/s_i}}, \quad A_i = \log(1 + e^{G_{i+1}/s_i}) - \frac{G_{i+1}}{s_i} \beta_i, \quad A_k = \log 2.$$

Then compute

$$T_1 = L_1 Z_1, \quad T_j = L_j Z_j + \beta_{j-1} T_{j-1}, \quad g_{0,j} = V_j - A_j T_j. \quad (\text{I.3})$$

Set $f_0 = J$, $g_0 = (g_{0,1}, \dots, g_{0,k})$. Equations (2)–(3) are the initial value/gradient evaluation. They contain $O(k)$ arithmetic steps and no optimization.

The strict concavity needed below follows on the full positive box. The perspective $s \log(1 + \exp(u/s))$ is convex and increases in u . Induction makes every G_i convex. In the Hessian of $\sum_i L_i \exp(G_i)$, the contribution

$$\sum_i L_i e^{G_i} \nabla G_i \nabla G_i^T \quad (\text{I.4})$$

is positive definite: the Jacobian of the G_i is upper triangular with diagonal $A_i > 0$. The other Hessian contribution is positive semidefinite. Thus J is smooth and strictly concave on this whole box. Direct differentiation gives $\partial_j G_i = A_j \prod_{i \leq q < j} \beta_q$ for $i \leq j$, which proves (3).

Theorem 4.1 implies that the original ordered saddle satisfies $s_k > 1/2$ and $s_1 < 1$. Therefore its maximum on the original ordered simplex is unchanged if restricted to the ordered box $1 \geq$

$s_1 \geq \dots \geq s_k \geq 1/4$. This justifies the fixed positive lower endpoint and places every evaluation away from the zero-product perspective singularities.

I.3 The finite recursive rule

At each stage we maximize the first remaining coordinate over the interval imposed by its successor. We record the maximizing value and the gradient of the resulting profiled objective. The correction for the moving lower endpoint is what allows plateau equalities to be handled by the recursion itself.

For $j = 1, \dots, k-1$, assume f_{j-1} and its gradient g_{j-1} are available as scalar-inverse expressions in $k-j+1$ variables on the full box $[a, b]^{k-j+1}$. For a tail $t = (t_1, \dots, t_{k-j})$ define

$$D_j(z; t) = g_{j-1,1}(z, t), \quad z_j(t) = \Re(t_1, b; D_j(\cdot; t)). \quad (\text{I.5})$$

Evaluate once more at that root, and set

$$f_j(t) = f_{j-1}(z_j(t), t),$$

$$g_{j,r}(t) = g_{j-1,r+1}(z_j(t), t) + \mathbf{1}_{r=1} \min\{0, g_{j-1,1}(z_j(t), t)\}, \quad 1 \leq r \leq k-j. \quad (\text{I.6})$$

The $\min(0, \dots)$ term is the contribution of the moving lower endpoint. It is present when the root clips there with negative derivative, and vanishes at an interior root or a fixed upper maximum. For $t_1 = b$, formula (6) gives the continuous left derivative of the profiled value, with the same sign rule.

The last scalar is

$$s_k^* = \Re(a, b; z \mapsto g_{k-1,1}(z)). \quad (\text{I.7})$$

Recover the other coordinates in reverse order:

$$s_j^* = z_j(s_{j+1}^*, \dots, s_k^*), \quad j = k-1, \dots, 1. \quad (\text{I.8})$$

For $k = 1$, (7) uses g_0 directly and there is no profiling step. It returns the usual scalar saddle $s_1 = \log(1/\log 2)/\log 2$.

For every k , the maximal runs of equalities in (8) give the plateaux. A lower clipping root can produce a plateau merger. The same branch handles a zero derivative at that endpoint, so the recursion specifies the exact equality face without an additional tolerance convention.

I.4 Monotonicity of the scalar equations

The recursion is valid because profiling preserves strict concavity and a continuous gradient. The following lemma also proves that the moving-boundary correction remains continuous at clipping transitions.

Lemma I.1 (Scalar profiling). *Let $f(z, u_1, \dots, u_d)$ be continuously differentiable and strictly concave on $[a, b]^{d+1}$, with continuous one-sided derivatives on its boundary. For each u , define*

$$z(u) = \Re(u_1, b; \partial_z f(\cdot, u)), \quad q(u) = f(z(u), u).$$

Then q is strictly concave and continuously differentiable on the full remaining box, with

$$\partial_{u_r} q = \partial_{u_r} f(z(u), u) + \mathbf{1}_{r=1} \min\{0, \partial_z f(z(u), u)\}. \quad (\text{I.9})$$

The root $z(u)$ is continuous. Each derivative $\partial_z f(z, u)$ is strictly decreasing in z on a nondegenerate interval, so all its scalar inverse branches are unique.

Proof. A continuously differentiable strictly concave scalar function has a strictly decreasing derivative. Otherwise two equal derivative values and concavity force a linear interval, contrary to strict concavity. Hence (1) returns its unique maximum on $[u_1, b]$. Compactness, continuity of f , and the continuous endpoints show that a limit of root values maximizes the limit interval; uniqueness proves continuity of $z(u)$.

For distinct u, v , let their two maximizing scalar values be $z(u), z(v)$. At $w = \theta u + (1 - \theta)v$, the convex combination $\theta z(u) + (1 - \theta)z(v)$ is feasible, because it is at least w_1 and at most b . The full input vectors are distinct even when their scalar roots coincide. Strict concavity of f gives

$$q(w) \geq f(\theta z(u) + (1 - \theta)z(v), w) > \theta q(u) + (1 - \theta)q(v).$$

Thus q is strictly concave; the remaining coordinates need not be ordered for this argument.

For a derivative in u_r , $r > 1$, the feasible interval is fixed; the elementary envelope inequalities compare the old and new maximizing values and give $\partial_{u_r} q = \partial_{u_r} f$ at the unique root. For u_1 , there are three strict cases. At a lower root with $\partial_z f < 0$, it remains the lower root locally, so the chain rule adds $\partial_z f$. At an upper root with $\partial_z f > 0$, that endpoint is fixed and adds nothing. At an interior root, $\partial_z f = 0$, and the envelope inequalities again add nothing.

At a clipping transition the derivative is zero, so these formulas agree. More explicitly, comparing an optimal point with its projection into a neighboring interval moves it by at most $|du_1|$. If the limit optimum is a boundary point with zero derivative, the resulting value error is $o(|du_1|)$ by uniform continuity of the first derivative. Ordinary parameter changes are expanded at the convergent optimizing point and give the displayed derivative. This also proves the formula without assuming the root itself differentiable. At $u_1 = b$, the same projection comparison from the left yields the extra derivative when $\partial_z f < 0$, and yields zero extra derivative when it is positive; this is precisely the min in (9). All formulas are continuous since the root and the derivatives are continuous. This proves the lemma. $\square$

Apply the lemma successively with $f = f_{j-1}$. It proves that g_j in (6) is its actual gradient and that f_j is C^1 and strictly concave on the entire remaining box. In particular D_j is continuous and strictly decreasing on every scalar interval in (5). Thus each scalar inverse used in the recursion has a unique branch.

I.5 Exact reconstruction of the ordered maximum

The recursion satisfies

$$f_j(s_{j+1}, \dots, s_k) = \max_{b \geq s_1 \geq \dots \geq s_j \geq s_{j+1}} J(s_1, \dots, s_k). \quad (\text{I.10})$$

Induction proves this: f_{j-1} has already eliminated the first $j - 1$ coordinates; maximizing the next scalar over $[s_{j+1}, b]$ is exactly the unique root in (5). The value and gradient recursion (3), (5)–(7) therefore evaluates the ordered maximum using scalar inverses.

The last root (7), followed by the reverse reconstruction (8), returns the ordered maximum. Strict concavity of J gives its uniqueness. The recursion covers all 2^{k-1} possible plateau patterns

and their equality boundaries. The lower endpoint $1/4$ leaves the maximum unchanged by the saddle compactness theorem.

I.6 The finite evaluator and the length threshold

The depth of the scalar-inverse recursion is k . The initial elementary computations have fixed dimension k . The stage definitions are a finite system of k explicit functions, each using one new scalar root and the previously constructed value/gradient functions. Expanding the definitions duplicates at most a number depending on k of earlier expressions per stage. For example the crude bound $[100(k+1)^3]^{k+1}$ bounds their expression-node count after full expansion, treating each independently defined scalar inverse as one operation. This bound is independent of all lengths. The number of scalar inverse functions depends only on k ; the numerical accuracy of each inverse can be increased by bisection.

AI disclosure

The author selected the problem and managed the project; the mathematical derivations were carried out by artificial intelligence.

References

- [1] A. D. Barbour, N. Ross and G. Zheng, *Stein's method, smoothing and functional approximation*, Electron. J. Probab. **29** (2024), Paper No. 20, 1–29. doi:10.1214/24-EJP1081.
- [2] H. J. Brascamp and E. H. Lieb, *On extensions of the Brunn–Minkowski and Prékopa–Leindler theorems, including inequalities for log concave functions, and with an application to the diffusion equation*, J. Funct. Anal. **22** (1976), no. 4, 366–389. doi:10.1016/0022-1236(76)90004-5.
- [3] R. H. Cameron and W. T. Martin, *Transformations of Wiener integrals under translations*, Ann. of Math. (2) **45** (1944), no. 2, 386–396. doi:10.2307/1969276.
- [4] E. A. Carlen, D. Cordero-Erausquin and E. H. Lieb, *Asymmetric covariance estimates of Brascamp–Lieb type and related inequalities for log-concave measures*, Ann. Inst. H. Poincaré Probab. Statist. **49** (2013), no. 1, 1–12. doi:10.1214/11-AIHP462. Author version: arXiv:1106.0709v2.
- [5] N. Castelle and F. Laurent-Bonvalot, *Strong approximations of bivariate uniform empirical processes*, Ann. Inst. H. Poincaré Probab. Statist. **34** (1998), no. 4, 425–480. https://www.numdam.org/item/AIHPB_1998__34_4_425_0/.
- [6] W.-K. Chen, N. Dafnis and G. Paouris, *Improved Hölder and reverse Hölder inequalities for Gaussian random vectors*, Adv. Math. **280** (2015), 643–689. doi:10.1016/j.aim.2014.09.029.
- [7] L. Dümbgen and A. Mösching, *On stochastic orders and total positivity*, ESAIM Probab. Statist. **27** (2023), 461–481. doi:10.1051/ps/2023005. Author version: arXiv:2209.07868v3.
- [8] S. Eberhard, K. Ford and B. Green, *Permutations fixing a k -set*, Int. Math. Res. Not. IMRN **2016** (2016), no. 21, 6713–6731. doi:10.1093/imrn/rnv371.

- [9] P. Erdős, *Some remarks on number theory*, Riveon Lematematika **9** (1955), 45–48 (Hebrew, with an English summary). https://www.renyi.hu/~p_erdos/1955-13.pdf.
- [10] P. Erdős, *An asymptotic inequality in the theory of numbers*, Vestnik Leningrad. Univ. **15** (1960), no. 13, 41–49 (Russian).
- [11] S. Fallat, S. Lauritzen, K. Sadeghi, C. Uhler, N. Wermuth and P. Zwiernik, *Total positivity in Markov structures*, Ann. Statist. **45** (2017), no. 3, 1152–1184. doi:10.1214/16-AOS1478.
- [12] K. Ford, *The distribution of integers with a divisor in a given interval*, Ann. of Math. (2) **168** (2008), no. 2, 367–433. doi:10.4007/annals.2008.168.367.
- [13] K. Ford, *Integers with a divisor in $(y, 2y]$* , in *Anatomy of Integers* (J.-M. De Koninck, A. Granville and F. Luca, eds.), CRM Proc. Lecture Notes, vol. 46, Amer. Math. Soc., Providence, RI, 2008, pp. 65–80. doi:10.1090/crmp/046/04. Original preprint: arXiv:math/0607473v1. Revised author version: <https://ford126.web.illinois.edu/wwwpapers/hxy2y.pdf>.
- [14] K. Ford, *Sharp probability estimates for generalized Smirnov statistics*, Monatsh. Math. **153** (2008), no. 3, 205–216. doi:10.1007/s00605-007-0516-y.
- [15] K. Ford, B. Green and D. Koukoulopoulos, *Equal sums in random sets and the concentration of divisors*, Invent. Math. **232** (2023), no. 3, 1027–1160. doi:10.1007/s00222-022-01177-y.
- [16] B. Green and M. Sawhney, *The proportion of permutations fixing a k -set*, preprint (2026), arXiv:2604.28116v1, 30 April 2026.
- [17] D. Koukoulopoulos, *Localized factorizations of integers*, Proc. London Math. Soc. (3) **101** (2010), no. 2, 392–426. doi:10.1112/plms/pdp056. Author version: <https://dms.umontreal.ca/~koukoulo/documents/publications/locfact.pdf>.
- [18] D. Koukoulopoulos, *Generalized and restricted multiplication tables of integers*, Ph.D. thesis, University of Illinois at Urbana–Champaign, 2010. <https://dms.umontreal.ca/~koukoulo/documents/publications/phdthesis.pdf>.
- [19] D. Koukoulopoulos, *On the number of integers in a generalized multiplication table*, J. Reine Angew. Math. **689** (2014), 33–99. doi:10.1515/crelle-2012-0064. Author version: arXiv:1102.3236v3.
- [20] B. Mallein and P. Miłoś, *Brownian motion and random walk above quenched random wall*, Ann. Inst. H. Poincaré Probab. Statist. **54** (2018), no. 4, 1877–1916. doi:10.1214/17-AIHP859.
- [21] L. D. Pitt, *Positively correlated normal variables are associated*, Ann. Probab. **10** (1982), no. 2, 496–499. doi:10.1214/aop/1176993872.
- [22] A. Prékopa, *On logarithmic concave measures and functions*, Acta Sci. Math. (Szeged) **34** (1973), 335–343. <https://acta.bibl.u-szeged.hu/14411/>.
- [23] J. B. Rosser and L. Schoenfeld, *Approximate formulas for some functions of prime numbers*, Illinois J. Math. **6** (1962), no. 1, 64–94. doi:10.1215/ijm/1255631807.

- [24] C. Sabuncu and C. Táfula, *The multiplication table problem in large dimensions*, preprint (2026), arXiv:2608.16163v1, 17 August 2026.
- [25] M. Sawhney (joint work with B. Green), $.086\dots$ and $1 + 2 \times 10^{-7}$ (*The Multiplication Table Problem*), in *Analytic Number Theory*, Oberwolfach Rep. **22** (2025), no. 4, 2725–2727. doi:10.4171/OWR/2025/51 (the full workshop report; published online 4 March 2026).
- [26] G. Tenenbaum, *Sur la probabilité qu'un entier possède un diviseur dans un intervalle donné*, Compositio Math. **51** (1984), no. 2, 243–263.
https://www.numdam.org/item/CM_1984__51_2_243_0/.